

What Follows from What:

Learning Inference Rules from Imitation, Coherence, and the World

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Abstract

Can a learner that acquires inference rules come to reason soundly, and with what justification? We consider a learner that imitates human inferences, is then conditioned on coherence (no good arguments for both P and $\neg P$), and receives occasional feedback from the world. We answer this precisely, for formal mathematics, informal mathematics, and the checking of physics solutions.

The answer is a division of labour among the learning signals, each with provable strengths and a provable residue.

- **Search forces worst-case soundness.** Accuracy on the distribution of human steps says nothing about a reasoner that searches: one tonk-like rule of arbitrarily small probability mass makes everything derivable. Cautious acceptance over schematic hypotheses is sound against every adaptive prover. Its price is a combinatorial dimension of the hypothesis class.
- **Imitation identifies rules.** Positive examples identify a schematic calculus exactly, at coupon-collector rates, and this generalizes to derivations of any size. But systematic human errors are indistinguishable from rules.
- **Coherence is eliminative negative data.** Its power is exactly Post-completeness. It pins down classical logic and complete theories, is blind for intuitionistic logic, and is Popperian for arithmetic. Carnap’s categoricity problem turns out to be Gold’s problem in the dual space of valuations.
- **Formal mathematics.** We prove an end-to-end theorem for a two-tier learner (audit, then assert; bold conjectures in a sandbox), and show that each ingredient is necessary.
- **Informal mathematics.** We model it as a latent formalization. A bounded-gap verifier is sound against adaptive provers and identifies validity exactly up to informal-validity equivalence. Counterexample objects are exponentially more informative than paradoxes.
- **Physics.** Contexts are given a filter semantics in which eternalism is provably impossible. Exports from idealized contexts need certificates of finite information radius. A structured-solution checker is sound against adversarial solvers relative to a judgment layer that we prove ineliminable.

We also settle two questions raised in the notes that motivated this work: a steeper simplicity penalty as a route from imitation to norms, and a “philosophical completeness theorem”.

Every main result passed adversarial verification. Core discrete results are machine-checked in Lean 4, and experiments illustrate the phenomena. We argue that principled justification beyond proof has a definite shape: proof within contexts, certified bridges between them, and

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an explicit, minimized, ineliminable judgment layer, with learning signals whose one-sided limits are known.

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1 Introduction

1.1 The question

Consider a learning process that learns *which inferences are valid*, and thereby learns to make valid inferences. The motivating design has six parts.

- It starts by imitating human inferences. These are the steps of formal or informal mathematical proofs, or of physics olympiad solutions.
- It treats an argument as valid when it is a sequence of valid inferences.
- It conditions the learner on *coherence*: it should not have good arguments for both P and $\neg P$.
- It also conditions the learner on practical success. Occasionally the world reports a truth value.
- On an inferentialist picture of meaning (Brandt, 1994), learning which inferences are valid is roughly learning what the words mean.
- Arguments happen in *contexts*. A physics problem says “assume air pressure is 0”, and from background knowledge one can prove that it is not. Arguments for P and for $\neg P$ in such contexts should not be penalized, yet some coherence constraint is clearly needed.

Does anything of this shape work, even with unlimited compute? Does it give rise to a reasoner that derives many correct conclusions? Can one *prove* that it does? The bar rises in three steps:

- a setup that provably works for formal mathematics;
- one that would have worked for mathematics *before* we knew how to formalize proofs;
- a principled system for checking physics olympiad solutions.

Behind all of these sits a broader question: what does principled justification look like beyond mathematical proof?

This report answers these questions as precisely as we could. The main results are summarized below and collected in Table 1. Each is stated as a theorem in the body, with a full proof in the appendix. Most theorems are paired with an impossibility result showing that its hypotheses cannot be dropped.

1.2 The answers in brief

(1) Search makes soundness a worst-case property (Section 3).

- A reasoner searches for derivations whose steps its learned checker accepts. It is sound iff every accepted step is derivable (Lemma 3.1). One accepted bad step is fatal.
- For every distribution over steps and every $\varepsilon > 0$, a single tonk-like schema with mass $< \varepsilon$ makes every formula derivable (Theorem 3.2). Every PAC learner can be modified to stay PAC while producing outputs that trivialize the reasoner (Corollary 3.3).
- Majority-vote ensembles of individually coherent verifiers can derive $\perp$ forever without any member being refuted (Theorem 3.12).

- Accuracy on human steps, which is what step-level reward models optimize, is therefore the wrong target. An experiment shows a step classifier with in-distribution AUC 0.995 that a search procedure exploits to prove most of a set of false equations (Section 12).
- (2) **Cautious acceptance is the canonical sound verifier, and structure makes it affordable** (Section 4).
- Accepting exactly what every surviving hypothesis accepts is sound against every adaptive prover. It is also the largest sound policy (Theorem 4.2).
 - Its worst-case cost in human escalations is exactly a positive-elasticity dimension of the class (Theorem 4.4).
 - For rule schemas learned by anti-unification this cost is linear in step size. Unstructured classes cost exponentially (Table 4).
 - A Bayesian version is time-uniformly sound via Ville’s inequality (Theorem 4.15).
- (3) **Imitation identifies rules but not norms** (Section 5).
- Positive examples identify a schematic calculus exactly, at coupon-collector rates (Theorem 5.7).
 - Exact identification generalizes to derivations of any size (Corollary 5.11). This is a precise sense in which finite usage fixes inferential role for unbounded use.
 - Systematic human errors are indistinguishable from rules at every rate (Corollary 5.20).
- (4) **Coherence is eliminative negative data, and its power is exactly Post-completeness** (Section 6).
- A derivation of $\perp$ from a context certified consistent is a *negative bag*: some step in it is invalid. Only bold learners ever receive this signal, with at most $\log_2(1/w^*)$ detections (Theorem 6.5).
 - Every structural extension of classical consequence is classical or trivial. So one coherence datum pins down classical logic with no simplicity bias (Theorems 6.13 and 6.15, machine-checked).
 - Coherence is blind among intermediate logics (Theorem 6.17). In arithmetic it is Popperian, detecting Π_1 truth but not Σ_1 truth, and runs out at Σ_2 (Theorem 6.22).
 - Carnap’s categoricity problem is Gold’s problem in the dual space of valuations. Single-conclusion data cannot fix the classical meanings, while finitely many bilateral data plus one coherence datum can (Theorems 6.27 and 6.29).
 - What survives everything is the set of coherent uniform alternatives: a precise “Kripkensteinian residue” (Theorem 6.38).
- (5) **An end-to-end theorem for formal mathematics** (Section 7).
- The naive two-tier design (cautious assertion plus a bold red team) is provably unsound, because practice contains fallacies (Proposition 7.5).
 - The corrected learner audits, then asserts. It assigns blame through the union of minimal conflicts, which is Reiter’s duality (Lemma 7.8), and states soundness relative to a refutation depth.

- It is sound against every adaptive prover and red team, and it converges to the target modulo an explicit residue (Theorem 7.14).
- Each ingredient is necessary (Table 9).
- For classical propositional logic and for complete decidable theories the residue is empty.

(6) Informal mathematics before formalization (Section 10).

- We model informal practice as the shadow of an unknown latent formalization. A bounded-gap conservative verifier is sound against adaptive provers. It must check links between contexts as well as steps, since Cauchy’s sum theorem is an equivocation (Proposition 10.7).
- Validity is identified exactly up to informal-validity equivalence, and no finer. Ontology is not identifiable: standard and nonstandard reals agree on the practice (Theorem 10.15 and Proposition 10.17).
- Counterexample objects need at most $\log_2 |\mathcal{H}|$ corrections, whereas paradoxes can need $|\mathcal{H}| - 1$; the sorites is the tight case (Theorems 10.22 and 10.23).
- The history Frege $\rightarrow$ Russell $\rightarrow$ Zermelo comes out as constrained repair selected by coverage of practice (Theorem 10.32).
- A robust-core theorem explains why informal mathematics survived formalization (Theorem 10.35).

(7) Contexts and the principled checking of physics (Section 11).

- Contexts denote filters of models from a declared deformation family. Suppositions are exact discharge. Idealizations may contradict the background.
- Eternalism is provably impossible (Theorem 11.8). Coherence belongs on the root and the anchored contexts only (Theorem 11.12).
- Designating “idealization plus full background” as coherent forces a globally paraconsistent logic (Proposition 11.10). This is a formal version of the air-pressure worry.
- Exporting a conclusion from an idealization needs information of finite radius about the de-idealized family: there is no free export (Theorem 11.21). Context choice is an attack surface (Theorem 11.27).
- A Structured Physics Solution checker is sound against adversarial solvers relative to a judgment layer (Theorem 11.37). That layer, consisting of the reading of the problem and the adequacy of the top model, is ineliminable (Theorem 11.38).
- “True in the context at hand” becomes supertruth over admissible completions (Theorem 11.39).
- The approach is worked through a EuPhO problem end to end (Section 11.7).

(8) Two questions from the motivating notes.

- A *steeper simplicity penalty* on function induction with private randomness does produce the hoped-for kink at the simple model (Theorem 8.6). But selection is a

rate threshold. Rare valid rules are lost and cheap fallacies kept, and the threshold cannot be calibrated from imitation data (Section 8).

- The “*philosophical completeness theorem*” (“a mode of talking is coherent iff there is something it could be talking about”) holds in a generalized weak sense, for positions, credences and context systems. It fails in the intended sense. Bounded-arity coherence constraints of the kind used in consistency-based probing do not characterize coherence (Section 9).

Three conclusions sit on top of these results (Section 13):

- every learning signal is *one-sided*;
- principled justification is a *set* of signals that jointly cover every direction of error, together with a declared residue;
- mathematical and physical justification differ in degree, measured by the size of an explicit judgment layer, not in kind.

Success criterion	Status
A setup that provably works for formal mathematics	Yes, with explicit scope. The two-tier learner (Theorem 7.14) is sound and exactly convergent for classical propositional logic and complete decidable theories (Corollaries 7.25 and 7.28). It is Popperian for arithmetic, with sharp impossibility results beyond Σ_1 (Theorem 7.29).
A setup that would have worked before formalization	Yes, relative to realizability. A bounded-gap verifier over latent formalizations is sound and identifies validity exactly up to informal-validity equivalence (Section 10). Open: affordable soundness when the learner must invent the formal language.
A principled system for checking physics olympiad solutions	Yes, as a checker with a declared residue. It is sound against adversarial solvers relative to the reading and the top model (Theorem 11.37). That residue is provably ineliminable (Theorem 11.38). We deliberately built a checker, not a solver.

Table 1: The three success criteria, and what this report establishes for each.

1.3 How the results were established

The work was carried out by many cooperating agent instances, coordinated by one orchestrating instance. It proceeded in four stages.

- Literature memos on eleven strands.
- Seven theory threads, which developed the mathematics.
- Adversarial verification. Every main result was attacked by two or three independent referees instructed to refute it, with brute-force computation on small cases where possible. The results were then repaired, and the repaired items were re-verified by fresh referees.
- Drafting, followed by a fidelity check of each section against its verified source.

In the first verification round the referees found four false statements and about twenty significant gaps. Each was repaired, weakened or retracted. One theorem needed a second repair round. The

full records are in `research/verification/` of the accompanying repository, and the repaired statements are the ones used here.

A core of discrete results is machine-checked in Lean 4 with Mathlib (Appendix K). It covers about fifty paper results in 14,600 lines with no `sorry`. Among them are:

- the Post-completeness and coherence theorems;
- Carnap’s problem in learning form, with the bilateral tell-tale;
- tonk beyond the horizon, the doctrinal paradox and the halving bounds;
- Reiter’s blame duality, and the separation between paradoxes and objects;
- bilateral completeness and the arity counterexample;
- the rate threshold and the rate-inversion counterexample;
- no free export from samples, and the Lipschitz certifier.

Experiments in a small symbolic toolkit illustrate the phenomena (Section 12). They concern school algebra with real systematic errors, propositional natural deduction, adversarial proof search against learned verifiers, and a prototype physics checker.

We have tried to mark scope honestly:

- Many results are “trivial once set up”. Their contribution is the setup, and we say so.
- Several ingredients are classical, and we credit them. Examples are Gold, Angluin, Plotkin, Natarajan, Rivest–Sloan, KWIK, Ville and the prior–posterior-ratio martingale, Post, Glivenko, Carnap and the bilateralists, chunk-and-permeate, information-based complexity, de Finetti and Gaifman.
- The soundness guarantees are relative to stated assumptions, chiefly realizability of the target in a structured hypothesis class and truthful designation of contexts. Computational cost is mostly not addressed.

1.4 Reader’s guide

The report is long because it treats three domains and two side questions in full. Readers can take the following routes.

- **Core argument:** Sections 2 to 7, about the formal case, then Section 13.
- **Informal mathematics and the history of rigor:** Section 10, after Section 2.
- **Contexts, idealization and physics checking:** Section 11. It is largely self-contained given Section 2.
- **Learning theory of imitation and simplicity:** Section 8.
- **Logic and the semantics of coherence:** Section 9.
- **Experiments and code:** Section 12.

Section 14 collects open problems. Proofs are in the appendices, one per section.

2 Setting: judgments, steps, rules, provers, and three channels

This section fixes the vocabulary of the paper. Nothing in it is deep; the point is a set-up in which the later theorems are short. Three questions are kept apart from the start: *what* is learned (a step set, a consequence relation, or a set of valuations; Section 2.3), *from what* (three channels of different logical form; Section 2.5), and *against whom* (an adaptive prover that searches for wrongly accepted steps; Section 2.6). Two running examples, propositional natural deduction and school algebra (Section 2.4), recur throughout.

2.1 Judgments, steps and closure

Fm is the set of formulas over atoms $p, q, r, p_0, p_1, \dots$ with $\neg, \wedge, \vee, \rightarrow, \perp, \top$; first-order languages are used for arithmetic and physics. A *context* Γ is a finite set of formulas.

Definition 2.1 (Judgments, positions). ^[T1 §1.1; T2 §1] A *judgment* is $\Gamma \vdash \varphi$ (single conclusion) or $\Gamma \vdash \Delta$ (multiple conclusion, Δ finite); J denotes a countable set of judgments. Contexts are part of judgments, so hypothetical reasoning, and “assume the air pressure is 0”, lives inside the step language. A *position* $[\Gamma : \Delta]$ asserts all of Γ and denies all of Δ . Following Restall (2005), $\Gamma \vdash \Delta$ says that *the position* $[\Gamma : \Delta]$ *is out of bounds*; a position is *coherent* if it is in bounds. When $\perp$ is present it counts as always denied, so $[\Gamma : \emptyset]$ is coherent iff $\Gamma \not\vdash \perp$. Without $\perp$, “coherent” means *non-trivial* (Γ does not entail every formula; for a structural relation, equivalently $\Gamma \not\vdash q$ for an atom q not occurring in Γ).

Restall’s reading gives inference data and coherence data opposite polarity: an inference says a position is *out of* bounds, a coherence datum certifies that one is *in* bounds. This is the bilateral picture of Smiley (1996) and Rumfitt (2000) and the multiple-conclusion logic of Shoesmith and Smiley (1978), anticipated in Carnap’s “full formalization” (Carnap, 1943). Section 6 uses this polarity to recast Carnap’s categoricity problem as a learning problem in the dual (valuation) space.

Definition 2.2 (Steps, rule sets, closure). ^[T1 §1.1] A *step* $P \triangleright j$ has a finite set $P \subseteq J$ of premises and a conclusion $j \in J$; S is the set of steps, and a *rule set* (step set) is any $R \subseteq S$. An R -*derivation* π of j from $B \subseteq J$ is a finite tree with root j whose nodes are leaves in B or conclusions of steps $P \triangleright j' \in R$ whose premises P label their children ($P = \emptyset$ allowed); $\text{Steps}(\pi)$ is the set of steps used. The *closure* $\text{Cl}_R(B)$ is the least $X \supseteq B$ such that $P \triangleright j \in R$ and $P \subseteq X$ imply $j \in X$; the *derived steps* are $\text{Sound}(R) = \{P \triangleright j : j \in \text{Cl}_R(P)\}$. The *target* is an unknown rule set R^* : its members are the *valid* steps, and the members of $\text{Sound}(R^*)$ the *derivable* ones.

Lemma 2.3 (Closure facts; trivial). ^[T1 §1.1] (a) $\text{Cl}_R(B)$ is the set of judgments with an R -derivation from B ; Cl_R is extensive, monotone, idempotent and finitary, and monotone in R . (b) $R \subseteq \text{Sound}(R)$ and $\text{Cl}_{\text{Sound}(R)} = \text{Cl}_R$; hence $\text{Sound}(R)$ is the largest step set A with $\text{Cl}_A = \text{Cl}_R$.

Proof in Appendix A.

Two formats recur. In the *formula format* (Hilbert style) $J = \text{Fm}$ and a step is $\varphi_1, \dots, \varphi_k \triangleright \psi$. In the *sequent format* (natural deduction in sequent style; Gentzen 1935; Prawitz 1965) judgments are sequents and discharge is an ordinary step, e.g. $\rightarrow\text{I}$ is $\{\Gamma, \varphi \vdash \psi\} \triangleright \Gamma \vdash \varphi \rightarrow \psi$. A rule set induces a relation on sets of formulas: $A \vdash_R \varphi$ iff $\varphi \in \text{Cl}_R(A)$ (formula format), or iff $(A_0 \vdash \varphi) \in \text{Cl}_R(\emptyset)$ for a finite $A_0 \subseteq A$ (sequent format). In the formula format $\vdash_R$ is a finitary consequence relation by Lemma 2.3(a): arguments chain, so cut is built in; for the natural-deduction calculi below, weakening and cut are admissible and the same holds. R is A -*coherent* if $A \not\vdash_R \perp$.

Reading. Section 3 shows that a verifier is safe under derivations of every length iff everything it accepts lies in $\text{Sound}(R^*)$ (T1 Lemma 1.1). So “learning which inferences are valid” has a sharp target: a set between R^* and $\text{Sound}(R^*)$.

2.2 Schemas and anti-unification

Steps are encoded as ground terms over a first-order signature Ω , e.g. $s(\text{and}(p_0, p_1), p_0)$ for an $\wedge E$ step; object-language atoms are *constants* of Ω .

Definition 2.4 (Schemas). ^[T1 §§1.3, 5; T7 §1.1] A *schema* σ is a term with metavariables $x, y, A, B, \dots$, and $\text{inst}(\sigma)$ is its set of ground instances. $\sigma \succeq \tau$ (σ is at least as general) iff $\tau = \sigma\theta$ for a substitution θ ; $\sigma \equiv \tau$ iff both (equality up to renaming). σ is *pure* if no atom constant occurs in it; its *purification* σ° replaces distinct atoms by distinct fresh metavariables. A *guarded schema* (σ, Ψ) carries guards $\Psi \subseteq \Phi$, a fixed finite family of decidable predicates on ground steps (freshness; “the facts entail $a \neq 0$ ”), and $\text{inst}(\sigma, \Psi)$ is the set of instances of σ satisfying every guard in Ψ . A *calculus* is a finite set Σ of (guarded) schemas, with $R_\Sigma = \bigcup_{\sigma \in \Sigma} \text{inst}(\sigma)$; a *tagged step* (i, s) records the schema i that it cites.

Anti-unification is the learning algorithm of this paper. For a tuple $\bar{u} = (u_1, \dots, u_n)$ of ground terms, $\mathcal{A}(\bar{u}) = f(\mathcal{A}(\bar{u}^1), \dots, \mathcal{A}(\bar{u}^r))$ if all u_j have root f of arity r ($\bar{u}^k$ is the tuple of k -th arguments); otherwise $\mathcal{A}(\bar{u}) = z_{\bar{u}}$, a metavariable indexed by the tuple, so equal tuples get equal metavariables.

Theorem 2.5 (Least general generalization; Plotkin 1970; Reynolds 1970). ^[T1 §1.3] (known) Every nonempty finite set $P = \{t_1, \dots, t_n\}$ of ground terms has a least general generalization $\text{lgg}(P)$, a schema with $P \subseteq \text{inst}(\text{lgg}(P))$ and $\text{lgg}(P) \preceq \tau$ whenever $P \subseteq \text{inst}(\tau)$. It is unique up to renaming and computed by $\mathcal{A}(t_1, \dots, t_n)$. Hence $\text{inst}(\text{lgg}(P)) = \bigcap \{\text{inst}(\tau) : P \subseteq \text{inst}(\tau)\}$: anti-unification is the closure algorithm for single-schema instance sets.

For a schema t let $\mu(t) := |t| - |\text{vars}(t)|$, where $|t|$ counts symbol occurrences (metavariables included) and $\text{vars}(t)$ is the set of distinct metavariables; so $\mu \geq 0$, $\mu(x) = 0$, and $\mu(t) = |t|$ for ground t .

Lemma 2.6 (Rank). ^[T1 Lemma 1.2] If $\tau = \sigma\theta$ then $\mu(\tau) \geq \mu(\sigma)$, with equality iff θ restricted to $\text{vars}(\sigma)$ is a renaming. Hence every strictly increasing generalization chain $\tau = \sigma_0 \prec \sigma_1 \prec \dots \prec \sigma_m$ has $m \leq \mu(\tau) - \mu(\sigma_m) \leq \mu(\tau)$.

Lemma 2.7 (When the lgg recovers the schema). ^[T1 Lemma 1.3] Let $t_j = \sigma\theta_j$ ($1 \leq j \leq n$) be ground instances of σ . Then $\text{lgg}(t_1, \dots, t_n) \preceq \sigma$, with $\equiv$ iff both witness events hold: (R) for each metavariable x of σ , the roots of $\theta_1 x, \dots, \theta_n x$ are not all equal; (D) for each $x \neq y$ there is j with $\theta_j x \neq \theta_j y$.

Proof idea. Substituting θx for the occurrences of x adds more symbols than distinct metavariables unless θx is a metavariable, which gives the rank lemma (folklore in the Plotkin–Reynolds lattice). Anti-unifying the t_j copies σ and puts $\mathcal{A}(\theta_1 x, \dots, \theta_n x)$ at x ; this is a metavariable iff (R) holds, and two of them coincide iff (D) fails. Full proofs in Appendix A. By the rank lemma, ascending generalization chains are finite, so lggs exist for infinite P as well. The positive-data theory of Sections 4 and 5 runs on counting witness events.

Lemma 2.8 (Guarded closure; trivial). *[T1 §5, side conditions] For nonempty finite $P \subseteq S$ let $\Psi_P = \{\phi \in \Phi : \phi \equiv 1 \text{ on } P\}$. Then $\text{inst}(\text{lbg}(P), \Psi_P)$ is the least set of the form $\text{inst}(\sigma, \Psi)$ containing P .*

So a learned guard is the strongest guard satisfied by *every* datum, and one datum that omits a side condition deletes it (Example 2.12). Proof in Appendix A.

2.3 Hypothesis classes

Definition 2.9 (Hypothesis classes). *[T2 §1, §4.1; T1 §1.2, §3] Hypotheses come in three guises. (i) Step sets $\mathcal{H} \subseteq 2^S$: e.g. single schemas $H_1 = \{\text{inst}(\sigma)\} \cup \{\emptyset\}$, tagged calculi $H_k^{\text{tag}} = \{\bigcup_{i \leq k} \{i\} \times \text{inst}(\sigma_i)\}$, untagged unions $H_k = \{\bigcup_{l \leq k} \text{inst}(\tau_l)\}$, finite unstructured classes. A learned step scorer (a process reward model) is a step set, and need not be closed under chaining. (ii) Consequence relations $\vdash$ (finitary, reflexive, monotone, cut-closed), equivalently finitary closure operators C ; $\vdash$ is *structural* if $\Gamma \vdash \varphi$ implies $s\Gamma \vdash s\varphi$ for every substitution s of formulas for atoms (Łoś and Suszko, 1958; Wójcicki, 1988). Each R induces $\vdash_R$; many R induce the same relation. (iii) Meanings as sets of valuations: a set V of admissible valuations $v : \text{Fm} \rightarrow \{0, 1\}$ (not necessarily compositional), with $\Gamma \models_V^m \Delta$ iff no $v \in V$ has $v[\Gamma] = 1$ and $v[\Delta] = 0$; dually, $\text{Val}(S')$ is the set of valuations violating no sequent in S' . The *version space* of data D is $\text{VS}(D) = \{h \in \mathcal{H} : h \text{ consistent with } D\}$ ($= \{h : D \subseteq h\}$ for positive data). $\mathcal{H}$ is *realizable* if it contains the target; w is a prior and $w^* = w(R^*)$.*

Structurality is uniformity. If σ is pure, $\text{inst}(\sigma)$ is closed under substitutions s for atoms, since the atoms of $\sigma\theta$ come from θ : $s(\sigma\theta) = \sigma\theta'$ with $\theta'x = s(\theta x)$. So a learner that generalizes human steps to pure schemas is structural. A schema that mentions an atom is not, and can be unsound without trivializing: added to a complete calculus for CPC, $x \vee p_0 \triangleright x$ derives $\neg p_0$ but stays inside $\text{Cn}(\neg p_0)$ (T1 Prop. 2.3, scope remark). Purification restores the link (Section 3).

Inference data are negative data about valuations. The sequent $\Gamma \vdash \Delta$ excludes exactly the valuations realizing $[\Gamma : \Delta]$; a coherence certification or a world report is *positive* data about valuations (T2 §4.1). Imitation and coherence inform the learner from opposite sides, and Section 6 makes this precise as “Carnap’s problem is Gold’s problem in the dual space”.

2.4 Practice and the running examples

Definition 2.10 (Practice). *[T7 Defs 1.1–1.2; T1 §6] The target is a finite calculus Σ^* (the genuine rules), with $R^* = R_{\Sigma^*}$. The systematic fallacies are a finite set F of schemas, $F \cap \Sigma^* = \emptyset$, each with an instance outside $\text{Sound}(R^*)$. The practice is $\Sigma^P = \Sigma^* \sqcup F$, with $R^P = R^* \cup R_F$. Human data are i.i.d. tagged steps: tag $i \in \Sigma^P$ has probability π_i , and given i the step is an instance of i with probability $1 - a_i$ and otherwise an arbitrary step outside $\text{inst}(i)$ tagged i (sporadic noise). Alternatively, the data are a text, an enumeration of R^P (Gold, 1967).*

Fallacy tags are idealized as *latent but distinct*: a student who affirms the consequent “cites” a rule that is in fact affirming the consequent. If fallacies are cited under a genuine rule’s name, the per-tag class becomes H_k (Sections 5 and 7).

Example 2.11 (Propositional natural deduction). Σ^* is a natural-deduction calculus for CPC ($\wedge\text{I}/\text{E}$, $\vee\text{I}/\text{E}$, $\rightarrow\text{I}$, $\rightarrow\text{E}$ = modus ponens (MP), $\neg\text{I}/\text{E}$, ex falso, double negation). In the formula format MP is $\varphi, \varphi \rightarrow \psi \triangleright \psi$. A student sometimes affirms the consequent, AC: $\psi, \varphi \rightarrow \psi \triangleright \varphi$; so $F = \{\text{AC}\}$. The data are steps such as $p \wedge q \triangleright p$ tagged $\wedge\text{E}$. The two instances $p \wedge q \triangleright p$ and

$(r \rightarrow p) \wedge p \triangleright r \rightarrow p$ already satisfy (R) and (D), so their lgg is $A \wedge B \triangleright A$ (Lemma 2.7). (In the sequent format contexts are sets, and anti-unification modulo set structure is finitary rather than unitary; we flag this where it matters.)

Example 2.12 (School algebra). Judgments are equations $s = t$ between terms built from $+$, $-$, $\cdot$, $/$, powers, $\sqrt{}$, numerals and variables, under finite sets of *facts* such as $x \neq 0$. Terms denote partial functions ($a/0$ is undefined), and $s = t$ is true under the facts iff s, t are Kleene-equal (both undefined, or both defined and equal) at every real assignment satisfying them. Steps are instances of rewrite schemas such as $(a + b)^2 \triangleright a^2 + 2ab + b^2$ and $a/a \triangleright 1$ with guard $a \neq 0$, plus trusted equational steps (symmetry, transitivity, congruence) and trusted numeral evaluation. The student commits the *freshman’s dream* $(a + b)^2 \triangleright a^2 + b^2$, an invalid schema, and *unguarded cancellation* $a/a \triangleright 1$, a genuine rule used without its guard (the instance $x/x \triangleright 1$ fails at $x = 0$). If the student cites “cancellation” for both uses, one datum such as $(y + 1)/(y + 1) \triangleright 1$ under no facts drops the guard from the learned rule (Lemma 2.8); with distinct latent tags the unguarded rule is learned as a schema of its own. Either way, positive data learn it as a rule.

The algebra example is experiment A and the natural-deduction example is experiment B of Section 12.

2.5 Three channels

Definition 2.13 (The channels). [T1 §1.2; T2 §1, §2.1–2.2; T7 Defs 1.2–1.4; L3 Lemma A]

- (P) *Practice/imitation* delivers human steps $X_1, X_2, \dots$, possibly tagged with rule names, possibly with sporadic noise (Definition 2.10). A noise-free datum lies in R^P ; no datum says that a step is invalid.
- (C) *Coherence* supplies a family $\mathcal{A}$ of *designated contexts*, finite premise sets certified coherent for the target ($A \not\vdash_{R^*} \perp$), or, bilaterally, *designated positions* $[\Gamma : \Delta]$ certified in bounds ($\perp \notin \text{Cl}_{R^*}(\Gamma)$ and $\Delta \cap \text{Cl}_{R^*}(\Gamma) = \emptyset$). By default $\mathcal{A}$ is fixed and known, independently of the target. A hypothesis h is $\mathcal{A}$ -*coherent* if $A \not\vdash_h \perp$ for all $A \in \mathcal{A}$. A *negative bag* is $\text{Steps}(\pi) \setminus T_{\text{trust}}$ for an argument π from some $A \in \mathcal{A}$ to $\perp$ (trusted steps T_{trust} as in (W)); it contains an invalid step, or chaining would give $\perp \in \text{Cl}_{R^* \cup T_{\text{trust}}}(A)$, which designation excludes (for $T_{\text{trust}} \neq \emptyset$ via (WS), Lemma 2.15).
- (W) The *world* is an evaluable fragment $J_W \subseteq J$ with truth function $W : J_W \rightarrow \{0, 1\}$, plus optional *trusted steps* T_{trust} . Its kinds: (W1) *truth values on a fragment* (closed propositional formulas, Δ_0 sentences, a decision procedure); (W2) *counterexample objects*, structures in which every line of a given argument can be evaluated; (W3) *computation with fresh randomness*, e.g. testing a polynomial identity at a point drawn after the claim is committed; (W4) *simulation*, a validated numerical solution of a less idealized model (Section 11).

Coherence is enforced *only* on designated contexts: deriving $\perp$ from an undesigned supposition is a reductio and is never penalized. A negative bag is a multiple-instance label (Dietterich et al., 1997), “not all of these steps are valid”; it collapses to an ordinary negative example for a learner that can test $A \triangleright \perp$ directly against cut-closed hypotheses, so the bag structure matters for step verifiers that are not themselves cut-closed. Designations that are data about the target carry more information (Section 6). Objects (W2) turn bags into singletons: if a structure makes the leaves of an argument true and its conclusion false, then starting at the conclusion and repeatedly moving to a false premise ends at a step with true premises and a false conclusion,

	(P) practice	(C) coherence	(W) world
datum	“this step is used”	an argument from a designated context to $\perp$	a truth value; an object; a random test; a simulation
logical form	positive instance	negative bag (“some step is invalid”)	singleton blame with objects; derivation-level otherwise
blind to	systematic errors	errors coherent with the target	what lies outside the evaluable fragment

Table 2: The three channels (Definition 2.13).

which alone is blamed (the *descent* lemma, Sections 7 and 10). Fresh randomness (W3) is sound against adaptive provers under sufficient conditions on commitment order, degree bounds and the field, and fails when the points are drawn before the prover searches (L3 Lemma A, Section 3); Schwartz–Zippel bounds its error (Schwartz, 1980; Zippel, 1979).

Assumption 2.14 ((WS), sound evaluation). ^[T7 §1.3] For each designated $[\Gamma : \Delta]$, the partial evaluation with $e(j) = 1$ if $j \in \Gamma$ or $W(j) = 1$, and $e(j) = 0$ if $j \in \Delta \cup \{\perp\}$ or $W(j) = 0$, is well defined and extends to a total $V : J \rightarrow \{0, 1\}$ under which no step of $R^* \cup T_{\text{trust}}$ has true premises and a false conclusion.

(WS) does not require $T_{\text{trust}} \subseteq \text{Sound}(R^*)$: in arithmetic the target may consist of non-logical axioms with first-order logic trusted (Section 7), and the relevant target closure is then $\text{Cl}_{R^* \cup T_{\text{trust}}}$. In formal mathematics V is truth in the intended structure $\mathfrak{M}$: (WS) holds when designated positions are true of $\mathfrak{M}$, W is truth in $\mathfrak{M}$ on its fragment, and target and trusted steps are $\mathfrak{M}$ -sound. When $J_W \neq \emptyset$ we assume the empty position $[\emptyset : \emptyset]$ is designated, so that world-only falsifications count (Definition 2.20).

Lemma 2.15 ((WS) and coherence; trivial). ^[T7 §1.3] Under (WS) every designated position is in bounds for $R^* \cup T_{\text{trust}}$, hence for R^* . Conversely, if $J_W = \emptyset$ and every designated position is in bounds for $R^* \cup T_{\text{trust}}$ (for instance, in bounds for R^* with $T_{\text{trust}} \subseteq \text{Sound}(R^*)$), then (WS) holds.

Proof in Appendix A.

Example 2.16 (Example 2.11 continued). (C) Designate $A_1 = \{q, p \rightarrow q, p \rightarrow \perp\}$: AC gives p , MP then gives $\perp$. The bag $\{q, p \rightarrow q \triangleright p; p, p \rightarrow \perp \triangleright \perp\}$ does not say which step is at fault, and the rival diagnosis “MP is the fallacy” reads $\rightarrow$ as $\leftarrow$ (Section 7). (W2) The valuation $v(p) = 0$, $v(q) = 1$ makes A_1 true and the line p false, so descent blames AC alone. (W1) The closed instance $\top, \perp \rightarrow \top \triangleright \perp$ of AC has true premises and a false conclusion.

Example 2.17 (Example 2.12 continued). (C) Designate the empty fact set, with $\perp$ any equation between distinct numerals. The freshman’s dream gives $(1 + 1)^2 = 1^2 + 1^2$, i.e. $4 = 2$ after trusted evaluation: a singleton bag. Unguarded cancellation gives $2 \cdot (0/0) = 2 \cdot 1 = 2$ and $2 \cdot (0/0) = (2 \cdot 0)/0 = 0/0 = 1$, so $1 = 2$; this bag also contains the genuine rules $a \cdot (b/c) \triangleright (a \cdot b)/c$, $a \cdot 0 \triangleright 0$ (for defined a) and $a \cdot 1 \triangleright a$. (W2) An evaluator that observes “undefined” blames $0/0 \triangleright 1$ alone. (W3) The freshman’s dream is off by $2ab$, so a uniformly random point of S^2 , $S \subseteq \mathbb{Z}$ finite, exposes it with probability at least $1 - 2/|S|$; over $\mathbb{F}_2$, where it is an identity, never: the field matters. The instance $x/x \triangleright 1$ fails only at $x = 0$, which random evaluation essentially never hits; the object $x = 0$ does. In experiment A, coherence without (W) removes value errors like the

freshman’s dream but almost never learns guards that concern only definedness (such as $a \neq 0$ for $0/a \triangleright 0$); almost all of what survives (96.5% of the surviving schemas for $N \geq 200$) is sound for a total reading of division over $\mathbb{C}$ with $1/0 = 0$, a coherent alternative meaning (computed; Section 12).

Reading. The channels are the user’s three proposed signals: imitation of human inferences, a coherence loss, and occasional feedback from the world. They are not three noisy versions of one signal. (P) can only confirm, (C) can only condemn sets, and (W) can condemn single steps when it supplies objects. Each later section says what one of them fixes and what it provably cannot.

2.6 Provers, verifiers, soundness and cost

Definition 2.18 (Protocol). ^[T1 §1.2; T2 §2.2] In rounds $t = 1, 2, \dots$ a *prover* submits $q_t \in S$ and the *verifier* answers $a_t \in \{\text{ACCEPT}, \text{REJECT}, \text{ESCALATE}\}$; on **ESCALATE** an oracle returns $\mathbf{1}[q_t \in R^*]$ (possibly noisy), which joins the history. The *acceptance region* $\hat{R}_t$ is the set of steps accepted outright at time t ; a reasoner chains them and derives $\text{Cl}_{\hat{R}_t}(B)$. The prover may be randomized, adaptive and computationally unbounded; it knows the target, the practice, the verifier’s code and the history, but not the verifier’s future coins, future human data, or the oracle’s future (fresh) noise. It models any proof search, the learner’s own included. It is *honest* if all its queries are valid.

Definition 2.19 (Soundness). ^[T1 Def 1.2; T7 §1.5] Fix a class $\mathcal{H}$ of targets and, per target, a set $S^\dagger$ of admissible steps (default $S^\dagger = R^*$). A verifier is δ -*sound* if for every target and every prover $\mathbb{P}[\exists t : a_t = \text{ACCEPT} \wedge q_t \notin S^\dagger] \leq \delta$ (a time-uniform guarantee); *deterministically* (or 0-) *sound* if $\delta = 0$; and *uniformly* δ -*sound* if some event G with $\mathbb{P}(G) \geq 1 - \delta$, depending only on the human data and the verifier’s coins, has no prover ever getting a step outside $S^\dagger$ accepted on G . *Target-soundness* takes $S^\dagger = R^*$ for the genuine Σ^* , *practice-soundness* $S^\dagger = R^P$. (*Truth-soundness*, in Section 7, means deriving only truths of the intended structure.)

Uniform δ -soundness implies δ -soundness, since every prover’s bad event lies outside G . Asking for $S^\dagger = R^*$ is stronger than the condition $\hat{R}_t \subseteq \text{Sound}(R^*)$ that chaining needs; the version-space verifiers of Section 4 meet it. Soundness is relative to the target calculus, not to truth: improving on R^* is the job of (C) and (W). Since coherence is Π_1 -complete (T2 Thm 5.3), no computable learner is, on every practice, both eventually complete and sound at all times relative to the full-depth residue of fallacies: some uniformly computable family of practices defeats it (T7 Thm 5.7, Section 7). Soundness is then stated relative to a refutation budget.

Definition 2.20 (Depth-relative soundness). ^[T7 Defs 1.5–1.6] Assume (WS). A *d-refutation* of a finite schema set B is a derivation of total size $\leq d$, relative to a designated position, whose leaves evaluate to 1, whose steps lie in $R_B \cup T_{\text{trust}}$, and whose conclusion evaluates to 0 ($\perp$, a denied judgment, or a W -false one). B is *d-clean* if it has none. The *depth-d residue* $F_{\text{res}}^{(d)} = \{\tau \in F : \Sigma^* \cup \{\tau\} \text{ is } d\text{-clean}\}$ decreases in d . A verifier is *depth-d sound* if it is (uniformly) δ -sound for $S^\dagger = R^* \cup R_{F_{\text{res}}^{(d)}}$.

Definition 2.21 (Cost). ^[T1 Def 1.3; T2 §2.2; T7 §1.4] (a) *Escalations*: the number of rounds in which a valid query is not accepted outright (in expectation, for randomized verifiers). $\text{Esc}(\mathcal{H} \mid D)$ is the infimum over 0-sound verifiers of the supremum over targets in $\mathcal{H}$ containing the data D , and

over honest query sequences, of this number; $\text{Esc}(\mathcal{H}) = \text{Esc}(\mathcal{H} \mid \emptyset)$ is the *escalation dimension*. Invalid queries are not charged. (b) *Detections*: rounds in which an argument π from some $A \in \mathcal{A}$ to $\perp$ with $\text{Steps}(\pi) \subseteq \hat{R}_t$ is exhibited; *incompleteness errors*: rounds presenting a valid step outside $\hat{R}_t$. (c) *Mind changes*: rounds with $\hat{R}_{t+1} \neq \hat{R}_t$; those that remove an accepted step are *retractions* (Kelly, 1996). (d) *Samples* N (human steps) and *calls* to a refutation oracle that returns a d -refutation of a queried set or reports none.

Example 2.22 (One bad step suffices). If AC is accepted, a prover derives any φ from no premises: $\top$ and $\varphi \rightarrow \top$ are theorems, and AC concludes φ . If unguarded cancellation is accepted, it derives $1 = 2$ (Example 2.17). A verifier whose errors are rare on the human distribution still accepts such a step every time a prover asks for it. This is Prior’s *tonk* (Prior, 1960) as an attack on a learned checker; Section 3 turns it into theorems.

Reading. This is how the set-up takes the user’s “compute is not a constraint, but it must actually work”. The prover is unbounded and adaptive, so soundness is a worst-case property: reliable learning (Rivest and Sloan, 1988a), perfect selective classification (El-Yaniv and Wiener, 2010a) and the “accept” side of KWIK (Li et al., 2008a), applied to steps. Its price is paid in escalations (human time), samples, detections and oracle calls, counted separately. Average-case accuracy is not enough (Section 3).

2.7 Contexts: a preview

Contexts let one argue inside a supposition or an idealization without being penalized for contradicting the world. Following McCarthy (1993), write $\text{ist}(c, \varphi)$ for “ φ holds in context c ”; contexts form a tree with root $@$, the actual or intended situation (T3 §1). A *supposition* $\text{SUP}(A)$ refines its parent by A , and discharge is exact: $\text{ist}(\text{SUP}(A), \psi)$ iff $A \rightarrow \psi$ holds in the parent (T3 Prop. 1.5); deriving $\perp$ there is a reductio of A , not an incoherence. An *idealization* DEF deforms named parameters of a declared model family toward an ideal value (“air pressure $\rightarrow 0$ ”). It may contradict the root, yet does not explode while it is proper: a context denotes a filter of parameter points of the model family, truth in it is truth on a set in that filter, and its theory is classically closed and consistent iff the filter is proper (T3 Def. 1.3, Prop. 1.4). Coherence is enforced on the root and on contexts anchored to it (certified, or exporting through an informative bridge into an anchored context), never on suppositions or idealizations as such; designating “idealization plus full background” as coherent would force every structural learner to give up a classical rule *globally* (T2 Prop. 7.1), so only consistent chunks are designated (Brown and Priest, 2004). Contexts, export certificates and the physics checker are the subject of Section 11.

3 Search makes soundness a worst-case property

The architecture studied in this paper has three parts: a learned relation $\hat{V}$ that says which steps are valid, a prover that searches for derivations all of whose steps $\hat{V}$ accepts, and a training signal that starts from human steps. This section isolates the first consequence of putting search on top of a learned checker. Once a prover searches, the only property of $\hat{V}$ that bears on the soundness of the resulting reasoner is a worst-case one: whether its acceptance region contains a non-derivable step. Accuracy on typical steps, on human steps, or on most instances of a rule does not bear on it, because search can seek out exactly the steps on which $\hat{V}$ is wrong.

None of the results below is deep, and several are trivial once set up; what they contribute is the setup and the quantifiers. Reasoner soundness is stepwise soundness (Lemma 3.1). Arbitrarily small average error is compatible with a reasoner that derives everything, and every PAC learner of calculi containing three basic introduction rules has a PAC variant, with essentially the same sample complexity, whose every output does so (Theorem 3.2 and Corollary 3.3). In classical propositional logic every unsound *pure* schema added to a complete calculus trivializes the reasoner, so errors are total but visible to coherence (Proposition 3.5 and Lemma 3.7). Simplicity is the wrong bias for positive data (Proposition 3.9). Average-case validity survives search in two regimes where its randomness is drawn after the rules or claims are fixed and outside the prover’s control, and fails in a fixed world with prover-chosen instances (Lemma 3.11). Voting ensembles of coherent verifiers can be incoherent in a way that coherence feedback cannot localize (Theorem 3.12). Except in Section 3.5, which measures truth in a scene, “sound” means sound relative to the human calculus R^* (or a target consequence relation), not relative to truth; going beyond R^* is the business of coherence and the world (Sections 6, 7 and 11).

3.1 Reasoner soundness is stepwise soundness

We recall the step language of Section 2. J is a countable set of judgments; contexts are part of judgments, so hypothetical reasoning (“assume air pressure is 0”) lives inside the step language. A *step* $\Pi \triangleright j$ has a finite set $\Pi \subseteq J$ of premises and a conclusion $j \in J$; $\mathcal{S}$ is the set of all steps. For $A \subseteq \mathcal{S}$ and $B \subseteq J$, $\text{Cl}_A(B)$ is the least $X \supseteq B$ such that $\Pi \subseteq X$ and $(\Pi \triangleright j) \in A$ imply $j \in X$, i.e. the set of judgments with a finite A -derivation from B . The derivable steps of the human calculus are $\text{Sound}(R^*) = \{\Pi \triangleright j : j \in \text{Cl}_{R^*}(\Pi)\}$. A deterministic verifier $\hat{V}$ has an *acceptance region* $A = \{s \in \mathcal{S} : \hat{V}(s) = \text{ACCEPT}\}$, and an unbounded prover that chains accepted steps from premises B reaches exactly $\text{Cl}_A(B)$.

Lemma 3.1 (Reasoner soundness is stepwise soundness). *[T1 Lemma 1.1] For every $A \subseteq \mathcal{S}$,*

$$\forall B \subseteq J : \text{Cl}_A(B) \subseteq \text{Cl}_{R^*}(B) \quad \Longleftrightarrow \quad A \subseteq \text{Sound}(R^*).$$

Lean: Cl_subset_Cl_iff_subset_Sound

Proof. ($\Leftarrow$) Let $(\Pi \triangleright j) \in A$ with $\Pi \subseteq \text{Cl}_{R^*}(B)$. As Cl_{R^*} is monotone and idempotent, $j \in \text{Cl}_{R^*}(\Pi) \subseteq \text{Cl}_{R^*}(\text{Cl}_{R^*}(B)) = \text{Cl}_{R^*}(B)$. So $\text{Cl}_{R^*}(B)$ is a superset of B closed under A , and $\text{Cl}_A(B) \subseteq \text{Cl}_{R^*}(B)$ by leastness. ($\Rightarrow$) For $(\Pi \triangleright j) \in A$ take $B = \Pi$: $j \in \text{Cl}_A(\Pi) \subseteq \text{Cl}_{R^*}(\Pi)$. $\square$

The lemma is trivial once set up; its content is that the right object is the *set* A , not a measure on it. There is no “small error rate” middle ground: one accepted step $\Pi \triangleright j \notin \text{Sound}(R^*)$ yields, from Π , a conclusion that R^* does not derive. The cautious verifiers of Section 4 satisfy the stronger $A \subseteq R^*$.

3.2 Average-case accuracy cannot give soundness

Fix a propositional language with atoms $p_0, p_1, \dots$, the constant $\top$ and connectives $\wedge, \vee$ (others optional); judgments are formulas. Let R^* contain the instances of

$$\top \text{I} : \triangleright \top, \quad \wedge \text{I} : A, B \triangleright A \wedge B, \quad \wedge \text{E}_{1,2}, \quad \vee \text{I}_1 : A \triangleright A \vee B, \quad \vee \text{I}_2 : B \triangleright A \vee B,$$

and possibly of other classically sound rules. Put $\top^{(1)} = \top$, $\top^{(j+1)} = \top \wedge \top^{(j)}$, and consider the pure one-metavariable schema $T_j := \{A \vee \top^{(j)} \triangleright A : A \in \text{Fm}\}$. Composed with $\vee \text{I}_2$, T_j is

the rule “from $\top^{(j)}$ infer anything”: a tonk (Prior, 1960; Belnap, 1962) with its trigger hidden beyond a size horizon. For a distribution Q on $\mathcal{S}$ and $h \subseteq \mathcal{S}$ write $\text{err}_Q(h) := Q(h \triangle R^*)$.

Theorem 3.2 (Tonk beyond the horizon). *[T1 Thm 2.1]*

- (i) Every formula C lies in $\text{Cl}_{R^* \cup T_j}(\emptyset)$, by a derivation of $j + 2$ steps of which only the last uses T_j . For non-tautologous C that step is unsound, hence not in $\text{Sound}(R^*)$.
- (ii) For every probability distribution Q on $\mathcal{S}$ (over valid steps, invalid steps, or both), $Q(T_j) \rightarrow 0$ as $j \rightarrow \infty$.

So for every Q and $\varepsilon > 0$ some $R^* \cup T_j$ has $\text{err}_Q(R^* \cup T_j) \leq Q(T_j) < \varepsilon$ and makes every formula derivable. If R^* is given by finitely many pure schemas, so is $R^* \cup T_j$.

Proof. (i) Derive $\top$ by $\top I$, then $\top^{(2)}, \dots, \top^{(j)}$ by $j - 1$ uses of $\wedge I$, then $C \vee \top^{(j)}$ by $\vee I_2$, and finally C by T_j . The premise $C \vee \top^{(j)}$ is a tautology, so for non-tautologous C the last step is unsound. (ii) The T_j are pairwise disjoint ($A \vee \top^{(j)} = A' \vee \top^{(j')}$ forces $A = A'$, $j = j'$), so $\sum_j Q(T_j) \leq 1$. Finally $(R^* \cup T_j) \triangle R^* \subseteq T_j$. $\square$

Corollary 3.3 (PAC learners can be maximally unsound). *[T1 Cor 2.2, as repaired]* In the setting of Theorem 3.2, let $\mathcal{L}$ be any learner mapping a sample $X_1, \dots, X_m$ of steps, drawn i.i.d. from Q and labelled by membership in R^* , to a step set. Let $|s|$ count the symbol occurrences in a step s , put $\hat{j} := 1 + \max_i |X_i|$, and

$$\mathcal{L}^+(X_{1..m}) := \mathcal{L}(X_{1..m}) \cup T_{\hat{j}} \cup \text{inst}(\top I, \wedge I, \vee I_2).$$

Then for every Q , $m \geq 1$ and $\varepsilon \in (0, 1)$: (a) $\text{err}_Q(\mathcal{L}^+) \leq \text{err}_Q(\mathcal{L}) + Q(T_{\hat{j}})$ and $\Pr[Q(T_{\hat{j}}) > \varepsilon] \leq (1 - \varepsilon)^m$. (b) Hence if $\mathcal{L}$ PAC-learns a class of targets R^* , each containing $\top I$, $\wedge I$ and $\vee I_2$, with sample complexity $m_{\mathcal{L}}(\varepsilon, \delta)$, then $\mathcal{L}^+$ PAC-learns the same class (improperly) with sample complexity at most $\max\{m_{\mathcal{L}}(\varepsilon/2, \delta/2), \lceil (2/\varepsilon) \ln(2/\delta) \rceil\}$. (c) Yet every output of $\mathcal{L}^+$ makes every formula derivable from $\emptyset$.

Proof idea. Steps of T_j have size $\geq j$, and $\hat{j}$ exceeds every sample’s size, so $T_{\hat{j}}$ lies in a tail of Q that the sample has not reached; that tail has mass $> \varepsilon$ only if all m samples missed a set of mass $> \varepsilon$. The three added schemas lie in R^* , so they cannot increase the error, and with $T_{\hat{j}}$ they carry the derivation of Theorem 3.2(i). Full proof in Appendix B. $\square$

The added schemas are needed for (c) (Remark B.1). The corollary is about what a PAC guarantee can certify, not a prediction that natural learners output tonks: the guarantee is satisfied equally well by a learner whose every output trivializes the reasoner.

Remarks. (a) Theorem 3.2 and Corollary 3.3 are trivial once set up. Their content is the quantifier: a soundness guarantee must be a $\sup_s$, not an $\mathbb{E}_{s \sim Q}$, because the prover chooses s after seeing the verifier. (b) *Selection effect.* If a search produces a derivation of a judgment that R^* does not derive from the premises used, then by Lemma 3.1 it contains an accepted step outside $\text{Sound}(R^*)$, whatever the verifier’s error rate on human steps. This is the mechanism behind reward-model overoptimization (Gao et al., 2023), and it limits what process reward models trained on human step labels (Lightman et al., 2023) can certify (Section 3.7). (c) The uniform requirement is reliable learning (Rivest and Sloan, 1988b), perfect selective classification (El-Yaniv and Wiener, 2010b), or the “accept” side of KWIK (Li et al., 2008b), applied to steps; Section 4 computes its price.

Reading. For the question “can a learner learn which inferences are valid?”, Theorem 3.2 separates two things a guarantee might mean. “Valid on typical steps” (PAC accuracy, held-out accuracy on human steps, a process reward model’s accuracy) is a statement about a measure, which can be arbitrarily close to perfect while the set it measures licenses every conclusion. “Valid on every step it accepts” is a statement about the set, and only it transfers to a reasoner. For justification the moral is the same: a conclusion reached by search inherits nothing from the checker’s average accuracy. A learned checker must certify an inclusion $A \subseteq \text{Sound}(R^*)$; the rest of the paper asks how imitation, coherence and the world can certify it.

3.3 In classical propositional logic, every unsound pure schema trivializes

One bad schema *can* trivialize a reasoner (Theorem 3.2). In classical propositional logic it always does, provided it mentions no specific atom. Errors are then total, but also maximally visible: the reasoner derives $\perp$ from no premises.

Definition 3.4 (Pure schemas; substitution-closed step sets). A *schema* τ is a step whose formulas may contain metavariables; $\text{inst}(\tau)$ is the set of its instances, obtained by substituting formulas for metavariables. Atoms are constants of the step language, so a schema may mention specific atoms, as in $x \vee p_0 \triangleright x$; least general generalization produces such schemas whenever a position of the data is constantly p_0 . A schema is *pure* if no atom occurs in it. A step set is *substitution-closed* if it is closed under uniform substitution of formulas for atoms; instance sets of pure schemas are. A step $\Pi \triangleright \varphi$ is *sound* if $\Pi \models \varphi$, and a schema or step set is sound if all its instances or steps are.

Proposition 3.5 (Every unsound pure schema is a tonk). *[T1 Prop 2.3, repaired scope] Let R_{CPC}^* be sound and complete for classical propositional consequence, i.e. $\text{Cl}_{R_{\text{CPC}}^*}(\Gamma) = \text{CnCPC}(\Gamma)$ for all Γ , over a language with $\top, \perp, \neg$. Let $A = R_{\text{CPC}}^* \cup A_1$ with A_1 substitution-closed. If A contains an unsound step, then $\text{Cl}_A(\emptyset) = \text{Fm}$.*

Proof. Let $\Pi \triangleright \varphi \in A$ be unsound, with $v(\Pi) = 1$, $v(\varphi) = 0$; it lies in A_1 since R_{CPC}^* is sound. Let σ_v send each atom true under v to $\top$ and each false one to $\perp$. Then $\sigma_v \Pi \triangleright \sigma_v \varphi \in A_1$ has variable-free formulas; each $\sigma_v \pi$ and $\neg \sigma_v \varphi$ is a tautology, so lies in $\text{Cl}_{R_{\text{CPC}}^*}(\emptyset) \subseteq \text{Cl}_A(\emptyset)$ by completeness. The step adds $\sigma_v \varphi$, and $\text{Cl}_A(\emptyset) \supseteq \text{CnCPC}(\{\sigma_v \varphi, \neg \sigma_v \varphi\}) = \text{Fm}$. $\square$

This is essentially Post (1921): classical propositional logic is Post-complete. The consequence-level form underlies the coherence results of Section 6.

Example 3.6 (Schemas that mention atoms). Without substitution-closure the proposition fails. $\tau = (x \vee p_0 \triangleright x)$ is unsound (instance $p_1 \vee p_0 \triangleright p_1$). Yet $\text{CnCPC}(\neg p_0)$ is closed under $R_{\text{CPC}}^* \cup \text{inst}(\tau)$, since $\neg p_0 \models B \vee p_0$ implies $\neg p_0 \models B$. So the reasoner derives the non-tautology $\neg p_0$ (from $\neg p_0 \vee p_0$) but never p_0 : unsound, not trivial. A ground error $p_0 \triangleright p_1$ never fires from $\emptyset$ at all.

The dichotomy is restored by *purification*: let τ° be τ with each atom occurring in it replaced by a fresh metavariable, distinct atoms by distinct metavariables.

Lemma 3.7 (Purification). *[T1 Prop 2.3, repair] In classical propositional logic, τ is sound if and only if τ° is sound.*

Corollary 3.8 (Coherence decides soundness of schema hypotheses). *[T1 Prop 2.3, repair] For any family of schemas τ_l , let $A = R_{\text{CPC}}^* \cup \bigcup_l \text{inst}(\tau_l)$ and $A^\circ = R_{\text{CPC}}^* \cup \bigcup_l \text{inst}(\tau_l^\circ)$. Then A is sound iff A° is sound iff $\perp \notin \text{Cl}_{A^\circ}(\emptyset)$.*

Proof idea. Every instance of τ is one of τ° . Conversely, an instance of τ° substitutes formulas only for τ ’s own atom occurrences; renaming those atoms inside the metavariable fillers to fresh atoms gives an instance of τ , which a uniform substitution maps back onto the given one, and uniform substitution preserves validity. For the corollary, the instances of the pure τ_l° form a substitution-closed set, so Proposition 3.5 applies to A° . Full proofs in Appendix B. $\square$

Reading. This is the positive face of Theorem 3.2. An unsound pure schema cannot hide: it makes the reasoner incoherent in the empty context, which is exactly the signal of the coherence channel (C). Coherence in the empty context is thus a complete soundness test for purified schema hypotheses over CPC; Section 6 turns it into identification of CPC and Section 7 into an end-to-end learner. Two limits. Purification is used as a *test*: one checks A° to decide A , without accepting the purified steps. And the test presupposes a structural target. A calculus with atom-specific principles (an axiom about one constant, a physical theory with named parameters) is not structural, and there coherence can miss errors as in Example 3.6. Beyond classical propositional logic the dichotomy fails too; Section 6 determines what coherence detects in intuitionistic logic and arithmetic.

3.4 Simplicity is the wrong bias for positive data

A natural response to Theorem 3.2 is to prefer simple hypotheses. On positive data alone this points the wrong way.

Proposition 3.9 (MDL picks “anything from anything”). *[T1 Prop 2.4] Encode steps as ground terms and schemas as terms with metavariables, of size equal to the number of symbol occurrences, and let $\mathcal{H}_1 = \{\text{inst}(\sigma) : \sigma \text{ a schema}\}$. For every nonempty set P of positive data, the single metavariable x is a shortest schema with $P \subseteq \text{inst}(x)$, and $\text{inst}(x) \supseteq \mathcal{S}$ accepts every step. It is the unique shortest one up to renaming unless P is a single step of size 1.*

Proof. x has the minimum size 1 and covers everything; any other size-1 schema is a constant c , with $\text{inst}(c) = \{c\}$. $\square$

More generally, an MDL learner without a likelihood term picks a shortest hypothesis containing the data. Positive data never refute an over-general hypothesis (Gold, 1967; Angluin, 1980), and over-general hypotheses are typically short, so it over-generalizes. The positive-data remedies are *least* generality, i.e. Plotkin’s least general generalization (Plotkin, 1970) and the version-space verifiers of Sections 4 and 5, and a likelihood with the size principle (Section 4). Even then, MAP selection is unsound whenever an over-general hypothesis is temporarily most probable, so Section 4 replaces it by conservative thresholding. Simplicity has a role in this paper (Section 8), but not as the acceptance criterion for steps.

3.5 When does average-case validity survive search? The commitment order

Theorem 3.2 seems to contradict a respectable tradition. In PAC-semantics (Valiant, 2000), learned rules that are $(1 - \varepsilon_i)$ -valid over a distribution of scenes are chained with additive error; learning to reason (Khaddon and Roth, 1997) and implicit learning (Juba, 2013) give guarantees

of the same average-case kind. There is no contradiction. What decides whether an average-case guarantee survives an adversarial prover is *what* is random and *when* it is drawn.

Definition 3.10 (Sequent-level validity in a scene). Fix a distribution D over *scenes* x (structures for a fixed language). A sequent $\Gamma \vdash \varphi$ is *true in x* if every assignment to its free variables that makes all of Γ true in x makes φ true in x . A rule instance (premise sequents, conclusion sequent) *holds in x* if its conclusion is true in x whenever all its premises are. A rule or schema ρ is *valid in x* if all its instances hold in x , and $(1 - \varepsilon)$ -*valid per scene* if $\Pr_{x \sim D}[\rho \text{ is valid in } x] \geq 1 - \varepsilon$.

The sequent level matters: rules that discharge assumptions ($\rightarrow$ I, *reductio*) or introduce eigenvariables ($\forall$ I) are not truth-preserving at formula level, but they are at sequent level. So hypothetical contexts are covered.

Lemma 3.11 (Commitment order). *[L3 Lemma A, as repaired]*

- (i) Random world, rules fixed first. Let $R = \{\rho_1, \dots, \rho_m\}$ be a finite library fixed before $x \sim D$ is drawn, each ρ_i $(1 - \varepsilon_i)$ -valid per scene. Then

$$\Pr_{x \sim D} [\text{some } R\text{-derivation from premises true in } x \text{ has a conclusion false in } x] \leq \sum_{i=1}^m \varepsilon_i,$$

where the derivation may be chosen after seeing x ; the bound holds simultaneously for all provers.

- (ii) Fixed world, prover-chosen instances. Fix a world x_0 , a schema ρ and a distribution μ on its instances, and call ρ $(1 - \varepsilon)$ -valid on average if $\mu\{\iota : \iota \text{ holds in } x_0\} \geq 1 - \varepsilon$. Suppose ρ has an instance failing in x_0 . It must have one if $\mu\{\iota : \iota \text{ fails in } x_0\} > 0$, and it may have one even when $\varepsilon = 0$, if the failing instance lies off the support of μ . Then a prover that submits it obtains, with probability 1, a one-step derivation from premises true in x_0 to a conclusion false in x_0 . No bound of the form (i) holds in this regime.
- (iii) Fresh randomness per check. Each claim is a polynomial identity $p \equiv q$ in n variables over a field F , committed by an adaptive prover. The verifier enforces $\deg(p - q) \leq d$, computing the degree from an explicit expression; only then does it draw a point uniformly from S_k^n , $S_k \subseteq F$ finite, independently of everything before, and accept iff p and q agree there. Call the claim false if $p - q$ is nonzero as a polynomial over F . A false k -th claim is accepted with probability at most $d/|S_k|$. So some false claim among the first N (N fixed in advance) is accepted with probability at most $\sum_{k \leq N} d/|S_k|$, uniformly over adaptive provers; for unboundedly many claims, $|S_k| \geq d2^k/\delta$ gives at most δ .

Proof idea. (i) Union bound over the whole library: with probability $\geq 1 - \sum_i \varepsilon_i$ every ρ_i is valid in x . This event does not depend on the prover, and on it induction on derivations preserves truth in x . (ii) Nothing is random. (iii) Schwartz–Zippel for each claim, conditional on everything committed before its point is drawn, then a union bound. Full proof in Appendix B. $\square$

Part (i) is the chaining lemma of PAC-semantics (Valiant, 2000) with the quantifier order made explicit; part (iii) is the Schwartz–Zippel lemma (Schwartz, 1980; Zippel, 1979; DeMillo and Lipton, 1978); the whole is folklore. The lemma gives two sufficient conditions and one failing regime. It is *not* a characterization: we do not claim that average-case guarantees survive search *only* in regimes (i) and (iii). Each hypothesis is load-bearing (examples in Appendix B.3). In (i), validity must hold *per scene*; a guarantee per random (scene, instance) pair does not give it, and a prover who picks the instance inside the scene is in regime (ii). In (iii), the verifier must

regime	what is random	drawn when	sound under search?
(i) random scene, fixed library	scene $x \sim D$	after the rules are fixed	yes, $\leq \sum_i \varepsilon_i$
(ii) one world, instance average	instance $\iota \sim \mu$	never: the prover chooses	no
(iii) fresh per-check randomness learned verifier	evaluation point training sample	after each claim before the search	yes, $\leq \sum_k d/ S_k $ no (Corollary 3.3)

Table 3: The commitment order (Lemma 3.11). In the two sound regimes the randomness is drawn after the rules or claims are fixed and outside the prover’s control.

enforce the degree bound (repeated squaring can hide degree 2^k), and $p - q$ must be nonzero over the field used: $(a + b)^p - a^p - b^p$ vanishes over $\mathbb{F}_p$, so evaluation mod p always accepts the freshman’s dream.

Reading. Table 3 sorts the channels of this paper. Mathematics has one world, so a learned checker’s in-distribution accuracy is randomness over instances that the prover chooses: regime (ii), of which Theorem 3.2 is an instance. Its training sample is drawn before the prover searches, so a learned checker is never a regime-(iii) oracle. Fresh per-check randomness, by contrast, is a legitimate world channel (W) in mathematics; part (iii) gives conditions under which such an oracle is worst-case sound. The random evaluation used in the experiments of Section 12 fits this setting only for value errors of polynomial identities, such as the freshman’s dream; a definedness error such as $x/x \triangleright 1$, which fails only at $x = 0$, lies outside it. In physics, the uncertainty that matters for an export rule (which carries a result from an idealized context back to the world) is over situations, and regime (i) makes PAC-semantics a worst-case-over-derivations semantics for bridges, with error at most the summed validity defects of the library. The caveat is distribution shift: olympiad problems are selected to stress idealizations, so the test distribution differs from the training one exactly where it matters (Section 11). For justification, the lemma gives sufficient conditions for a statistical warrant to compose with search: the chance event is drawn after the reasoner has committed, and outside its control.

3.6 Voting ensembles: incoherence without information

A common response to an exploitable verifier is an ensemble that accepts a step if most members do. With coherence feedback this looks attractive: a derivation of $\perp$ from premises certified coherent shows that something accepted is invalid. Stepwise voting, however, can produce an incoherence that implicates no member.

We use the online protocol of Section 6 in minimal form. Hypotheses are step sets in a countable class $\mathcal{H}$ with prior w ; the target $R^* \in \mathcal{H}$ is unknown. A family $\mathcal{A}$ of *designated contexts*, finite premise sets with $\perp \notin \text{Cl}_{R^*}(A)$, is fixed. An *argument* π from A is a finite derivation tree each of whose nodes is either a leaf labelled by an element of A or the conclusion of a step whose premises label its children (possibly none); $\text{Steps}(\pi)$ is the set of its steps. In round t the learner holds $\text{VS}_t \subseteq \mathcal{H}$ (initially $\mathcal{H}$) and announces an accepted set $\hat{R}_t$, whose steps the reasoner may chain. The environment then either (P) presents some $s \in R^*$, and the learner deletes every $R \in \text{VS}_t$ with $s \notin R$; or (N) exhibits a *detected incoherence*, an argument π from some $A \in \mathcal{A}$ to $\perp$ with $\text{Steps}(\pi) \subseteq \hat{R}_t$, and the learner deletes every $R \in \text{VS}_t$ with $\text{Steps}(\pi) \subseteq R$; or does nothing. Since chaining steps of R^* cannot reach $\perp$ from A , $\text{Steps}(\pi) \not\subseteq R^*$ and the target is never deleted (the negative-bag lemma of Section 6). *Majority aggregation* announces

$$\hat{R}_t = \{s : w(\{R \in \text{VS}_t : s \in R\}) > \frac{1}{2}w(\text{VS}_t)\}.$$

Theorem 3.12 (Doctrinal paradox: uninformative incoherence). *[T2 Thm 2.4, as repaired] Let $\mathcal{H} = \{h_1, h_2, h_3\}$ with uniform prior, where $h_i = \{\Gamma \triangleright \varphi : T_i \cup \Gamma \models \varphi\}$ for*

$$T_1 = \{p, q\}, \quad T_2 = \{p, \neg q\}, \quad T_3 = \{\neg p, q\},$$

and let $\mathcal{A} = \{\emptyset\}$. Every h_i is coherent, and any of them may be the target. Under majority aggregation, for every target there is an environment under which the prover exhibits the same detected incoherence in every round and no hypothesis is ever deleted. The environment presents only that incoherence, or interleaves it with uninformative positive data such as $\triangleright p \vee \neg p$, which lie in every h_i .

Proof idea. While $\text{VS}_t = \mathcal{H}$, the majority set is the set of steps in at least two h_i . The argument

$$\triangleright p, \quad \triangleright q, \quad p, q \triangleright p \wedge q, \quad \triangleright \neg(p \wedge q), \quad p \wedge q, \quad \neg(p \wedge q) \triangleright \perp$$

uses only majority steps: $\triangleright p \in h_1, h_2$; $\triangleright q \in h_1, h_3$; $\triangleright \neg(p \wedge q) \in h_2, h_3$; the last two are in all three. Yet h_1 lacks $\triangleright \neg(p \wedge q)$, h_2 lacks $\triangleright q$, and h_3 lacks $\triangleright p$. So the detection deletes nothing, nor do tautologous data, and the situation repeats forever. Full proof in Appendix B. $\square$

The environment must withhold discriminating data. If the target is h_1 and $\triangleright p$ is presented, h_3 is deleted, majority over $\{h_1, h_2\}$ is $h_1 \cap h_2$, and the argument is unavailable ($\triangleright q \notin h_2$). As the h_i are pairwise incomparable, on a text enumerating the target every wrong hypothesis is eventually refuted. So the theorem concerns what coherence feedback can guarantee round by round against an unhelpful environment, not convergence on complete data.

This is Kornhauser and Sager’s doctrinal paradox (Kornhauser and Sager, 1986) and Pettit’s discursive dilemma (Pettit, 2001), transposed from judges to verifiers. List and Pettit (2002) prove that no aggregation function satisfies universal domain, anonymity and systematicity while producing consistent and complete collective judgments, for agendas with at least two atoms and their conjunction (or disjunction, or material conditional). For oligarchic aggregation under mere deductive closure, see Gärdenfors (2006) and Dietrich and List (2008).

Reading. Each h_i is a coherent calculus, yet the vote derives $\perp$ from no premises, so by Lemma 3.1 the majority acceptance region is unsound relative to *every* candidate target: voting produces an accepted set that no member endorses. Worse, the incoherence does not say which member is wrong, so the coherence channel, which should supply the negative data that imitation lacks, is silenced. The repair (Section 6): against a worst-case prover, every possible detection is guaranteed to remove at least half of the remaining prior weight only if the accepted set lies inside every member of a coalition of at least half the weight (*oligarchic* acceptance). Each detection then costs at least one bit, which bounds detections only, not incompleteness. Oligarchy also requires truth maintenance: lemmas certified by an earlier coalition must be re-validated when the coalition changes, or the union over coalitions reproduces the paradox across time.

3.7 Search in practice

Experiment C (Section 12) runs a black-box adversarial prover (bidirectional best-first search, 3000 verifier queries per false goal) against learned verifiers for partial field algebra, trained on simulated human derivations with noise and systematic fallacies (computed). A gradient-boosting step

classifier, given oracle-labelled positive *and* negative examples, reaches test AUC 0.995 on held-out human-like steps. Calibrated to 99% in-distribution balanced precision $\text{TPR}/(\text{TPR} + \text{FPR})$, it still lets the prover prove 16.3 of 23 false goal equations (mean over three seeds). The learned calculus, least-general-generalization schemas repaired by step-level world feedback, proves 0 of the 23 false goals and 32.3 of 33 true identities, as does the ground-truth calculus. That learned verifiers are exploited by search is well known in machine-learning practice: reward-model overoptimization follows smooth scaling laws (Gao et al., 2023), and verifier-reranked accuracy peaks and then falls as more solutions are sampled (Cobbe et al., 2021). This section explains why in-distribution accuracy cannot rule it out. In-distribution accuracy is a statement about a measure fixed before the prover moves (Lemma 3.11(ii)); soundness of the reasoner is a statement about a set (Lemma 3.1); and no bound on the former controls the latter (Theorem 3.2). The theorems do not predict the *rate* at which a given search finds errors; for neural verifiers it is graded, not all-or-nothing, as the overoptimization curves show.

4 Cautious verification and its price

Section 3 reduced the soundness of a search-driven reasoner to a set inclusion: everything the verifier accepts must lie in $\text{Sound}(R^*)$, and average-case accuracy does not certify this. This section shows how a learner can certify the inclusion, and what it costs. The answer is caution. Accept a step only if every hypothesis consistent with the evidence accepts it, reject it only if none does, and otherwise escalate it to a human or an oracle. This is reliable learning (Rivest and Sloan, 1988a), perfect selective classification (El-Yaniv and Wiener, 2010a) and the “accept” side of KWIK (Li et al., 2008a), applied to steps. It is sound against every prover from the first datum on, and it is the most permissive 0-sound verifier (Theorem 4.2).

Its price is paid in escalations, never in soundness. The worst case is exactly a bounded, target-relative form of Wright’s elasticity (Theorem 4.4). Table 4 gives it for the main classes. It is linear in the step size for rules cited by name, and at least polynomial of degree k for k rules without citations. It is exponential in description length for unstructured classes, and super-exponential (over growing signatures) if invalidity must be certified too. A Bayesian version adds time-uniform soundness from noisy data, with no union bound over queries (Section 4.4), but its escalation cost is exponential in description length unless structure intervenes. In one line: *structure, not the prior, makes sound learned verification affordable*.

The results are elementary, and several are trivial once set up. As far as we know, four are new: the exact single-schema budget, the tagged/untagged separation, the Bell-number bound for two-sided checking over growing signatures, and the tightness of the Ville constant.

Conventions. Protocol, soundness and cost are as in Definitions 2.18, 2.19 and 2.21. Human data $P_0 \subseteq R^*$ are fixed unless said otherwise. The cost of a run is the (expected) number of rounds in which a *valid* query is not accepted outright; invalid queries are not charged. $\text{Esc}(\mathcal{H} \mid P_0)$ is the infimum, over 0-sound verifiers, of the worst-case cost over targets in $\mathcal{H}$ containing P_0 and over honest query sequences. $\text{Esc}(\mathcal{H}; N)$ restricts queries to steps of size (number of symbols) at most N .

Two facts from Section 2 do most of the combinatorial work. First, the *rank* $\mu(t) = |t| - |\text{vars}(t)|$ strictly decreases along strict generalization (Lemma 2.6; T1 Lemma 1.2). Second, the lgg of instances of σ recovers σ iff two *witness events* hold (Lemma 2.7; T1 Lemma 1.3): (R) no metavariable is always instantiated by terms with the same root, and (D) no two metavariables

are always instantiated alike.

4.1 The version-space verifier is sound and optimal

Definition 4.1 (Version space; VS verifier). ^[T1 §3] Let $\mathcal{H} \subseteq 2^S$ be realizable. For sets P (positive) and P^- (negative) of labelled steps, $\text{VS}(P, P^-) = \{R \in \mathcal{H} : P \subseteq R, R \cap P^- = \emptyset\}$, and $\text{VS}(P) = \text{VS}(P, \emptyset)$. The *VS verifier* starts with $P = P_0$, $P^- = \emptyset$, and answers a query q with **ACCEPT** if $q \in \bigcap \text{VS}(P, P^-)$, with **REJECT** if $q \notin \bigcup \text{VS}(P, P^-)$, and with **ESCALATE** otherwise; an escalated q joins P or P^- according to the oracle’s label. For a history h , VS_h is the version space computed from P_0 and the labels in h .

In the terms of [El-Yaniv and Wiener \(2010a\)](#), this is the consistent selective strategy: it escalates exactly on the disagreement region of the version space.

Theorem 4.2 (Soundness and optimality of caution). ^[T1 Thm 3.1, as repaired]

- (a) *The VS verifier is uniformly 0-sound. For every target $R^* \in \mathcal{H}$ with $P_0 \subseteq R^*$ and every prover, $R^* \in \text{VS}(P, P^-)$ at all times. So everything it accepts lies in R^* , and a reasoner that chains accepted steps is sound (Lemma 3.1).*
- (b) *Let V be any (possibly randomized) δ -sound verifier for $\mathcal{H}$ with human data P_0 . Fix a target $R^* \in \mathcal{H}$ with $P_0 \subseteq R^*$, a history h of t rounds (queries $q_1, \dots, q_t$, answers, oracle labels) whose labels agree with R^* , and a query $q \notin \bigcap \text{VS}_h$. Let p be the joint (unconditional) probability, over V ’s coins, that against R^* the prover issuing $q_1, \dots, q_t, q$ produces h and then has q accepted. Then $p \leq \delta$.*
- (c) *In particular, a deterministic δ -sound verifier with $\delta < 1$ accepts no $q \notin \bigcap \text{VS}_h$ at any history h that it produces against a target in $\mathcal{H}$. History by history, its acceptance region lies inside $\bigcap \text{VS}_h$, the set the VS verifier accepts.*

Lean: (a) `thm_3_1_a`; (c) `thm_3_1_b`

Proof. (a) All labels are truthful, so the target never leaves the version space, and hence $\bigcap \text{VS} \subseteq R^*$.

(b) Pick $R' \in \text{VS}_h$ with $q \notin R'$. Then $R' \in \mathcal{H}$, $P_0 \subseteq R'$, and R' agrees with every label in h . Run the same prover against target R' . In each round the verifier’s coins have the same law, and on $q_1, \dots, q_t$ the oracle returns the same labels as under R^* . So the event “ h , then q accepted” has the same probability p under R' as under R^* ; this is a coupling through the coins (Appendix C.1). Since $q \notin R'$, δ -soundness for the target R' gives $p \leq \delta$.

(c) A deterministic verifier answering fixed queries has $p \in \{0, 1\}$, so $p \leq \delta < 1$ forces $p = 0$. □

The joint form in (b) is essential. Read conditionally on h , (b) is false. A verifier that accepts everything with probability ε , and otherwise runs the VS verifier, is ε -sound. Yet it reaches some histories only in that reckless mode, and after them it accepts steps outside $\bigcap \text{VS}_h$ with conditional probability 1 (Appendix C.1). With random human data the coupling must also transport the data, giving only $\mathbb{P}_{R'}[\text{data}] \cdot p \leq \delta$; this is where Section 4.4 takes over. Part (a) and case (c) are formalized in Lean (`lean/InfLearn/StepSoundness.lean`). The theorem is trivial once set up; the set-up shows that the VS verifier is the *largest* 0-sound acceptance rule, so its escalations are forced on every 0-sound verifier.

Reading. A learner can *guess* the calculus (the simplest or most probable hypothesis), which is unsound under search (Section 3). Or it can *certify* steps, accepting only what every calculus consistent with the evidence accepts, which is sound from the first datum without negative data. Gold’s problem (Gold, 1967) does not threaten certification. An over-general hypothesis, tonk included, never licenses a step by itself, because the target stays in the version space as a guardian. Over-generality costs only completeness. Since contexts are part of judgments, this covers hypothetical steps verbatim ($\rightarrow I$; “assume the air pressure is 0”). As a model of justification: a step is accepted when no hypothesis left open by the evidence defeats it, relative to $\mathcal{H}$ and the human calculus, not to truth.

4.2 The escalation dimension is a positive elasticity

Definition 4.3 (Elastic chain; positive elasticity). ^[T1 §3] Let $R \in \mathcal{H}$ with $P_0 \subseteq R$. An *elastic chain in R* (relative to P_0) is a sequence $s_1, \dots, s_m \in R$ with $s_i \notin \bigcap \text{VS}(P_0 \cup \{s_1, \dots, s_{i-1}\})$ for every i . Equivalently, there are *witnesses* $R_i \in \mathcal{H}$ with $P_0 \cup \{s_1, \dots, s_{i-1}\} \subseteq R_i \not\supseteq s_i$. The *positive elasticity* $\text{el}(\mathcal{H}, R \mid P_0)$ is the supremum of the lengths of such chains, and $\text{el}(\mathcal{H}, R \mid P_0; N)$ restricts the s_i to size at most N .

Wright’s finite elasticity (Wright, 1989; Motoki et al., 1991) forbids infinite sequences of this kind and suffices for identification in the limit from text. Positive elasticity is its bounded version, with the points in the target.

Theorem 4.4 (Escalation dimension = positive elasticity). ^[T1 Thm 3.2, proof as repaired] For every class $\mathcal{H}$ and human data P_0 ,

$$\text{Esc}(\mathcal{H} \mid P_0) = \sup_{R \in \mathcal{H}, R \supseteq P_0} \text{el}(\mathcal{H}, R \mid P_0).$$

The VS verifier attains it: against an honest prover for target R^ , it never rejects and escalates at most $\text{el}(\mathcal{H}, R^* \mid P_0)$ times. Conversely, if $s_1, \dots, s_m$ is an elastic chain in some $R \in \mathcal{H}$, the honest prover for R that queries $s_1, \dots, s_m$ in order forces every δ -sound (possibly randomized) verifier to expected cost at least $(1 - \delta)m$, and every 0-sound verifier to cost m almost surely. The same holds with all steps restricted to size at most N .*

Proof idea. Against an honest prover, the VS verifier escalates q exactly when $q \notin \bigcap \text{VS}(P_0 \cup E)$, where E is the set of earlier escalated queries. So the escalated queries form an elastic chain in R^* . Conversely, the targets R and R_i label $s_1, \dots, s_{i-1}$ alike, so the transcripts up to round i have the same law under both. Under R_i the step s_i is invalid, so it is accepted with probability at most δ ; hence the same holds under R . This is the joint form of Theorem 4.2(b). Full proof in Appendix C.2. $\square$

So the cost of caution depends only on the class, the target and the human data P_0 , not on the prover, the human distribution or a prior. It is KWIK’s adversarial abstention count (Li et al., 2008a, 2011), made one-sided. Rivest and Sloan (1988a) and El-Yaniv and Wiener (2010a) bound instead the distributional mass of abstention, a usefulness guarantee (Section 5).

Proposition 4.5 (Intersection-closed classes). ^[T1 Prop 3.3] (known in substance) Suppose $\mathcal{H} \cup \{\emptyset\}$ is closed under arbitrary intersections. For P contained in some member, let $\text{cl}(P)$ be the intersection of all members of $\mathcal{H}$ that contain P . Then:

- (i) $\bigcap \text{VS}(P) = \text{cl}(P)$, the least member of $\mathcal{H} \cup \{\emptyset\}$ containing P . So, against an honest prover, the VS verifier is the closure algorithm: it accepts iff $q \in \text{cl}(P_0 \cup E)$, where E is the set of steps escalated so far.
- (ii) $\text{el}(\mathcal{H}, R \mid P_0)$ is the largest m for which there is a strict chain $\text{cl}(P_0) = C_0 \subsetneq C_1 \subsetneq \dots \subsetneq C_m \subseteq R$ of members of $\mathcal{H} \cup \{\emptyset\}$.

Proof idea. An elastic chain gives the strict chain $C_i = \text{cl}(P_0 \cup \{s_1, \dots, s_i\})$; conversely, pick $s_i \in C_i \setminus C_{i-1}$ (Appendix C.3). This is the closure algorithm of [Helmhold et al. \(1990\)](#); its use for one-sided learning from positive examples goes back to [Natarajan \(1987\)](#). We claim nothing new here beyond reading the chain length as an escalation count.

Lemma 4.6 (Single schemas are intersection-closed). *[T1 §3, after Prop 3.3] $H_1 = \{\text{inst}(\sigma) : \sigma \text{ a schema}\} \cup \{\emptyset\}$ is closed under arbitrary intersections. For nonempty P , $\text{cl}(P) = \text{inst}(\text{lbg}P)$.*

Proof idea. Two instance sets meet in the instance set of a most general common instance, or not at all. A nonempty intersection contains a ground term, which has only finitely many generalizations up to renaming. Full proof in Appendix C.3.

Reading. Caution climbs a lattice from below. Each escalation moves the accepted set one strict step up, and the price is the longest climb from what the human data force to the target. For schemas the lattice is Plotkin’s ([Plotkin, 1970](#); [Reynolds, 1970](#)), the closure algorithm is anti-unification, and each step up costs a unit of rank. A learner that acquires a connective’s inferential role from its uses climbs in this way. The elasticity bounds how often it must ask before its accepted inferences cover that role.

4.3 What caution costs: from named rules to unstructured classes

Theorem 4.7 (One schema, learned by anti-unification). *[T1 Thm 3.4, as restated] Let the target be $\text{inst}(\sigma^*) \in H_1$, written σ^* inside el .*

- (i) For nonempty $P_0 \subseteq \text{inst}(\sigma^*)$: $\text{el}(H_1, \sigma^* \mid P_0) \leq \mu(\text{lbg}P_0) - \mu(\sigma^*)$.
- (ii) From no data: $\text{el}(H_1, \sigma^* \mid \emptyset; N) \leq 1 + N - \mu(\sigma^*)$. Hence $\text{Esc}(H_1; N) \leq N + 1$.
- (iii) This is tight. If the signature contains two constants a, b and a unary symbol g , then for the universal schema $\sigma^* = x$, $\text{el}(H_1, x \mid \emptyset; N) = N + 1$. So $\text{Esc}(H_1; N) = N + 1$.

Proof idea. (i) With $g_i = \text{lbg}(P_0 \cup \{s_1, \dots, s_i\})$, Lemma 4.6 gives $s_i \notin \text{inst}(g_{i-1})$, so $\text{lbg}P_0 = g_0 \prec \dots \prec g_m \preceq \sigma^*$ strictly (by the lbg property; no assumption on the signature is needed), and the rank lemma bounds m . (ii) From no data the first query escalates, and it has rank at most N . (iii) Query $g^{N-1}(a)$, $g^{N-1}(b)$, $g^{N-2}(a)$, $g^{N-3}(a)$, $\dots$, a . The successive lggs are $g^{N-1}(a)$, $g^{N-1}(x)$, $g^{N-2}(x)$, $\dots$, x , of ranks $N, N-1, \dots, 0$, and each query lies outside the previous lgg. Full proof in Appendix C.4. $\square$

Exhaustive search gives exactly $N + 1$ with two constants, and N with one (computed). After a corpus P_0 , the residual budget is at most the *generality gap* $\mu(\text{lbg}P_0) - \mu(\sigma^*)$. It is zero once the corpus satisfies the witness events (Section 5). These numbers are small only because the protocol is one-sided.

Proposition 4.8 (Certifying invalidity is expensive). *[T1 Prop 3.5, as revised] Call a verifier two-sided if it is 0-sound and rejects only steps outside $\bigcup \text{VS}$, and count all its escalations, of valid and invalid queries alike. Over a signature with constants $a, b_1, \dots, b_n$ and an n -ary symbol f , there are a target in H_1 and a prover that force every two-sided verifier for H_1 to escalate at least $\text{Bell}(n)$ times on steps of size $n + 1$. Over a fixed finite signature, by contrast, the two-sided VS verifier escalates on steps of size at most N at most as often as there are such steps, $2^{O(N)}$.*

Proof idea. Let target and P_0 be $\{f(a, \dots, a)\}$. For a partition π of $[n]$, s_π puts b_j and σ_π a metavariable x_j on the j -th block; $s_{\pi'} \in \text{inst}(\sigma_\pi)$ iff π refines π' . Query the s_π finest first: each is invalid, yet σ_π is still consistent when s_π is queried. Full proof in Appendix C.5. $\square$

Brute force gives $\text{Bell}(2), \text{Bell}(3), \text{Bell}(4) = 2, 5, 15$ forced escalations (computed). Certifying validity costs at most $N + 1$ escalations over any signature. Certifying invalidity costs at least $\text{Bell}(N - 1)$ over a signature with N constants and an $(N - 1)$ -ary symbol; compare the exponential KWIK cost of conjunctions (Li et al., 2011) and the negative results for two-sided reliable learning (Kivinen, 1995). A step checker never needs it: *one-sidedness is what makes learned verification cheap.*

Theorem 4.9 (k rules cited by name). *[T1 Thm 3.6, as repaired] A tagged step (i, s) records the rule $i \in [k]$ that it cites, as formal proofs do: Lean lemma names, Metamath labels, natural-deduction rule names. Let*

$$H_k^{\text{tag}} = \left\{ \bigcup_{i \leq k} \{i\} \times \text{inst}(\sigma_i) \right\},$$

with target schemas $\sigma_1^*, \dots, \sigma_k^*$ and human data P_0 whose tag- i part is $P_0^{(i)}$. Then

$$\text{el}(H_k^{\text{tag}}, R^* \mid P_0; N) = \sum_{i=1}^k \text{el}_i, \quad \text{el}_i \leq \begin{cases} \mu(\lg P_0^{(i)}) - \mu(\sigma_i^*) & \text{if } P_0^{(i)} \neq \emptyset, \\ 1 + N - \mu(\sigma_i^*) & \text{if } P_0^{(i)} = \emptyset, \end{cases}$$

where el_i is the positive elasticity of $\text{inst}(\sigma_i^*)$ among single-schema instance sets, relative to $P_0^{(i)}$, on steps of size at most N . In particular $\text{Esc}(H_k^{\text{tag}}; N) \leq k(N + 1)$, with equality under the hypothesis of Theorem 4.7(iii).

Proof idea. H_k^{tag} is not literally intersection-closed (a componentwise intersection can empty one component only). What the proof uses is that the version space *factors*, $\text{VS}(P) = \prod_i \text{VS}_i(P^{(i)})$ with nonempty single-schema factors, so $\bigcap \text{VS}(P) = \bigcup_i \{i\} \times \bigcap \text{VS}_i(P^{(i)})$. Elastic chains are thus interleavings of per-rule chains, each bounded by Theorem 4.7. Full proof in Appendix C.6. $\square$

Without citations the picture changes, even when only one rule is in use.

Theorem 4.10 (k rules without citations). *[T1 Thm 3.7, scope as repaired] Let $H_k = \{\bigcup_{l \leq k} \text{inst}(\tau_l)\}$ be the class of untagged unions of at most k schemas.*

- (i) Lower bound, even for one rule. Suppose the signature contains a k -ary p , a unary g and a constant c . Then an honest sequence for the single-rule target $\text{inst}(p(x_1, \dots, x_k)) \in H_1 \subseteq H_k$ forces $\lfloor (N - 1)/k \rfloor^k$ escalations on steps of size at most N . So $\text{Esc}(H_k; N) \geq \lfloor (N - 1)/k \rfloor^k$, whereas $\text{Esc}(H_1; N) \leq N + 1$.
- (ii) Upper bound. For $k \geq 2$, $\text{Esc}(H_k; N) \leq (k^{N+1} - 1)/(k - 1)$.

(iii) Linear flat schemas. Let the steps be the points of $\{a, b\}^n$, $n \geq k$, encoded as $f(c_1, \dots, c_n)$. Let $\mathcal{H}$ be the unions of at most k linear flat schemas $f(t_1, \dots, t_n)$: each t_j is a , b or a metavariable, and no metavariable repeats. These are the unions of at most k subcubes. Then

$$\sum_{w=0}^k \binom{n}{w} \leq \text{Esc}(\mathcal{H}) \leq 1 + (2^k - 1) \binom{n}{k},$$

which is $\Theta(n^k)$. If metavariables may repeat, the lower bound persists, and $\text{Esc} \leq 1 + (2^{2k} - 1) \binom{n}{2k}$ for $n \geq 2k$. So Esc then lies between $\Omega(n^k)$ and $O(n^{2k})$; only the linear case is pinned down to $\Theta(n^k)$.

Proof idea. (i) Query $p(g^{a_1}c, \dots, g^{a_k}c)$, $a \in \{0, \dots, n\}^k$, $n+1 = \lfloor (N-1)/k \rfloor$, by non-increasing $\sum_j a_j$. Each earlier step lies in the union of the k schemas $p(x_1, \dots, g^{a_j+1}(x_j), \dots, x_k)$, which misses the current one. (ii) Indices whose prefixes always sit in one block of the current witness yield a strictly increasing lgg chain, so at most $N+1$ of them; a Ramsey-type recursion $g(d) \leq 1 + k g(d-1)$ finishes. (iii) A subcube avoiding p lies in a half-cube $\{x_j \neq p_j\}$, so p escapes iff it shows a new pattern on some k coordinates; count patterns, and order points of weight $\leq k$ by weight. Full proof in Appendix C.7. $\square$

Exact values (computed): for subcubes with $k = 2$, $n = 2, 3, 4$: 4, 7, 11 = $\sum_{w \leq 2} \binom{n}{w}$. For deep schemas over $\{c, g/1, p/2\}$ with $k = 2$, $N = 3, 4, 5$: 4, 8, 13. The partition abstraction of (ii) attains $2^d - 1$ ($k = 2$, $d = 3, 4$), so a polynomial bound must use more than this abstraction, such as intersection-closure.

Conjecture 4.11 (Binomial bound). *[T1 Conj 3.8] (conjecture) Let $\mathcal{C}$ be a family of subsets of U that is closed under intersections and contains U . Let h be its height: the number of inclusions in the longest strict chain of members from $\bigcap \mathcal{C}$ to U . Then the positive elasticity of the unions of 1 to k members of $\mathcal{C}$, relative to $P_0 = \emptyset$ and with target U , is at most $\binom{h+k-1}{k}$.*

Evidence (as corrected after verification). The bound is exact for $k = 1$ (Proposition 4.5) and for $h = 2$ (after removing $\bigcap \mathcal{C}$, the members other than U are $\emptyset$ and pairwise disjoint atoms, each holding at most one chain element, so the witness of the last element covers at most k earlier ones; singletons attain $k+1$). If true, it is tight at $k = 2$ for every height tested: the flats of the graphic matroid $M(K_{h+1})$ have height h , and for $h = 2, \dots, 5$ their 2-union elasticity is exactly $\binom{h+1}{2}$, namely 3, 6, 10, 15 (computed). The case $(k, h) = (2, 3)$ is proved (Appendix C.8). Hill-climbing with $k = 2$ found exactly the conjectured maxima 6 and 10 for $h \leq 3$ and $h \leq 4$, on 9 and 12 points. A run with $h \leq 5$ used only 14 points. Elasticity never exceeds the number of points, so its maximum of 14 is a ceiling artifact: it is not evidence of slack below 15, and it does not test the conjecture, since a counterexample needs at least 16 points. An independent search on 11–12 points with $h \leq 4$ found no counterexample (computed). For schemas $h \leq N+1$, so the conjecture would give $\text{Esc}(H_k; N) \leq \binom{N+k}{k}$. This would match (i) up to $e^{O(k)}$, making the truth $\Theta_k(N^k)$.

Theorem 4.12 (Unstructured classes). *[T1 Thm 3.9] (known: KWIK enumeration) For finite $\mathcal{H}$, $\text{Esc}(\mathcal{H}) \leq |\mathcal{H}| - 1$. The class $\{U \setminus \{u\} : u \in U\}$ attains $|U| - 1$.*

Proof. Each escalation of an honest step $q \notin \bigcap \text{VS}$ removes from the version space a hypothesis that misses q , and the target is never removed. For the class, query the elements of $R^* = U \setminus \{u^*\}$ in any order. When s is queried, the witness $U \setminus \{s\}$ contains all earlier queries and misses s . $\square$

class (steps of size $\leq N$)	worst-case escalations	result
intersection-closed class	longest strict chain of members from $\text{cl}(P_0)$ into the target	Proposition 4.5
one schema, after a corpus P_0	$\leq \mu(\text{lgg}P_0) - \mu(\sigma^*)$ (generality gap)	Theorem 4.7(i)
one schema, no data	$\leq 1 + N - \mu(\sigma^*)$; worst case $N + 1$, tight	Theorem 4.7(ii,iii)
k rules cited by name (tagged)	$\sum_i \text{el}_i \leq k(N + 1)$, tight	Theorem 4.9
k rules without citations	$\geq \lfloor (N - 1)/k \rfloor^k$ even for a one-rule target; $\leq (k^{N+1} - 1)/(k - 1)$; conjecturally $\leq \binom{N+k}{k}$	Theorem 4.10(i,ii), Conjecture 4.11
k linear flat schemas on $\{a, b\}^n$	$\Theta(n^k)$; between $\Omega(n^k)$ and $O(n^{2k})$ with repeated metavariables	Theorem 4.10(iii)
unstructured finite class	$ \mathcal{H} - 1$, tight; $2^L - 1$ for L -bit hypotheses	Theorem 4.12
two-sided, one schema	$\geq \text{Bell}(N - 1)$ over a growing signature; $\leq 2^{O(N)}$ over a fixed one	Proposition 4.8
Bayes-conservative, VS	$\leq \ln(1/w^*)/\delta$; $\tilde{\Theta}(2^L)$ on unstructured classes with $w^* = 2^{-L}$	Theorem 4.17, Corollary 4.18
posterior, $\delta \leq w^*$		
structure plus Bayes	$\min\{\text{el}(\mathcal{H}, R^* \mid P_0), \ln(1/w^*)/\delta\}$	Corollary 4.19

Table 4: The price of caution: escalations (human interventions on valid steps) needed by a sound verifier against an honest prover. “Tight” means attained over a signature with two constants and a unary symbol. All entries are proved except the conjectured bound.

This is KWIK’s enumeration bound (Li et al., 2008a): 2^L hypotheses of L bits can need $2^L - 1$ escalations. Single schemas of size $\leq N$ (with a binary symbol) also number $2^{\Omega(N)}$, yet need at most $N + 1$. The price is measured by lattice height, not cardinality.

Reading. Table 4 quantifies one way in which “inventing a language” with named rules, the user’s description of how mathematics was formalized, makes inference learnable. Formal proofs cite their rules, and then caution costs linearly in step size, additively over rules. Informal proofs and physics solutions cite none. The verifier must then pay at least $\lfloor (N - 1)/k \rfloor^k$ even when one rule is in use, since it must rule out that the steps seen so far are covered by k narrower rules. The proved upper bound is exponential, and $\Theta_k(N^k)$ is conjectured. Informal steps are moreover derived, multi-step elements of $\text{Sound}(R^*)$, and Section 10 treats them separately.

4.4 Bayes-conservative verification and Ville’s inequality

A posterior relaxes the version space’s demand of truthful data, and makes very large classes usable, which eases (but does not remove) the demand of realizability. MAP acceptance is unsound whenever an over-general hypothesis is temporarily most probable (Section 3.4). Caution instead thresholds the posterior probability of invalidity.

Definition 4.13 (Bayes-conservative verifier). ^[T1 §4] Let $\mathcal{H}$ be countable with prior w and $w^* = w(R^*) > 0$. Each $R \in \mathcal{H}$ carries a law p_R on S for human data (which may model errors, and may follow the size principle, spreading mass more thinly over larger R) and an oracle kernel $\ell_R(y \mid q)$, $y \in \{0, 1\}$, which is *deterministic* if $\ell_R(y \mid q) = \mathbf{1}[y = \mathbf{1}[q \in R]]$. The model is *well specified*: human data are i.i.d. p_{R^*} , and oracle answers follow $\ell_{R^*}(\cdot \mid q_t)$. After the observations of rounds $\leq t$, the likelihood is $L_t(R)$, the posterior is $w_t(R) \propto w(R)L_t(R)$, and

$$Z_t := \sum_{R \in \mathcal{H}} w(R) \frac{L_t(R)}{L_t(R^*)}, \quad \text{so that} \quad w_t(R^*) = \frac{w^*}{Z_t}.$$

The verifier V_δ answers q with **ACCEPT** iff $w_t(\{R : q \notin R\}) < \delta$, optionally with **REJECT** iff $w_t(\{R : q \in R\}) < \delta_r$, and with **ESCALATE** otherwise. The *VS posterior* is $w_t^{\text{VS}}(R) \propto w(R) \mathbf{1}[R \text{ is consistent with all data}]$.

Theorem 4.14 (VS posterior: deterministic soundness). *[T1 Thm 4.1] Suppose all data are truthful: human positives lie in R^* , and oracle labels are correct. If $\delta \leq w^*$, then V_δ with the VS posterior never accepts an invalid step, for any prover and at any time. The guarantee is per target: it covers every target with $w(R^*) \geq \delta$. Conversely, for every $\delta > w^*$ there is a two-hypothesis class on which V_δ accepts an invalid step at time 0.*

Proof. Truthful data never contradict the target, so $w_t^{\text{VS}}(R^*) = w^*/w(\text{VS}_t) \geq w^*$. For an invalid q , $w_t(\{R : q \notin R\}) \geq w_t(R^*) \geq w^* \geq \delta$, so q is not accepted. For the converse, take $\mathcal{H} = \{R^*, R'\}$ with $R' \supsetneq R^*$, $w(R^*) = w^*$ and $q \in R' \setminus R^*$. Then $w_0(\{R : q \notin R\}) = w^* < \delta$. $\square$

The truth is a *guardian*: an invalid step can be accepted only once the target’s own posterior has fallen below δ . With noisy data this can happen, but only on one global event.

Theorem 4.15 (Time-uniform soundness). *[T1 Thm 4.2, as repaired] Assume the model is well specified and $\delta \leq w^* \delta'$.*

- (a) *Suppose that, conditional on $\mathcal{F}_t$, the next observation has its model law: a fresh $X \sim p_{R^*}$, or an answer $y \sim \ell_{R^*}(\cdot \mid q_{t+1})$. Here $\mathcal{F}_t$ contains the history and all of the prover’s information, including its internal randomness and its choice of q_{t+1} . So the prover cannot foresee human data or oracle noise, and the noise is fresh at every query. Then for every such prover*

$$\mathbb{P}[\exists t : V_\delta \text{ accepts some } q_t \notin R^*] \leq \delta'.$$

- (b) *If the oracle is deterministic, the guarantee is uniform. There is an event of probability at least $1 - \delta'$, depending only on the human data, on which no prover strategy ever gets an invalid step accepted. This holds even for a prover that knows the target and all future human data. Only the human data need to be i.i.d. p_{R^*} .*

Proof idea. Under (a), each observation multiplies the R -term of Z_t by a likelihood ratio of conditional mean $\sum_{o: \ell_{R^*}(o) > 0} \ell_R(o) \leq 1$. So Z is a nonnegative supermartingale with $Z_0 = 1$, and Ville’s inequality (Ville, 1939) gives $\mathbb{P}[\sup_t Z_t \geq 1/\delta'] \leq \delta'$; off this event $w_t(R^*) = w^*/Z_t > \delta$ for all t . In (b), deterministic answers multiply each term by at most 1, so Z_t is bounded pathwise by a supermartingale of the human data alone. Full proof in Appendix C.9. $\square$

Credit. Part (a) is the prior–posterior–ratio martingale of Waudby-Smith and Ramdas (2020), applied to validity relations: the ratio of prior to posterior at the true parameter is a nonnegative martingale. Part (b) is an easy corollary, since truthful constraints multiply Z_t by at most 1. See Ville (1939); Shafer et al. (2011); Ramdas et al. (2023) on Ville’s inequality and test martingales. The closest existing analogues are conservative Bayesian agents that defer to a mentor (Cohen and Hutter, 2020; Cohen et al., 2022). New here is only the tightness of the constant (Proposition 4.16); the application to step validity is routine.

No union bound over queries. A prover may try 10^{100} steps, and one event of probability at least $1 - \delta'$ covers them all, because accepting *any* invalid step requires the posterior of the truth to fall below δ . A per-step scorer has no such single event, and its errors can be found one at a time (Theorem 3.2).

The hypothesis of (a) is needed. Take $\mathcal{H} = 2^{\{0,1\}}$ with uniform prior, $R^* = \{0\}$, symmetric flip noise of rate η , and $\delta' = 0.05$. A prover that queries the invalid step 1 exactly when it knows that the answer will be flipped to “valid” gets 1 accepted with probability 1.0, at both $\eta = 0.1$ and $\eta = 0.3$ (computed; Appendix C.9).

Proposition 4.16 (The constant is tight). ^[T1 Prop 4.3] (proved; computed) Let $\mathcal{H} = \{R^*, R'\}$ with $R^* = \{a, b\}$, $p_{R^*}(a) = 1 - u$, $p_{R^*}(b) = u$, $R' = \{a, c\}$, $p_{R'}(a) = 1$, prior $w(R^*) = w^*$, a deterministic oracle, and $\delta = w^* \delta'$. Let t^* be the least t with $(1 - u)^{-t} > \theta := (1 - w^* \delta') / ((1 - w^*) \delta')$. A prover that waits for t^* human data and then queries the invalid step c gets c accepted with probability

$$(1 - u)^{t^*} \in [(1 - u)/\theta, 1/\theta), \quad \text{where } 1/\theta = \frac{(1 - w^*) \delta'}{1 - w^* \delta'} \leq \delta'.$$

This tends to δ' as $u, w^* \rightarrow 0$. The exact values are $0.989 \delta'$ at $u = 0.05$, $w^* = \delta' = 0.01$, and $0.991 \delta'$ at $u = 0.01$, $w^* = 10^{-3}$, $\delta' = 0.05$.

Proof idea. After t copies of a , the likelihood ratio of R' to R^* is $(1 - u)^{-t}$, and R' dies at the first b . If c is queried earlier, it is escalated, and the answer “invalid” kills R' . Appendix C.10.

Theorem 4.17 (Escalation bound). ^[T1 Thm 4.4] Let the oracle be deterministic, let the human data satisfy the hypothesis of Theorem 4.15(a), and let $\delta_m = \min(\delta, \delta_r)$. Write $\#ESC_t$ for the number of escalations up to time t .

- (i) With probability at least $1 - \delta''$, simultaneously for all t , $\#ESC_t \leq (\ln(1/w^*) + \ln(1/\delta'')) / \delta_m$.
- (ii) With the VS posterior, deterministically, $\#ESC_t \leq \ln(1/w^*) / \delta_m$.

For an honest prover, or when only escalations of valid queries are counted, δ_m may be replaced by δ . Without a reject option ($\delta_r = 0$), the bound on escalations of invalid queries is vacuous.

Proof idea. Z_t evolves multiplicatively. A human datum multiplies it by a factor of conditional mean at most 1. Escalating a valid (invalid) q multiplies it by at most $1 - \delta$ ($1 - \delta_r$), since q was not accepted (rejected). Combine with $Z_t = w^* / w_t(R^*) \geq w^*$ and Ville’s inequality for the data factors. Appendix C.11. $\square$

Corollary 4.18 (Bayes is essentially optimal on unstructured classes). ^[T1 Cor 4.5] On $\mathcal{H} = \{U \setminus \{u\} : u \in U\}$ with the uniform prior ($w^* = 1/|U|$), every δ' -sound verifier has worst-case honest cost at least $(1 - \delta')(1/w^* - 1)$, while $V_{w^* \delta'}$ with the VS posterior is 0-sound and pays at most $\ln(1/w^*) / (w^* \delta')$. The ratio of the two bounds is $\ln(1/w^*) / (\delta'(1 - \delta')(1 - w^*))$, which is $O(\ln(1/w^*) / \delta')$ for $w^*, \delta' \leq 1/2$. With a description-length prior $w^* = 2^{-L}$, the cost is $\tilde{\Theta}(2^L)$.

Proof. The lower bound is Theorem 4.4 applied to the chain of Theorem 4.12, which has length $1/w^* - 1$. The upper bound is Theorem 4.17(ii), and soundness is Theorem 4.14. $\square$

The factor $1/\delta'$ is an artifact of the threshold. With $\delta = w^*$ the VS posterior accepts exactly $\cap$ VS on this class, since every consistent hypothesis has posterior at least w^* . It is then the VS verifier itself, which is exactly optimal.

Corollary 4.19 (Structure plus Bayes). ^[T1 Cor 4.6, as repaired] With truthful data, the VS-posterior verifier with threshold δ never accepts an invalid step, for any prover and every target with $w(R^*) \geq \delta$. Its escalations of valid queries number at most

$$\min \{ \text{el}(\mathcal{H}, R^* \mid P_0), \ln(1/w^*) / \delta \}.$$

Proof. Soundness is Theorem 4.14. An escalated valid q had $w(\{R : q \notin R\}) \geq \delta > 0$, so $q \notin \bigcap \text{VS}(P, P^-) \supseteq \bigcap \text{VS}(P)$. Here P is P_0 together with the earlier escalated valid queries; human data that arrive later only enlarge $\bigcap \text{VS}$ further. So these queries form an elastic chain in R^* . The second term is Theorem 4.17(ii), counting valid queries only. $\square$

This guarantee is not 0-soundness in the sense of Definition 2.19 unless every target has prior at least δ , which fails for infinite $\mathcal{H}$.

Structure, not the prior. For single schemas with a description-length prior, $\ln(1/w^*)/\delta \geq \ln(1/w^*)/w^*$ is exponential in $|\sigma^*|$, while $\text{el} \leq N + 1$. The prior earns its keep elsewhere. If $\mathcal{H}$ contains every finite step set (calculi with unboundedly many ground rules, so exception lists are hypotheses), the bare version space never generalizes: $\bigcap \text{VS}(P) = P$. Under the hypotheses of Theorem 4.15, the Bayesian verifier is sound with probability at least $1 - \delta'$ for every target with $w(R^*) \geq \delta/\delta'$, i.e. under a description-length prior, every rule system of description length at most $\log_2(\delta'/\delta)$. Hence a three-layer recipe:

- a structured version-space core, for cheap escalations and exact soundness when realizable;
- a Bayesian hierarchy over it, as insurance against unrealizability;
- conservative thresholds, never MAP.

One caveat: Theorem 4.15 needs well-specification. Systematic human errors are exactly a misspecification the model does not anticipate, and the posterior then concentrates on “rule plus fallacy” as a rule (Section 5).

Reading. From noisy practice and a noisy oracle, Theorem 4.15 gives a warrant that search cannot erode, as long as the searcher cannot foresee the data or the noise. Its error probability δ' is fixed in advance and does not grow with the number of steps tried, the length of derivations, or the running time. That is what “it must actually work if run” asks of a checker that a strong prover searches against. The warrant is fallibilist and reliabilist, conditional on a well-specified model. The prior does not make it cheap: *structure, not the prior, makes sound learned verification affordable.*

4.5 What caution buys, and what it cannot do

Learning which inferences are valid. A cautious learner learns exactly what its evidence forces and never adopts a rule it cannot certify. It pays for completeness in escalations, and the positive elasticity is the exact worst case. With rules cited by name and sufficiently varied human data, positive data alone bring it to exactly the human calculus, after coupon-collector many steps, and it is sound at every moment along the way (Section 5).

Coherence. A cautious verifier gets no signal from coherence. Its accepted steps lie in R^* , so no argument from a designated context to $\perp$ uses only accepted steps (T2 Prop. 2.9(a); Section 6). Coherence informs only *bold* learners. Hence the two tiers of Section 7: cautious assertion, and a bold sandbox attacked by a red team.

Limits. Realizability is load-bearing. If the true calculus lies outside the class, for example two untagged rules while $\mathcal{H} = H_1$, the lgg over-generalizes and soundness fails. The Bayesian hierarchy repairs this only probabilistically, at a cost exponential in the complexity of what is missing. Soundness is relative to the *human* calculus, naive comprehension included if humans use

it, and improving on it is the job of coherence and the world (Sections 6, 7 and 10). The first-order term model leaves out associative-commutative contexts, binders and variable-arity rules, where the bounds of Table 4 would need re-proof. Steps inside suppositions and idealizations are certified like any others. Which contexts may be designated, and how results are exported, is left to Section 11.

Justification. Certified acceptance models a warrant that survives search. A step is warranted when every hypothesis the evidence leaves open licenses it, and no amount of adversarial search erodes such a warrant, unlike a statistical one (Section 3). The Bayesian version trades certainty for a single bounded chance of error, which also does not grow with search. Both are conditional reliabilist vindications, relative to explicit assumptions (realizability; truthful or well-specified data), and later sections show that each assumption is needed (Section 13).

5 Imitation: what positive examples fix

The first channel is imitation. The learner sees human steps, possibly with the names of the rules they cite, and nothing else; no step is ever labelled invalid. This section asks what such data fix when acceptance must be cautious, as search requires (Sections 3 and 4). The answer has two halves.

With the right hypothesis class (schemas, learned by least general generalization), positive examples fix the rules *exactly*. A cautious verifier fed only human steps is sound at every time. It becomes exactly complete after finitely many examples iff the target has a finite *anchor* (Theorem 5.4). For rules that humans cite by name this happens at a coupon-collector rate (Theorem 5.7); without citations it needs a diversity condition (Theorem 5.9). Once exact, the verifier generalizes systematically: it accepts exactly the target calculus, on steps and derivations of any size, whatever the distribution of the human data (Corollary 5.11). This is the formal sense in which finite usage fixes inferential role for unbounded use.

The other half is that positive data cannot tell what humans do from what is valid. One mislabelled step collapses a learned rule into “anything follows” (Proposition 5.13). Sporadic errors can be trimmed when they are rarer than the evidence for each rule’s generality, and this threshold is necessary up to a factor of two (Theorems 5.14 and 5.15 and Corollary 5.19). Systematic errors are indistinguishable from rules at every rate (Corollary 5.20). Removing them needs negative information (Sections 6 and 7).

Most results below are trivial once set up, or known in substance, and we say which. New here, as far as we know, are three things: anchors read as the completeness condition of *certified* verification; the witness-event sample complexities and their noise versions, with necessity up to a factor of two; and the exact statement that schema-generated errors are rules. None is deep. Together they pin down when imitation works. Throughout, the signature Ω in which steps are encoded (Section 2.2) has at least two constants. Then a proper specialization of a schema has strictly fewer instances, and the instance sets of all schemas have empty intersection (Appendix D.1).

5.1 Cautious learning from positive data

Let $\mathcal{H} \subseteq 2^S$ be a hypothesis class of step sets (Definition 2.9) containing the target R^* , and let the data D be a finite multiset of human steps. Unlike in Section 4, there is no oracle. Until Section 5.6 the data are clean, $D \subseteq R^*$.

Definition 5.1 (Cautious verifier). ^[T1 §5; L1 §9] The (positive-data) version space is $VS(D) = \{R \in \mathcal{H} : D \subseteq R\}$. The *cautious verifier* accepts exactly $\hat{R}(D) := \bigcap VS(D)$ (with $\bigcap \emptyset := S$).

Lemma 5.2 (Soundness and maximality; trivial). ^[T1 Thm 3.1, Prop 3.3, positive data] (a) If $D \subseteq R^*$ then $\hat{R}(D) \subseteq R^*$. So the reasoner that chains accepted steps derives only Cl_{R^*} -consequences, against every prover, at every time. (b) Every acceptance set contained in all $R \in VS(D)$ (i.e. sound for every target compatible with D) is contained in $\hat{R}(D)$. (c) $D \subseteq D'$ implies $\hat{R}(D) \subseteq \hat{R}(D')$. (d) If $\mathcal{H} \cup \{\emptyset\}$ is closed under arbitrary intersections and $VS(D) \neq \emptyset$, $\hat{R}(D)$ is the least member of $\mathcal{H} \cup \{\emptyset\}$ containing D ; for $\mathcal{H} = H_1$ and $D \neq \emptyset$ it is $\text{inst}(\text{lbg}(D))$.
^{Lean: thm_3_1_a_static Lean: thm_3_1_b_static}

Proof. (a) $R^* \in VS(D)$; apply Lemma 3.1. (b), (c) are immediate. (d) The intersection of all members containing D is a member; for H_1 use Theorem 2.5. $\square$

So $\hat{R}$ is the pointwise largest acceptance region that is sound for every target compatible with the evidence. For intersection-closed classes it is the closure algorithm (Natarajan, 1987; Helmbold et al., 1990), and it coincides with El-Yaniv and Wiener’s consistent selective strategy and with the “accept” side of KWIK (El-Yaniv and Wiener, 2010a; Li et al., 2008a). Its soundness is deterministic and distribution-free. Every question below is about *completeness*: when does $\hat{R}(D)$ reach R^* ?

5.2 Anchors: when positive data suffice

Definition 5.3 (Anchor). ^[T1 Thm 5.1] A finite $T \subseteq R^*$ is an *anchor* (for R^* in $\mathcal{H}$) if every $R \in \mathcal{H}$ with $T \subseteq R$ satisfies $R \supseteq R^*$.

Theorem 5.4 (Eventual completeness iff finite anchor). ^[T1 Thm 5.1, as repaired] (a) For finite $D \subseteq R^*$: $\hat{R}(D) = R^*$ iff D contains an anchor. (b) Let the data be a text for R^* (finite $D_1 \subseteq D_2 \subseteq \dots \subseteq R^*$ with $\bigcup_t D_t = R^*$), or i.i.d. draws from a law with support R^* . If R^* has an anchor, $\hat{R}(D_t) = R^*$ for all large t (a.s. in the i.i.d. case); if not, $\hat{R}(D_t) \neq R^*$ for every t .

Proof. (a) If $D \supseteq T$ for an anchor T , every $R \in VS(D)$ contains R^* , so $\hat{R}(D) \supseteq R^*$; the reverse inclusion is Lemma 5.2(a). Conversely, if $\hat{R}(D) = R^*$, then D is itself an anchor, since every $R \in \mathcal{H}$ containing D lies in $VS(D)$. (b) A text eventually contains any finite $T \subseteq R^*$, and then (a) and monotonicity give exactness forever. S is countable, so i.i.d. data with support R^* form a text a.s. $\square$

“Finite” matters in (a): in the class of Proposition 5.5(c), $\hat{R}(2\mathbb{N}) = 2\mathbb{N}$ but $2\mathbb{N}$ has no anchor.

Proposition 5.5 (Anchors versus tell-tales). ^[T1 Prop 5.2, as repaired] Call a finite $T \subseteq R^*$ an (Angluin) tell-tale if no $R \in \mathcal{H}$ satisfies $T \subseteq R \subsetneq R^*$. (a) Every anchor is a tell-tale. (b) If $\mathcal{H} \cup \{\emptyset\}$ is closed under intersections, every nonempty tell-tale is an anchor; the empty one need not be ($\mathcal{H} = \{\{1\}, \{2\}\}$, $R^* = \{1\}$). As existence conditions the two coincide when $R^* \neq \emptyset$. (c) In general the converse fails: $\mathcal{H} = \{2\mathbb{N}\} \cup \{R_n\}_{n \in \mathbb{N}}$ with $R_n = \{0, 2, \dots, 2n, 2n+1\}$ is identifiable in the limit from text, but $2\mathbb{N}$ has no anchor, so on a text for $2\mathbb{N}$ the cautious verifier never becomes complete.

Proof in Appendix D.2. *Credit.* Identification in the limit is due to Gold (1967), and tell-tales characterize it for indexed families (Angluin, 1980). Anchors are the “ $\subseteq$ -tell-tales” of strong-monotonic learning, where every conjecture is contained in the target (Lange and Zeugmann,

1992; Lange et al., 2008). As we recall their characterization (we have not re-checked its exact form), strong-monotonic learnability is, up to effectivity, the existence of anchors. What is new here is only the reading of anchors as the completeness condition of certified verification. For intersection-closed classes, anchors exist iff R^* is the closure of a finite set (L1 §2.2).

Anchors exist in the schema classes H_1 , H_k^{tag} and H_k (fixed k), by finite elasticity (Wright, 1989; Motoki et al., 1991); infinite targets have none in superfinite classes such as unions of unboundedly many schemas (Section 5.4).

Proposition 5.6 (Finite elasticity gives anchors). *[L1 §2.3, Lemma] Say $\mathcal{H}$ has infinite elasticity if there are $w_0, w_1, \dots \in S$ and $R_1, R_2, \dots \in \mathcal{H}$ with $\{w_0, \dots, w_{n-1}\} \subseteq R_n \not\supseteq w_n$ for all $n \geq 1$, and finite elasticity otherwise. If $\mathcal{H}$ has finite elasticity, every $R^* \in \mathcal{H}$ has an anchor.*

The proof (Appendix D.2) builds an infinite elasticity sequence from the failure of each finite subset of R^* to be an anchor. A ground step is an instance of only finitely many schemas up to renaming (a generalization is never larger than its instances), so H_1 has finite thickness, hence finite elasticity. Finite elasticity survives unions of at most k members (Wright, 1989, as corrected by Motoki et al. 1991). So H_1 , H_k^{tag} and H_k all have anchors, as do richer classes such as Shinohara’s length-bounded elementary formal systems with at most k clauses (Shinohara, 1994), which model learning theorem sets rather than steps (L1 §2.5).

Reading. “Learning which inferences are valid” can mean *guessing* R^* in the limit, as in Gold’s paradigm, or *certifying* a step: every hypothesis compatible with the evidence licenses it. Under search only the second protects. A learner that guesses $2\mathbb{N}$ in Proposition 5.5(c) converges, but it is unsound whenever the truth is a large R_n , and a prover can find the difference. An anchor is a finite body of usage that, given $\mathcal{H}$, justifies a rule for all of its uses. This is the kind of justification by imitation that survives adversarial search. It is relative to $\mathcal{H}$, a point we return to after Corollary 5.11.

5.3 Rates: cited rules and the coupon collector

Formal proofs cite their rules (natural-deduction rule names, Lean lemma names, Metamath labels). Take tagged calculi H_k^{tag} (Definition 2.9) with target rules $\sigma_1, \dots, \sigma_k$, and let $v_i = |\text{vars}(\sigma_i)|$. The version space factors over tags (T1 Thm 3.6; Appendix D.1). So the cautious verifier is the *per-rule lgg verifier* $\hat{R}(D) = \bigcup_i \{i\} \times \text{inst}(\text{lgg}(D^{(i)}))$, where $D^{(i)}$ is the tag- i data (a term with $D^{(i)} = \emptyset$ contributes nothing), and rule i is identified once its data satisfy the witness events (R) and (D) of Lemma 2.7. Human data are i.i.d. pairs $(I, \sigma_I \Theta)$ with $\mathbb{P}(I = i) = \pi_i$ and $\Theta \sim \Lambda_i$ given $I = i$. For distinct metavariables x, y of σ_i let

$$\rho_{i,x} := \max_{G \subseteq \Omega} \min(\mathbb{P}[\text{root}(\Theta x) \in G], \mathbb{P}[\text{root}(\Theta x) \notin G]), \quad r_{i,xy} := \mathbb{P}[\Theta x \neq \Theta y],$$

let ρ_i be the minimum of all these numbers, and $c_i := 2v_i + \binom{v_i}{2}$. For a ground rule ($v_i = 0$) put $\rho_i := c_i := 1$.

Theorem 5.7 (Exact identification at a coupon-collector rate). *[T1 Thm 5.3] For N i.i.d. tagged human steps, $\mathbb{P}[\hat{R}(D) \neq R^*] \leq \sum_{i=1}^k c_i e^{-N\pi_i \rho_i}$. So $N \geq \max_i \frac{1}{\pi_i \rho_i} \ln \frac{k c_i}{\delta}$ suffices for exact identification with probability at least $1 - \delta$. Order $1/(\pi_i \rho_i)$ samples are also necessary: if $\mathbb{P}[\text{root}(\Theta x) = f] = 1 - \rho$ for some rule i , metavariable x , symbol f and $\rho \in (0, 1]$, then $\mathbb{P}[\hat{R}(D) \neq R^*] \geq (1 - \pi_i \rho)^N$.*

N	12	24	48	96	192
empirical exact-identification rate	.003	.41	.98	1.00	1.00
bound $1 - 30e^{-N/12}$ of Theorem 5.7	vacuous	vacuous	.45	.99	.999997

Table 5: Toy natural deduction with six tagged rules ($\wedge I$, $\wedge E_{1,2}$, $\vee I_{1,2}$, MP), random formulas over four atoms, 400 trials per N ; $\pi_i = 1/6$, $\rho_i = 1/2$, $c_i \leq 5$ (computed; T1 §5, T1-code/toy_nd.py).

Proof idea. Given n rule- i samples, (R) fails at x only if all n roots fall on one side of the best split G , which has probability at most $2(1-\rho_i)^n$. (D) fails at x, y with probability $(1-r_{i,xy})^n \leq (1-\rho_i)^n$. A union bound over the c_i witness events, averaged over $n \sim \text{Bin}(N, \pi_i)$, gives $c_i(1 - \pi_i\rho_i)^N$. A ground rule fails only if it is never sampled. Full proof in Appendix D.3.

Table 5 compares the bound with simulation. The coupons are the witness events of each rule; in spirit this is stochastic finite learning, as Rossmanith and Zeugmann (2001) studied for pattern languages.

Remark 5.8 (Teaching sets, corpora, and distribution dependence). [L1 §4.2, §9] Given $2v_i$ distinct constants, two instances of σ_i whose metavariables are replaced by two disjoint sets of distinct constants satisfy (R) and (D). So a textbook with two well-chosen examples per rule is an anchor, while a corpus needs about $\ln(kc_i/\delta)/(\pi_i\rho_i)$ steps. No distribution-free completeness rate exists, even for single schemas. For every $m = 2^h$, the complete binary trees of depth $\log_2 m$ with one distinguished leaf carry a law under which every sound learner has completeness error above $1/2$ after fewer than $m/2$ samples (Appendix D.4). Soundness is distribution-free; completeness cannot be, and $\pi_i\rho_i$ is what the distribution must supply.

Reading. A rule used in a fraction π_i of human steps, with variability ρ_i , is pinned down after about $\ln(kc_i/\delta)/(\pi_i\rho_i)$ steps; further examples add nothing. What makes this cheap is the citation of rules, which lets the version space factor. The user’s note that mathematics was formalized by *inventing a language* into which proofs can be translated has a precise counterpart here: a language with named rules makes inference learnable from imitation at coupon-collector cost. Without names the problem becomes a union problem, with a diversity condition (Theorem 5.9) and, with escalation, worst-case cost at least $\lfloor (n-1)/k \rfloor^k$ instead of at most $k(n+1)$ for steps of size at most n (T1 Thms 3.6–3.7, Section 4).

5.4 Uncited rules: a diversity condition

Now let steps carry no citations: $R^* = \bigcup_{i \leq k'} \text{inst}(\sigma_i)$, in the class $H_k = \{\bigcup_{l \leq k} \text{inst}(\tau_l)\}$ with $k \geq k'$. Some bound k is unavoidable. With unbounded k every finite set of ground steps is a hypothesis, the class is superfinite in Gold’s sense, and $\hat{R}(D) = D$: the cautious verifier never generalizes (Gold, 1967). For rule i let $\text{Sub}_i(D) := \{\theta|_{\text{vars}(\sigma_i)} : \sigma_i\theta \in D\}$, with *failure sets*

$$\mathfrak{F}_i := \{\{\theta : \text{root}(\theta x) = f\} : x \in \text{vars}(\sigma_i), f \in \Omega\} \cup \{\{\theta : \theta x = \theta y\} : x \neq y\}.$$

By Lemma 2.7, a nonempty set Q of substitutions has $\text{lgg}\{\sigma_i\theta : \theta \in Q\} \equiv \sigma_i$ (call Q *generic*) iff Q lies in no single failure set.

Theorem 5.9 (Untagged unions: anchors and sample complexity). [T1 Thm 5.4] (a) *If for each $i \leq k'$ the set $\text{Sub}_i(D)$ is nonempty and not contained in a union of k failure sets from $\mathfrak{F}_i$, then*

$\hat{R}_{H_k}(D) = R^*$. In particular, $k + 1$ pairwise-generic instances of each rule form an anchor of size $k'(k + 1)$. (b) Let the rule- i data be i.i.d. $\sigma_i\Theta$, $\Theta \sim \Lambda_i$. For $v_i \geq 1$ let

$$\zeta_i := \inf_{F_1, \dots, F_k \in \mathfrak{F}_i} \Lambda_i(\overline{F_1 \cup \dots \cup F_k}), \quad d_i := 2k \log_2(4k(v_i^2 + 1)).$$

If every $\zeta_i > 0$ and the data contain, for each rule with $v_i \geq 1$, at least

$$n_i \geq \frac{8d_i}{\zeta_i} \log_2 \frac{13}{\zeta_i} + \frac{4}{\zeta_i} \log_2 \frac{2k'}{\delta}$$

independent rule- i samples, and one sample of each ground rule, then $\hat{R}_{H_k}(D) = R^*$ with probability at least $1 - \delta$.

Proof of (a). Let $R = \bigcup_{l \leq k} \text{inst}(\tau_l) \in \text{VS}(D)$. Partition $\text{Sub}_i(D)$ by the least l with $\sigma_i\theta \in \text{inst}(\tau_l)$. If no class were generic, each would lie in a failure set, and k failure sets would cover $\text{Sub}_i(D)$. So some class Q is generic, and its $\tau_l \succeq \text{lgg}\{\sigma_i\theta : \theta \in Q\} \equiv \sigma_i$. Hence $R \supseteq R^*$. A failure set contains at most one member of a pairwise-generic set, so $k + 1$ such members cannot be covered. $\square$

Part (b) bounds the VC dimension of complements of k -unions of failure sets by d_i and applies the ε -net theorem (Haussler and Welzl, 1987; Blumer et al., 1989) with $\varepsilon = \zeta_i$ (Appendix D.5). The condition $\zeta_i > 0$ asks for more instantiation variety than k rules can absorb, and some such condition is needed:

Proposition 5.10 (When variety is too low). *[T1 Remark after Thm 5.4, as repaired] (a) Suppose that for some i the data in $\text{inst}(\sigma_i)$ not covered by the other $k' - 1$ rules are covered by $k - k' + 1$ specializations of σ_i of the forms $\sigma_i[x \mapsto f(\bar{z})]$ ($\bar{z}$ fresh) or $\sigma_i[y \mapsto x]$, and that some instance of σ_i lies outside these specializations and the other rules. Then $\hat{R}_{H_k}(D) \neq R^*$. (b) For $k' \geq 2$, low variety need not block identification: for $k = k' = 2$, $R^* = \text{inst}(p(x)) \cup \text{inst}(q(y))$ and $D = \{p(a), p(b), q(a), q(b)\}$, $\hat{R}_{H_2}(D) = R^*$, although x takes only 2 = k roots.*

For $k' = 1$, (a) says that if a metavariable is only ever instantiated with at most k root symbols (and others are possible), identification fails while this persists. Proofs in Appendix D.6; (b) is also checked by brute force (T1-code/union_remark.py). For $k' \geq 2$ the exact threshold, between “no cover by k failure sets” and “a cover by $k - k' + 1$ plus an uncovered instance”, is open. So “allow up to k rules” is a real assumption about the diversity of usage: without rule names, imitation must see each rule used in more ways than the $k - k' + 1$ spare schemas of the class can absorb (for $k' = 1$, e.g. more than k root symbols at each metavariable). This is the positive-data face of the escalation lower bound for untagged unions (Section 4).

5.5 Systematic generalization: finite usage fixes inferential role

Corollary 5.11 (Systematic out-of-distribution generalization). *[T1 Cor 5.5] If $\hat{R}(D) = R^*$, the verifier accepts exactly R^* and the reasoner derives exactly $\text{Cl}_{R^*}(B)$ for every B , including conclusions that need derivations far longer, and judgments far larger, than any in the data, independently of the distribution that produced the data.*

Proof. $\hat{R}(D) = R^*$ gives $\text{Cl}_{\hat{R}(D)} = \text{Cl}_{R^*}$. $\square$

The corollary is trivial; Theorems 5.7 and 5.9 supply its hypothesis, and Lemma 5.2 keeps the verifier sound before that. The result is a learner of inference rules from imitation that is sound at all times against adaptive search, exact after about $\max_i \ln(kc_i/\delta)/(\pi_i\rho_i)$ cited steps, and *systematic*: correctness on terms of size 40 is the same event as on terms of size 10. A learned step scorer comes with none of these guarantees (Sections 3 and 5.8).

Reading. Under an inferentialist picture of meaning (Brandson, 1994), the meaning of a connective is its inferential role, the steps in which it may figure. Corollary 5.11 says that a finite set of uses (an anchor; for cited natural-deduction rules, two generic instances per rule) fixes that role for unbounded use. Three qualifications make this precise. (i) The role is fixed *relative to* $\mathcal{H}$. An anchor leaves open, within $\mathcal{H}$, only hypotheses that contain R^* , so the cautious verifier accepts exactly R^* ; it says nothing about hypotheses outside $\mathcal{H}$. Quus-like alternatives (Kripke, 1982) are excluded by the choice of hypothesis class, not by the data. Which of simplicity, coherence and convention does that work is the subject of Sections 6, 8 and 13. (ii) On clean data, overgeneralizations such as *tonk* are never adopted, because the lgg is the *least* generalization covering usage. The learner adds to a connective’s role only what usage forces. This is a learning-theoretic analogue of Belnap’s conservativeness requirement (Belnap, 1962), and it is why simplicity is the wrong bias here (Proposition 3.9). (iii) The role fixed is that of *practice*: a fallacy used systematically becomes part of it (Section 5.6). *Contexts.* Contexts are part of judgments (Definition 2.1), so steps inside suppositions are data like any other, and the theorems apply to them unchanged as long as contexts are encoded as first-order terms (Remark 5.12). But they must *be* in the data. Theorems determine a structural consequence relation only up to the interval between derivable and admissible rules, which is non-degenerate for intuitionistic logic (the Kreisel–Putnam rule is admissible but not derivable). So, unless the hypothesis class enforces a deduction theorem, assertions alone cannot in general teach reasoning under hypotheses (L1 §§8.2–8.3).

Remark 5.12 (Side conditions). ^[T1 §5, extensions] With guards from a finite family Φ (Definition 2.4), the guarded single-schema class stays intersection-closed. Its closure is the lgg with the strongest guard true on all data (Lemma 2.8), anchors exist, and the escalation budget of Section 4 grows by at most $|\Phi|$ (Appendix D.7). But a positive example shows that a guard *held*, never that it was *needed*. A guard is learned only from the absence of violating uses. It is lost as soon as one human use omits it (Example 2.12), and it cannot be learned at all if Φ cannot express it. Set-valued contexts and binders lie outside first-order anti-unification; modulo AC it is finitary rather than unitary, and the bounds above would need re-proof (T1 §5).

5.6 Noise: sporadic and systematic errors

Human data contain errors, and the two kinds behave completely differently.

Proposition 5.13 (One error collapses the lgg). ^[T1 Prop 6.1] *Let D consist of $\wedge E_1$ instances $s(\text{and}(A_j, B_j), A_j)$ whose A_j do not all have the same root, plus one invalid step $s(\varphi, \psi)$ tagged $\wedge E_1$, with φ not a conjunction. Then $\text{lgg}(D) = s(z, w)$ with $z \neq w$: the learned rule accepts every one-premise step, and from any derivable formula it derives everything.*

Proof. The premise column mixes the root **and** with the root of φ ; the conclusion column has mixed roots because the A_j do. Both become metavariables, distinct because the columns differ ($\text{and}(A_1, B_1) \neq A_1$). $\square$

(computed; T1 §6) A mis-tagged “ $p_0 \vee p_1 \triangleright p_0$ ” does exactly this. A 15% rate of affirming the consequent, cited as MP, turns MP’s lgg into $s_2(\text{imp}(v_0, v_1), v_2, v_3)$: from $A \rightarrow B$ and anything, infer anything. This is the *tonk* of Theorem 3.2, produced by learning: the lgg is least among generalizations of *all* the data, errors included.

The remedy for sporadic errors is to let hypotheses miss a few data. Counting with multiplicity, let $\text{VS}_e(D) = \{R \in \mathcal{H} : |D \setminus R| \leq e\}$, and with per-tag budgets $\text{VS}_{(e_i)}(D) = \{R \in \mathcal{H} :$

$|D^{(i)} \setminus R^{(i)}| \leq e_i$ for all i }, where $D^{(i)}, R^{(i)}$ are the tag- i parts. The *trimmed verifiers* accept $\bigcap \text{VS}_e(D)$, resp. $\bigcap \text{VS}_{(e_i)}(D)$.

Theorem 5.14 (Trimmed version space). *[T1 Thm 6.2, as repaired] (a) For any $\mathcal{H}$: if D contains at most e invalid steps, then $\bigcap \text{VS}_e(D) \subseteq R^*$; likewise $\bigcap \text{VS}_{(e_i)}(D) \subseteq R^*$ if at most e_i invalid steps are tagged i , for each i . The verifier is then sound against every prover. (b) For tagged schemas and the per-tag verifier, suppose that for each i : (i) at most e_i invalid steps are tagged i ; and (ii) the valid rule- i samples are e_i -robustly generic: there are more than e_i of them; for every x, f , more than e_i of them have $\text{root}(\Theta x) \neq f$; and for every $x \neq y$, more than e_i of them have $\Theta x \neq \Theta y$. Then $\bigcap \text{VS}_{(e_i)}(D) = R^*$.*

Proof idea. (a) R^* lies in the trimmed version space. (b) A hypothesis in $\text{VS}_{(e_i)}$ covers all but at most e_i valid rule- i samples. By (ii) the rest is nonempty and still satisfies (R) and (D), so its tag- i schema generalizes σ_i . Full proof in Appendix D.8.

Two features of (b) were added in verification, with counterexamples in Appendix D.8. The clause “more than e_i of them” is the only content of (ii) for ground rules, and without it the theorem fails for them. The budgets must be *per tag*: one global budget lets a hypothesis spend everything on one rule. (computed; T1 §6) With $e = 1$ (resp. 2), one (resp. two) mis-tagged errors are removed exactly; with $e = 1$ and two errors the lgg collapses again.

Under i.i.d. noise the budgets can be set from the rates. Suppose each datum independently is a valid rule- i step with probability β_i (with $\Theta \sim \Lambda_i$), or an invalid step tagged i with probability α_i (with arbitrary law); in terms of Definition 2.10, $\beta_i = \pi_i(1 - a_i)$ and $\alpha_i = \pi_i a_i$.

Theorem 5.15 (i.i.d. noise: witness frequency must beat error frequency). *[T1 Thm 6.3, as repaired] Let ρ_i, c_i be as in Theorem 5.7 ($\rho_i = c_i = 1$ for ground rules). If every margin $\Delta_i := (\rho_i \beta_i - \alpha_i)/2$ is positive and $e_i := \lfloor (\alpha_i + \Delta_i)N \rfloor$, then $\mathbb{P}[\bigcap \text{VS}_{(e_i)}(D) = R^*] \geq 1 - \sum_i (1 + c_i) e^{-2N\Delta_i^2}$.*

Proof idea. The number of invalid steps tagged i is $\text{Bin}(N, \alpha_i)$ and exceeds e_i with probability at most $e^{-2N\Delta_i^2}$ (Hoeffding). Robust genericity follows from c_i counts (valid rule- i samples with root in G and in G^c for the best split G at each x ; with $\Theta x \neq \Theta y$ for each pair), each $\text{Bin}(N, p)$ with $p \geq \rho_i \beta_i = \alpha_i + 2\Delta_i$. Appendix D.9.

The budgets grow linearly with N ; a fixed support threshold is not a soundness guarantee under sporadic noise, because coincidences between random errors grow with N (Section 5.8). The threshold $\rho_i \beta_i > \alpha_i$ is necessary up to a factor of two, because a noisy law for one target is a clean law for another.

Definition 5.16 (Robust soundness). *[T1 Thm 6.4] A (δ, α) -robustly sound verifier maps the data so far (and private coins) to an acceptance set Acc_t . It is (δ, α) -robustly sound for $\mathcal{H}$ if for every $R^* \in \mathcal{H}$ and every law Q on S with $Q(S \setminus R^*) \leq \alpha$, i.i.d. data from Q give $\mathbb{P}[\exists t : \text{Acc}_t \not\subseteq R^*] \leq \delta$.*

Theorem 5.17 (Indistinguishability). *[T1 Thm 6.4] Let the verifier be (δ, α) -robustly sound for $\mathcal{H}$, let $R^* \subsetneq R'$ both lie in $\mathcal{H}$, and let Q be a law on R' with $Q(R' \setminus R^*) \leq \alpha$. Under i.i.d. data from Q , every $q \in R' \setminus R^*$ has $\mathbb{P}[\exists t : q \in \text{Acc}_t] \leq \delta$. So if $\delta < 1$, the verifier is not complete for the target R' under its clean law Q .*

Proof. Q is an admissible noisy law for the target R^* , and $q \in \text{Acc}_t$ implies $\text{Acc}_t \not\subseteq R^*$; the data law is the same in both scenarios. $\square$

This is trivial once set up, in the spirit of Kearns and Li (1993). For schemas it yields a matching necessity. Let $m_{i,x} := \max_f \mathbb{P}[\text{root}(\Theta x) = f]$.

Lemma 5.18 (Balanced splits). *[T1 §6 after Thm 6.4, repaired] $\rho_{i,x} \leq 1 - m_{i,x} \leq 2\rho_{i,x}$, and the constant 2 is attained by the uniform law on an odd number of symbols.*

Corollary 5.19 (Witness frequencies must exceed the noise rate). *[T1 §6 after Thm 6.4, repaired] Let $\mathcal{H} = H_k^{\text{tag}}$ and let Q be a clean law for R^* under which a datum is a rule- i instance $\sigma_i\Theta$, $\Theta \sim \Lambda_i$, with probability β_i . If a (δ, α) -robustly sound verifier with $\delta < 1$ has $\mathbb{P}_Q[R^* \subseteq \text{Acc}_t] > \delta$ for some t , then for all x, f and all $x \neq y$,*

$$\beta_i \mathbb{P}[\text{root}(\Theta x) \neq f] > \alpha \quad \text{and} \quad \beta_i \mathbb{P}[\Theta x \neq \Theta y] > \alpha.$$

With per-tag noise budgets, α may be read as α_i .

Proof idea. If a witness frequency is at most α , then Q is an α -noisy law for the target in which σ_i is replaced by $\sigma_i[x \mapsto f(\bar{z})]$ (resp. $\sigma_i[y \mapsto x]$), and Theorem 5.17 applies. Proofs in Appendices D.10 and D.11.

So Theorem 5.15 asks for $\rho_{i,x}\beta_i > \alpha_i$ and $r_{i,xy}\beta_i > \alpha_i$ (plus margins), while necessity asks for $(1 - m_{i,x})\beta_i > \alpha_i$ and $r_{i,xy}\beta_i > \alpha_i$. By Lemma 5.18 the two differ by at most a factor of two in the root frequencies and agree on the equality frequencies; the factor two in the lemma is attained. The trade-off is unavoidable. A positive-data verifier that tolerates error rate α must refuse every generalization supported by less than a fraction α of the data. With human error rates around 1%, the generality of a rule used in less than about 1% of steps needs negative information.

Corollary 5.20 (Systematic errors are rules). *[T1 Cor 6.5] Let human errors be schema-generated: the data law is $Q = (1 - \alpha)Q_0 + \alpha Q_{\text{err}}$, with Q_0 supported on R^* and Q_{err} on $\text{inst}(\tau)$ for a schema τ with $\text{inst}(\tau) \not\subseteq R^*$. Suppose $R' := R^* \cup \text{inst}(\tau) \in \mathcal{H}$, as holds whenever $\mathcal{H}$ is closed under adding a schema (tagged calculi with a fresh tag for τ ; untagged unions with room for one more schema). Then Q is a clean law for R' at every error rate α . Hence a (δ, α) -robustly sound verifier with $\delta < 1$ accepts each $q \in \text{inst}(\tau) \setminus R^*$ with probability at most δ under Q . So with probability at least $1 - \delta$ it never accepts all of R' , a calculus in which τ is a genuine rule used at the same rate.*

Proof. $Q(S \setminus R') = 0$; apply Theorem 5.17 to $R^* \subsetneq R'$. □

So no positive-data method, with any amount of data, can both tolerate schema-generated errors at rate α and learn genuine rules used at rate α . This also removes the protection of the Bayes-conservative verifier (Section 4). Its time-uniform guarantee assumes a well-specified likelihood. A model of errors as sporadic is misspecified for Q , and the clean hypothesis R' fits Q exactly.

Remark 5.21 (Imitation identifies practice). *[T7 Prop 2.2; T1 §6] With latent but distinct fallacy tags (Definition 2.10), the theorems above apply to the practice $\Sigma^P = \Sigma^* \sqcup F$ tag by tag. On the event of Theorem 5.15 the trimmed verifier accepts exactly $R^P = R^* \cup R_F$, including an instance of every fallacy outside $\text{Sound}(R^*)$, which a prover will query. In classical logic with a complete Σ^* and a pure fallacy schema, one closed instance with true premises and a false conclusion trivializes the reasoner (T7 Prop. 2.2; cf. Proposition 3.5). If fallacies are cited under genuine rules' names, a learner with one schema per tag collapses the rule instead (Proposition 5.13).*

Reading. The line between sporadic and systematic errors is exact. Positive data remove an error only if it is rarer than the witness frequencies of the genuine rules. Rule-like errors, at any rate, need *negative* information: escalation (Section 4), coherence (Section 6), or the world (Sections 7 and 11). For the user’s program, imitation learns which inferences are *used*, and which are *valid* only insofar as use is valid. Getting from the first to the second is what the coherence and world channels are for, and positive data provably cannot do it. In the language of justification, imitation can certify a step as licensed by the practice, never as licensed by anything beyond it. This is why the cautious tier of Section 7 is sound only relative to practice, and why coherence must act on a separate, bold tier.

5.7 An exactly solvable case: dimensions as inferential roles

Remark 5.22 (Dimension inference). [L9 §3.1, TC1] (proof sketch) The dimension of a physical quantity is its inferential role: which sums and equations it may enter. Here learning meaning from positive examples is solvable in closed form. Let $q_1, \dots, q_n$ be quantity symbols, and let observed equations be polynomial (or Laurent) equations among them; each monomial has an exponent vector in $\mathbb{Z}^n$. A *grading* is a homomorphism $g : \mathbb{Z}^n \rightarrow A$ into an abelian group ($\deg q_i = g(e_i)$), and an equation is *g -homogeneous* if all its monomials have the same degree. For observed equations E , let $K(E) \leq \mathbb{Z}^n$ be generated by the differences of exponent vectors of monomials in a common equation. (i) The gradings making E homogeneous are the homomorphisms vanishing on $K(E)$, so $\mathbb{Z}^n \rightarrow \mathbb{Z}^n / K(E)$ is the *finest* one: every consistent grading factors through it. It is computed by Smith normal form (modulo torsion for rational exponents). The cautious verifier for “dimensionally admissible” accepts exactly the equations whose differences lie in $K(E)$ (in its saturation, if gradings must be torsion-free). (ii) The admissible sets form an intersection-closed class indexed by subgroups of $\mathbb{Z}^n$. Since these are finitely generated, every target has a finite anchor, and Theorem 5.4 gives exact identification once the data exhibit generators of the target’s relations. This is an instance of learning algebraic closure systems from text, where Noetherianity is finite generation (Stephan and Ventsov 2001; L1 §2.5). (iii) Gold’s limitation appears in miniature: positive data never refute a coarser grading (“everything dimensionless” is always consistent), but the cautious learner never adopts one. The collapse of Proposition 5.13 appears too: one wrong equation adds a spurious generator and merges dimensions. (iv) The residue is conventional. The finest grading is fixed only up to isomorphism, i.e. up to the choice of base dimensions, which no inference can see. Whether a constant such as c is dimensional is a choice between data sets: natural units add $c = 1$ and coarsen the grading, and Gaussian units absorb a dimension that SI keeps. These are coherent, empirically equivalent alternative meanings, settled by convention (Section 13). Dimension types implement the type-checking side (Kennedy, 1994); Buckingham’s Π theorem is the invariant-theoretic face (Buckingham, 1914). Sketch of (i)–(ii) in Appendix D.12. This remark has not been through the project’s adversarial verification.

5.8 Imitation in practice

Experiment A (Section 12) learns school algebra over partial terms (Example 2.12) from simulated human derivations (computed; `code/results/algebra_learning.md`). The target has 41 rewrite schemas, 11 of them guarded. The corpus has sporadic noise (5% random corruptions), five systematic fallacies (each committed with probability 0.3 per opportunity), or both. The learner is a heuristic relative of the cautious verifier (MDL-clustered anti-unification within rule-name buckets, support threshold two); by default it infers no guards, and a variant takes the most

specific guard true on all instances (Lemma 2.8). Means are over three seeds; soundness is measured, not proved. Positive data alone gave the following. *Shapes are recovered where Lemma 2.7 predicts*: a rule’s shape was recovered in 0% of (rule, run) pairs with one human instance, 67% with two, 92% with three or four and 100% with five or more; on clean data, all 41 shapes for $N \geq 200$ derivations. *Generalization is systematic*: on clean data at $N = 500$, valid out-of-distribution steps (terms of about 40 nodes, against about 10 in human steps) are accepted as often as in-distribution ones (1.00 and 1.00), whereas a gradient-boosted step classifier trained on the noisy corpus *with* validity labels accepts 89% of valid in-distribution steps, 73% of valid out-of-distribution ones, and about half of the invalid ones. *Guards are not displayed*: on clean data at $N = 500$, the default learner ends with 11.7 ± 0.5 unsound active schemas (essentially the 11 guarded rules without guards) and accepts 34% of invalid out-of-distribution steps; most-specific guards reduce this to 0.3 ± 0.5 unsound schemas (those whose guard the guard language cannot express), but with fallacies present the count rises to 5.3 ± 0.5 , with 22% false accepts. *Fallacies reach support*: at $N = 500$ each of the five fallacies is learned as an active schema in 6/6 runs, with and without guard inference (Corollary 5.20). *A fixed support threshold is no soundness guarantee*: under sporadic noise, spurious unsound schemas grow with N , reaching 102 ± 11 at $N = 500$ with threshold one and 18.5 ± 3.6 for the untagged learner with threshold two (3.0 on clean data); hence the linear budgets of Theorem 5.15. With noise and fallacies together at $N = 500$, the cited-rule learner recovers all 41 shapes, yet 15.7 ± 0.5 active schemas are unsound, 51% of invalid out-of-distribution steps are accepted, and a white-box prover finds 65 ± 2 accepted invalid steps. Experiment B shows the same pattern for propositional natural deduction (Section 12).

In short, positive data alone recover rule shapes and generalize out of distribution, but they never display a guard, and systematic fallacies reach support. What they fix is: rule shapes, given an anchor (at coupon-collector cost for cited rules, under a diversity condition for uncited ones); their extension to unbounded use; side conditions only as “strongest guard true on all data”; and sporadic errors rarer than every witness frequency. What they cannot fix is: the choice of $\mathcal{H}$, the necessity of a guard, generalizations rarer than the error rate, and any schema-generated error. Sections 6 and 7 show what coherence and world feedback do about the latter.

6 Coherence: what contradictions fix

Imitation cannot tell a systematic human error from a rule (Corollary 5.20). The user’s second ingredient is a coherence loss: penalize having good arguments for both P and $\neg P$. In our setting this is channel (C) of Definition 2.13: a family $\mathcal{A}$ of *designated contexts* is certified coherent, and a derivation of $\perp$ from one of them shows that one of its steps is invalid. This section determines what that signal fixes. In brief: coherence is negative data, but only for *bold* learners, and it controls only one of two error types (Section 6.1); it pins the target exactly when the target sits at the top of its coherent version space, as classical propositional logic does by Post-completeness, and it is blind for intuitionistic logic (Section 6.2); in arithmetic it is Popperian (Section 6.3); Carnap’s categoricity problem is Gold’s problem in the dual space, and is solved by data with two denials, not by coherence (Section 6.4); coherence catches tonk but not non-conservativity (Section 6.5); what survives is the set of coherent uniform alternatives, the Kripkensteinian residue (Section 6.6); and the use of coherence here is eliminative, which the coherentist impossibility theorems do not touch (Section 6.7).

Most results are known in substance or trivial once set up. Post-completeness (Post, 1921;

(Wójcicki, 1988), Glivenko’s theorem, Gold’s limit-point argument, halving and weighted majority (Littlestone, 1988; Littlestone and Warmuth, 1994), the multiple-conclusion answer to Carnap (Carnap, 1943; Shoesmith and Smiley, 1978), Belnap’s conservativity (Belnap, 1962) and the Σ_1/Π_1 asymmetry of limiting inquiry (Putnam, 1965; Gold, 1965; Kelly, 1996) are classical. New, as far as we know, are the reduction of contradictions to coalitions and the two-sided trade-off, the denial-rank trichotomy and twelve-datum tell-tale as learning results, the rule-level analysis of complete theories, and the systematic reading of all of these as statements about what a coherence loss can teach.

Cn_2 is classical propositional consequence ($\varphi \in \text{Cn}_2(X)$ iff every Boolean valuation making X true makes φ true), $\text{Taut} = \text{Cn}_2(\emptyset)$, and $C_{\text{Fm}}(X) = \text{Fm}$ is the trivial operator. Hypotheses are step sets (Definition 2.2) or finitary consequence relations h , with $h(A) = \{\varphi : A \vdash_h \varphi\}$; h is $\mathcal{A}$ -coherent if $A \not\vdash_h \perp$ for all $A \in \mathcal{A}$, read as non-triviality $h(A) \neq \text{Fm}$ in languages without $\perp$. Unless stated otherwise $\mathcal{A}$ is *fixed and known*, independently of the target.

6.1 Coherence as negative data

Here hypotheses are step sets R in a countable class $\mathcal{H}$ with prior w , $\sum_R w(R) = 1$, and target R^* ; every $A \in \mathcal{A}$ is target-coherent. An *argument* π from A to $\perp$ is a derivation of $\perp$ from A ; $\text{Steps}(\pi)$ is its finite set of steps.

Definition 6.1 (Online coherence protocol). [T2 §2.2] In round t the learner holds a version space $\text{VS}_t \subseteq \mathcal{H}$ ($\text{VS}_0 = \mathcal{H}$) and announces an *accepted set* $\hat{R}_t$, whose steps the reasoner may chain. The environment, possibly adversarial, possibly the learner’s own proof search, then either (P) presents some $s \in R^*$, and the learner deletes every $R \in \text{VS}_t$ with $s \notin R$; or (N) exhibits an argument π from some $A \in \mathcal{A}$ to $\perp$ with $\text{Steps}(\pi) \subseteq \hat{R}_t$, and the learner deletes every $R \in \text{VS}_t$ with $\text{Steps}(\pi) \subseteq R$; or does nothing. A (P)-round with $s \notin \hat{R}_t$ is an *incompleteness error*; an (N)-round is a *detection*.

Lemma 6.2 (Negative bags; trivial). [T2 Lemma 2.1, as repaired] Let π be an argument from a designated A to $\perp$. (a) $\text{Steps}(\pi) \not\subseteq R^*$. (b) Every $R \supseteq \text{Steps}(\pi)$ has $\perp \in \text{Cl}_R(A)$, witnessed by π . (c) The designation alone excludes every h with $A \vdash_h \perp$; deletion by the witness removes only $\{h : \text{Steps}(\pi) \subseteq h\}$, which can be strictly smaller even for consequence relations (target Cn_2 , $A = \{p\}$, π with steps $p \triangleright q$, $q \triangleright \perp$, and h the least consequence relation containing $p \triangleright \perp$). Lean: *negative_bag*

Proof. (a) Otherwise chaining gives $\perp \in \text{Cl}_{R^*}(A)$. (b) Chain in R . (c) $h(X) = X \cup \{\perp\}$ if $p \in X$, else $h(X) = X$. $\square$

So a contradiction is a multiple-instance label (Dietterich et al., 1997): “not all of these steps are valid”. For a learner that can test $A \triangleright \perp \in h$ by membership it collapses to an ordinary negative example and π is only a checkable witness; the bag structure matters for step verifiers that are not cut-closed, such as process reward models and ensembles.

Proposition 6.3 (Caution receives no signal). [T2 Prop 2.9(a)–(c), as repaired] (a) If $\hat{R}_t \subseteq R^*$, no detection can occur; since deletions never remove R^* , a certified-sound learner ($\hat{R}_t \subseteq \bigcap \text{VS}_t$) never sees one. (b) Certified soundness can be expensive. For $k \geq 1$ and distinct atoms a_i, p_i ($1 \leq i \leq k$), $\ell_i^1 = p_i$, $\ell_i^0 = \neg p_i$, let $h_b = \{\Gamma \triangleright \varphi : \{a_i \rightarrow \ell_i^{b_i}\}_{i \leq k} \cup \Gamma \models \varphi\}$ for $b \in \{0, 1\}^k$, and $\mathcal{A} = \{\emptyset\}$. Every certified-sound learner makes $2^k - 1 = |\mathcal{H}| - 1$ incompleteness errors on some text. (c) If $\mathcal{A}$ is fixed and known and membership in and $\mathcal{A}$ -coherence of each $h \in \mathcal{H}$ are decidable, coherence is prior knowledge (a restriction of $\mathcal{H}$), not data.

(a) and (c) are immediate; for (b), $\triangleright \bigvee_i (a_i \rightarrow \ell_i^{1-c_i})$ lies in h_b iff $b \neq c$, so presenting it while h_c survives is an error that deletes only h_c (Appendix E.2). So coherence feedback is a dividend paid on boldness, and with $\mathcal{A}$ fixed and coherence decidable it is not even data. It is an informative *signal* when consistency is undecidable (Π_1 ; Theorem 6.33), so that a prover must find contradictions for the learner, and when designations are data about the target (Proposition 6.12).

Oligarchic acceptance. How should a bold learner accept, so that each contradiction teaches it something? Not by voting: stepwise majority over coherent verifiers can derive $\perp$ while implicating no member (Theorem 3.12).

Definition 6.4 (Oligarchic halving). ^[T2 §2.2] OH chooses a *coalition* $S_t \subseteq \text{VS}_t$ with $w(S_t) \geq \frac{1}{2}w(\text{VS}_t)$ and announces $\hat{R}_t = \bigcap_{R \in S_t} R$.

Theorem 6.5 (Oligarchic halving). ^[T2 Thm 2.2] Under OH with $w(R^*) > 0$: (i) R^* is never deleted; (ii) there are at most $\log_2(1/w(R^*))$ detections (at most $\log_2 |\mathcal{H}|$ for the uniform prior), and the bound says nothing about incompleteness errors; (iii) if $R^* \in S_t$, then $\text{Cl}_{\hat{R}_t}(B) \subseteq \text{Cl}_{R^*}(B)$ for all B ; (iv) always $\text{Cl}_{\hat{R}_t}(B) \subseteq \text{Cl}_R(B)$ for all $R \in S_t$, so chaining never leaves what each coalition member accepts. *Lean: thm_2_2 (finite classes)*

Proof. (i) (P)-data lie in R^* , and (N)-deletions spare R^* by Lemma 6.2(a). (ii) In a detection every $R \in S_t$ contains $\hat{R}_t \supseteq \text{Steps}(\pi)$, so all of S_t is deleted and $w(\text{VS}_{t+1}) \leq \frac{1}{2}w(\text{VS}_t)$; (P)-rounds never add weight. So $w(R^*) \leq 2^{-D}$ after D detections. (iii), (iv) $\hat{R}_t \subseteq R$ for $R \in S_t$. $\square$

The halving argument is that of Barzdin and Freivalds (1972) and Littlestone (1988); new is only the reduction of a bag to a coalition. The choice of S_t is the learner’s style (“simplest half” is MDL; “boldest half” is the boldest oligarchic choice). Oligarchy needs *truth maintenance*: lemmas certified by an old coalition must be re-validated, or the union over coalitions reproduces the doctrinal paradox across time.

Proposition 6.6 (Per-round halving forces oligarchy). ^[T2 Prop 2.3, as repaired] Let VS be countable with $w(\text{VS}) > 0$, and assume every finite $P \subseteq \hat{R}$ with $P \not\subseteq \bigcap \text{VS}$ (i.e. legal for some target in VS) can occur as $\text{Steps}(\pi)$. Then every possible detection deletes at least $\frac{1}{2}w(\text{VS})$ iff $\hat{R} \subseteq \bigcap S$ for some $S \subseteq \text{VS}$ with $w(S) \geq \frac{1}{2}w(\text{VS})$. *Lean: prop_2_3 (finite version spaces)*

Proof in Appendix E.1. The scope is per-round halving against a worst-case prover; coalitions of weight $\frac{1}{3}$ still give $\log_{3/2}(1/w(R^*))$. Under universal domain, independence and unanimity, and for suitably rich agendas, aggregation rules yielding deductively closed collective judgments are oligarchic (Gärdenfors, 2006; Dietrich and List, 2008); Proposition 6.6 is the learning-theoretic face of that result. Designations can be wrong (“air pressure is 0” plus full background knowledge is inconsistent), so we also need a robust version.

Theorem 6.7 (Mis-designated contexts). ^[T2 Thm 2.5] On a detection multiply $w(R)$ by $\beta \in [0, 1]$ for every $R \supseteq \text{Steps}(\pi)$; choose S_t with $w_t(S_t) \geq \frac{1}{2}W_t$ (W_t the total weight, $W_0 = 1$) and announce $\bigcap S_t$; positive data are noise-free. If m of the D detections are false alarms (their context is target-incoherent), then $D \leq (\ln(1/w_0(R^*)) + m \ln(1/\beta)) / \ln(2/(1 + \beta))$, with $0 \cdot \ln(1/0) := 0$. *Lean: thm_2_5*

Proof. Each detection penalizes a superset of S_t , so $W_{t+1} \leq \frac{1+\beta}{2}W_t$; (P)-rounds never raise W_t and never delete R^* . R^* is penalized only in false alarms, so $\beta^m w_0(R^*) \leq w_T(R^*) \leq (\frac{1+\beta}{2})^D$. $\square$

This is the bound of Littlestone and Warmuth (1994). A structural learner that trusts a false alarm fully fares worse: it must become sub-classical everywhere (Proposition 11.10).

Proposition 6.8 (Boldness is a two-sided trade-off). *[T2 Prop 2.9(d),(e), added after verification]* (a) On the class of Proposition 6.3(b) with uniform prior, every learner with $\hat{R}_t \subseteq R$ for some $R \in \text{VS}_t$ in every round, in particular every OH learner, makes $2^k - 1$ incompleteness errors on some text for some target. The non-oligarchic union learner $\hat{R}_t = \bigcup \text{VS}_t$ makes no error of either type there, and is silently unsound. (b) On the class of Theorem 3.12, no accepted set guarantees that every possible incompleteness error and every possible detection deletes half the weight.

Proof in Appendix E.2. *Reading.* Oligarchy makes each contradiction worth a bit, robustly. It does not make boldness cheap: on this class every oligarchic learner, including the boldest, pays exactly the price of caution in missed valid steps, and a learner that avoids this price must at some round accept a set inside no surviving hypothesis, hence a target-invalid step. The two-sided minimax is open (Section 14); Section 7 resolves the trade-off architecturally, with cautious assertion and a bold sandbox under attack.

What coherence cannot see. Now let hypotheses be finitary consequence relations with target $\vdash^*$, let $h \oplus D$ be the least consequence relation containing $h \cup D$, and let the environment be complete. Call h *text-refutable* if $\vdash^* \not\subseteq h$, *coherence-refutable* if it is $\mathcal{A}$ -incoherent, and *refutable by coherence with data* if $h \oplus D$ is $\mathcal{A}$ -incoherent for some finite $D \subseteq \vdash^*$ (accepted human conclusions used as premises).

Theorem 6.9 (Silent over-generalizations; trivial). *[T2 Thm 2.6]* (i) h is refutable by coherence with data iff $h \oplus \vdash^*$ is $\mathcal{A}$ -incoherent. (ii) The hypotheses refuted in none of the three ways are exactly $U(\vdash^*) = \{h : \vdash^* \subseteq h, h \text{ } \mathcal{A}\text{-coherent}\}$. (iii) Each $h \in U(\vdash^*) \setminus \{\vdash^*\}$ is unsound, and if it is a possible target, every finite stage of every data stream for $\vdash^*$ is one for h .

Proof. (i) $h \oplus \vdash^*$ is the directed union of the $h \oplus D$, and a finite derivation of $\perp$ lies in one of them. (ii) If $\vdash^* \not\subseteq h$, h is text-refutable; otherwise $h \oplus D = h$ for all $D \subseteq \vdash^*$. (iii) $h \supsetneq \vdash^*$ accepts an invalid sequent; data for $\vdash^*$ are data for h , and $\mathcal{A}$ is the same. $\square$

Proposition 6.10 (Coherent chains have coherent unions). *[T2 Prop 2.7]* The union of a chain of $\mathcal{A}$ -coherent finitary consequence relations is one: cut and $A \triangleright \perp$ involve finitely many sequents, which lie in one member.

Corollary 6.11 (Fixed designations do not remove limit points). *[T2 Cor 2.8, as repaired]* Let $\mathcal{A}$ be fixed and known independently of the target. If the possible targets include a strictly increasing chain $h_1 \subsetneq h_2 \subsetneq \dots$ of $\mathcal{A}$ -coherent hypotheses together with $h_\omega = \bigcup_n h_n$, no learner, computable or not, identifies them in the limit (EX or BC) from text plus coherence information.

Proof. The coherence information is target-independent, so fold it into the learner. A learner for h_ω has a locking sequence σ (Blum and Blum, 1975), or its BC analogue; its finite content lies in some h_k , and on a text for h_k extending σ the learner conjectures h_ω forever. $\square$

This is Gold’s limit-point theorem (Gold, 1967). The hypothesis “ $\mathcal{A}$ fixed” is essential: designations that are data about the target, as when a problem setter certifies a setup consistent with the true theory, are genuine negative data.

Proposition 6.12 (Target-dependent designations are negative data; trivial). *[T2 Prop 2.8', repair]* Suppose the environment designates only target-coherent finite contexts, and eventually all of them. If every hypothesis satisfies classical reductio ($\Gamma \vdash_h \varphi$ iff $\Gamma \cup \{\neg\varphi\} \vdash_h \perp$), text plus designations is an informant, and every countable such class is EX-identifiable by enumeration. Example: for $h_n = \text{Cn}_2\{\neg p_i : i < n\}$ and $h_\omega = \text{Cn}_2\{\neg p_i : i \in \omega\}$, $\{p_j\}$ is h_n -coherent iff $j \geq n$, so the limit point is identified.

Proof in Appendix E.3. So with fixed designations coherence prunes only over-generalizations that are not limit points, and positive-data structure (Section 5) decides the rest. Text plus coherence *refutes* every wrong hypothesis exactly when $U(\vdash^*) \cap \mathcal{H} = \{\vdash^*\}$ (a condition for refutation, not identifiability: $\{\vdash^*, h\}$ with $h \supsetneq \vdash^*$ is identifiable from text). *Reading*: a coherence loss removes over-acceptance that *conflicts*; the silent over-generalizations $U(\vdash^*)$ are the precise content of “coherent but wrong”, and whether they exist depends on the target, not on the learner.

6.2 Post-completeness: when coherence suffices

A consequence operator C is *structural* if $\sigma C(X) \subseteq C(\sigma X)$ for every substitution σ : rules are uniform schemas, so a learner that generalizes steps to schemas (Section 2.2) is a structural learner.

Theorem 6.13 (Structural Post-completeness; known). *[T2 Thm 3.1]* If Fm contains $\neg$ and one of $\wedge, \vee, \rightarrow$, all interpreted by their truth functions, every structural closure operator $C \supseteq \text{Cn}_2$, finitary or not, is Cn_2 or C_{Fm} . The same holds for pure $\{\rightarrow\}$. *Lean: Post.eq_Cn2_or_eq_trivial*
Lean: Post.impFrag_postComplete

Proof (main case). Let $\varphi \in C(X) \setminus \text{Cn}_2(X)$, with $v[X] = 1$, $v(\varphi) = 0$. Fix a tautology $\top_0$ in one atom and $\perp_0 = \neg\top_0$, and let $\sigma_v(p) = \top_0$ or $\perp_0$ as $v(p) = 1$ or 0 . Every Boolean u has $u(\sigma_v\psi) = v(\psi)$, so $\sigma_v X \subseteq \text{Taut} \subseteq C(\emptyset)$ and $\sigma_v\varphi$ is a contradiction. By structurality $\sigma_v\varphi \in C(\sigma_v X) \subseteq C(C(\emptyset)) = C(\emptyset)$, hence $\text{Fm} = \text{Cn}_2(\{\sigma_v\varphi\}) \subseteq C(\emptyset)$. For $\{\rightarrow\}$ see Appendix E.4. $\square$

At the level of theorems this is Post (1921); the consequence-level form is standard too (Pogorzelski, 1971; Wójcicki, 1988; Pogorzelski and Wojtylak, 2008), and Proposition 3.5 is the same substitution trick for step sets. One falsifying row of one bad rule yields a derivation of $\perp$ from no premises. Without a theorem in the language there is one more extension: in the $\{\wedge, \vee\}$ fragment ($\text{Cn}_2(\emptyset) = \emptyset$) let $C_{\text{ai}}(\emptyset) = \emptyset$ and $C_{\text{ai}}(X) = \text{Fm}$ for $X \neq \emptyset$ (*almost inconsistent*).

Proposition 6.14 (The theorem-free fragment; presumably folklore). *[T2 Prop 3.2]* In the $\{\wedge, \vee\}$ language the structural closure operators extending Cn_2 are exactly Cn_2 , C_{ai} and C_{Fm} .

Proof in Appendix E.4. So there coherence must be enforced on a non-empty designated context. For sequents D let $\langle D \rangle$ be the least structural consequence relation containing D , and for a designated Cn_2 -consistent A_0 let $\mathcal{S}(D, A_0) = \{h \text{ structural} : D \subseteq h, A_0 \not\vdash_h \perp\}$.

Theorem 6.15 (Coherence pins classical logic). *[T2 Thm 3.3, as repaired]* Let the language be as in Theorem 6.13. (a) For $D \subseteq \text{Cn}_2$: $\mathcal{S}(D, A_0) = \{\text{Cn}_2\}$ iff $\langle D \rangle = \text{Cn}_2$, and a finite D suffices (with $\rightarrow$ in the language, Hilbert-style bases with modus ponens as the only rule; without $\rightarrow$, by Rautenberg 1981; Appendix E.4). (b) Let $\mathcal{H} = (h_i)$ be any uniformly decidable family of structural consequence relations with $\text{Cn}_2 \in \mathcal{H}$. The learner outputting the least i such that h_i contains the data so far and $A_0 \not\vdash_{h_i} \perp$ identifies Cn_2 in the limit from every text, with at most i^* mind changes

(i^* the least index of Cn_2), for every ordering of $\mathcal{H}$: no simplicity or minimality bias is needed.
(c) Each assumption is needed: $\text{Cn}_2 + \{\triangleright p\}$ (p not in A_0) is non-structural, A_0 -coherent and contains every text; C_{Fm} contains every text; in the theorem-free $\{\wedge, \vee\}$ fragment, C_{ai} survives the empty context.
Lean: `Post.versionSpace_eq_singleton_iff` Lean: `Post.learner_identifies`

Proof idea. (a) Members of $\mathcal{S}$ extend $\langle D \rangle = \text{Cn}_2$, so are Cn_2 or C_{Fm} , and A_0 excludes C_{Fm} .
(b) For $j < i^*$, either the text refutes h_j , or $h_j \supsetneq \text{Cn}_2$ is C_{Fm} and A_0 rejects it. Details in Appendix E.4. $\square$

The form of the basis matters: if the $\wedge$ clauses are sequent rules ($p \wedge q \triangleright p, \dots$) while modus ponens is the only rule for $\rightarrow$, then $\triangleright p \rightarrow (q \rightarrow p \wedge q)$ is underivable (countermodel in Appendix E.4); giving them as $\rightarrow$ -axioms repairs this.

Reading. This is a setup of the user’s shape that provably works for formal propositional mathematics. The algorithm is trivial; the content is Theorem 6.13, whose proof explains why: an invalid rule has a falsifying row, and substituting $\top/\perp$ along it turns the rule into “from tautologies infer a contradiction”. A uniform learner cannot hide an invalid schema from coherence. It must generalize schematically, though: closure under *renaming* atoms is not enough, since the least renaming-invariant extension of Cn_2 containing $p \triangleright q$ is $\emptyset$ -coherent with theorems Taut ([L5 Prop 2.5], Appendix E.4). In Experiment B (Section 12) a bold learner over classical natural deduction accepted all 13 valid and refuted all 44 invalid structural candidate schemas, each by an explicit derivation of $\vdash \perp$ (median 2.5 steps for generalizations of target rules), while the non-structural $\vdash p$ survived all eight designated contexts (computed; quick run).

Proposition 6.16 (Theorems versus consequence). *[T2 Props 3.4, 3.5, as repaired] For a substitution-closed theorem set Th let $C_{\text{adm}}(X) = \{\varphi : \forall \sigma (\sigma X \subseteq \text{Th} \Rightarrow \sigma \varphi \in \text{Th})\}$. (a) C_{adm} is the largest structural closure operator with theorems Th; on finite premise sets it is the structural completion (admissible rules). (b) For $\text{Th} = \text{Taut}$ in a language as in Theorem 6.13, $C_{\text{adm}} = \text{Cn}_2$ (structural completeness); in the $\{\wedge, \vee\}$ fragment $C_{\text{adm}} = C_{\text{ai}}$. (c) For IPC, $C_{\text{adm}} \supsetneq \vdash_{\text{IPC}}$ (Harrop’s rule $\neg p \rightarrow q \vee r / (\neg p \rightarrow q) \vee (\neg p \rightarrow r)$ is admissible, not derivable; Harrop 1960; Rybakov 1997; Iemhoff 2001), and the two have the same theorems and the same coherence profile, $A \vdash_{\text{adm}}^{\text{IPC}} \perp$ iff $A \vdash_{\text{CPC}} \perp$.*
Lean: `Post.le_Cn2_of_empty_eq_Taut`

Proof in Appendix E.4. So a learner can learn the theorems and then take the structural completion; a learner that is not bold here may settle on $X \mapsto X \cup \text{Taut}$, which has the right theorems but cannot chain inferences from premises. And no learner fed theorems and coherence data can tell IPC-derivability from IPC-admissibility.

Theorem 6.17 (Intuitionistic logic: coherence is blind). *[T2 Thm 3.6] (a) For every intermediate logic L and finite A: $A \vdash_L \perp$ iff $A \vdash_{\text{CPC}} \perp$. (b) A learner outputting a maximal coherent structural hypothesis never outputs IPC. (c) There is a strictly increasing chain of finitely axiomatizable intermediate logics $L_1 \subsetneq L_2 \subsetneq \dots$ whose union L_ω is intermediate, and no learner identifies $\{L_k : k \leq \omega\}$ from text plus coherence information, for any A, even with target-dependent designations. (d) The conservative learner outputting $\langle D_n \rangle$ BC-identifies every finitely axiomatizable structural target.*

Proof idea. (a) is Glivenko’s theorem (Glivenko, 1929). (c) uses Jankov formulas of an infinite antichain of finite Heyting algebras (Jankov, 1968; Chagrov and Zakharyashev, 1997); by (a) all designations are the same for every member, so Corollary 6.11 applies. Appendix E.5.

So for IPC the work is done by *minimality* (Section 5), not coherence. Minimality is right when the target is at the bottom of its coherent version space, coherence plus boldness when it is at the top; classical logic is at both ends. In Experiment B a bold learner over intuitionistic natural deduction accepted excluded middle, double-negation elimination, reductio, Peirce’s law, weak excluded middle and Dummett’s linearity, all IPC-underivable (computed; quick run).

Complete theories. Much of the algebra in a physics calculation is real-closed-field reasoning, and RCF is complete. Is it pinned like CPC?

Proposition 6.18 (Complete theories: theorems, and rules under $\exists$ -closure). *[T2 Prop 3.11, repaired]*
Let T be a complete consistent first-order theory. (a) The only consistent closed theory in the language of T containing T is T ; if T is decidable, the learner of Theorem 6.15(b), run on a uniformly decidable family of theories containing T , identifies T from a text and the coherence datum $\emptyset$. (b) Add parameters (fresh constants); the target is $\vdash_T$ ($\Gamma \vdash_T \varphi$ iff $T \cup \Gamma \vdash \varphi$). If $h \supseteq \vdash_T$ is closed under substituting terms for parameters and under $\exists$ -elimination (from $\Gamma \vdash_h \exists x\psi$ and $\Gamma, \psi(d) \vdash_h \chi$, d fresh, infer $\Gamma \vdash_h \chi$), and $\emptyset \not\vdash_h \perp$, then $h = \vdash_T$. (c) Without $\exists$ -closure (b) fails, even for a non-empty context. For $T = \text{RCF}$, the least parameter-structural $h_2 \supseteq \vdash_{\text{RCF}}$ containing all instances of $t^3 - 4t + 1 = 0 \triangleright t > 1$ has $h_2(\emptyset) = \text{Cn}_{\text{RCF}}(\emptyset)$ and $h_2(\{c^3 - 4c + 1 = 0\}) = \text{Cn}_{\text{RCF}}(\{c^3 - 4c + 1 = 0, c > 1\})$: coherent on both, yet unsound (the cubic has three real roots and Galois group S_3). The context $\{c^3 - 4c + 1 = 0, c < 0\}$ catches it.

Proof idea. (b) An invalid $\Gamma(\bar{c}) \vdash_h \varphi(\bar{c})$ gives $T \vdash \exists \bar{x}(\wedge \Gamma \wedge \neg \varphi)$; a fresh instance derives $\perp$ in h , and $\exists$ -elimination discharges the parameters one at a time. (c) The rule fires only on terms the context forces to be roots, and $\mathbb{Q}(r)$ contains no other root. Appendix E.6, which also gives a simpler example ($t \cdot t = 2 \triangleright t > 0$) caught by $\{c \cdot c = 2\}$.

Reading. At the level of theorems, RCF, Presburger arithmetic and Tarski’s geometry are pinned as cleanly as CPC. At the level the user cares about, rules applied in contexts with parameters, a step verifier that is merely a set of uniform rules is *not* pinned by one coherence datum; it is pinned if its hypotheses are closed under $\exists$ -elimination, a discharge meta-rule rather than a step, or if the designated contexts are rich enough. (Whether a fixed finite family of contexts suffices for RCF is open. With sin on $\mathbb{R}$, which defines $\mathbb{Z}$, incompleteness enters; for exp the question is open: $\text{Th}(\mathbb{R}, \text{exp})$ is model complete (Wilkie, 1996), and decidable if Schanuel’s conjecture holds (Macintyre and Wilkie, 1996).) Designation is soundness-critical in the other direction too: designating an idealization together with the background it contradicts forces every structural coherent learner to lose a classical schema *everywhere* (Proposition 11.10), which is the formal content of the user’s air-pressure worry and the subject of Section 11.

6.3 Arithmetic: coherence is Popperian

Hypotheses are r.e. theories $T \supseteq \text{PA}$, world feedback is a Δ_0 -oracle, and $\mathbb{N}$ is the intended model; $\mathbb{Q}$ is Robinson arithmetic.

Lemma 6.19 (Δ_0 feedback is subsumed by coherence; standard). *[T2 Lemma 3.7]* A consistent $T \supseteq \mathbb{Q}$ proves every true Δ_0 sentence and no false one, so no Δ_0 feedback refutes it.

Proof. $\mathbb{Q}$ is Σ_1 -complete (Boolos et al., 2007); if T proved a false δ it would also prove $\neg \delta$. $\square$

Theorem 6.20 (No maximal r.e. hypothesis; cited). *[T2 Thm 3.8, as repaired]* Every consistent r.e. $T \supseteq \mathbb{Q}$ has a ρ with $T + \rho$ and $T + \neg \rho$ consistent (Gödel, 1931; Rosser, 1936; Tarski et al., 1953).

So a bold learner with r.e. hypotheses and $\mathcal{A} = \{\emptyset\}$ has no maximal choice. (If $\mathcal{A}$ contains all finite sets of true sentences, T is $\mathcal{A}$ -coherent iff $T \subseteq \text{Th}(\mathbb{N})$, and at most one refinement survives.)

Theorem 6.21 (Turing-chain impossibility). *[T2 Thm 3.9, as repaired] Let $T_0 = \text{PA}$, $T_{k+1} = T_k + \text{Con}(T_k)$, $T_\omega = \bigcup_k T_k$ (Turing, 1939; Feferman, 1962), and let $\mathcal{A}$ be any fixed family of finite sets of true sentences. No learner identifies a class containing $\{T_k : k \leq \omega\}$ from text, $\mathcal{A}$ -coherence and a Δ_0 -oracle.*

Proof. All T_k are sound, so all side information is target-independent; the chain is strict by Gödel II; apply Corollary 6.11. $\square$

The Σ_1 -unsound $T_k + \neg \text{Con}(T_k)$ are natural *hypotheses*: consistent, unrefuted by Δ_0 data, and refuted by coherence only relative to accepted premises proving $\text{Con}(T_k)$. Restricting designations to true contexts is essential: if $\{\neg \text{Con}(T_k)\}$ could be designated as data about the target, the chain would be identifiable (Proposition 6.12).

Theorem 6.22 (Coherence as a Π_1 -truth detector; the Σ_2 barrier). *[T2 Thm 3.10, as repaired] (a) For Π_1 π , $\text{PA} + \pi$ is consistent iff π is true. (b) For Σ_1 σ , truth implies consistency of $\text{PA} + \sigma$, not conversely ($\sigma = \neg \text{Con}(\text{PA})$). (c) The Popperian learner (“true” on a Π_1 sentence until PA refutes it; “false” on a Σ_1 sentence until PA proves it) converges to the truth on $\Sigma_1 \cup \Pi_1$ with at most one mind change per sentence. (d) The naively bold learner (accept φ_n iff consistent with PA plus the earlier accepted sentences) converges to a Lindenbaum completion that depends on the enumeration, and on every enumeration beginning with $\neg \text{Con}(\text{PA})$ it is Σ_1 -unsound. (e) Every computable learner using a computable text, coherence searches and a Δ_0 -oracle fails to converge to the truth value of some Σ_2 sentence.*

Proof idea. (a) A false Π_1 sentence has a true, hence PA-provable, Σ_1 negation. (e) Convergent computable guesses define a Δ_2 set (Shoenfield, 1959), and the true Σ_2 sentences are Σ_2 -complete. Appendix E.7.

This is Popper’s asymmetry in its standard learning-theoretic form (Putnam, 1965; Gold, 1965; Kelly, 1996), read for coherence-learners. The same boundary appears for credences, in a self-contained form of a theorem of Sawin and Demski (2013). Quantifier blocks may be empty, so Π_1 sentences also count as Π_2 and Σ_2 .

Theorem 6.23 (The coherence/ Π_1 / Π_2 trilemma). *[L3 Thm D, as repaired] Let $P_t(\varphi) \in \mathbb{Q} \cap [0, 1]$ be computable uniformly in t, φ , with (i) $\lim_t P_t(\pi) = 1$ for every true Π_1 π , and (ii) $\limsup_t (P_t(\pi) + P_t(\varphi)) \leq 1$ whenever π is Π_1 and $\text{PA} \vdash \pi \rightarrow \neg \varphi$. Then infinitely many true Π_2 sentences have $\liminf_t P_t(\varphi) = 0$.*

Proof. A false Π_2 $\varphi = \forall x \exists y R$ has a true Π_1 refuter $\forall y \neg R(\bar{x}_0, y)$, so $\limsup_t P_t(\varphi) = 0$ by (i), (ii). Thus $\{\varphi \in \Pi_2 : \liminf_t P_t(\varphi) > 0\}$, a Σ_2 set, contains no false Π_2 sentence; if it contained all but finitely many true ones, the Π_2 -complete set of true Π_2 sentences would be Σ_2 , contradicting Post’s hierarchy theorem. $\square$

So one can have at most three of: computability, weak coherence, Π_1 -convergence, and non-dogmatism on true Π_2 sentences; logical induction drops Π_1 -convergence. Coherence does *not*, however, confine credences to Δ_2 :

Proposition 6.24 (What weak coherence costs). *[L3 Thm D', as repaired] Say computable credences gradually verify C if $P_t(\varphi) \rightarrow 1$ iff φ is true, for $\varphi \in C$. (a) Convergent credences, coherent*

or not, have a Π_2 success set, so never gradually verify the Σ_2 sentences. (b) Credences as in Theorem 6.23(ii) never gradually verify all Π_2 sentences. (c) Some computable credences, weakly coherent for all pairs, gradually verify the Σ_2 sentences (necessarily without converging on all of them).

Proof in Appendix E.8. Incoherent computable credences can gradually verify the Π_2 sentences convergently (L3 Thm G(f), a standard construction given there as a sketch); so among convergent credences, and among the classes $\Sigma_1, \Pi_1, B(\Sigma_1), \Sigma_2, \Pi_2, \Pi_3$ of that success table, weak coherence costs exactly the Π_2 class, and part of what incoherent credences reach beyond that is paid for by giving up convergence, not coherence.

Reading. In arithmetic, computation adds nothing beyond coherence with a Σ_1 -complete base. What matters is how boldness is applied: boldly on universal claims, which coherence can refute, cautiously on existential ones, which need a witness. Beyond Δ_2 no computable channel decides truth in the limit (Theorem 6.22(e)); only the weaker gradual verification, with credence tending to 1 exactly on truths, reaches further (Proposition 6.24). The Gödel–Rosser alternatives are a permanent residue, removed only by accepting stronger principles (reflection, Con(PA), a stronger theory): a justification that is not a PA-proof (Section 13).

6.4 Carnap’s problem is Gold’s problem

Under inferentialism, learning valid inferences should be learning meanings. Carnap (1943) observed that the classical rules do not fix the truth tables: the all-true valuation, and “true iff tautologous”, respect every classical inference.

Definition 6.25 (Meanings). ^[T2 §4.1] A *valuation* is any $v : \text{Fm} \rightarrow \{0, 1\}$, identified with its true-set in Cantor space 2^{Fm} ; a *meaning* is a set V of valuations; BV is the set of Boolean ones. For finite Γ, Δ , $\Gamma \models_V^1 \varphi$ iff every $v \in V$ with $v[\Gamma] = 1$ has $v(\varphi) = 1$, and $\Gamma \models_V^m \Delta$ iff no $v \in V$ has $v[\Gamma] = 1$, $v[\Delta] = 0$. $\text{Val}(S)$ is the set of valuations satisfying the sequents S ; $V^\cap$ closes V under intersections of true-sets, including the empty one $v_\top \equiv 1$; v_{Taut} is the characteristic function of Taut . V is *structural* if $v \in V$ implies $v \circ \sigma \in V$ for all substitutions σ .

The sequent $\Gamma \triangleright \Delta$ excludes exactly the clopen set of valuations realizing the position $[\Gamma : \Delta]$. So *inference data are negative data about valuations*, while *coherence certifications* (“ $[A : D]$ is in bounds”) and world feedback are *positive data*: Restall’s reading of consequence (Restall, 2005) turned into a statement about data.

Lemma 6.26 (Duality). ^[T2 Lemma 4.1] (a) $\text{Val}(\models_V^m) = \overline{V}$. (b) For closed V , $C_V(X) = \bigcap \{v \in V : X \subseteq v\}$ is finitary and the valuations satisfying every valid single-conclusion sequent form $V^\cap$. (c) $\text{BV}^\cap$ is the set of characteristic functions of classical theories, including Fm . Lean: Val_mrel
Lean: lemma_4_1_b Lean: intClosure_BV_eq

Theorem 6.27 (Carnap’s problem in learning form). ^[T2 Thm 4.2, as repaired] $\models_{\text{BV}}^1 = \models_W^1$ whenever $\text{BV} \subseteq W \subseteq \text{BV}^\cap$. So no data that are a function of the single-conclusion relation (text, informant, or metarules with discharge read globally, as closure conditions) distinguish BV from W . The non-Boolean members of $\text{BV}^\cap$ are the v_T for theories T that are not maximal consistent (including $T = \text{Fm}$): $v_\top$ violates the table for $\neg$; v_{Taut} and every non-maximal consistent v_T violate those for $\neg, \vee, \rightarrow$; all respect $\wedge$. $\text{BV}^\cap$ is closed and structural. Lean: carnap_single

exclusion data (sequents)	rank	excludes
<i>non-contradiction</i> , no denial: $p, \neg p \triangleright$ (“not both P and $\neg P$ ”)	0	$v_\top$, inconsistent valuations
imitation of single-conclusion inferences	1	non-closed valuations
<i>exhaustiveness</i> : $\triangleright p, \neg p$ (“not deny both”), $p \vee q \triangleright p, q$	2	v_{Taut} , all gappy theories

Table 6: What each shape of data excludes (Theorem 6.28). Coherence *certifications* are positive data about meanings; they exclude only meanings realizing no certified position, such as $V = \emptyset$, and never $v_\top$.

Proof. By Lemma 6.26(b) every $w \in W$ satisfies $\models_{\text{BV}}^1$, and $W \supseteq \text{BV}$ gives the converse. For consistent non-maximal T , some φ has $\varphi, \neg\varphi \notin T \ni \varphi \vee \neg\varphi$. Lemma 6.26 is proved in Appendix E.9. $\square$

Under the *local* reading of rule validity (Garson, 2013), $\rightarrow\text{I}$ does exclude v_{Taut} , but only by imposing the two-conclusion sequent $\triangleright\varphi, \varphi \rightarrow \psi$. The right measure is the number of denials.

Theorem 6.28 (Denial-rank trichotomy). ^[T2 Thm 4.3] *With $\neg$ in the language, let the denial rank $d(v)$ be the least $|\Delta|$ over classically valid $\Gamma \triangleright \Delta$ (finite) that v violates (∞ if none). Then $d(v) = 0$ iff v ’s true-set is inconsistent; $d(v) = 1$ iff it is consistent but not closed; $d(v) = 2$ iff it is a consistent, closed, non-maximal theory; $d(v) = \infty$ iff $v \in \text{BV}$.* *Lean: denialRank_eq_two_iff*

Proof. $d = 0$ is a finite inconsistent subset; for consistent v , $d \leq 1$ says some $\Gamma \subseteq v$ entails some $\varphi \notin v$; a consistent closed non-maximal T omits φ and $\neg\varphi$, violating $\triangleright\varphi, \neg\varphi$; maximal theories are Boolean. $\square$

Reading. “Coherence” covers two kinds of data (Table 6): certifications that a position is in bounds (positive about V), and 0-denial constraints such as $p, \neg p \triangleright$ (negative about V), which remove $v_\top$. Neither removes v_{Taut} . A learner trained on imitation plus a coherence loss, in either sense, is provably free to settle on gappy valuations: consistent, inferentially competent, never committing on open questions. These are van Fraassen’s supervaluations (van Fraassen, 1966), the honest semantics of “settled by the premises”. Pinning classical meaning needs a loss with *two* denials. Bilateral signed sequents (Smiley, 1996; Rumfitt, 2000) carry the same rank-2 information as multiple conclusions. A testable prediction for language-model training: imitation plus a contradiction penalty yields consistent but noncommittal reasoners.

With rank-2 data finitely many data suffice. Let TT be the eleven truth-table schemata: $\varphi, \neg\varphi \triangleright$ and $\triangleright\varphi, \neg\varphi$; $\varphi \wedge \psi \triangleright \varphi, \varphi \wedge \psi \triangleright \psi$, $\varphi, \psi \triangleright \varphi \wedge \psi$; $\varphi \triangleright \varphi \vee \psi, \psi \triangleright \varphi \vee \psi, \varphi \vee \psi \triangleright \varphi, \psi$; $\triangleright\varphi, \varphi \rightarrow \psi, \psi \triangleright \varphi \rightarrow \psi, \varphi, \varphi \rightarrow \psi \triangleright \psi$. TT(p, q) are their instances in distinct atoms.

Theorem 6.29 (Bilateral determinacy with a twelve-datum tell-tale). ^[T2 Thm 4.4, as repaired] *Let the language be $\{\neg, \wedge, \vee, \rightarrow\}$ (with $\perp, \top$ add $\perp \triangleright$ and $\triangleright \top$; the Lean version has $13 + 1$ data). (a) $\text{Val}(\text{TT}) = \text{BV}$, which is closed. (b) If V is structural, every $v \in V$ satisfies the eleven sequents TT(p, q), and $V \neq \emptyset$ (one coherence datum), then $V = \text{BV}$. (c) So among structural meanings BV is identified finitely by twelve data; without the coherence datum $V = \emptyset$ survives, the Gold over-generalization. (d) Without structurality BV is not identifiable in the limit at all, even with the coherence datum: every finite set of valid sequents is valid in a closed meaning $\text{BV} \cup \{v'_\chi\}$, v'_χ a Boolean valuation flipped at a formula χ the set does not mention. And the dense non-structural $\text{BV} \setminus \{u_0\}$ has the same valid sequents and in-bounds positions as BV.* *Lean: structural_eq_BV*
Lean: BV_not_BC_learnable

	Gold (sequent space)	Carnap (valuation space)
hypothesis	consequence relation	set V of admissible valuations
positive data	valid sequents (text)	coherence certifications; world feedback
negative data	coherence certifications	sequents, incl. 0-denial ones
over-generalization	trivial logic	$v_{\top}, v_{\text{Taut}}, \dots$
failure	<i>learnability</i> : finite data never exclude supersets	<i>identifiability</i> : complete single-conclusion data fix V only up to $(\overline{V})^{\cap}$
remedy	one coherence datum plus Post-completeness	two-denial data, or a compositional prior

Table 7: Gold’s and Carnap’s problems as two instances of one duality [T2 §4.5; L4 §3].

Proof idea. (a) Each schema enforces one table row. (b) v satisfies $\sigma(s)$ iff $v \circ \sigma \in V$ satisfies s ; and any Boolean u is $v \circ \sigma$ for σ sending atoms to $p_0 \vee \neg p_0$ or its negation. (d) No tell-tale, then the locking-sequence argument. Appendix E.9.

The three two-conclusion schemata in TT are exactly the rank-2 data that v_{Taut} violates. Brute force over the sixteen formulas of depth ≤ 1 in two atoms confirms (a), Theorem 6.27 and all 2^{16} denial ranks; the T2 verifier found, over all structural meanings of two- and three-element matrices, no non-Boolean survivor of the twelve data and each datum necessary (computed). A second remedy restricts the hypothesis class, as in Gold’s setting:

Proposition 6.30 (A compositional prior suffices). *[T2 Prop 4.5] Exactly one two-element matrix with designated value 1 validates explosion, $\neg\neg p \triangleright p$, $p \triangleright \neg\neg p$, the three $\wedge$ rules, $\vee$ -introduction, disjunctive syllogism, modus ponens, $q \triangleright p \rightarrow q$ and $\neg p \triangleright p \rightarrow q$: the standard one.*

Proof in Appendix E.9; compare Bonnay and Westerståhl (2016), whose compositionality is more general. World feedback alone fixes nothing, since the actual valuation is Boolean and never refutes $V \supseteq \text{BV}$; under the compositional prior it identifies the tables once every row has been observed [T2 Prop 4.6].

Reading. Carnap’s problem is a semantic twin of Gold’s only in part (Table 7). Gold’s trivial logic ($V = \emptyset$) is removed by a coherence certification, Carnap’s $v_{\top}$ by a 0-denial sequent, and v_{Taut} , the Horn blindness of single-conclusion data, by neither. So imitation plus coherence fixes inferential role, but not classical truth conditions. That does not matter for deriving; it matters for interpreting world feedback (“ A is false” licenses $\neg A$ only through the bilateral bridge) and for “verification from truth”. And rules can be identifiable when truths are not: Theorem 6.29 identifies classical meaning finitely, while no computable learner decides arithmetical truth (Theorem 6.22(e)).

6.5 Tonk, harmony and conservativity

Prior (1960)’s tonk ($\varphi \triangleright \varphi \text{ tonk } \psi$, $\varphi \text{ tonk } \psi \triangleright \psi$) is the paradigm of a rule search exploits (Theorem 3.2); Belnap (1962) demanded that new rules be *conservative*.

Proposition 6.31 (Tonk). *[T2 Prop 5.1] Under cut, tonk gives $\varphi \vdash \psi$ for all φ, ψ , and coherence refutes it in two steps ($a \triangleright a \text{ tonk } \perp \triangleright \perp$) if $\mathcal{A}$ has a non-empty context or the logic has a theorem. In a theorem-free base with $\mathcal{A} = \{\emptyset\}$ it survives: classical $\{\wedge, \vee\}$ plus tonk is C_{ai} . Without cut tonk need not trivialize (Cook, 2005; Ripley, 2015).*

Proof. One cut gives the first claim. In the theorem-free case every rule has premises, so nothing is derivable from $\emptyset$, while from any premise tonk yields everything. $\square$

Transitivity is double-edged: it lets one bad rule poison everything, and it lets coherence catch it. In Experiment B a bold learner offered the two tonk rules in either order accepted the first and rejected the second (each alone has a truth-table reading): boldness is order-dependent when individually coherent rules are jointly incoherent (computed; quick run).

Proposition 6.32 (Coherence is not conservativity). *[T2 Prop 5.2, as repaired] Intuitionistic $\rightarrow$ plus $\neg$ -introduction, $\neg$ -elimination and $\neg\neg$ -elimination is the $\{\rightarrow, \neg\}$ fragment of CPC: coherent, but not conservative over $\text{IPC}_{\rightarrow}$ (it proves Peirce’s law). $\neg\neg$ -elimination alone would be conservative. Conservativity implies coherence when coherence is non-triviality on the old vocabulary, or $\perp$ with *ex falso* is in the base, but not in general ($\text{IPC}_{\rightarrow} + \{\triangleright\perp\}$, with $\perp$ new and without *ex falso*).*

Proof in Appendix E.10. So Belnap’s learner rejects classical negation over intuitionistic positive logic, and the coherence learner accepts it. Harmony *with normalization* gives conservativity (Prawitz, 1965; Dummett, 1991), but local harmony alone does not: Read’s bullet (Read, 2000), introduced from a derivation of $\perp$ from $[\bullet]$ and eliminated from $\bullet, \bullet$ to $\perp$, has local reductions yet derives $\perp$ from nothing. Over a Post-complete base the two notions coincide ^[L5 Cor 2.4]: if C' is structural on $L' \supseteq L$ with $C'|_L \supseteq \text{Cn}_2(L)$ (L as in Theorem 6.13), then $C'|_L$ is Cn_2 or C_{Fm} , so C' is conservative iff it is coherent on one classically consistent context. Belnap’s existence condition is, over classical logic, non-triviality; harmony matters only over sub-classical bases.

Theorem 6.33 (Coherence is Π_1 , conservativity Π_2). *[T2 Thm 5.3] For finite rule systems (elementary formal systems; Post 1943; Smullyan 1961): (a) $\{R : R \not\vdash \perp\}$ is Π_1 -complete; (b) for some fixed finite base B , $\{N : B + N \text{ conservative over } B\}$ is Π_2 -complete; (c) so coherence is decidable in the limit with one mind change, while conservativity is not decidable in the limit by any computable learner, only refutable in the limit (Kelly, 1996); (d) for decidable B (e.g. CPC, IPC) conservativity is Π_1 .*

Proof idea. (a) Simulate $\varphi_e(e)$ by rules and add “halt $\triangleright\perp$ ”. (b) A new symbol c with $\triangleright c \bar{e}$ and $c y, \text{Num}(x) \triangleright T(x, y)$ over a universal base is conservative iff φ_e is total. Appendix E.10.

Reading. A coherence filter can be built into a convergent learner (Theorem 6.15(b)); a conservativity filter cannot in general (over the base of (b)). Coherence plus decidable local checks such as harmony are the learner-friendly norms. Where non-conservativity is the point, as for theoretical terms in physics, a concept whose rules smuggle in a substantive inference (Dummett’s *Boche*) is coherent, and only world feedback can test it.

6.6 Fallacies and the Kripkensteinian residue

A learner with a *generalization operator* $\mathcal{G}$ (e.g. structural closure of least generalizations) turns finitely many fallacious human steps F into rules $\mathcal{G}(F)$. Let h^* be the target consequence relation.

Theorem 6.34 (Elimination criterion; trivial). *[T2 Thm 6.1] Coherence (with data) eventually refutes the generalized fallacy iff $h^* \oplus \mathcal{G}(F)$ is $\mathcal{A}$ -incoherent; for OH the refutation is charged to the budget of Theorem 6.5. If h^* is sound for a class of intended structures and each $A \in \mathcal{A}$ holds in one that also validates $\mathcal{G}(F)$, the fallacy survives.*

Proof. $h^* \oplus \mathcal{G}(F)$ contains every positive datum; apply Theorem 6.9. A structure validating h^* , $\mathcal{G}(F)$ and A shows coherence on A . $\square$

fallacy	rule	witness
affirming the consequent	$q, p \rightarrow q / p$	$p := \perp, q := \top$
denying the antecedent	$\neg p, p \rightarrow q / \neg q$	$p := \perp, q := \top$
conversion	$p \rightarrow q / q \rightarrow p$	$p := \perp, q := \top$
illicit contraposition	$p \rightarrow q / \neg p \rightarrow \neg q$	$p := \perp, q := \top$
“or” read as “xor”	$p \vee q, p / \neg q$	$p := q := \top$

Table 8: Propositional fallacies and their Post witnesses (computed; T2-checks).

Corollary 6.35 (All structural propositional fallacies die, quickly). *[T2 Cor 6.2] If $\mathcal{G}$ is structural closure and $h^* \supseteq \text{Cn}_2$ is structural and coherent on a designated context (so $h^* = \text{Cn}_2$), every classically invalid rule is eliminated: substituting $\top/\perp$ along a falsifying row makes its premises tautologies and its conclusion a contradiction (Table 8). Given the row, the $\perp$ -derivation has polynomial size in any Frege system (proof sketch for the size bound); finding the row is an NP search.*

Proof. $h^* \oplus \mathcal{G}(F)$ is a structural proper extension of Cn_2 , hence C_{Fm} (Theorem 6.13). $\square$

Experiment B shows the negative-data role end to end (computed; quick run). From 50 simulated human proofs with systematic fallacies (affirming the consequent, denying the antecedent, and an illegitimate discharge that was never learned) and 5% corrupted steps, imitation ended with 22 unsound active rules, 13 of them tonk-like, and a budgeted adversarial prover derived $\vdash \perp$ and all 34 of its false goals. Coherence pruning on five designated contexts, with blame assigned by a minimum-weight hitting set over the bags found, removed every unsound rule and the two fallacies; all 13 target rules were recovered (exactly or with a stronger guard), completeness matched the target calculus under the same prover budget, and the adversary proved none of its goals. From only 10 proofs affirming the consequent survived, plausibly because the learned calculus lacked the rules needed to derive the contradiction: coherence is only as strong as the calculus that searches for it. Outside a Post-complete target, fallacies can survive.

Example 6.36 (Survivors). *[T2 §6, as repaired] (1) Quantifier swap $\forall x \exists y R / \exists y \forall x R$ is sound in one-element structures, hence $\emptyset$ -coherent; its alternative meaning is “the domain is a singleton”, and any designated context with $0 \neq 1$ kills it. (2) The gambler’s fallacy is incoherent with independence, but coherent without it: an anti-persistent stationary Markov chain with switching probability 0.6 has fair marginals and validates it. Only frequencies or an accepted independence premise remove it. (3) Base-rate neglect as a uniform schema $P(E|H) = a / P(H|E) = a$ yields $P(H|E) + P(\neg H|E) = 1.8$ when $P(E|H) = P(E|\neg H) = 0.9$, and dies with additivity. (4) Conditional perfection (from q and a law $p \rightarrow q \in K$, infer p ; Geis and Zwicky 1971) is coherent whenever the per-law converse $\{q \rightarrow p : (p \rightarrow q) \in K\}$ is consistent with $K \cup A$. This is sufficient, not necessary ($K = \{p_1 \rightarrow q_1, p_2 \rightarrow q_2\}$, $A = \{q_1 \vee q_2, \neg p_1, \neg p_2\}$: the converse is inconsistent with $K \cup A$, yet neither q_i is derivable, so the step rule never fires and the fallacy survives, for step-set hypotheses without proof by cases). It dies when two exclusive laws share a consequent ($\text{rain} \rightarrow \text{wet}, \text{sprinkler} \rightarrow \text{wet}, \neg(\text{rain} \wedge \text{sprinkler})$, $A = \{\text{wet}\}$). The per-law converse is not Clark’s completion (Clark, 1978), which is consistent there.*

Proposition 6.37 (Without uniformity, coherence is toothless; trivial). *[T2 Prop 6.3] If $h^* \oplus F$ is $\mathcal{A}$ -coherent, the non-structural $h^* \oplus F$ survives forever; e.g. $\text{Cn}_2 + \{\triangleright p_{17}\}$, the “quus” of propositional logic, when every designated context is consistent with p_{17} .*

Theorem 6.38 (The Kripkensteinian residue). ^[T2 Thm 6.4, as repaired] $\text{Alt}_{\mathcal{G}}(h^*, \mathcal{A}) = \{h \text{ } \mathcal{G}\text{-closed} : h \supseteq h^*, h \text{ } \mathcal{A}\text{-coherent}\}$ is, in the limit of complete data, the set of $\mathcal{G}$ -closed hypotheses never refuted by positive data and coherence (Theorem 6.9). (i) $\mathcal{G}$ structural, $h^* = \text{Cn}_2$, $\mathcal{A} \neq \emptyset$: $\text{Alt} = \{\text{Cn}_2\}$. (ii) $\mathcal{G}$ structural, $h^* = \text{IPC}$, $\mathcal{A}$ a non-empty family of classically consistent contexts: $\text{Alt} = \{h \text{ structural} : \text{IPC} \subseteq h \subseteq \text{CPC}\}$, all intermediate logics. (iii) $\mathcal{G}$ deductive closure, $h^* = \text{PA}$, $\mathcal{A} = \{\emptyset\}$: Alt is the set of all consistent closed extensions of PA, e.g. $\text{PA} + \neg \text{Con}(\text{PA})$ and its non-standard completions. Designating all finite sets of true sentences removes exactly the unsound ones; the sound T_k of Theorem 6.21 remain, and cannot be discriminated.

Proof idea. (i) is Theorem 6.13. For (ii) $\subseteq$, a classically invalid sequent in a structural $h \supseteq \text{IPC}$ becomes, under σ_v , an IPC-provable premise set with an IPC-refutable conclusion. Appendix E.11.

Division of labour. *Community practice* (positive data) eliminates under-generalizations. *Uniformity* (structurality, least generalization) eliminates gerrymanders such as Proposition 6.37; it is vocabulary-relative (Goodman’s point), and it converts an instance-level fallacy into a schema that coherence can refute. *Coherence* eliminates uniform alternatives that conflict with accepted uniform rules: tonk; affirming the consequent; “quus” as a second rule for $+$ beside $x + Sy = S(x + y)$. *Nothing* in the setup eliminates coherent uniform alternatives that agree with all data. For classical propositional logic there are none; for arithmetic they are the Gödel–Rosser and non-standard alternatives, a precise mathematical counterpart of Kripkenstein for arithmetic (Kripke, 1982), removable only by accepting reflection or stronger theories on other grounds (Section 13).

6.7 A remark on eliminative versus confirmational coherence

Coherentist epistemology has impossibility results: no informative coherence measure is truth-conducive ceteris paribus (Bovens and Hartmann, 2003b,a; Olsson, 2005a,b). They do not threaten the use of coherence in this section, and they do refute a tempting other use ^[L11 §3.4].

Remark 6.39 (Three uses of coherence). (U1) Eliminative: “ h derives $\perp$ from a context certified consistent, so h has an invalid rule.” This is modus tollens, not a measure. Conditioning a prior on “ h is $\mathcal{A}$ -coherent” multiplies every coherent hypothesis by $1/\mathbb{P}(\text{coherent})$: under truthful designation the target’s posterior weakly rises and odds among coherent rivals are unchanged (L11 TC4). The impossibility theorems, about degrees of coherence of consistent information sets, do not apply; the prices are Theorem 6.9 and the need for truthful designation. (U2) Probabilistic consistency of credences: de Finetti constraints, satisfied by the noncommittal assignment, the credal analogue of gappy valuations. (U3) Agreement as evidence: annotators or sampled derivations agree. This is the witness scenario of Lewis (1946) that the impossibility theorems address, and they bite in full.

Remark 6.40 (Consensus does not amplify under a common cause). ^[L11 TC3] Let m annotators judge a step. With probability $\pi > 0$, independently of validity, they all copy one shared judgment that says “valid” with probability a for valid and $b > 0$ for invalid steps; otherwise they judge independently with rates $p > q$. Then

$$\frac{\mathbb{P}(\text{all say valid} \mid \text{valid})}{\mathbb{P}(\text{all say valid} \mid \text{invalid})} = \frac{\pi a + (1 - \pi)p^m}{\pi b + (1 - \pi)q^m} \leq \frac{1}{\pi b}$$

for every m , against $(p/q)^m$ when $\pi = 0$. So unanimity among annotators sharing an error source, or self-consistency across samples of one model, moves the odds by a bounded factor. Human

errors in mathematics are often systematic in this sense, which is why corroboration must come from channels whose errors are independent of the human ones, Daniels’ independence constraint (Daniels, 1979).

The eliminative use has a quantitative safety guarantee whose converse shows what it depends on. Let $\mathcal{W} \subseteq \{0, 1\}^S$ be the worlds allowed by constraints on a finite set S of sentences, K their convex hull, and score credences x in world w by $\|x - w\|^2$.

Lemma 6.41 (Coherentizing is safe iff the constraints are sound). *[L3 Lemma E', as repaired] For $x \notin K$ with Euclidean projection x^* onto K , every $w \in K$ has $\|x - w\|^2 \geq \|x - x^*\|^2 + \|x^* - w\|^2$. Conversely, if the actual world’s indicator v is not in $\mathcal{W}$, then $v \notin K$, and projecting $x = v$ raises the actual loss from 0 to $\|v - v^*\|^2 > 0$.*

Proof. $\langle x - x^*, w - x^* \rangle \leq 0$. For the converse, a 0/1 vector is an extreme point of the cube, so it lies in the convex hull of 0/1 vectors only if it is one of them. $\square$

This is the accuracy-dominance argument of de Finetti (1974) and Joyce (1998) (Bregman projections for other proper scores; Predd et al. 2009). It belongs to the projection, not to arbitrary arbitrage removal: for $x = (0.6, 0.6)$ on $\{\varphi, \neg\varphi\}$, moving to $(1, 0)$ costs 2.0 in world $(0, 1)$, against 0.52 before and 0.5 for the projection. The converse is the probabilistic form of the air-pressure worry: coherence pressure from an idealized context the world violates, or from untrusted learned rules, can make credences strictly worse.

Reading. The setup’s use of coherence is eliminative, hence immune to the coherentist impossibility theorems, but weak: it never confirms, it only removes what conflicts. Corroboration must come from channels with independent errors: computation, randomized evaluation, measurement (Section 2.5). Where the target is coherence-pinned, uniqueness of the coherent extension, not truth-conduciveness of coherence, does the work. Section 13 reads this division as vindicated wide reflective equilibrium (Goodman, 1955; Daniels, 1979).

7 An end-to-end theorem for formal mathematics: the two-tier learner

The user’s first success criterion is “a setup of this shape that provably works for formal math”: imitate human inferences, then condition on coherence and on occasional feedback from the world. Sections 4 to 6 supply the parts. The natural way to assemble them is a statement of roughly this form. For every target calculus with finite anchors, human data with sporadic noise below the margin and a systematic fallacy set F , and every adaptive prover: (i) a cautious assertion tier is sound at all times with probability at least $1 - \delta$; (ii) the number of detected incoherences is at most $\log_2(1/w^*)$ plus a noise cost; (iii) after N human steps the tier equals the target plus the fallacies coherent with it; and (iv) that residue is empty for classical propositional logic and for complete decidable theories.

This statement is false as stated, and the way it fails is the main finding of this section. Three corrections are needed, and each is forced by a matching lower bound.

1. *Caution is relative to a class, and fallacies break realizability.* Positive data cannot tell a systematic fallacy from a rule (Corollary 5.20), so the realizable object is the practice, not the target. A tier that is cautious over the positive-data version space asserts every frequent fallacy

(Proposition 7.5). A tier that is cautious over hypotheses allowed to declare practice rules fallacious asserts nothing (Proposition 7.6). What works is *presumption of validity applied only after an audit*, with abstention until then, and the waiting is necessary (Theorem 7.20).

2. *Coherence condemns sets, and the blame problem is real.* The practice schemas that can be soundly condemned as schemas are exactly those in the union of the minimal conflicts, which is the hitting-set duality of model-based diagnosis (Lemma 7.8). At stable depth this collateral damage is unavoidable for every sound learner (Theorem 7.21). The extremal witness is almost too clean: in the practice {modus ponens, affirming the consequent} the rival diagnosis is the hypothesis that “ $\rightarrow$ ” means “ $\leftarrow$ ”. World feedback with descent, or a single denial, turns bags into singletons (Lemma 7.9).
3. *Soundness must be relativized to refutation size.* No computable learner is both sound at all times relative to full coherence and eventually complete (Theorem 7.22).

With these corrections the theorem holds (Theorem 7.14). We call the resulting learner TTL, the *two-tier learner*: a cautious, audited assertion tier and a bold sandbox that a red team attacks.

Depth, honestly. Most individual steps are trivial once set up, or are results of Sections 3 to 6 transported. Lemma 7.8 is Reiter’s diagnosis theory, Lemma 7.9 is Shapiro’s contradiction backtracing, Theorem 7.20 is a standard two-point rare-event bound, and Theorem 7.22 is a routine reduction. What is new is diagnostic: where the natural statement breaks (Propositions 7.5 and 7.6); the audit principle; the blame theorem with its $\rightarrow/\leftarrow$ witness and its stable-depth form; the depth relativization; the observation that world feedback earns its keep by *localization*, not refutation; and the assembly into one statement with explicit constants. None of it is deep mathematics. Its value is that it pins down, with matching lower bounds, what “provably works for formal math” can and cannot mean. All results come from theory thread T7 and survived two rounds of adversarial verification; several statements below are the repaired ones, and we say so where it matters.

7.1 Set-up: practice, evaluation, refutations

We use the practice of Definition 2.10: a target calculus Σ^* of genuine rules, a finite set F of systematic fallacies, the practice $\Sigma^P = \Sigma^* \sqcup F$ with $|\Sigma^P| \leq K$, and i.i.d. human steps tagged by practice schemas, with sporadic noise. Fallacy tags are latent but distinct. For tag i write $\beta_i = \pi_i(1 - a_i)$ and $\alpha_i = \pi_i a_i$, let ρ_i, c_i be the variability and witness count of Theorem 5.7 ($\rho_i = c_i = 1$ for ground rules, so $c_i \geq 1$ for every tag), and let $\Delta_i = (\rho_i \beta_i - \alpha_i)/2$ be the margin of Theorem 5.15.

Assumption 7.1 ((Floor)). [T7 §1.2, Def 1.2] Constants $\Delta > 0$, $c \geq 1$, K and $\bar{\alpha}_i \geq \alpha_i$ are known, with $|\Sigma^P| \leq K$, $c_i \leq c$ and $(\rho_i \beta_i - \bar{\alpha}_i)/2 \geq \Delta$ for every tag i , fallacy tags included.

The evaluation sources are those of Definition 2.13: a finite family $\mathcal{A}$ of designated positions $[A : D]$, a world $W : J_W \rightarrow \{0, 1\}$, and trusted steps T_{trust} (for instance, first-order logic when only non-logical axioms are learned). Each position induces the partial evaluation $e_{[A:D]}$ of (WS) (Assumption 2.14): value 1 on A and on W -true judgments, value 0 on $D \cup \{\perp\}$ and on W -false ones. When $J_W \neq \emptyset$, the empty position $[\emptyset : \emptyset]$ is designated. A step is *falsified* under e if its premises are e -true and its conclusion e -false. (WS) is assumed for every designated position unless stated otherwise.

Definition 7.2 (Derivations, refutations, conflicts, oracle). [T7 Def 1.5, size convention of Round 2]

- (a) A *derivation* is a finite sequence of *distinct* judgments, each a leaf or the conclusion of one designated step whose premises occur earlier. Its *size* $|\pi|$ is the total number of symbols in its distinct judgments. A tree derivation (Definition 2.2) compresses to such a sequence, using only its steps and leaves and no larger: keep, for each judgment, the step at an occurrence of least height, and order the judgments by that height.
- (b) For a finite schema set B (or a step set S) a *d-refutation* is a derivation of size at most d , relative to some $[A : D] \in \mathcal{A}$, whose leaves have e -value 1, whose steps lie in $R_B \cup T_{\text{trust}}$ (resp. $S \cup T_{\text{trust}}$), and whose conclusion has e -value 0. This is Definition 2.20 with the size convention made explicit. A single falsified instance is a one-step refutation. B is *d-clean* if it has no d -refutation, and *clean at every depth* if it is d -clean for all d .
- (c) $\text{Conf}_d = \{C \subseteq \Sigma^P : C \text{ is not } d\text{-clean}\}$ is upward closed. Its minimal members $\mathcal{C}_d$ are the *minimal d-conflicts*, and $\mathcal{C}_d(B)$ denotes those contained in B . Since Σ^P is finite there is d_0 with $\text{Conf}_d = \text{Conf}_{d_0}$ for $d \geq d_0$; we write $d = \infty$ for these stable values.
- (d) The *refutation oracle* $\text{Ref}_d(B)$ returns a d -refutation of B or NONE, deterministically. By default it returns the lexicographically least among the smallest refutations. The *prefer-unblocked* variant returns one whose descent (Lemma 7.9) is unblocked if one exists, then the smallest, then the lexicographically least.

Ref_d is computable by brute force when instance membership and W are decidable and there are finitely many derivations of size at most d up to renaming the symbols that occur in neither $\mathcal{A}$ nor B and on which W does not depend. Its cost is $\exp(O(d))$, and it is accounted separately from samples. As $d \rightarrow \infty$ refutation search is only semi-decidable (Theorem 6.33).

Definition 7.3 (Residue, survivors, collateral). [T7 Def 1.6] The *depth-d residue* is $F_{\text{res}}^{(d)} = \{\tau \in F : \Sigma^* \cup \{\tau\} \text{ is } d\text{-clean}\}$ (Definition 2.20), and $F_{\text{res}} = \bigcap_d F_{\text{res}}^{(d)}$; for $\tau \in F$, $d_\tau = \min\{d : \Sigma^* \cup \{\tau\} \text{ is not } d\text{-clean}\}$. The *survivors* are $F_d^{\text{surv}} = F \setminus \bigcup \mathcal{C}_d$, and the *collateral* is $\text{Coll}_d = \Sigma^* \cap \bigcup \mathcal{C}_d$, the genuine rules lying in some minimal conflict; $\text{Coll}_d(B) = \Sigma^* \cap \bigcup \mathcal{C}_d(B)$. *Singleton blame* (SB_d) holds if every member of $\mathcal{C}_d$ is a singleton.

The residue is the finite-depth version of the Kripkensteinian residue of Theorem 6.38: the fallacies that are coherent with the target on the designated positions, and world-adequate, up to size d . Provers are as in Definition 2.18: adaptive, unbounded, and knowing Σ^* , F , the learner's code and the history, but not future data. A tier is *uniformly δ -sound relative to a step set $S^\dagger$* (Definition 2.19) if some event G with $\mathbb{P}(G) \geq 1 - \delta$, depending only on the human data, has no prover ever getting a step outside $S^\dagger$ accepted on G .

Lemma 7.4 (Refutational evidence is one-sided and target-blind; trivial). [T7 Lemma 2.1 and §1.3, as repaired] Assume (WS). (a) Every step set $S \subseteq \text{Sound}(R^*)$ is clean at every depth; in particular every subset of Σ^* is. (b) Every member of Conf_d contains a fallacy. (c) Conf_d and every answer of Ref_d are functions of $(\mathcal{A}, W, T_{\text{trust}}, d)$ and the queried set alone; they do not depend on which schemas are genuine. (d) If $e_{[A:D]}$ is well defined, then (WS) holds for $[A : D]$ iff Σ^* is clean at every depth relative to $[A : D]$. The same holds with any calculus Σ in place of Σ^* .

Proof idea. The total valuation V of (WS) extends e and preserves truth along $R^* \cup T_{\text{trust}}$, hence along $\text{Sound}(R^*)$; so every node of a derivation from e -true leaves is V -true, and its conclusion is not e -false. For $(d, \Leftarrow)$ take V to be the indicator of $\text{Cl}_{R^* \cup T_{\text{trust}}}(A \cup W^{-1}(1))$. Appendix F.1.

So refutations can never condemn a subset of the target, and two targets with the same practice receive identical refutational evidence on every query. Coherence and the world can say that a *set* of practice rules contains a fallacy; they can never say that a rule is genuine. Everything positive that is ever learned about a rule comes from imitation.

7.2 Why the naive two-tier claim is false

Proposition 7.5 (Caution over practice asserts the fallacies; trivial). *[T7 Prop 2.2, as repaired] Let the positive-data tier be the per-tag trimmed version-space verifier (Theorem 5.14) over the practice tags. On the event of Theorem 5.15 its accepted set is exactly $R^P = R^* \cup R_F$. So it accepts an instance outside $\text{Sound}(R^*)$ of every $\tau \in F$, and an adaptive prover gets that instance accepted by querying it. In CPC with a complete Σ^* and a pure τ , the prover can query the closed $\top/\perp$ instance of τ with true premises and a false conclusion (Lemma 7.24), and that single step makes every formula derivable from $\emptyset$.*

Proof. Theorem 5.15 applies tag by tag, and fallacy tags are tags like any other. In the CPC case, Σ^* derives the closed true premises and the negation of the closed false conclusion j , both tautologies; the step adds j , and then $\perp$ and everything follow. $\square$

Purity matters for the last clause: adding the single invalid step $p \triangleright q$ to a complete calculus for CPC leaves $\text{Cl}(\emptyset)$ unchanged, and for an impure τ even all of $\text{inst}(\tau)$ need not trivialize (Example 3.6). The proposition is the precise sense in which “the cautious assertion tier is sound” is *false* without further structure. Caution is soundness relative to the realizable class (Theorem 4.2), and systematic fallacies make the practice, not the target, the realizable object (Remark 5.21).

The obvious repair is to be cautious over hypotheses that may declare practice rules fallacious. Suppose the practice is known, and let $\mathcal{V} = \{\Sigma : \Sigma \subseteq \Sigma^P\}$, where a candidate Σ reads “ Σ is the target and the rest of the practice is fallacious”. All candidates explain the positive data equally well. Let $\mathcal{V}_d$ be the d -clean candidates and $\mathcal{V}'_t$ those consistent with the refutations found by time t .

Proposition 7.6 (The caution–presumption dilemma). *[T7 Prop 2.4; (c) as repaired]*

- (a) Caution without presumption is vacuous. $\emptyset \in \mathcal{V}_d$ by Lemma 7.4(a), so $\bigcap_{\Sigma \in \mathcal{V}_d} R_\Sigma = \emptyset$, and the version-space verifier over $\mathcal{V}$ never asserts anything.
- (b) Presumption without audit is unsound. The verifier asserting $\bigcap\{R_\Sigma : \Sigma \text{ maximal in } \mathcal{V}'_t\}$ asserts all of R^P until the first refutation is found: an instance outside $\text{Sound}(R^*)$ of every fallacy, including those that a d -refutation will condemn when $F \not\subseteq F_{\text{res}}^{(d)}$.
- (c) Presumption after audit is sound. At the level of schemas, asserting the schemas common to all maximal members of $\mathcal{V}_d$ asserts

$$R_{\bigcap\{\Sigma : \Sigma \text{ maximal in } \mathcal{V}_d\}} = R_{\Sigma^P \setminus \bigcup \mathcal{C}_d} = R_{(\Sigma^* \setminus \text{Coll}_d) \cup F_d^{\text{surv}}}, \quad F_d^{\text{surv}} \subseteq F_{\text{res}}^{(d)},$$

which is sound relative to $R^* \cup R_{F_{\text{res}}^{(d)}}$. At the level of instances, $\bigcap\{R_\Sigma : \Sigma \text{ maximal in } \mathcal{V}_d\}$ contains this set, can be strictly larger, and is still contained in $R^* \cup R_{F_{\text{res}}^{(d)}}$ and d -clean.

Proof idea. (a) and (b) are immediate. In (c) the schema-level equality is Lemma 7.8(b) below; a non-residual fallacy τ has $\Sigma^* \cup \{\tau\} \supseteq C'$ for some $C' \in \mathcal{C}_d$, and $\tau \in C'$ because $C' \not\subseteq \Sigma^*$.

For the instance level, Σ^* lies in a maximal clean candidate, all of whose fallacies are residual. Appendix F.4.

The strictness is real (computed). In the practice $\{\text{MP}, \text{AC}\}$ on the designated context $A_1 = \{q, p \rightarrow q, p \rightarrow \perp\}$ of Example 2.16, the maximal candidates are $\{\text{MP}\}$ and $\{\text{AC}\}$, so the schema-level set is empty. But $\text{MP} = (A, A \rightarrow B \triangleright B)$ and $\text{AC} = (B, A \rightarrow B \triangleright A)$ unify, with most general common instance $(A, A \rightarrow A \triangleright A)$, which is valid and may be asserted (`overlap.py`).

Remark 7.7 (A Bayesian middle way). [T7 Remark 2.4'] (proof sketch) Put a prior w on $\mathcal{V}$ and accept s iff $w\{\Sigma \in \mathcal{V}'_t : s \notin R_\Sigma\} < w_0$. By the argument of Theorem 4.14 this is sound at all times for every target with $w(\Sigma^* \cup F_{\text{res}}^{(d)}) \geq w_0$, provided that set is clean, and it waits for the audit automatically. It cannot break the symmetry of Theorem 7.21, where the rival diagnosis has the same prior mass. We do not develop it; the audit-then-assert design below is simpler and has explicit constants.

Reading. The dilemma is the formal shape of a familiar norm. A practice is presumed valid, but the presumption is applied only after the hearing at which it could have been impeached. A learner that presumes too early asserts the community’s fallacies; a learner that never presumes asserts nothing, because refutational evidence cannot certify a rule. “Audit, then assert” is the only stable design, and it is the design of TTL.

7.3 Blame: from bags to culprits

A contradiction from a designated context condemns a set of steps (Lemma 6.2). Which members should a sound learner withhold?

Lemma 7.8 (Hitting-set duality). [T7 Lemma 2.3] *Lean: Blame.iInter_maxClean_eq* Let U be finite, let $\mathcal{K} \subseteq 2^U$ be upward closed with $\emptyset \notin \mathcal{K}$, and let $\mathcal{C}$ be its minimal members; call $M \subseteq U$ clean if $M \notin \mathcal{K}$. (a) The maximal clean sets are exactly the complements of the minimal transversals (hitting sets) of $\mathcal{C}$. (b) An element lies in some minimal transversal iff it lies in some member of $\mathcal{C}$. Hence $\bigcap \{\text{maximal clean sets}\} = U \setminus \bigcup \mathcal{C}$, and this set is clean. (c) The minimal transversal is unique iff every member of $\mathcal{C}$ is a singleton.

Proof of (a) and (b). (a) By upward closure, M is clean iff it contains no member of $\mathcal{C}$, iff $U \setminus M$ meets every member of $\mathcal{C}$. (b) If $x \in C \in \mathcal{C}$, then $(U \setminus C) \cup \{x\}$ is a transversal (a member of $\mathcal{C}$ inside C equals C), and any minimal transversal inside it must contain x to meet C . Conversely, if x lies in a minimal transversal T , some $E \in \mathcal{C}$ has $E \cap T = \{x\}$. The intersection of clean sets is clean, since cleanness is closed downward. Part (c) is in Appendix F.3. $\square$

This is Reiter’s theory of diagnosis from first principles (Reiter, 1987; de Kleer and Williams, 1987), and (b) is the standard fact that the vertices of the transversal hypergraph are those of the minimal edges (Berge, 1989). The dictionary: components are practice schemas, “abnormal” means fallacious, conflicts are the negative bags of Section 6, and diagnoses are hypotheses about which practice rules are fallacies. All three parts are formalized in Lean for a finite universe (`Blame.isMaxClean_iff`, `Blame.iInter_maxClean_eq`, `Blame.existsUnique_minTransversal_iff`), and `duality.py` checks them on 4000 random families with no failure. With $U = \Sigma^P$ and $\mathcal{K} = \text{Conf}_d$, every maximal candidate keeps $\Sigma^P \setminus \bigcup \mathcal{C}_d$. So $\bigcup \mathcal{C}_d$ is exactly what a learner that presumes validity, and is cautious across the remaining ambiguity *at the level of schemas*, must withhold. At the level of instances it can do better (Proposition 7.6(c)), and that the withholding

is forced for *every* sound learner is true at stable depth (Theorem 7.21(b)) but can fail at finite depth (Theorem 7.21(c)).

Descent. Bags shrink to culprits when the argument can be evaluated line by line. Fix a derivation π relative to $[A : D]$ and let e_π^+ be the partial map on the judgments of π obtained from $e_{[A:D]}$ by closing under two rules that use only the *trusted steps* of π : *forward*, if all premises of a trusted step of π have value 1 then its conclusion gets 1; *backward*, if a trusted step of π has conclusion value 0 and all but one premise at 1, then the remaining premise gets 0. A worklist computes this fixed point with $O(|\pi|\varphi)$ evaluations of e , where φ is the maximal fan-in. (Closing under *all* trusted steps globally is not computable in general: in arithmetic, forward closure from the true Δ_0 sentences reaches exactly the true Σ_1 sentences. It is also not needed.) To *descend* on a refutation, start at its conclusion; at a node of value 0 produced by a step s , output s if all premises of s have value 1, and otherwise move to a premise of value 0. The descent is *unblocked* if at every visited node either all premises have value 1 or some premise has value 0.

Lemma 7.9 (Descent with trusted steps). *[T7 Lemma 2.5, as repaired] Lean: Blame.LineArg.descent Assume (WS) for $[A : D]$, with total valuation V . Then e_π^+ agrees with V wherever it is defined; in particular, a judgment receiving both values certifies that $[A : D]$ violates (WS). Let π be a d -refutation of B relative to $[A : D]$ whose descent is unblocked. Then: (a) the descent outputs an untrusted step $s \in R_B$ with V -true premises and a V -false conclusion, so $s \notin R^*$; (b) computing e_π^+ and descending costs $O(|\pi|\varphi) \leq O(d\varphi)$ evaluations of e ; (c) every $\sigma \in B$ with $s \in \text{inst}(\sigma)$ is a fallacy, so removing all of them removes no genuine rule (certified singleton blame); (d) for each such σ , $\{\sigma\}$ is a d -conflict: s together with the trusted steps of π that justify the values of its premises and conclusion forms a d -refutation of $\{\sigma\}$.*

Proof idea. Both propagation rules are V -sound because trusted steps preserve V -truth. The descent follows nodes of value 0 strictly downward and leaves have value 1, so it stops at a step with V -true premises and a V -false conclusion, which by (WS) is neither trusted nor in R^* . Part (d) extracts the refutation from the judgments of π ; this is where size must count distinct judgments (Definition 7.2(a)). Appendix F.5.

This is Shapiro’s *contradiction backtracing* from algorithmic debugging and model inference (Shapiro, 1981, 1983), the counterexample-to-blame step of Lemma 10.20, and Lakatos’s conversion of a global counterexample into a local one (Lakatos, 1976). Two features are specific to this setting. Trusted steps carry values *backward* through unevaluable nodes, which arithmetic needs because $\forall x \theta$ is not Δ_0 (Theorem 7.29(a)). And by (c) and (d), blame lands on *schemas*, with no collateral, and every blamed schema is a singleton conflict. When the descent is blocked, all we have is the bag $\{\sigma \in B : \sigma \text{ used in } \pi\}$. The Lean formalization covers descent for a total evaluation of a line-list argument, including the final clause that the output step lies outside any truth-preserving rule set (Blame.LineArg.descent_not_mem); the trusted-step propagation and part (d) are not formalized.

Reading: localization. In classical propositional logic and in arithmetic, world feedback refutes nothing that coherence does not already refute: by Post-completeness an invalid schema is incoherent in the empty context (Theorem 6.13), and a consistent extension of $\mathbf{Q}$ is Δ_0 -sound (Lemma 6.19). Its whole added value is that descent turns its refutations into *singletons*. World feedback earns its keep by localization, not by refutation. Example 2.16 shows this in miniature: the context A_1 yields the bag $\{AC, MP\}$, and the valuation $p = 0, q = 1$ blames AC alone.

7.4 The learner $\text{TTL}(d, \delta)$

Definition 7.10 (The two-tier learner). ^[T7 §3.1, as repaired] *Inputs:* the floor constants $(\Delta, c, K, \bar{\alpha})$; an oracle Ref_d (for $\mathcal{A}, W, T_{\text{trust}}$) with either tie-breaking rule; a sandbox class $\mathcal{H}_S$ of step sets with prior w . The *burn-in* is

$$N_1 = \left\lceil \frac{1}{2\Delta^2} \ln \frac{(1+c)K}{\delta} \right\rceil, \quad \text{or, in noise-free mode,} \quad N_1 = \left\lceil \frac{1}{\lambda} \ln \frac{cK}{\delta} \right\rceil \quad \text{with a known } \lambda \leq \min_i \pi_i \rho_i.$$

0. *Phase 0* ($t < N_1$). The assertion tier *abstains*. A query may be escalated to a human, or, if all its judgments lie in J_W , answered by a separate *world channel* that accepts it iff it is not W -falsified; world-channel answers are not acceptances of the tier. Human steps are recorded; human answers to escalated queries are not added to the sample.
1. *Practice identification* (at $t = N_1$). For each tag i with $n_i > e_i$ samples, where $e_i = \lfloor (\bar{\alpha}_i + \Delta)N_1 \rfloor$ ($e_i = 0$ in noise-free mode), let $\hat{\sigma}_i = \text{mgu}\{\text{lgg}(P_i \setminus E) : |E| = e_i\}$, the most general common instance of the trimmed lggs of the tag- i data P_i ; its instance set is the accepted set of the tag's trimmed version space (Theorem 5.14). Tags with $n_i \leq e_i$ contribute nothing. Let $\hat{P} = \{\hat{\sigma}_i\}$.
2. *Audit*. Set $B := \hat{P}$ and repeat: call $r = \text{Ref}_d(B)$; if $r = \text{NONE}$, return B ; descend on r ; if the descent outputs s , set $B := B \setminus \{\sigma \in B : s \in \text{inst}(\sigma)\}$ and continue; if it is blocked, return $B \cup \mathcal{C}_d(B)$ (the *fallback*).
3. *Freeze* the returned set A . In *Phase 1* ($t \geq N_1$) the tier accepts q iff $q \in R_A$, decided by matching; other queries may be escalated.
4. *Sandbox* (all times). Oligarchic halving (Definition 6.4) over $\mathcal{H}_S$, for example over candidate generalizations of asserted schemas or candidate new rules: announce $B_t = \bigcap S_t$ with $w(S_t) \geq \frac{1}{2}w(\text{VS}_t)$. A red team, which may be the prover or the learner's own search, exhibits refutations π of B_t , and each deletes every $h \supseteq \text{Steps}(\pi) \setminus T_{\text{trust}}$. *Isolation*: nothing from the sandbox enters R_A as a primitive step. A conjecture κ is promoted only when a schematic R_A -derivation of it is found, and then as a *macro*: a κ -instance is answered by expanding it into that derivation.
5. *Truth maintenance*. If the audit is re-run at a larger depth, every cached lemma or macro that uses a removed schema is invalidated.

Two details were found by verification and simulation. In noise-free mode the budget must be $e_i = 0$: with $\bar{\alpha}_i = 0$ the noisy formula still gives $e_i = \lfloor \Delta N_1 \rfloor > 0$, and then the event of Theorem 5.7 does not imply identification. With one tag, one metavariable, two equiprobable roots ($\lambda = \frac{1}{2}$) and $\delta = 0.001$, $N_1 = 16$; identification fails with probability 0.077 for $e = 4$ and $3 \cdot 10^{-5}$ for $e = 0$ (computed; `noisefree.py`). And with $n_i \leq e_i$ samples the trimmed version space contains the empty rule, so its intersection is $\emptyset$, not “everything”; coding it as “accept all” made TTL unsound in 12 of 20 simulated runs at $N = 60$.

Lemma 7.11 (Audit correctness). ^[T7 Lemma 3.1] *Assume (WS) and $\hat{P} = \Sigma^P$.* (a) *If no descent is blocked, the audit returns a d -clean A with $\Sigma^* \subseteq A \subseteq \Sigma^* \cup F_{\text{res}}^{(d)}$, using at most $|F| + 1$ calls to Ref_d ; every successful call removes at least one fallacy and no genuine rule.* (b) *If the fallback triggers at stage B_0 , then $\Sigma^* \subseteq B_0$, and $A = B_0 \setminus \bigcup \mathcal{C}_d(B_0)$ is d -clean with $\Sigma^* \setminus \text{Coll}_d(B_0) \subseteq A \subseteq \Sigma^* \cup F_{\text{res}}^{(d)}$. If $B_0 = \Sigma^P$, then $A = (\Sigma^* \setminus \text{Coll}_d) \cup F_d^{\text{surv}}$.*

Proof. By Lemma 7.9(a),(c) each successful descent removes at least one schema, all of them fallacies, so $\Sigma^* \subseteq B$ throughout, and there are at most $|F|$ removals. (a) The returned B is d -clean; for $\tau \in B \setminus \Sigma^*$, $\Sigma^* \cup \{\tau\} \subseteq B$ is clean because cleanness is closed downward, so $\tau \in F_{\text{res}}^{(d)}$. (b) By Lemma 7.8 with $U = B_0$, A is the intersection of the maximal clean subsets of B_0 , hence clean. If $\tau \in A \cap F$ were not residual, $\Sigma^* \cup \{\tau\} \subseteq B_0$ would contain a minimal conflict, which contains τ by Lemma 7.4(a), contradicting $\tau \in A$. The last claim is Proposition 7.6(c). $\square$

Cost. The descent path costs at most $|F| + 1$ oracle calls and $O((|F| + 1)d\varphi)$ evaluations of e . The fallback needs $\bigcup \mathcal{C}_d(B_0)$: at most $2^{|B_0|}$ oracle calls by brute force. From a cleanness (membership) oracle, the minimal conflicts and the maximal clean sets of a monotone property can be generated jointly in time quasi-polynomial in their total number (Bioch and Ibaraki, 1995; Gurvich and Khachiyan, 1999), with the dualization algorithm of Fredman and Khachiyan (1996) as the engine; the second family can be exponentially larger than the first, and both can be exponential in $|F|$. At stable depth the fallback’s schema-level incompleteness is unavoidable (Theorem 7.21(b)); its cost is the price of bags.

Mis-designated positions. Theorem 7.14 below assumes (WS) for *every* designated position. When some positions may be wrong, as when an idealization is designated together with background it contradicts, the audit can vote.

Proposition 7.12 (Voting audit). *[T7 Prop 3.2, as repaired] Suppose at most m designated positions violate (WS), it is not known which, and $\hat{P} = \Sigma^P$. For each position p run the descent loop of the audit relative to p alone, from $B_p = \Sigma^P$ and $\text{Blamed}_p = \emptyset$, with the prefer-unblocked oracle restricted to p . Stop when the oracle returns NONE (p is finished), when the descent is blocked, or when e_p is ill defined or e_π^+ gives a judgment both values (then p is certified bad and its blame is discarded); otherwise move every $\sigma \in B_p$ with $s \in \text{inst}(\sigma)$ into Blamed_p . Let $\text{votes}(\sigma) = \#\{p : \sigma \in \text{Blamed}_p\}$, counting blame from $[\emptyset : \emptyset]$ as $m + 1$ votes if the world is known to be truthful, and return $A = \Sigma^P \setminus \{\sigma : \text{votes}(\sigma) \geq m + 1\}$. Then (a) $\Sigma^* \subseteq A$; (b) at most $|\mathcal{A}|(|F| + 1) + m|\Sigma^*|$ oracle calls are made; (c) call p complete if it is good and finished; then $A \cap F \subseteq \{\tau \in F : \#\{p \text{ complete} : \Sigma^* \cup \{\tau\} \text{ is not } d\text{-clean relative to } p\} \leq m\}$. For $m = 0$ and all positions complete this is Lemma 7.11(a).*

Proof in Appendix F.6. Per-position loops matter: one smallest refutation per position can hide a fallacy refutable at $m + 1$ positions behind another one. With $m = 1$, truthful positions $[q, p \rightarrow q, \neg p : p, \neg q]$ and $[\neg r, r \rightarrow t : \neg t]$ and practice $\{\text{MP}, \text{AC}, \text{DA}\}$, the per-position loops remove denying the antecedent (refutable at both positions) and may keep AC (refutable at one), while one-query voting kept both (computed; prop32_voting.py). The soft version of this is Theorem 6.7.

Proposition 7.13 (Sandbox isolation and budget). *[T7 Prop 3.3, as repaired; transport of T2 Thms 2.2, 2.5]*

(a) *The assertion tier is a function of the first N_1 human steps and of the audit’s oracle answers alone. No sandbox state, red-team action or prover action affects it, and promotion by derivation leaves Cl_{R_A} unchanged.* (b) *If $\mathcal{H}_S$ contains some h_0 that is clean at every depth with $w(h_0) > 0$ (for instance $h_0 = R^*$, by Lemma 7.4(a)), then for every red team the number of detections against the coalition is at most $\log_2(1/w(h_0))$, and with m false alarms and penalty factor β at most $(\ln \frac{1}{w(h_0)} + m \ln \frac{1}{\beta}) / \ln \frac{2}{1+\beta}$. Without such an h_0 the bound is vacuous.* (c) *Coalition unanimity is decidable for finite S_t : $s \in \bigcap_{h \in S_t} h$ iff s matches a schema of every h , and for schema-valued hypotheses the intersection of instance sets is the instance set of the most general unifier (Robinson, 1965; Plotkin, 1970; Reynolds, 1970).*

Proof idea. (a) holds by construction. For (b), the proofs of Theorems 6.5 and 6.7 use only that the protected hypothesis is never deleted, and a detection by π deletes only hypotheses containing the untrusted steps of π , of which π is then a refutation. Appendix F.7, which also shows what fails if promoted rules were primitive: a rule packaging the first half of a residual fallacy’s refutation of size $2d$ yields one of size at most d .

Reading. How does cautious acceptance interact with the bold coalition? *It does not.* The coalition is bold, and its errors are bounded by halving. The assertion tier is cautious, and its soundness is certified by the audit. The only channel from the sandbox to the tier is derivation, which cannot enlarge the closure (Lemma 3.1). Re-deriving sandbox lemmas inside R_A is what makes the separation safe; truth maintenance is the same point made across time. This resolves the tension left by Sections 4 and 6: caution receives no signal from coherence (Proposition 6.3), and boldness is unsound; so both tiers exist, and only the cautious one asserts.

7.5 The main theorem

Theorem 7.14 (Two-tier coherent inferential learning). *[T7 Thm 4.1, as repaired] Assume (Floor) (Assumption 7.1); realizability, i.e. the practice consists of tagged single schemas as in Definition 2.10, with fallacy tags latent but distinct; (WS) for every designated position; and Ref_d exact for refutations of size at most d , with either tie-breaking rule. Run $\text{TTL}(d, \delta)$. There is an event G with $\mathbb{P}(G) \geq 1 - \delta$, depending only on the first N_1 human steps, on which the following hold for every adaptive prover and every red team.*

- (i) Uniform soundness at all times. *For $t < N_1$ the tier accepts nothing. For $t \geq N_1$ every accepted step lies in $R^* \cup R_{F_{\text{res}}^{(d)}}$, so for every premise set B*

$$\text{Cl}_{R_A}(B) \subseteq \text{Cl}_{R^* \cup R_{F_{\text{res}}^{(d)}}}(B), \quad \text{Cl}_{R_A \cup T_{\text{trust}}}(B) \subseteq \text{Cl}_{R^* \cup T_{\text{trust}} \cup R_{F_{\text{res}}^{(d)}}}(B).$$

If $F_{\text{res}}^{(d)} = \emptyset$, the reasoner is target-sound against arbitrary search. The schema set A is d -clean: no prover can ever exhibit a refutation of R_A of size at most d .

- (ii) Refutation budget. *On its descent path the audit uses at most $|F| + 1$ oracle calls and $O((|F| + 1)d\varphi)$ evaluations, and removes no genuine rule. If $\mathcal{H}_S$ contains an h_0 clean at every depth with $w(h_0) > 0$, the sandbox suffers at most $\log_2(1/w(h_0))$ detections (with m false alarms, the bound of Proposition 7.13(b)).*
- (iii) Convergence modulo the residue. *For $t \geq N_1$ the tier accepts exactly R_A , where $\Sigma^* \subseteq A \subseteq \Sigma^* \cup F_{\text{res}}^{(d)}$ if no descent is blocked, and $\Sigma^* \setminus \text{Coll}_d(B_0) \subseteq A \subseteq \Sigma^* \cup F_{\text{res}}^{(d)}$ if the fallback triggers at B_0 , with $A = (\Sigma^* \setminus \text{Coll}_d) \cup F_d^{\text{surv}}$ when $B_0 = \Sigma^P$. So under (SB_d) , or whenever no descent is blocked, the tier is complete for the target: it accepts every genuine step, and the only fallacies it accepts are residual.*
- (iv) Depth schedule. *Re-run the audit on the frozen $\hat{P}$ at depths $d_1 < d_2 < \dots$, with truth maintenance. Then A changes only finitely often, and is constant for $d \geq d_1^*$, the largest depth at which some answer $\text{Ref}_d(B)$, $B \subseteq \Sigma^P$, changes. At each stage (i) holds with the current depth, and from d_1^* on the tier is sound relative to $R^* \cup R_{F_{\text{res}}}$. The changes need not be retractions: A is not monotone in d .*

- (v) Uniformity. G is defined by the human data alone, and Ref_d and W are deterministic, so (i)–(iv) hold simultaneously for all provers and red teams, with no union bound over queries.

Proof idea. Identification: G is the event of Theorem 5.15 at N_1 , with α_i replaced by $\bar{\alpha}_i$ and Δ_i by Δ . Hoeffding and a union bound over the $1 + c_i$ events per tag give $\mathbb{P}(G^c) \leq (1 + c)Ke^{-2N_1\Delta^2} \leq \delta$. On G , Theorem 5.14(b) makes each tag’s trimmed verifier accept exactly $\text{inst}(\sigma_i)$, so $\hat{P} = \Sigma^P$. In noise-free mode Theorem 5.7 gives $\mathbb{P}(G^c) \leq \sum_i c_i e^{-N_1\pi_i\rho_i} \leq cKe^{-N_1\lambda} \leq \delta$. *Audit:* Lemma 7.11. *Soundness:* $R_A \subseteq R^* \cup R_{F_{\text{res}}^{(d)}}$, and Lemma 3.1 with that calculus as target. *Sandbox:* Proposition 7.13. *Depth:* each $\text{Ref}_d(B)$ changes at most twice as d grows, and there are $2^{|\Sigma^P|}$ sets B . Full proof in Appendix F.8. Given Theorems 5.15 and 6.5, the proof is about a page: the theorem is an assembly, and its interest lies in the hypotheses it needs (Section 7.6), not in its proof.

The non-monotonicity in (iv) is concrete (computed; `depth_nonmono.py`). Take the practice $\{\text{MP}, \text{AC}\}$, no world, and the positions $[A_1 : \emptyset]$ and $[r, s \rightarrow r : s]$ with r, s formulas of size 7. Then $A = \{\text{MP}, \text{AC}\}$ for $d < 9$; $A = \emptyset$ for $9 \leq d < 29$, by fallback on the conflict $\{\text{MP}, \text{AC}\}$; and $A = \{\text{MP}\}$ for $d \geq 29$, where the new conflict $\{\text{AC}\}$ makes the old one non-minimal and restores MP. Restored schemas are genuine or residual at the larger depth, so soundness is unaffected.

Corollary 7.15 (Truth-soundness). *[T7 Cor 4.2, as repaired] Suppose R^* is $\mathfrak{M}$ -sound, every member of $F_{\text{res}}^{(d)}$ is $\mathfrak{M}$ -valid (vacuous if $F_{\text{res}}^{(d)} = \emptyset$), and trusted steps, if used, are $\mathfrak{M}$ -sound. Then on G the reasoner derives only $\mathfrak{M}$ -consequences, at all times and against all provers. This also holds for a reasoner that chains world-channel steps with R_A -steps, provided W is truth in $\mathfrak{M}$ on J_W .*

World-channel steps are truth-sound but need not be target-sound: for $\Sigma^* = \{\wedge I, \wedge E_1, \wedge E_2\}$ the step $\top \triangleright \top \vee \perp$ preserves truth but is not R^* -derivable. That is why the world channel is kept outside the tier, and why Theorem 7.14(i) concerns the tier only.

Remarks. (1) *What is and is not certified.* Genuine rules are never certified valid by anything in the architecture: their assertion rests on positive data (anchoring with margin) plus presumption after audit. Fallacies are certified invalid by descent, or implicated as a set by conflicts. Lemma 7.4 forces this asymmetry. (2) *Realizability is load-bearing.* If the human calculus has a rule outside the class, identification fails, and nothing downstream repairs it; at best the audit removes an over-general identified schema, which restores soundness but not completeness. Quantifier rules and the induction schema are such rules (Corollary 7.28 and Theorem 7.29). If fallacies are cited under genuine rules’ names, the per-tag class must be unions of schemas (Theorem 5.9). (3) *Why freeze.* Freezing $\hat{P}$ keeps the accounting to one event. Learning can continue at the cost of a union bound over $N \geq N_1$ (an additive $\approx \ln \frac{1}{2\Delta^2}$ inside the logarithm) and a re-audit whenever $\hat{P}$ changes. (4) *Before N_1 .* Abstention can be softened by the world channel or by a trusted base. Both are truth-sound (Corollary 7.15) but outside (i) unless they lie in $\text{Sound}(R^*)$, and Theorem 7.20 shows that no learner of the relevant kind can do much better on the schemas that might be fallacies.

Reading. For the user’s question, the theorem says what “a learner of inference rules, conditioned on coherence and the world” can deliver. It identifies the target calculus exactly, modulo a residue that no refutational channel at depth d can see. It is sound at every moment against any amount of search. Its costs are explicit: N_1 human steps, $|F| + 1$ refutations on the descent

path, $\log_2(1/w(h_0))$ detections in the sandbox. And it never asserts a rule because the rule is coherent, only because the community uses it robustly *and* it survived the audit. Coherence is a filter on imitation, not a source of rules.

7.6 Every ingredient is necessary

We now show that each ingredient of TTL is forced. Here a *learner* is any randomized procedure that receives the same kinds of information as TTL, possibly with one channel removed: human samples, the queries of a prover and red team, answers of Ref_d and W , its own coins and, optionally, escalation answers. Lower bounds use a *fixed* prover and red team, i.e. strategies mapping the history and their own coins to queries, the same in every scenario. Escalation answers, if used, *follow the practice*: they are drawn from a law depending only on the query and on the practice (for instance $\mathbf{1}[q \in R^P]$), so they have the same law in scenarios with the same practice. That covers Propositions 7.17 and 7.18 and Theorems 7.21 and 7.22; Theorem 7.20 compares different practices and needs an explicit assumption instead. A target-truthful labeller ($\mathbf{1}[q \in R^*]$, the labelled setting of Section 4) is a different and stronger channel, and defeats Theorems 7.20 and 7.21.

Fix the practice with its frequencies, instance laws and noise law, and fix $\mathcal{A}, W, T_{\text{trust}}$. A target $\Sigma \subseteq \Sigma^P$ is *admissible* if it satisfies (WS), equivalently is clean at every depth (Lemma 7.4(d)), and each member of $\Sigma^P \setminus \Sigma$ has an instance outside $\text{Sound}(R_\Sigma)$. All admissible targets share the data law and the oracle answers (Lemma 7.4(c)) and satisfy (Floor) with the same constants. $F_{\text{res}}^{(d)}(\Sigma)$ denotes the residue when Σ is the target.

Proposition 7.16 (Positive data are needed; trivial). *[T7 Prop 5.1, as repaired] Let $\Sigma_1 \subsetneq \Sigma_2$ be calculi such that every subset of Σ_2 is clean at every depth and $R_{\Sigma_2} \not\subseteq \text{Sound}(R_{\Sigma_1})$. A learner that receives only refutation evidence (oracle answers on schema sets of its choice, but no human steps and no escalations) has the same view under the targets Σ_1 and Σ_2 . So it is unsound for Σ_1 or incomplete for Σ_2 on the nonempty set $R_{\Sigma_2} \setminus \text{Sound}(R_{\Sigma_1})$. Example: $\Sigma_1 = \{\wedge I, \wedge E_1, \wedge E_2\}$, $\Sigma_2 = \Sigma_1 \cup \{\forall I_1\}$ with classical evaluation. The non-derivability hypothesis is needed: for $\Sigma_2 = \Sigma_1 \cup \{A \wedge B \triangleright B \wedge A\}$, accepting R_{Σ_2} is both sound for Σ_1 and complete for Σ_2 .*

The proof is Lemma 7.4(c). This concerns soundness relative to the *human* calculus. In CPC, structurality and closed evaluation decide schema validity (Lemma 7.24), so a learner wanting only truth-soundness could enumerate schemas and assert the valid ones without human data; positive data then only say *which* valid rules the practice uses. In arithmetic, positive data are needed even for truth-completeness, because no refutational evidence ever certifies a true Π_1 axiom.

Proposition 7.17 (Caution is needed). *[T7 Prop 5.2; (a) as repaired]*

(a) *Let $\mathcal{T}$ be the admissible targets. For any learner, a fixed prover, a finite history h and a step q , the joint probability $p(h, q)$ that the run produces h and then accepts q is the same under every $\Sigma \in \mathcal{T}$. Hence if $q \notin \bigcap_{\Sigma \in \mathcal{T}} \text{Sound}(R_\Sigma \cup R_{F_{\text{res}}^{(d)}(\Sigma)})$, the learner is unsound with probability at least $p(h, q)$ under some admissible target, and a learner that is δ -sound under every admissible target has $p(h, q) \leq \delta$. TTL passes: on G , $R_A \subseteq R_\Sigma \cup R_{F_{\text{res}}^{(d)}(\Sigma)}$ for every $\Sigma \in \mathcal{T}$.*

(b) *The following accept a falsified instance of some $\tau \in F \setminus F_{\text{res}}^{(d)}$ before the refutation implicating τ is found: presumption without audit (Proposition 7.6(b)); the bold coalition used as the*

assertion tier, under priors for which every coalition member contains such an instance, for instance a prior concentrated on R^P (Theorem 6.5(iii) gives soundness only while $R^* \in S_t$, while a prior with more than half its mass on R^* gives a sound coalition); and any MAP or MDL learner over practice (Proposition 3.9). A prover that knows F queries such an instance at once, and in CPC with a complete Σ^* and a pure τ that one step trivializes the reasoner (Proposition 7.5).

Part (a) is the joint-probability form of Theorem 4.2(b), relativized to the residue; proof in Appendix F.9. Both qualifications were added in verification. The conditional probability given h would be false for randomized learners: a learner that with probability ε accepts everything from time 1 on, and otherwise runs TTL, is $(\delta + \varepsilon)$ -sound, yet given the history “ q_1 accepted at time 1” it accepts every fallacy instance. And plain $\text{Sound}(R_\Sigma)$ would make the intersection trivial, since typically $\emptyset \in \mathcal{T}$, convicting TTL itself.

Proposition 7.18 (Negative evidence is needed). ^[T7 Prop 5.3] *Let the practice be $\{\wedge I, AC\}$ with fixed frequencies. In world W_1 the connectives have their classical tables, AC is a fallacy and the target is $\{\wedge I\}$. In W_2 the tables are classical except that $\rightarrow$ is read as $\leftrightarrow$; then AC is valid (from B and $A \leftrightarrow B$ infer A) and the target is $\{\wedge I, AC\}$. A learner without refutation evidence sees identically distributed data in both, so it is unsound under W_1 or incomplete under W_2 . The closed instance $(\top, \perp \rightarrow \top \triangleright \perp)$ separates them: under W_1 its premises are true and its conclusion false, while under W_2 the premise $\perp \leftrightarrow \top$ is false.*

This is Corollary 5.20 in two-world form.

Proposition 7.19 (Structurality is needed; trivial). ^[T7 Prop 5.4] (a) *If hypotheses are instance sets rather than schemas, each refutation removes finitely many instances, so a fallacy with infinitely many invalid instances keeps some invalid instance asserted at every finite time by any learner that asserts the practice minus the refuted instances.* (b) *Without uniformity coherence is toothless (Proposition 6.37: $Cn_2 + \{\triangleright p_{17}\}$).* (c) *Without structure positive data cannot generalize: for classes of finite step sets the cautious verifier accepts exactly the data (Section 5.4).*

Theorem 7.20 (Burn-in is necessary: the rare refuter). ^[T7 Thm 5.5, as repaired in two rounds] *Let $0 < \pi < 1$, and let σ, τ, μ be schemas such that $\{\sigma, \tau\}$ and $\{\sigma, \mu\}$ are clean at every depth and $\{\mu, \tau\}$ is a d -conflict. Compare, with the same $\mathcal{A}, W, T_{\text{trust}}$ and a fixed prover and red team: scenario 1, practice $\{\sigma, \tau, \mu\}$, target $\{\sigma, \mu\}$, $F = \{\tau\}$, tag frequencies $(1 - \pi)q_\sigma$, $(1 - \pi)q_\tau$, π ; and scenario 2, practice $\{\sigma, \tau\}$, target $\{\sigma, \tau\}$, $F = \emptyset$, frequencies q_σ, q_τ , where $q_\sigma + q_\tau = 1$ and the per-tag laws of σ and τ are the same in both. Assume*

(OE) at each round, every input to the learner other than the human sample (prover and red-team queries, answers of Ref_d and W , escalation answers if any, the learner’s coins) is generated from the history by the same kernel in both scenarios.

Then some τ -instance lies outside $\text{Sound}(R_{\{\sigma, \mu\}})$, and for every learner satisfying (OE), every t and every such instance s : if in scenario 1 the learner accepts s at time t with probability at most δ , then in scenario 2 it does so with probability at most $\delta(1 - \pi)^{-t}$. Hence, if $\delta + \delta' < 1$, accepting s at time t in scenario 2 with probability at least $1 - \delta'$ requires

$$t \geq \frac{\ln \frac{1 - \delta'}{\delta}}{\ln \frac{1}{1 - \pi}} \geq \frac{1 - \pi}{\pi} \ln \frac{1 - \delta'}{\delta}.$$

The first bound is attained.

Proof idea. On samples whose tag is not μ , the scenario-1 law is $(1 - \pi)$ times the scenario-2 law, so on the event E_t that no μ -tag appears among the first t samples, $P_1^{\otimes t} = (1 - \pi)^t P_2^{\otimes t}$. By (OE) the learner's whole view has law $P_j^{\otimes t}(dx)K_t(x, \cdot)$ with one kernel K_t for both scenarios; integrate the acceptance probability over E_t . The learner that accepts τ -instances at time t iff no μ -tag has occurred has scenario-1 error exactly $(1 - \pi)^t$ and scenario-2 completeness 1. Appendix F.10.

(OE) holds for Ref_d and W (Lemma 7.4(c)), for the fixed prover and red team, for learners that do not escalate, and for TTL's tier (Proposition 7.13(a); escalation answers are not added to the sample). It fails for practice-following escalation, and then the conclusion is false: some μ -instance lies outside $\text{Sound}(R_{\{\sigma, \tau\}})$, a human following practice $\{\sigma, \tau, \mu\}$ confirms it and one following $\{\sigma, \tau\}$ rejects it, so one escalation separates the scenarios. Linear programming over all tag sequences of length $t \leq 6$ confirms both facts: the best learner under (OE) attains exactly $\min(1, \delta(1 - \pi)^{-t})$ in all 108 cases, and with one practice-following escalation the cap 0 becomes 1 (computed; `two_point_bounds.py`). A lower bound for learners with membership queries to the human would need the refuter to be unknown among many candidates; that is open.

A concrete instance: A_1 designated, no world, $\sigma = \wedge I$, $\tau = AC$, $\mu = MP$. The closure of A_1 under AC alone is $\{q, p \rightarrow q, p \rightarrow \perp, p\}$; under $\{\wedge I, AC\}$ it is infinite but contains no $\perp$, because $\wedge I$ creates no implications and AC needs an implication with consequent q or $\perp$; $\{\wedge I, MP\}$ is classically sound; and AC followed by MP derives $\perp$.

Reading. Suppose modus ponens is rare, with frequency π . No learner that is sound whether or not affirming the consequent is a fallacy, and that learns about the practice only from the stream, can accept AC in the world where it is genuine before about $\pi^{-1} \ln(1/\delta)$ samples: roughly the time by which MP , the rule that would expose AC if it were a fallacy, has been seen with confidence. TTL's noise-free burn-in $\lambda^{-1} \ln(cK/\delta)$, with $\lambda \leq \pi \rho_{MP}$, matches this up to the factor $\ln K$ and the variability ρ_{MP} . This is the standard two-point rare-event bound behind PAC sample-complexity lower bounds; the only new point is that the relevant rare event is the absence of the *refuter*. Waiting is not timidity: a rule can be presumed valid only once the rules that could impeach it have had their chance.

Theorem 7.21 (Blame: world or bilateral evidence is needed for completeness). *[T7 Thm 5.6, as repaired]*

- (a) Two-point core. *Let $\Sigma_0 \cup \{\mu\}$ and $\Sigma_0 \cup \{\tau\}$ be clean at every depth and $\{\mu, \tau\}$ a d -conflict. Compare scenario 1 (target $\Sigma_0 \cup \{\mu\}$, $F = \{\tau\}$) and scenario 2 (target $\Sigma_0 \cup \{\tau\}$, $F = \{\mu\}$), with equal tag frequencies and a fixed prover. The data and all oracle answers have identical laws, and some μ -instance s , genuine in scenario 1, is accepted at any given time in scenario 1 with probability at most δ by every learner that is δ -sound in both scenarios. So μ , which lies in Coll_d in scenario 1, is collateral for every sound learner.*
- (b) Stable depth: the fallback is optimal among schema-level policies. *Let $d \geq d_0$ and suppose that on G the audit falls back at B_0 . Then for every $g \in \bigcup \mathcal{C}_d(B_0)$ there are an admissible target $M \subseteq B_0$ with $g \notin M$ and $F_{\text{res}}^{(d)}(M) = \emptyset$, and an instance $s_g \in \text{inst}(g) \setminus \text{Sound}(R_M)$. Consequently every learner that is δ -sound under every admissible target accepts s_g at any fixed time with probability at most δ , under every admissible target, the true one included; every genuine $g \in \text{Coll}_d(B_0)$ is withheld on some instance by every such learner; and a policy that asserts $R_{A'}$ for a possibly random $A' \subseteq \Sigma^P$, and is δ -sound under every admissible target, includes each schema outside TTL's A with probability at most δ . The same holds on the descent path, with $M := A$.*

(c) At finite depth the fallback can be strictly suboptimal (*computed; thm56_finite_d.py*). Take no world, the practice $\{\text{MP}, \text{AC}\}$, and $\mathcal{A} = \{[A_1 : \emptyset], [A_2 : \emptyset]\}$ with $A_2 = \{\varphi, \perp \rightarrow \varphi\}$, $\varphi = q \wedge (q \wedge q)$. The minimal refutation sizes are 9 for $\{\text{MP}, \text{AC}\}$ (on A_1) and 13 for $\{\text{AC}\}$ (on A_2 , where AC infers $\perp$ from φ and $\perp \rightarrow \varphi$), and $\{\text{MP}\}$ is clean at every depth. For $9 \leq d \leq 12$ the only minimal conflict is $\{\text{MP}, \text{AC}\}$, the descent is blocked, and TTL returns $A = \emptyset$. Yet the admissible targets are only $\{\text{MP}\}$ and $\emptyset$, and asserting R_{MP} is sound under both (under $\emptyset$, MP is residual). For $d \geq 13 = d_0$, TTL returns $\{\text{MP}\}$.

Proof idea. (a) A μ -instance outside $\text{Sound}(R_{\Sigma_0 \cup \{\tau\}})$ exists, or the refutation of $\{\mu, \tau\}$ would transfer to the clean set $\Sigma_0 \cup \{\tau\}$ (Lemma 3.1); in scenario 2 it is unsound and non-residual, and the scenarios are indistinguishable. (b) Pick, by Lemma 7.8(b), a maximal clean $M \subseteq B_0$ avoiding g ; at stable depth it is clean at every depth, hence admissible. Appendix F.11.

The witness: $\rightarrow$ versus $\leftarrow$. Take $\mathcal{A} = \{A_1\}$, no world, $\Sigma_0 = \emptyset$, $\mu = \text{MP}$ and $\tau = \text{AC}$. $\{\text{MP}\}$ is classically sound and A_1 satisfiable; $\{\text{AC}\}$ is clean (the closure above); and AC followed by MP derives $\perp$. Scenario 2 is literally the hypothesis that “ $\rightarrow$ ” denotes converse implication. Under that reading AC is modus ponens for $\leftarrow$, MP is affirming the consequent for $\leftarrow$, and A_1 reads $\{q, q \supset p, \perp \supset p\}$, which is consistent. *Coherence alone cannot tell which way the arrow points.* Either of two things breaks the symmetry: world feedback, since the closed instance $(\perp \rightarrow \perp, \perp \rightarrow (\perp \rightarrow \perp) \supset \perp)$ of AC is falsified; or a single denial, since in the designated position $[A_1 : p]$ AC alone derives the denied p , and the conflict becomes $\{\text{AC}\}$. Both make scenario 2 violate (WS) (*computed; hilbert_blame.py*).

A Hilbert illustration (*computed; hilbert_blame.py*, formulas with at most 6 leaves over $\{p, q, \perp\}$). Take the practice $\{K, S, \text{MP}, \text{DN}, \text{AC}\}$ with $\mathcal{A} = \{[\emptyset : \emptyset], [A_1 : \emptyset]\}$. The minimal conflicts are $\{\text{AC}, K\}$ and $\{\text{AC}, \text{MP}\}$, the diagnoses are $\{\text{AC}\}$ and $\{K, \text{MP}\}$, and the collateral is $\{K, \text{MP}\}$. The conflict $\{\text{AC}, K\}$ is a four-step derivation from no premises: K gives $\perp \rightarrow (p \rightarrow \perp)$ and $(p \rightarrow \perp) \rightarrow (\perp \rightarrow (p \rightarrow \perp))$, AC gives $p \rightarrow \perp$, and AC again gives $\perp$. Both conflicts are minimal at every depth beyond their sizes, and with no world and no trusted steps every descent is blocked; so by (b), K and MP are *necessarily* collateral at full depth. Whether $\{S, \text{DN}, \text{AC}\}$ is clean at every depth, i.e. whether the collateral also contains S or DN, is open; no matrix with at most four values witnesses its cleanness (exhaustive backtracking, 5.0M nodes for four values with three designated). A prior that prefers fewer fallacies, Quine’s maxim of minimum mutilation, picks $\{\text{AC}\}$ over $\{K, \text{MP}\}$; in the core $\{\text{MP}, \text{AC}\}$ case it is silent unless it uses frequency, which does not track validity (Section 8).

Reading. This is the blame problem in its sharpest form. A community whose practice contains both modus ponens and affirming the consequent has an incoherent practice, and coherence correctly detects it. But coherence does not say which half of the practice is the meaning of “if”. A learner that is sound whatever the answer must give up both as rules, keeping at most their shared instances (Proposition 7.6(c)). What settles the direction is the world, or a denial: the bilateralists’ rejection (Smiley, 1996; Rumfitt, 2000; Restall, 2005) does syntactically what a counterexample object does semantically.

Theorem 7.22 (Depth relativization is necessary). *[T7 Thm 5.7, as repaired] There is a uniformly computable family of practices $(\Sigma_e^P)_{e \in \mathbb{N}}$, with $\mathcal{A} = \{[\emptyset : \emptyset]\}$, no world, and frequencies and instance laws computable uniformly in e , such that for the fixed prover that queries $\vdash \text{start}_e$ in every round*

ingredient dropped	failure mode	witness
positive data	refutational evidence is target-blind; nested sound calculi are indistinguishable	Proposition 7.16
caution (audit, then assert)	an adaptive prover exploits an unaudited fallacy; one step trivializes CPC	Proposition 7.17
negative evidence	fallacy and rule are indistinguishable ($\rightarrow$ versus $\leftrightarrow$)	Proposition 7.18
structurality	instance-level refutation never converges; coherence is toothless	Proposition 7.19
burn-in	rare refuter: $t \gtrsim \pi^{-1} \ln(1/\delta)$	Theorem 7.20
world or bilateral blame	MP/AC ($\rightarrow$ versus $\leftarrow$) symmetry; collateral for every sound learner	Theorem 7.21
depth relativization	Π_1 barrier for anytime soundness	Theorem 7.22
truthful designation	global loss of classical schemas	Proposition 7.23

Table 9: Each ingredient of the two-tier learner is necessary.

no computable learner achieves both of the following for all e , for any $\delta < \frac{1}{2}$: (a) δ -soundness at all times relative to the full-depth residue, $\mathbb{P}[\exists t : a \text{ step outside } R^* \cup R_{\text{res}} \text{ is accepted at } t] \leq \delta$; and (b) *eventual completeness*, $\mathbb{P}[\exists t \forall t' \geq t : R^* \subseteq \text{accepted at } t'] \geq 1 - \delta$.

Proof idea. A finite elementary formal system Σ_0 simulates the universal machine on input e from start_e and contains $\text{halt} \triangleright \perp$ (Theorem 6.33); add the axiom $\tau_e = (\triangleright \text{start}_e)$. If $\varphi_e(e)$ diverges the target is $\Sigma_0 \cup \{\tau_e\}$; if it converges, τ_e is a non-residual fallacy. A learner meeting (a) and (b) would make $\{e : \varphi_e(e) \text{ diverges}\}$ a Σ_1 set. Appendix F.12.

This is the Π_1 -completeness of coherence turned into a statement about *anytime* soundness. TTL’s form of soundness (relative to $F_{\text{res}}^{(d_t)}$ at time t , and to F_{res} only in the limit, with finitely many changes) cannot be improved by any computable procedure.

Proposition 7.23 (Truthful designation is needed). *[T7 Prop 5.8; T2 Prop 7.1, Thm 2.5] If a designated position violates (WS), descent can blame a genuine rule: with target $\{\text{MP}\}$ and the bad position $[a, a \rightarrow b : b]$, MP is blamed. Under structurality, one classically inconsistent designated context forces the loss of a classical schema everywhere (Proposition 11.10). Proposition 7.12 bounds the damage when at most m positions, not known which, violate (WS), and Theorem 6.7 is the soft version. Nothing bounds the damage of an unknown number of mis-designations.*

7.7 Specializations: what provably works for formal mathematics

Classical propositional logic. The language has $\neg, \wedge, \vee, \rightarrow, \top, \perp$. Judgments are formulas, or sequents whose contexts are *lists* with explicit structural rules (set or multiset contexts would need anti-unification modulo associativity and commutativity, for which the bounds of Section 5 are not proved). All schemas are *pure* (Definition 3.4); by Proposition 7.19 this cannot be dropped. Σ^* is any finite set of classically valid pure schemas that is complete for classical consequence, F any finite set of classically invalid pure schemas, the world is $J_W =$ closed formulas with $W =$ truth value (“computation”), and $\mathcal{A}$ consists of truthful positions and contains $[\emptyset : \emptyset]$.

Lemma 7.24 (Closed-instance refutability; Post’s substitution). *[T7 Lemma 6.1] Lean: Post.schema_invalid_iff_closedRefutation*

A pure schema τ is classically invalid iff some substitution of $\top/\perp$ for its metavariables yields a closed instance whose premises are W -true and whose conclusion is W -false. As a one-step

derivation this instance has size at most $|\tau|$, and there are at most $2^{v(\tau)}$ candidates, $v(\tau)$ the number of metavariables.

Proof. ($\Leftarrow$) A falsified instance is invalid. ($\Rightarrow$) Let a valuation v make the premises of an instance $\tau\theta$ true and its conclusion false, and put $c_x = \top$ if $v(\theta x) = 1$, else $c_x = \perp$. By induction on τ , every subformula of $\tau[c_x/x]$ has the value of the corresponding subformula of $\tau\theta$ under v , and closed formulas have the same value under every valuation. $\square$

This is the substitution step of Proposition 3.5 and Theorem 6.13, and essentially Post (1921). For sequents with a context metavariable Γ , substitute the one-element list $[\top]$ or $[\perp]$ according to $v(\wedge \Gamma\theta)$. A brute-force check on 30,000 random pure schemas (16,309 invalid) found no mismatch (computed; lemma61.py), and the formula case is formalized in Lean.

Corollary 7.25 (TTL on classical propositional logic is exact). *[T7 Cor 6.2, as repaired] Let $d \geq \max_{\tau \in F} |\tau|$, and let Ref_d return the first closed $\top/\perp$ instance, in a fixed enumeration, of a schema of B with W -true premises and a W -false conclusion, and NONE if there is none. This oracle is exact for refutations of size at most d and deterministic; its refutations are one-step and fully evaluable, so every descent is unblocked; and its answers do not depend on $d \geq \max_{\tau} |\tau|$. Then on G , with $\mathbb{P}(G) \geq 1 - \delta$: (a) the minimal conflicts are exactly the singletons $\{\tau\}$, $\tau \in F$, so (SB_d) holds and $\text{Coll}_d = F_{\text{res}}^{(d)} = \emptyset$; (b) the tier abstains before N_1 and accepts exactly R^* afterwards, so for every premise set B the reasoner derives exactly $\text{Cl}_{R^*}(B) = \text{Cn}_{\text{CPC}}(B)$, for derivations of any length and formulas of any size (Corollary 5.11), against any prover; (c) the audit makes at most $|F| + 1$ oracle calls and at most $|F|$ successful descents, and at most $(|F| + 1) \sum_{\sigma \in \Sigma^P} 2^{v(\sigma)}$ closed-instance checks ($\sum_{\sigma \in \Sigma^P} 2^{v(\sigma)}$ with caching); (d) the tier is truth-sound at all times (Corollary 7.15).*

Proof. If every schema of B is valid, a classical valuation agreeing with the truthful position and with W falsifies no step of R_B , so there is no refutation at any size; otherwise Lemma 7.24 gives a one-step refutation of size at most $|\tau| \leq d$. Hence each $\{\tau\}$ is a conflict, and since every conflict contains a fallacy (Lemma 7.4(b)) it contains a singleton conflict. So (SB_d) holds and $F_{\text{res}}^{(d)} = \emptyset$, and Theorem 7.14(iii) gives $A = \Sigma^*$; completeness of Σ^* gives (b). Each descent removes the schemas of which the falsified step is an instance, all fallacies and at least one. $\square$

The counts are upper bounds: one falsified step can be an instance of several fallacies, as $(\top, \perp \rightarrow \top \triangleright \perp)$ is of both AC and $(\top, A \rightarrow \top \triangleright A)$, and one descent removes both. The instance checks include those of the genuine schemas, which certify the final NONE. The oracle need not return a smallest refutation; exactness and determinism are all that the audit uses, and part (iv) of Theorem 7.14 is trivial here.

Reading. This is the user's first success criterion in its strongest form. From imitation of human proofs containing systematic fallacies and sporadic slips, plus truth-table evaluation of closed formulas, the learner identifies exactly the classical calculus the community uses, after N_1 human steps, with at most $|F|$ refutations. It is sound against every prover at every moment, and its reasoner derives exactly classical consequence, out of distribution.

Proposition 7.26 (Coherence alone on CPC: sound but lossy; one denial repairs it). *[T7 Prop 6.3; (b) as repaired] Let $J_W = \emptyset$ and $\mathcal{A}$ be truthful. (a) If $[\emptyset : \emptyset] \in \mathcal{A}$, every $\tau \in F$ lies in some minimal conflict once $d \geq d_\tau$, because $\Sigma^* \cup \{\tau\}$ derives $\perp$ from $\emptyset$ (Proposition 3.5). Hence $F_{\text{res}}^{(d)} = \emptyset$ for $d \geq \max_{\tau} d_\tau$,*

and TTL is truth-sound. (b) The collateral can be nonempty, and then TTL’s fallback is incomplete, necessarily so at stable depth (Theorem 7.21(b)). Let Σ^* be a complete Hilbert system for the full language containing K and MP , $F = \{AC\}$ and $\mathcal{A} = \{[\emptyset : \emptyset]\}$ or $\{[\emptyset : \emptyset], [A_1 : \emptyset]\}$. Then $\{AC, K\}$ is a minimal conflict at every depth at least the size of the four-step derivation above, and with A_1 designated so is $\{AC, MP\}$. Every descent is blocked, the fallback runs at Σ^P , and K (with A_1 , also MP) lies in Coll_d for all large d : every sound learner using coherence alone must withhold it as a rule. (c) Bilateral repair. If for each $\tau \in F$, $\mathcal{A}$ contains a counterexample position $[\Pi' : \varphi']$, where $\Pi' \triangleright \varphi'$ is a closed falsified instance of τ , and $d \geq \max_\tau |\tau|$, then (SB_d) holds and $A = \Sigma^*$.

Proof in Appendix F.13. *Reading.* In CPC closed evaluation refutes exactly the schemas that coherence refutes; by Post, both refute all invalid ones. The whole difference between the channels is localization: computation names the culprit, a contradiction only a list of suspects. A counterexample position is a syntactic object, a partial valuation, so “bilateral coherence with denials” and “world feedback with descent” are the same resource in two notations.

Complete decidable theories. Let T be complete and decidable, such as real closed fields (Tarski, 1951), Presburger arithmetic (Presburger, 1929), ACF_p or dense linear orders, with $\mathfrak{M} \models T$. Judgments are sequents $\Gamma \vdash \varphi$ of first-order formulas, with $W(\Gamma \vdash \varphi) = [\mathfrak{M} \models \forall \bar{x} (\bigwedge \Gamma \rightarrow \varphi)]$, decidable by quantifier elimination, so $J_W = J$. Σ^* is a finite schema calculus, sound and complete for T -consequence on sequents, including $\rightarrow$ -introduction and $\forall$ -introduction.

Lemma 7.27 (Validity is truth preservation). *[T7 Lemma 6.4]* (a) $\text{Sound}(R^*)$ is exactly the set of W -truth-preserving steps. (b) A schema is a fallacy iff it has a W -falsified instance, so each $\tau \in F$ has a smallest falsified instance, of some size $f_\tau \geq d_\tau$.

Proof idea. If all premises are true they are derivable, since T is complete, so Cl_{R^*} of them is the set of true sequents; if one is false, Σ^* derives its false universal closure, whose negation T proves, and then everything. Appendix F.14.

Corollary 7.28 (TTL on complete decidable theories, conditional on realizability). *[T7 Cor 6.5, conditional after verification]* Suppose in addition that Σ^* and F lie in a schema class for which the identification of Theorem 5.15 holds, for instance first-order patterns, possibly with decidable side conditions such as the eigenvariable condition of $\forall I$. Let $d \geq \max_\tau f_\tau$, $[\emptyset : \emptyset] \in \mathcal{A}$, and let Ref_d search the instances of size at most d of the schemas of B for a W -falsified one. Then everything in Corollary 7.25 holds verbatim: (SB_d) , $F_{\text{res}}^{(d)} = \emptyset$, $A = \Sigma^*$, truth-soundness at all times on G , and at most $|F| + 1$ oracle calls.

The realizability hypothesis is not established for the listed theories. The rule $\forall E$, from $\forall x \varphi$ infer $\varphi[t/x]$, is not a first-order pattern, because its conclusion is a substitution result; with t arbitrary it is not a higher-order pattern in the sense of Miller (1991) either. The lgg of three $\forall E$ instances is $(\forall x (z_0 + z_1 = z_2) \triangleright z_3 + z_4 = z_5)$, with the invalid instance $(\forall x (0 + 0 = 0) \triangleright 0 + 0 = S0)$, and more data only generalize it further (computed; `ind_lgg.py`). The usual axiomatizations of RCF and ACF_p contain degree-indexed families, and Presburger arithmetic an induction or congruence schema; none is a single first-order pattern. Dense linear orders are finitely axiomatized, so there only the quantifier rules matter. Without realizability, the identified schema for a non-pattern rule is in general an over-general lgg. If it has an instance outside $\text{Sound}(R^*)$, as for $\forall E$ and induction, the total oracle W falsifies one (Lemma 7.27(b)), and an

audit of sufficient depth removes it with singleton blame. So TTL stays sound, eventually so under the depth schedule, but A lacks the rule and exactness fails. Two repairs are open: an encoding with auxiliary decidable judgments $\text{Sub}(\varphi, x, t, \psi)$ (“ ψ is $\varphi[t/x]$ ”) evaluated by W , carried out below for induction; or an identification theorem for higher-order-pattern schemas.

An honest caveat: why learn at all? Here W is already a sound and complete step checker, so learning is not needed for epistemic access. TTL adds two things. It identifies *which* calculus the practice uses. And it makes deployed checking fast: schema matching takes linear time, while real quantifier elimination is doubly exponential in the worst case (Davenport and Heintz, 1988), deciding RCF sentences is possible in exponential space (Ben-Or et al., 1986), the first-order theory of $(\mathbb{R}, +)$ already needs exponential time, and Presburger arithmetic needs $2^{2^{\Omega(n)}}$ time (Fischer and Rabin, 1974). The audit calls W only on the few instances needed to falsify each fallacy. This dividend is real but modest. A restricted oracle shows the division of labour more sharply. If W evaluates only quantifier-free sentences with rational constants and the logic is trusted, the fallacy $\tau = (\triangleright \neg \exists x (x \cdot x = 1 + 1))$ has no instance that this oracle can evaluate, so the world alone never refutes it and Lemma 7.27(b) fails; designating the true sentence $\exists x (x \cdot x = 1 + 1)$ catches it, since τ and the trusted $\neg E$ derive $\perp$, and descent backward through $\neg E$ blames τ alone. Over $\mathbb{R}$, coherence with a designated truth catches what this computation misses; over $\mathbb{N}$ the converse holds (Lemma 6.19). The real-closed-field core of olympiad algebra and geometry would be covered by Corollary 7.28 once its realizability hypothesis is met; trigonometric steps are not, since $(\mathbb{R}, +, \cdot, \sin)$ defines $\mathbb{Z}$.

Arithmetic. The language is that of PA and $\mathfrak{M} = \mathbb{N}$. T_{trust} is first-order logic: the logic is fixed, and non-logical axiom schemas are learned (the logic can be learned first, at the propositional level, by Corollary 7.25). The world is $J_W = \Delta_0$ sentences with $W = \text{computation}$, and $\mathcal{A}$ consists of truthful positions. A practice axiom schema is genuine iff all its instances are true in $\mathbb{N}$, so Σ^* may contain PA together with true extras such as $\text{Con}(\text{PA})$, and (WS) holds with $V = \text{truth in } \mathbb{N}$. Since R^* consists of axiom steps, the relevant target closure is $\text{Cl}_{R^* \cup T_{\text{trust}}}$ (Theorem 7.14(i)). Ground axioms (those of $\mathbf{Q}$, each $\text{Con}(T_j)$, $\neg \text{Con}(\text{PA})$) are realizable; the induction schema is not (part (e)). Parts (a)–(d) and (f) concern the audit and hold for any $\hat{P} = \Sigma^P$.

Theorem 7.29 (Arithmetic is Popperian). *[T7 Thm 6.6, as repaired in two rounds]*

- (a) False Π_1 axioms get singleton blame. *Let τ have a false instance $\forall \bar{x} \theta(\bar{x})$ with $\theta \in \Delta_0$ and least counterexample $\bar{n}$. The refutation “ $\vdash \forall \bar{x} \theta$ (by τ); $\vdash \theta(\bar{n})$ (by trusted $\forall E$)”, with $W(\theta(\bar{n})) = 0$, has an unblocked descent that blames τ alone, by the backward rule through $\forall E$. With the prefer-unblocked oracle, τ is removed and no PA axiom is put at risk. For one quantified variable with m free occurrences in θ , $\tau \notin F_{\text{res}}^{(d)}$ once $d \geq 2|\theta| + m(n+1) + O(1)$; for k variables $(k+1)(|\theta| + m(n+1) + O(k))$ suffices, n the largest component of $\bar{n}$.*
- (b) Under presumption of validity, false Σ_1 axioms survive. *For $\tau = (\triangleright \neg \text{Con}(\text{PA}))$, $\Sigma^* = \text{PA}$ and positions with Δ_0 assertions and denials, $\Sigma^* \cup \{\tau\}$ is clean at every depth, so $\tau \in F_{\text{res}}$ is asserted forever.*
- (c) Popperian Σ_1 -caution removes syntactically Σ_1 residue. *Fix a computable prenex normal form with negations pushed into the Δ_0 matrix, and assert an instance of a practice axiom with Σ_1 normal form $\exists \bar{x} \theta$ only once W has verified a witness. Then false axioms with Σ_1 normal form, $\neg \text{Con}(\text{PA})$ included, are never asserted, and true ones are asserted after a witness search, with no uniform time bound; for schematic axioms, acceptance becomes a*

semi-decision. The policy is syntactic: $\forall y \exists p (\text{Prf}_{\text{PA}}(p, \ulcorner \perp \urcorner) \vee y \neq y)$, equivalent to $\neg \text{Con}(\text{PA})$ but with a Π_2 normal form, escapes it and is residual.

- (d) Beyond Σ_1 there is a permanent residue. *False Π_2 or Σ_2 axioms τ with $\Sigma^* \cup \{\tau\}$ clean at every depth (for example, $\Sigma^* \supseteq \mathbf{Q}$ and $\Sigma^* \cup \{\tau\} \cup A \cup \neg D$ consistent for every designated $[A : D]$) admit no finite refutation and survive, and no computable policy sorts the true from the false ones in the limit (Theorem 6.22(e)). The same holds for randomized learners that are correct in the limit with probability above $\frac{1}{2}$ (proof sketch).*
- (e) Finite reflection-extended practices are learned under the floor, given a realizable encoding of induction. *The induction schema is not a first-order pattern: its lgg has false instances, and the audit refutes it with singleton blame, so the asserted calculus would lack induction. Add auxiliary decidable judgments $\text{Sub}(\varphi, x, t, \psi)$ to J_W and write induction as $\text{Sub}(\varphi, x, 0, a), \text{Sub}(\varphi, x, Sx, b) \triangleright \vdash a \wedge \forall x(\varphi \rightarrow b) \rightarrow \forall x\varphi$, with the human data recording the Sub premises. This is a first-order pattern, recovered by the lgg of generic instances. Then under (Floor) a practice has at most $\min(K, \lfloor 1/(2\Delta) \rfloor)$ axiom schemas, and $T_k = \text{PA} + \text{Con}(\text{PA}) + \dots + \text{Con}(T_{k-1})$ is identified from positive data for every k within that bound. Without a floor there are no uniform sample bounds: comparing the targets T_{k-1} and T_k , where the tag $\text{Con}(T_{k-1})$ has frequency π , under (OE), a learner that accepts $\vdash \text{Con}(T_{k-1})$ at time t with probability at most δ under T_{k-1} accepts it under T_k with probability at most $1 - (1 - \delta)(1 - \pi)^t$; so if $\delta + \delta' < 1$, acceptance with probability at least $1 - \delta'$ needs $t \geq \ln \frac{1-\delta}{\delta'} / \ln \frac{1}{1-\pi}$, unbounded as $\pi \rightarrow 0$. This bound is tight.*
- (f) Reflection as designation. *If $\mathcal{A}$ asserts $\text{Con}(\text{PA})$, then $\neg \text{Con}(\text{PA})$ becomes a singleton conflict.*

Proof idea. (a) is Lemma 7.9. (b) $\text{PA} + \neg \text{Con}(\text{PA})$ is consistent by Gödel’s second theorem (Gödel, 1931) and proves every true Δ_0 sentence and no false one (Lemma 6.19). (d) reduces to Shoenfield’s limit lemma (Shoenfield, 1959) and Post’s hierarchy theorem. (e) is the mirror image of Theorem 7.20. Appendix F.15.

Three computations support (a) and (e) (`computed`; `ind_lgg.py`, `ind_sub_lgg.py`, `two_point_bounds.py`). For $\theta(x) = x < S^5 0$ the refutation’s judgments $\forall x (x < S^5 0)$ and $S^5 0 < S^5 0$ have sizes 10 and 13. The lgg of four induction instances has the false instance $(0 + 0 = 0) \wedge \forall x (0 + 0 = x \rightarrow 0 + S0 = Sx) \rightarrow \forall x (0 + 0 = x)$, while the lgg of five generic Sub-encoded instances equals the pattern up to renaming. And the floor bound in (e) is attained, by linear programming over all tag sequences, in all 108 cases checked. The bound in (e) differs from that of Theorem 7.20 because the rare tag now lives in the scenario where acceptance is *sound*: the learner must see the tag at least once, so the bound scales with $\ln(1/\delta')$, not $\ln(1/\delta)$. Over classes containing T_ω , identification in the limit fails altogether (Theorem 6.21); T_ω has infinitely many axioms, so it lies outside the finite practices considered here.

Reading. In arithmetic TTL achieves exact, localized, Popperian falsification of false Π_1 axioms; exclusion of syntactically Σ_1 errors under Σ_1 -caution; identification of finite reflection-extended practices, given a realizable induction schema; and soundness at all times relative to the stated residue. It does not achieve, and no computable procedure can achieve, the elimination of false Σ_2 axioms consistent with the practice, or uniform sample bounds without a frequency floor; and under presumption of validity it keeps false Σ_1 axioms such as $\neg \text{Con}(\text{PA})$. The Gödel–Rosser alternatives are the Kripkensteinian residue of arithmetic (Theorem 6.38(iii)), and TTL leaves it exactly where Section 6 located it, now with explicit sample sizes, refutation budgets and

N	positive-only cautious tier unsound	TTL unsound	TTL exact ($A = \Sigma^*$)	untrimmed lgg + audit exact
60	12/20	0/20	0/20	6/20
120	20/20	0/20	10/20	8/20
250	20/20	0/20	20/20	1/20
500	20/20	0/20	20/20	0/20

Table 10: TTL against an exhaustive adversarial prover on a propositional practice with four systematic fallacies and 1% sporadic noise, 20 trials per N (computed; `t1l_sim.py`, re-run for this draft). At $N = 500$ one trial had more than e noisy steps, on the fallacy tag “xor”, and was still exact.

soundness against adaptive provers. Reflection enters as a *designation*, a justification beyond PA-proof that is made explicit rather than smuggled in as a rule.

7.8 Simulation evidence

The two-tier learner on a propositional practice. The script `t1l_sim.py` runs TTL in the setting of Corollary 7.25. The practice has nine genuine tagged natural-deduction rules ($\wedge$ I, $\wedge$ E_{1,2}, $\vee$ I_{1,2}, MP, modus tollens, disjunctive syllogism, double-negation elimination) and four fallacies with their own tags (affirming the consequent, denying the antecedent, conversion, “or” as “xor”). Sporadic noise at rate $\alpha = 0.01$ is mis-tagged at random, and the trim budget is $e = 2$. The audit is the per-schema closed-instance oracle. An adversarial prover enumerates every step over a pool of ten formulas and checks it by truth table. Table 10 shows the result of re-running the script (seed 0; it is deterministic across hash seeds).

The positive-only cautious tier asserts the fallacies as soon as they are anchored (Proposition 7.5): it is unsound in 72 of 80 runs. TTL is unsound in none of the 80, and exact in all 20 runs at $N = 250$ and at $N = 500$. Excess noise on a *fallacy* tag is harmless, since whatever schema the trimmed lgg produces for that tag is world-refuted and dropped anyway. The untrimmed learner is kept sound by the audit, which refutes the collapsed lgg of Proposition 5.13, but it loses whole tags, and loses more of them as N grows. The script’s scope is limited: it tests the singleton-conflict case of CPC with a fixed budget and a per-schema world audit. It does not exercise descent on multi-step refutations, the fallback, or the N_1 formula; a simulation of the full audit is future work.

The propositional experiment. Experiment B (Section 12) runs a related but distinct pipeline on sequent-style natural deduction with 13 target rules: a support-thresholded version-space learner, coherence pruning on five designated contexts with blame by a minimum-weight hitting set, and optional sparse world feedback (two observed valuations). It is not an implementation of TTL, but it exercises the phenomena of this section (computed; `code/results/prop_learning.md`; final numbers in Section 12). Over the 72 runs of its main learner, the adversary derived $\vdash \perp$ in 32 runs after imitation alone, and in none after coherence pruning or after world feedback. A bold learner over classical natural deduction, searching with *Post probes* (substituting $\perp/\neg\perp$ along a falsifying row, as in Lemma 7.24) and blind search, refuted every one of 118 invalid structural candidate schemas by an explicit derivation of $\vdash \perp$, and accepted all 59 valid ones. The non-structural rules $\vdash p$ and $p \vdash q$ survived, as Proposition 7.19 predicts; most unsound rules that survive pruning in any run mention specific atoms. One observed valuation already pinpoints the guilty rule for schematic rules, so world feedback needed fewer pruning rounds

than coherence alone (5.0 versus 9.1 on average): localization, not refutation. Over intuitionistic natural deduction the bold learner accepted excluded middle, double-negation elimination, reductio, Peirce’s law, weak excluded middle and Dummett’s linearity, each underivable in IPC (Kripke countermodels), and rejected the classically invalid candidates. That is Theorem 6.17 in action: coherence pins CPC from above and is blind for IPC, so a bold extension of IPC drifts to CPC-like logics, and only the cautious, imitation-based tier keeps the intuitionistic calculus.

7.9 Reading in the user’s terms

“Learning which inferences are valid” is learning the *audited practice*, in three acts done by different channels. (1) *Imitation finds the practice’s rules*. Generalizing human steps to the least general schemas that cover them, with trimming against sporadic slips, finds the schemas the practice uses, its systematic fallacies included. It is the only channel that says anything positive about a rule (Lemma 7.4 and Proposition 7.16). (2) *Confrontation sorts the rules: an audit, not a loss*. The learner presumes the practice valid and searches for derivations, from accepted positions, of $\perp$, of a denied claim, or of a computably false claim. Each shows that *some* rule used is wrong; the sound response is to withhold every rule in a minimal such set (Lemma 7.8), and when the derivation can be checked step by step against the world the set shrinks to the culprit (Lemma 7.9). A “coherence loss” that merely penalizes having arguments for P and $\neg P$ does the first half of this, and only the first half. (3) *Assertion comes last*: after the audit, and after enough practice has been seen to rule out a rare rule that would expose a fallacy (Theorem 7.20). The resulting checker cannot be fooled by any prover (Theorem 7.14(i)). In propositional logic it accepts exactly the valid steps (Corollary 7.25); in complete decidable theories it does so too, if the target calculus is realizable in the schema class (Corollary 7.28); in arithmetic it works in Popper’s sense and no further (Theorem 7.29).

“An argument is valid when it is a sequence of valid inferences.” This is Lemma 3.1, and it cuts both ways. One accepted bad step poisons every argument; and chaining is also what lets a contradiction be traced back to a step. Descent is the user’s own “go back to what caused the contradiction”, made into an algorithm with a guarantee.

“Learning meanings” is identification up to the residue. On an inferentialist reading (Brand, 1994), the asserted calculus is the learned inferential role. Imitation fixes which role the community *uses*. Coherence fixes it only up to the choice of *which* member of each conflicting set to give up; in the MP/AC example that choice is literally whether “ $\rightarrow$ ” means $\rightarrow$ or $\leftarrow$, and no sound learner using coherence alone can assert modus ponens as a rule there, although it can accept those instances of MP that are also instances of AC. This ambiguity is not a gappy-valuation phenomenon (Section 6.4); it is a *blame* phenomenon: the practice is incoherent, and coherence does not say which half of it is the meaning. The world (computation, counterexample objects) or a denial settles the direction. That is a precise sense in which the user’s “hooking of a model onto the world” is needed beyond inferential role: not to make the role coherent, but to make it the *intended* one when practice conflicts with itself. What remains after all three channels is the residue F_{res} , the fallacies coherent with the rules and invisible to the available world. It is empty for propositional logic and, given realizability, for complete decidable theories. In arithmetic it contains the non-standard alternatives, such as $\neg\text{Con}(\text{PA})$ under presumption of validity, which only a designated reflection commitment or a Popperian asymmetry removes, and past Σ_1 nothing computable removes.

Justification. A step accepted by TTL carries a three-part warrant: it instantiates a schema the practice uses robustly; that schema survived an audit of stated depth against stated designated positions and a stated world; and the argument is a chain of such steps. None of the three parts is a proof of validity. Together they give a guarantee relative to explicit parameters $(d, \mathcal{A}, W, \delta)$. Raising d changes the warrants only finitely often, sometimes restoring a rule withheld as collateral (Theorem 7.14(iv)), and no computable procedure can avoid the possibility of change (Theorem 7.22). This matches the user’s view that justification is an infinite endeavour, in a form where every intermediate stage is sound relative to what has been checked. It is also Goodman’s mutual adjustment of rules and inferences (Goodman, 1955), with each adjustment licensed by a theorem; Section 13 takes this up. The open problems this section leaves (the query complexity of sound blame, the full-depth Hilbert rival, blame with priors, the size of coherence witnesses in first-order logic, and TTL with anchored physical contexts as designated positions) are collected in Section 14.

8 Simplicity and normativity from imitation

Imitation identifies practice, errors included: for positive data, a systematic error generated by a schema *is* a rule (Corollary 5.20). This section examines a different remedy, proposed in K. Hänni’s notes, and answers the question it raises.¹ The note asks how one can understand “what someone means when using a word. as opposed to becoming a predictor of what they would say”. It proposes

“an even stronger simplicity prior than solomonoff. eg if there are explainable mistakes on a simple model, you want the simple model that doesn’t predict the mistakes. [...] let’s just do a version of the simple hypothesis with noise, and then penalize the likelihood term less.”

The note anticipates an objection: in sequence prediction, a steeper penalty favours a universal hypothesis that absorbs everything. It argues that this pathology “is not present in the function case, if [...] each input has to sample its own random bits”. It conjectures that the intended model is “a vertex of the convex hull of the set of attainable (hypothesis complexity, expected neg log likelihood) tuples”. And it adds: “I consider it very much plausible that there is some other pathology”.

Answers in brief. The function-case intuition and the convex-hull argument are both right: private randomness makes every hypothesis pay for mixing per input, the product-model frontier has the hoped-for kink, and selection is a threshold on *rates* (risk reduction per bit) along the lower hull (Theorems 8.3, 8.6 and 8.11 and Lemma 8.8). But there are other pathologies. Universality returns at block granularity (Proposition 8.5); selection is by rate, not validity, so cheap fallacies out-rate rare valid rules and guards erode (Theorem 8.15 and Propositions 8.16 and 8.18); and the right steepness is not identifiable from imitation data (Theorem 8.17). Steeper simplicity is a good *third* filter beside coherence and the world, and the note’s own alternative, “you specify properties of the thing or notion”, is a separate channel that removes cheap fallacies at every steepness (Theorem 8.20 and Proposition 8.23). Section 8.7 gives the verdict in the note’s terms.

¹The note ‘beating solomonoff induction at grokking a notion’ (unpublished). Quotations are verbatim.

Most results are standard or trivial once set up. As far as we know, three things are new: the product-model frontier and its kink, the rate threshold for arbitrary hypotheses, and the precise forms of the pathologies and of the division of labour.

8.1 The proposal, formalized

Inputs $x \in X$ are drawn from μ , and labels lie in a finite set Y . The teacher is a channel $P^*(y | x)$. In the note’s picture, P^* arises from an intended $norm\ h^\circ$ through sporadic noise and systematic error. Data are i.i.d. from $\mu \times P^*$. *Normativity* asks the learner to output h° , or its deterministic core, rather than P^* . In Section 8.6, x is a proof context and y the next human step; then P^* is the practice R^P of Definition 2.10, viewed as a channel.

Definition 8.1 (Randomness and objective). ^[T5 Def 1.1, §1.3] A *product hypothesis* (private randomness) is a conditional distribution $h(\cdot | x)$, with likelihood $\prod_i h(y_i | x_i)$. A *shared-randomness* hypothesis is a distribution $P(y_{1:N} | x_{1:N})$; sequence prediction is its causally conditioned case. In a *block* hypothesis, the input at position i of a block (e.g. a document) includes the block’s earlier pairs. The note’s loss is $\lambda \ell(h) + \sum_{i \leq N} -\log_2 h(y_i | x_i)$, with $\lambda > 1$ growing with N . Writing $\lambda = cN$ and dividing by N gives $\hat{J}_c = c\ell + \hat{\mathcal{R}}_N$, and its population version $J_c = c\ell + \mathcal{R}$, where $\mathcal{R}(h) = \mathbb{E}_{\mu \times P^*}[-\log_2 h(y | x)]$. Let $\Phi(c) = \inf_h J_c(h)$. In the *idealized* version, $\ell = K$ (prefix complexity) over total programs with rational outputs, and p_h is a shortest program for h . In the *computable* version, ℓ is a prefix code on a countable class. A *component* costing k bits that lowers $\mathcal{R}$ by r has *rate* r/k .

The steepness c is measured in bits per sample per bit of description. Where a result compares the class with a semimeasure or an infinite mixture, the class is enlarged to lower-semicomputable semimeasures and computable real-valued predictors.

Lemma 8.2 (Empirical selection tracks population selection). ^[T5 Lemma 1.2] *In the computable version, suppose $h(y | x) \geq 2^{-B}$ for all h . With probability at least $1 - \delta$, simultaneously for all h , $|\hat{\mathcal{R}}_N(h) - \mathcal{R}(h)| \leq B\sqrt{(\ell(h) \ln 2 + \ln(2/\delta))/(2N)}$. On this event, each minimizer $\hat{h}$ of $\hat{J}_c$ satisfies, for every fixed h_0 ,*

$$J_c(\hat{h}) \leq \Phi(c) + 2B\sqrt{((\ell(h_0) + B/c) \ln 2 + \ln(2/\delta))/(2N)}.$$

In particular, a unique population minimizer with gap γ is recovered once this bound is below γ .

This is a standard Occam-type bound; the proof is Hoeffding’s inequality with a Kraft union bound (Appendix G.1).

Fixed versus growing steepness. With fixed λ , $c \rightarrow 0$, so every component of positive rate is eventually absorbed, error or not (T4 Prop 7.1(i); L11 Prop 2.2). Imitation with a fixed simplicity weight therefore converges to the teacher’s practice. This is the learner’s version of Cohen’s view that normative standards are a systematization of ordinary judgments (Cohen, 1981a). With $\lambda \propto N$, as the note proposes, selection is asymptotically stable. So the obstruction found below is *rate inversion*, not sample size; this corrects the finite-window reading of L11.

Relation to the literature. The tempered posterior $\propto 2^{-\ell} \prod_i h(y_i | x_i)^\eta$ has its MAP at the minimizer of $\ell/\eta + N\hat{\mathcal{R}}_N$, so the note’s λ is $1/\eta$: a *learning rate* $\eta < 1$ (Grünwald, 2012; Grünwald and van Ommen, 2017; Zhang, 2006; Bissiri et al., 2016; Bhattacharya et al., 2019). With $\lambda = cN$ the posterior tends to the non-concentrating Gibbs measure $\propto 2^{-(\ell + \mathcal{R}/c)}$ (Catoni,

2007), whose mode the proposal selects. SafeBayes learns η from data to repair misspecification; Theorem 8.17 suggests why no such criterion finds the note’s η , since an imitation model is well specified *for the teacher* (our reading, not checked against the papers).

Kolmogorov’s structure function $h_x(\alpha) = \min\{\log |S| : x \in S, K(S) \leq \alpha\}$ lies, to logarithmic precision, above the sufficiency line $K(x) - \alpha$, and essentially every compatible shape occurs (Vereshchagin and Vitányi, 2004) (exact side conditions unchecked by us); see also Gács et al. (2001); Vereshchagin and Shen (2017). Set and probability models share randomness. The product-model structure function $\rho^\times(k) = \min\{\mathcal{R}(h) : \ell(h) \leq k\}$ need not have slope -1 and can kink (Theorem 8.6): that is the note’s insight in this language. All sufficient statistics tie at $\lambda = 1$, so $\lambda = 1 + \epsilon$ selects roughly the minimal one and larger λ selects *insufficient* models (proof sketch), as the note wants, since the errors are structure. The objective $c\ell + \mathcal{R}$ is the algorithmic rate–distortion Lagrangian with log-loss distortion (Vereshchagin and Vitányi, 2010); classical two-part MDL (Rissanen, 1978; Barron and Cover, 1991; Grünwald, 2007) has $\lambda = 1$ (whether Barron and Cover also use $\lambda > 1$ we have not checked).

8.2 No universal hypothesis in the function case

The note asks whether it is “just failing to construct the right ‘universal hypothesis’”. The answer is no as long as randomness is private and inputs carry no other labels, and yes as soon as they do.

Theorem 8.3 (Per-input versus once-only pricing). *[T5 Thm 2.1, Prop 2.2, as repaired] Let $\mathcal{C}$ be a finite class of deterministic labelers, $x_1, \dots, x_N$ inputs, and $\mathcal{C}(x) = \{f(x) : f \in \mathcal{C}\}$.*

(i) *For every product h and every distribution Q on $\mathcal{C}$,*

$$\max_f \sum_i -\log h(f(x_i) \mid x_i) \geq \sum_i H(Q_{x_i}),$$

where Q_x is the law of $f(x)$. The minimax value over product h is exactly $\max_Q \sum_i H(Q_{x_i}) \leq \sum_i \log |\mathcal{C}(x_i)|$. Equality holds iff some Q has uniform marginals on every $\mathcal{C}(x_i)$. Such a Q (the uniform one) exists when a group acts on $\mathcal{C}$ by $(g \cdot f)(x) = \sigma_{g,x}(f(x))$, with permutations $\sigma_{g,x}$ of Y , transitively on each $\mathcal{C}(x)$; parities are an example.

(ii) *For shared randomness, $\min_P \max_f -\log P(f(x_{1:N})) = \log |\{f(x_{1:N})\}| \leq \log |\mathcal{C}|$ (Shtarkov, 1987).*

(iii) *A mixture with weights w_q pays $\log(1/w_q)$ at every input if it mixes per input, and once if it mixes sequences. Both bounds are tight.*

The core of the proof is one line: $\max_f \geq \mathbb{E}_{f \sim Q}$, and $\mathbb{E}_Q[-\log h(f(x) \mid x)] = H(Q_x) + \text{KL}(Q_x \| h(\cdot \mid x))$. The minimax claim (via Sion’s theorem) and the group condition are proved in Appendix G.2.

For parities on $\{0, 1\}^{10}$ with $N = 200$, the per-input mixture pays 200.0 bits and the Bayes mixture 10.0; the uniform-marginal condition is needed, since $\{(a, a), (a, b), (b, a), (c, a)\}$ has value $2.5431 < \log_2 3 + 1$ (computed; T5 c2(a), repair_checks).

Proposition 8.4 (The sequence pathology, exactly). *[T5 Prop 2.3, threshold corrected] Let $\mathbf{M}$ be a universal lower-semicomputable conditional semimeasure on label sequences, with $\mathbf{M} \geq 2^{-K(q) - c_0} q$ for every computable q (Li and Vitányi, 2008), and let $c_{\mathbf{M}} = K(\mathbf{M})$. For every $\lambda > 1$,*

$$\lambda K(\mathbf{M}) - \log \mathbf{M}(y_{1:N} \mid x_{1:N}) \leq \lambda c_{\mathbf{M}} + K(q) + c_0 - \log q(y_{1:N} \mid x_{1:N}).$$

This beats q 's own objective whenever $K(q) > c_{\mathbf{M}} + (c_{\mathbf{M}} + c_0)/(\lambda - 1)$.

Proof. Take logarithms in the dominance inequality and add $\lambda c_{\mathbf{M}}$. $\square$

As $\lambda \rightarrow \infty$ the threshold tends to $c_{\mathbf{M}}$, not 0. In the function case, even the universal conditional semimeasure $\mathbf{m}(y \mid x) \asymp 2^{-K(y \mid x)}$ is a fixed predictor [T5 Thm 2.4]. For binary Y we have $K(y \mid x) = O(1)$, so $\mathcal{R}(\mathbf{m}) \geq \log_2 \frac{1}{1-2^{-\kappa_U}} > 0$ for every teacher, where κ_U is a machine constant.

Proposition 8.5 (Per-block pricing: in-context universality). [T5 Prop 2.6, as repaired] Suppose the input at position i of block b contains the block's earlier pairs. Let (q_θ) be a uniformly computable family of “persona” models $q_\theta(y \mid x, \text{history})$, with computable weights $\sum_\theta w_\theta \leq 1$. The in-context Bayes mixture h_w , with weights $w(\theta \mid \text{history}) \propto w_\theta \prod_{k < i} q_\theta(y_{b,k} \mid \cdot)$, is a single function hypothesis with $K(h_w) \leq K(w, (q_\theta)) + O(1)$. On every block,

$$\sum_{i \in b} -\log h_w(y_{b,i} \mid \cdot) \leq \min_\theta \left[\log \frac{1}{w_\theta} + \sum_{i \in b} -\log q_\theta(y_{b,i} \mid \cdot) \right].$$

Proof. Let $Z_i = \sum_\theta w_\theta \prod_{k < i} q_\theta(y_{b,k} \mid \cdot)$. Then $h_w(y_{b,i} \mid \cdot) = Z_{i+1}/Z_i$, so the block likelihood telescopes to $Z_{|b|+1}/Z_1 \geq w_\theta \prod_i q_\theta(y_{b,i} \mid \cdot)$. This holds for history-dependent personas and inputs too, because everything is conditioned on the realized history. $\square$

So h_w pays $\log(1/w)$ per block, not per input. Suppose that 20% of humans each have an idiosyncratic error, “in error contexts play wrong move j ”, with j uniform among 2^{16} moves. Each such rule has rate about 0. Yet an $O(1)$ -bit copier beats the norm model once a block has two error contexts (computed; T5 c5). With two error contexts, the copier pays 3.03 bits per error context; with sixteen, it pays 1.33; the norm model pays exactly 4.635 at any block length.

Reading. An LLM imitating whole proof documents is a block model: it learns each author's systematic errors in context. A step checker whose input is a single step (Π, j) (Section 2) is a function model. The step is therefore the granularity at which the note's defence holds. This gives a second, independent argument for the step-level architecture of Sections 3 and 4.

8.3 The kink: shared versus private randomness

In sequence prediction, the note says, “it's pretty much just 1 bit paying for 1 bit”. In the function case one might instead “see some point at which the derivative changes”. We prove this for the note's example, made rigorous.

Setting. Let $X = \{0, 1\}^d$, $n = 2^d$ and μ uniform. The simple model is a parity $f_a(x) = \langle a, x \rangle \bmod 2$ with $K(a) \geq d$. The error set E has $|E| = pn$, $p < 1/4$ and $K(p) = O(\log d)$ (the last assumption was added after verification), and the teacher is $y^* = f_a \oplus 1_E$. Let $L_E = \log_2 \binom{n}{pn} \approx nH(p)$, and let $\delta_E = L_E - K(E \mid a, K(a))$ be the randomness deficiency of E , which is $O(\log n)$ for most E . Write $\hat{s}_h(a')$ for the correlation of h 's vote $h(0 \mid x) - h(1 \mid x)$ with $(-1)^{\langle a', x \rangle}$. The good model is f_a with flip rate p ; it has $K \leq d + \kappa'$ with $\kappa' = O(\log d)$, and risk $H(p)$. The bad model y^* has risk 0.

Theorem 8.6 (Kink). [T5 Thm 2.5, as repaired] There is a machine constant κ such that the following hold.

- (a) Shared randomness has (almost) no kink. Let P_u be the uniform shared-randomness mixture of the 2^u noisy parities that agree with a on its first $d - u$ coordinates. Then $|K(P_u) - (d - u)| \leq \kappa'$ and $-\log P_u(y^*) = u + nH(p) - o(1)$, where the $o(1)$ term is nonnegative and super-exponentially small in n . So the total $K(P_u) - \log P_u(y^*)$ is $d + nH(p)$ up to $\pm(\kappa' + o(1))$ for every u , and every minimizer u^* of $\lambda K(P_u) - \log P_u(y^*)$ has $d - u^* \leq \frac{\lambda}{\lambda - 1}(2\kappa' + o(1))$.
- (b) Private randomness has a kink. For every product h and $t \in \mathbb{N}$ with $K(h) \leq d - 2t - 2\log_2(t + 1) - 2\log_2 d - \kappa$,

$$\mathcal{R}(h) \geq 1 - \log_2(1 + \beta_t) \geq 1 - \beta_t / \ln 2,$$

where $\beta_t = \min\{2^{-t} + 2p, (1 - 2p)2^{-t} + \varepsilon_E\}$, with $\varepsilon_E = \sqrt{32 \ln 2 \cdot p(\delta_E^+ + d + \kappa_n)/n}$, $\delta_E^+ = \max(\delta_E, 0)$ and $\kappa_n = \log_2 n + 2\log_2(\log_2 n + 2) + \kappa$.

- (c) The error segment. For every product h , $K(h) + n\mathcal{R}(h) \geq d + L_E - \delta_E - 2\log_2 d - \kappa$.

Proof idea. In (a), the true component has weight 2^{-u} and likelihood exactly $2^{-nH(p)}$. Every other component disagrees with y^* on at least $n/2 - pn$ points. In (b), Parseval shows that at most 2^{2t} parities correlate at least 2^{-t} with h . This is the standard list-size bound for the Hadamard code (Goldreich and Levin, 1989; Kushilevitz and Mansour, 1993). So a short h that correlated with f_a would encode a in fewer than d bits. Jensen's inequality turns small correlation into risk near one bit. The slack $2p$ arises because h may be right on E for free. The slack ε_E removes it, using the randomness of E relative to h , via Hoeffding's inequality for sampling without replacement (Hoeffding, 1963) and counting. Part (c) is the coding theorem. Full proofs, with the lemmas, are in Appendix G.3.

Reading. Part (a) is the note's "1 bit paying for 1 bit": the shared frontier is the sufficiency line, so a steeper penalty selects a mixture that knows almost nothing of a and then learns the errors as any Bayes mixture does. Part (b) makes the note's hope quantitative. A product hypothesis missing $u = d - K(h)$ bits gains at most about $2^{-u/2 + O(\log d)}$ bits per sample over the coin, plus slack: exponentially little, though not nothing (list mixtures missing $u = 1, 2, 3$ bits gain 0.454, 0.221, 0.110 bits (computed)). So on the scale of d bits the product frontier is flat near the coin and drops at the simple model; that drop is the change of derivative.

Two caveats. First, the kink is resolved only up to $O(\log d)$ bits next to the simple model. Second, the unconditional slack $2p$ gives a kink only for $p < 0.1421$, where $1 - 2p/\ln 2 > H(p)$; the slack ε_E gives one for every $p < 1/4$ once n is large and $\delta_E = O(\log n)$. For example, at $p = 0.2$ and $d = 20$ (taking $\delta_E = \kappa = 0$), $\varepsilon_E = 0.014$, against $2p = 0.4$.

8.4 Selection traces the hull: rate thresholds

Let $z_h = (\ell(h), \mathcal{R}(h))$, $Z = \{z_h\}$ and $\hat{C} = \text{conv}(Z + \mathbb{R}_{\geq 0}^2)$. Only finitely many h have $\ell \leq L$, and $\mathcal{R} \geq 0$, so each J_c with $c > 0$ attains its minimum.

Definition 8.7 (Lower hull). [T5 §3.1, as repaired] $\text{hull}^-(Z)$ is the set of boundary points of $\hat{C}$ supported by a line of slope $-c$ with $0 < c < \infty$. A *vertex* is an extreme point of $\hat{C}$ that lies in $\text{hull}^-(Z)$; vertices belong to Z .

Lemma 8.8 (Selection traces the lower hull; standard). *[T5 Lemma 3.1, as repaired]* (i) For $c > 0$, the minimizers of J_c are the h with $z_h \in \text{hull}^-(Z)$ on a supporting line of slope $-c$, and every point of $\text{hull}^-(Z)$ in Z arises this way. (ii) If $c < c'$, and h and h' minimize J_c and $J_{c'}$, then $\ell(h) \geq \ell(h')$ and $\mathcal{R}(h) \leq \mathcal{R}(h')$. (iii) A point $z \in Z$ is the unique minimizing point for all c in some nonempty open interval iff it is a vertex. The interval is then (s_+, s_-) , given by the absolute slopes of the adjacent edges. A hypothesis is the unique minimizer there iff, in addition, no other hypothesis has the same point.

Lean: RateThreshold.onLowerHull_iff Lean: RateThreshold.lemma_3_1_ii Lean: RateThreshold.lemma_3_1_iii_hyp

Proof of (i), (ii). For (i), $J_c(h) = \langle (c, 1), z_h \rangle$ has the same minimum over Z as over $\hat{C}$. For (ii), adding the two optimality inequalities gives $(c' - c)(\ell(h') - \ell(h)) \leq 0$. Part (iii) is proved in Appendix G.4. $\square$

The Lean proofs of (i) and (iii) cover finite classes.

Reading. The note’s convex-hull argument is correct, with two refinements. The intended model must be a *vertex*, and the only hypothesis at its point. And λ must lie in (s_+N, s_-N) . The note’s alternative, to “set abs bound on model complexity”, yields the same family of selections, parametrized by the Lagrange dual.

Theorem 8.9 (Rate threshold, separable classes; trivial once set up). *[T5 Thm 3.2; T4 Prop 7.1]* Partition X into finitely many regions X_j , and let $\mathcal{H} = \prod_j \mathcal{H}_j$ with $\ell = \sum_j \ell_j$. Then $J_c = \sum_j [c\ell_j + \mathcal{R}_j]$, and the minimizers are exactly the products of per-region minimizers. In particular, let $\mathcal{H}_j = \{\text{default}, \text{component } j\}$, where the component costs $k_j > 0$ extra bits and lowers $\mathcal{R}_j$ by r_j . Then component j is selected iff $r_j/k_j > c$; at equality both choices are optimal.

Lean: RateThreshold.prod_isMin_iff Lean: RateThreshold.rate_threshold

Proof. The objective is a sum of functions of disjoint coordinates. This is the standard hard-thresholding fact for separable penalties. $\square$

Corollary 8.10 (Separating rules from fallacies). *[T4 Cor 7.2]* Split finitely many separable components into valid ones and fallacies. Some $c > 0$ makes the valid set the unique minimizer iff $\min_v \text{valid } r_v/k_v > \max_e \text{fallacy } r_e/k_e$, where $\max \emptyset := 0$.

Lean: RateThreshold.selectable_iff

Proof. If the inequality holds, take c between the two sides. If instead $r_e/k_e \geq r_v/k_v$, any c that strictly selects v has $c < r_e/k_e$, and so also selects e . $\square$

Arbitrary hypotheses. With $\ell = K$ and all hypotheses allowed, a hypothesis might share information across regions, or learn components only partially. For independent parities neither helps, up to explicit slack.

Take regions $X_j = \{j\} \times \{0, 1\}^{d_j}$, $j \leq m$, with $\mu(j, z) = \pi_j 2^{-d_j}$, and teacher $y^*(j, z) = \langle a_j, z \rangle \bmod 2$, with $K(a_1, \dots, a_m) \geq \sum_j d_j$. The rate of component j is π_j/d_j , and $\hat{s}_{h,j}$ is the correlation of h with a_j on region j . The function $\omega(t) = 1 - \log_2(1 + 2^{1-t})$ is the exact per-region risk floor (Lemma G.2 with $p = 0$). It is increasing and concave, with $\omega(2) = 0.415$.

Theorem 8.11 (Rate threshold, idealized). *[T5 Thm 3.3, as repaired]* For all $c > 0$,

$$\left| \Phi(c) - \sum_j \min(c d_j, \pi_j) \right| \leq \Delta_0(c) := c \left(\sum_j (2 \log_2 d_j + 2 \log_2 m + 4 + \kappa) + \kappa \right) + \sum_j \pi_j 2^{1-d_j/4}.$$

Moreover, suppose every $d_j \geq 8$. Let $J_c(h) \leq \Phi(c) + \xi$ and $\Delta = 2\Delta_0(c) + \xi$. Then for each j :

- (a) if $\pi_j < c d_j - \Delta - 4c$, then $\hat{s}_{h,j} < \frac{1}{2}$, so h has not learned component j ;
- (b) if $\pi_j \omega(d_j/4) > c d_j + \Delta$ and $\pi_j \omega(2) > \Delta + 8c$, then $\hat{s}_{h,j} \geq \frac{1}{2}$, so h has learned it.

The two conditions in (b) are implied by $\pi_j > c d_j + \Delta + \pi_j 2^{1-d_j/4}/\ln 2$ and $\pi_j > 2.41(\Delta + 8c)$.

Proof idea. The upper bound is witnessed by a hypothesis that knows a_j when $c d_j < \pi_j$, and flips coins elsewhere.

For the lower bound, on each region we either list-decode a_j from p_h or write it out literally. Joint randomness then gives

$$K(h) \geq \sum_j (d_j - s(t_j))^+ - O\left(\sum_j \log d_j + m \log m\right),$$

where $s(t) = 2t + 2\log_2(t+1)$ and 2^{-t_j} is h 's correlation scale on region j . The risk on region j is at least $\omega(t_j)$. Since ω is concave, each per-region minimization reduces to its endpoints. See Appendix G.5.

The repair. The original clause (b) assumed only $\pi_j > c d_j + \Delta + \pi_j 2^{1-d_j/4}$ and $\pi_j > 2\Delta + 16c$, and was false: $m = 1$, $\pi_1 = 1$, $d_1 = 40$, $c \approx 10^{-8}$ and $h(f_a(z) | z) = 0.745$ give $\hat{s} = 0.49$ although the old hypotheses hold with $\xi := J_c(h)$. The constant $1/\omega(2) \approx 2.41$ is the best this method gives.

Corollary 8.12 (Every convex frontier is realizable, after rescaling). *[T5 Cor 3.4, as repaired] Let $s_1 > \dots > s_m > 0$ be rational and $d_j \geq 1$ integers, with $\sum_j s_j d_j \leq 1$. Set $\pi_j = s_j d_j$, and add a fair-coin region of mass $\pi_0 = 1 - \sum_j \pi_j$. Let P be the convex polygon from $(0, 1)$ whose successive edges have slope $-s_j$ and horizontal length d_j , and which is flat at height π_0 afterwards. Then the lower boundary $\check{p}$ of the closure of $\hat{C}$ satisfies*

$$P(k + C') - \epsilon \leq \check{p}(k) \leq P(k - C') + \epsilon,$$

where $C' = O(\sum_j \log d_j + m \log m + m\kappa)$ and $\epsilon = \sum_j \pi_j 2^{1-d_j/4}$.

The proof is Legendre duality (Appendix G.6). Scaling every d_j up by L and every s_j down by L makes the bands negligible. So every convex, decreasing, piecewise-linear frontier that drops by at most one bit is realized after rescaling. This is the product-model analogue of the “all shapes” theorem of Vereshchagin and Vitányi (2004), with the same moral: whether a good model is a vertex is a property of the data, not a theorem.

The note’s example, verified. Return to the setting of Theorem 8.6. Let $\tau_t = 2t + 2\log_2(t+1) + 2\log_2 d + \kappa$ and $K_{\text{tot}} = d + L_E - \delta_E - 2\log_2 d - \kappa$.

Theorem 8.13 (Parity version of the note’s example). *[T5 Thm 3.5, as repaired] For every $c > 0$ and $t \in \mathbb{N}$,*

$$\Phi(c) \leq \min\{1 + c\kappa, c(d + \kappa') + H(p), c(d + L_E + \kappa \log n)\}$$

and

$$\Phi(c) \geq \min\{1 - \beta_t/\ln 2, c(d - \tau_t) + (L_E - \delta_E - \kappa)/n, c K_{\text{tot}}\}.$$

The terms correspond to the coin, the good model and the bad model, which switch near $c = (1 - H(p))/d$ and $c = H(p)/L_E \approx 1/n$.

Corollary 8.14 (Identification in the parity example). *[T5 Cor 3.5a, added after verification] If $J_c(h) \leq \Phi(c) + \xi$, then $\hat{s}_h(a) \geq 2^{1-c(d+\kappa')-H(p)-\xi} - 1 - 2p$. If moreover $cn > 1$, then $\mathcal{R}(h) \geq H(p) - (\xi + c\Lambda)/(cn - 1)$, where $\Lambda = \log_2(n + 1) + \delta_E + 2\log_2 d + \kappa + \kappa'$.*

Both are proved in Appendix G.7. So for c in the window ($\approx 1/n, \approx (1 - H(p))/d$), every near-optimal hypothesis correlates strongly with f_a and memorizes almost none of the errors. For example, at $d = 20$, $p = 2^{-10}$ and $c = 10^{-4}$, we get $\hat{s} \geq 0.980$ (taking $\kappa = \kappa' = \xi = 0$).

The note's own numbers (computed; T5 c1). These are a 100-bit simple model (plus 10 bits for the flip rate), and a 10 000-bit error on a 2^{-10} fraction of inputs, with $H(2^{-10}) = 0.0112$ bits per sample (about 1/100, as the note says). The rates are $8.99 \cdot 10^{-3}$ for the model and $1.12 \cdot 10^{-6}$ for the error. So the good model is selected iff c lies between these, a window spanning a factor of about 8000. At $N = 10^6$ this means $\lambda \in (1.12, 8990)$. Standard MDL ($\lambda = 1$) switches to the error model at $N \approx 8.9 \cdot 10^5$.

Reading. The error's rate is about $1/n$: a K-random error set costs one bit per $1/n$ of risk, the slowest possible rate. So *the note's example is the most favourable case for the proposal*. Normativity is achieved there because the errors are incompressible relative to the norm.

8.5 Pathologies

Theorem 8.15 (Rate blindness; trivial once set up). *[T5 Thm 4.1, as repaired]*

- (i) $c \mapsto \arg \min J_c$ depends on the data only through (μ, P^*) ; validity enters nowhere.
- (ii) Let v be valid and e an error with $r_v/k_v \leq r_e/k_e$. In the setting of Theorem 8.9, if v is selected then so is e , for every c other than a common tie value. In the setting of Theorem 8.11, the same holds for near-optimal h wherever clauses (a)–(b) decide both v and e ; that is, outside $O(\Delta)$ bands around their rates.
- (iii) h° is the unique minimizer on an open c -interval iff z_{h° is a vertex and no other hypothesis shares it.
- (iv) Take two worlds with the same (μ, P^*) , in which some component is an error in one and a rare valid rule in the other. Every imitation-only procedure gives the same output in both, so it is wrong in one of them.

Lean: RateThreshold.rate_blindness

Proof. Parts (i) and (iv) hold because J_c is a functional of (μ, P^*) . Part (iii) is Lemma 8.8.

For (ii) in the separable case: if v is selected and c is not a tie, then $c < r_v/k_v \leq r_e/k_e$; and if $c = r_v/k_v < r_e/k_e$, then e is strictly selected. The idealized case is proved in Appendix G.8. $\square$

The Lean proof covers (ii) in the separable case only. Part (iv) is Corollary 5.20 again. What is new is the quantitative criterion: the penalty separates errors from rules *by rate, not validity*.

For instance, a 200-bit valid rule used in 10^{-4} of steps, gaining at most 2.2 bits per use, is discarded at every c that rejects the note's 10 000-bit error (the comparison mixes two settings; only the rates are comparable); rarely used irregular verbs likewise regularize first (Lieberman et al., 2007).

Cheap fallacies [T5 Prop 4.2]. In step imitation (Section 8.6), a schema used in a fraction π of steps has rate $\pi g / \ell$. Affirming the consequent (AC: $q, p \rightarrow q / p$) has the size of modus ponens. The freshman’s dream (FD: $(a + b)^2 \rightarrow a^2 + b^2$) is shorter than the correct expansion.

Take a toy with $M = 64$ candidate steps and slip rate $\varepsilon = 0.05$, so that the gain is $g = 5.42$ bits (computed; T5 c4). AC, at frequency 0.02 and 20 bits, then has rate $5.4 \cdot 10^{-3}$. That beats proof by cases ($1.4 \cdot 10^{-3}$), reductio ($1.6 \cdot 10^{-3}$), ex falso ($7.2 \cdot 10^{-4}$) and induction ($1.2 \cdot 10^{-4}$). So no steepness keeps every valid rule while removing every error.

We conjecture that this is no accident (conjecture; empirical). A fallacy becomes systematic because it is a short, natural variant of a valid pattern.

Proposition 8.16 (Rate inversion: the vertex conjecture fails). [T4 Prop 7.3] *In Theorem 8.9, take $k_r = \ell(r)$ and $r_r = \pi_r g_r$, with three rules: a common valid rule A , with $\ell = 10$, $\pi = 0.5$, $g = 8$; a rare valid rule B , with $\ell = 40$, $\pi = 0.01$, $g = 8$; and the freshman’s dream F , with $\ell = 8$, $\pi = 0.05$, $g = 8$. The rates are 0.4, 0.002 and 0.05. As c grows, the selection passes through $\{A, B, F\}$, $\{A, F\}$, $\{A\}$ and $\emptyset$, so no c selects the valid set $\{A, B\}$. In fact, in (bits, residual bits per datum), $\{A, B\} = (50, 0.4)$ is Pareto-dominated by $\{A, F\} = (18, 0.08)$. So no criterion strictly increasing in both coordinates selects it.*

Lean: `RateThreshold.valid_not_minimizer` Lean: `RateThreshold.pareto_dominated`

Proof. Apply Theorem 8.9. The residual of a set R is $\sum_{r \notin R} \pi_r g_r$. □

This answers the note’s conjecture. The hull mechanism is right, but the intended model is a vertex only if every valid component out-rates every error (Corollary 8.10). That holds for rare, expensive, idiosyncratic errors. It fails for cheap systematic fallacies, which are common in student algebra and in Euler-style manipulation.

Theorem 8.17 (The steepness is not identifiable from imitation data). [T5 Thm 4.3, as repaired]

- (i) *Along the population path, $\mathcal{R}(h_c)$ is non-increasing as c decreases. So, for a fixed finite grid in the computable version with a noise floor, choosing c by held-out log-loss converges as $N \rightarrow \infty$ to a grid value of least risk; the smallest grid value always qualifies. At finite N this can fail, through the B/c term of Lemma 8.2.*
- (ii) *For every strictly proper scoring rule, the teacher channel strictly maximizes the expected score at every x (Gneiting and Raftery, 2007). So every validation criterion that is an empirical proper score on fresh imitation data targets P^* , errors included.*
- (iii) *No imitation-data statistic picks a good c in both worlds of Theorem 8.15(iv).*

The proof is in Appendix G.9. In the note’s example, cross-validation picks the error model: held-out log-loss is 1.0 at $c = 10^{-1}$, 0.0112 on $[10^{-5}, 10^{-3}]$, and 0 at $c = 10^{-7}$ (computed). A non-proper “knee” heuristic (the interior vertex with the widest log c -interval) picks the good model in this example. By (iii), though, no such rule is right in both twin worlds.

Reading. Choosing c is a normative act, not an estimate. The note says that “is this a chair?” is “much closer to ‘should the person who taught me the chair notion consider it a chair?’” than to what that person would say. Part (iii) makes literal the claim that this target is not a function of P^* .

Proposition 8.18 (Guard erosion). [T5 Prop 4.4, hypotheses added after verification] *Let $\sigma|_G$ guard a schema σ (e.g. $x/x \rightarrow 1$ if $x \neq 0$). Humans apply σ in G -contexts (frequency π_G , per-step gain*

g). In contexts where σ applies but G fails (frequency π_{-G}), they play a non- σ step. In the σ -applicable region a hypothesis has three options: none: uniform over the M candidates, at cost 0; unguarded: mass $1 - \varepsilon + \varepsilon/M$ on the σ -step, at cost k_σ ; and guarded: the same as unguarded in G -contexts, and uniform on the non- σ steps otherwise, at cost $k_\sigma + k_G$. Define

$$\rho_\sigma = \frac{\pi_G g - \pi_{-G} \log_2(1/\varepsilon)}{k_\sigma}, \quad \rho_G = \frac{\pi_{-G} \Delta_G}{k_G}, \quad \Delta_G = \log_2(M/\varepsilon) - \log_2(M-1).$$

If $\rho_G < \rho_\sigma$, then for $c \in (\rho_G, \rho_\sigma)$ the unguarded, unsound σ is selected, although the teacher never errs. For $c < \rho_G$ the guarded rule is selected, and for $c > \rho_\sigma$ neither is. If $\rho_G \geq \rho_\sigma$, then apart from ties the path goes from guarded to none, and the learner stays sound.

Proof. Relative to none, the three options are the points $(0, 0)$, $(k_\sigma, -k_\sigma \rho_\sigma)$ and $(k_\sigma + k_G, -k_\sigma \rho_\sigma - k_G \rho_G)$ in (bits, risk). The middle point is a vertex iff $\rho_\sigma > \rho_G$. Apply Lemma 8.8 and Theorem 8.9. The rates are derived in Appendix G.10. $\square$

Take $\pi_G = 0.05$, $\pi_{-G} = 5 \cdot 10^{-4}$, $k_\sigma = 22$, $k_G = 12$, $M = 64$ and $\varepsilon = 0.05$ (computed; T5 c4). Then $\rho_G = 1.8 \cdot 10^{-4}$ and $\rho_\sigma = 1.22 \cdot 10^{-2}$, so the unguarded rule is selected on $(1.8 \cdot 10^{-4}, 1.22 \cdot 10^{-2})$, a range that includes steepnesses at which ex falso (rate $7.2 \cdot 10^{-4}$) is kept. With $\pi_G = \pi_{-G} = 0.002$ instead, $\rho_\sigma < \rho_G$ and there is no unsound phase.

Reading. What a careful teacher exhibits most rarely is what goes first: side conditions, eigenvariable conditions, edge cases. One unsound schema can trivialize a classical calculus (Proposition 3.5). With field rules, $0/0 = 1$ and $0/0 = 0 \cdot 0^{-1} = 0$ give $1 = 0$, a negative bag (Lemma 6.2). So coherence must restore what simplicity erodes.

Further pathologies (T5 §4.4). *Error envelopes.* Errors concentrated in a cheaply describable region are learned as an envelope (“flip rate θ on S ”), at the envelope’s own rate, which can far exceed that of the full error. Where the error density there exceeds $\frac{1}{2}$, the learner predicts the error label throughout the region. Real systematic errors are low-complexity patterns of exactly this kind. *Parasitic errors.* If $K(e \mid v) \ll K(e)$, as with FD, a one-symbol mutation of $(ab)^n = a^n b^n$, then bundles enter the hull together. A valid rule may be kept only because of the error it subsidizes. An earlier claim, that conditional rates fix the order of entry, was false and is retracted. *Distribution dependence.* A valid rule that is rare in training stays discarded as $N \rightarrow \infty$. *Language relativity* [T5 §4.4(d6), corrected]: for any separable gains r_j , target set S° with $r_j > 0$ on it, and $c > 0$, some prefix code makes S° the unique selection (give h_S length $m + \sum_{j \in S} k_j$ with integers $k_j < r_j/c$ on S° and $k_j > r_j/c$ off it; the Kraft sum is $2^{-m} \prod_j (1 + 2^{-k_j}) \leq 1$). Between universal machines marginal costs move by at most $2c_{UV}$: negligible for 10 000-bit errors, decisive for 20-bit schemas. So the note’s “simplicity prior defined in terms of existing understanding” cuts both ways: a language shaped by human concepts gives human-natural errors short codes.

8.6 Inference rules: the division of labour

In *situation-typed step imitation*, a context x is a premise set and a goal (Section 2), and y is the next human step among M_τ candidates. Contexts fall into computable situation types τ of frequency π_τ . In type τ the human applies a schema $\sigma_\tau \in \Sigma^P$, genuine or fallacious, with probability $1 - \varepsilon$, and otherwise moves uniformly.

Modelling type τ costs $\ell(\sigma_\tau) + O(1)$ bits and gains $g_\tau = \log_2 M_\tau - H_\tau$ per step. Its rate is therefore $\rho_\tau = \pi_\tau g_\tau / (\ell(\sigma_\tau) + O(1))$, and by Theorem 8.9 it is selected iff $\rho_\tau > c$. The induced verifier accepts the instances of the selected schemas. It is sound (Definition 2.19) iff no error schema is selected and no guard has eroded.

Definition 8.19 (Admissible family, witnesses). [T5 §5.2, as repaired] The components are $V \sqcup F$ (valid, erroneous), with net values $\nu_j = r_j - c k_j$. The *admissible family* $\mathfrak{A} \subseteq 2^{V \sqcup F}$ is the intersection of the filters in use: $\mathcal{A}$ -coherence (Definition 2.13), W -adequacy, and coherence with designated meaning postulates Ax . It is down-closed and contains 2^V . A *witness* for $e \in F$ is a minimal $W \subseteq V$ with $W \cup \{e\} \notin \mathfrak{A}$. It is the negative bag of Lemma 6.2, seen from the rules. Let $V_+ = \{v : \nu_v > 0\}$ and $F_+ = \{e : \nu_e > 0\}$. At a given c we impose two conditions. *Error-independence*: for $U \subseteq V$, $U \cup F' \in \mathfrak{A}$ iff $U \cup \{e\} \in \mathfrak{A}$ for each $e \in F'$. *Valid dominance*: every $e \in F_+$ with a witness inside V_+ has $\beta(e) > \sum_{F_+} \nu$. Here $\beta(e)$ is the least value of $\sum_B \nu$ over sets $B \subseteq V_+$ meeting all of e 's witnesses inside V_+ .

Theorem 8.20 (Division of labour). [T5 Thm 5.2, proof tightened after verification] *Fix c , and assume separability, error-independence and valid dominance at c . Then the optimal admissible sets are exactly the admissible sets $S^*(c) \cup Z$ with $Z \subseteq \{\nu = 0\}$, where*

$$S^*(c) = V_+ \cup \{e \in F_+ : V_+ \cup \{e\} \in \mathfrak{A}\}.$$

So, up to zero-value ties, simplicity removes exactly the errors of rate at most c , coherent or not; coherence, the world and postulates remove exactly the errors that clash with the valid rules actually kept (V_+ , plus Ax), whatever their rate; the residue is the high-rate errors that pass every filter; and valid rules of rate at most c are lost, whatever the other filters do.

Proof idea. An optimal set has no negative-value component. Suppose it contained an error that is inadmissible together with V_+ . Then it would have to omit some $B \subseteq V_+$ meeting that error's witnesses. Valid dominance makes restoring B and dropping the error better, and error-independence keeps the result admissible.

Appendix G.11 gives the full proof, including the zero-value case on which the original last step failed. A brute-force check of 17 806 instances found no violation (computed).

Corollary 8.21 (Best steepness given the other filters). [T5 Cor 5.2a, as repaired] *Assume error-independence and valid dominance at every c considered, and $k_j > 0$. Then every optimal admissible set contains V and no error iff*

$$\max\{\rho_e : V \cup \{e\} \in \mathfrak{A}\} < c < \min_{v \in V} \rho_v, \quad \text{with } \max \emptyset := 0.$$

The proof is in Appendix G.11. Valid dominance is essential. For a counterexample without it, take $V = \{v\}$ with rate 10^{-3} , and an error e of rate $2 \cdot 10^{-3}$, both with $k = 10$, whose witness is $\{v\}$. No error survives relative to V , yet for $c < 10^{-3}$ the optimum is $\{e\}$.

So coherence and the world *widen the window*: the rates of incoherent or refutable errors stop mattering. Simplicity is needed only against coherent, adequate, low-rate errors, such as idiosyncratic false lemmas and quus-like patches (Proposition 6.37).

Proposition 8.22 (Repair by sacrifice). [T5 Prop 5.3, as repaired] *Assume error-independence. Let $e \in F_+$ have a witness inside V_+ , and let $B \subseteq V_+$ be a cheapest set meeting all of e 's witnesses inside V_+ , with $\beta(e) = \sum_B \nu < \nu_e$. Then the clean set $S_{\text{clean}} = V_+ \cup \{e' \in F_+ : V_+ \cup \{e'\} \in \mathfrak{A}\}$ is not optimal. Indeed, $(V_+ \setminus B) \cup \{e\} \cup (S_{\text{clean}} \cap F)$ is admissible, with value $\text{val}(S_{\text{clean}}) + \nu_e - \beta(e)$.*

The proof is in Appendix G.11. Here is a computed instance (computed; T5 c4). AC’s Post witness (Corollary 6.35), with $p := \perp$ and $q := A \rightarrow A$, needs only $\rightarrow I$. If $\rightarrow I$ is rare, with rate $8.7 \cdot 10^{-4}$, then at $c = 5 \cdot 10^{-4}$ the coherent optimum is $\{AC, MP, \top I\}$. *The learner drops $\rightarrow I$ to keep the fallacy.*

This instance assumes $\mathcal{A} = \{\emptyset\}$. With a designated context such as $\{p_0 \rightarrow q_0, q_0, \neg p_0\}$, MP is itself a witness partner, and AC is dropped instead. The phenomenon is Lakatos’s monster-barring (Lakatos, 1976) run backwards: the repair is chosen by description cost, not by validity. It refines Theorem 6.34, whose criterion must be read relative to the rules the learner keeps.

Meaning postulates. The note lists a separate route: “you specify properties of the thing or notion sometimes”. For example, “ $1+1=2$ won’t be true if you accidentally assign 1->rabbit and 2->chicken from a demonstration (for any reasonable meaning of plus)”. Formally, a finite set Ax of sentences endorsed as constitutive of a notion enters as designated premises. These are Carnap’s meaning postulates (Carnap, 1952).

Proposition 8.23 (Postulates are rate-independent). *[T5 Prop 5.4, as repaired]*

- (a) Assume the hypotheses of Theorem 8.20 at every c , with Ax designated, and let $\rho_e > 0$. Up to zero-value ties, $e \notin S^*(c)$ iff $\rho_e \leq c$ or some witness of e lies inside $V_+(c) = \{v : \rho_v > c\}$. Hence e is excluded at every c iff some witness W of e has $\rho_w \geq \rho_e$ for all $w \in W$.
- (b) Let $Ax_{\text{num}} = \{1 + 1 = 2, 1^2 = 1, 2^2 = 4, 4 \neq 2\}$, and keep substitution of equals. Then $(1 + 1)^2 = 1^2 + 1^2$ rewrites to $4 = 2$, so FD is excluded however frequent it is. Without postulates, FD together with the ring rules derives $2 = 0$. That is coherent, since FD holds in characteristic 2.
- (c) The demonstrations (“1”, one rabbit) and (“2”, two chickens) fit both numeral $\mapsto$ cardinality and numeral $\mapsto$ kind. With $+$ read as disjoint union, the postulate $1 + 1 = 2$ holds under the first reading only.
- (d) Postulates eliminate exactly the readings outside $\text{Mod}(Ax)$. For first-order arithmetic, the non-standard models remain (Theorem 6.38).

Part (a) is proved in Appendix G.11; (b)–(d) are direct. Here “rate-independent” means that exclusion needs only that the witness rules be kept. For FD the witness rule is substitution of equals, of rate $4.07 \cdot 10^{-2}$, against $\rho_{FD} = 2.46 \cdot 10^{-3}$ (computed). Random numerical evaluation through the world channel also refutes FD (Schwartz, 1980; Zippel, 1979). Part (c) is Quine’s gavagai (Quine, 1960), with a cure.

Reading: avowal versus performance. Postulates are the teacher’s *avowals* (“ $1+1=2$ ”, “MP is valid, AC is not”, “a chair is for sitting”), not the teacher’s *performance*. The steeper penalty sees only performance. This is where the debate between Stich and Nisbett (1980) and Cohen (1981a) (L11 §2) lands.

Cohen grounds norms in a systematization of ordinary judgments; Stich and Nisbett reply that systematic fallacies, such as the gambler’s fallacy, survive it. Performance data indeed cannot separate a fallacy from a rule with the same profile (Theorem 8.15(iv)). Avowals are a second channel: few bits that exclude whole high-rate classes of errors. Where an avowal is itself wrong, as with naive comprehension, coherence catches it (Section 10).

The note’s abstract/concrete example is a cross-notion postulate: “if something is abstract, then it usually helps a lot to study examples to understand it”. It ties the notion to another

error	rate	simplicity alone removes it at	removed by
idiosyncratic false lemma (300 bits, freq. $5 \cdot 10^{-4}$)	$9 \cdot 10^{-6}$	$c > 9 \cdot 10^{-6}$	simplicity (coherent; outside W)
affirming the consequent	$5.4 \cdot 10^{-3}$	$c > 5.4 \cdot 10^{-3}$ (loses 4 valid rules)	coherence , given $\rightarrow I$ kept
freshman’s dream	$2.5 \cdot 10^{-3}$	$c > 2.5 \cdot 10^{-3}$ (loses 4 valid rules)	postulates or world
quantifier swap	$5.4 \cdot 10^{-4}$	$c > 5.4 \cdot 10^{-4}$ (loses induction)	postulates /designation ($0 \neq 1$)
gambler’s fallacy	$9.0 \cdot 10^{-4}$	$c > 9 \cdot 10^{-4}$ (loses ex falso, induction)	world (frequencies)
conditional perfection	$2.2 \cdot 10^{-3}$	$c > 2.2 \cdot 10^{-3}$ (loses 4 valid rules)	nothing : the residue

Table 11: Which channel removes which error, in the situation-typed toy ($M = 64$, $\varepsilon = 0.05$, illustrative frequencies). With all filters, the window that removes the false lemma and keeps every valid rule is $c \in (9 \cdot 10^{-6}, 1.2 \cdot 10^{-4})$. With simplicity alone, no window works (computed; T5 c4).

notion that has independent evidence, which turns a labelling error into a W-refutation. This makes “should the person who taught me [...] consider it a chair” operational. The norm is whatever fits the notion’s avowed role, checked against the other channels.

Table 11 puts the pieces together. The residue is the alternative meanings of Theorem 6.38 *that humans actually use*, such as “if” read as “iff” in closed worlds. Calling them errors is itself a normative decision outside the data, and it may be wrong: conditional perfection is arguably part of everyday “if”.

8.7 Verdict

Where the proposal is vindicated. Function induction with private randomness has no universal-hypothesis pathology, by exactly the mechanism the note names: “you have to pay the likelihood term separately at every x ” (Theorem 8.3). Shared randomness pays for partial knowledge once, product models gain exponentially little from it, so the product frontier kinks (Theorem 8.6). The convex-hull argument is correct; the change of derivative is the vertex condition, plus uniqueness at the vertex (Lemma 8.8). The note’s example works, asymptotically, on a window of more than three orders of magnitude (Corollary 8.14).

Where it fails. The penalty is validity-blind: it sorts by rate, betting that norms are high-rate structure and errors low-rate structure (Theorem 8.15). The bet is right for idiosyncratic slips and wrong for cheap, frequent fallacies (Proposition 8.16). It costs rare valid rules and, worse, rarely exercised guards, so the learner can be less sound than its teacher (Proposition 8.18). The good c cannot be estimated from imitation data (Theorem 8.17), and private randomness fails whenever inputs carry the same person’s earlier labels (Proposition 8.5).

What does the normativity work. Of the note’s other ideas, “you specify properties of the thing or notion” is decisive: postulates are avowal rather than performance, and they remove high-rate fallacies at every c (Proposition 8.23). The idea “what is it that this person is trying to teach me” calls for a teacher model with a competence/performance split. As we recall their result, [Armstrong and Mindermann \(2018\)](#) show that simplicity alone does not identify such splits. That is Theorem 8.15(iv) in another guise.

The division of labour comes to this: simplicity removes the *expensive* errors; coherence removes the *incoherent* ones; the world removes the *refutable* ones; postulates remove those that *violate avowed properties*. What survives is the meanings humans actually use. Steeper simplicity still has a job no other channel does: removing idiosyncratic, coherent, unrefutable

errors (Table 11, first row). It is a good third filter, not a route to normativity on its own.

In the terms of Section 13, a learner fitted to practice alone is in narrow reflective equilibrium with its teacher, with or without a simplicity penalty. Stich’s objection applies to it exactly. What removes coherent systematic fallacies is information that is not a function of performance: avowed postulates, and an independent world channel. That world channel is what makes the equilibrium wide.

Open problems (Section 14) include kinks beyond parities, a product-model algorithmic statistics, a data-free choice of c , competence/performance models in which joint simplicity identifies the competence, and the best block size for a step imitator. The simulations are in Section 12.

9 Coherence and existence: a “philosophical completeness theorem”

Sections 6 and 7 used coherence as a learning signal. This section asks what coherence *means*: if a learned mode of talking is coherent, is there something it is talking about? The user’s notes pose exactly this question (logic/a ‘philosophical version’ of gödel’s completeness theorem.md):

“a mode of talking makes sense (is coherent) iff there is something it could be talking about”
as long as the ‘mode of talking’ is a first-order theory and ‘something’ is a model of it, then
this is literally the completeness theorem
is a generalization of the completeness thm available that justifies this philosophical proposition
more broadly?

The note suggests further instances (“probabilities describe world iff coherent”, after Christiano et al. 2013; “maybe something could be said about vnm stuff also?”) and records a remark of Sam’s: “completeness is an adjunction between sentences and models . . . if you are in image, then round trip is just identity. what is this image?” Elsewhere the user adds the caveat that “there might only be such a thing in the same sense that there is a ‘proof’ of the Gödel sentence G ” (L10 §1.5).

Answer in brief. There is a whole family of “coherent iff there is something it could be talking about” theorems with one anatomy: completeness holds iff the *points* of a calculus are realized by models (§9.1). Sam’s slogan is strong completeness and the user’s is weak completeness; they differ. Every calculus is complete for models built from its own syntax, so the proposition has content only for an independently specified semantics. Instances: positions (Stone duality, §9.2), credences (de Finetti with *counting sequents*; no bounded-arity, CCS-style constraint family characterizes coherence, §9.3), and contexts (the somethings are belief states, not worlds, §9.4). A coherent learned calculus has a model, but a syntactic one; the Kripkensteinian residue of Section 6 becomes a set of non-isomorphic things talked about; and a compactness and a computability barrier show that coherence cannot in general deliver the *intended* model (§9.5). The verdict is in §9.6.

Credit. Almost everything here is a standard result, reorganized; the organization is the contribution. The Galois connection is Birkhoff (1940) and Ore (1944); bilateral completeness is Scott (1974) and Shoesmith and Smiley (1978); probabilistic coherence is de Finetti (1937) and Gaifman (1964); local-models semantics is Ghidini and Giunchiglia (2001). New as far

as we know, and all elementary: the counting-sequent form of de Finetti’s theorem (probably folklore; cf. Paris 1994), the strictness results for CCS-style constraints (Theorems 9.14 and 9.15), the multi-context completeness theorem in exactly this form (Theorem 9.22), and the corollary on logical inductors (Corollary 9.19). Theorems 9.25 and 9.35 are standard arguments, newly applied. Apart from the semantic half of Theorem 9.10, which is Lemma 6.26(a) and is formalized for the propositional language, nothing in this section is formalized in Lean; sanity checks are in `research/theory/T6-checks/`.

9.1 The adjunction, and the anatomy of a completeness theorem

Definition 9.1 (Semantic frames and completeness). ^[T6 Def 1.1, §1.1] A *semantic frame* $(S, M, \models)$ consists of a set S of “sentences”, a set or class M of “models” and a relation $\models \subseteq M \times S$. Put $\text{Mod}(\Sigma) = \{m : m \models \sigma \ \forall \sigma \in \Sigma\}$ and $\text{Th}(K) = \{\sigma : m \models \sigma \ \forall m \in K\}$. A *deductive system* is a closure operator C on $\mathcal{P}(S)$ (extensive, monotone, idempotent), with closed sets (*theories*) $\text{Fix}(C)$; it is *finitary* if $C(X)$ is the union of the $C(X_0)$ over finite $X_0 \subseteq X$. C is *sound* if $C \leq \text{Th} \circ \text{Mod}$ pointwise, *strongly complete* if $C = \text{Th} \circ \text{Mod}$, and *weakly complete* if every *coherent* X (i.e. $C(X) \neq S$) has $\text{Mod}(X) \neq \emptyset$.

Lemma 9.2 (Galois connection). ^[T6 Fact 1.2] (known) $K \subseteq \text{Mod}(\Sigma)$ iff $\Sigma \subseteq \text{Th}(K)$. Hence $\text{Th} \circ \text{Mod}$ and $\text{Mod} \circ \text{Th}$ are closure operators, and Mod, Th are mutually inverse inclusion-reversing bijections between their fixed points (Birkhoff, 1940; Ore, 1944).

Read as an adjunction between $(\mathcal{P}(S), \subseteq)$ and $(\mathcal{P}(M), \supseteq)$, this is $\text{Mod} \dashv \text{Th}$: the precise content of “completeness is an adjunction between sentences and models”. The adjunction exists for *every* frame; a completeness theorem is a statement about its fixed points.

Proposition 9.3 (Sam’s image). ^[T6 Prop 1.3] For every closure operator C : $\{\text{Th}(K) : K \subseteq M\} = \text{Fix}(C)$ iff C is strongly complete.

Proof. The image of Th is $\text{Fix}(\text{Th} \circ \text{Mod})$, as $\text{ThModTh} = \text{Th}$; and a closure operator is determined by its fixed points, $C(X) = \bigcap \{T \in \text{Fix}(C) : X \subseteq T\}$. $\square$

Trivial once set up (and soundness is not needed): the round trip fixes exactly the sets closed under the rules iff the rules are complete.

Theorem 9.4 (Tautological completeness). ^[T6 Thm 1.4] Every closure operator C is strongly complete for its canonical frame $(S, \text{Fix}(C), \ni)$, whose models are the theories T , with $T \models \sigma$ iff $\sigma \in T$.

Proof. $\text{Th}(\text{Mod}(X)) = \bigcap \{T \in \text{Fix}(C) : X \subseteq T\} = C(X)$. $\square$

Reading. If “something it could be talking about” may be *any* object, the user’s proposition is trivially true: every mode of talking is about its own theories. Real completeness theorems have content because their models (truth tables, structures, points of $\bar{k}^n$, measures) are specified *before and independently* of the calculus. For a learner of inference rules this is a warning in advance: that a learned calculus is “about something” is cheap; what matters is whether it is about something we can access independently.

Theorem 9.5 (Completeness is realization of points). *[T6 Thm 1.5, as repaired]* Call a theory T a point if $T \neq S$ and, for some $\sigma \notin T$, T is maximal among theories omitting σ ; write $\text{Pt}(C)$. Let C be finitary. (a) Lindenbaum. Every theory is the intersection of the points containing it. (b) If C is sound, it is strongly complete iff every point is the theory of a model: $\text{Pt}(C) \subseteq \{\text{Th}(\{m\}) : m \in M\}$. (c) If C is sound, some finite F has $C(F) = S$, and $\text{Th}(\{m\}) \neq S$ for all m , then C is weakly complete iff every maximal proper theory is the theory of a model. Without soundness, $(\Rightarrow)$ fails.

Proof idea. (a) Zorn, using finitariness; (b) realize each point in the decomposition (a), and conversely use complete meet-irreducibility. Full proof and the counterexample for (c) in Appendix H.1. So every completeness proof has two steps: (i) points exist, which is Zorn and free; (ii) points are realized by models of the independently given kind, which is where content lives (Henkin witnesses, Zariski’s lemma, Carathéodory extension). Categorical logic has the same slogan: Deligne’s theorem that coherent toposes have enough points is equivalent to completeness for coherent logic (Makkai and Reyes, 1977).

Proposition 9.6 (Weak $\neq$ strong). *[T6 Prop 1.6]* IPC is sound and weakly complete, but not strongly complete, for the Boolean valuations BV.

Proof. $p \vee \neg p$ holds in all of BV but is not IPC-provable. For weak completeness use Theorem 9.5(c) with $F = \{\perp\}$: a maximal IPC-consistent T contains φ or $\neg\varphi$ for each φ (deduction theorem), so it contains excluded middle, so it is a maximal classical theory. Alternatively, use Glivenko’s theorem (Glivenko, 1929). $\square$

Lemma 9.7 (Classical points are maximal). *[T6 reading of Prop 1.6, as repaired]* Every point of Cn_2 is a maximal consistent theory. (Proof by cases; Appendix H.1.)

Reading. Weak completeness needs the *maximal* theories to be realized, strong completeness *all* points; they coincide when every point is maximal, as in classical logic. IPC has non-maximal points, realized only by Heyting or Kripke models. This mirrors Theorem 6.17: intuitionistic and classical logic have the same coherence data, every coherent intuitionistic position *is* about a classical world, and yet those worlds do not determine intuitionistic consequence. Having somethings for every coherent position does not mean the somethings fix what follows from what.

Table 12 lists the family. The Nullstellensatz row is Sam’s other pointer: the weak Nullstellensatz (“ $1 = 0$ is not derivable from $f_1 = \dots = f_m = 0$ by ideal manipulations iff the f_i have a common zero in $\bar{k}^n$ ”) is weak completeness; the strong one, $I(V(J)) = \sqrt{J}$, is strong completeness, so Sam’s image is the radical ideals; the points are the maximal ideals, realized by Zariski’s lemma. The rows typically carry a *non-standard caveat*, existence only in a closure of the naive model class: $x^2 + 1 = 0$ is coherent over $\mathbb{Q}$ with no rational zero; $\{x \geq n : n \in \mathbb{N}\}$ has no finite refutation and no real solution; $2x = 1$ is LP-coherent with no integer solution. Cauchy’s (1847) treatment of complex numbers as real polynomials modulo $x^2 + 1$ is the anatomy at work: a coherent way of talking about $\sqrt{-1}$, with the Lindenbaum quotient as its object.

9.2 Positions: bilateral completeness is Stone duality

Fix any set Fm ; until Theorem 9.11 connectives play no role. A sequent $\Gamma \triangleright \Delta$ has finite $\Gamma, \Delta \subseteq \text{Fm}$; a *valuation* is any $v : \text{Fm} \rightarrow \{0, 1\}$, and $v \models \Gamma \triangleright \Delta$ iff v falsifies some $\gamma \in \Gamma$ or verifies some $\delta \in \Delta$. A *position* $[X : Y]$ asserts all of X and denies all of Y ; it is *incoherent* for a set $\vdash$ of

mode of talking (calculus)	something	realization of points	theorem
finite sequents (Ov, Wk, Cut) first-order sentences	valuations $\text{Fm} \rightarrow 2$ structures	trivial Henkin witnesses	Scott (1974); §9.2 Gödel (1930); Henkin (1949)
polynomial equations over k (ideal, radical)	points of $\bar{k}^n$	Zariski's lemma	Hilbert (1893); Zariski (1947)
finite linear inequalities	points of $\mathbb{R}^n$	LP duality	Farkas (1902)
real polynomial (in)equalities	points of $\mathbb{R}^n$	real closure	Stengle (1974)
credences, finite agenda (counting sequents)	mixtures of valuations	separating hyperplane	de Finetti (1937); §9.3
first-order credences (Gaifman coherence)	measures on complete theories	Carathéodory	Gaifman (1964); §9.3
finite preference data	utility functions	Motzkin	von Neumann and Morgenstern (1944); §9.3
labelled claims (local rules, bridges)	belief-state families	canonical chain	§9.4

Table 12: One anatomy, many completeness theorems [T6 §1.5]. Typically coherence buys existence only in a *closure* of the naive model class (see text).

sequents if some $\Gamma \triangleright \Delta \in \vdash$ has $\Gamma \subseteq X$, $\Delta \subseteq Y$ (Restall 2005; Definition 6.25). A *Scott relation* is a set of sequents closed under (Ov) $\Gamma \triangleright \Delta$ if $\Gamma \cap \Delta \neq \emptyset$; (Wk) from $\Gamma \triangleright \Delta$ infer $\Gamma \cup \Gamma' \triangleright \Delta \cup \Delta'$; (Cut) from $\Gamma, \varphi \triangleright \Delta$ and $\Gamma \triangleright \varphi, \Delta$ infer $\Gamma \triangleright \Delta$. $\langle S \rangle$ is the least Scott relation containing S , and $\text{Val}(\vdash)$ the set of valuations satisfying $\vdash$.

Theorem 9.8 (Bilateral Lindenbaum). [T6 Thm 2.2] (known) *Let $\vdash$ be a Scott relation. (a) Every coherent position $[X : Y]$ is realized by some $v \in \text{Val}(\vdash)$: $v[X] = 1$, $v[Y] = 0$. (b) The maximal coherent positions are exactly the $[v^{-1}(1) : v^{-1}(0)]$ with $v \in \text{Val}(\vdash)$.*

Corollary 9.9 (Strong bilateral completeness). [T6 Cor 2.3] *For every set S of sequents, the sequents valid in every valuation that satisfies S are exactly those in $\langle S \rangle$: $\text{Th}(\text{Mod}(S)) = \langle S \rangle$.*

Proof idea (Scott, 1974; Shoesmith and Smiley, 1978). Zorn gives a maximal coherent extension, since sequents are finite; (Ov) makes it disjoint, and (Cut) makes it exhaustive: if φ can be added to neither side, weaken the two witnessing sequents to a common context and cut on φ . Full proofs in Appendix H.2.

Theorem 9.10 (The duality). [T6 Thm 2.4] (known) *Give 2^{Fm} the product topology. The Galois connection between sets of sequents and sets of valuations has syntactic closure $S \mapsto \langle S \rangle$ and semantic closure $V \mapsto \bar{V}$, the topological closure. Hence $\vdash \mapsto \text{Val}(\vdash)$ is a dual lattice isomorphism from Scott relations onto closed subsets of 2^{Fm} , with inverse Th . Lean: Carnap.Val_mrel (semantic half, propositional Fm)*

The syntactic half is Corollary 9.9, the semantic half is Lemma 6.26(a): each sequent excludes a basic clopen set. This is Stone duality (Stone, 1936) for the free Boolean algebra B on Fm , every formula an independent generator: $\Gamma \triangleright \Delta$ is the clause $\bigvee_{\Gamma} \neg \gamma \vee \bigvee_{\Delta} \delta$ of B , and Scott relations, closed sets of valuations and filters of B correspond. So *at this level the connectives play no role in the duality*; they enter only through *which* closed set the rules carve out. This is

Suszko’s thesis seen as a duality: a Tarskian consequence is determined by the characteristic functions of its theories, which are bivaluations that need not be truth-functional (Suszko 1977 stated it for structural logics).

Theorem 9.11 (Structural calculi and Lindenbaum bundles). *[T6 Thm 2.6] Let Fm be a term algebra and call $\vdash$ structural if it is closed under substitutions σ . (a) A Scott relation $\vdash$ is structural iff $\text{Val}(\vdash)$ is closed under $v \mapsto v \circ \sigma$. (b) If $\vdash$ is structural, each matrix $L_v = \langle \text{Fm}, v^{-1}(1) \rangle$, $v \in \text{Val}(\vdash)$, validates $\vdash$ under every assignment; $\vdash$ is exactly the set of sequents valid in all L_v ; and every coherent position is realized in some L_v by the identity assignment.*

The proof (Appendix H.2) rests on $v \circ \sigma \models \Gamma \triangleright \Delta$ iff $v \models \sigma\Gamma \triangleright \sigma\Delta$.

Reading. Theorem 9.8 is the cleanest form of the user’s proposition: a position is coherent iff it partially describes an admissible valuation, “some way the sentences could all come out”. Theorem 9.11 upgrades the something to a compositional model whose algebra is the language itself, the multiple-conclusion analogue of Łoś and Suszko (1958) and Wójcicki (1988). Whether the *intended* semantics is reached depends on the rules: with the truth-table sequents and one coherence datum, $\text{Val}(\vdash) = \text{BV}$ (Theorem 6.29); with single-conclusion data only, the admissible valuations include the “gappy” $\text{BV}^\cap$ (Theorem 6.27), so a single-conclusion coherence loss licenses talk about undecided worlds.

9.3 Credences: de Finetti, counting sequents, and why local constraints fail

Let F be a finite agenda that is a union of *complementary pairs* $\{\varphi, \varphi^*\}$, $\varphi^* \equiv \neg\varphi$ (a finite agenda cannot be closed under $\neg$); BV the Boolean valuations of its atoms; $\bar{v} = (v(\varphi))_{\varphi \in F}$. A credence $P : F \rightarrow [0, 1]$ is *negation-coherent* if $P(\varphi^*) = 1 - P(\varphi)$, and *coherent* if $P = \sum_v \mu(v) \bar{v}$ for a probability μ on BV , i.e. P lies in the *coherence polytope* $\Pi_F = \text{conv}\{\bar{v}\}$. A *counting sequent* (Φ, m) is a finite multiset Φ of members of F with an integer *threshold* m ; it is *valid* if every $v \in \text{BV}$ makes at least m members of Φ true (with multiplicity), and P *satisfies* it if $\sum_{\varphi \in \Phi} P(\varphi) \geq m$. An ordinary sequent $\Gamma \triangleright \Delta$ is the case $m = 1$, $\Phi = \{\gamma^* : \gamma \in \Gamma\} \cup \Delta$.

Theorem 9.12 (de Finetti, with counting sequents). *[T6 Thm 3.1, as repaired] For negation-coherent P the following are equivalent: (i) P is coherent; (ii) there is no Dutch book, i.e. no $\lambda \in \mathbb{R}^F$ with $\lambda \cdot (\bar{v} - P) < 0$ for all $v \in \text{BV}$; (iii) for rational λ, c , if $\lambda \cdot \bar{v} \geq c$ for all v then $\lambda \cdot P \geq c$; (iv) P satisfies every valid counting sequent. Negation coherence is needed: $P \equiv 1$ satisfies (iv) but is incoherent.*

Proof idea. Π_F is a rational polytope, so a point outside it violates one of finitely many rational defining inequalities, which gives a Dutch book; clearing denominators and moving negative coefficients from φ to φ^* turns the inequality into a counting sequent. Appendix H.3.

Reading. The probabilistic completeness theorem says a credence is coherent iff it is a mixture of worlds. Its proof-theoretic side is not ordinary sequents but *counting sequents*, graded positions such as “at least two of these six claims hold”. An ordinary sequent gives only the union bound $\sum_\Gamma (1 - P(\gamma)) + \sum_\Delta P(\delta) \geq 1$, the multiple-conclusion form of Adams’s uncertainty-sum bound (Adams, 1966).

Which consistency constraints suffice? The user’s notes on consistency-based probing (ai/DLK/) impose negation coherence $P(\neg Q) = 1 - P(Q)$, the identity $P(A \wedge B) + P(A \vee B) = P(A) + P(B)$, and modus ponens read as the tautology $\neg P \vee \neg(P \rightarrow Q) \vee Q$, i.e. the union bound $P(Q) \geq P(P) + P(P \rightarrow Q) - 1$. Contrast-consistent search (CCS) trains a probe with exactly a negation-consistency loss plus a confidence loss (Burns et al., 2023).

Example 9.13 (Small agendas). [T6 Prop 3.2, Ex 3.3](computed) On $\{p, q, p \wedge q\}$ plus complements, $P(p) = P(q) = 0.2$, $P(p \wedge q) = 0.9$ is negation-coherent but violates the union bound of $p \wedge q \triangleright p$. For $(P(p), P(q), P(p \rightarrow q)) = (a, b, c)$ the coherence polytope is the tetrahedron with facets $a + c \geq 1$ (from $\triangleright p, p \rightarrow q$), $b \geq a + c - 1$ (modus ponens), $b \leq c$ (from $q \triangleright p \rightarrow q$) and the trivial $c \leq 1$: on this tiny agenda the user’s MP constraint, two other union bounds and a trivial bound are coherence.

This does not scale. For $k \geq 3$ let $F_k = \{A_i, \neg A_i : i \leq k\} \cup \{A_i \wedge A_j, \neg(A_i \wedge A_j) : i < j \leq k\}$, and $P_k(A_i) = \frac{1}{k-1}$, $P_k(A_i \wedge A_j) = 0$, with complements on negations.

Theorem 9.14 (No bounded-arity family of constraints suffices). [T6 Thm 3.4] (a) *For every $F' \subseteq F_k$ whose formulas jointly mention at most $k-1$ atoms, $P_k|_{F'}$ extends to a coherent credence on F_k .* (b) *P_k is incoherent. Hence no family of necessary coherence constraints each depending only on sub-agendas over fewer than k atoms characterizes coherence; in particular pairwise (CCS-style) constraints fail, and so does every bounded arity.*

Proof. (a) With J the mentioned atoms, put mass $\frac{1}{k-1}$ on each world making exactly one A_j , $j \in J$, true, and the rest, $1 - \frac{|J|}{k-1} \geq 0$, on the all-false world. (b) A representing μ would make the A_i pairwise disjoint, so $\mu(\bigvee_i A_i) = \frac{k}{k-1} > 1$. $\square$

The same holds if F_k is enlarged to all formulas in at most two atoms, valued by the two-atom marginals of the μ in (a), which do not depend on J : P_k is “pairwise coherent” in the strongest sense, every pair of atoms having a genuine joint distribution, and still globally incoherent. This is the frustration in Specker’s parable of the three boxes (Specker, 1960).

Theorem 9.15 (The counting-sequent hierarchy by threshold is strict). [T6 Thm 3.5, as repaired] (a) *P_k violates the valid counting sequent $\Phi = \{\neg A_1, \dots, \neg A_k\} \cup \{A_i \wedge A_j : i < j\}$, $m = k-1$, which repeats no formula.* (b) *P_k satisfies every valid counting sequent over F_k with $m \leq k-2$, with arbitrary repetitions; in particular ($m = 1$) the union bound of every valid ordinary multiple-conclusion sequent.*

Proof. (a) In a world with t true atoms, $(k-t) + \binom{t}{2} \geq k-1$ members of Φ hold, as $(t-1)(t-2) \geq 0$; but $\sum_{\Phi} P_k = \frac{k(k-2)}{k-1} < k-1$. (b) uses only the all-false and singleton worlds plus integrality (Appendix H.3). An exhaustive enumeration for $k = 3$ (503,270 valid ordinary sequents, none violated) and an exact MILP for $k = 3, 4, 5$ agree. $\square$

Scope. For any *single* finite agenda the finitely many rational inequalities defining Π_F (its facets, plus its affine hull when it is not full-dimensional) give finitely many counting sequents that characterize coherence; what Theorems 9.14 and 9.15 rule out is a bound on arity or threshold that works *uniformly across agendas* (F_k needs arity k and threshold $k-1$). Moreover, deciding coherence on arbitrary agendas is NP-hard ($P(\varphi) = 1$, $P(\varphi^*) = 0$ is coherent iff φ is satisfiable), so no efficiently checkable constraint family characterizes it unless $P = NP$. Closing the agenda changes the picture:

Proposition 9.16 (On logically closed agendas, local additivity suffices). *[T6 Prop 3.6] If F represents a finite Boolean subalgebra of the Lindenbaum algebra and P respects equivalence, then P is coherent iff $P \geq 0$, $P(\top) = 1$ and $P(\varphi) = P(\varphi \wedge \psi) + P(\varphi \wedge \neg\psi)$ for all $\varphi, \psi \in F$; each constraint mentions at most three formulas, and the user’s identity follows from them.*

A caution for consistency-based probing. A CCS-style probe reads credences off a model on an *unclosed* list of statements and enforces negation consistency, perhaps with further pairwise or local constraints. By Example 9.13 and Theorems 9.14 and 9.15, no fixed stock of constraint *types* (not pairwise, not k -wise for fixed k , not even all union bounds of ordinary sequents) guarantees that the probe’s credences are a mixture of worlds, i.e. that they are “about” anything at all. Global coherence needs the agenda’s facets, or an agenda closed under the connectives (Proposition 9.16). And even global coherence buys only *some* distribution over *some* worlds (Theorem 9.17), not the actual one: coherence is eliminative, not confirmational (Section 6.7). A natural experiment, which we have not run, is to train probes with negation-only, pairwise and counting-sequent losses on agendas with k -way exclusive alternatives and measure their distance from the coherence polytope.

First-order credences. For a countable first-order language L , $P : \text{Sent}_L \rightarrow [0, 1]$ is (*Gaifman*-)coherent if (G1) $P(\varphi) = 1$ whenever $\vdash \varphi$, and (G2) $P(\varphi \vee \psi) = P(\varphi) + P(\psi)$ whenever $\vdash \neg(\varphi \wedge \psi)$ (Gaifman, 1964); this is the notion of the draft cited by the user (Christiano et al., 2013).

Theorem 9.17 (Coherent credences are mixtures of complete theories). *[T6 Thm 3.7] (known) Let S_L be the Stone space of complete consistent L -theories, with clopens $[\varphi] = \{T : \varphi \in T\}$. Then P is coherent iff $P(\varphi) = \mu([\varphi])$ for a (unique) Borel probability μ on S_L .*

Proof idea. P is a finitely additive probability on the clopen algebra; by compactness this is automatically countably additive there, and Carathéodory extends it (Appendix H.3). By Gödel completeness, a coherent credence is the expected truth value of φ in a random model: the user’s proposition holds verbatim. So does the caveat: for complete $T \supseteq \text{PA} + \neg\text{Con}(\text{PA})$, δ_T is coherent, gives 1 to every true quantifier-free sentence, and “believes” there is a proof of $0 = 1$.

Theorem 9.18 (The Gaifman condition is a probabilistic ω -rule). *[T6 Thm 3.8, as repaired] In the language of arithmetic, let P be coherent, give probability 1 to every true quantifier-free sentence, and satisfy the Gaifman condition $P(\exists x \varphi(x)) = \sup_n P(\bigvee_{i \leq n} \varphi(i))$. Then $P = \delta_{\text{Th}(\mathbb{N})}$. No arithmetic axioms are needed.*

Proof idea. Induction on logical symbols: coherence forces $\{0, 1\}$ -values through $\neg, \wedge$; numerals carry no logical symbols, and every number is a numeral’s value (Appendix H.3). This is essentially a consequence of Gaifman (1964): a probability with the Gaifman property is determined by its quantifier-free values, here the true diagram of $\mathbb{N}$.

Corollary 9.19 (Logical inductors’ limits are non-standard). *[T6 Cor 3.9, as repaired] Let $\mathbb{P}$ be a logical inductor (Garrabrant et al., 2016) over a consistent r.e. $\Gamma \supseteq \mathbb{Q}$ in the language of arithmetic. Then $\mathbb{P}_\infty$ violates the Gaifman condition. More generally, no limit-computable coherent credence giving 1 to every true quantifier-free sentence satisfies it; the quantifier-free hypothesis cannot be dropped.*

Proof idea. By Limit Coherence, $\mathbb{P}_\infty$ is coherent and gives 1 to true quantifier-free sentences, so Theorem 9.18 would make it $\delta_{\text{Th}(\mathbb{N})}$. Non-Dogmatism refutes this (the false member of a Gödel–Rosser pair gets positive probability), and so, independently, does Tarski’s theorem, as $\text{Th}(\mathbb{N})$ would be Δ_2 . Appendix H.3.

Reading. The probabilistic “something” is always available; a *standard* something is fixed by an infinitary condition that no limit-computable coherent credence of this kind meets. A Gaifman-style probe of a language model’s arithmetic would compare $P(\exists x \varphi(x))$ with $\max_{i \leq n} P(\varphi(i))$: a large gap marks belief in a witness that is never named.

Proposition 9.20 (Finite preference data). *[T6 Prop 3.10] Finitely many strict comparisons $p_i \succ q_i$ and weak comparisons $r_j \succsim s_j$ of lotteries over a finite set X have an expected-utility representation $u \in \mathbb{R}^X$ ($u \cdot (p_i - q_i) > 0$, $u \cdot (r_j - s_j) \geq 0$) iff no $\lambda, \kappa \geq 0$ with $\lambda \neq 0$ satisfy $\sum_i \lambda_i (p_i - q_i) + \sum_j \kappa_j (r_j - s_j) = 0$.*

Proof. Motzkin’s transposition theorem (Motzkin, 1936). □

This answers “vnm stuff”: the certificate is a derivation of $L \succ L$ by mixing (independence) and transitivity (von Neumann and Morgenstern, 1944), money-pump-like (an interpretive gloss; no trading is formalized), and the something is a real-valued utility.

9.4 Contexts: a completeness theorem for “true in the context at hand”

What is a coherent *system of contexts* (Section 11) about? We use a monotone multi-context calculus whose labelled formulas $i:\varphi$ are McCarthy’s $\text{ist}(i, \varphi)$ (McCarthy, 1993), with a simplified local-models semantics (Ghidini and Giunchiglia, 2001).

Definition 9.21 (Multi-context systems, MC, belief states). *[T6 Def 4.0, as repaired; §4.1] For each context $i \in I$ fix a language L_i , local models M_i with $C_i = \text{Th}_i \circ \text{Mod}_i$ (e.g. classical logic; for a learned logic C_i , the proper theories $M_i = \text{Fix}(C_i) \setminus \{L_i\}$ with $T \models_i \varphi$ iff $\varphi \in T$), local axioms K_i , and a falsum $\perp_i$ with $\text{Mod}_i(\perp_i) = \emptyset$. A *bridge rule* is $j_1:\varphi_1, \dots, j_n:\varphi_n \Rightarrow i:\psi$. For labelled premises Γ , $\text{Der}(\Gamma) = (T_i)_i$ is the least family with $T_i \supseteq K_i \cup \Gamma_i$, each T_i C_i -closed, and closed under bridges; $\Gamma \vdash_{\text{MC}} i:\varphi$ iff $\varphi \in T_i$. A *belief state* is $c = (c_i)$, $c_i \subseteq M_i$, with $c \models i:\varphi$ iff every $m \in c_i$ satisfies φ . It is a *model* if it satisfies every K_i and is *bridge-compatible* (if $c \models j_k:\varphi_k$ for all premises then $c \models i:\psi$); $\Gamma \models_{\text{LMS}} i:\varphi$ iff every model of Γ satisfies $i:\varphi$.*

Theorem 9.22 (Soundness and completeness; the canonical chain). *[T6 Thm 4.1] Let $c_i^\Gamma = \text{Mod}_i(T_i)$ for $(T_i) = \text{Der}(\Gamma)$. (a) c^Γ is a model of Γ . (b) Every model c of Γ has $c_i \subseteq c_i^\Gamma$: c^Γ is the least committal model. (c) $\Gamma \vdash_{\text{MC}} i:\varphi$ iff $\Gamma \models_{\text{LMS}} i:\varphi$.*

Proof idea. $c^\Gamma \models i:\varphi$ iff $\varphi \in \text{Th}_i \text{Mod}_i(T_i) = T_i$; and the theories $\text{Th}_i(c_i)$ of any model form a closed, bridge-closed family containing $K \cup \Gamma$, hence contain $\text{Der}(\Gamma)$. Appendix H.4.

Corollary 9.23 (Context coherence iff existence). *[T6 Cor 4.2] Let $D \subseteq I$ be the designated contexts (Definition 2.13). $\Gamma \not\vdash_{\text{MC}} i:\perp_i$ for every $i \in D$ iff some model c of Γ has $c_i \neq \emptyset$ for every $i \in D$.*

Proof. $\perp_i \in T_i$ iff $c_i^\Gamma = \emptyset$ by Theorem 9.22(a), and c^Γ is the largest model by (b). □

Remark 9.24 (Learned local logics). *[T6 remark after Cor 4.2, added after verification] With the semantics $M_i = \text{Fix}(C_i) \setminus \{L_i\}$, $c_i^\Gamma \neq \emptyset$ iff $T_i \neq L_i$, so for learned local logics designated coherence must be read as *non-triviality* of T_i ; this coincides with $\perp_i \notin T_i$ exactly when $\perp_i$ is C_i -explosive.*

Non-designated contexts, such as suppositions under reductio, may have $c_i = \emptyset$: they are about nothing, which is what a successful reductio shows. This is the designation discipline of Section 6 as an existence theorem. The canonical chain is also the least equilibrium in the sense

of [Brewka and Eiter \(2007\)](#); with non-monotone (default) bridges equilibria can fail to exist even when every local knowledge base K_i is consistent, so coherence alone does not guarantee existence there (Proposition [H.2](#), Appendix [H.4](#)).

Rules of proof versus conditionals between worlds. A world family $w = (w_i)$ picks a single model $w_i \in M_i$ of K_i for each i and satisfies each bridge as a *material conditional*: if $w_{j_k} \models \varphi_k$ for all k then $w_i \models \psi$. Write $\models_W$ for consequence over world families.

Theorem 9.25 (Belief states, not worlds). *[T6 Thm 4.5, as repaired] (a) $\models_{\text{LMS}} \subseteq \models_W$. (b) The inclusion is strict: with classical local logics, $K_j = \{p \vee q\}$, $K_i = \{\neg r\}$ and bridges $j:p \Rightarrow i:r$, $j:q \Rightarrow i:r$, MC does not derive $i:\perp$, but there is no world family. (c) For classical propositional local logics, $\models_W$ is classical consequence in the disjoint union of the local languages from the tagged axioms $K_i^{(i)}$, the tagged premises and the bridge axioms $\bigwedge_k \varphi_k^{(j_k)} \rightarrow \psi^{(i)}$ (the “eternalist” translation).*

Proof. (a) A world family is a model chain of singletons. (b) $T_j = \text{Cn}_2(p \vee q)$ contains neither p nor q , so no bridge fires and $c_i^\Gamma = \text{Mod}(\neg r) \neq \emptyset$; but any w_j satisfies p or q , forcing $w_i \models r$. (c) World families are the Boolean valuations of the tagged language satisfying the tagged axioms. $\square$

This is the context-logic instance of a standard distinction, rule of proof versus conditional (global versus local consequence): $c \models j:\varphi$ is a box over c_j , and (b) is the failure of $\Box(p \vee q) \models \Box p \vee \Box q$; it is why [Ghidini and Giunchiglia \(2001\)](#) use *sets* of local models. New is only its use as a criterion for physics bridges. `mcs_completeness.py` confirms Theorem [9.22\(c\)](#) on 400 random three-context systems and finds 2,348 pairs where $\models_W$ strictly exceeds $\models_{\text{LMS}}$.

Reading. MC reads a bridge as “export what context j establishes”; the eternalist translation reads it as “whatever world j is about, if φ holds there, ψ holds in i ’s world”. Only the first is right for idealizations: a physics bridge (“if the idealized model establishes $Q = q$, then in the world $Q \approx q \pm \varepsilon$ ”, Section [11.4](#)) is certified for what the idealization establishes, not for case splits over its undetermined possibilities, none of which is actual. So *a context system makes sense iff there is something it could be about* holds with belief states as the somethings (Corollary [9.23](#)) and fails with worlds (Theorem [9.25\(b\)](#)).

Local coherence versus a global world. The user worries that frames “needn’t be easily reconcilable” (L10 §1.3). Even under the world reading this has a precise form.

Theorem 9.26 (Local-to-global on trees). *[T6 Thm 4.6] Let the contexts be the vertices of a finite tree, context i classical propositional over atoms A_i with a consistent closed theory T_i . Assume running intersection ($\{i : a \in A_i\}$ is connected for each atom a) and mutual conservativity on edges ($T_i \cap L(A_i \cap A_j) = T_j \cap L(A_i \cap A_j)$). Then every model of every T_i extends to a valuation satisfying all T_j .*

This is the join-tree, or iterated Robinson-consistency, argument (cf. [Beeri et al. 1983](#)): conservativity makes model sets agree on overlaps, and models glue breadth-first (Appendix [H.4](#)).

Proposition 9.27 (The frustrated triangle). *[T6 Prop 4.7] On atoms $\{a, b\}, \{b, c\}, \{c, a\}$ let $T_{ab} = \text{Cn}_2(a \leftrightarrow \neg b)$, $T_{bc} = \text{Cn}_2(b \leftrightarrow \neg c)$, $T_{ca} = \text{Cn}_2(c \leftrightarrow \neg a)$. Each is consistent, each pair is mutually conservative on its shared atom, and the union is inconsistent; likewise pair marginals uniform on $\{01, 10\}$ agree on all single marginals but have no joint distribution.*

Proof. An odd cycle of negations has no Boolean solution; each theory has only tautologies in a single atom. $\square$

This is Specker’s parable in logical form (Specker, 1960) and the simplest instance of *contextuality*: local sections agreeing on overlaps with no global section (Abramsky and Brandenburger, 2011). Acyclicity of the cover is what rules it out (Beeri et al., 1983). So an olympiad solution’s contexts need not describe one world, nor be reconcilable into one. What coherence of the designated contexts guarantees is local belief states, nonempty where designated (Corollary 9.23; cf. Remark 11.14); one world for all contexts is strictly stronger and cover-dependent.

Definition 9.28 (True in the context at hand). [T6 Def 4.8, as repaired] φ is *true in context* c iff $\Gamma \vdash_{MC} c:\varphi$, i.e. (Theorem 9.22) φ holds in every local model that c ’s canonical belief state leaves open. φ is *true enough in* c for a world claim ε iff (i) φ is true in c ; (ii) a bridge $c:\varphi \Rightarrow @:\varepsilon$, possibly with side conditions σ , is in the system and σ holds in the actual world (for a checker, $\Gamma \vdash_{MC} @:\sigma$); and (iii) that bridge is *sound*: whenever c establishes φ and σ holds in the actual world, so does ε .

This answers the user’s “maybe you just check that a claim is true in the context at hand. but wtf is that???” (**verification from truth.md**): truth in the context at hand is supervaluational truth over the context’s admissible local models, and a checker accepting exactly MC-derivations accepts exactly what is true in this sense (compare Theorem 11.39). It does not justify bridges: bridge soundness is world-relative and is certified by stability arguments or world feedback (Section 11.4), never by coherence, which can only demand $c_i^\Gamma \neq \emptyset$ at designated contexts. The context-tree calculus of the literature memo L7 is an MC system, and its own semantics agrees with ours when every step and export instance is sound (proof sketch; Appendix H.4).

9.5 What coherence of a learned calculus buys, and what pins the model down

Let R be a learned set of rule schemas, C_R the least structural finitary consequence operation containing it, and $\langle R \rangle$ the least structural Scott relation. The designated contexts $A \in \mathcal{A}$ are certified coherent: $\perp \notin C_R(A)$, or $[A : D]$ is $\langle R \rangle$ -coherent. As in Section 6, this is all the learner knows.

Theorem 9.29 (What coherence buys). [T6 Thm 5.1] For each designated A : (a) some C_R -theory $T \supseteq A$ is maximal among theories omitting $\perp$; (b) the Lindenbaum matrix $\langle \text{Fm}, T \rangle$ validates every rule of C_R under every assignment, and the identity assignment designates all of A and not $\perp$ (Łoś and Suszko, 1958; Wójcicki, 1988); (c) bilaterally, each designated coherent $[A : D]$ is realized by some $v \in \text{Val}\langle R \rangle$, and L_v is a matrix model of $\langle R \rangle$.

Proof. (a) is the Zorn step in the proof of Theorem 9.5(a), applied to the closed set $C_R(A)$ and $\sigma = \perp$ (Appendix H.1). (b) If $\varphi \in C_R(\Gamma)$ and $\sigma\Gamma \subseteq T$, structurality gives $\sigma\varphi \in C_R(\sigma\Gamma) \subseteq C_R(T) = T$. (c) is Theorems 9.8 and 9.11. $\square$

(The proof of (b) works for any C_R -theory $T \supseteq A$ omitting $\perp$.) So if a learned mode of talking is coherent on a context, there is something it is talking about there. But the model’s universe is the language: the formal content of the user’s worry about the “proof” of the Gödel sentence. When is the syntactic something the intended one? For a matrix $\langle \mathbf{A}, D \rangle$ the *Leibniz congruence* $\Omega_{\mathbf{A}}(D)$ is the largest congruence compatible with D ($a \theta b$ and $a \in D$ imply $b \in D$),

and the reduction $\langle \mathbf{A}/\Omega, D/\Omega \rangle$ identifies elements no formula context can tell apart (Blok and Pigozzi, 1989; Font, 2016); write $\Omega(T)$ for $\Omega_{\mathbf{Fm}}(T)$.

Theorem 9.30 (The syntactic model collapses to the intended one). ^[T6 Thm 5.2] (a) If $h : \mathbf{Fm} \rightarrow \mathbf{A}$ is a surjective homomorphism and $T = h^{-1}(D)$, then

$$\langle \mathbf{Fm}/\Omega(T), T/\Omega(T) \rangle \cong \langle \mathbf{A}/\Omega_{\mathbf{A}}(D), D/\Omega_{\mathbf{A}}(D) \rangle;$$

if $\langle \mathbf{A}, D \rangle$ is reduced, the reduced Lindenbaum matrix of T is $\langle \mathbf{A}, D \rangle$. (b) Every maximal consistent theory of CPC has reduced Lindenbaum matrix $\cong \langle \mathbf{2}, \{1\} \rangle$.

Proof idea. $\Omega(T) = h^{-1}(\Omega_{\mathbf{A}}D)$, because a congruence compatible with T , joined with $\ker h$, pushes forward along h to one compatible with D (Appendix H.5). Surjectivity matters: in general one gets the reduction of the *generated* submatrix.

Proposition 9.31 (What the points of a finite-matrix logic talk about). ^[T6 Prop 5.3, as repaired] Let C be the consequence of a finite matrix $\langle \mathbf{A}, D \rangle$. Every point of C has reduced Lindenbaum matrix isomorphic to the reduction of a generated submatrix $\langle h(\mathbf{Fm}), D \cap h(\mathbf{Fm}) \rangle$ (a strict homomorphic image of a submatrix, not necessarily a submatrix); up to isomorphism there are finitely many.

Proof. C is sound and strongly complete for the homomorphisms $h : \mathbf{Fm} \rightarrow \mathbf{A}$, and finitary by compactness of $\mathbf{A}^{\text{Var}}$ (the standard argument: each condition $h(\gamma) \in D$ is clopen there, as $\mathbf{A}$ is finite). So every point is some $h^{-1}(D)$ by Theorem 9.5(b); apply Theorem 9.30(a) to h onto its image. $\square$

So for classical logic everything a coherent position can be about is, after identifying indiscernibles, the two truth values, because every point of CPC is maximal (Lemma 9.7) and maximal theories are Boolean. The reason is *not* Post-completeness: IPC is not Post-complete, yet its maximal consistent theories are classical (Proposition 9.6) and reduce to $\mathbf{2}$. What can differ between calculi is the non-maximal points.

Example 9.32 (The residue as a set of things talked about). ^[T6 Ex 5.4] (i) Map $\mathbf{Fm}$ onto the three-element Heyting chain $\mathbf{H}_3 = \{0 < \frac{1}{2} < 1\}$ with $h(p) = \frac{1}{2}$. Then $T = h^{-1}(1)$ is a non-maximal point of IPC (maximal among theories omitting p) whose reduced matrix is $\langle \mathbf{H}_3, \{1\} \rangle$: one coherent intuitionistic position talks about the three-element chain while the maximal ones talk about $\mathbf{2}$. (ii) First-order logic with equality plus the fallacy $\forall x \exists y R \triangleright \exists y \forall x R$ is coherent (Example 6.36); its models are exactly the one-element structures, which is what the learned fallacy is “about”.

In general, for multiple-conclusion hypotheses, Theorems 9.10 and 9.11 put the residue $\text{Alt}(h^*, \mathcal{A})$ of coherent uniform alternatives (Theorem 6.38) in exact correspondence with the closed, substitution-invariant $V \subseteq \text{Val}(h^*)$ realizing every designated position. *Non-identifiability of meaning is non-uniqueness of the thing talked about.*

First-order term models. For a consistent first-order theory, the Lindenbaum construction is Henkin’s (Henkin, 1949): witnesses, a complete Henkin theory, closed terms modulo provable equality (the Leibniz reduction). The user’s DLK note says that “any assignment of truth-values to sentences which does not violate inference rules can be extended to a full model (this is cool because a model also needs to assign actual functions and objects ...)”. This is correct for complete consistent assignments: the objects are classes of witness terms. If the assignment makes $\exists x \varphi(x)$ true and every named instance false, the model needs unnamed objects, the Gaifman failure of Corollary 9.19.

Theorem 9.33 (Prime models). *[T6 Thm 5.5](known) For complete consistent $T \supseteq \text{PA}$ the definable elements of any model of T form the prime model $\mathcal{K}(T)$, unique up to isomorphism and elementarily embeddable in every model of T (Kaye, 1991); $\mathcal{K}(\text{Th}(\mathbb{N})) = \mathbb{N}$; if $\neg\text{Con}(\text{PA}) \in T$, $\mathcal{K}(T)$ is non-standard.*

For the last clause: “the least code of a PA-proof of $0 = 1$ ” is definable, while T refutes each standard instance $\text{Prf}(\underline{n}, \ulcorner 0=1 \urcorner)$, whose negation is a true Δ_0 sentence (PA being consistent) and so provable in $\text{Q} \subseteq T$. This is the precise form of the user’s footnote: a complete coherent theory has a canonical thing it talks about, the objects it can define, and the “proof” is one of them, a definable non-standard number.

Two layers of non-standardness. (i) *Coherent but false:* $\text{PA} + \neg\text{Con}(\text{PA})$ is consistent, false, even Π_1 -sound (L4 Prop 3.1); this is finitary and comes from incompleteness (coherence is not truth, Theorem 6.22(b)). (ii) *True but not categorical:* $\text{Th}(\mathbb{N})$ itself has non-standard models, a limit phenomenon from compactness. Pinning comes in levels: P0 coherent (a model exists); P1 complete (all models elementarily equivalent; for logics, Post-complete, Theorem 6.13); P2 κ -categorical; P3 categorical (unique model up to isomorphism; for calculi, all *points* reduce to one matrix, Proposition 9.31).

Theorem 9.34 (Compactness barrier). *[T6 Thm 5.6](known) A first-order theory with an infinite model has models of every infinite cardinality $\geq |L|$ (Löwenheim–Skolem–Tarski), so no first-order mode of talking reaches P3 for an infinite structure.*

Every finite structure in a finite language, by contrast, is characterized up to isomorphism by one sentence. In the user’s icosahedron example (`math is a mere string game iff everything is.md`), after the physical icosahedron explodes the mode of talking stays coherent and its object is still pinned up to isomorphism; the caveat bites only for infinite objects.

Theorem 9.35 (Computability barrier). *[T6 Thm 5.7, as repaired] Let S be a decidable set of finite syntactic objects and K an intended class of models. If some calculus has an r.e. set of incoherent finite sets, and a finite $X \subseteq S$ is incoherent iff it has no model in K , then $\{X : X \text{ has a } K\text{-model}\}$ is co-r.e. (Π_1). Consequently no such calculus exists for (a) arithmetic with $K = \{\mathbb{N}\}$ ($\text{Th}(\mathbb{N})$ is not arithmetical); (b) Diophantine equations with $K = \{\mathbb{Z}\}$ (solvability is Σ_1 -complete, Matiyasevich 1970); (c) first-order sentences in a vocabulary with a binary relation symbol, with K the finite structures (finite satisfiability is Σ_1 and undecidable, Trakhtenbrot 1950; for monadic vocabularies it is decidable). (d) For K all structures, satisfiability is Π_1 , as Gödel’s theorem requires; for polynomial equations over $\bar{k}$ it is decidable.*

Proof. “No K -model” is “incoherent”, which is r.e. In (b), (c) the set is also Σ_1 , hence would be decidable; in (a) $\text{Th}(\mathbb{N})$ would be Π_1 , contradicting Tarski’s undefinability theorem (e.g. Boolos et al. 2007). $\square$

This is the textbook argument “a complete r.e. calculus makes semantic consequence r.e.”; what is new is the framing. *For any computable notion of coherence, “coherent iff there is something it could be talking about” can hold only if having such a something is a Π_1 property.* “Having some model” is Π_1 ; “having the standard model”, “having an integer solution” and “having a finite model” (in a vocabulary with a binary relation symbol) are not. The proposition is true exactly to the extent that “something” is construed liberally.

Proposition 9.36 (Anchoring pins exactly the definable vocabulary). *[T6 Prop 5.8, reading as repaired]*
 Let $L_o \subseteq L$ be the anchored (world-hooked) vocabulary and T an L -theory. Any two models of T with the same universe and the same interpretation of L_o coincide iff every symbol of L is explicitly T -definable from L_o (Beth, 1953).

Reading. Beth’s theorem quantifies over all models of T , i.e. over every possible anchoring structure: anchoring determines the rest of the language *uniformly, whatever the anchoring structure* iff the rest is definable. Over the *actual* structure pinning can hold without definability: if T says that R is a nonempty initial segment of $<$ without a largest element, then over $(\mathbb{N}, <)$ the only expansion is $R = \mathbb{N}$, while over $\omega + \omega$ both $R = \omega$ and $R = \omega + \omega$ work. Undefinable theoretical terms are multiply realizable over *some* anchoring, though perhaps not over the actual one (the Ramsey–Lewis situation, Lewis 1970).

How are the natural numbers pinned down? The user asks this, adding “is there a way they are the ‘lowest-complexity-structure’ which satisfy some axioms??”. Coherence and first-order completeness cannot do it (Theorems 9.34 and 9.35). The known pins each add something: *second-order induction* (Dedekind, 1888) adds quantification over all subsets, and no complete r.e. calculus exists for its consequence, as Theorem 9.35(a) predicts; internal categoricity (Parsons, 1990; McGee, 1997; Button and Walsh, 2018) pins the structure relative to the available language, not the theory. *Initiality* ($\mathbb{N}$ is the initial $(0, S)$ -algebra, the least Herbrand model of $\{N(0), N(x) \rightarrow N(Sx)\}$, van Emden and Kowalski 1976, and an initial segment of every model of PA) adds a universal property, “nothing else is a number”, which is not first-order. The ω -rule, or Gaifman condition, adds an infinitary rule that no limit-computable credence of the relevant kind obeys (Corollary 9.19). World anchoring by computation adds nothing beyond coherence with Q (Lemma 6.19). And:

Theorem 9.37 (Tennenbaum). *[T6 §5.5] (known)* In no countable non-standard model of PA is addition computable, and in none is multiplication computable (Tennenbaum, 1959; Kaye, 1991). Hence $\mathbb{N}$ is the unique computable model of PA up to isomorphism.

This is a theorem-level “yes” to the “lowest-complexity structure” question, with two caveats: “computable” is defined using the meta-language’s natural numbers, as the user himself notes (logic/confusions/soundness.md); and the axioms matter, since weaker theories such as open induction have computable non-standard models (Shepherdson, 1964). Every pin uses something beyond finitary computable coherence: second-order quantification, a universal property, a complexity bound measured in $\mathbb{N}$, or an infinitary rule. Theorems 9.34 and 9.35 say this is forced.

9.6 Verdict

Where yes. In each row of Table 12, “coherent iff there is something it could be talking about” is a theorem, with a realization step that is non-trivial except for positions, where points already are valuations; for the modes of talking that matter to the learner, see Theorems 9.8, 9.12, 9.17 and 9.29, Proposition 9.20, and Corollary 9.23. Sam’s slogan holds in each strongly complete case (Proposition 9.3 and Theorem 9.10).

Where no. (1) The something may be made of syntax (Theorems 9.4 and 9.29); the proposition is substantive only for an independently specified semantics, and which semantics is right is

answered by which models we can independently access: the truth values, $\mathbb{N}$ through computation, the world through observation. (2) Weak $\neq$ strong (Proposition 9.6). (3) Intended existence needs more than coherence: layer (i) is defeated only by truth-tracking principles (reflection, stronger theories, Popperian policies, Theorem 6.22), layer (ii) only by infinitary, second-order or meta-level ingredients, and both limits are forced (Theorems 9.34 and 9.35). (4) The something for a context system is a family of belief states, not of worlds (Theorem 9.25), and a global world exists automatically only on tree-like covers with conservative overlaps (Theorem 9.26 and Proposition 9.27). (5) Probabilistic coherence is global: local constraints on an unclosed agenda need not make a probe about any mixture of worlds (Theorems 9.14 and 9.15).

What this means for the learner. The user’s split between coherence and “hooking onto the world” becomes two precise questions. Coherence answers “is there *any* world this way of talking could be about?”: yes, canonically, but partly syntactically. Hooking answers “is it *this* world?”: world feedback is positive data in valuation space (Section 6.4), and uniformly over anchorings it pins exactly the definable vocabulary (Proposition 9.36). Neither does the other’s job. This is the division of labour of Sections 6, 7 and 11 seen semantically: coherence removes what is incoherent with the target, its residue Alt is a set of non-isomorphic somethings, and only world channels or extra pins select among them. The user’s “C-model without an L-model” (`logical models as distinct from mental models.md`) becomes an incoherent calculus used only through coherent chunks: local belief states (Corollary 9.23) without a global section (Proposition 9.27). And the computability barrier is a formal reason why there is no final formula for $\mathbb{N}$: pins such as reflection or stronger theories must be added on other grounds, matching the open-ended picture of justification in Section 13.

Verdict. The philosophical proposition is *true in a precise generalized sense*: for positions, credences, preferences, context systems and learned structural calculi, coherence is equivalent to the existence of a model of the kind the semantics allows. It is *not vindicated in the intended sense*: standard or intended existence needs categoricity, anchoring or an infinitary condition, and no computable coherence notion can certify it beyond Π_1 .

10 Informal mathematics: learning a latent formalization

This section is the paper’s answer to the second success criterion: a setup that *would have worked for mathematical proofs before anyone knew how to formalize them*. The user asks it in two sharper forms. “How come mathematicians ended up with such a nice notion of proof? ... this nice structure of a formal proof is sorta there in their activities. how did it come to be there?” And, as a hypothesis: pre-formal mathematicians used only the part of their concepts that later survived formalization, because “the ‘rest of the concept’ ... was untrustworthy/messy ... and for this reason unusable whenever people were proving things” (the *robust-core hypothesis*).

The model: a pre-formal practice is the shadow of an unknown *latent formalization* $h^* = (L^*, R^*, \rho^*, \mathbb{M}^*)$, consisting of a formal language, a calculus with a library of definitions and lemmas, a partial, context-dependent *reading* of informal sentence occurrences into the language, and an intended semantics. A human step is the ρ^* -image of an R^* -derivation with at most g rule applications (an operational de Bruijn factor), plus noise. The learner sees accepted steps (P), paradoxes derived from designated contexts (C), and objects (O): examples and counterexamples on which informal sentences can be evaluated. We show that a conservative verifier which checks

steps *and* re-uses of sentences across contexts against every surviving latent formalization is sound against adaptive provers (Section 10.2); that closure-level data identify the latent formalization up to sameness of informal consequence on the practice’s domain and no finer, that bounded-gap data identify its short steps but only bracket its meaning, and that ontology, and the latent language beyond granularity, are unidentifiable (Section 10.3); that counterexample objects can be exponentially more informative than paradoxes, whose price grows linearly with the number of suspect steps (Section 10.4); that Frege $\rightarrow$ Russell $\rightarrow$ Zermelo is a theorem about simplicity, coherence and coverage in a comprehension toy (Section 10.5); and that a robust core of steps valid under every admissible sharpening survives every formalization, while selection by objects plus Gödel completeness explains why surviving steps are derivable, though not why they are short (Section 10.6).

Two caveats govern the section. Every positive result assumes *realizability*, that the true latent formalization lies in the learner’s fixed class, whereas the decisive historical steps (ε - δ , uniform convergence, ideals) were inventions of language; *affordable soundness with language invention* is the hardest open problem we leave (Section 10.7). And most results are trivial once set up; their value lies in definitions under which history becomes theorems. The genuine mathematical content is the paradox side of the object–paradox separation (Theorem 10.23 and Proposition 10.25; its novelty relative to the Littlestone-dimension literature is uncertain) and the comprehension-toy facts (Lemma 10.31); the deep ingredients are cited (Gödel completeness, Robinson’s transfer, Nelson’s conservativity, Incurvati and Murzi’s theorem).

10.1 The model

Definition 10.1 (Occurrences, steps, arguments, links). ^[T4 §1.1] Ξ is a countable set of informal sentence strings and C a set of *contexts* (definitions, conventions, notation and local suppositions in force). An *occurrence* is $x = (c, \xi) \in X := C \times \Xi$, with $\text{str}(x) := \xi$. An *informal step* is $\Gamma \triangleright y$ with $\Gamma \subseteq_{\text{fin}} X$, $y \in X$. An *informal argument* α is a list of lines $y_1, \dots, y_m \in X$, each a *premise*, an *inference* $\Gamma_j \triangleright y_j$ with Γ_j among earlier lines, or a *link* $y_i \rightsquigarrow y_j$ with $i < j$ and $\text{str}(y_i) = \text{str}(y_j)$ (“the same claim, used in another context”). $\text{Prem}(\alpha)$ is its set of premises, y_m its conclusion, and its depth the length of its longest dependency path.

Links are the formal locus of *equivocation*: a sentence proved in one context is re-used in another, where it may be read differently.

Definition 10.2 (Latent formalization). ^[T4 Def 1.1, §1.2] A *latent formalization* is a quadruple $h = (L_h, R_h, \rho_h, \mathbb{M}_h)$ consisting of a countable language L_h with a sentence $\perp$; a set R_h of formal steps $\Pi \triangleright \varphi$ over $\text{Sent}(L_h)$ (instances of finitely many schemas plus a library of definitions and derived lemmas); a partial *reading* $\rho_h : X \rightarrow \text{Sent}(L_h)$ with domain D_h ; and a class $\mathbb{M}_h$ of L_h -structures for which R_h is sound and $\perp$ is false.

- (i) $\Gamma \triangleright y$ is *valid under* h , written $(\Gamma \triangleright y) \in \models_h$ or $\Gamma \models_h y$, iff $\Gamma \cup \{y\} \subseteq D_h$ and $\rho_h(y) \in \text{Cl}_{R_h}(\rho_h \Gamma)$.
- (ii) It is *g-valid*, $(\Gamma \triangleright y) \in \text{St}_h^g$, iff moreover $\rho_h(y)$ has an R_h -derivation from $\rho_h \Gamma$ with at most g rule applications and formulas of size bounded by a fixed function of g and the input. We assume ρ_h computable with decidable domain, R_h decidable, and derivations restricted to the input’s symbols plus a finite, computably given part of L_h depending on g ; then St_h^g is decidable and $\text{St}_h^g \subseteq \models_h$, which is all that is used.

- (iii) A link $y_i \rightsquigarrow y_j$ is *valid under h* iff $(\{y_i\} \triangleright y_j) \in \text{St}_h^g$ (typically $\rho_h(y_i) = \rho_h(y_j)$: zero applications).
- (iv) The *admissible valuations* are $\mathfrak{A}h := \{v_M \circ \rho_h : M \in \mathbb{M}_h\}$, with $v_M(\varphi) = [M \models \varphi]$.
- (v) h is *first-order complete* if L_h is first-order, $\mathbb{M}_h = \text{Mod}(T_h)$, and $\text{Cl}_{R_h}(\Gamma) = \{\varphi : T_h \cup \Gamma \models \varphi\}$.

De Bruijn’s ratio of formal to informal proof size is roughly constant (intrinsically about 4, Wiedijk 2000) only relative to a definitional library, as Mathias’s 4.5×10^{12} -symbol term for Bourbaki’s “1” shows (Mathias, 2002); so g is measured over R_h with its library.

Assumption 10.3 ((BG), bounded gap). ^[T4 §1.3] There is a target $h^* = (L^*, R^*, \rho^*, \mathbb{M}^*)$, with $D^* := D_{h^*}$, such that (BG1) $h^* \in \mathcal{H}$ (*realizability*); (BG2) each accepted human step is, with probability $1 - \eta$, drawn from a distribution over $\text{St}_{h^*}^g$, and otherwise noise, sporadic or systematic (instances of a fallacy schema); (BG3) human links are h^* -valid up to the same noise.

Steps that are valid but not g -short are routed out of (P): they are escalated with a request to expand them (Proposition 10.11), or count as noise.

Definition 10.4 (The three channels). ^[T4 §1.4] (P) *Practice*: the stream of (BG2); escalation answers are practice data labelled by $\text{St}_{h^*}^g$. (C) *Coherence*: a family $\mathcal{A}$ of finite *designated contexts* $A \subseteq X$ with the promise $\perp \notin \text{Cl}_{R^*}(\rho^* A)$; a *paradox for h* is an R_h -derivation of $\perp$ from $\rho_h A$, $A \in \mathcal{A}$, and refutes h . Suppositions are not designated, so reductio is never penalized. (O) *Objects*: a finite partial valuation $v : X \rightarrow \{0, 1\}$ with the promise $v \subseteq u$ for some $u \in \mathfrak{A}h^*$; it refutes h iff $v|_{D_h}$ extends to no member of $\mathfrak{A}h$. Interactively, the learner holds an object and queries its values (“checks the step on an example”). *Computation* is the special case of an intended object $M_0 \in \mathbb{M}^*$ queried on a decidable fragment, like Euler’s six-decimal value of $\sum 1/n^2$. *Convention*: $\rho_h A := \rho_h(A \cap D_h)$, so a sentence h cannot read asserts nothing under h .

(O) is the world channel (W) of Definition 2.13 inside mathematics: objects (W2), with computation (W1, W3) as a special case. The model fixes ρ^* (Section 10.6 makes it vague instead) and the class of languages.

10.2 Sound bounded-gap verification

At time t , h is *consistent with the data* if it labels every practice step and escalation answer correctly by St_h^g , agrees with every link answer, and has not been refuted by a paradox or an object datum found within the search budget so far. VS_t is the set of such h . Every refutation is a certificate, so budgeted search never refutes an unrefutable hypothesis.

Definition 10.5 (Bounded-gap conservative verifier). ^[T4 Def 2.1] On a query q (an informal step or a link), V_t^g answers **ACCEPT** iff q is g -valid, resp. link-valid, under *every* $h \in \text{VS}_t$, with a certificate for each; otherwise **ESCALATE** (ask the oracle, or ask the prover to expand q).

V_t^g is computable for finite $\mathcal{H}$; for infinite $\mathcal{H}$ unanimity is a Π_1 condition, and the effective version is the Bayesian verifier below, truncated to finitely many hypotheses. Any budgeted proof search that counts “not found” as “not valid” only shrinks the accepted set and so preserves the soundness results below (T4 Prop 2.6), though not the escalation bounds.

Lemma 10.6 (Argument soundness). ^[T4 Lemma 2.2, repaired] If every line of α lies in D_h and every inference and link of α (a link as the step $\{y_i\} \triangleright y_j$) lies in $\models_h$, then $\rho_h(y_j) \in \text{Cl}_{R_h}(\rho_h \text{Prem}(\alpha))$ for every line y_j . Hence every line is true in every $M \in \mathbb{M}_h$ that makes the premises true.

Proof. Induction on j : $\rho_h y_j \in \text{Cl}_{R_h}(\rho_h \Gamma_j) \subseteq \text{Cl}_{R_h}(\text{Cl}_{R_h}(\rho_h \text{Prem})) = \text{Cl}_{R_h}(\rho_h \text{Prem})$ by monotonicity and idempotence (Lemma 2.3); links are the case $\Gamma_j = \{y_i\}$. Then use soundness of R_h . $\square$

Proposition 10.7 (Equivocation: step-checking alone is unsound; trivial). *[T4 Prop 2.3] There are h and α such that every inference of α is h -valid and every link joins equal strings, yet $\text{Prem}(\alpha) \triangleright y_m$ is h -invalid.*

Proof (Cauchy’s sum theorem). Let ρ_h read “ $\sum f_n$ converges” as uniform convergence in context c_1 (Cauchy’s proof) and as pointwise convergence in c_2 (the statement). Lines: (1), (2) premises (c_2 , “each f_n is continuous”), (c_2 , “ $\sum f_n$ converges”); (3), (4) links of (2), (1) into c_1 ; (5) the valid inference to (c_1 , “the sum is continuous”); (6) a link of (5) back into c_2 . Links (4) and (6) are valid; link (3), pointwise to uniform, is not; and $\{1, 2\} \triangleright 6$ fails on Abel’s series $\sum_n (-1)^{n+1} \sin(nx)/n$, whose sum is $x/2$ on $(-\pi, \pi)$, extended periodically. $\square$

So the verifier checks links. Lakatos’s “hidden lemma” in Cauchy’s proof (Lakatos, 1976, App. 1) is, here, an invalid link.

Theorem 10.8 (Deterministic soundness; trivial once set up). *[T4 Thm 2.4] Assume (BG1) and noise-free truthful data: practice and escalation answers lie in and are labelled by $\text{St}_{h^*}^g$, link answers agree with h^* , designated contexts are h^* -coherent, object data are h^* -admissible. Then for every prover (randomized, adaptive, unbounded, knowing h^* , the verifier and the history) and every t , everything V_t^g accepts lies in $\text{St}_{h^*}^g \subseteq \models_{h^*}$; so every argument assembled from accepted items has an h^* -valid conclusion (unused premises outside D^* dropped), true in every $M \in \mathbb{M}^*$ satisfying its premises.*

Proof. No datum can refute h^* , so $h^* \in \text{VS}_t$, and unanimity implies g -validity under h^* ; then apply Lemma 10.6. $\square$

There is *no union bound over queries* (as in Section 4): one structural fact, $h^* \in \text{VS}_t$, covers every attempt. Incoherent survivors only shrink the intersection: Frege’s inconsistent 1903 “way out” (Quine, 1955) would have made a conservative verifier timid, never unsound. With noisy practice h^* can be refuted, so we use a posterior; the threshold then involves the prior mass of the target’s observational-equivalence class $[h^*]$ (Definition 10.12).

Theorem 10.9 (Time-uniform soundness with a class prior). *[T4 Thm 2.5; T1 Thm 4.2] Let $\mathcal{H}$ be countable with prior w and assume (BG1). Each h has a practice likelihood p_h (including noise); practice data are i.i.d. from p_{h^*} , each independent of everything before it (the prover does not foresee them); likelihoods are shadow-measurable ($p_h = p_{h'}$ whenever $h \equiv_g h'$); all other data (escalation and link answers, paradoxes, objects) are truthful and multiply the weight of h by 1 if consistent with h , by 0 otherwise. Let $V_t^{g,\delta}$ accept q iff the posterior gives $\{h : q \notin \text{St}_h^g\}$ mass below δ , and put $W^* := w([h^*])$. If $\delta \leq W^* \delta'$, there is an event of probability at least $1 - \delta'$, depending only on the practice data, on which no prover ever gets a query outside $\text{St}_{h^*}^g$ accepted.*

Proof idea. With $Z_t := \sum_h w(h) L_t(h) / L_t(h^*)$, shadow-measurability gives $w_t([h^*]) = W^* / Z_t$; constraints only shrink Z_t , which is dominated by a practice-only nonnegative supermartingale with initial value 1. Ville’s inequality (Ville, 1939) keeps it below $1/\delta'$ forever with probability $\geq 1 - \delta'$, and then $w_t([h^*]) > \delta$. This is the prior-posterior-ratio martingale of Waudby-Smith and Ramdas (2020). Full proof in Appendix I. $\square$

W^* counts the $\equiv_g$ -class (renamings; the ontological variants of Proposition 10.17), not the meaning class: at gap g , soundness is paid for at the level of g -shadows, between formalizations and meanings. Its price is escalations. Let $\mathcal{H}^g := \{\text{St}_h^g : h \in \mathcal{H}\}$ and let $\text{el}(\mathcal{H}^g, \text{St}_{h^*}^g)$ be the positive elasticity of Section 4: the supremum of lengths of sequences $s_1, s_2, \dots \in \text{St}_{h^*}^g$ with each s_i outside the intersection of the members of $\mathcal{H}^g$ containing $s_1, \dots, s_{i-1}$.

Theorem 10.10 (Escalation costs). *[T4 Thm 2.7, repaired; T1 §§3–4] Let the prover be honest and count only escalations answered by the oracle (expansion requests yield no label). (a) With noise-free data V_t^g escalates at most $\text{el}(\mathcal{H}^g, \text{St}_{h^*}^g) \leq |\mathcal{H}/\equiv_g| - 1$ times. (b) In the setting of Theorem 10.9, with probability $\geq 1 - \delta''$, $V_t^{g,\delta}$ escalates at most $(\ln(1/W^*) + \ln(1/\delta''))/\delta$ times in all. (c) If the n hypotheses' g -step relations are $B \cup (U \setminus \{u\})$, $u \in U$, then (a) equals $n - 1$ and is attained, and with the uniform prior every δ' -sound verifier fails to accept (escalates or rejects), in expectation, at least $(1 - \delta')(n - 1)$ queries of some honest prover, so (b) with $\delta = W^*\delta'$ is tight up to a factor $O((\ln(1/W^*) + \ln(1/\delta''))/\delta')$.*

Proof in Appendix I, by reduction to the escalation theorems of Section 4. Expansion closes the gap. Call h^* *step-expressive* if every $(\Gamma \triangleright y) \in \models_{h^*}$ is the conclusion of an informal argument from Γ whose inferences lie in $\text{St}_{h^*}^1$ (one rule application each) and whose links join occurrences with equal ρ^* -readings.

Proposition 10.11 (Expansion makes the verifier complete; trivial). *[T4 Prop 2.8] If h^* is step-expressive and $\bigcap_{h \in \text{VS}_t} \text{St}_h^1 \supseteq \text{St}_{h^*}^1$, then every h^* -valid step has an argument all of whose steps and links V_t^1 accepts.*

Proof. Each inference of the expansion lies in $\text{St}_{h^*}^1$, and each link is in $\text{St}_{h^*}^1$ with zero applications; all lie in every survivor's St_h^1 . $\square$

Under complete presentation, the hypothesis on VS_t is what Theorem 10.15(d) delivers at $g = 1$.

Reading. The architecture the user sketches, a checker that asks the arguer to “explain/justify/derive” what it does not recognize, is *sound at all times* and *complete via expansion* once the atomic level is learned. Learning which informal inferences are valid becomes learning a set of latent formalizations and accepting what all of them validate; no measure over steps enters the guarantee. For contexts, Proposition 10.7 says that a sentence true in one context is not thereby usable in another: the re-use is an inference to be checked, the zero-length case of the export rules of Section 11.

10.3 What practice can identify

Definition 10.12 (Equivalences). *[T4 Def 3.1] $h \approx h'$ iff $\models_h = \models_{h'}$ (this fixes $D_h = \{x : \{x\} \models_h x\}$); $h \approx_D h'$ iff they agree on steps with all occurrences in D . $h \equiv_g h'$ iff $\text{St}_h^g = \text{St}_{h'}^g$, $\mathfrak{A}h = \mathfrak{A}h'$, and h, h' are coherent on the same $A \in \mathcal{A}$; $\equiv_\infty$ is the same with $\models$ for St^g .*

Proposition 10.13 (No learner sees past the shadow; trivial). *[T4 Prop 3.2, repaired] Suppose the law of the practice data is a function of $\text{St}_{h^*}^g$ and the history, escalation and link answers are $\text{St}_{h^*}^g$ -labels, the law of the object data (which occurrences are shown or queried, and their values) is a function of $\mathfrak{A}h^*$ and the history, $\mathcal{A}$ does not depend on the target, and queries come from the learner or from provers that depend on the target only through its shadow. If $h \equiv_g h'$, every data*

sequence has the same probability under target h as under h' , so no learner distinguishes them. The closure-level version replaces St^g by $\models$ and $\equiv_g$ by $\equiv_\infty$.

Proof in Appendix I. Which shadow determines which? Call D_h *negation-closed* if an informal operation $x \mapsto \sim x$ (“it is not the case that ...”) preserves D_h with $T_h \vdash \rho_h(\sim x) \leftrightarrow \neg \rho_h(x)$, and call $v : D_h \rightarrow \{0, 1\}$ a *coherent valuation* of $\models_h$ if $v(\sim x) = 1 - v(x)$ and $v(y) = 1$ whenever $\Gamma \models_h y$ for a finite $\Gamma \subseteq v^{-1}(1)$.

Lemma 10.14 (Informal completeness). *[T4 Lemma 3.3, repaired] (standard corollary of compactness and completeness) Let h be first-order complete with negation-closed $D_h \neq \emptyset$. (i) $\mathfrak{A}h$ is exactly the set of coherent valuations of $\models_h$. (ii) $\Gamma \models_h y$ iff $\Gamma \cup \{y\} \subseteq D_h$ and every $v \in \mathfrak{A}h$ with $v(\Gamma) = 1$ has $v(y) = 1$. (iii) A finite $A \subseteq D_h$ is h -coherent iff $v(A) = 1$ for some $v \in \mathfrak{A}h$, iff $A \not\models_h y$ for some $y \in D_h$. So $\models_h$ determines $\mathfrak{A}h$ and coherence; conversely $\mathfrak{A}h$ determines $\models_h$ if $\mathfrak{A}h \neq \emptyset$, and coherence of all finite subsets of D_h determines it, but coherence on a fixed family $\mathcal{A}$ does not.*

Proof idea. A coherent v makes $T_h \cup \{\rho_h(x) : v(x) = 1\}$ finitely satisfiable (a finite inconsistent part would force $v(x) = v(\sim x) = 1$), so by compactness some model induces v ; the rest is completeness. Full proof in Appendix I. $\square$

Reading. This is the user’s “philosophical completeness theorem” (“a mode of talking makes sense (is coherent) iff there is something it could be talking about”) for a practice’s informal shadow: *the coherent ways of evaluating the practice’s sentences are exactly its admissible objects*. It is a direct corollary of Lindenbaum, compactness and completeness (Gödel, 1930). His caveat is visible in $\mathbb{M}_h = \text{Mod}(T_h)$: if only standard objects are presented, strengthenings such as $T + \text{Con}(T)$ survive, over-generalizations of the practice’s validity relation though not errors about truth (Section 9).

For a step set S and $D \subseteq X$, let $\text{Ch}_D(S)$ be the steps $\Gamma \triangleright y$ derivable from Γ by an argument whose lines lie in D and whose inferences and links lie in S . h^* is *step-expressive on D^* at gap g* if $\text{Ch}_{D^*}(\text{St}_{h^*}^g) = \models_{h^*}$ (implied by step-expressiveness, for $g \geq 1$).

Theorem 10.15 (Identification up to $\approx$ on the practice’s domain). *[T4 Thm 3.4, (d) repaired] Let every $h \in \mathcal{H}$ be first-order complete with negation-closed nonempty domain (one fixed $\sim$), let T^* be consistent, and let $\mathcal{A}$ consist of subsets of D^* .*

- (a) $h \equiv_\infty h'$ iff $h \approx h'$.
- (b) Under the closure-level version of Proposition 10.13, no learner identifies the target more finely than $\approx$.
- (c) Let $\mathcal{H}$ be finite with $h^* \in \mathcal{H}$ (BG1), and the presentation complete (every element of $\models_{h^*}$ appears in (P) , every finite restriction of every member of $\mathfrak{A}h^*$ in (O)). After finitely many data the survivors are exactly $U^* := \{h : h \approx_{D^*} h^*\}$, and unanimity on $\models_h$ accepts on D^* exactly $\models_{h^*}$. (This concerns the ideal version space: refutation by a practice datum needs a certificate of non-validity, which is co-r.e.)
- (d) If instead (P) presents $\text{St}_{h^*}^g$ completely, the survivors are eventually exactly

$$S^g := \{h \in \mathcal{H} : \text{St}_h^g \supseteq \text{St}_{h^*}^g \text{ and } \models_h|_{D^*} \subseteq \models_{h^*}\},$$

V^g accepts exactly $\text{St}_{h^*}^g$, and every survivor’s meaning is bracketed: $\text{Ch}_{D^*}(\text{St}_{h^*}^g) \subseteq \models_h|_{D^*} \subseteq \models_{h^*}$. If h^* is step-expressive on D^* at gap g , then $S^g = \{h \in U^* : \text{St}_h^g \supseteq \text{St}_{h^*}^g\}$; without it,

survivors can have strictly weaker meaning. Here all refutations are semi-decidable, so the budgeted VS_t reaches S^g .

Proof idea. The key is an *upper-bracket lemma*: for $D_h \supseteq D^*$, some object datum refutes h iff $\models_h|_{D^*} \not\subseteq \models_{h^*}$, and if $\models_h|_{D^*} \subseteq \models_{h^*}$ no paradox refutes h . A step valid under h but not h^* has an h^* -countermodel (Lemma 10.14(ii)) whose restriction refutes h ; conversely a datum with no h -admissible extension yields a finite h -inconsistent, hence h^* -inconsistent, set of literals, contradicting its origin. Practice removes hypotheses that miss a presented step. The lower bracket is Lemma 10.6 under h . Full proof in Appendix I. $\square$

Example 10.16 (A weaker rival survives bounded-gap data). [T4 Thm 3.4(d), referee A] (computed) Read D^* as $p_0, \neg p_0, p_3, \neg p_3$; let $T^* = \{p_0 \rightarrow p_1, p_1 \rightarrow p_2, p_2 \rightarrow p_3\}$ with axiom steps plus one-step tautological consequence, and let the rival have $T_h = \emptyset$. For $g \leq 3$ the g -step relations agree on D^* , but $(p_0 \triangleright p_3) \in \models_{h^*} \setminus \models_h$, and neither objects nor paradoxes refute the rival ($\text{Mod}(T^*) \subseteq \text{Mod}(\emptyset)$); at $g = 4$ it is refuted. On 2604 random (target, rival) pairs the original statement of (d), “survivors = $\{h \in U^* : \text{St}_h^g \supseteq \text{St}_{h^*}^g\}$ ”, mismatches the true survivors 527 times, the revised one never, and the original never on the 1545 step-expressive targets (T4-checks/verification_checks.py, V1). Verified in Appendix I.

Reading. Bounded-gap data identify the *g-step relation* and *bracket* meaning: practice bounds it from below, objects from above (they refute a hypothesis for validating too much, never too little). *Coarser* rivals, same inferences in larger steps, survive by Gold’s asymmetry (Gold, 1967), now about obviousness; *weaker* rivals miss inferences of h^* that are not chains of short steps through the practice’s own sentences. So a practice is guaranteed to fix the meaning of its sentences when its long inferences decompose into short steps through them, and otherwise it may not; expansion is the remedy.

The equivalence classes are large: practices fix inferential role and leave ontology and latent language open.

Proposition 10.17 (Ontology is unidentifiable at every gap). [T4 Prop 3.5, repaired] Let $h = (L, R, \rho, \mathbb{M})$ and $h' = (L, R, \rho, \mathbb{M}')$ be latent formalizations such that every member of $\mathbb{M}'$ agrees with some member of $\mathbb{M}$ on all sentences in $\rho(D_h)$, and vice versa. Then $h \equiv_g h'$ for every g , and $h \equiv_\infty h'$.

Proof. R, ρ are shared, so St^g , $\models$ and paradoxes are; and $v_M \circ \rho$ depends only on the truth values of $\rho(D_h)$ in M . $\square$

Instance (modulo the cited transfer principle). Let ρ read informal analysis into *bounded* sentences of a countable sublanguage of the $\in$ -language of the superstructure $V(\mathbb{R})$, with constants for the standard objects the practice can name, and let R be first-order logic with bounded axioms true in $V(\mathbb{R})$. With $\mathbb{M} = \{V(\mathbb{R})\}$ and $\mathbb{M}' = \{^*V(\mathbb{R})\}$, Robinson’s transfer principle (Robinson, 1966), which rests on Łoś’s theorem, gives the hypothesis. So *whether the intended reals of practice contain infinitesimals* cannot be learned from any data in this language. Boundedness matters: the two structures are not elementarily equivalent in the full $\in$ -language (“iterated power sets of $\mathbb{R}$ exist for every finite length” fails at nonstandard lengths).

Proposition 10.18 (ε - δ versus internal set theory). [T4 Prop 3.6, repaired] (modulo cited theorems of Nelson) Let ρ_{IST} read informal analysis literally in Nelson’s IST (Nelson, 1977), with every contextual constant (“let f be continuous”) read as standard ($T_{\text{IST}} := \text{IST} + \{\text{st}(c)\}$) and

infinitesimal suppositions folded into bound variables (ξ under “ dx is infinitesimal” becomes $\forall dx (dx \simeq 0 \wedge dx \neq 0 \rightarrow \xi)$). Let D_{st} be the occurrences read as sentences with standard parameters only, and let $h_W = (\text{ZFC} + \text{constants}, \rho_W)$, where $\rho_W(x)$ is the internal sentence Nelson’s reduction algorithm assigns to $\rho_{\text{IST}}(x)$. Then $h_W \approx_{D_{\text{st}}} h_{\text{IST}}$.

Proof idea. Nelson’s reduction algorithm and conservativity of IST over ZFC, with transfer for the standard parameters. Full proof in Appendix I. $\square$

Outside D_{st} nothing is claimed: $f(x) = Nx$ with N unlimited is ε - δ continuous at 0 but fails the literal reading $\forall x (x \simeq 0 \rightarrow f(x) \simeq f(0))$ at $x = 1/N$. Nor is agreement of the g -step relations claimed: if infinitesimal proofs are shorter, noise-free bounded-gap practice generated by h_{IST} refutes h_W as a granularity hypothesis, leaving meaning unaffected.

Proposition 10.19 (What numbers could not be). ^[T4 Prop 3.7] (modulo standard facts of ZF) Let D_{ar} be the occurrences of informal arithmetic sentences, read into the $\in$ -language via von Neumann numerals ($\rho_{\text{vN}}, n + 1 = n \cup \{n\}$) or Zermelo numerals ($\rho_{\text{Z}}, n + 1 = \{n\}$), with $+, \times$ defined by recursion. Then $(\text{ZF}, \rho_{\text{vN}}) \approx_{D_{\text{ar}}} (\text{ZF}, \rho_{\text{Z}})$, although the readings differ on “ $1 \in 3$ ”, which lies outside D_{ar} .

Proof in Appendix I: transport along the recursion-defined isomorphism given by Dedekind’s categoricity theorem (Dedekind, 1888). Benacerraf (1965) thus becomes a non-identifiability theorem: practice fixes $\approx$ on its domain and nothing else.

Reading: $\approx$ is inferentialist meaning. Inferential role, incompatibility (here incoherence, Lemma 10.14(iii)) and Sellarsian language-entry transitions (here object evaluations, Lemma 10.14(i)) are all functions of $\models_h$ on D^* , and soundness refers only to $\models_{h^*}$. Everything finer (ontology, latent language, junk like “ $1 \in 3$ ”) never changes a practice inference and is unidentifiable from closure-level data. Granularity is finer and partly identifiable, but it is pragmatics (which steps are *acceptable*), not semantics (which are *valid*). Whether ε - δ and infinitesimals differ in what they are about the practice cannot settle; the verifier needs only $\approx$.

10.4 Counterexample objects versus paradoxes

Lemma 10.20 (Descent along false lines). ^[T4 Lemma 4.1] If v is defined on the lines of α , true on the premises and false on the conclusion, then α contains an inference with true antecedents and false conclusion, or a link with true source and false target, found within $\text{depth}(\alpha)$ moves evaluating only antecedents of visited lines. If v is h -admissible, the item is h -invalid.

Proof. From the false conclusion, repeatedly move to a false antecedent (or link source); stop when there is none. Premises are true, so the walk stops at an inference or link. If $v = v_M \circ \rho_h$ with $M \in \mathbb{M}_h$, a valid item would have a true conclusion in M . $\square$

This is Shapiro’s contradiction backtracing (Shapiro, 1983) for informal arguments, Lakatos’s conversion of a global counterexample into a local one (Lakatos, 1976), and Easwaran’s demand that proofs be fine enough to convert rebutting into undercutting defeaters (Easwaran, 2015). Descent can end at a link: the blame then falls on an equivocation, not a rule (Lakatos’s “global but not local” case).

Definition 10.21 (Correction game). ^[T4 Def 4.2] Let $\mathcal{H} \subseteq 2^S$ be finite, S a finite step universe, $h^* \in \mathcal{H}$. Each round the learner announces $A_t \subseteq S$; if $A_t \neq h^*$ the environment returns (+) some $s \in h^* \setminus A_t$, or (−obj) some $s \in A_t \setminus h^*$, or (−bag_r) some $B \subseteq A_t$ with $|B| \leq r$, $B \not\subseteq h^*$; the learner deletes the hypotheses $h \not\ni s$, $h \ni s$, $h \supseteq B$ respectively. A *correction* is a round with feedback. $M_{\text{obj}}(\mathcal{H})$ (feedback (+) or (−obj)) and $M_{\text{bag}}^{(r)}(\mathcal{H})$ (feedback (+) or (−bag_r)) are the optimal worst-case numbers of corrections; $M_{\text{bag}} := M_{\text{bag}}^{(|S|)}$. The positive elasticity $\text{el}(\mathcal{H}, h^*)$ is the supremum of lengths of sequences $s_1, s_2, \dots \in h^*$ with $s_i \notin \bigcap \{h \in \mathcal{H} : h \supseteq \{s_1, \dots, s_{i-1}\}\}$, and $\text{el}^*(\mathcal{H}) := \max_{h^*} \text{el}(\mathcal{H}, h^*)$.

In reality a negative channel can be silent (objects need presentable countermodels; coherent-but-wrong hypotheses yield no paradox, Section 6); the bounds below then still bound corrections, but unsound rounds may pass silently.

Theorem 10.22 (Objects: logarithmic). ^[T4 Thm 4.3] (*classical*) (a) $M_{\text{obj}}(\mathcal{H}) \leq \lfloor \log_2 |\mathcal{H}| \rfloor$; with a prior w , weighted halving makes at most $\log_2(1/w(h^*))$ corrections, so with $w = 2^{-\ell}$ the number of counterexamples needed is at most the description length of the target. (b) $M_{\text{obj}}(\mathcal{H}) \leq M_{\text{bag}}^{(r)}(\mathcal{H})$ for all $r \geq 1$, with equality at $r = 1$. (c) $M_{\text{obj}}(\mathcal{H}) = \text{Ldim}(\mathcal{H})$, the Littlestone dimension.

Proof of (a). Announce the set of steps carried by more than half of the surviving weight. Then (+) on $s \notin A_t$ and (−obj) on $s \in A_t$ each keep at most half of it, and h^* always survives. (b) and (c) are in Appendix I. $\square$

This is classical halving (Barzdin and Freivalds, 1972; Littlestone, 1988; Angluin, 1988); the object game is the mistake-bound model, hence (c) (Littlestone, 1988) (confirmed on 248 random classes, V4 (computed)).

Theorem 10.23 (Paradoxes: linear, tightly). ^[T4 Thm 4.4] (a) $M_{\text{bag}}^{(r)}(\mathcal{H}) \leq \text{el}^*(\mathcal{H}) \leq |\mathcal{H}| - 1$. (b) Single culprit: for $S = \{b_1, \dots, b_n\}$ and $\mathcal{H}_n = \{S \setminus \{b_j\} : j \leq n\}$, $M_{\text{obj}}(\mathcal{H}_n) = 1$ and $M_{\text{bag}}^{(r)}(\mathcal{H}_n) = \min(r, n - 1)$; so $M_{\text{bag}}(\mathcal{H}_n) = |\mathcal{H}_n| - 1$ against $\log_2 |\mathcal{H}_n|$, and (a) is tight. (c) The chain paradox: take sentences $q_0, \dots, q_n$, suspect steps $b_i = (q_{i-1} \triangleright q_i)$, and the designated context $\{q_0, \sim q_n\}$, e.g. $q_k = \text{“a pile of } n - k \text{ grains is a heap”}$. Let h_j validate every b_i with $i \neq j$ (axioms $p_{i-1} \rightarrow p_i$, $i \neq j$). Every h_j is coherent on the designated context; every paradox uses all the b_i , so every bag is S ; every learner using paradoxes suffers $n - 1$ corrections; and one admissible object pins down the culprit by descent.

Proof idea. (a) The cautious learner $A_t = \bigcap \text{VS}_t$ receives only (+) items, forming an elastic chain. (b) Against bags, an adversary answers a rejected candidate with (+) and an accepted candidate set C with a bag of r candidates, or of all of C if $|C| \leq r$ (deleting nothing). (c) The admissible valuation of h_{j^*} with q_0 true and q_n false makes q_i true iff $i < j^*$. Full proof in Appendix I. $\square$

Exact minimax values (T4-checks/bag_vs_object_game.py) (computed): for the single-culprit class with $n = 6$, $M_{\text{obj}} = 1$ and $M_{\text{bag}}^{(r)} = 1, 2, 3, 4, 5, 5$ for $r = 1, \dots, 6$; for “two culprits of six” ($|\mathcal{H}| = 15$), $M_{\text{obj}} = 2$ and $M_{\text{bag}}^{(r)} = 2, 4, 4, 4, 4, 4$. On 295 random classes $M_{\text{obj}} \leq M_{\text{bag}} \leq \text{el}^* \leq |\mathcal{H}| - 1$ always held, with $M_{\text{obj}} < M_{\text{bag}}$ in 43 (V3). “Paradoxes are worthless” ($M_{\text{bag}} = \text{el}^*$) is false in general: $\mathcal{H} = \{\emptyset, \{0\}, \{0, 1\}\}$ has $M_{\text{bag}} = 1 < 2 = \text{el}^*$ (announce $\{0\}$). Real paradoxes have few *suspect* steps, since steps every survivor accepts drop out of the bag.

Theorem 10.24 (Bags of size at most r). ^[T4 Thm 4.5] The super-majority learner $A_t = \{s : w(\{h \in \text{VS} : s \in h\}) > \frac{r}{r+1}w(\text{VS})\}$ makes at most $\ln(1/w(h^*))/\ln(1+1/r)$ corrections. With the uniform prior, $M_{\text{bag}}^{(r)}(\mathcal{H}) \leq \ln|\mathcal{H}|/\ln(1+1/r) \leq (r+1)\ln|\mathcal{H}|$.

Proof. Each $s \in A_t$ is missed by less than $\frac{1}{r+1}w(\text{VS})$, so a bag $B \subseteq A_t$ of size $\leq r$ leaves (union bound) less than $\frac{r}{r+1}w(\text{VS})$; a (+) item $s \notin A_t$ leaves at most that. So $w(h^*) \leq (\frac{r}{r+1})^M$, and $\ln(1+1/r) \geq \frac{1}{r+1}$. $\square$

Proposition 10.25 (Linear dependence on r is necessary). ^[T4 Prop 4.6] For the product of k single-culprit blocks of size $b \geq 2$ ($|\mathcal{H}| = b^k$), $M_{\text{obj}} = k$ and $M_{\text{bag}}^{(r)} \geq k \min(r, b-1)$. With $b = r+1$: $M_{\text{bag}}^{(r)} \geq r \log|\mathcal{H}|/\log(r+1)$ while $M_{\text{obj}} = \log|\mathcal{H}|/\log(r+1)$.

Proof in Appendix I, via an adversary with potential $\sum_X \min(r, |C_X| - 1)$ over the blocks' candidate sets; equality was confirmed exhaustively for $(b, k) \in \{(2, 2), (3, 2), (2, 3)\}$ (computed). So adversarial multiple-instance feedback costs $\tilde{\Theta}(r \log|\mathcal{H}|)$ over classes of a given size, whereas in the i.i.d. multiple-instance setting (Dietterich et al., 1997) bag size enters far more mildly (Sabato and Tishby, 2012); and the form “Ldim $\times$ polylog(r)” is refuted, as products have Ldim = k .

Proposition 10.26 (Bag price when $M_{\text{obj}} = 1$). ^[T4 Prop 4.6'] If $M_{\text{obj}}(\mathcal{H}) = 1$ then $M_{\text{bag}}^{(r)}(\mathcal{H}) \leq r$.

Conjecture 10.27 (Bag price). ^[T4 §4.2] $M_{\text{bag}}^{(r)}(\mathcal{H}) \leq r \cdot \text{Ldim}(\mathcal{H})$ for every finite $\mathcal{H}$.

Proposition 10.26 (proof in Appendix I) is the case Ldim = 1. The conjecture is tight on single-culprit classes ($r \leq n-1$), products ($r \leq b-1$) and some two-culprit classes, and no violation was found on 519 (class, r) pairs (bag_size_conjecture.py) or in a referee's search, exhaustive for $|S| \leq 4$ and covering about 4.7 million random pairs with $|S| \leq 6$.

Objects can also be neutralized. A step blamed by an object can be repaired by changing R (lemma-incorporation), changing ρ so that the monster no longer satisfies the premise (monster-barring by redefinition), rejecting or re-interpreting the datum (monster-barring by exclusion, monster-adjustment), or extending L (a proof-generated concept naming the hidden lemma) (Lakatos, 1976).

Proposition 10.28 (Free barring nullifies objects; trivial). ^[T4 Prop 4.7] If barring object data is free, objects constrain nothing: every hypothesis consistent with (P) and (C) can survive forever. If objects are truthful and each barred datum costs $c > 0$ bits, a hypothesis that must bar m data loses to h^* in description length once $m > (\ell(h^*) - \ell(h))/c$.

Proof. If barring is free, (O) refutes nothing, and only (P) and (C) remain. With cost c , h scores $\ell(h) + mc$ against $\ell(h^*)$, since truthful objects never force h^* to bar. $\square$

So objects are powerful only if certified independently (by the community, or by computation) or if revisions of ρ are paid for, which is Lakatos's own preference ordering.

Reading. Coherence says that *some* step of a paradox is bad; an object says *which*. The sorites is the canonical uninformative paradox. The oligarchic halving of Section 6 bounds *detected incoherences* by $\log_2(1/w(h^*))$, yet total corrections, counting the rejections of valid steps that caution forces, can still be $|\mathcal{H}| - 1$: coherence bounds the price of boldness, not of learning; objects bound both. Russell's paradox, by contrast, is a bag of size one; what it could not do is choose the *generalization* (Section 10.5). A learner of inference rules must generate and evaluate objects, not only search for $\perp$.

name	$\varphi(x, a, b)$	role	pos.	strat.	$x \in a \wedge \psi$
V	$x = x$	Frege's extension of "object"	✓	✓	
EMP	$x \neq x$	empty set		✓	
S	$x \in x$		✓		
R	$x \notin x$	Russell's set			
INT	$x \in a \wedge x \in b$	intersection (Dedekind)	✓	✓	✓
UNI	$x \in a \vee x \in b$	union	✓	✓	
PAIR	$x = a \vee x = b$	pairs	✓	✓	
DIFF	$x \in a \wedge x \notin b$	difference (Cantor)		✓	✓
CMP	$x \notin a$	complement		✓	
ZR	$x \in a \wedge x \notin x$	Zermelo's 1908 subset			✓

Table 13: The comprehension menu Φ [T4 §5.2]. ZR is Cantor's diagonal set $\{x \in a : x \notin f(x)\}$ (Cantor, 1891) for $f = \text{id}$; identifying diagonal practice with ZR is a modelling simplification.

10.5 Frege $\rightarrow$ Russell $\rightarrow$ Zermelo

Definition 10.29 (Repair setting). [T4 §5.1] I is a set of rule instances; $\text{Coh} \subseteq \mathcal{P}(I)$ is a coherence predicate, downward closed and of finite character; $\mathcal{R} \subseteq \mathcal{P}(I)$ is a finite class of *uniformly specified restrictions* with description lengths ℓ . Practice at time t is a finitely supported weight $w_t : I \rightarrow [0, \infty)$, with coverage $\text{cov}_t(c) := \sum_{i \in c} w_t(i)$. Budget- t $\perp$ -search refutes c iff c has a refutation of size $\leq t$; incoherent c has one of size $k(c) < \infty$. $\text{MDL}_t := \arg \min \{\ell(c) : c \in \mathcal{R}, \text{supp}(w_t) \subseteq c\}$, and REP_t is the lexicographic $\arg \max$ of $(\text{cov}_t, -\ell)$ over the members of $\mathcal{R}$ not refuted with budget t .

Theorem 10.30 (Abstract repair; trivial). [T4 Thm 5.1, repaired] (a) If $I \in \mathcal{R}$ is strictly shortest, $\text{MDL}_t = I$ for all t . (b) Coherent members are never refuted; incoherent c is refuted for $t \geq k(c)$. (c) For $t \geq t_0 := \max\{k(c) : c \in \mathcal{R} \text{ incoherent}\}$, REP_t maximizes $(\text{cov}_t, -\ell)$ over the coherent members of $\mathcal{R}$. (d) If $w_t/\|w_t\|_1 \rightarrow w_\infty$ in ℓ_1 and the coherent maximizer of cov_∞ is unique, REP_t is eventually equal to it. (e) If $c_1, c_2 \in \mathcal{R}$ are coherent and $c_1 \cup c_2$ is not, no coherent set contains both, and while $\text{supp}(w_t) \subseteq c_1 \cap c_2$ only ℓ chooses between them.

Proof in Appendix I. The content is in an instance that makes every step of the story checkable. In first-order logic with $\in, =$, let $\text{Comp}(\varphi) := \forall a \forall b \exists y \forall x (x \in y \leftrightarrow \varphi(x, a, b))$, with φ from the menu Φ of Table 13. A formula is *stratified* (as in Quine's NF, Quine 1937) if it has a typing t with $t(v) = t(u) + 1$ for $u \in v$ and $t(u) = t(v)$ for $u = v$; $x \in x$ has none. $\mathcal{R}$ consists of NC (the whole menu), POS (positive formulas), STRAT (stratified; NF-like), SEP (of the form $x \in a \wedge \psi$), and $Z := \text{SEP} \cup \{\text{EMP}, \text{PAIR}, \text{UNI}\}$ (Zermelo-like). The description lengths are *stipulated*: $\ell(\text{NC}) < \ell(\text{POS}) < \ell(\text{SEP}) < \ell(\text{STRAT}) < \ell(\text{Z})$, and outcomes depend on this choice.

Lemma 10.31 (Comprehension facts). [T4 §5.2, F1–F6] (F1) NC is incoherent ($\text{Comp}(\text{R})$ with $x := y$ refutes it in four lines). (F2) POS is coherent, even for all positive formulas. (F3) $\text{STRAT} \cap \Phi$ is coherent. (F4) $\text{Z} \cap \Phi$ is coherent. (F5) POS, STRAT, Z are pairwise jointly incoherent. (F6) $\text{SEP} \subsetneq \text{Z}$, and POS, STRAT, Z are pairwise $\subseteq$ -incomparable.

Proof idea. (F2) one element $u \in u$; (F3) $\mathbb{N}$ with $x \in y$ iff $x \in e(y)$ for a bijection e onto the finite and cofinite sets; (F4) the hereditarily finite sets; (F5) $\{\text{V}, \text{ZR}\}$ and $\{\text{S}, \text{CMP}\}$ each yield Russell's set. Full proof in Appendix I. $\square$

Theorem 10.32 (The Frege–Russell–Zermelo trajectory). *[T4 Thm 5.2, (iv) repaired] (i) For every practice $\text{MDL} = \text{NC}$: Frege’s Basic Law V (Frege, 1893) as the compression of “every concept has an extension”. (ii) Coherence refutes it with a four-line budget. (iii) The maximal coherent members of $\mathcal{R}$ are exactly POS, STRAT and Z, pairwise jointly incoherent: there is no minimal repair. (iv) Coverage selects ($t \geq t_0$; $\text{DC} := \{\text{INT}, \text{UNI}, \text{PAIR}, \text{DIFF}, \text{EMP}\}$, the Dedekind–Cantor operations, all supports with positive weights):*

- support DC : STRAT and Z cover everything and ℓ selects **STRAT**. For supports $\subseteq \text{DC}$: STRAT iff the support meets both $\{\text{DIFF}, \text{EMP}\}$ and $\{\text{UNI}, \text{PAIR}, \text{EMP}\}$; otherwise POS if it avoids DIFF and EMP (a positive practice selects positive set theory); otherwise SEP;
- support $\text{DC} \cup \{\text{ZR}\}$: only Z covers everything, and Z is selected; for $\text{ZR} \in \text{support} \subseteq \text{DC} \cup \{\text{ZR}\}$, Z iff the support meets $\{\text{UNI}, \text{PAIR}, \text{EMP}\}$, else SEP;
- support $\text{DC} \cup \{\text{ZR}, \text{V}\}$: Z iff $w(\text{ZR}) > w(\text{V})$, ties to STRAT; with partial supports POS can tie STRAT and win by ℓ (e.g. $\{\text{INT}, \text{UNI}, \text{PAIR}, \text{ZR}, \text{V}\}$ with $w(\text{V}) > w(\text{ZR})$).

Proof idea. Theorem 10.30(c) and the membership table: with total weight W , $\text{cov}(\text{STRAT}) = W - w(\text{ZR}) - w(\text{S}) - w(\text{R})$ and $\text{cov}(\text{Z}) = W - w(\text{V}) - w(\text{S}) - w(\text{R}) - w(\text{CMP})$. Full proof in Appendix I; all 93 sub-supports agree with the stated conditions (V2) (computed), and `comprehension_toy.py` reproduces the dynamics (experiment G, Section 12). $\square$

Historical reading (“a nice thing shadowed in” practice, not a claim about what Zermelo did). Zermelo (1908) proves from Separation that every set M has a subset $\{x \in M : x \notin x\}$ not in M , so the domain is not a set: ZR at arbitrary a , the diagonal argument as a theorem. So practice containing Zermelo-style diagonal subsets selects Separation over stratification, and Frege’s universal extension is the price. Without diagonal practice, and under the stipulated ℓ , simplicity would have favoured NF (on its consistency see Holmes 2015) only if the practice used difference or $\emptyset$ and also union, pairs or $\emptyset$; a purely positive practice would have favoured positive set theory, and one using only intersection and difference, SEP. A second separating datum lies outside the toy: NF refutes choice (Specker, 1953), which Zermelo defended by its uses (Moore, 1982). His stated method, to restrict “sufficiently to exclude all contradictions” yet keep “all that is valuable” (Zermelo, 1908), is REP’s constrained optimization; the theorem adds that the optimum exists only in a uniform class and is chosen by coverage. Without uniformity it is far worse.

Proposition 10.33 (Continuum many maximal repairs). *[T4 Prop 5.4] The instances of naive comprehension have exactly $2^{\aleph_0}$ maximal consistent subsets; even $\text{Comp}(x \neq x)$ alone has $2^{\aleph_0}$ pairwise incompatible maximal consistent extensions.*

Proof idea. With $p_n = \text{“some object has exactly } n \text{ elements”}$ and $J(q) := \text{Comp}(x \notin x \wedge q)$, which refutes q , each $A \subseteq \mathbb{N}_{\geq 1}$ gives a consistent set deciding every p_n according to A ; extend by Zorn. Full proof in Appendix I. $\square$

Incurvati and Murzi (2017) prove more: under minimal assumptions, multiple incompatible maximal consistent sets of instances of naive comprehension exist and *none is recursively axiomatizable*, generalizing McGee’s theorem for the T-schema (McGee, 1992). So “the maximal repair” does not exist and no learner can output a recursive axiomatization of one (greedy limit approximation with a halting oracle does reach Δ_2 maximal sets). *Uniformity of the restriction class is what makes the learning problem well posed.* The toy omits Replacement (coverage of

a *conclusion*), the cumulative-hierarchy conception (a choice of ℓ), and undecidable coherence, which by Theorem 10.8 costs escalations, never soundness.

Reading. Imitation plus simplicity produces the over-general rule (Section 3); coherence refutes it, here with sharp blame; but coherence cannot choose the repair, since the maximal coherent generalizations are several. Coverage of what the practice does chooses: Lakatos’s “heuristic falsifiers”, Zermelo’s “all that is valuable”. What coverage leaves open is a Kripkensteinian residue (Section 6): coherent meanings of “set” that agree on everything the practice has used.

10.6 The robust core and the emergence of a notion of proof

Definition 10.34 (Sharpenings; robust core). ^[T4 Def 6.1] Fix a first-order L , a base theory T_0 and vague words W . A *sharpening* σ gives each $w \in W$ an explicit L -definition; Σ is the set of *admissible* joint sharpenings. Each σ gives a first-order complete $h_\sigma = (L, R_\sigma, \rho_\sigma, \text{Mod}(T_\sigma))$, with $T_\sigma = T_0$ plus σ ’s definitions and ρ_σ reading each w by its definition. *Local supervaluational validity* is $\models_{\text{SV}} := \bigcap_\sigma \models_\sigma$; put $\text{St}_{\text{SV}}^g := \bigcap_\sigma \text{St}_\sigma^g$. A precise object $M \models T_0$ expands uniquely to $M_\sigma \models T_\sigma$; an occurrence is *super-determinate* at M if all M_σ agree on it, and object data are *SV-truthful* if reported only for super-determinate occurrences. The *robust-core hypothesis* RCH_g : every non-noise practice step lies in St_{SV}^g and every practice link is valid under every σ .

Local validity (truth-preservation at each sharpening) differs from the *global* validity (preservation of supertruth) usually called supervaluational (Fine, 1975; Williamson, 1994; Keefe, 2000; Varzi, 2007); local validity is the right notion for “survives every formalization”.

Theorem 10.35 (Robust core). ^[T4 Thm 6.2, repaired]

- (a) *Survival.* Under RCH_g , an argument whose inferences and links are non-noise practice items has, for every σ , its conclusion in $\text{Cl}_{R_\sigma}(\rho_\sigma \text{Prem})$; a theorem so proved from premises every T_σ proves is a theorem of every admissible sharpening, whichever a later formalization adopts.
- (b) *SV-truthful data never refute a sharpening.* Let $\{h_\sigma\} \subseteq \mathcal{H}$, and let practice lie in St_{SV}^g , escalation and link answers be given only when all σ agree (“yes” on St_{SV}^g , “no” outside $\bigcup_\sigma \text{St}_\sigma^g$), designated contexts be coherent under every σ , and objects be SV-truthful. Then every h_σ survives forever, and V_t^g accepts only steps in St_{SV}^g : it is sound for every sharpening at once. If moreover $\mathcal{H}$ is finite and every element of St_{SV}^g eventually appears in (P) , then V_t^g eventually accepts exactly St_{SV}^g .
- (c) *Nothing more is soundly acceptable.* Under these data, (c1) an acceptance set sound for every possible target h_σ lies in $\models_{\text{SV}}$; (c2) one that is g -sound for every target lies in St_{SV}^g , which V_t^g attains when $\mathcal{H} = \{h_\sigma\}$. The gap $\models_{\text{SV}} \setminus \text{St}_{\text{SV}}^g$ is in general nonempty.
- (d) *Permanent abstention.* Steps in $\bigcup_\sigma \text{St}_\sigma^g \setminus \text{St}_{\text{SV}}^g$ are never accepted. (d1) Those outside $\models_{\text{SV}}$ are invalid under some admissible sharpening, and only a definition (shrinking Σ) or lemma-incorporation removes them; (d2) those in $\models_{\text{SV}}$ are valid but not short under every sharpening, and expansion or a library lemma removes them.

Proof idea. (a) is Lemma 10.6 for each σ ; (b) checks that each datum is consistent with each h_σ (a super-determinate report at M agrees with $v_{M_\sigma} \circ \rho_\sigma$); (c) holds because every h_σ remains a possible target. Full proof in Appendix I, with an example of the gap in (c) and one showing that for infinite Σ the robust core can be undecidable. $\square$

Theorem 10.36 (Steps invalid under a sharpening have monsters; trivial). *[T4 Thm 6.3, repaired]*
 If $\Gamma \cup \{y\} \subseteq D_{h_\sigma}$ and $(\Gamma \triangleright y) \notin \models_\sigma$ for an admissible σ , some $M \models T_\sigma$ makes $\rho_\sigma \Gamma$ true and $\rho_\sigma y$ false: a σ -monster.

Proof. By definition of $\models_\sigma$, since Cl_{R_σ} is consequence over T_σ ; Gödel completeness is what makes such R_σ exist. $\square$

Steps in $\models_{\text{SV}} \setminus \text{St}_{\text{SV}}^g$ have no monster, and a monster need not be presentable under SV-truthful reporting. *Cauchy, again:* the sum-theorem step is valid under the uniform sharpening of “converges” only, so it lies outside the robust core, and Abel’s series is a monster relative to the pointwise one. Lakatos’s responses change Σ or the language: monster-barring drops the pointwise sharpening; lemma-incorporation adds “uniformly”, making the step robust; a proof-generated concept introduces a word whose only sharpening is uniform convergence. The robust-core hypothesis predicts *where* pre-formal mathematics broke: at the non-robust steps.

Proposition 10.37 (Only unanimity chains; the sorites is tight). *[T4 Prop 6.4] (known; re-derived)*
 (a) $\models_{\text{SV}}$ is a consequence relation on $\bigcap_\sigma D_{h_\sigma}$. (b) For a probability μ on Σ (with $\{\sigma : s \in \models_\sigma\}$ measurable), if each of the n steps of an argument (links included) is valid with probability $\geq 1 - \varepsilon$, the argument is valid with probability $\geq 1 - n\varepsilon$; this is attained for every $\varepsilon \leq 1/n$, and at $\varepsilon = 1/n$ by an argument valid under no sharpening.

Proof idea. (b) is Lemma 10.6 per σ plus the union bound. Tightness: $q_k = “k \text{ grains make a heap}”$, sharpenings σ_c reading q_k as $k \geq c$ ($1 \leq c \leq n$); $q_k \triangleright q_{k-1}$ fails under σ_c iff $c = k$, and $q_n \triangleright q_0$ fails under every σ_c . Full proof in Appendix I. $\square$

The bound is Adams’s (Adams, 1966), its use for the sorites over precisifications is Edgington’s (Edgington, 1997), and the disjoint-failure tightness is the lottery paradox’s (Kyburg, 1961). “Valid on most readings” is to vagueness what majority vote is to verifier ensembles (the doctrinal paradox of Section 3): it does not compose. Among properties “valid with probability $\geq 1 - \varepsilon$ ” only $\varepsilon = 0$ composes without loss, and among properties guaranteeing validity whatever the admissible sharpening turns out to be, unanimity is the weakest.

Proposition 10.38 (How the nice notion of proof came to be there). *[T4 Prop 6.5, repaired]* Let T be an r.e. first-order theory, ρ^* a reading, and A_0 the practice’s initial repertoire of steps (within D^*). Suppose the community drops any step refuted by a presented monster (a model of T), and every step outside $\models_T$ is eventually refuted.

- (a) Derivability: the survivors are $A_\infty = A_0 \cap \models_T$, and the ρ^* -images of any argument chained from them interpolate into a derivation in any complete calculus for T .
- (b) No gap bound follows: for A_0 the instances of the single schema “from χ infer θ ” (χ, θ arithmetic sentences) and $T = \text{PA}$, A_∞ is undecidable and no g has $A_\infty \subseteq \text{St}^g$ for any complete calculus for PA meeting the effectivity assumptions of Definition 10.2.
- (c) A bounded gap needs two extra hypotheses: if A_0 is the instance set of finitely many schemas, selection acts on schemas (a schema is dropped when any instance is refuted), and each surviving schema is uniformly derivable (a derivation template with at most g rule applications, in the schema’s metavariables, instantiates to a derivation of each instance), then all surviving steps are in St^g of a complete calculus for T , the shadow of finitely many derived rules. Uniform derivability is automatic for pure propositional schemas and fails in general: in PA, “infer $\theta(\bar{n})$ ” with $\theta(n) = “n \text{ codes no PA-proof of } 0 = 1”$ has all instances provable, but a uniform template would yield a PA-proof of $\text{Con}(\text{PA})$.

Proof idea. (a) Survival iff no model of T refutes iff semantic consequence iff derivability (completeness). (b) $A_\infty \subseteq \text{St}^g$ would make $A_\infty = A_0 \cap \text{St}^g$ decidable. (c) Instantiate the templates. Full proof in Appendix I. $\square$

In the vague case, (a) applied to each (T_σ, ρ_σ) gives survivors $A_0 \cap \models_{\text{SV}}$, though complete monster presentation is then in tension with SV-truthful reporting.

Reading: how the nice notion of proof came to be there. The model’s partial answer has three ingredients. (1) *Local checkability*: steps are checked on objects line by line (Lemma 10.20), so monster pressure selects steps that preserve truth in every object. (2) *Completeness*: Gödel’s theorem turns “truth-preserving in every object of an elementary class” into “derivable in a finite calculus”, with no bound on length. (3) *Schema-level selection of uniformly derivable schemas*: an extra hypothesis, not derived here, which alone makes surviving practice the shadow of finitely many derived rules with a bounded gap. So the formal structure is “sorta there” in the activity because the activity was *selected by objects*, and completeness says that being so selected is being shadowed by a calculus. The model explains why surviving steps are *derivable*, and why they are *short* only under (3); what separates the calculus from “the set of truths” is that T is r.e. Caveats: complete monster presentation is an idealization (the selection took centuries); for intended-model semantics such as $\{\mathbb{N}\}$, (2) fails and selection by standard objects heads toward true arithmetic, which no calculus captures; and completeness of the surviving calculus for T does not follow, since selection only removes steps. With Theorem 10.35 this also answers the robust-core question: *if* pre-formal practice used only steps valid under every admissible sharpening, its proofs survive every formalization among them, and the conservative verifier never accepts beyond that core and, for a finite class under complete practice, converges to it.

Remark 10.39 (Euclid: an analogy). ^[T4 §6.2] The user suggests looking at Euclid, who “didn’t have a full set of axioms”. If a “sharpening” of a diagram is any configuration near it, Manders’s *co-exact* properties (Manders, 2008) are the super-determinate ones, his thesis that Euclid reads off only co-exact properties is the analogue of RCH, and the system E of Avigad et al. (2009), sound and complete for a class of sequents relative to ruler-and-compass semantics, plays the role of the robust core of Theorem 10.35(c2). Perturbations are not sharpenings in the sense of Definition 10.34, so this is an analogy only.

10.7 Historical reading, and what a learner before formalization needed

How informal mathematics stayed reliable (Table 14). (i) The dominant negative datum was an *object*, not a contradiction; only the set-theoretic antinomies were pure derivations of $\perp$. Objects localize blame, at logarithmic rather than linear cost (Theorems 10.22 and 10.23); mathematicians still check how particular steps apply to specific examples (Weber and Mejía-Ramos, 2011), and Euler accepted $\pi^2/6$ on six-decimal agreement. (ii) The faulty rules were *average-case* sound: right on the tame objects practitioners had in mind, wrong on degenerate ones (Steiner’s 7776 fails for *every* input, because of an excess component of double lines). Practice survived because its search was guided by intended models; rigour arrived when mathematicians *became* adversarial over objects (Weierstrass, Peano, Russell). Rigour is deliberate monster-hunting, and this is Section 3 in history: a checker trained on human steps inherits the tameness assumption, and a strong prover finds the monsters. (iii) Lags ran from weeks to decades, so guarantees must be time-uniform (Theorem 10.9). (iv) Labels are noisy both ways; some “known errors” are not errors (Hales, 2007), and whether Cauchy’s theorem was false depends on the reading

episode	faulty informal rule	negative evidence	repair
Cauchy 1821	convergent series of continuous functions has continuous sum	object: Fourier series (Abel 1826); hidden lemma (Seidel, Stokes 1847)	uniform convergence (L)
Ampère 1806	continuous functions differentiable except at isolated points	object: Weierstrass's function (1872)	function concept broadened (ρ)
Riemann 1851/57	Dirichlet's principle	object: Weierstrass's variational problem (1870)	side conditions (Hilbert) (R)
Steiner 1848	7776 conics	excess component of double lines	3264; genericity
Lamé 1847	unique factorization in cyclotomic integers	failure known (Kummer 1844)	ideal numbers (L)
Poincaré 1900	homology characterizes S^3	object: homology sphere (1904)	fundamental group (L)
Frege 1893	Basic Law V	paradox: Russell (1902)	restricted comprehension (R)
Zermelo 1908	no Replacement	coverage: $\aleph_\omega$ not provable	Replacement, 1922 (R)

Table 14: Negative evidence in informal mathematics, and the component of $(L, R, \rho, \mathbb{M})$ revised (from the project's literature memo L6, §1, where dates are checked).

(Proposition 10.7 and Theorem 10.36). Modularity (Avigad, 2021), a track record of heavy use, and a social process good at locating weak points (Thurston, 1994) tell the object search where to look. Kreisel's squeezing argument (Kreisel, 1967) pins informal *logical* validity to derivability; the axioms cannot be squeezed, and they are what coherence and coverage leave non-unique.

What a learner before formalization would have needed: (1) a class of latent formalizations with a growing definitional library, so that practice has a bounded gap; (2) conservative acceptance by unanimity, applied to links as well as steps, with a time-uniform Bayesian version (Theorems 10.8 and 10.9); (3) an expansion protocol (Proposition 10.11); (4) an object generator, adversarial over objects, with descent (Lemma 10.20 and Theorem 10.22); (5) paid revisions of readings (Proposition 10.28); (6) coverage of practice to choose among coherent repairs within a uniform class (Theorem 10.32 and Proposition 10.33); (7) acceptance restricted to the robust core when concepts are vague, with non-robust steps flagged as where monsters live (Theorems 10.35 and 10.36); (8) a human corpus treated as noisy and corrigible.

The answer to success criterion 2 is yes, in a precise and conditional sense. A learner with (1)–(8) whose class contains the practice's latent formalization is sound at all times against adaptive provers, including its own search; for a finite class under complete presentation, identifies the practice's step relation and brackets its meaning (identifying it under step-expressiveness), and converges to the robust core and nothing beyond; and in the comprehension toy reproduces the structure of Frege–Russell–Zermelo, including that coverage, not coherence, chooses the repair. But *soundness is relative to realizability of the latent formalization class* (BG1), and history violated realizability at every decisive step: ε - δ , uniform convergence, ideals and the fundamental group were inventions of language. The user's own note says as much: in mathematics “we created a language in which formal proofs can be written”.

The hardest open problem: affordable soundness with language invention. A universal class (all computable (L, R, ρ) with a description-length prior) restores realizability: Theorem 10.9 is sound with $W^* \geq 2^{-K(h^*)}$, and, information-theoretically, objects need at most $K(h^*)$ corrections

(Theorem 10.22(a), weighted halving). But corrections are not soundness, since halving is bold; the sound verifier pays escalations, $\tilde{\Theta}(2^{K(h^*)})$ on unstructured parts of the class (Theorem 10.10(c), Section 4), and paradoxes cannot rescue it (Theorem 10.23). The problem is to find a class that is closed under the definitional extensions history made (say, predicates defined by formulas of bounded quantifier rank over the old language) and has escalation dimension, or correction complexity with objects, polynomial in the target’s description length; or to prove that such closure forces exponential antichains like the single-culprit class. Definitions are schemas, so the anti-unification bounds of Section 5 may transfer to definition learning; whether proof-generated concepts can be learned with $O(\text{depth})$ objects each is the concrete sub-question. Further open problems (Conjecture 10.27; misspecified practice, such as Leibnizian infinitesimals; moving readings; a corpus test of RCH; identifiability of meaning without step-expressiveness) are collected in Section 14.

11 Contexts, idealization, and checking physics

Physics is where the user’s programme meets its hardest test, and where the problem of contexts cannot be postponed. An olympiad solution is, in his words, “a dance with many clear setups, with some relations to the physical problem of interest”. Some setups are suppositions made for *reductio*. Some are idealizations that contradict known facts (“assume the air pressure is 0”). Some complete a problem that leaves something open (the sun’s elevation is not given). A coherence loss that penalizes good arguments for both P and $\neg P$ would penalize all three. A checker that asks whether each claim is true fails at once, because good solutions are full of literally false claims; the user’s note asks what else it could check: “check that a claim is true in the context at hand. but wtf is that???” (L10 §1.7).

This section answers with a semantics (contexts as filters of models, §11.2), a theory of what coherence should constrain (§11.3), of export (§11.4) and of tolerances (§11.5), and a checker that is sound against every solver relative to a provably ineliminable judgment layer, with a worked run (§§11.6–11.7).

We build a *checker*, not a solver. Gold-level solving of IPhO-style problems under rubric grading has been reported for current AI systems (L9 §5.2); the unmet target is principled checking, and capability work is the part the user’s notes flag as risky. Mathematically, almost everything here is trivial once set up or standard. In T3’s own accounting, Propositions 11.3 to 11.5, 11.40 and J.2, Lemma 11.7, Theorems 11.8, 11.16, 11.23, 11.26, 11.28, 11.30 and 11.37 to 11.39, and Corollary 11.17 are trivial once set up (the content is in the definitions); Łoś’s theorem, Gronwall’s inequality, split conformal prediction, the radius of information, the adversary argument of information-based complexity behind Lemma 11.20 and Theorems 11.21 and 11.30, Lipschitz covering bounds and interval inclusion monotonicity are imported; and Theorem 11.18(c) is an elementary illustration. The small but real contributions (new as far as we know) are the frame-free realizability criterion with anchoring and the necessity of informativeness (Theorem 11.12(a)), the two-sided analysis of adversarial stipulation including the well-posedness certificate (Theorem 11.27), the table of one-sided signals (whose reading entry, Proposition 11.40, is only an observation), and a rigorous thin-leg certificate with a complete worked checker run. Results are from T3 unless marked otherwise; proofs not given inline are in Appendix J.

11.1 The problem: eternalism, contexts, and three import regimes

The simplest coherence constraint is *eternalist*: each proposition is true or false independently of context, and a learner is penalized whenever it can argue for both P and $\neg P$. Physics reasoning violates this constantly and correctly. A hydrostatics problem stipulates $p_{\text{atm}} = 0$, while the background derives $p_{\text{atm}} \approx 1.0 \times 10^5$ Pa. A reductio derives $\perp$ under a supposition. One exam treats the same physics in incompatible models: at IPhO 2025, Theory 2 excludes surface tension and lets a gas phase appear exactly at the equilibrium threshold, while Theory 3 makes surface tension the mechanism of bubble growth (L9 §8.4). The user’s notes record why the two obvious repairs fail. Conjoining every sentence with the whole background framework “leads to just assigning probability 0 to everything”; and proofs by contradiction inside a framework known to be inadequate “might ... really involve subverting the background framework” (L10 §1.8).

Three import regimes. “Suppose” covers three operations that differ in what the new context imports from its parent (L7 §4):

Regime	Example	Imports	$\perp$ inside means	What leaves the context
supposition	“suppose $\sqrt{2} = p/q$ ”	everything	success (reductio)	$\neg A$, or $A \rightarrow B$
counterfactual	“if the atmosphere vanished, the water would boil”	a maximal compatible part	the revision was done badly	$A \Box \rightarrow B$
idealization	“neglect drag”, “ $p_{\text{atm}} = 0$ ”, “massless rope”	a designated minimal fragment	the model is ill-posed	query answers, through a certified bridge

Physics idealizations are *not* counterfactuals. “If the air pressure were 0, the water in the beaker would boil” is a true counterfactual and irrelevant: the problem asks what follows in a designated model that imports incompressible hydrostatics and the data, and nothing else. Similarity semantics (Stalnaker, 1968; Lewis, 1973) or AGM revision (Alchourrón et al., 1985) imports exactly the background (the vapour pressure of water) that makes the context degenerate. Idealization is designated minimal import, as in chunk-and-permeate (Brown and Priest, 2004), McCarthy’s $\text{ist}(c, \varphi)$ (McCarthy, 1993), multi-context systems with local-models semantics (Ghidini and Giunchiglia, 2001), Cartwright’s models (Cartwright, 1983), and compositional modelling with explicit assumptions (Falkenhainer and Forbus, 1991). We keep their shape, “classical inside, filtered between”; what none of them supplies, as far as we know, is different coherence obligations for suppositions and idealizations, and rigorous certificates on the bridges (accuracy reasoning in compositional modelling is mostly qualitative; L7 §7).

Anatomy of a solution. The olympiad solutions decomposed in L9 (§2.1, §8) share one structure. A *root* c_P holds the problem text, figures and conventions; for grading, the “world” is c_P , not the actual world. A *reading* ρ maps c_P to an admissible model family, possibly with free completion parameters, and types the query (“express in terms of a, I_0, r, φ ”). *Local models* are clear mathematical problems, joined by *bridges*: exact ones (symmetry, decomposition, isomorphism) and approximate ones (a small parameter set to its limit, with an error claim), ending in an *export*. Approximate bridges are stipulated by the problem, regime-stated by it, quantified by it, introduced by the solver with an estimate or silently, or conventional; only the

solver-introduced ones need a certificate from the solver. The user’s hypothesis that “there’s a local setting up of a clear thing” holds in all three problems: the unclear part concentrates in the reading and in a few approximate bridges.

11.2 Filter semantics for contexts

Definition 11.1 (Frames, worlds, readings). ^[T3 Def 1.1] Let L be a many-sorted first-order language with a sort $\mathbb{R}$ and parameter constants $p_1, \dots, p_n$ of that sort. (i) A *deformation frame* is a partial map $\mathfrak{M} : \Lambda \rightarrow \text{Str}(L)$, $\Lambda \subseteq \mathbb{R}^n$, with $p_i^{\mathfrak{M}(\lambda)} = \lambda_i$; $\text{dom } \mathfrak{M}$ is where the model exists (is well-posed). In every $\mathfrak{M}(\lambda)$ the sort $\mathbb{R}$ is the ordered field of reals with standard numerals; $\mathcal{K}$ is the class of L -structures with this standard real sort. Ideal values at infinity are reparametrized ($m \rightarrow \infty$ as $1/m \rightarrow 0$). (ii) A *world* is $w = (\mathfrak{M}_w, \lambda_w^*)$ with $\lambda_w^* \in \text{dom } \mathfrak{M}_w$: a top-model family and the actual (or intended) parameter point. (iii) A *reading* ρ yields a set W_ρ of worlds, the *admissible completions* (given parameters fixed, measured ones in intervals, free ones in ranges); a query term Q ; and a set Ax_ρ of sentences true at $\mathfrak{M}_w(\lambda_w^*)$ for every $w \in W_\rho$. The *laws* of a frame are the sentences true at every $\mathfrak{M}_w(\lambda)$, $\lambda \in \text{dom } \mathfrak{M}_w$.

For a projectile, $\mathfrak{M}(\lambda)$ solves $\ddot{\mathbf{x}} = -g\hat{z} - k|\dot{\mathbf{x}}|\dot{\mathbf{x}}$ with parameters (g, v_0, θ, k) , and “neglect air” deforms k toward 0.

Definition 11.2 (Contexts; filter semantics). ^[T3 Defs 1.2–1.3] A *context tree* has root $@$ and parent map π ; each other context is $\text{SUP}(A)$, a supposition of a sentence A ; or $\text{DEF}(D, \lambda^\circ, \mathcal{G})$, a *declared deformation* with deformed parameters $D \subseteq \{1, \dots, n\}$, ideal values $\lambda^\circ \in \mathbb{R}^D$, and a proper filter $\mathcal{G}$ on $\mathbb{R}^D$ converging to λ° (the *path*); or AUX , a definitional extension that shares its parent’s semantics. A DEF context uses *limit semantics* if $\mathcal{G}$ is principal at λ° , and *germ semantics* if $\mathcal{G}$ is the punctured neighbourhood filter, possibly restricted to a curve. Fix a world w , write $\|\varphi\|_w = \{\lambda \in \text{dom } \mathfrak{M}_w : \mathfrak{M}_w(\lambda) \models \varphi\}$ and $\lambda[D:=\mu]$ for λ with its D -coordinates replaced by μ . Each context c gets a (possibly improper) filter $\mathcal{F}_c(w)$ on $\text{dom } \mathfrak{M}_w$: $\mathcal{F}_@(w)$ is principal at λ_w^* ; for $c = \text{SUP}(A)$ with parent p , $\mathcal{F}_c(w)$ is generated by $\mathcal{F}_p(w) \cup \{\|A\|_w\}$; for $c = \text{DEF}(D, \lambda^\circ, \mathcal{G})$, it is generated by the sets $\{\lambda[D:=\mu] : \lambda \in F, \mu \in G\} \cap \text{dom } \mathfrak{M}_w$, $F \in \mathcal{F}_p(w)$, $G \in \mathcal{G}$. *Truth in a context* is $c \models_w \varphi$ iff $\|\varphi\|_w \in \mathcal{F}_c(w)$, and the context-free proposition $\text{ist}(c, \varphi)$ is true at w iff $c \models_w \varphi$. A context is *proper* at w if $\emptyset \notin \mathcal{F}_c(w)$. A claim is *supertrue* under ρ if it is true at every $w \in W_\rho$.

Below a root parent, a germ context makes φ true iff φ holds for all deformed parameters close enough to the ideal along the path, at the actual values of everything else; a limit context is evaluated at the single point $\lambda^*[D:=\lambda^\circ]$, and is improper if no model exists there.

Proposition 11.3 (Classicality without explosion; Łoś). ^[T3 Prop 1.4] Let $\text{Th}_w(c) = \{\varphi : c \models_w \varphi\}$. (a) $\text{Th}_w(c)$ is closed under first-order consequence. (b) It is consistent iff c is proper at w . (c) $\text{Th}_w(c) = \bigcap_U \text{Th}(\prod_U \mathfrak{M}_w)$, where U ranges over the ultrafilters on $\text{dom } \mathfrak{M}_w$ extending $\mathcal{F}_c(w)$ and $\prod_U \mathfrak{M}_w$ is the ultraproduct of the structures $\mathfrak{M}_w(\lambda)$.

Proof. (a) If $\text{Th}_w(c) \models \psi$, compactness gives $\varphi_1, \dots, \varphi_k \in \text{Th}_w(c)$ entailing ψ , and $\|\psi\| \supseteq \bigcap_i \|\varphi_i\| \in \mathcal{F}_c(w)$. (b) If c is proper, every finite subset of $\text{Th}_w(c)$ has nonempty truth set, hence a model; use compactness. If not, $\|\perp\| = \emptyset \in \mathcal{F}_c(w)$. (c) A set lies in a filter iff it lies in every ultrafilter extending it, and Łoś’s theorem (Łoś, 1955) gives $\prod_U \mathfrak{M}_w \models \varphi$ iff $\|\varphi\| \in U$. $\square$

Under limit semantics $\mathcal{F}_c(w)$ is principal or improper, so the ultraproducts in (c) are just the standard model at the context’s point ($\lambda^*[D:=\lambda^\circ]$ for a child of the root). Under germ

semantics with a root parent the ultraproducts are nonstandard models in which the deformed parameters are infinitely close to the ideal along the path and the undeformed ones take *exactly* their actual values. This gives McCarthy’s ist a concrete semantics, and in the germ case it is the sense in which Robinson vindicated Leibnizian infinitesimals (Robinson, 1966). $\text{Th}_w(c)$ may be incomplete (germs can oscillate); that is underdetermination the reading must settle.

Proposition 11.4 (Supposition is exact discharge). *[T3 Prop 1.5] For $c = \text{SUP}(A)$ with parent p : $c \models_w \psi$ iff $p \models_w A \rightarrow \psi$. In particular, c is improper at w iff $p \models_w \neg A$.*

Proof. $X \in \mathcal{F}_c(w)$ iff $X \supseteq F \cap \|A\|$ for some $F \in \mathcal{F}_p(w)$, and $\|\psi\| \supseteq F \cap \|A\|$ iff $\|A \rightarrow \psi\| \supseteq F$. Take $\psi = \perp$. $\square$

Proposition 11.5 (What idealized contexts may do). *[T3 Prop 1.6, rev.] (a) Contradicting the root. In a hydrostatics frame whose laws do not mention phase change (so the model exists at $p_{\text{atm}} = 0$), let c deform p_{atm} toward 0 (limit semantics). Then $c \models p_{\text{atm}} = 0$ and $@ \models p_{\text{atm}} = 1.0 \times 10^5$, both contexts are proper, and no sentence is both true and false in one context. (b) Inconsistent limit, consistent germ. A rope of mass m_r is pulled by a force $\Phi \neq 0$, with $\text{dom} = \{m_r > 0\}$; deform m_r toward 0 below the root. The limit context is improper; the germ context is proper and proves $|a| > K$ for every numeral K . (c) Norton’s distinction. Let $c_{\text{lim}}, c_{\text{germ}}$ be the limit and germ children of the root for the same D, λ° , with $\lambda^*[D := \lambda^\circ] \in \text{dom}$, and let $q(\mu)$ be the value of Q at $\lambda^*[D := \mu]$. Then $c_{\text{germ}} \models Q \in I$ for every open interval $I \ni q(\lambda^\circ)$ iff $q(\mu) \rightarrow q(\lambda^\circ)$ along $\mathcal{G}$ (where the deformed models exist).*

Part (c) is Norton’s “limit property versus property of the limit system” (Norton, 2012): reasoning inside the limit model justifies a value exactly when the query is continuous along the path. D’Alembert’s paradox is the failure case (potential-flow drag is 0; drag as viscosity $\rightarrow 0$ is not). Part (b) is the shape of idealizations that are inconsistent at the limit but consistent along it, such as Painlevé’s rigid bodies with Coulomb friction (Stewart, 2000) or a uniform isothermal gas ball in vacuum (L9 §2.3). *Germ semantics* handles them: well-posedness is required along the family, not at the limit. Different filters on one family give different contexts, so the path to the limit is part of the reading.

Imports are fixed by the deformation. L7 treated the import filter of an idealized context as a third learned object; L9 observed that it is problem-relative.

Definition 11.6 (Stability). *[T3 Def 1.7] φ is D -stable in w if $\mathfrak{M}_w(\lambda) \models \varphi \iff \mathfrak{M}_w(\lambda') \models \varphi$ for all $\lambda, \lambda' \in \text{dom } \mathfrak{M}_w$ differing only in coordinates in D . Laws are D -stable, and so are undeformed data: arithmetic formulas in the $p_i, i \notin D$.*

Lemma 11.7 (Import soundness). *[T3 Lemma 1.8, rev.: (b), (c) added] (a) If φ is D -stable in w and $p \models_w \varphi$, then $c \models_w \varphi$ for every DEF child c of p deforming D , for any parent filter and path. (b) Uniform converse. Fix $\mathfrak{M}_w$ and D . If for every actual point $\lambda^* \in \text{dom } \mathfrak{M}_w$ and every ideal value $\lambda^\circ \in \mathbb{R}^D$ the limit child c of the root satisfies $@ \models \varphi \Rightarrow c \models \varphi$, then φ is D -stable. (c) Scope. For one fixed deformation the converse of (a) fails: with $D = \{k\}$, $\lambda^\circ = 0$, limit semantics and $\varphi := (k \leq 1)$, importing φ is sound at every world, yet φ is not D -stable if dom contains points differing only in k with $k = 0.5$ and $k = 2$.*

Proof idea. (a) For $\lambda \in \|\varphi\|$, every $\lambda[D := \mu] \in \text{dom}$ differs from λ only on D , so the generator built from $F = \|\varphi\|$ lies in $\|\varphi\|$. (b) Put the actual point at a φ -point λ and the ideal value at λ'_D . (c) The child always satisfies its stipulation $k = 0$. Details in Appendix J. $\square$

Reading. The imports that are sound *uniformly* in the actual point and the ideal value are exactly what the declared top-model family makes stable. “Water’s vapour pressure is 2.3 kPa” is a law of a frame with phase equilibrium; there it is importable, and the $p_{\text{atm}} \rightarrow 0$ context correctly boils. In incompressible hydrostatics without phases it is not in the language. So learning uniformly sound import filters reduces to learning which top-model family the problem intends: part of the reading (§11.6).

11.3 What coherence should constrain

Theorem 11.8 (No truth-functional eternal reading). *[T3 Thm 2.1, rev.] Let c be a DEF context, proper at w , with a stipulation S_c such that $c \models_w S_c$ and S_c is false at $\mathfrak{M}_w(\lambda_w^*)$ (e.g. $p_{\text{atm}} = 0$). Let $\tau(c, \varphi)$ translate “ φ holds in c ” into a world sentence whose truth value at w is $f(S_c^w, \varphi^w)$ for some $f : \{0, 1\}^2 \rightarrow \{0, 1\}$. Then τ does not track in-context truth at w : $c \models_w \top$ and $c \not\models_w \neg S_c$, but $\tau(c, \top)^w = \tau(c, \neg S_c)^w$.*

Proof. $\top$ and $\neg S_c$ are both true at w , so both translations have value $f(0, 1)$. And $c \models_w S_c$ with $\text{Th}_w(c)$ consistent (Proposition 11.3(b)). $\square$

Since S_c is false at w , τ depends only on $f(0, \cdot)$. *Stripping* ($f = \wedge$) makes every in-context claim false at w , hence incoherent with the root; *conditionalizing* ($f = \Rightarrow$) makes every one true, hence vacuous; the other two choices collapse “in c ” into “at w ” or its negation. (Properness is needed: for the improper rope limit of Proposition 11.5(b), conditionalizing tracks in-context truth perfectly.)

Corollary 11.9 (Eternalist coherence penalizes correct reasoning). *[T3 Cor 2.2, rev.] Call a check eternalist if it flags a set J of context-indexed judgments (c, Γ, φ) whenever $\{\bigwedge \Gamma \rightarrow \varphi : (c, \Gamma, \varphi) \in J\}$ is unsatisfiable (context indices deleted, suppositions not added). It flags (i) every sound reductio whose internal judgment $(\text{SUP}(A), \emptyset, \perp)$ is recorded; (ii) every sound idealization whose stipulation contradicts a recorded root judgment (Proposition 11.5(a)); (iii) every pair of sound contexts whose different answers to one query are both recorded (IPhO 2025 T2 versus T3). If only the discharged $(\text{@}, \emptyset, \neg A)$ is recorded, nothing is flagged: what is penalized depends on how much of the reasoning is recorded.*

A learner trained to keep such sets coherent loses classical logic, and not only in the offending context. This is T2’s formal version of the user’s air-pressure worry.

Proposition 11.10 (Mis-designation forces global sub-classicality). *[T2 Prop 7.1] Let C be a structural consequence relation that is $\mathcal{A}$ -coherent ($A \not\vdash_C \perp$ for $A \in \mathcal{A}$), where some designated $A \in \mathcal{A}$ is classically inconsistent. (a) $C \not\supseteq \text{Cn}_{\text{CPC}}$; moreover, for every classical derivation of $\perp$ from A in a Hilbert or sequent-step format (no discharge metarules), some rule schema used in it is not C -valid: its most general instance is not in C . (b) If $A \supseteq \{\alpha, \neg\alpha\}$, then $p, \neg p \not\vdash_C \perp$ for every atom p : C is atomically paraconsistent everywhere.*

Proof. (a) If every step instance were in C , chaining would give $A \vdash_C \perp$; the missing instance is a substitution instance of its schema’s most general instance, which by structurality is then missing too. (b) Substitute α for p and weaken to A . $\square$

Reading. Designate “ $p_{\text{atm}} = 0$ plus the full background” as coherent, and a coherence loss teaches a sub-classical logic for *all* reasoning: if the background contains $p_{\text{atm}} \neq 0$ outright, the learner must reject $p, \neg p \vdash \perp$ for atoms, in every context; if it only derives it, the learner must reject atomic explosion or some classical schema used in that derivation. The fix has three parts. Designate only consistent chunks (Brown and Priest, 2004). Keep the root *phenomenological* (interval claims about measurables, observations, exports), with laws living in model contexts (Cartwright, 1983); physics is a patchwork of locally valid inferential practices (Wilson, 2006), and a root holding unrestricted laws may itself be inconsistent (L7 §12). And adopt *meta-level eternalism*: $\text{ist}(c, \varphi)$ is an eternal proposition about a finitely specified context (McCarthy, 1993), by Theorem 11.8 necessarily non-truth-functional in the world’s facts. Which contexts, then, must be consistent?

11.3.1 The right constraint: realizability

Definition 11.11 (Judgment sets, bridges, realizations). [T3 Def 2.3, rev.] “Model” and “satisfiable” refer to $\mathcal{K}$; for a set M of structures, $M \models \chi$ means $N \models \chi$ for all $N \in M$. A *judgment set* J is a finite set of triples (c, Γ, φ) over a context tree in which SUP contexts have only SUP descendants; a set C_{cert} of DEF contexts is *certified* (claimed well-posed). A *bridge* β has a pattern $\phi_\beta(v)$, a side condition σ_β and an export formula $\varepsilon_\beta(v)$, $v \in V$. It is a *value-export bridge* for Q if $\phi_\beta(v)$ is $Q = \bar{v}$ with standard names (distinct values denote distinct objects in $\mathcal{K}$) and σ_β does not depend on v ; it is *informative* if $\{\sigma_\beta\} \cup \{\varepsilon_\beta(v) : v \in V_0\}$ has no model for some *finite* $V_0 \subseteq V$ (e.g. interval exports over unbounded V). A *bridge use* is $(c, \beta, t) \in U$ with c a DEF context and $(c, \emptyset, \phi_\beta(t)), (\pi c, \emptyset, \sigma_\beta), (\pi c, \emptyset, \varepsilon_\beta(t)) \in J$. A (*frame-free*) *realization* assigns each context a set $M_c \subseteq \mathcal{K}$ with (R0) $M_{@} \neq \emptyset$; (R1) $M_c = M_{\pi c} \cap \|A\|$ for $c = \text{SUP}(A)$; (R2) M_c arbitrary, possibly empty, for DEF contexts; (R3) every judgment holds: for $N \in M_c$, $N \models \bigwedge \Gamma$ implies $N \models \varphi$; (R4) every used bridge is sound: for all v , $M_c \models \phi_\beta(v)$ implies $M_{\pi c} \models \sigma_\beta \rightarrow \varepsilon_\beta(v)$; (R5) $M_c \neq \emptyset$ for $c \in C_{\text{cert}}$. The *SUP-collapse* J^* replaces each (c, Γ, φ) with c a SUP context by $(b, \Gamma \cup \text{Asm}(c), \varphi)$, where b is the nearest non-SUP ancestor and $\text{Asm}(c)$ the suppositions on the path; $\text{Th}^*(c) := \{\bigwedge \Gamma \rightarrow \varphi : (c, \Gamma, \varphi) \in J^*\}$. The *anchored set* Anc is the least set of non-SUP contexts containing $@$ and C_{cert} and containing c whenever $(c, \beta, t) \in U$ with β informative and $\pi c \in \text{Anc}$.

Informativeness is defined in finite form because $\mathcal{K}$ is not compact: $\{Q \neq v : v \in \mathbb{R}\}$ is finitely satisfiable but has no model with a standard real sort.

Theorem 11.12 (Realizability criterion). [T3 Thm 2.4, rev.: (a) frame-free; (b), (c) new] Suppose every used bridge is an informative value-export bridge. (a) Frame-free. J has a realization satisfying (R0)–(R5) iff $\text{Th}^*(c)$ is satisfiable for every $c \in \text{Anc}$. (b) Necessity in the frame semantics. Call J frame-realizable if for some world w , with the filters of Definition 11.2, every judgment holds ($c \models_w \bigwedge \Gamma \rightarrow \varphi$), every $c \in C_{\text{cert}}$ is proper, and every bridge use is sound at w ($c \models_w \phi_\beta(v)$ implies $\pi c \models_w \sigma_\beta \rightarrow \varepsilon_\beta(v)$). If J is frame-realizable, the criterion of (a) holds, for limit or germ paths. (c) Non-sufficiency. The converse of (b) fails. CE1: a certified limit child c of the root deforms $D = \{2\}$ toward 0, and $J = \{(@, \emptyset, p_1 = 1), (c, \emptyset, p_1 = 0)\}$; the criterion holds, but the root forces $\lambda_1^* = 1$, and c , proper because certified, sits at $\lambda^*[2:=0]$, so $c \models p_1 = 1$. CE2: two certified siblings with the same declaration have equal filters, so $Q = 1$ in one and $Q = 2$ in the other cannot both hold.

Proof idea. By (R1), SUP contexts impose on their base exactly the constraints Th^* . An *anchoring lemma* says every anchored context has $M_c \neq \emptyset$: from an empty M_c an informative

bridge would export every $\varepsilon_\beta(v)$, forcing a model of $\{\sigma_\beta\} \cup \{\varepsilon_\beta(v) : v \in V_0\}$ in $M_{\pi c}$. Conversely, put one model of $\text{Th}^*(c)$ at each anchored context and $\emptyset$ elsewhere; standard names make (R4) bite only at $v = t$. Part (b) is the same argument with properness for nonemptiness. Full proof in Appendix J. $\square$

Brute force agrees with (a) on 6,500 random instances (including 2,000 from a referee’s deeper stress test) with 0 mismatches; with uninformative bridges (a) fails on 117/600, so the hypothesis is needed; and over all frames on $\Lambda = \{0, 1\}^2$, 8,000 instances show 0 violations of (b) and 598 in the gap of (c) (computed). The locality of (a) is a *consequence of (R2)*, not a fact about the frame semantics, where rigid parameters and shared filters force cross-context constraints. Under limit semantics these are characterized exactly (Proposition J.2, Appendix J): the theories of all contexts forced to be proper *at one parameter point* must be jointly satisfiable with that point’s values. That is pointwise adjunction, the structural core of discussive and preservationist logics (Jaśkowski, 1969; Schotch and Jennings, 1980). Germ semantics is open.

Corollary 11.13 (Reductio immunity; well-posedness where it matters). *[T3 §2.2 Cors, rev.] Under the hypotheses of Theorem 11.12: (i) a SUP context deriving $\perp$ contributes exactly $\neg \wedge$ Asm to its base, in both semantics; (ii) if an idealized context deriving $\perp$ is anchored, J is frame-free unrealizable, and if it is not, that judgment imposes no constraint on frame-free realizability; in the frame semantics the first half holds, the second can fail; (iii) eternalist coherence is strictly stronger than frame-free realizability.*

Reading. “Arguments for P and $\neg P$ in contradictory contexts should not be penalized” is correct, and Theorem 11.12(a) says exactly which contexts *must* be consistent: the root, and idealizations that are certified or export informatively. This is L7’s “ill-posed idealizations must not export”, now an iff. In the frame semantics the criterion is necessary but not sufficient, and the remaining constraints (CE1, CE2) are enforced in the checker by a declaration discipline, not by coherence (§11.6).

Remark 11.14 (Belief states, worlds, contextuality). Section 9 proves a completeness theorem for monotone multi-context calculi with local-models semantics: truth in a context is truth in every local model of the canonical belief state, and coherence on designated contexts is equivalent to a model nonempty there (T6 Thm 4.1, Cor 4.2). It also shows that bridges must be read as *rules of proof* over belief states, since reading them as conditionals between single worlds strictly strengthens entailment (T6 Thm 4.5); and that locally coherent contexts need not glue into one world, as in the odd cycle of three pairwise-conservative contexts, a logical form of contextuality (Abramsky and Brandenburger, 2011) (T6 Prop 4.7).

11.3.2 A calculus, its soundness, and blame localization

Definition 11.15 (Context calculus). *[T3 §2.3, rev.]* Judgments are $c \Vdash \varphi$; $\hat{R}$ is the (learned) rule set and Imp_c the declared import set of c . (Ax) $c \Vdash \varphi$ for $\varphi \in \text{Ax}(c)$, where $\text{Ax}(@) = \text{Ax}_\rho$, $\text{Ax}(\text{SUP}(A)) = \{A\}$, and $\text{Ax}(\text{DEF}(D, \lambda^\circ, \mathcal{G}))$ is the set of *canonical stipulations*: $p_i = \lambda_i^\circ$ ($i \in D$) under limit semantics, $p_D \in U$ for definable $U \in \mathcal{G}$ under germ semantics; stipulations about undeclared parameters are not axioms. (Step) From $c \Vdash \varphi_1, \dots, c \Vdash \varphi_k$ and an instance $\varphi_1, \dots, \varphi_k \triangleright \psi$ of $\hat{R}$, infer $c \Vdash \psi$. (Imp) From $\pi c \Vdash \varphi$ with $\varphi \in \text{Imp}_c$, infer $c \Vdash \varphi$. (Dis) For $c = \text{SUP}(A)$: from $c \Vdash \psi$ infer $\pi c \Vdash A \rightarrow \psi$. (Exp) For a DEF context c : from $c \Vdash \phi_\beta(t)$, $\pi c \Vdash \sigma_\beta$ and $\pi c \Vdash \omega_c$, infer $\pi c \Vdash \varepsilon_\beta(t)$; here the *well-posedness certificate* ω_c is a sentence such that for

every w and $\lambda \in \|\omega_c\|_w$, $\{\mu : \lambda[D:=\mu] \in \text{dom } \mathfrak{M}_w\} \in \mathcal{G}$. An instance is *sound at w* if it preserves truth there (for Step: $c \models_w \bigwedge_i \varphi_i \rightarrow \psi$), and *sound* if sound at every $w \in W_\rho$.

Step soundness is *context-relative*: time reversal is sound in a drag-free DEF context and unsound at the root.

Theorem 11.16 (Soundness). *[T3 Thm 2.5] If every Ax in a derivation of $c \Vdash \varphi$ is true at w and every Step, Imp and Exp instance in it is sound at w , then $c \models_w \varphi$.*

Proof. Induction: Ax and Step by hypothesis and filter closure (Proposition 11.3(a)); Imp and Exp by soundness of the instance; Dis by Proposition 11.4. $\square$

Corollary 11.17 (Blame localization). *[T3 Cor 2.6, rev.] Let $W_\rho \neq \emptyset$. Call d designated if it is the root, or a certified DEF context with a properness witness $w_d \in W_\rho$ at which d is proper. Let $\mathcal{D}$ derive $d \Vdash \perp$, and let its cone be all instances in its derivation tree, including those in sub-contexts whose conclusions reach d by Dis or Exp. (a) At every w at which d is proper (every $w \in W_\rho$ if $d = @$), the cone contains an Ax false at w or an instance unsound at w . (b) Items outside the cone are never implicated: an undischarged SUP contradiction, or a pair of contradictory ist-claims. (c) If $\pi d \models_w \omega_d$ and πd is proper at w , then d is proper at w ; so parent-derived certificates are properness witnesses.*

Proof idea. (a) is the contrapositive of Theorem 11.16; (b) follows. (c) In each parent filter set pick $\lambda \in \|\omega_d\|$; its deformed models exist for $\mathcal{G}$ -almost all μ . Proposition J.3 adds how $\perp$ in an anchored DEF context propagates to a designated one, and where the blame then falls. $\square$

The cone is a *negative bag* in the sense of Section 6, with context-indexed items (context and Step, Imp or Exp instance; Ax judgments). The halving argument there uses only that the target never contains a whole detected bag, so with the root and the certified contexts as the *only* designated contexts, an oligarchic-halving learner makes at most $\log_2(1/w^*)$ detections and no false alarms while the reading is correct (T2 Thm 2.2). What root coherence cannot see is silent unsoundness: exports that are false but consistent with everything else (§11.5).

11.3.3 Reductio hygiene, and 100 = 99.9

The user warns that a proof by contradiction inside a known-inadequate framework may “really involve subverting the background framework ... and there’s a chance such an argument could equally be provided starting from the statement itself” (L10 §1.8). Let c have an explicit *imported fragment* K_c , a finite set of premises. A *reductio of A in c* is an $\hat{R}$ -derivation of $\perp$ from $K_c \cup \{A\}$ in $\text{SUP}(A)$; accepting it yields $c \Vdash \neg A$. An instance is *c -sound at w* if $c \models_w \bigwedge \varphi_i \rightarrow \psi$.

Theorem 11.18 (Reductio hygiene). *[T3 Thm 1.9, rev.: (c)–(e)] (a) Soundness. If $c \models_w K_c$ and every instance used is c -sound at w , every accepted reductio of A yields $c \models_w \neg A$. (b) Inconsistent imports refute everything. If $K_c \vdash_{\hat{R}} \perp$, every sentence, A and $\neg A$ alike, has a reductio in c , whatever the world. (c) Derivation-local tests fail. Let $K = \{A \rightarrow q, \neg A \rightarrow q, \neg q\}$. K is unsatisfiable, but the reductio of A uses only the satisfiable $\{A \rightarrow q, \neg q\}$ and that of $\neg A$ only the satisfiable $\{\neg A \rightarrow q, \neg q\}$; any test that is a function of the derivation alone and accepts every sound reductio from a consistent premise set accepts both. (d) A consistency certificate suffices. If a structure N satisfies K_c and every instance used ($N \models \bigwedge \varphi_i \rightarrow \psi$), every reductio-refuted A is false in N , so no A and $\neg A$ are both refuted. The condition on instances is automatic for logically sound $\hat{R}$ and needed otherwise (if $c \models q$, the instance $\triangleright q$ is c -sound, and $K_c = \{\neg q\}$, which has*

a model, refutes everything). If c is proper at w , $c \models_w K_c$ and the instances are c -sound at w , a suitable $N = \mathfrak{M}_w(\lambda)$ exists: a properness witness is a certificate. (e) Learning consequence. A coherence loss fed with reductios from a context whose fragment is $\hat{R}$ -inconsistent receives “ A false” and “ $\neg A$ false” for every A : jointly unsatisfiable, systematic, world-independent labels.

Proof idea. (a) $\text{SUP}(A)$ refines c ’s filter, so c -sound instances and K_c stay true there; induction gives $\text{SUP}(A) \models_w \perp$, and Proposition 11.4 gives $\neg A$. (b) Monotonicity in premises. (c) Truth tables; the reductio of A is literally the sound reductio from the consistent $\{A \rightarrow q, \neg q\}$ (hygiene.py). (d) Induction inside N . Details in Appendix J. $\square$

Reading. The user is right that a reductio in an inadequate framework can be spurious, and the remedy is not to abandon reductio. Hygiene is a *global* certificate on the imported fragment, not a syntactic relevance condition on the derivation, which (c) shows cannot work. In physics the certificate is a well-posedness witness, an explicit or validated-numerical solution of the local model; L9’s solution format demands one for every local model.

Proposition 11.19 (The continuity asymmetry). ^[T3 Prop 1.10] Let premises $x_i = m_i$ hold only to tolerance, $x_i \in [m_i \pm r_i]$, with box B . (a) A direct derivation of $y = g(x)$, g continuous on B , errs by at most g ’s modulus of continuity on B . (b) Exact-equality reasoning derives $\perp$ from approximately true premises read as exact: $x = 100\text{ m}$ (flat Earth) and $x = 99.9\text{ m}$ (sphere), both true to $\pm 0.2\text{ m}$, give $0 = 0.1$ by substitution, while as intervals they give $x \in [99.8, 100.1] \neq \emptyset$. (c) Sound replacement. If A with the exact laws implies $f(x) = 0$, and F is an inclusion-monotone interval extension of f with $0 \notin F(B)$, the interval premises refute A soundly; the condition is sufficient, not necessary.

Proof. (a) Definition. (b) Arithmetic. (c) Inclusion monotonicity (Moore, 1966) gives $f(B) \subseteq F(B) \not\ni 0$. $\square$

Reading. The user asks: “how come we usually avoid this sort of catastrophic reasoning [100 = 99.9]? i don’t think any nice ‘formal criterion’ is going to be right?” (L10 §1.8). Here is a partial answer, a sufficient condition rather than a definition of good reasoning: “ $x = y$ ” defines a closed set of data with empty interior, so derivability of $\perp$ is a *discontinuous* function of the data, while derived values are continuous ones. This is why his “constructive” reasoning is more trustworthy in messy domains. The fix is interval typing with a *margin*; Theorem 11.26(c) shows the catastrophe as a budget violation.

11.4 Export: what can justify leaving a context

Fix a one-parameter deformation (the multi-parameter case is identical) with ideal value 0 and actual value $\lambda^* \in (0, \Lambda]$. A family is represented by its query function $Q : [0, \Lambda] \rightarrow \mathbb{R}$; a *family class* is a set $\mathcal{C}$ of them. An *information map* I gives what an export rule may use (the jet at 0, the germ, a restriction $Q|_S$, finitely many values). An *export rule* E maps $(I(Q), \lambda^*)$ to “abstain” or a finite interval $[q \pm \epsilon]$, and is *sound on* $\mathcal{C}$ if $Q(\lambda^*) \in E(I(Q), \lambda^*)$ whenever it does not abstain.

Lemma 11.20 (Invisible perturbation). ^[T3 Lemma 3.2] If some f with $f(\lambda^*) \neq 0$ has $Q + Af \in \mathcal{C}$ and $I(Q + Af) = I(Q)$ for all $Q \in \mathcal{C}$, $A \in \mathbb{R}$, then every export rule sound on $\mathcal{C}$ abstains at λ^* on every $Q \in \mathcal{C}$.

Proof. An output $[q \pm \epsilon]$ on $I(Q) = I(Q + Af)$ would have to contain $Q(\lambda^*) + Af(\lambda^*)$ for every A . $\square$

This is the standard adversary argument of information-based complexity (Traub et al., 1988).

Theorem 11.21 (No free export). *[T3 Thm 3.3, rev.: closure hypothesis] Every rule sound on $\mathcal{C}$ abstains at every $\lambda^* > 0$, on every $Q \in \mathcal{C}$, if (a) $\mathcal{C} + C^\infty[0, \Lambda] \subseteq \mathcal{C}$ and I is the full jet at 0; or (b) $\mathcal{C} + C^\infty[0, \Lambda] \subseteq \mathcal{C}$ and I is the germ at 0, or $Q|_S$ for any S whose closure omits λ^* ; or (c) $\mathcal{C}$ consists of restrictions of entire functions and I is the jet up to a finite order N ; or (d) $\mathcal{C} + C^\infty[0, \Lambda] \subseteq \mathcal{C}$ and I is any finite set of values $Q(\lambda_j)$, $\lambda_j \neq \lambda^*$. With only $\mathcal{C} \supseteq C^\infty$ in (a), (b), (d), the conclusion holds for $Q \in C^\infty$ but can fail outside it.*

Proof idea. Lemma 11.20 with the flat e^{-1/λ^2} in (a), a C^∞ bump at λ^* vanishing near S in (b) and (d), and λ^{N+1} in (c). For the weaker hypothesis, $\mathcal{C} = C^\infty \cup \{\sqrt{\lambda}\}$ with the germ admits a sound non-abstaining rule on $\sqrt{\lambda}$. Details in Appendix J. $\square$

For analytic families the full jet determines $Q(\lambda^*)$, but a sound use of a series needs a remainder bound (e.g. a Cauchy bound on a disc of radius $R > \lambda^*$), which is information *away* from 0: “an expansion without a remainder bound is uncontrolled”. This vindicates McMullin’s and Laymon’s demand for de-idealization information (McMullin, 1985; Laymon, 1987).

Proposition 11.22 (Minimax export is the radius of information). *[T3 Prop 3.4; Traub et al. 1988] For an information value i let $\mathcal{C}_i = \{Q \in \mathcal{C} : I(Q) = i\}$ and $r(i) = \frac{1}{2}(\sup_{\mathcal{C}_i} Q(\lambda^*) - \inf_{\mathcal{C}_i} Q(\lambda^*))$. A sound rule with tolerance ϵ at i exists iff $r(i) \leq \epsilon$; the optimal one outputs the midrange.*

Proof. A sound interval must contain $\{Q(\lambda^*) : Q \in \mathcal{C}_i\}$, and the midrange interval of radius $r(i)$ does. $\square$

So an *export certificate is information of small radius*: a uniform derivative bound, a Gronwall bound, an invariance that pins Q , a monotone sandwich between the ideal and a computed less-ideal value, or data near λ^* .

Theorem 11.23 (Certified-neighbourhood and rate-qualified export). *[T3 Thm 3.5] (a) If the parent proves $\lambda^* \in U$ and a frame theorem gives $\sup_{\lambda \in U} |Q(\mathfrak{M}(\lambda)) - q| \leq \epsilon$, the parent may soundly assert $Q \in [q \pm \epsilon]$. (b) A germ claim $Q \sim q$ exports nothing at a fixed λ^* ; with a rate $|Q/q - 1| \leq C\|\lambda\|^k$ on U it exports $|Q/q - 1| \leq C \sup_U \|\lambda\|^k$. (c) If the problem declares a regime, the root is a germ context and asymptotic equivalence is the correctness relation itself; for exp-log expressions in one parameter it is decidable modulo zero-equivalence of constants (Gruntz, 1996; Richardson, 1968).*

Part (a) fixes the logical form of every side condition: a parent-certified set plus a uniform bound over it, not a value at a guessed point. (At IPhO 2025 T2 C.3 the official premise “ X is a few cm” is false, but its conclusion is robust over the whole apparatus range; L9 §8.3.)

Theorem 11.24 (Gronwall finite-horizon export). *[T3 Thm 3.6, rev.: (b) new; Gronwall 1919] Let $\dot{x} = f(x)$ and $\dot{y} = f(y) + g(t, y)$ have solutions on $[0, T]$, $L > 0$, and $B(t) = |x_0 - y_0|e^{Lt} + \frac{\eta}{L}(e^{Lt} - 1)$. (a) If f is L -Lipschitz on a set K containing both trajectories and $|g(t, y(t))| \leq \eta$, then $|x(t) - y(t)| \leq B(t)$. (b) Checkable form. Let K_r be the closed tube of radius r around $x([0, T])$. If f is L -Lipschitz on K_r , $|g| \leq \eta$ on $[0, T] \times K_r$, and $B(T) < r$, then y stays in K_r and the bound of (a) holds.*

Every hypothesis of (b) concerns the computed idealized trajectory and explicit bounds, so the parent can certify it; those of (a) concern the unknown trajectory. The bound grows with T : qualitative long-time claims from structurally unstable idealizations do not export, finite-horizon quantitative ones do.

Theorem 11.25 (Rigorous projectile certificate). *[T3 Thm 3.7] A point mass is launched with speed v_0 at angle $\theta \in (0, \pi/2)$ under gravity g , with neglected force per unit mass $a_n = -\kappa(t)v$, $0 \leq \kappa(t) \leq k|v(t)|$ (quadratic drag with any $C_d \leq C_{d,\max}$, or any antiparallel force dominated by it, time-varying ones included; lift is not covered). With $x := kv_0^2/g < 1$ and $R_0 = v_0^2 \sin 2\theta/g$,*

$$R_0 \left[\frac{1}{1+x} - \frac{x \tan \theta}{(1+x)^2} \right] \leq R \leq \frac{R_0}{1-x}.$$

Proof idea. Energy decreases, so $|v| \leq v_0$ and $|a_n| \leq xg$; this sandwiches the vertical acceleration, hence the flight time, and the horizontal velocity only decreases. Details in Appendix J. $\square$

Numbers (computed). Steel ball ($r = 1$ cm, $m = 32.7$ g, $C_d = 0.47$, $v_0 = 10$ m/s, 45°): $x = 0.0276$, certified $R \in [9.653, 10.483]$ m, simulated 9.978 m. Ping-pong ball: $x = 1.42$; the certificate is silent and the export blocked, while the simulated range is 49% short of R_0 , although the in-context derivation is word for word the same. Random ($x \in [0, 0.9], \theta$): 0/300 violations; a referee’s bang-bang $\kappa(t)$: 0/400. At 45° the 5% validity region is $x \leq (1/0.95)^{1/2} - 1 = 0.0260$ certified, against $x \leq 0.0679$ true: sound and about $2.6\times$ conservative.

Theorem 11.26 (Error propagation through chains). *[T3 Thm 3.8, rev.] (a) For a chain $\lambda^\circ = \lambda^{(0)}, \dots, \lambda^{(k)} = \lambda^*$ with certified $|Q(\lambda^{(i)}) - Q(\lambda^{(i-1)})| \leq \delta_i$, $|Q(\lambda^*) - Q(\lambda^\circ)| \leq \sum \delta_i$. (b) If the answer is $F(q_1, \dots, q_m)$ from exports with $|Q_j - q_j| \leq \delta_j$, and F is differentiable on the box $\prod_j [q_j \pm \delta_j]$ with $L_j = \sup |\partial_j F|$ there, then $|F(Q) - F(q)| \leq \sum_j L_j \delta_j$. (c) If $\sum_j L_j \delta_j \geq |F(q)|$, not even the sign of $F(Q)$ is certified by this budget. For “100 – 99.9” with tolerances 0.2 the budget is $0.4 > 0.1$, and indeed $Q_1 - Q_2 \in [-0.3, 0.5]$.*

(a) is the triangle inequality, (b) the mean value theorem on the convex box. A direct range enclosure can still certify a sign the budget cannot: $F(q) = q^2$ at $q = 1$ with $\delta = 0.6$ has budget 1.92, yet $F(Q) \in [0.16, 2.56]$.

Adversarial stipulation. Context choice is an attack surface. Take a schema “neglect p_D ”: pattern $Q = \bar{v}$ in the child, side condition $s(\bar{p}) \leq \eta$ for a dimensionless smallness parameter (e.g. kv_0^2/g), conclusion $Q \in [v \pm \epsilon]$ in the parent.

Theorem 11.27 (Where side conditions live, and what contexts may be). *[T3 Thm 3.9, rev.: (c)] (a) Child-evaluated side conditions are unsound. If Exp accepts $c \Vdash s \leq \eta$ in place of $\pi c \Vdash s \leq \eta$, the export fires on every world (at $p_D = 0$, $s = 0$); on any C^∞ -closed class this is unsound for every ϵ . The ping-pong ball gets the drag-free range $\pm 5\%$, while its true range is 49% short. (b) Free stipulations are unsound even with parent evaluation. If contexts may set undeclared parameters, a solver stipulates $p_D = 0$ and $g_c = g'$; the parent condition holds for the steel ball, and the child’s $Q = R_0(g')$ is any desired value. (c) Sufficiency. Suppose contexts are declared deformations with only canonical stipulations as axioms; σ_β and a well-posedness certificate ω_c are derived in the parent; and the schema is a theorem of the frame: for all w , all $\lambda \in \text{dom } \mathfrak{M}_w$ with $\mathfrak{M}_w(\lambda) \models \sigma_\beta$ and all v , if $\{\mu : \lambda[D := \mu] \in \text{dom } \mathfrak{M}_w, Q(\mathfrak{M}_w(\lambda[D := \mu])) = v\} \in \mathcal{G}$ then $\mathfrak{M}_w(\lambda) \models \varepsilon_\beta(v)$. Then every accepted export is sound, for every solver. Without ω_c , (c) is false: in the rope frame of Proposition 11.5(b) under limit semantics, the improper child proves $a = 0$, the schema holds vacuously, the parent proves $m_r \leq \eta$, and Exp would accept $|a| \leq \epsilon$ while $|a| = |\Phi|/m_r \geq |\Phi|/\eta$.*

Proof idea. (a), (b) are the constructions. For (c), the child-filter generator witnessing $Q = \bar{v}$, intersected with the parent sets for σ_β and ω_c , puts the schema’s antecedent at every point of a parent filter set. Details in Appendix J. $\square$

Theorem 11.25 is such a schema: σ is $x \leq \eta$ with $\eta < 1$, $\omega = \top$, and $\epsilon(v, \theta) = v \max(\frac{\eta}{1-\eta}, \frac{\eta}{1+\eta} + \frac{\eta \tan \theta}{(1+\eta)^2})$. Germ contexts naturally conclude $Q \simeq v$ (standard part v); the analogous asymptotic schema is sound below a principal parent given ω_c (Lemma J.4), and under a nested germ parent it also needs uniform convergence on a parent filter set.

Reading. This is the search problem of Section 3 transplanted to idealizations: a solver chooses contexts as a prover chooses steps. The remedy is the analogue of cautious acceptance: contexts restricted to declared deformations with canonical stipulations (Strevens’s “default values” made into a syntactic discipline; Strevens 2008), side conditions evaluated in the parent, and a parent-derived well-posedness certificate.

11.5 Learning tolerances and validity regions

A bridge schema s has a dimensionless regime parameter $\pi \in \Pi$ and an error function $e_s(\pi)$, the relative difference between top model and idealization. How can one learn *where* $e_s \leq \tau$, soundly against a solver who chooses π ?

Theorem 11.28 (Coherence cannot calibrate tolerances). *[T3 Thm 4.1] Schemas $s = 1, \dots, m$ export $[q_{s,i} \pm \epsilon_s]$ for true values V_i ; a coherence-only calibrator chooses $\hat{\epsilon}$ from the discrepancies $(q_{s,i} - q_{s',i})$ alone. If the environment class is closed under common shifts $((q, V)$ admissible implies $(q, V + b)$ admissible for $|b| \leq \beta$: all models may share an omitted effect), every $\hat{\epsilon}$ sound on the class has $\hat{\epsilon}_s \geq \beta$; with unbounded shifts, $\hat{\epsilon} = \infty$.*

Proof. $(q, V \pm \beta)$ give the same data, hence the same $\hat{\epsilon}$, and soundness for both signs gives $2\beta \leq 2\hat{\epsilon}_s$. $\square$

Worse, coherent tolerance vectors form an up-set, so a minimal coherent repair only ever widens tolerances (T3 Prop 4.2): exporting a pendulum’s T/T_0 by the small-angle schema and the exact one at amplitudes up to 179.9° drives the small-angle tolerance from 0.0077 to 4.4, and coherence never restricts the schema’s regime (computed). Two small-angle-based schemas never disagree, so both export ratio 1 at 120° , where the truth is 1.373: common-mode error, which is why agreement among models sharing an idealization is not independent evidence (Levins, 1966; Odenbaugh and Alexandrova, 2011).

Theorem 11.29 (Split conformal, and its failure under selection). *[T3 Thm 4.3] (a) If $e_1, \dots, e_{n+1}$ are exchangeable and $\hat{\epsilon}$ is the k -th smallest of $e_1, \dots, e_n$, $k = \lceil (n+1)(1-\alpha) \rceil$ ($\hat{\epsilon} = \infty$ if $k > n$), then $\mathbb{P}(e_{n+1} \leq \hat{\epsilon}) \geq 1 - \alpha$ (Vovk et al., 2005; Lei et al., 2018). (b) If the solver picks the instance after $\hat{\epsilon}$ is fixed, from a regime where $\sup e > \hat{\epsilon}$, coverage on it is 0.*

Calibrated on $\theta_0 \sim U[0^\circ, 30^\circ]$ ($n = 200$, $\alpha = 0.1$), the small-angle schema gets coverage 0.913; the adversary’s 60° has error 7.3×10^{-2} against $\hat{\epsilon} = 1.44 \times 10^{-2}$ (computed). Mondrian (per-regime) calibration narrows the gap but cannot close it: this is the commitment problem of Section 3 again.

Theorem 11.30 (No certification without regularity). *[T3 Thm 4.4] Let $\mathcal{C}$ be closed under $e \mapsto e + Af$ for every continuous bump $f \geq 0$ and $A > 0$. A deterministic certifier that makes finitely many oracle queries and outputs a region V with $e \leq \tau$ on V , soundly on $\mathcal{C}$, has V inside the set of queried points. A randomized certifier that is sound almost surely satisfies this inclusion almost surely.*

Proof idea. A bump at an unqueried $\pi \in V$, vanishing at the queries, leaves the adaptive transcript unchanged and lifts $e(\pi)$ above τ . Details in Appendix J. $\square$

No free export and no free certification rest on one argument: simulation is an oracle for the top model, but without a regularity guarantee it certifies only the points queried. With regularity, sound certification is possible at a tight price:

Theorem 11.31 (Conservative Lipschitz certifier). *[T3 Thm 4.5] Let $\Pi = [0, 1]^d$, e be L -Lipschitz in $\|\cdot\|_\infty$, and a validated oracle return $e_{hi}(\pi) \in [e(\pi), e(\pi) + \eta]$ (interval ODE enclosures; Tucker 2011; Immler 2018). With G_h the centres of the $\lceil 1/h \rceil^d$ cubes of side $\leq h$ tiling Π , output $V = \Pi \cap \bigcup_{\pi_j \in G_h} \{\pi : \|\pi - \pi_j\|_\infty \leq (\tau - e_{hi}(\pi_j))/L\}$. (a) Soundness: $e \leq \tau$ on V for every L -Lipschitz e , deterministically, against any solver. (b) Completeness: if $\gamma > \eta$ and $h \leq (\gamma - \eta)/L$, then $V \supseteq \{e \leq \tau - \gamma\}$, with $\lceil L/(\gamma - \eta) \rceil^d$ calls. (c) Lower bound: every deterministic certifier sound on the L -Lipschitz class and complete at margin γ makes at least $\lfloor L/(2\gamma') \rfloor^d$ queries for every $\gamma' > \gamma$, even with an exact oracle.*

Proof idea. (a) Lipschitz extension from each centre. (b) Every point is within $h/2$ of a centre. (c) Against the constant $\tau - \gamma$, an unqueried packing ball of radius γ'/L hides a cone of height γ' . Details in Appendix J. $\square$

The bounds match up to 2^d and the oracle slack: the standard $\Omega((L/\gamma)^d)$ barrier for Lipschitz problems (Nemirovsky and Yudin, 1983). The constant L must be proved in the top model; an L estimated from simulations makes the projectile's region $x \in [0, 0.066]$ (true: $x \leq 0.0679$) a calibrated guess, not a certificate.

Theorem 11.32 (Monotone certifier). *[T3 Thm 4.6, rev.: query count, adversary] Let e be non-decreasing on $[0, 1]$ (no continuity needed), with a validated upper oracle. Lazy bisection starts with $lo = 0$, $hi = 1$ unqueried, queries $m = (lo + hi)/2$, sets $lo = m$ (“accepted”) if $e_{hi}(m) \leq \tau$ and $hi = m$ otherwise, k times, and outputs $V = [0, lo]$ if some query was accepted, $\emptyset$ otherwise. (a) It is sound. (b) With an exact oracle, V misses at most an interval of length 2^{-k} of $\{e \leq \tau\}$ (a variant that first queries both endpoints guarantees only $2^{-(k-2)}$). (c) Missing at most length h on every non-decreasing e needs $\lceil \log_2(1/h) \rceil$ queries, even with an exact oracle. (d) For down-sets in dimension $d \geq 2$, $\Theta(h^{1-d})$ queries are necessary and sufficient (proof sketch).*

Proof idea. (a) Monotonicity below an accepted point. (b) $hi - lo$ halves per query. (c) The step functions $2\tau \mathbf{1}[\pi \geq t]$ give two-valued transcripts, so q queries separate at most 2^q thresholds. Details in Appendix J. $\square$

For the small-angle schema, $e(\theta_0) = \frac{2}{\pi} K(\sin^2 \frac{\theta_0}{2}) - 1$ is provably increasing (a power series in $k^2 = \sin^2 \frac{\theta_0}{2}$ with positive coefficients), with rigorous enclosure $[k^2/4, k^2/4 + \frac{9}{64}k^4/(1 - k^2)]$. Twenty queries on $[0, 0.9\pi]$ certify $\theta_0 \leq 7.243^\circ, 16.168^\circ, 22.810^\circ, 49.946^\circ$ for $\tau = 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 5 \times 10^{-2}$ (exact thresholds $7.2441^\circ, 16.1687^\circ, 22.8138^\circ, 50.1040^\circ$), each truncated downward and re-checked in interval arithmetic (computed).

Proposition 11.33 (Laymon monotonicity can fail). ^[T3 Prop 4.7] (computed) For a pendulum with quadratic drag β at amplitude 6.75° , the signed relative period shift ($\times 10^{-6}$) is $-3.0, -5.3, -7.7, -3.9, +10.9, +36.2$ at $\beta = 0.0125, 0.025, 0.05, 0.1, 0.15, 0.2$: amplitude decay shortens the period, damping lengthens it at second order. With $\tau = 6 \times 10^{-6}$, a monotone certifier querying $\beta = 0.1$ ($|e| = 3.9 \times 10^{-6}$) certifies $[0, 0.1]$, which contains $\beta = 0.05$ with $|e| = 7.7 \times 10^{-6} > \tau$.

Reading. “Better data, better predictions”, the monotonicity Laymon (1987) uses to explicate idealized data, is a property of a model–query pair, to be proved, not assumed. The division of labour for tolerances is then: coherence catches tolerances that are too narrow, never ones too wide; exchangeable feedback gives coverage on average and nothing against selection; sound worst-case validity regions need a *proved* regularity property of the error plus validated evaluation, and then simulation of the top model is a sound oracle. World feedback proper is needed only to calibrate the top model itself, and for olympiad grading, where the root is c_P , not even that (L9 §10).

11.6 The Structured Physics Solution checker

Definition 11.34 (Structured Physics Solution). ^[T3 Def 5.1; L9 §7.1] An *SPS* is a tuple $(\rho, \mathcal{T}, \mathcal{D}, X)$: a reading ρ (the frame class with parameter roles: given, measured with intervals, free with ranges, deformable; the query and its type, i.e. dimension and allowed symbols; conventions); a context tree $\mathcal{T}$; a derivation $\mathcal{D}$ in the calculus of Definition 11.15 whose Exp instances cite schemas from a catalogue $\mathcal{B}$; and a final judgment $X = (@ \Vdash Q \in [q \pm \delta])$. A declared-regime problem uses a top-level germ context d whose declaration belongs to the trusted specification, and $X = (d \Vdash \chi)$.

Definition 11.35 (The checker). ^[T3 Def 5.2, rev.: Ax legality, schema indexing] The checker accepts iff all of the following hold. *Ax legality*: every Ax at the root is in Ax_ρ , every Ax in $SUP(A)$ is A , every Ax in a DEF context is a canonical stipulation of its declaration. *Kernel and laws*: every Step is accepted by the kernel, and every law it cites is in the trusted law catalogue. *Schema indexing*: every Exp cites a $\beta \in \mathcal{B}$ whose catalogue entry is indexed by *exactly* the context’s declaration $(D, \lambda^\circ, \mathcal{G})$; σ_β and ω_c are derived *in the parent*; asymptotic schemas are cited only below a principal parent. *Imports*: every Imp into a DEF context is a catalogue law or an undeformed datum for its D (imports into SUP children are unrestricted). *Shape*: every non-root context is SUP, DEF or AUX, every AUX bridge is a kernel-checked theorem, and X is derived at the root (or the declared germ root). All checks are deterministic.

A first version of this definition lacked Ax legality and schema indexing, and accepted the attack of Theorem 11.27(b); the verification of T3 caught it. Which parameters are “deformable” need not be trusted: if a solver declares g deformable and uses $DEF(\{k, g\}, (0, g'))$, the “neglect k ” schema, indexed by $DEF(\{k\}, 0)$, cannot be cited. The deformable list matters only for fidelity to the intended problem.

Assumption 11.36 (Trusted base). ^[T3 §5, rev.] (TB1) The kernel and the law catalogue are sound: every accepted Step instance and every catalogue law is valid in every $\mathfrak{M}_w(\lambda)$, $\lambda \in \text{dom } \mathfrak{M}_w$, $w \in W_\rho$ (including top-model theorems used as certified-neighbourhood bridges). (TB2) Every $\beta \in \mathcal{B}$ is a theorem of the frame for its indexed declaration, in the pointwise form of Theorem 11.27(c) or the asymptotic form of Lemma J.4. (TB3) Validated numerics return correct enclosures.

Theorem 11.37 (Relative soundness of the SPS checker). *[T3 Thm 5.3, rev.] Assume (TB1)–(TB3). For every solver whatsoever, if the checker accepts $(\rho, \mathcal{T}, \mathcal{D}, X)$ with a root judgment X , then $Q^{\mathfrak{M}_w(\lambda_w^*)} \in [q \pm \delta]$ for every $w \in W_\rho$ (for a germ root, $d \models_w \chi$). Hence if the intended reading satisfies $W_{\rho^*} \subseteq W_\rho$ the export is supertrue for the intended problem, and if the actual system is in W_{ρ^*} it is true of the world.*

Proof idea. Theorem 11.16 at each $w \in W_\rho$: Step by (TB1); root axioms by the definition of Ax_ρ ; DEF axioms because canonical stipulations hold in the child filter (the check that blocks Theorem 11.27(b)); imports by Lemma 11.7(a); exports by Theorem 11.27(c) with (TB2), (TB3) and schema indexing. Nothing depends on how $\mathcal{D}$ was produced. Details in Appendix J. $\square$

The residual *judgment layer* is exactly (J1) *reading containment*, $W_{\rho^*} \subseteq W_\rho$; (J2) *top-model adequacy*, that the actual system is a world in W_{ρ^*} (needed only for world claims, not for olympiad grading, where c_P is the root); and (J3) the *trusted base* (TB1)–(TB3). (J3) is the trusted core of any proof checker. (J1) and (J2) cannot be removed:

Theorem 11.38 (The judgment layer is ineliminable). *[T3 Thm 5.4, rev.] (a) Reading. Let $\mathcal{R}$ be the readings that the checker’s inputs (text, figures, SPS) do not exclude. A checker sound for every intended $\rho^* \in \mathcal{R}$ accepts X only if X holds on $\bigcup_{\rho \in \mathcal{R}} W_\rho$. (b) Open world. Extend the frame by an unmodelled effect μ , with $\mathfrak{M}'(\lambda, 0) = \mathfrak{M}(\lambda)$. If the admissible extensions include worlds with actual $\mu^* \neq 0$ and are closed under $Q \mapsto Q + Af(\mu)$ for all $A \in \mathbb{R}$ and $C^\infty f$ with $f(0) = 0$, then every finite-tolerance world claim accepted by a checker that sees only the slice $\mathfrak{M}'(\cdot, 0)$ and the SPS is false in some admissible extension.*

Proof. (a) If X fails at some $w \in W_{\rho_0}$, $\rho_0 \in \mathcal{R}$, the checker is unsound for $\rho^* = \rho_0$. (b) Lemma 11.20 in the μ direction, with $f(\mu^*) = 1$. $\square$

If $\mu^* = 0$, or if laws bound Q , (b) can fail: then the open world has been closed by a modelling assumption, i.e. moved into (J2).

Theorem 11.39 (Supervaluational correctness; completion invariance). *[T3 Thm 5.5, rev.] Write $Q^w = Q^{\mathfrak{M}_w(\lambda_w^*)}$, with W_ρ ranging over all admissible completions (free parameters $\theta_f \in \Theta_f$, measured ones in their intervals). (a) “ $Q \in [q \pm \delta]$ ” is supertrue iff $\sup_{w \in W_\rho} |Q^w - q| \leq \delta$; a supertrue answer at precision δ exists iff $\text{osc}_{W_\rho} Q \leq 2\delta$, and the midrange is one. (b) A completion-invariance certificate is a root derivation of $Q \in [q \pm \delta]$ from Ax_ρ in which θ_f enters only through “ $\theta_f \in \Theta_f$ ”. By Theorem 11.37 the conclusion of every accepted root derivation holds at every completion, so in the formal checker completion invariance is not a separate obligation but the content of deriving from Ax_ρ rather than from a narrowed axiom set. Informal forms are $\partial Q / \partial \theta_f = 0$ on a connected Θ_f where Q is differentiable in θ_f , or a variation bound; in §11.7, E is constant in α only while the point stays lit for every admissible α , and jumps to 0 at $s = L \tan \alpha$. (c) Supertruth over $\bigcup \mathcal{R}$ is the largest acceptance criterion sound under reading uncertainty.*

Reading. This answers “check that a claim is true in the context at hand. but wtf is that???”: a claim is true in the context at hand iff it holds on the context’s filter in every admissible completion of the reading. It is supervaluation (Fine, 1975) over completions of an underspecified problem. Checking it decomposes into kernel steps, stable imports, certified bridges with parent-evaluated side conditions, and a reading; only the reading is not mechanizable. Hence the strongest recommendation: *the problem setter should supply a formal specification of the*

admissible completions, after which the checker is sound relative to the top model alone. The user’s episode on the problem of §11.7 (“i suspected [the angle doesn’t matter] but then i messed up a calculation”) is a failed completion-invariance obligation that a checker would flag at once.

Proposition 11.40 (Gricean determinacy is one-sided). *[T3 Prop 5.6] Call ρ determinate at δ if $\text{osc}_{W_\rho} Q \leq 2\delta$. If $W_{\rho'} \subseteq W_\rho$ then $\text{osc}_{\rho'} Q \leq \text{osc}_\rho Q$. So “the problem is well-posed” can refute only (some) over-wide readings, never over-narrow ones; and silent narrowing (fixing the sun’s angle, choosing a specular mirror, importing an extra law, deciding “the finger is horizontal”) never decreases determinacy, while it violates (J1).*

Proof. A supremum over a subset is at most the supremum over the set; dually for infima. $\square$

The free internal signal for readings thus pushes *toward* the unsafe error: silent narrowing is the solver’s natural attack. With Section 6 and Theorem 11.28 this completes a table of three one-sided signals: The opposite signal must come from outside: a formal specification by the

Table 15: Three objects, three free signals, three blind spots.

Object	Free internal signal	Error it catches	Error it misses
rules	coherence (Section 6)	over-acceptance	under-acceptance
tolerances	coherence across exports (Theorem 11.28)	too narrow	too wide
readings	Gricean determinacy (Proposition 11.40)	(some) too wide	too narrow

setter (which moves (J1) into the checked part); intended-completion data (official solutions, marking schemes; the user “peeked at the solution and it seems to assume that’s not the case”); and round-trip checks such as rendering the model’s observable and comparing it with the figure. These are also one-sided: passing a round trip does not certify a reading. Section 13 returns to the table.

11.7 Worked example: EuPhO 2025 Theory 1(a), the polished chair leg

The setup is reconstructed in L9 §8.2 from the user’s notes and has not been checked against the official text; the answer $E = \frac{aI_0}{2} |\sin \frac{\varphi}{2}|/r$ is the user’s, reported by him to match the official one. *Specification:* a vertical specular cylinder of radius a , lit over height L , stands on a flat floor; the sun is a parallel beam of irradiance I at zenith angle α , travelling in $+x$; the floor’s direct illuminance is $I_0 = I \cos \alpha$. The query is the reflected illuminance (“surplus”) $E(r, \varphi)$, with φ measured from the anti-sun direction $+x$. Given a, I_0, L ; free completion $\alpha \in [30^\circ, 60^\circ]$, which the photograph does not fix; conventions: perfect specular reflectance, geometric optics, incoherent addition, sun radius $\delta_s = 4.65$ mrad, leg vertical; answer type: a function of (a, I_0, r, φ) with the dimension of I_0 . Write $e(\vartheta) = (\cos \vartheta, \sin \vartheta)$.

Proposition 11.41 (Exact reflected illuminance). *[T3 Prop 6.1] (SymPy-verified) Parametrize the lit side by $\vartheta \in (\pi/2, 3\pi/2)$ and the reflection height by $h \in [0, L]$; let $\psi = 2\vartheta - \pi$, $s = h \tan \alpha$ and $c = \sin \frac{\psi}{2}$. (a) The reflected ray lands at $P = a e(\vartheta) + s e(\psi) = (s + ac) e(\psi) + a \cos \frac{\psi}{2} e(\psi + \frac{\pi}{2})$. (b) $(\vartheta, s) \mapsto P$ is injective with $|\det \partial P / \partial(\vartheta, s)| = 2s + ac$. (c) $E(P) = I_0 ac / (2s + ac)$ for $s \leq L \tan \alpha$, and $E = 0$ off the image and where the preimage has $s > L \tan \alpha$; the total reflected flux $2aLI \sin \alpha$ equals the intercepted flux. (d) At a fixed floor point E does not depend on α ; only the lit domain $s \leq L \tan \alpha$ does.*

Injectivity (the caustic of a convex cylindrical mirror is virtual) is proved by intersecting two reflected rays explicitly (Appendix J). A 40-million-ray Monte Carlo gives MC/exact = 0.9997–1.0099 on five test bins, consistent with $\approx 0.6\%$ noise (computed). Part (d) is the completion-invariance certificate, *exact* in the top model. The formula is elementary catoptrics; we have not found it in the sources checked, but the official solution may contain it, so its novelty is uncertain.

Proposition 11.42 (Thin-leg bridge, rigorous). *[T3 Prop 6.2, rev.: lit domain] Let $E_0 = \frac{aI_0}{2}\sigma/r$, $\sigma = |\sin \frac{\varphi}{2}|$, $\epsilon = a/r$, $m = \frac{1}{2} \arcsin \epsilon$. If $\epsilon < 2/\sqrt{5}$, $\sigma > m$ and $r \leq L \tan \alpha$ (or $L = \infty$), then $1 - \frac{m}{\sigma} \leq \frac{E}{E_0} \leq \frac{1+m/\sigma}{\sqrt{1-\epsilon^2}-\epsilon/2}$.*

Without the lit-domain hypothesis the lower bound fails ($L = 60a$, $\alpha = 30^\circ$, $r = 40a$, $\varphi = \pi$ gives $E = 0$). The bound holds on 2.5×10^6 test points and is rigorous but conservative (computed): it certifies 5% from $r = 21a, 31a, 44a, 111a$ for $\sigma = 1, 0.5, 0.3, 0.1$, while the exact error exceeds 5% only for $r \leq 10.5a, 10.4a, 27.6a, 98a$; near $\sigma = 1/\sqrt{2}$ the gap reaches $23\times$, and it is unbounded as the tolerance shrinks, because the exact error $\approx \epsilon(2\sigma^2 - 1)/(2\sigma)$ vanishes to first order there. The export domain is an annulus $a \ll r < L \tan \alpha$ away from the shadow-side kink: intermediate asymptotics (Barenblatt, 1996).

Proposition 11.43 (Finite-sun bridge). *[T3 Prop 6.3, rev.: hypotheses] Let the admissible zenith angles be $[\alpha_{\min}, \alpha_{\max}] \subseteq (0, \pi/2]$, the sun’s angular radius $\delta_s < \alpha_{\min}$ with $\alpha_{\max} + \delta_s \leq \pi/2$ (the whole disc above the horizon), and $\Delta = \arcsin(\sin \delta_s / \sin \alpha_{\min})$. If $r \leq L \tan(\alpha_{\min} - \delta_s)$, $\epsilon < 2/\sqrt{5}$ and $\sigma - \Delta/2 > m$, then for any radiance profile over the solar disc, $\frac{E_\odot}{E_0} \in [(1 - \frac{\Delta}{2\sigma}) \text{lo}(\epsilon, \sigma - \frac{\Delta}{2}), (1 + \frac{\Delta}{2\sigma}) \text{hi}(\epsilon, \sigma - \frac{\Delta}{2})]$, with lo, hi the bounds of Proposition 11.42.*

The proof writes $E_\odot/I_0$ as a convex combination of the α -free geometric factor over the sun’s azimuth spread, bounded by Δ via Napier’s rule. A referee’s test with a limb-darkened disc found 0/859 violations (computed).

The checker run. A mini-checker (T3-checks/sps_checker.py; Experiment E in Section 12) encodes the specification with $L = 60a$ and runs concrete versions of the checks of Definition 11.35: (C0) reading containment against the formal specification; (C1) answer type: symbols (a, I_0, r, φ) only, so no free-completion symbol, and dimension; (C2) every declared import is in the specification; (C3) completion invariance: the answer must coincide identically with the certified α -free form; (C4) side conditions recomputed from the root’s declared domain, every hypothesis of Propositions 11.42 and 11.43 asserted, and the composed error budget evaluated at the worst corner. All admissible α light the disc $r \leq 60 \tan(29.73^\circ)a = 34.3a$. SymPy stands in for a proof kernel.

Residual defects of the prototype. The first version accepted V6–V9: C1 allowed α in the answer, C3 checked only the oscillation over α of answer over certified form, never its offset from 1, and C4 never enforced $\epsilon < 2/\sqrt{5}$ or $\sigma_{\min} - \Delta/2 > m$ (beyond $2/\sqrt{5}$ the upper bound turns negative and was silently dropped). The repaired version is still a toy. (i) It hard-codes one top model and two bridge schemas. (ii) SymPy and floating-point evaluation of the budget at a corner stand in for a kernel and interval arithmetic; that the corner is worst rests on monotonicity of the bounds, argued, not machine-checked. (iii) C3 accepts only answers *symbolically identical* to the certified form, so it is incomplete; a fuller checker would add a rigorously enclosed offset

Table 16: Mini-checker run on EuPhO 2025 T1(a), repaired version. “First version” marks attacks the original, syntactic checker accepted.

Submitted SPS	Outcome
Honest: $\frac{aI_0}{2} \sin \frac{\varphi}{2}/r$, $r \in [20a, 34a]$, $\sigma \geq 0.5$, tolerance 10%	ACCEPT, certified error $\leq 8.88\%$ (exact point-sun error $\leq 2.56\%$; sun term $\leq 0.93\%$)
V0: same, tolerance 5%	REJECT: budget $0.0888 > 0.05$ (completeness gap: the truth is within 5%)
V1: overclaimed domain $r \geq 5a$	REJECT: budget $0.380 > 0.10$
V2: thin-leg side condition evaluated in the child ($a := 0$), declared	REJECT (Theorem 11.27(a)); a silent one cannot change the budget, which C4 recomputes
V3: $\frac{aI_0}{2 \cos \alpha} \sin \frac{\varphi}{2}/r$ (the user’s slip: I for I_0)	REJECT by C1 (α in answer), C3 (half-oscillation 0.42)
V4: reading narrowed to $\alpha = 45^\circ$	REJECT by C0, only because the specification is formal; else (J1)
V5: declared extra import “10% Lambertian”	REJECT: import not in the specification
V6: $2E_{\text{thin}} \cos(\alpha)^{1/1000}$ (true error ≈ 0.5)	REJECT by C1, C3; first version accepted (“ ≤ 0.0888 ”)
V7: $\frac{23}{20} E_{\text{thin}} \cos(\alpha)^{1/50}$ (true error 0.159)	REJECT by C1, C3; first version accepted
V8: $1.5E_{\text{thin}} e^{\alpha/1000}$	REJECT by C1, C3; first version accepted
V9: $r \geq 1.05a$, $\sigma \geq 1$, tolerance 0.7 (true error 0.909)	REJECT by C4 ($\epsilon = 0.952 > 2/\sqrt{5}$); first version accepted (“ ≤ 0.635 ”)
V10: $1.09E_{\text{thin}}$ (control)	REJECT by C3: differs from the certified form
V11: undeclared 10% diffuse correction $1.1E_{\text{thin}}$	REJECT by C3: differs from the certified form

$|\text{answer/certified} - 1|$ to the budget. (iv) V2 and V5 are rejected because the SPS *declares* the violation; the semantic protection is that C4 recomputes side conditions from root data and C3 catches answer-changing undeclared imports (V11). (v) V0 is a completeness gap, and V4 is caught only because the specification is formal. The episode is a small instance of the lesson of Section 3: plausible syntactic checks accepted four simple attacks, which were rejected only once every check was made semantic.

What remains judgment. The photo-to-model reading (a *specular* cylinder, lit over its full height, its unsupplied reflectance, the meaning of “illuminance surplus”, the α range) and the open world (taper, tilt, floor roughness, the seat’s shadow). Each effect could be added as a declared deformation with its own bridge, moving it from judgment to checking; what cannot be moved is the claim that *these* are all the effects the setters intend (Theorem 11.38). The one checkable test of the reading is the round trip: render the predicted floor pattern and compare it with the photograph. A second example: IPhO 2025 T2 C.4 is a germ-root problem whose leading-order $W^* = 19.8 \text{ mJ}$ differs from the exact optimum 20.1 mJ by 1.3%, inside the marking interval $[19, 21] \text{ mJ}$, so the tolerance-aware check of Theorem 11.26 passes (L9 §8.3) (computed).

11.8 The four logical roles of physicists’ checks

The brief listed dimensional analysis, limiting cases, symmetry and conservation as “coherence checks”. They play at least four logical roles (L9 §3), and only the first two are coherence in the sense of Section 6:

Dimensions are a type system: dimensions form a free abelian group and inference is

Table 17: Logical roles of the physicist’s checks (after L9 §3.7).

Role	Checks	What it certifies	Soundness status
1. invariance of the rule set	dimensions (unit group); declared context symmetries	homogeneity, equivariance of derived equations	worst case, decidable; necessary only
2. in-model theorem	conservation laws	a computation inside one local model	worst case within the model; silent on exports
3. regression against another context	limiting cases	agreement with solved sub-models	sound only for regular limits (Norton)
4. export side condition	order of magnitude; “the neglected term is small”	a bridge at a stated tolerance	sound with explicit budgets; heuristic otherwise

unification modulo abelian groups (Kennedy, 1994); the Buckingham Π theorem (Buckingham, 1914) turns “express in terms of” into a type signature (in IPhO 2025 A.1 the force must be $mg\Phi(\rho Sh/m, P_0 S/(mg))$, and the official $(m + \rho Sh)g$ is $\Phi = 1 + \pi_1$). Dropping a small Π group assumes complete similarity, which fails in intermediate asymptotics (Barenblatt, 1996), so “ $\Pi_2 \ll 1$, ignore it” is a bridge (role 4), not a type fact. *Symmetry* needs uniqueness, as the user notes: a symmetric problem implies only a symmetric *set* of solutions. Context symmetries are context-indexed (time reversal holds without drag), and a declared symmetry gives free negative data: a non-equivariant candidate rule is refuted by a symmetry image of one of its own accepted instances (L9 TC2; dimension inference as learning meaning from positive examples is discussed in Section 5). *Limiting cases* are cross-context regression tests whose validity is Norton’s continuity (Proposition 11.5(c)). *Order-of-magnitude* checks are export side conditions (Theorem 11.26). Qualitative checks are one-sided refutations, as in qualitative simulation, which is sound but incomplete (Kuipers, 1986). The closest precedent for the in-model layer is the Andes tutor (VanLehn et al., 2005), which checks each equation for units and against an author-supplied model, and never checks bridges, because the author fixes the idealizations.

What is and is not established. Contradictory contexts are fine, and realizability says which must be consistent. “True in the context at hand” is truth on the context’s filter in every admissible completion, and checking it reduces to kernel steps, stable imports, certified bridges and a reading. Reductio in inadequate frameworks needs a consistency certificate, and with approximate premises a margin. Untrusted solvers are handled for everything except the reading, whose characteristic attack, silent narrowing, no internal signal sees. Not established (Section 14): any theorem about *learning* readings (the deep part, J1); frame realizability under germ semantics; idealizations inside suppositions and non-value exports; automated proofs of the regularity constants of Theorems 11.31 and 11.32; a checker beyond one worked problem. The mini-checker illustrates the checks on one problem and twelve adversarial variants; it is not a system, and its own soundness is argued, not proved.

12 Experiments

(placeholder: being written)

13 Justification beyond proof

The investigation began with a question about principled justification. What does it look like, once we go beyond mathematical proof? This section steps back from the theorems and says what they show about that question. It also says what they do not show. Throughout, “we” means the authors of this report. The view argued for is deliberately modest. The theorems do not deliver a theory of justification. They make precise one shape that justification can take, and they locate exactly where that shape runs out.

13.1 The setup is Goodman’s circle, made into a learning problem

The setup we studied has three parts:

- learn inference rules from accepted inferences;
- revise them when they generate unacceptable conclusions;
- revise accepted inferences when they conflict with rules one is unwilling to give up.

This is, almost verbatim, Goodman’s account of the justification of deduction and induction:

“Rules and particular inferences alike are justified by being brought into agreement with each other. *A rule is amended if it yields an inference we are unwilling to accept; an inference is rejected if it violates a rule we are unwilling to amend.*” (Goodman, 1955, ch. III)

In our vocabulary, the channels correspond to Goodman’s account as follows:

- Imitation (P) supplies the accepted particular inferences.
- Rule schemas, learned by least generalization, are the rules.
- Coherence (C) is the mechanism by which a rule “yields an inference we are unwilling to accept”. A derivation of $\perp$ from a context certified consistent is a negative bag, and some rule in it must go.
- World feedback (W) adds a channel whose errors are independent of human errors.

Without (W), the process is *narrow* reflective equilibrium. With (W) it is *wide* reflective equilibrium in the sense of Daniels (1979). Daniels’ “independence constraint” is, formally, the conditional-independence assumption that the coherence impossibility theorems show to be necessary (Section 6.7).

What our theorems add to Goodman’s picture is a set of *vindications with explicit conditions*. Under stated channel assumptions, mutual adjustment is *sound against every adaptive arguer*, with explicit costs:

- realizability of the target in a structured hypothesis class;
- truthful designation of the contexts that are certified consistent;
- a world channel of the right kind.

For each assumption there is a matching impossibility result showing that dropping it breaks the guarantee (Section 7). This is the sense in which the theorems are *conditional reliabilist meta-justifications* of reflective equilibrium. They are not a non-circular foundation for logic. As Dummett (1973) observed, any justification of deduction is explanatory rather than suasive: it

uses deduction. What they provide is an explicit list of what the process relies on, together with proofs that it relies on nothing else.

Two features of the guarantees matter philosophically.

- First, they are *anytime*, not merely in the limit. Limit theorems are Reichenbachian vindications. They fall to the objection of Salmon (1967) that infinitely many methods share the same limit while disagreeing at every finite stage. Our soundness guarantees hold at every time, with probability over the data (Theorem 4.15). Our costs are explicit finite quantities: escalations, detections, samples. This also matches a recurring theme in the notes that prompted this work: suspicion of limit framings, and the view that thinking and mathematics are “infinite endeavors” with no final formula.
- Second, they are *worst case over arguers*. A step checker that is accurate on the distribution of human steps can be exploited by search (Theorem 3.2). Justification that is to survive an untrusted or adversarial generator must be uniform over what the generator may propose. This is the case that motivates the question in the first place.

13.2 What each signal can do, and the residue it leaves

The main results combine into a division of labour. Table 18 summarizes it.

Two observations about this table seem to us to be the real philosophical content of the investigation.

Every signal is one-sided. Each learning signal we studied can refute errors in only one direction:

- Positive data never refute an over-generalization (Gold).
- Coherence never refutes over-caution: a cautious learner never sees a contradiction (Proposition 6.3). Nor does it refute a coherent over-generalization (Theorem 6.9).
- Coherence among exported tolerances never refutes a tolerance that is too wide. Widening an interval cannot create a contradiction (Theorem 11.28).
- The demand that a problem be well posed refutes readings that are too wide, but it rewards readings that are too narrow (Proposition 11.40).

Each one-sided signal is an attack surface for an adversarial arguer, and it calls for a complementary signal that refutes errors in the other direction. Principled justification, on this picture, is not one master criterion. It is a set of signals that *together* cover every direction of error, plus an explicit statement of the directions that remain uncovered. Table 15 lists the pairs for physics. The same structure appears in mathematics. Imitation proposes and caution guards. Coherence condemns, but only bold hypotheses. Objects localize.

Each residue has a name, and different philosophies disagree only about the residue. Outside the residue of coherent, uniform, world-adequate alternatives, every position we know of counts the learner as correct:

- the realist;
- the Peircean, for whom truth is the limit of inquiry;
- the Dummettian, for whom truth is warranted assertibility;

- Carnap’s tolerance;
- Quine’s pragmatism.

Inside the residue, they disagree. The realist sees undetectable error. The Peircean or Dummettian sees no fact of the matter. Carnap sees a free choice of language. Quine sees a choice made on pragmatic grounds. The theorems make the residue precise in several cases:

- It is empty for classical propositional logic.
- It contains all intermediate logics for an intuitionistic practice.
- It contains $\text{PA} + \neg\text{Con}(\text{PA})$ for arithmetic.
- In the algebra experiment it contains the total semantics with $1/0 = 0$, which pure coherence converged to (Section 12). This is, as it happens, exactly the convention adopted by Lean’s Mathlib and by Isabelle/HOL, where $x/0 = 0$.

A theorem can say exactly where these philosophies part ways. It cannot settle who is right inside the residue. We think that is the correct division of labour between mathematics and philosophy here.

One consequence deserves to be stated on its own. *Rules can be identifiable when truths are not.* The bilateral tell-tale identifies classical meanings from finitely many data (Theorem 6.29). Yet no computable learner on the same channels decides Σ_2 truth in the limit (Theorem 6.22). This undercuts the simple form of Dummett’s acquisition argument. That argument holds that a meaning one could acquire must be manifestable in use, so meaning cannot outrun decidability. But a learner can acquire the meaning-constitutive *rules* exactly, while the truth values they determine remain beyond any computable reach.

13.3 Rules are not premises

The tortoise of Carroll (1895) refused to infer Z from A , B and “if A and B then Z ” until that inference was itself added as a premise, and so on forever. For a learner the lesson has a precise form:

- A learned *rule* is a schema. Its validity is a universally quantified claim over all instances, of Π form.
- No finite amount of world feedback verifies such a claim. Feedback can only refute it.
- The justification for *using* a learned rule therefore cannot be further premises. It has to be a guarantee about the process that accepted it: cautious acceptance, and an anchor that has been observed.

This is why the right acceptance policy for learned rules is Popperian. Accept tentatively. Assert only what every surviving hypothesis licenses. Treat each refutation as permanent. It is also why self-certification must never feed back into acceptance:

- a learned verifier that “proves its own soundness” has provided no evidence that distinguishes it from an unsound verifier sharing the certifying rule;
- in arithmetic, by Löb’s theorem a theory that proves the reflection schema $\text{Prov}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi$ for its *own* provability predicate, for all φ , proves everything. By contrast, the corresponding reflection *rule* is admissible for Σ_1 -sound theories.

Reflection is a principled source of genuinely new inferences, beyond imitation, that a learner who has accepted a calculus is committed to. Where it is used, it must enter as a stratified rule, never as a self-referential premise.

13.4 What principled justification beyond proof looks like

Putting the pieces together, the results suggest the following shape for a principled justification of a claim about something other than pure mathematics. A physics olympiad answer is the running case.

1. **Proof inside contexts.** Each step inside a context is checked against a calculus. The calculus is formal where possible. Where it is learned, acceptance is cautious, so the check is sound against an arguer who searches for weaknesses (Sections 4 and 10).
2. **Certified bridges between contexts.** Moving a conclusion from an idealized context to the claim of interest is not a logical step. It needs a certificate carrying information of finite radius about how the quantity behaves away from the idealization. Facts about the idealized model alone never suffice (no free export, Theorem 11.21). Side conditions are evaluated where they are true, in the parent context, and contexts are declared deformations. Otherwise an arguer can “stipulate” its way to anything (Theorem 11.27).
3. **An explicit judgment layer.** What remains is provably ineliminable (Theorem 11.38):
 - that the problem was read correctly, i.e. the admissible worlds were not narrowed;
 - that the top model is adequate for the world claims;
 - that the trusted base is trustworthy.

A principled justification does not hide this layer. It minimizes it and declares it.

4. **Correctness as supertruth.** For underspecified questions, “true in the context at hand” means true in every admissible completion of the intended model, to the demanded precision. This is the largest acceptance criterion that stays sound under uncertainty about the reading (Theorem 11.39). Answer-invariance under unspecified features, which experienced solvers notice and rely on, is exactly a certificate of supertruth.
5. **Learning signals with known limits.** Every learned component comes with a statement of the signal that trained it and the residue that signal leaves. The learned components are the calculus, the bridge validity regions and the tolerances.

Measured against this shape, mathematical proof is the limiting case in which:

- the judgment layer has shrunk to the reading of the formal statement and trust in the axioms;
- every bridge is exact;
- the calculus is fixed rather than learned.

Informal mathematics before formalization sits in between. It was a bounded-gap practice over a latent calculus, as the robust-core hypothesis suggests. In it, counterexample objects did most of the corrective work, and paradoxes were rare and uninformative (Section 10). Physics adds a nonempty, explicit judgment layer and non-logical bridges, but the shape is the same. On this

view the difference between mathematical and physical justification is one of degree, measured by the size of the declared residue, not one of kind.

13.5 What this says about meaning

The inferentialist premise of the question was that learning which inferences are valid is roughly learning meanings. The results support a qualified version:

- **Inferential roles are learnable, exactly, from finite use.** A finite anchor fixes the inferential role of a vocabulary for unbounded use (Corollary 5.11). This is a precise, if modest, answer to the rule-following question of how finite training fixes an infinite norm. The answer is: through structure (schemas) plus a small set of telling instances. The skeptic’s “quus” survives only as a non-uniform patch, which uniformity rules out, or as a coherent uniform alternative, which is a named residue.
- **More than role is not identifiable.** Practice, coherence and object data identify a latent formalization exactly up to informal-validity equivalence, and no further (Theorem 10.15). Ontology is not identifiable: standard and nonstandard reals are observationally equivalent on the practice (Proposition 10.17). Neither is the choice of set-theoretic reduction (Proposition 10.19).
- **Coherence buys existence, but not intended existence.** Every coherent mode of talking has *something* it could be talking about. That is a generalized completeness theorem, true for positions, credences and context systems alike (Section 9). But that something may be a syntactic or nonstandard model. Intended existence needs one of three additional things:
 - categoricity;
 - anchoring of the vocabulary to the world (Beth definability);
 - an infinitary condition, such as the probabilistic ω -rule.

None of these can be certified computably beyond Π_1 .

Inferential role is one aspect of meaning, and the project captures that aspect precisely. Other aspects are not captured by anything studied here:

- the “hooking” of a conception onto the world;
- the activities a concept supports;
- the questions it lets one ask.

We do not claim otherwise.

13.6 Answers to the questions that prompted the investigation

The investigation was prompted by a set of questions. Most of them are recorded in notes that are publicly available. Table 19 lists the main ones with pointers to where they are answered or partly answered.

14 Limits and open problems

We collect the main limitations of the results and the open problems they leave. The hardest of them is the first.

1. Affordable soundness when the learner must invent language. Every soundness guarantee here is relative to *realizability*: the target calculus, or the latent formalization, must lie in the hypothesis class. History did not respect any fixed class. It invented uniform convergence, ideals, ε - δ and the fundamental group as proof-generated concepts.

- A universal, language-inventing class restores realizability, but sound verification then costs on the order of $2^{K(h^*)}$ escalations (Theorems 4.12 and 10.10).
- What is needed is a class that is closed under the definitional extensions mathematics actually made, and that still has small escalation dimension. Equivalently, we need a theory of when concept invention, for example by anti-unification over definitions or by naming the hidden lemma found by descent, is cheap.

2. Computational cost. Our bounds count samples, escalations, detections and refutation-oracle calls, not computation.

- Membership in the version-space intersection for untagged unions is likely NP-hard.
- Consistency with negative bags is a multiple-instance problem. Its hardness is the formal shadow of the obfuscated-arguments problem in debate.
- Refutation search is only semi-decidable. Our depth-relative soundness (Theorem 7.22) is the honest form of what can be had.

A complexity-theoretic version of the two-tier theorem would be valuable. It would identify which audits are polynomial and for which classes.

3. The reading layer. In physics the irreducible judgment is the *reading*: which admissible worlds the problem text determines (Theorem 11.38).

- Determinacy refutes only over-wide readings (Proposition 11.40).
- What positive data identify the intended admissible class? We conjecture that official solutions together with marking schemes form a tell-tale within a class of “minimal completions consistent with the intended imports”.
- Learning readings soundly is, we think, the central open problem for principled checking beyond mathematics.

4. Sharp combinatorics. Several bounds have gaps:

- The binomial conjecture for k -unions of intersection-closed classes (Conjecture 4.11) would close the gap for untagged schema learning.
- The bag-price conjecture $M_{\text{bag}}^{(r)} \leq r \cdot \text{Ldim}$ (Conjecture 10.27) would give the exact exchange rate between paradoxes and counterexample objects.
- The query complexity of sound blame, i.e. computing the union of the minimal conflicts with a conflict oracle, is open.

5. Coherence beyond Post-completeness.

- For which first-order or modal targets does every invalid uniform rule have a short coherence witness in a natural designated context?
- Is there a hierarchy of coherence-witness complexity?
- Which logics and theories are *coherence-pinned*, meaning that the residue is empty for natural families of designated contexts?
- For arithmetic: is there a principled learner over reflection progressions that converges to “the right amount of reflection” under a natural feedback notion?

6. Simplicity and normativity.

- For which classes of simple models does the product-model frontier kink, beyond parities?
- Is there a principled, data-free choice of the simplicity coefficient, for example one tied to the cheapest postulate-violating error?
- Which assumptions on a performance model, such as errors being time-bounded shortcuts, make joint description length identify competence despite the known non-identifiability of preferences from behaviour?

7. Probabilistic and contextual coherence.

- Find a finitely presented proof system for valid counting sequents that is complete for de Finetti coherence.
- Prove a probabilistic local-to-global theorem for context systems on tree covers.
- Decide whether contextual frustration between idealized contexts should be penalized or tolerated.

8. Scale.

- Our experiments are illustrations in small symbolic domains with realizable targets, cheap world feedback and few seeds.
- Whether learned calculi with cautious acceptance remain affordable at the scale of real mathematical corpora (for example Lean or Metamath step data with injected fallacies at measured human rates) is an empirical question we have not addressed.
- We have deliberately not built a strong prover, and we would recommend that any such work keep the checker separate from the generator.

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A Proofs for Section 2

Theorem 2.5 is due to Plotkin (1970) and Reynolds (1970). All other statements of Section 2 are proved here. The rank lemma and the recovery lemma are T1 Lemmas 1.2–1.3, which the referees reported correct with no change needed; the other three are routine.

A.1 Proof of Lemma 2.3 (closure facts)

(a) Let D be the set of judgments with an R -derivation from B . D contains B (one-node derivations) and is closed under R : if $P \triangleright j \in R$ and $P \subseteq D$, graft derivations of the finitely many members of P under a new root j . So $\text{Cl}_R(B) \subseteq D$. Conversely, if $X \supseteq B$ is R -closed, induction on the height of derivations shows that every node of every derivation from B lies in X ; so $D \subseteq \text{Cl}_R(B)$. Extensivity and monotonicity in B and in R hold because a derivation from B using steps of R is also one from any $B' \supseteq B$ using steps of any $R' \supseteq R$. Idempotence: a derivation from $\text{Cl}_R(B)$ becomes a derivation from B after grafting a derivation from B above each leaf. Finitarity: a derivation has finitely many leaves, so it is a derivation from a finite $B_0 \subseteq B$.

(b) A single step $P \triangleright j \in R$ is a derivation of j from P , so $R \subseteq \text{Sound}(R)$, and monotonicity in the rule set gives $\text{Cl}_R(B) \subseteq \text{Cl}_{\text{Sound}(R)}(B)$. Conversely, $\text{Cl}_R(B)$ is closed under $\text{Sound}(R)$: if $P \subseteq \text{Cl}_R(B)$ and $j \in \text{Cl}_R(P)$, then by (a) $j \in \text{Cl}_R(\text{Cl}_R(B)) = \text{Cl}_R(B)$; it contains B , so it contains $\text{Cl}_{\text{Sound}(R)}(B)$. Finally, $\text{Sound}(A) = \{P \triangleright j : j \in \text{Cl}_A(P)\}$ depends only on Cl_A , so $\text{Cl}_A = \text{Cl}_R$ gives $A \subseteq \text{Sound}(A) = \text{Sound}(R)$; and $\text{Sound}(R)$ itself has the closure Cl_R . $\square$

A.2 Proof of Lemma 2.6 (rank)

Throughout, $|t|$ is the number of symbol occurrences in t , metavariable occurrences included, and $\text{vars}(t)$ is the set of distinct metavariables of t . A *renaming* of a set X of metavariables is a substitution whose restriction to X is an injective map into metavariables. If $\tau = \sigma\theta$ with θ a renaming of $\text{vars}(\sigma)$, then $\sigma = \tau\theta^{-1}$, so $\sigma \equiv \tau$; conversely, $\sigma \equiv \tau$ means equality up to such a renaming, and then $\mu(\sigma) = \mu(\tau)$.

Let $\tau = \sigma\theta$. Let $x_1, \dots, x_k$ be the distinct metavariables of σ , and let x_i occur $m_i \geq 1$ times in σ .

Size. Replacing each of the m_i occurrences of x_i (one symbol) by θx_i (with $|\theta x_i|$ symbols) gives

$$|\tau| = |\sigma| + \sum_{i=1}^k m_i(|\theta x_i| - 1).$$

Metavariables. Every metavariable of τ occurs in some θx_i , so

$$|\text{vars}(\tau)| \leq \sum_{i=1}^k |\text{vars}(\theta x_i)|,$$

with equality iff the sets $\text{vars}(\theta x_i)$ are pairwise disjoint.

Difference. Since $|\text{vars}(\sigma)| = k$,

$$\mu(\tau) - \mu(\sigma) \geq \sum_{i=1}^k \left[m_i(|\theta x_i| - 1) + 1 - |\text{vars}(\theta x_i)| \right]. \quad (1)$$

We bound each summand.

- If θx_i is a metavariable, then $|\theta x_i| = 1 = |\text{vars}(\theta x_i)|$ and the summand is 0.
- Otherwise θx_i contains at least one function or constant symbol and at least one occurrence of each of its metavariables, so $|\theta x_i| \geq |\text{vars}(\theta x_i)| + 1$. Using $m_i \geq 1$, the summand is at least $(|\theta x_i| - 1) + 1 - |\text{vars}(\theta x_i)| \geq 1$.

Hence $\mu(\tau) \geq \mu(\sigma)$.

Equality. If $\mu(\tau) = \mu(\sigma)$, then every summand in (1) is 0 and the inequality there is tight. The first forces every θx_i to be a metavariable; the second forces the sets $\text{vars}(\theta x_i) = \{\theta x_i\}$ to be pairwise disjoint, i.e. the θx_i to be pairwise distinct. So θ restricted to $\text{vars}(\sigma)$ is a renaming. Conversely, if it is a renaming, then $|\tau| = |\sigma|$ and $|\text{vars}(\tau)| = |\text{vars}(\sigma)|$, so $\mu(\tau) = \mu(\sigma)$.

Chains. Suppose $\sigma_{l+1} \succ \sigma_l$ strictly, i.e. $\sigma_l = \sigma_{l+1}\theta_l$ and $\sigma_l \not\equiv \sigma_{l+1}$. Then θ_l is not a renaming of $\text{vars}(\sigma_{l+1})$ (otherwise the two would be equivalent), so $\mu(\sigma_l) > \mu(\sigma_{l+1})$, i.e. $\mu(\sigma_l) \geq \mu(\sigma_{l+1}) + 1$. Summing along $\tau = \sigma_0 \prec \sigma_1 \prec \dots \prec \sigma_m$ gives $\mu(\tau) \geq \mu(\sigma_m) + m$, and $\mu(\sigma_m) \geq 0$. $\square$

A.3 Proof of Lemma 2.7 (when the lgg recovers the schema)

Let σ have metavariables $x_1, \dots, x_v$, let $\theta_1, \dots, \theta_n$ be ground substitutions ($n \geq 1$), and let $t_j = \sigma\theta_j$. For each metavariable x of σ write $c_x = (\theta_1 x, \dots, \theta_n x)$ for its *column*, and let η be the substitution $\eta(x) := \mathcal{A}(c_x)$.

Claim. For every subterm σ' of σ , $\mathcal{A}(\sigma'\theta_1, \dots, \sigma'\theta_n) = \sigma'\eta$.

By induction on σ' . If $\sigma' = x$ is a metavariable, both sides are $\mathcal{A}(c_x)$. If $\sigma' = f(\sigma'_1, \dots, \sigma'_r)$ (with $r = 0$ for a constant), then every $\sigma'\theta_j = f(\sigma'_1\theta_j, \dots, \sigma'_r\theta_j)$ has root f , so by definition $\mathcal{A}$ outputs f applied to $\mathcal{A}$ of the argument columns $(\sigma'_k\theta_1, \dots, \sigma'_k\theta_n)$, which by the induction hypothesis are $\sigma'_k\eta$. Hence the output is $f(\sigma'_1\eta, \dots, \sigma'_r\eta) = \sigma'\eta$.

Taking $\sigma' = \sigma$ and using Theorem 2.5,

$$\text{lgg}(t_1, \dots, t_n) = \mathcal{A}(t_1, \dots, t_n) = \sigma\eta \preceq \sigma.$$

When η is a renaming. By the definition of $\mathcal{A}$:

- $\eta(x) = \mathcal{A}(c_x)$ is a metavariable iff the entries of c_x do not all have the same root, i.e. iff (R) holds at x . If (R) fails at x , $\eta(x)$ is a compound term or a constant.
- If (R) holds at x and at y , then $\eta(x) = z_{c_x}$ and $\eta(y) = z_{c_y}$, and these coincide iff $c_x = c_y$, i.e. iff (D) fails for the pair x, y .

So η restricted to $\text{vars}(\sigma)$ is a renaming iff (R) holds for every x and (D) holds for every pair $x \neq y$.

Conclusion. If (R) and (D) hold, η is a renaming and $\text{lgg}(t_1, \dots, t_n) = \sigma\eta \equiv \sigma$. If either fails, η is not a renaming of $\text{vars}(\sigma)$, so by Lemma 2.6 $\mu(\sigma\eta) > \mu(\sigma)$; since equivalent schemas have equal rank, $\text{lgg}(t_1, \dots, t_n) \not\equiv \sigma$, and it is strictly more specific. $\square$

A.4 Proof of Lemma 2.8 (guarded closure)

$P \subseteq \text{inst}(\text{lgg}(P))$ by Theorem 2.5, and every $\phi \in \Psi_P$ holds on P , so $P \subseteq \text{inst}(\text{lgg}(P), \Psi_P)$. Let $P \subseteq \text{inst}(\sigma, \Psi)$. Then $P \subseteq \text{inst}(\sigma)$, so $\sigma \succeq \text{lgg}(P)$ and $\text{inst}(\text{lgg}(P)) \subseteq \text{inst}(\sigma)$. Every $\phi \in \Psi$ holds on all of P , so $\Psi \subseteq \Psi_P$ and $\bigcap_{\phi \in \Psi_P} \phi^{-1}(1) \subseteq \bigcap_{\phi \in \Psi} \phi^{-1}(1)$. Intersecting, $\text{inst}(\text{lgg}(P), \Psi_P) \subseteq \text{inst}(\sigma, \Psi)$. $\square$

A.5 Proof of Lemma 2.15 ((WS) and coherence)

This is the remark following (WS) in T7 §1.3, in the form repaired after verification (without the clause $T_{\text{trust}} \subseteq \text{Sound}(R^*)$). Write $R^+ = R^* \cup T_{\text{trust}}$. Let $[\Gamma : \Delta]$ be designated and V as in (WS). V makes every member of Γ true, and no step of R^+ has true premises and a false conclusion, so the set of V -true judgments is an R^+ -closed superset of Γ and contains $\text{Cl}_{R^+}(\Gamma)$. Since V makes $\perp$ and every member of Δ false, $\perp \notin \text{Cl}_{R^+}(\Gamma)$ and $\Delta \cap \text{Cl}_{R^+}(\Gamma) = \emptyset$. As $\text{Cl}_{R^*}(\Gamma) \subseteq \text{Cl}_{R^+}(\Gamma)$ (Lemma 2.3(a), monotonicity in the rule set), the position is also in bounds for R^* .

Conversely, let $J_W = \emptyset$ and let $[\Gamma : \Delta]$ be a designated position that is in bounds for R^+ . Then e is well defined, because $\Gamma \subseteq \text{Cl}_{R^+}(\Gamma)$ is disjoint from $\Delta \cup \{\perp\}$. Let V be the indicator function of $\text{Cl}_{R^+}(\Gamma)$. It extends e : it is 1 on Γ and 0 on $\Delta \cup \{\perp\}$. The set $\text{Cl}_{R^+}(\Gamma)$ is closed under R^+ , so no step of R^+ has V -true premises and a V -false conclusion, and (WS) holds. For the parenthetical instance: if $T_{\text{trust}} \subseteq \text{Sound}(R^*)$, then $R^* \subseteq R^+ \subseteq \text{Sound}(R^*)$, so by Lemma 2.3(a),(b) $\text{Cl}_{R^+} = \text{Cl}_{R^*}$, and being in bounds for R^* is the same as being in bounds for R^+ . $\square$

B Proofs for Section 3

Lemma 3.1, Theorem 3.2, Proposition 3.5 and Proposition 3.9 are proved in full in the main text. This appendix proves the remaining results of Section 3 and verifies the claims of Example 3.6. Notation is as in Section 3. Throughout, $\models$ is classical propositional consequence, and a step $\Pi \triangleright \varphi$ over formulas is *sound* if $\Pi \models \varphi$.

B.1 PAC learners can be maximally unsound

Proof of Corollary 3.3. For $N \geq 0$ let $\mathcal{S}_{\geq N} := \{s \in \mathcal{S} : |s| \geq N\}$.

Step 1: $T_j \subseteq \mathcal{S}_{\geq j}$. Every step of T_j has the premise $A \vee \top^{(j)}$, and $\top^{(j)}$ contains j occurrences of $\top$. So every step of T_j has size at least j .

Step 2: the tail bound. The sets $\mathcal{S}_{\geq N}$ decrease in N and have empty intersection, so $Q(\mathcal{S}_{\geq N}) \downarrow 0$ by continuity from above. Fix $\varepsilon \in (0, 1)$ and let $N_\varepsilon := \min\{N \geq 0 : Q(\mathcal{S}_{\geq N}) \leq \varepsilon\}$.

- If $\hat{j} \geq N_\varepsilon$, then $Q(T_{\hat{j}}) \leq Q(\mathcal{S}_{\geq \hat{j}}) \leq Q(\mathcal{S}_{\geq N_\varepsilon}) \leq \varepsilon$, by Step 1 and monotonicity.
- If $\hat{j} < N_\varepsilon$, then $N_\varepsilon \geq 2$, because $\hat{j} \geq 1$. By minimality of N_ε , $Q(\mathcal{S}_{\geq N_\varepsilon-1}) > \varepsilon$. Moreover $\hat{j} < N_\varepsilon$ means $|X_i| \leq N_\varepsilon - 2$ for every i , i.e. no sample lies in $\mathcal{S}_{\geq N_\varepsilon-1}$. Since the X_i are i.i.d. from Q , this event has probability $(1 - Q(\mathcal{S}_{\geq N_\varepsilon-1}))^m < (1 - \varepsilon)^m$.

Hence $\Pr[Q(T_{\hat{j}}) > \varepsilon] \leq \Pr[\hat{j} < N_\varepsilon] \leq (1 - \varepsilon)^m$.

Step 3: the error decomposition. Write $h = \mathcal{L}(X_{1..m})$ and $Y = \text{inst}(\top I, \wedge I, \vee I_2) \subseteq R^*$. An element of $(h \cup T_{\hat{j}} \cup Y) \setminus R^*$ lies in $h \setminus R^*$ or in $T_{\hat{j}}$, because $Y \subseteq R^*$. An element of $R^* \setminus (h \cup T_{\hat{j}} \cup Y)$ lies in $R^* \setminus h$. So $(h \cup T_{\hat{j}} \cup Y) \triangle R^* \subseteq (h \triangle R^*) \cup T_{\hat{j}}$, and $\text{err}_Q(\mathcal{L}^+) \leq \text{err}_Q(\mathcal{L}) + Q(T_{\hat{j}})$ pointwise in the sample. This proves (a).

Step 4: PAC. Let $m \geq \max\{m_{\mathcal{L}}(\varepsilon/2, \delta/2), \lceil (2/\varepsilon) \ln(2/\delta) \rceil\}$. Then $\Pr[\text{err}_Q(\mathcal{L}) > \varepsilon/2] \leq \delta/2$, since as usual the guarantee of $\mathcal{L}$ holds for every sample size $m \geq m_{\mathcal{L}}(\varepsilon, \delta)$. By Step 2 applied with $\varepsilon/2$, $\Pr[Q(T_{\hat{j}}) > \varepsilon/2] \leq (1 - \varepsilon/2)^m \leq e^{-m\varepsilon/2} \leq \delta/2$. By Step 3 and a union bound, $\Pr[\text{err}_Q(\mathcal{L}^+) > \varepsilon] \leq \delta$. Both events are computed from the same sample, so no extra samples beyond the maximum are needed. The outputs of $\mathcal{L}^+$ need not lie in the hypothesis class of $\mathcal{L}$, so the guarantee is improper. This proves (b).

Step 5: trivialization. Every output of $\mathcal{L}^+$ contains $Y \cup T_{\hat{j}}$. The derivation in the proof of Theorem 3.2(i), with $j = \hat{j}$, uses only steps of Y (namely $\top I$, $\wedge I$ and $\vee I_2$) and one step of $T_{\hat{j}}$. So $\text{Cl}_{\mathcal{L}^+(X_{1..m})}(\emptyset) = \text{Fm}$ for every sample. This proves (c). $\square$

Remark B.1. Step 5 is where the three added schemas are needed. If $\mathcal{L} \equiv \emptyset$ and they are omitted, the output is $T_{\hat{j}}$ alone. Every step of $T_{\hat{j}}$ has a premise, so $\emptyset$ is closed under it and $\text{Cl}_{T_{\hat{j}}}(\emptyset) = \emptyset$.

B.2 Schemas that mention atoms, and purification

Verification of Example 3.6. Let $\tau = (x \vee p_0 \triangleright x)$ and $A = R_{\text{CPC}}^* \cup \text{inst}(\tau)$.

- τ is *unsound*. Its instance $p_1 \vee p_0 \triangleright p_1$ fails under $v(p_0) = 1, v(p_1) = 0$.
- A is *not trivial*. Let $X := \text{Cn}_{\text{CPC}}(\neg p_0)$. Then $X \supseteq \emptyset$. X is closed under R_{CPC}^* , because $\text{Cl}_{R_{\text{CPC}}^*}(X) = \text{Cn}_{\text{CPC}}(X) = X$. X is also closed under $\text{inst}(\tau)$: if $B \vee p_0 \in X$, every valuation with $v(p_0) = 0$ satisfies $B \vee p_0$ and hence B , so $\neg p_0 \models B$ and $B \in X$. By leastness, $\text{Cl}_A(\emptyset) \subseteq X$. Since $p_0 \notin X$ (take $v(p_0) = 0$), $\text{Cl}_A(\emptyset) \neq \text{Fm}$.
- A is *unsound*. The tautology $\neg p_0 \vee p_0$ lies in $\text{Cl}_{R_{\text{CPC}}^*}(\emptyset)$, and the instance of τ with $x := \neg p_0$ yields $\neg p_0$, which is not a tautology.
- $\text{inst}(\tau)$ is *not substitution-closed*. The substitution $p_0 \mapsto p_1$ maps the instance $p_2 \vee p_0 \triangleright p_2$ to $p_2 \vee p_1 \triangleright p_2$, which is not an instance of τ .
- A *ground error does not fire*. For the single step $p_0 \triangleright p_1$, the set $\text{Cn}_{\text{CPC}}(\emptyset)$ is closed under $R_{\text{CPC}}^* \cup \{p_0 \triangleright p_1\}$, because $p_0 \notin \text{Cn}_{\text{CPC}}(\emptyset)$. So the closure from $\emptyset$ is just the set of tautologies.

Proof of Lemma 3.7. Let τ have metavariables $\bar{x} = x_1, \dots, x_k$, and let $p_{i_1}, \dots, p_{i_l}$ be the distinct atoms occurring in τ . Let $y_1, \dots, y_l$ be fresh metavariables and $\tau^\circ := \tau[p_{i_t} := y_t]_{t \leq l}$. Write $\tau[\bar{x} := \bar{A}]$ for the instance of τ obtained by substituting the formulas $\bar{A}$ for $\bar{x}$, and similarly for τ° .

($\Leftarrow$) Every instance of τ is an instance of τ° : $\tau[\bar{x} := \bar{A}] = \tau^\circ[\bar{x} := \bar{A}, \bar{y} := (p_{i_1}, \dots, p_{i_l})]$. So if τ° is sound, so is τ .

($\Rightarrow$) Assume τ is sound, and let $\iota = \tau^\circ[\bar{x} := \bar{A}, \bar{y} := \bar{B}]$ be an instance of τ° . Then ι is τ with each occurrence of p_{i_t} in τ itself replaced by B_t , and each x_s replaced by A_s . Occurrences of p_{i_t} inside the A_s are left alone. There are infinitely many atoms, so we can choose distinct atoms

$r_1, \dots, r_l$ that occur nowhere in $\bar{A}$, $\bar{B}$ or τ . Let ρ be the renaming $p_{i_t} \mapsto r_t$ ($t \leq l$), identity on all other atoms, and put $A'_s := \rho(A_s)$. Then $\iota' := \tau[\bar{x} := \bar{A}']$ is an instance of τ , hence sound.

Let σ be the simultaneous uniform substitution with $\sigma(p_{i_t}) = B_t$, $\sigma(r_t) = p_{i_t}$ ($t \leq l$), and identity on all other atoms. We claim $\sigma(\iota') = \iota$. The formulas of ι' are built from the skeleton of τ , whose atoms are among the p_{i_t} and include no r_t , with the fillers A'_s plugged in for the metavariables.

- On the skeleton, σ replaces each p_{i_t} by B_t , exactly as the substitution $\bar{y} := \bar{B}$ does in τ° .
- Each filler $A'_s = \rho(A_s)$ contains no p_{i_t} , since ρ renamed them all. It contains r_t exactly where A_s contains p_{i_t} , because A_s contains no r_t . All its other atoms are fixed by both ρ and σ . Hence $\sigma(A'_s) = A_s$.

So $\sigma(\iota') = \tau^\circ[\bar{x} := \bar{A}, \bar{y} := \bar{B}] = \iota$.

Finally, uniform substitution preserves classical validity. Suppose $\Pi' \models \varphi'$ and let v be any valuation. Define $v'(p) := v(\sigma(p))$. By induction on formulas, $v'(\psi) = v(\sigma\psi)$ for every ψ . So if v satisfies $\sigma\Pi'$, then v' satisfies Π' , hence φ' , and v satisfies $\sigma\varphi'$. Applied to ι' , this shows that $\iota = \sigma(\iota')$ is sound. $\square$

Proof of Corollary 3.8. R_{CPC}^* is sound, so A is sound iff every τ_l is sound. By Lemma 3.7, this holds iff every τ_l° is sound, i.e. iff A° is sound.

Next, $A_1^\circ := \bigcup_l \text{inst}(\tau_l^\circ)$ is substitution-closed. Let $\tau^\circ\theta$ be an instance, where θ maps metavariables to formulas, and let σ be a uniform substitution of formulas for atoms. Since τ° contains no atom, $\sigma(\tau^\circ\theta) = \tau^\circ(\sigma \circ \theta)$, which is again an instance.

If A° is unsound, Proposition 3.5 gives $\text{Cl}_{A^\circ}(\emptyset) = \text{Fm} \ni \perp$. If A° is sound, then every step of A° is in $\text{Sound}(R_{\text{CPC}}^*)$ by completeness: $\Pi \models \varphi$ implies $\varphi \in \text{Cn}_{\text{CPC}}(\Pi) = \text{Cl}_{R_{\text{CPC}}^*}(\Pi)$. By Lemma 3.1, $\text{Cl}_{A^\circ}(\emptyset) \subseteq \text{Cl}_{R_{\text{CPC}}^*}(\emptyset) = \text{Cn}_{\text{CPC}}(\emptyset)$, which does not contain $\perp$. $\square$

B.3 The commitment-order lemma

Recall Definition 3.10. An *R-derivation* is a finite tree of sequents in which each node is either a leaf marked as a *premise* (chosen freely by the prover) or, together with its children, forms an instance of some rule of R : the children are the instance's premise sequents (none, for a premise-free rule) and the node is its conclusion sequent. The *conclusion* of the derivation is the sequent at the root.

Proof of Lemma 3.11(i). Let $G := \{x : \rho_i \text{ is valid in } x \text{ for every } i \leq m\}$. By the union bound over the whole library,

$$\Pr_{x \sim D} [x \notin G] \leq \sum_{i=1}^m \Pr_{x \sim D} [\rho_i \text{ is not valid in } x] \leq \sum_{i=1}^m \varepsilon_i.$$

Fix $x \in G$ and any R -derivation whose premises are all true in x . We show by induction on height that every sequent in the tree is true in x . Premise nodes are true by assumption. Every other node is the conclusion of an instance ι of some ρ_i , whose premise sequents are its children (possibly none) and are true in x by the induction hypothesis. Since $x \in G$, ρ_i is valid in x , so ι holds in x and its conclusion is true in x . In particular the conclusion of the derivation is true in x .

So the event “some R -derivation from premises true in x has a conclusion false in x ” is contained in the complement of G . Its probability (its outer probability, if it is not measurable)

is therefore at most $\sum_i \varepsilon_i$. The event G is determined by x and the library R alone. Because R is fixed before x is drawn, G does not depend on the prover, and the bound holds simultaneously for every prover, including one that sees x and chooses any number of derivations of any length. The sum runs over the whole library, not over the rules a particular derivation uses. $\square$

Remark B.2 (Per-pair guarantees do not give per-scene validity). Suppose each scene in the support of D falsifies exactly one of M instances of a rule ρ , and the instance is drawn uniformly from these M . Then $\Pr_{(x,\iota)}[\iota \text{ holds in } x] = 1 - 1/M$, but ρ is valid in no scene, so its per-scene validity is 0. A prover that sees x submits the falsified instance; this is regime (ii) inside each scene.

Proof of Lemma 3.11(ii). Let ι_0 be an instance of ρ that fails in x_0 : its premise sequents are true in x_0 and its conclusion sequent is false in x_0 . The tree consisting of ι_0 alone (its conclusion at the root, its premises as leaves) is a $\{\rho\}$ -derivation from premises true in x_0 with a conclusion false in x_0 . A prover that knows x_0 submits it, and nothing in this regime is random, so this happens with probability 1. If $\mu\{\iota : \iota \text{ fails in } x_0\} > 0$, such an ι_0 exists. If μ is supported on instances that hold in x_0 , the average validity is 1, i.e. $\varepsilon = 0$, but a failing instance off the support of μ may still exist, and the same prover uses it. So no bound in terms of ε holds. $\square$

For part (iii) we use the Schwartz–Zippel lemma (Schwartz, 1980; Zippel, 1979; DeMillo and Lipton, 1978): if $f \in F[z_1, \dots, z_n]$ is nonzero of total degree at most d and $S \subseteq F$ is finite, then $\Pr_{a \sim \text{Unif}(S^n)}[f(a) = 0] \leq d/|S|$.

Proof of Lemma 3.11(iii). Formalize the protocol as follows. In round k the prover, which may be randomized, adaptive and computationally unbounded, commits to explicit expressions for $p_k, q_k \in F[z_1, \dots, z_n]$. The verifier computes an upper bound on $\deg(p_k - q_k)$ from the expressions and rejects if it exceeds d . Otherwise it draws a_k uniformly from S_k^n , independently of everything so far, and accepts iff $p_k(a_k) = q_k(a_k)$.

Let $\mathcal{H}_k$ be the σ -algebra generated by everything that happens before a_k is drawn: earlier claims, points and answers, and the prover’s coins. The claim (p_k, q_k) is $\mathcal{H}_k$ -measurable, and a_k is independent of $\mathcal{H}_k$. Let E_k be the event that claim k passes the degree check and $p_k - q_k$ is nonzero in $F[z_1, \dots, z_n]$, and Acc_k the event that it is accepted. By the Schwartz–Zippel lemma, applied conditionally on $\mathcal{H}_k$,

$$\Pr[E_k \cap \text{Acc}_k \mid \mathcal{H}_k] = \mathbf{1}_{E_k} \cdot \Pr_{a \sim \text{Unif}(S_k^n)}[(p_k - q_k)(a) = 0] \leq d/|S_k|.$$

Taking expectations, $\Pr[E_k \cap \text{Acc}_k] \leq d/|S_k|$. A false claim (one with $p_k - q_k \neq 0$ over F) that fails the degree check is rejected, so a false claim is accepted only on $E_k \cap \text{Acc}_k$. By the union bound,

$$\Pr[\exists k \leq N : \text{claim } k \text{ is false and accepted}] \leq \sum_{k \leq N} d/|S_k|,$$

which is $Nd/|S|$ when $S_k = S$ for all k . If $|S_k| \geq d2^k/\delta$ for all $k \geq 1$, then $\sum_{k \geq 1} d/|S_k| \leq \sum_{k \geq 1} \delta 2^{-k} = \delta$, which bounds the probability that any false claim in an unbounded sequence is accepted. The prover’s strategy enters only through the $\mathcal{H}_k$, so the bounds hold uniformly over provers. $\square$

Why the conditions of (iii) are needed.

- *Degree enforcement.* Over $F = \mathbb{F}_{2^k}$ with $S = F$, the claim $z^{2^k} \equiv z$ can be written with k squarings. The polynomial $z^{2^k} - z$ is nonzero of degree 2^k , but it vanishes on all of F , so random evaluation always accepts. The bound $d/|S|$ is vacuous here, since $d = 2^k = |S|$. A verifier that read the degree off the length of the expression, roughly k , would wrongly believe its error to be about $k/2^k$.
- *The field.* For prime p , $f = (a + b)^p - a^p - b^p = \sum_{0 < i < p} \binom{p}{i} a^i b^{p-i}$ is nonzero over $\mathbb{Z}$, but p divides every $\binom{p}{i}$ with $0 < i < p$, so $f = 0$ in $\mathbb{F}_p[a, b]$. Evaluation mod p accepts this false integer identity with probability 1. Over $\mathbb{Q}$ with $S \subseteq \mathbb{Z}$, the lemma applies with $d = p$.
- *Order of commitment.* If the evaluation points were drawn before the prover committed to its claims, and visible to it, the prover could choose a nonzero $p - q$ that vanishes at those points. This is regime (ii) again.

B.4 The doctrinal paradox

Proof of Theorem 3.12. Step 1: each h_i is a coherent, cut-closed calculus. The sets T_1, T_2, T_3 are satisfied by the valuations $(v(p), v(q)) = (1, 1), (1, 0), (0, 1)$ respectively. For every set Γ , $\text{Cl}_{h_i}(\Gamma) = \text{Cn}_{\text{CPC}}(T_i \cup \Gamma)$. For $\supseteq$: if $T_i \cup \Gamma \models \varphi$, compactness gives a finite $\Gamma_0 \subseteq \Gamma$ with $T_i \cup \Gamma_0 \models \varphi$, so $\Gamma_0 \triangleright \varphi \in h_i$ and $\varphi \in \text{Cl}_{h_i}(\Gamma)$. For $\subseteq$: $\text{Cn}_{\text{CPC}}(T_i \cup \Gamma)$ contains Γ and is closed under h_i , since $\Pi \subseteq \text{Cn}_{\text{CPC}}(T_i \cup \Gamma)$ and $T_i \cup \Pi \models \varphi$ imply $T_i \cup \Gamma \models \varphi$. In particular $\text{Cl}_{h_i}(\emptyset) = \text{Cn}_{\text{CPC}}(T_i)$, which does not contain $\perp$ because T_i is satisfiable. So every h_i is $\mathcal{A}$ -coherent and is a possible target.

Step 2: memberships. The following table records which h_i contain each step of π .

step	$h_1 (p, q)$	$h_2 (p, \neg q)$	$h_3 (\neg p, q)$
$\triangleright p$	✓	✓	–
$\triangleright q$	✓	–	✓
$p, q \triangleright p \wedge q$	✓	✓	✓
$\triangleright \neg(p \wedge q)$	–	✓	✓
$p \wedge q, \neg(p \wedge q) \triangleright \perp$	✓	✓	✓

The checkmarks are immediate: $p \in T_1, T_2$; $q \in T_1, T_3$; $\neg q \models \neg(p \wedge q)$ and $\neg p \models \neg(p \wedge q)$; and the third and fifth steps are valid outright. For the dashes: $T_3 \models \neg p$, $T_2 \models \neg q$ and $T_1 \models p \wedge q$, and each T_i is satisfiable, so $T_3 \not\models p$, $T_2 \not\models q$ and $T_1 \not\models \neg(p \wedge q)$.

Step 3: π is a legal detection that deletes nothing. The five steps form an argument π from $\emptyset$ to $\perp$: the first two yield p and q , the third yields $p \wedge q$, the fourth yields $\neg(p \wedge q)$, and the fifth yields $\perp$. With uniform weights, s is in the majority set over $\mathcal{H}$ iff s lies in at least two of the h_i , since $2/3 > 1/2 > 1/3$. By the table, every step of π lies in at least two h_i , so $\text{Steps}(\pi) \subseteq \hat{R}_{\text{maj}}$, the majority set over $\mathcal{H}$. Also by the table, $\text{Steps}(\pi) \not\subseteq h_i$ for each i . So the deletion set $\{R \in \mathcal{H} : \text{Steps}(\pi) \subseteq R\}$ is empty.

Step 4: induction on rounds. Fix any target h_{i^*} . Consider the environment that, in each round, exhibits π or presents the positive datum $\triangleright p \vee \neg p$, in any order. We show $\text{VS}_t = \mathcal{H}$ for all t . Initially $\text{VS}_1 = \mathcal{H}$. If $\text{VS}_t = \mathcal{H}$, then $\hat{R}_t = \hat{R}_{\text{maj}} \supseteq \text{Steps}(\pi)$. So π is a legal (N)-datum, since $\emptyset \in \mathcal{A}$, and by Step 3 it deletes nothing. The datum $\triangleright p \vee \neg p$ lies in every h_i , so it is a legal (P)-datum for the target and deletes $\{R : \triangleright p \vee \neg p \notin R\} = \emptyset$. Hence $\text{VS}_{t+1} = \mathcal{H}$. So $\text{VS}_t = \mathcal{H}$ and $\hat{R}_t = \hat{R}_{\text{maj}}$ for all t , and π is available in every round. $\square$

Consequences recorded in the main text.

- *The vote is unsound relative to every target.* By Step 3, $\perp \in \text{Cl}_{\hat{R}_{\text{maj}}}(\emptyset)$. By Step 1, $\perp \notin \text{Cl}_{h_i}(\emptyset)$ for each i . By Lemma 3.1 with $B = \emptyset$, $\hat{R}_{\text{maj}} \not\subseteq \text{Sound}(h_i)$ for each i .
- *Discriminating data break the paradox.* Let the target be h_1 and present $\triangleright p \in h_1$. This deletes h_3 , since $\triangleright p \notin h_3$. Over $\text{VS} = \{h_1, h_2\}$, with weight $1/3$ each, a step is a strict majority iff it lies in both, so $\hat{R} = h_1 \cap h_2$. Since $\triangleright q \notin h_2$, π is no longer available. The h_i are pairwise incomparable: $\triangleright p \wedge q$ lies only in h_1 , $\triangleright p \wedge \neg q$ only in h_2 , and $\triangleright \neg p \wedge q$ only in h_3 . So on a text enumerating all of the target, each wrong hypothesis is eventually deleted by a datum it lacks, and from then on the uninformative detections stop.

C Proofs for Section 4

The following are proved in full in the main text: Theorem 4.2(a) and (c), Theorem 4.12, Theorem 4.14, and Corollaries 4.18 and 4.19. This appendix proves the remaining results, checks the examples of Section 4, and records the evidence for Conjecture 4.11. All statements are the versions repaired after adversarial verification (T1 Verification log). Where a repair changed a statement or a proof, this is noted.

Throughout, a verifier’s internal randomness is a single draw $\omega \sim \Lambda$ on a space Ω_V , made before the first round. Nothing is lost by this, since coins drawn later can be drawn up front. Steps are ground terms, $|t|$ is the number of symbol occurrences of t , and $\mu(t) = |t| - |\text{vars}(t)|$. We use the rank lemma (Lemma 2.6) and the characterization of the lgg (Theorem 2.5).

C.1 Optimality of the VS verifier

Fix human data P_0 and a deterministic, non-adaptive prover issuing $q_1, q_2, \dots$. For a step set T used as the target, the transcript after i rounds is $\tau_i^T(\omega) = (e_1, \dots, e_i)$ with $e_j = (q_j, a_j, y_j)$. It is defined recursively by $a_j = V(P_0, \tau_{j-1}^T(\omega), q_j, \omega)$, and $y_j = \mathbf{1}[q_j \in T]$ if $a_j = \text{ESCALATE}$, while $y_j = *$ (no label) otherwise. A history $h = (e_1, \dots, e_t)$ is *consistent* with a step set R if $y_j = \mathbf{1}[q_j \in R]$ whenever $y_j \neq *$.

Lemma C.1 (Coupling through the coins). *Let h be a history of t rounds that is consistent with two step sets T and T' , with queries $q_1, \dots, q_t$. Then for every ω , $\tau_t^T(\omega) = h$ iff $\tau_t^{T'}(\omega) = h$.*

Proof. We show by induction on $i \leq t$ that $\tau_i^T(\omega) = h_{\leq i}$ iff $\tau_i^{T'}(\omega) = h_{\leq i}$. The case $i = 0$ is trivial. Suppose the claim holds for $i - 1$. If neither transcript equals $h_{\leq i-1}$, neither extends to $h_{\leq i}$. If both equal $h_{\leq i-1}$, the verifier’s answer $a_i = V(P_0, h_{\leq i-1}, q_i, \omega)$ is the same under both targets. If a_i differs from the answer recorded in h , neither transcript equals $h_{\leq i}$. If it agrees and is not **ESCALATE**, both transcripts equal $h_{\leq i}$. If it agrees and is **ESCALATE**, the labels are $\mathbf{1}[q_i \in T]$ and $\mathbf{1}[q_i \in T']$. Both equal the label recorded in h , by consistency, so both transcripts equal $h_{\leq i}$. $\square$

Proof of Theorem 4.2(b). Since $q \notin \bigcap \text{VS}_h$, there is $R' \in \text{VS}_h$ with $q \notin R'$. Then $R' \in \mathcal{H}$ and $P_0 \subseteq R'$, and R' contains every step labelled positive in h and no step labelled negative, so h is consistent with R' . By hypothesis, h is consistent with R^* . Let the prover issue $q_1, \dots, q_t, q_{t+1} := q$. For $T \in \{R^*, R'\}$ put

$$E_T := \{\omega : \tau_t^T(\omega) = h \text{ and } V(P_0, h, q, \omega) = \text{ACCEPT}\}.$$

By Lemma C.1, $E_{R^*} = E_{R'}$ as subsets of Ω_V , so $p = \Lambda(E_{R^*}) = \Lambda(E_{R'})$. On $E_{R'}$, with target R' , the invalid query $q \notin R'$ is accepted in round $t + 1$. V is δ -sound, and R' is an admissible target, so $\Lambda(E_{R'}) \leq \mathbb{P}_{R'}[\exists s : a_s = \text{ACCEPT} \wedge q_s \notin R'] \leq \delta$. $\square$

This joint-probability form replaces the original statement, which bounded the acceptance probability at h and is false when read conditionally (referee report A-1). The deterministic form (c) follows because $\Lambda(E) \in \{0, 1\}$ when Λ is a point mass. Any history that a deterministic verifier produces against a target is consistent with that target, since the oracle is truthful.

The reckless verifier. Let $\mathcal{H} = \{R_1 = \{a, c\}, R_2 = \{a, b, c\}, R_3 = \{a, b\}\}$ and $P_0 = \emptyset$. The verifier draws $B \sim \text{Bernoulli}(\varepsilon)$. If $B = 1$ it accepts every query; otherwise it runs the VS verifier.

- It is ε -sound. When $B = 0$ no invalid query is ever accepted, by Theorem 4.2(a), so the probability of an invalid acceptance is at most $\mathbb{P}[B = 1] = \varepsilon$.
- Take target R_2 , and let the prover query c and then b . Initially $\cap \text{VS} = \{a\}$, so when $B = 0$ the query c is escalated. The history $h = ((c, \text{ACCEPT}, *))$ therefore has probability exactly ε . It carries no label, so it is consistent with every hypothesis, and $\text{VS}_h = \mathcal{H}$ with $\cap \text{VS}_h = \{a\} \not\ni b$.
- Given h (that is, $B = 1$), b is accepted with probability 1. The joint probability is $\varepsilon \leq \delta = \varepsilon$, as (b) says.

A simulation reproduces this (`T1-code/thm31b_counterexample.py`).

Random human data. Suppose instead that P_0 is random, with a countable range and a target-dependent law, and independent of ω . Let d be an observed value with $d \subseteq R'$, where R' is as above with VS_h computed from $P_0 = d$. The coupling, applied conditionally on $P_0 = d$, gives

$$p(d) := \mathbb{P}_{R^*}[h, \text{ then } q \text{ accepted} \mid P_0 = d] = \mathbb{P}_{R'}[h, \text{ then } q \text{ accepted} \mid P_0 = d].$$

Hence $\mathbb{P}_{R'}[P_0 = d] \cdot p(d) \leq \delta$, which is the weaker bound stated in the main text.

C.2 The escalation dimension

Proof of Theorem 4.4. Upper bound. Let V be the VS verifier, and let $R^* \in \mathcal{H}$ with $P_0 \subseteq R^*$. Consider any honest prover, which may be adaptive.

- (1) V never rejects. By Theorem 4.2(a), $R^* \in \text{VS}$, so every query $q \in R^*$ lies in $\cup \text{VS}$.
- (2) Every label is 1. So $P^- = \emptyset$ throughout, and $P = P_0 \cup E$, where E is the set of queries escalated so far.
- (3) Let $s_1, s_2, \dots$ be the escalated queries in order. When s_i is escalated, $s_i \notin \cap \text{VS}(P_0 \cup \{s_1, \dots, s_{i-1}\})$, and $s_i \in R^*$.

So the escalated queries form an elastic chain in R^* , and since nothing is rejected, the cost equals their number, which is at most $\text{el}(\mathcal{H}, R^* \mid P_0)$. As V is 0-sound, $\text{Esc}(\mathcal{H} \mid P_0) \leq \sup_R \text{el}(\mathcal{H}, R \mid P_0)$.

Lower bound. Let V be δ -sound. Let $s_1, \dots, s_m$ be an elastic chain in some $R \in \mathcal{H}$ with $P_0 \subseteq R$, with witnesses R_i . The prover queries $s_1, \dots, s_m$ in this order, which is honest for target R . Fix i . For $j < i$ we have $s_j \in R$ and $s_j \in R_i$, so the oracle's label on s_j , if requested, is 1 under both targets. Induction on rounds, as in Lemma C.1, gives $\tau_{i-1}^R(\omega) = \tau_{i-1}^{R_i}(\omega)$ for every ω . Hence the answers in round i coincide, $a_i^R(\omega) = a_i^{R_i}(\omega)$. Under target R_i , which is admissible since $R_i \in \mathcal{H}$ and $P_0 \subseteq R_i$, the query s_i is invalid. So $\Lambda(a_i^{R_i} = \text{ACCEPT}) \leq \delta$, and therefore $\Lambda(a_i^R = \text{ACCEPT}) \leq \delta$. Every s_i is valid for R , so the expected cost under R is at least

$\sum_{i=1}^m \Lambda(a_i^R \neq \text{ACCEPT}) \geq (1 - \delta)m$. If $\delta = 0$, each $a_i^R \neq \text{ACCEPT}$ almost surely, so the cost is m almost surely. Every 0-sound verifier thus pays at least m for every finite $m \leq \sup_R \text{el}(\mathcal{H}, R \mid P_0)$, so $\text{Esc}(\mathcal{H} \mid P_0) \geq \sup_R \text{el}(\mathcal{H}, R \mid P_0)$.

Size restriction. The same argument applies verbatim when all queries and chain elements have size at most N . $\square$

The original proof of the lower bound cited the conditional reading of Theorem 4.2(b). The direct coupling above replaces it (referee report A-4). The cost of a randomized verifier is its expected cost (Definition 2.21).

C.3 Intersection-closed classes; single schemas

Proof of Proposition 4.5. (i) By definition, $\bigcap \text{VS}(P)$ is the intersection of the members of $\mathcal{H}$ that contain P , which is $\text{cl}(P)$. Intersection-closure puts it in $\mathcal{H} \cup \{\emptyset\}$. It contains P , and every member containing P contains it. Against an honest prover there are no negative labels, and $P = P_0 \cup E$, so the VS verifier accepts iff $q \in \text{cl}(P_0 \cup E)$.

(ii) Let $s_1, \dots, s_m$ be an elastic chain in R , and put $C_i = \text{cl}(P_0 \cup \{s_1, \dots, s_i\})$. Then $C_0 = \text{cl}(P_0)$, and $C_{i-1} \subseteq C_i$ because cl is monotone. The inclusion is strict: $s_i \in C_i$, while $s_i \notin \bigcap \text{VS}(P_0 \cup \{s_1, \dots, s_{i-1}\}) = C_{i-1}$ by (i). Finally $C_m \subseteq R$, because R is a member containing $P_0 \cup \{s_1, \dots, s_m\}$.

Conversely, let $\text{cl}(P_0) = C_0 \subsetneq \dots \subsetneq C_m \subseteq R$ be a chain of members of $\mathcal{H} \cup \{\emptyset\}$, and pick $s_i \in C_i \setminus C_{i-1}$. Then $s_i \in R$. For $i = 1$, $\text{cl}(P_0) = C_0 \not\ni s_1$. For $i \geq 2$, $C_{i-1} \supsetneq C_0$ is nonempty, hence a member of $\mathcal{H}$. It contains $P_0 \subseteq C_0$ and each $s_j \in C_j \subseteq C_{i-1}$ for $j < i$, so $\text{cl}(P_0 \cup \{s_1, \dots, s_{i-1}\}) \subseteq C_{i-1} \not\ni s_i$. By (i), $s_i \notin \bigcap \text{VS}(P_0 \cup \{s_1, \dots, s_{i-1}\})$, so the s_i form an elastic chain in R . $\square$

Proof of Lemma 4.6. (1) *Two schemas.* Let σ, τ be schemas, renamed apart. A ground term t lies in $\text{inst}(\sigma) \cap \text{inst}(\tau)$ iff $t = \sigma\theta = \tau\theta'$ for ground substitutions θ, θ' . Since the variables are disjoint, $\theta \cup \theta'$ then unifies σ and τ . If σ and τ are not unifiable, the intersection is empty. Otherwise let η be a most general unifier. Every unifier factors through η , so $t = \sigma\eta\rho$ for some ρ , and $t \in \text{inst}(\sigma\eta)$. Conversely, every ground instance $\sigma\eta\rho = \tau\eta\rho$ lies in both sets. So $\text{inst}(\sigma) \cap \text{inst}(\tau) = \text{inst}(\sigma\eta) \in H_1$.

(2) *Arbitrary families.* Let $\mathcal{F} \subseteq H_1$. The empty family has intersection $S = \text{inst}(x) \in H_1$. If $\bigcap \mathcal{F} = \emptyset$, it lies in H_1 . Otherwise $\bigcap \mathcal{F}$ contains a ground term t , and every member of $\mathcal{F}$ is $\text{inst}(\sigma)$ with $\sigma\theta = t$ for some θ . Then $|\sigma| \leq |t|$, and every function or constant symbol of σ occurs in t . So, up to renaming, there are finitely many such σ , and $\mathcal{F}$ has finitely many distinct members. Its intersection is a finite intersection, which lies in H_1 by (1).

(3) *Lggs of arbitrary nonempty sets.* Let $P \neq \emptyset$. For finite $F \subseteq F' \subseteq P$ we have $\text{lgg}F \preceq \text{lgg}F'$, since $F \subseteq F' \subseteq \text{inst}(\text{lgg}F')$. By the rank lemma, a strictly increasing chain of lggs has length at most $\mu(\text{lgg}\{t\}) = |t|$ for any $t \in P$. So there is a finite $F_0 \subseteq P$ such that $\text{lgg}F \equiv \text{lgg}F_0$ for every finite F with $F_0 \subseteq F \subseteq P$. Put $g := \text{lgg}F_0$. For each $s \in P$, $\text{lgg}(F_0 \cup \{s\}) \equiv g$, so $s \in \text{inst}(g)$. If $P \subseteq \text{inst}(\tau)$, then $F_0 \subseteq \text{inst}(\tau)$, so $g \preceq \tau$. Thus g is the lgg of P , and $\text{inst}(g) = \bigcap \{\text{inst}(\tau) : P \subseteq \text{inst}(\tau)\}$.

(4) *The closure.* For nonempty P , the members of H_1 that contain P are the sets $\text{inst}(\tau)$ with $P \subseteq \text{inst}(\tau)$ (the empty set does not contain P). So $\text{cl}(P) = \text{inst}(\text{lgg}P)$ by (3), or by Theorem 2.5 when P is finite. $\square$

C.4 One schema

Proof of Theorem 4.7. (i) Let $s_1, \dots, s_m$ be an elastic chain in $\text{inst}(\sigma^*)$ relative to a nonempty $P_0 \subseteq \text{inst}(\sigma^*)$. Put $g_i := \text{lgg}(P_0 \cup \{s_1, \dots, s_i\})$ for $0 \leq i \leq m$. By Lemma 4.6, $\bigcap \text{VS}(Q) = \text{inst}(\text{lgg}Q)$ for nonempty Q . So the elasticity condition says $s_i \notin \text{inst}(g_{i-1})$, while $s_i \in \text{inst}(g_i)$. Now $P_0 \cup \{s_1, \dots, s_{i-1}\} \subseteq \text{inst}(g_i)$, so $g_{i-1} \preceq g_i$ by the lgg property, and the instance sets differ, so $g_{i-1} \prec g_i$ strictly. This uses the lgg property and no assumption on the signature; mere inclusion of instance sets would not give generality over a poor signature (referee report A-6). Also $g_m \preceq \sigma^*$, since $P_0 \cup \{s_1, \dots, s_m\} \subseteq \text{inst}(\sigma^*)$. By the rank lemma $\mu(g_{i-1}) \geq \mu(g_i) + 1$ for each i , and $\mu(g_m) \geq \mu(\sigma^*)$. Summing, $\mu(\text{lgg}P_0) = \mu(g_0) \geq \mu(\sigma^*) + m$.

(ii) Let $s_1, \dots, s_m$, $m \geq 1$, be an elastic chain in $\text{inst}(\sigma^*)$ relative to $\emptyset$, with $|s_i| \leq N$. Then $s_2, \dots, s_m$ is an elastic chain relative to $\{s_1\}$, because the version spaces involved are the same. By (i), $m-1 \leq \mu(s_1) - \mu(\sigma^*)$. Since s_1 is ground, $\mu(s_1) = |s_1| \leq N$. So $m \leq 1 + N - \mu(\sigma^*) \leq N + 1$, as $\mu \geq 0$. Taking the supremum over targets and applying Theorem 4.4 with the size restriction gives $\text{Esc}(H_1; N) \leq N + 1$.

(iii) Let $N \geq 1$, and consider the steps

$$t_0 = g^{N-1}(a), \quad t_1 = g^{N-1}(b), \quad t_j = g^{N-j}(a) \quad (2 \leq j \leq N),$$

all of size at most N and all in $\text{inst}(x)$. Let $L_j = \text{lgg}(t_0, \dots, t_j)$. Then $L_0 = t_0$. For $j \geq 1$, every t_i with $i \leq j$ has the form $g^{N-j}(u_i)$. The u_i have mixed roots: they are a, b when $j = 1$, and include $u_0 = g^{j-1}(a)$ and $u_j = a$ when $j \geq 2$. So anti-unification copies the prefix g^{N-j} and places a metavariable at the bottom, giving $L_j \equiv g^{N-j}(z)$ and $\mu(L_j) = N - j$. Each t_j escapes:

- t_0 escapes because $\emptyset \in H_1$, so $\bigcap \text{VS}(\emptyset) = \emptyset$.
- $t_1 \notin \text{inst}(L_0) = \{g^{N-1}(a)\}$.
- For $j \geq 2$, $t_j = g^{N-j}(a) \notin \text{inst}(g^{N-j+1}(z))$, whose instances begin with $N - j + 1$ copies of g .

By Lemma 4.6, $t_0, \dots, t_N$ is therefore an elastic chain of length $N + 1$ in $\text{inst}(x)$. Together with (ii) and $\mu(x) = 0$, $\text{el}(H_1, x \mid \emptyset; N) = N + 1$, and $\text{Esc}(H_1; N) = N + 1$ by Theorem 4.4. $\square$

Computation. Exhaustive search over all ground terms of size at most N (`T1-code/elast.py`) gives a maximal elasticity of 4, 5 at $N = 3, 4$ over $\{a, b, g, p\}$, which is $N + 1$. With a single constant, over $\{c, g, p\}$ at $N = 3, 4, 5$, it gives $3, 4, 5 = N$. We re-ran both for this paper.

C.5 Two-sided checking

Proof of Proposition 4.8. Let the signature have constants $a, b_1, \dots, b_n$ and an n -ary symbol f . Take the target $R^* = \text{inst}(f(a, \dots, a)) = \{f(a, \dots, a)\}$ and $P_0 = R^*$. For a partition π of $[n]$ with blocks indexed $1, \dots, |\pi|$, let $\pi(j)$ be the index of the block containing j , and put

$$s_\pi := f(b_{\pi(1)}, \dots, b_{\pi(n)}), \quad \sigma_\pi := f(x_{\pi(1)}, \dots, x_{\pi(n)}).$$

Three facts:

- (1) $f(a, \dots, a) \in \text{inst}(\sigma_\pi)$.
- (2) $s_{\pi'} \in \text{inst}(\sigma_\pi)$ iff π refines π' . Indeed, $s_{\pi'}$ is an instance iff the assignment $x_{\pi(j)} \mapsto b_{\pi'(j)}$ is well defined, i.e. iff $\pi(j) = \pi(k)$ implies $\pi'(j) = \pi'(k)$.

- (3) Each s_π has size $n + 1$ and is invalid.

List the $\text{Bell}(n)$ partitions as $\pi_1, \pi_2, \dots$ along a linear extension of refinement with the finest first, so that if π_u properly refines π_v then $u < v$. The prover queries $s_{\pi_1}, s_{\pi_2}, \dots$ in this order.

Let V be two-sided, and consider the query s_{π_v} . Every label received so far is negative, except for the positives in P_0 , and it concerns some s_{π_u} with $u < v$. Such a step is not an instance of σ_{π_v} : by (2) that would require π_v to refine π_u , which for $u < v$ and $\pi_u \neq \pi_v$ contradicts the order. With (1), $\text{inst}(\sigma_{\pi_v})$ is in the current version space and contains s_{π_v} . So $s_{\pi_v} \in \bigcup \text{VS}$, and V cannot reject it. V is 0-sound and s_{π_v} is invalid, so V cannot accept it. Hence V escalates every one of the $\text{Bell}(n)$ queries.

Fixed signature. The two-sided VS verifier escalates q only if $q \in \bigcup \text{VS} \setminus \bigcap \text{VS}$. Once escalated, q is labelled and joins P or P^- . From then on $q \in \bigcap \text{VS}$ (accepted) or $q \notin \bigcup \text{VS}$ (rejected), so no step is escalated twice. A ground term of size n is a string of n symbols in Polish notation, so over a finite signature Ω there are at most $\sum_{n \leq N} |\Omega|^n \leq (|\Omega| + 1)^N$ steps of size at most N . $\square$

The proposition originally claimed super-exponential cost without qualification. That needs a signature that grows with N , and the fixed-signature bound was added (referee report A-7). *Computation.* A brute-force search over all depth-one schemas $f(t_1, \dots, t_n)$ with $t_i \in \{a, b_j, \text{metavariables}\}$, together with x , queried in order of decreasing number of blocks, gives 2, 5 and 15 forced escalations for $n = 2, 3, 4$. These are $\text{Bell}(2), \text{Bell}(3), \text{Bell}(4)$ (T1-code/checks37.py, re-run).

C.6 Tagged rules

Proof of Theorem 4.9. Let $G = \{\text{inst}(\sigma) : \sigma \text{ a schema}\}$ be the class of nonempty single-schema instance sets. For a set Q of steps let $\text{VS}^G(Q) = \{C \in G : Q \subseteq C\}$. For a set P of tagged steps let $P^{(i)} = \{s : (i, s) \in P\}$.

(1) *Factorization.* A hypothesis $\bigcup_i \{i\} \times \text{inst}(\sigma_i)$ contains P iff $P^{(i)} \subseteq \text{inst}(\sigma_i)$ for every i . So $\text{VS}(P)$ corresponds bijectively to $\prod_i \text{VS}^G(P^{(i)})$, via $(\sigma_i)_i \mapsto (\text{inst} \sigma_i)_i$. If $P \subseteq R^*$, each factor contains $\text{inst}(\sigma_i^*)$ and is nonempty.

(2) *Intersection.* $(i, s) \in \bigcap \text{VS}(P)$ iff $s \in C_i$ for every tuple $(C_1, \dots, C_k)$ in the product. Because the other factors are nonempty, this holds iff $s \in \bigcap \text{VS}^G(P^{(i)})$. So $\bigcap \text{VS}(P) = \bigcup_i \{i\} \times \bigcap \text{VS}^G(P^{(i)})$.

(3) *Chains.* Let $(i_1, s_1), \dots, (i_m, s_m) \in R^*$. By (2), (i_u, s_u) escapes relative to P_0 and the earlier steps iff $s_u \notin \bigcap \text{VS}^G(P_0^{(i_u)} \cup \{s_v : v < u, i_v = i_u\})$. So the sequence is elastic iff, for each i , its tag- i subsequence is an elastic chain for $\text{inst}(\sigma_i^*)$ in G relative to $P_0^{(i)}$. Splitting an elastic sequence by tags gives $\leq$, and concatenating maximal per-tag chains gives $\geq$. Hence $\text{el}(H_k^{\text{tag}}, R^* \mid P_0; N) = \sum_i \text{el}_i$.

(4) *Per-rule bounds.* The classes G and H_1 differ only by $\emptyset$, which contains no nonempty set. So for nonempty data they have the same version spaces. If $P_0^{(i)} \neq \emptyset$, Theorem 4.7(i) gives $\text{el}_i \leq \mu(\text{lgg} P_0^{(i)}) - \mu(\sigma_i^*)$. If $P_0^{(i)} = \emptyset$, drop the first element of a chain, and the rest is a chain relative to a singleton. The argument of Theorem 4.7(ii) then gives $\text{el}_i \leq 1 + N - \mu(\sigma_i^*)$. Hence $\text{Esc}(H_k^{\text{tag}}; N) = \text{supel} \leq k(N + 1)$ by Theorem 4.4.

(5) *Tightness.* Assume constants a, b and a unary g , and take $\sigma_i^* = x$ for all i and $P_0 = \emptyset$. In G , $\bigcap \text{VS}^G(\emptyset) \subseteq \text{inst}(a) \cap \text{inst}(b) = \emptyset$, so the chain $t_0, \dots, t_N$ of Theorem 4.7(iii) is elastic in G . Concatenate its k tagged copies $(i, t_0), \dots, (i, t_N)$ for $i = 1, \dots, k$. By (3) this is an elastic chain of length $k(N + 1)$, so $\text{Esc}(H_k^{\text{tag}}; N) = k(N + 1)$. $\square$

The original justification asserted that H_k^{tag} is intersection-closed. It is not: for example, $(\{1\} \times \text{inst}(a) \cup \{2\} \times \text{inst}(c)) \cap (\{1\} \times \text{inst}(b) \cup \{2\} \times \text{inst}(c)) = \{2\} \times \text{inst}(c)$ is neither a member nor empty. The factorization (1)–(2) is what the proof uses (referee report A-8).

C.7 Untagged rules

Proof of Theorem 4.10(i). Assume $N \geq k + 1$, and let $n + 1 := \lfloor (N - 1)/k \rfloor \geq 1$. For $a \in \{0, \dots, n\}^k$ put $s_a := p(g^{a_1}c, \dots, g^{a_k}c)$, which has size $1 + \sum_j (a_j + 1) \leq 1 + k(n + 1) \leq N$. All s_a lie in the target $\text{inst}(p(x_1, \dots, x_k))$. Order the $(n + 1)^k$ vectors by non-increasing $\sum_j a_j$, breaking ties arbitrarily, and query the s_a in this order. Let a be the current vector and $a' \neq a$ an earlier one. Then $\sum a' \geq \sum a$, so $a'_j \geq a_j$ for some j ; otherwise $a' \leq a$ componentwise, and with $\sum a' \geq \sum a$ this would give $a' = a$. Put

$$R'_a := \bigcup_{j=1}^k \text{inst}(p(x_1, \dots, x_{j-1}, g^{a_j+1}(x_j), x_{j+1}, \dots, x_k)) \in H_k.$$

Every earlier $s_{a'}$ lies in the j -th schema for a j with $a'_j \geq a_j + 1$. The current s_a lies in none, since its j -th argument has only a_j copies of g . So R'_a witnesses that s_a escapes, and the sequence is an elastic chain of length $(n+1)^k$ relative to $P_0 = \emptyset$. By Theorem 4.4, $\text{Esc}(H_k; N) \geq \lfloor (N-1)/k \rfloor^k$. $\square$

Proof of Theorem 4.10(ii). Let $k \geq 2$, and let $s_1, \dots, s_m$ be an elastic chain relative to $\emptyset$, with $|s_i| \leq N$ and witnesses $R_i = \bigcup_{l \leq k} \text{inst}(\tau_{i,l})$. Unions of fewer than k schemas are padded by repetition (only the witness of s_1 can be empty, and it plays no role, since π_1 partitions $\emptyset$). Let π_i be the partition of $[i - 1]$ that assigns j to block l when l is the least index with $s_j \in \text{inst}(\tau_{i,l})$. It has at most k blocks.

Chains of indices. Call $i_0 < i_1 < \dots < i_r$ a *chain* if, for each $b \geq 1$, the indices $i_0, \dots, i_{b-1}$ all lie in one block of π_{i_b} , say block l . Then $\tau_{i_b,l}$ covers $s_{i_0}, \dots, s_{i_{b-1}}$, and it misses s_{i_b} because $s_{i_b} \notin R_{i_b}$. So $s_{i_b} \notin \text{instlgg}(s_{i_0}, \dots, s_{i_{b-1}}) \subseteq \text{inst}(\tau_{i_b,l})$, and hence

$$\text{lgg}(s_{i_0}) \prec \text{lgg}(s_{i_0}, s_{i_1}) \prec \dots \prec \text{lgg}(s_{i_0}, \dots, s_{i_r})$$

strictly. The first term is ground, with $\mu \leq N$. By the rank lemma, μ drops by at least one at each step and stays nonnegative. So $r + 1 \leq N + 1$: every chain has length at most $N + 1$. This is the argument of Theorem 4.7, and it needs no common target schema.

The recursion. Let $g(d)$ be the largest m for which there is a sequence $\pi_1, \dots, \pi_m$, with π_i a partition of $[i - 1]$ into at most k blocks, all of whose chains have length at most d . A two-element index set $\{i_0 < i_1\}$ is always a chain, so $g(1) = 1$. For $d \geq 2$, split $[m - 1]$ into the blocks of π_m . Fix a block B , and restrict each π_j with $j \in B$ to $B \cap [j - 1]$. This gives a sequence of the same kind indexed by B . Its chains are chains of the original sequence, and appending m to any of them gives a chain again, because B is one block of π_m . So its chains have length at most $d - 1$, and $|B| \leq g(d - 1)$. Hence $m - 1 \leq k g(d - 1)$. So $g(d) \leq 1 + k g(d - 1)$, and $g(d) \leq \sum_{u < d} k^u = (k^d - 1)/(k - 1)$. With $d = N + 1$, $m \leq (k^{N+1} - 1)/(k - 1)$, and Theorem 4.4 gives the bound on $\text{Esc}(H_k; N)$. $\square$

Proof of Theorem 4.10(iii). Identify a, b with $0, 1$, so that steps are points $x \in \{0, 1\}^n$. A linear flat schema denotes a subcube, which fixes some coordinates and leaves the others free. For points y, p and a set J of coordinates, say y agrees with p on J if $y_j = p_j$ for all $j \in J$. The pair $(J, p|_J)$ with $|J| = k$ is a *pattern*, and a set P of points realizes it if some $y \in P$ agrees with p on J .

Escape criterion. Let P be the set of earlier chain elements. A subcube Q with $p \notin Q$ fixes some coordinate j to $1 - p_j$, since otherwise $p \in Q$. So Q lies in the half-cube $\{x_j = 1 - p_j\}$, which is itself a subcube avoiding p . Hence p escapes iff some union of at most k half-cubes $\{x_j \neq p_j\}$, $j \in J$, covers P . That holds iff no $y \in P$ agrees with p on J . Padding J to size k (possible since $n \geq k$) preserves this. So p escapes iff some pattern of p is not realized by P .

Upper bound. There are $2^k \binom{n}{k}$ patterns. The first chain element realizes $\binom{n}{k}$ of them. Each later element escapes, so it realizes at least one pattern not realized before, and realized patterns stay realized. So the chain has length at most $1 + (2^k - 1) \binom{n}{k}$.

Lower bound. The target is the full cube $f(x_1, \dots, x_n)$. Query all points of weight at most k in order of non-decreasing weight. Let x have weight $w \leq k$, and choose $J \supseteq \text{supp}(x)$ with $|J| = k$. An earlier point y that agrees with x on J has $y_j = 1$ on $\text{supp}(x)$, so $\text{supp}(y) \supseteq \text{supp}(x)$. Its weight is at most w , so $y = x$, which is impossible for an earlier, distinct point. So every such point escapes, and the chain has length $\sum_{w=0}^k \binom{n}{w}$.

Repeated metavariables. A flat schema with repeated metavariables denotes the points that satisfy its constants and the equalities $x_i = x_j$ for positions sharing a metavariable. If such a schema avoids p , then either a constant position j has $p_j \neq$ the constant, so the schema lies in $\{x_j \neq p_j\}$, or all constants agree with p and some pair i, j sharing a metavariable has $p_i \neq p_j$, so the schema lies in $\{x_i = x_j\}$. Each such set is determined by at most two coordinates of p , and each of its points disagrees with p on one of them. If p escapes, there is therefore a set J of at most $2k$ coordinates on which no earlier point agrees with p . Padding J to size $2k$ (for $n \geq 2k$) and counting patterns as above gives a chain length of at most $1 + (2^{2k} - 1) \binom{n}{2k}$. The class contains the linear one, and elastic chains stay elastic in a larger class, so the lower bound persists. $\square$

The original summary said “ $\Theta(n^k)$ for flat schemas”. That is proved only for linear flat schemas, and it needs $n \geq k$ (referee report A-9). *Computation.* The exact values quoted in Section 4.3 come from `T1-code/elast.py` (deep schemas), `subcube.py` (subcubes) and `abstr2.py` (the partition abstraction), and were re-run for this paper. Separately, explicit elastic sequences of length $\lfloor (N-1)/k \rfloor^k$ were checked for $(k, n) \in \{(2, 1), (2, 2), (2, 3), (3, 1), (3, 2)\}$ (`checks37.py`).

C.8 The binomial conjecture: the case $k = 2$, $h = 3$, and the evidence

Proposition C.2 (Conjecture 4.11 for $(k, h) = (2, 3)$). *[T1 Conj 3.8, case proved during verification] Let $\mathcal{C}$ be a family of subsets of U , closed under arbitrary intersections, with $U \in \mathcal{C}$ and of height 3. Then every elastic sequence for the class of unions of one or two members of $\mathcal{C}$, relative to $\emptyset$, has length at most $6 = \binom{4}{2}$.*

Proof. Reduction. Let $C_0 = \bigcap \mathcal{C}$. Its points lie in every member, so they never escape and never occur in an elastic sequence. Removing C_0 from U and from every member preserves intersection-closure, height and elastic sequences. So we may assume $C_0 = \emptyset$.

Ranks. Let the rank $r(X)$ of a member be the length of the longest strict chain of members from $\emptyset$ to X . If $X \subsetneq Y$ are members, then $r(X) < r(Y)$. Since $r(U) = 3$, every member $X \neq U$ has $r(X) \leq 2$. The only member of rank 0 is $\emptyset$. Members of rank 1 are *atoms* and members of rank 2 are *planes*.

Fact 1. If X, Y are members with $X \not\subseteq Y$, then $X \cap Y$ is a member with $r(X \cap Y) < r(X)$. So the intersection of a plane with a member not containing it is $\emptyset$ or an atom, and the intersection of an atom with a member not containing it is $\emptyset$.

Let $s_1, \dots, s_m$ be elastic, with witnesses $Y_i^1 \cup Y_i^2$ (taking $Y_i^1 = Y_i^2$ for a single member) that cover $s_1, \dots, s_{i-1}$ and miss s_i .

Claim A: each atom contains at most one s_i . Suppose $s_i, s_j \in A$ with $i < j$. Then $s_i \in Y_j^l$ for some l , and $s_j \notin Y_j^l$, so $A \not\subseteq Y_j^l$. By Fact 1, $A \cap Y_j^l = \emptyset$, contradicting $s_i \in A \cap Y_j^l$.

Claim B: each plane contains at most three s_i . Let X be a plane and s_j the last sequence element in X . The earlier elements of X are covered by $Y_j^1 \cup Y_j^2$. Since $s_j \notin Y_j^l$, $X \not\subseteq Y_j^l$, so each $X \cap Y_j^l$ is $\emptyset$ or an atom. By Claim A, each contains at most one sequence element. So X has at most two elements before s_j .

By Claims A and B, every member other than U contains at most three sequence elements.

The contradiction. Suppose $m \geq 7$. The members Y_7^1, Y_7^2 miss s_7 , so they differ from U , and together they cover $s_1, \dots, s_6$. Each covers at most three, so each covers exactly three, disjointly. Atoms cover at most one, so both are planes; call them X and Y . Moreover, X contains no element covered only by Y , since it would then contain four sequence elements, contrary to Claim B; and likewise for Y . Say $s_6 \in Y$. The witness $Z_1 \cup Z_2$ of s_6 misses s_6 and covers the three X -elements, which all lie among $s_1, \dots, s_5$.

Let Z be a member with $s_6 \notin Z$. If $Z \not\supseteq X$, then $Z \cap X$ is $\emptyset$ or an atom, so Z covers at most one X -element. If $Z \supsetneq X$, then $r(Z) \geq 3$, so $Z = U \ni s_6$, which is impossible. Hence a member that covers two or more X -elements equals X . One of Z_1, Z_2 covers at least two X -elements; say $Z_1 = X$. Then X contains neither of the two Y -elements among $s_1, \dots, s_5$, so Z_2 must cover both. But $s_6 \in Y \setminus Z_2$, so $Z_2 \cap Y$ is $\emptyset$ or an atom, which covers at most one of them by Claim A. This is a contradiction, so $m \leq 6$. $\square$

This argument was sketched by a referee and written out in full by the author of T1 (referee report A-5). The same verification pass made three corrections to the evidence:

- An earlier claim that “a natural one-step decomposition proof fails on some extremal sequences” had no surviving script. It is withdrawn and plays no role here.
- The run with height $h \leq 5$ used a universe of 14 points. The elasticity cannot exceed the number of points, so “14 versus the conjectured 15” is a ceiling artifact, not evidence.
- The graphic-matroid data were added. The ground set of $M(K_{h+1})$ is the edge set of K_{h+1} , with $\binom{h+1}{2}$ points. Its flats are the edge sets that are unions of complete graphs on the blocks of a partition of the vertices. Flats are closed under intersection, and the rank of the matroid is h , so the family has height h . `T1-code/graphic.py` computes the 2-union elasticity as $3, 6, 10, 15 = \binom{h+1}{2}$ for $h = 2, 3, 4, 5$, so every edge can be placed in an elastic sequence. For $k = 3$ it computes 6 and 10 for K_4 and K_5 , which is the number of edges. We re-ran these computations for this paper.

C.9 Time-uniform soundness

Lemma C.3 (Ville’s inequality (Ville, 1939)). *If $(M_t)_{t \geq 0}$ is a nonnegative supermartingale with $\mathbb{E}M_0 \leq 1$, then $\mathbb{P}[\exists t : M_t \geq 1/\delta'] \leq \delta'$ for every $\delta' \in (0, 1]$.*

Proof of Theorem 4.15. Observations and likelihoods. In round u there is at most one observation O_u : a human datum $X \in S$, an oracle answer $y \in \{0, 1\}$ to an escalated query q_u , or nothing. For each R let $\lambda_R(O_u)$ be $p_R(X)$, $\ell_R(y \mid q_u)$ or 1 accordingly, and let $L_t(R) = \prod_{u \leq t} \lambda_R(O_u)$. Under the hypothesis of (a), conditional on $\mathcal{F}_u$ the law of O_{u+1} is λ_{R^*} restricted to the type of round $u + 1$. That type is $\mathcal{F}_u$ -measurable: V_δ is deterministic, and q_{u+1} and the schedule

of human data are $\mathcal{F}_u$ -measurable. In particular $\lambda_{R^*}(O_{u+1}) > 0$ almost surely, so $L_t(R^*) > 0$ almost surely.

(a) *Supermartingale.* Observation spaces are countable. For each R ,

$$\mathbb{E}\left[\frac{\lambda_R(O_{u+1})}{\lambda_{R^*}(O_{u+1})} \mid \mathcal{F}_u\right] = \sum_{o: \lambda_{R^*}(o) > 0} \lambda_R(o) \leq 1.$$

The quantity $L_u(R)/L_u(R^*)$ is $\mathcal{F}_u$ -measurable, and the sum defining Z_{u+1} has nonnegative terms. So conditional monotone convergence gives

$$\mathbb{E}[Z_{u+1} \mid \mathcal{F}_u] = \sum_R w(R) \frac{L_u(R)}{L_u(R^*)} \mathbb{E}\left[\frac{\lambda_R(O_{u+1})}{\lambda_{R^*}(O_{u+1})} \mid \mathcal{F}_u\right] \leq Z_u,$$

and $Z_0 = \sum_R w(R) = 1$. By Lemma C.3, the event $B = \{\exists t : Z_t \geq 1/\delta'\}$ has probability at most δ' . Off B , for every t ,

$$w_t(R^*) = w^*/Z_t > w^*\delta' \geq \delta.$$

If the query in round $t+1$ is some $q \notin R^*$, then $w_t(\{R : q \notin R\}) \geq w_t(R^*) > \delta$, so V_δ does not accept it. Hence $\mathbb{P}[\exists t : V_\delta \text{ accepts an invalid } q_t] \leq \mathbb{P}(B) \leq \delta'$.

(b) *Deterministic oracle.* The answer to an escalated q is $y = \mathbf{1}[q \in R^*]$, and $\lambda_R(y)/\lambda_{R^*}(y) = \mathbf{1}[\mathbf{1}[q \in R] = \mathbf{1}[q \in R^*]] \leq 1$. Let $X_1, X_2, \dots$ be the human data in order of arrival, and let $n(t)$ be the number that have arrived by round t . Then, pathwise,

$$Z_t \leq Z_{n(t)}^H, \quad Z_n^H := \sum_R w(R) \prod_{j \leq n} \frac{p_R(X_j)}{p_{R^*}(X_j)}.$$

Since the X_j are i.i.d. p_{R^*} , the computation of (a), in the filtration $\mathcal{G}_n = \sigma(X_1, \dots, X_n)$, shows that Z^H is a nonnegative supermartingale with $Z_0^H = 1$. Let $G = \{\sup_n Z_n^H < 1/\delta'\}$. It depends only on the human data, and $\mathbb{P}(G) \geq 1 - \delta'$ by Lemma C.3. On G , for every prover strategy and every arrival schedule, $Z_t < 1/\delta'$ for all t , and the argument of (a) shows that no invalid step is ever accepted. The prover may know R^* , the verifier and all of $X_1, X_2, \dots$ in advance. It can choose only the timing of the data, and the bound holds for every timing. $\square$

The hypothesis of (a) was implicit in the original statement. The referees showed that it is needed, and it was made explicit (referee report B-2).

A prover that foresees the noise. Let $\mathcal{H} = 2^{\{0,1\}}$ with the uniform prior, so $w^* = 1/4$, and let $R^* = \{0\}$. There are no human data. The oracle flips the true label with probability η , independently at each query, so $\ell_R(y \mid q) = 1 - \eta$ if $y = \mathbf{1}[q \in R]$ and η otherwise. Set $\delta' = 0.05$ and $\delta = w^*\delta'$. The prover knows in advance whether the next answer will be flipped. If it will be, the prover queries the invalid step 1, and the answer reads “valid”. If not, it queries the valid step 0, and the answer again reads “valid”. So every answer is “valid”, and each answer to a query of 1 multiplies the likelihood ratio of $\{0,1\}$ to $\{0\}$ by $(1 - \eta)/\eta > 1$. No answer ever lowers that ratio. Flips occur infinitely often, so the posterior mass of the hypotheses that exclude 1 eventually falls below δ , and the next query of 1 is accepted. The simulation `T1-code/prescient.py` gives acceptance probability 1.0 at $\eta = 0.1$ and at $\eta = 0.3$, against the bound 0.05. This prover violates the hypothesis of (a), because $\mathcal{F}_t$ would have to contain the future noise. In (b) no such prover exists, because there is no noise to foresee.

C.10 Tightness of the constant

Proof of Proposition 4.16. Let $\mathcal{H} = \{R^*, R'\}$ with $w(R^*) = w^*$ and $w(R') = 1 - w^*$. While all human data seen are a , after t of them $L_t(R^*) = (1 - u)^t$ and $L_t(R') = 1$. Write $r_t = (1 - u)^{-t}$. Then

$$w_t(\{R : c \notin R\}) = w_t(R^*) = \frac{w^*}{w^* + (1 - w^*)r_t}.$$

V_δ with $\delta = w^*\delta'$ accepts c iff this is less than $w^*\delta'$, i.e. iff $w^* + (1 - w^*)r_t > 1/\delta'$, i.e. iff $r_t > \theta = (1 - w^*\delta')/((1 - w^*)\delta')$. Note that $\theta \geq 1$ since $\delta' \leq 1$, so $t^* \geq 1$.

- Once a b arrives, $L(R') = 0$, and c is never accepted.
- If c is queried at some $t < t^*$ while only a 's have arrived, it is not accepted. If it is escalated, the deterministic answer “invalid” sets $L(R') = 0$, and again c is never accepted.

So the prover's best strategy is to query c exactly once, after the t^* -th datum. It succeeds iff the first t^* data are all a , which has probability $(1 - u)^{t^*}$.

By the choice of t^* , $r_{t^*} > \theta$, so $(1 - u)^{t^*} < 1/\theta$. By minimality, $r_{t^*-1} \leq \theta$, so $(1 - u)^{t^*} \geq (1 - u)/\theta$. Also

$$1/\theta = (1 - w^*)\delta'/(1 - w^*\delta') \leq \delta',$$

and $(1 - u)/\theta \rightarrow \delta'$ as $u, w^* \rightarrow 0$. The exact values $0.989\delta'$ and $0.991\delta'$ quoted in the statement come from `T1-code/ville2.py`, which computes $(1 - u)^{t^*}$ exactly; we re-ran it. It also gives, for example, $0.781\delta'$ at $u = 0.5$, $w^* = \delta' = 0.01$. $\square$

C.11 The escalation bound

Proof of Theorem 4.17. Let $\mathcal{F}_u$ be as in Theorem 4.15(a). Z changes only in data rounds and in escalation rounds.

Data round. The update is $Z_u = Z_{u-1}F_u$, where $F_u = \sum_R w_{u-1}(R) p_R(X_u)/p_{R^*}(X_u)$. This uses $w_{u-1}(R) = w(R)L_{u-1}(R)/(Z_{u-1}L_{u-1}(R^*))$. Since w_{u-1} is $\mathcal{F}_{u-1}$ -measurable and X_u is fresh,

$$\mathbb{E}[F_u | \mathcal{F}_{u-1}] = \sum_R w_{u-1}(R) \sum_{x: p_{R^*}(x) > 0} p_R(x) \leq 1.$$

Escalation round. The deterministic answer is $y = \mathbf{1}[q \in R^*]$, and the update is $Z_u = Z_{u-1} \sum_R w_{u-1}(R) \mathbf{1}[\mathbf{1}[q \in R] = y]$.

- If q is valid, the factor is $w_{u-1}(\{R : q \in R\}) = 1 - w_{u-1}(\{R : q \notin R\}) \leq 1 - \delta$, because q was not accepted.
- If q is invalid, the factor is $w_{u-1}(\{R : q \notin R\}) = 1 - w_{u-1}(\{R : q \in R\}) \leq 1 - \delta_r$, because q was not rejected. In every case it is at most 1.

Counting. Let $M_t = \prod_{\text{data rounds } u \leq t} F_u$. This is a nonnegative supermartingale with $M_0 = 1$. Let E_t^+ and E_t^- be the numbers of escalations of valid and of invalid queries up to time t . Since $w_t(R^*) \leq 1$,

$$w^* \leq \frac{w^*}{w_t(R^*)} = Z_t \leq M_t (1 - \delta)^{E_t^+} (1 - \delta_r)^{E_t^-}.$$

By Lemma C.3, with probability at least $1 - \delta''$ we have $M_t < 1/\delta''$ for all t . On that event, $(1 - \delta_m)^{E_t^+ + E_t^-} > w^*\delta''$, and using $\ln \frac{1}{1-x} \geq x$,

$$\#\text{ESC}_t = E_t^+ + E_t^- \leq \frac{\ln(1/w^*) + \ln(1/\delta'')}{\delta_m}.$$

Dropping the factor $(1 - \delta_r)^{E_t^-} \leq 1$ gives $E_t^+ \leq (\ln(1/w^*) + \ln(1/\delta''))/\delta$ against every prover. For an honest prover $E_t^- = 0$.

VS posterior. Here a datum's likelihood is $\mathbf{1}[X_u \in R]$, and $X_u \in R^*$. So $F_u = w_{u-1}(\{R : X_u \in R\}) \leq 1$ and $M_t \leq 1$ deterministically. The same computation then gives $\#ESC_t \leq \ln(1/w^*)/\delta_m$, and $E_t^+ \leq \ln(1/w^*)/\delta$. $\square$

T1 states the escalation bound with $\delta_m = \delta$ for honest provers. The bound on E_t^+ against every prover is read off the same computation: invalid escalations multiply Z by at most 1. Corollary 4.19 uses it in this form.

D Proofs for Section 5

Lemma 5.2, Theorem 5.4, Theorem 5.9(a), Corollary 5.11, Proposition 5.13, Theorem 5.17 and Corollary 5.20 are proved in full in the main text. This appendix proves the remaining results of Section 5. The proofs follow T1 §§5–6 in the form repaired after verification; the repairs are noted where they occur. The last subsection is a sketch and is labelled as such.

Notation is as in Sections 2 and 5. Steps are ground terms over a signature Ω with at least two constants $a \neq b$. For a schema σ , $\text{vars}(\sigma)$ is its set of metavariables and $v = |\text{vars}(\sigma)|$. A substitution θ for σ maps $\text{vars}(\sigma)$ to ground terms. Anti-unification is the map $\mathcal{A}$ of Section 2.2, and lgg is computed by it (Theorem 2.5).

D.1 Three elementary facts

Lemma D.1. (F1) *If $\tau = \sigma\theta$ and θ restricted to $\text{vars}(\sigma)$ is not a renaming, then $\text{inst}(\tau) \subsetneq \text{inst}(\sigma)$.* (F2) *The intersection of the instance sets of all schemas is empty.* (F3) *For tagged calculi H_k^{tag} and clean data D , $\bigcap \text{VS}(D) = \bigcup_{i \leq k} \{i\} \times \hat{R}_i$, where $\hat{R}_i = \text{inst}(\text{lgg}(D^{(i)}))$ if $D^{(i)} \neq \emptyset$ and $\hat{R}_i = \emptyset$ otherwise.*

Proof. (F1) Inclusion holds because instances of τ are instances of σ . For strictness there are two cases. First, suppose some θx is not a metavariable, say with root g . Choose a constant $c \neq g$; it exists because there are two constants. Let $t = \sigma\theta'$ with $\theta'x = c$ and θ' ground elsewhere. At a position of σ where x occurs, t carries c , while every instance of τ carries a term with root g . So $t \notin \text{inst}(\tau)$. Second, suppose every θx is a metavariable. Then θ is not injective on $\text{vars}(\sigma)$, say $\theta x = \theta y$ with $x \neq y$. Let $\theta'x = a$ and $\theta'y = b$. Every instance of τ has equal subterms at the positions of x and y , and $\sigma\theta'$ does not. So $\sigma\theta' \notin \text{inst}(\tau)$.

(F2) The ground schemas a and b have disjoint instance sets $\{a\}$ and $\{b\}$.

(F3) Write I for the family of single-schema instance sets. Then $H_k^{\text{tag}} = \{\bigcup_i \{i\} \times A_i : A_i \in I\}$, and $D \subseteq \bigcup_i \{i\} \times A_i$ iff $D^{(i)} \subseteq A_i$ for every i . So $\text{VS}(D)$ is the set of all such unions with $(A_1, \dots, A_k) \in \prod_i \text{VS}_i$, where $\text{VS}_i := \{A \in I : D^{(i)} \subseteq A\}$. Each VS_i is nonempty, since it contains $\text{inst}(\sigma_i)$. Hence $(i, s) \in \bigcap \text{VS}(D)$ iff $s \in A_i$ for every choice in the product, iff $s \in \bigcap \text{VS}_i$. Finally, $\bigcap \text{VS}_i = \text{inst}(\text{lgg}(D^{(i)}))$ for nonempty $D^{(i)}$ by Theorem 2.5, and $\bigcap \text{VS}_i = \emptyset$ for empty $D^{(i)}$ by (F2). $\square$

(F3) is the positive-data case of the factorization used for T1 Thm 3.6. The verification noted that H_k^{tag} is not literally intersection-closed, and the factorization is what the argument actually uses.

D.2 Proofs of Propositions 5.5 and 5.6

Proof of Proposition 5.5. (a) If T is an anchor and $T \subseteq R \in \mathcal{H}$, then $R \supseteq R^*$, so $R \not\subsetneq R^*$.

(b) Let $T \neq \emptyset$ be a tell-tale, and suppose $T \subseteq R \in \mathcal{H}$ with $R \not\supseteq R^*$. Then $R \cap R^* \supseteq T$ is nonempty, so it is a member of $\mathcal{H}$ by intersection-closure of $\mathcal{H} \cup \{\emptyset\}$. And $T \subseteq R \cap R^* \subsetneq R^*$, which contradicts the tell-tale property. For $\mathcal{H} = \{\{1\}, \{2\}\}$ and $R^* = \{1\}$: no member lies strictly inside $\{1\}$, so $\emptyset$ is a tell-tale; $\{2\} \supseteq \emptyset$ does not contain R^* , so $\emptyset$ is not an anchor. Here $\mathcal{H} \cup \{\emptyset\}$ is intersection-closed. For the existence claim, if T is a tell-tale and $r \in R^*$, then $T \cup \{r\}$ is a nonempty tell-tale (fewer members contain it), hence an anchor; anchors are tell-tales by (a).

(c) The class is an indexed family of uniformly recursive sets. $\{0\}$ is a tell-tale for $2\mathbb{N}$, since every R_n contains the odd number $2n + 1$ and so no member lies strictly inside $2\mathbb{N}$. R_n is a tell-tale for itself. By Angluin's theorem (Angluin, 1980) the class is identifiable in the limit from text. Every finite $T \subseteq 2\mathbb{N}$ lies in R_n for any n with $2n \geq \max T$, and $R_n \not\supseteq 2\mathbb{N}$. So $2\mathbb{N}$ has no anchor, and by Theorem 5.4(b) the cautious verifier is never complete on a text for $2\mathbb{N}$. The same example shows that “finite” is needed in Theorem 5.4(a): every member containing $2\mathbb{N}$ is $2\mathbb{N}$, so $\hat{R}(2\mathbb{N}) = 2\mathbb{N}$. $\square$

Proof of Proposition 5.6. If $R^* = \emptyset$, then $\emptyset$ is an anchor. Otherwise suppose R^* has no anchor, and pick $w_0 \in R^*$. Given $w_0, \dots, w_{n-1} \in R^*$, the finite set $\{w_0, \dots, w_{n-1}\} \subseteq R^*$ is not an anchor. So some $R_n \in \mathcal{H}$ contains it with $R_n \not\supseteq R^*$; pick $w_n \in R^* \setminus R_n$. Then $\{w_0, \dots, w_{n-1}\} \subseteq R_n \not\supseteq w_n$ for all $n \geq 1$, which is infinite elasticity. $\square$

This is the lemma of L1 §2.3 in a direct form. The facts used after the proposition in the main text are standard: finite thickness implies finite elasticity, and the union theorem of Wright (1989) with the corrected definition of Motoki et al. (1991).

D.3 Proof of Theorem 5.7

By (F3), $\hat{R}(D) = R^*$ iff every rule i is *identified*, i.e. $\hat{R}_i = \text{inst}(\sigma_i)$. Let E_i be the event that rule i is not identified; we bound each $\mathbb{P}(E_i)$ and take a union bound over i .

Rules with $v_i \geq 1$. Let $N_i = |D^{(i)}|$, so $N_i \sim \text{Bin}(N, \pi_i)$. Given $N_i = n$, the rule- i substitutions $\Theta_1, \dots, \Theta_n$ are i.i.d. Λ_i . Say that (R) *fails at x* if all n roots $\text{root}(\Theta_j x)$ equal one symbol (vacuously so if $n = 0$), and that (D) *fails at $\{x, y\}$* if $\Theta_j x = \Theta_j y$ for all j . If $n \geq 1$ and no witness event fails, Lemma 2.7 gives $\text{lgg}(D^{(i)}) \equiv \sigma_i$, so $\hat{R}_i = \text{inst}(\sigma_i)$. Hence E_i is contained in the union of the $v_i + \binom{v_i}{2}$ failure events.

Fix x . The function $G \mapsto \min(F(G), 1 - F(G))$, with F the law of $\text{root}(\Theta x)$, attains its maximum $\rho_{i,x}$. Indeed $G \mapsto F(G) = \sum_f \mathbf{1}_G(f) F(f)$ is continuous on the compact space $\{0, 1\}^\Omega$, because the series converges uniformly. Let G attain the maximum. If all roots equal f , then all lie in G (if $f \in G$) or all lie in G^c . So

$$\mathbb{P}[(\text{R}) \text{ fails at } x \mid N_i = n] \leq F(G)^n + (1 - F(G))^n \leq 2(1 - \rho_{i,x})^n \leq 2(1 - \rho_i)^n,$$

since $\min(F(G), 1 - F(G)) = \rho_{i,x}$ gives $\max(F(G), 1 - F(G)) \leq 1 - \rho_{i,x}$. For a pair,

$$\mathbb{P}[(\text{D}) \text{ fails at } \{x, y\} \mid N_i = n] = (1 - \rho_{i,xy})^n \leq (1 - \rho_i)^n.$$

By the union bound, $\mathbb{P}(E_i \mid N_i = n) \leq c_i(1 - \rho_i)^n$. For $n = 0$ this is trivially true, since $c_i \geq 2$. Averaging over N_i ,

$$\mathbb{P}(E_i) \leq c_i \mathbb{E}[(1 - \rho_i)^{N_i}] = c_i \sum_{n=0}^N \binom{N}{n} \pi_i^n (1 - \pi_i)^{N-n} (1 - \rho_i)^n = c_i (1 - \pi_i \rho_i)^N \leq c_i e^{-N \pi_i \rho_i}.$$

Ground rules. If $v_i = 0$ and $N_i \geq 1$, then $D^{(i)}$ consists of copies of σ_i and $\hat{R}_i = \{\sigma_i\}$. So $E_i = \{N_i = 0\}$, and $\mathbb{P}(E_i) = (1 - \pi_i)^N \leq e^{-N\pi_i} = c_i e^{-N\pi_i \rho_i}$ under the conventions $\rho_i = c_i = 1$. The verification noted that the general formula gives $c_i = 0$ for ground rules; hence the separate convention.

Sample size. If $N \geq \frac{1}{\pi_i \rho_i} \ln \frac{kc_i}{\delta}$ for every i , each term is at most δ/k , and the sum is at most δ .

Necessity. Fix i , x , f and $\rho \in (0, 1]$ with $\mathbb{P}[\text{root}(\Theta x) = f] = 1 - \rho$. Let A be the event that no datum is a rule- i instance with $\text{root}(\Theta x) \neq f$. The data are i.i.d., and each datum is such an instance with probability $\pi_i \rho$, so $\mathbb{P}(A) = (1 - \pi_i \rho)^N$. On A , either $N_i = 0$, and then $\hat{R}_i = \emptyset \neq \text{inst}(\sigma_i)$; or $N_i \geq 1$, and then every column that anti-unification meets at an occurrence of x has root f throughout. In the second case $\text{lgg}(D^{(i)})$ carries a term with root f there. Since $\rho > 0$, some ground term t with root $\neq f$ is in the support. The instance of σ_i with $x \mapsto t$ lies in $\text{inst}(\sigma_i)$ but not in $\hat{R}_i$. So $\mathbb{P}[\hat{R}(D) \neq R^*] \geq (1 - \pi_i \rho)^N$. $\square$

D.4 Proof of the claims in Remark 5.8

Teaching sets. Let θ_1, θ_2 map the v_i metavariables of σ_i to $2v_i$ pairwise distinct constants. For each x , $\theta_1 x \neq \theta_2 x$ are distinct constants, so their roots differ and (R) holds. For $x \neq y$, $\theta_1 x \neq \theta_1 y$, so (D) holds. By Lemma 2.7, $\text{lgg}(\sigma_i \theta_1, \sigma_i \theta_2) \equiv \sigma_i$. In a tagged class this set (one per rule) is an anchor, by (F3).

No distribution-free completeness rate (the example is from L1 §9). Let g be binary, $h \geq 1$, $m = 2^h$, and let u_j ($1 \leq j \leq m$) be the complete g -tree of depth h with leaf j equal to a and all other leaves equal to b . Let $J \subseteq [m]$ with $|J| \geq 2$, and anti-unify $\{u_j : j \in J\}$. All internal positions carry g and are copied. At leaf $\ell \in J$ the column contains a (from u_ℓ) and b (from any other u_j , $j \in J$), so it becomes a metavariable z_ℓ . Columns at distinct leaves $\ell, \ell' \in J$ differ (at index ℓ one has a , the other b), so $z_\ell \neq z_{\ell'}$. At leaves $\ell \notin J$ the column is constantly b . Hence the lgg has distinct metavariables exactly at the leaves in J and b elsewhere. Its instances among the u 's are exactly $\{u_j : j \in J\}$, because u_k with $k \notin J$ has a at a leaf where the lgg has b . For $|J| = 1$ the lgg is u_j itself.

Take the target $\text{inst}(x) \in H_1$ and the uniform law Q on $\{u_1, \dots, u_m\}$. A sample of $n < m/2$ draws hits a set J of at most n indices, and by the above $\hat{R}(D) \cap \{u_1, \dots, u_m\} = \{u_j : j \in J\}$. So $Q(R^* \setminus \hat{R}(D)) \geq (m - n)/m > 1/2$ for every such sample. Any learner whose acceptance set is sound for every target in H_1 compatible with its data accepts a subset of $\hat{R}(D)$ (Lemma 5.2(b)), so its completeness error is at least as large. Since h is arbitrary, no bound on the number of samples that is independent of Q can guarantee completeness error at most $1/2$. $\square$

D.5 Proof of Theorem 5.9(b)

Fix a rule i with $v_i \geq 1$. Write $M := v_i + \binom{v_i}{2}$ and $M' := v_i^2 + 1 \geq M$. Consider the range space on substitutions (a countable set) whose ranges are

$$\mathcal{C}_i := \{ \overline{F_1 \cup \dots \cup F_k} : F_1, \dots, F_k \in \mathfrak{F}_i \}.$$

Step 1: traces of $\mathfrak{F}_i$. Let $\theta_1, \dots, \theta_n$ be substitutions. For fixed x , the sets $\{\theta : \text{root}(\theta x) = f\}$ for different f are pairwise disjoint. So their traces on $\{\theta_1, \dots, \theta_n\}$ are pairwise disjoint subsets, of which at most n are nonempty, giving at most $n + 1$ distinct traces. Each pair $x \neq y$ contributes one set, hence one trace. So $\mathfrak{F}_i$ has at most $v_i(n + 1) + \binom{v_i}{2} \leq M(n + 1)$ distinct traces on n points.

Step 2: traces of $\mathcal{C}_i$. The trace of $\overline{F_1 \cup \dots \cup F_k}$ is determined by the traces of $F_1, \dots, F_k$. So $\mathcal{C}_i$ has at most $(M(n+1))^k \leq (M'(n+1))^k$ traces on n points.

Step 3: VC dimension. If $\mathcal{C}_i$ shatters d points, then $2^d \leq (M'(d+1))^k$, i.e. $\phi(d) := d - k \log_2(M'(d+1)) \leq 0$. Let $L := \log_2(4kM') \geq 2$ and $d_i = 2kL$. Then $\phi(d_i) > 0$ iff $(4kM')^2 > M'(2kL+1)$, iff $16k^2M' > 2kL+1$. This holds because $L < 4kM'$ gives $2kL+1 < 8k^2M'+1 \leq 16k^2M'$. Moreover $\phi'(d) = 1 - k/((d+1) \ln 2) > 0$ for $d \geq d_i \geq 4k$. So $\phi(d) > 0$ for all $d \geq d_i$, and the VC dimension of $\mathcal{C}_i$ is less than d_i .

Step 4: the ε -net theorem (Haussler and Welzl, 1987; Blumer et al., 1989). Let a range space have VC dimension at most d , let μ be a probability measure, and let $0 < \varepsilon, \delta' < 1$. If

$$m \geq \max \left\{ \frac{4}{\varepsilon} \log_2 \frac{2}{\delta'}, \frac{8d}{\varepsilon} \log_2 \frac{13}{\varepsilon} \right\},$$

then with probability at least $1 - \delta'$ an i.i.d. μ -sample of size m meets every range of μ -measure at least ε . We use the theorem in this “measure $\geq \varepsilon$ ” form. The symmetrization proof gives it with the same constants. Alternatively, apply the standard “measure $> \varepsilon$ ” form with $\varepsilon = \zeta_i/2$, at the cost of a constant factor. (This boundary point was noted in verification.)

Step 5: conclusion. Apply Step 4 with $\mu = \Lambda_i$, $\varepsilon = \zeta_i$, $d = d_i$ and $\delta' = \delta/k'$. The sum in the statement exceeds the maximum. By definition of ζ_i , every range in $\mathcal{C}_i$ has Λ_i -measure at least ζ_i . So with probability at least $1 - \delta/k'$ the n_i rule- i substitutions meet every range in $\mathcal{C}_i$, i.e. they are not contained in any union of k failure sets. They are a subset of $\text{Sub}_i(D)$, so neither is $\text{Sub}_i(D)$. For a ground rule, one sample makes $\text{Sub}_i(D)$ nonempty, and $\mathfrak{F}_i = \emptyset$. A union bound over the k' rules and part (a) finish the proof. $\square$

D.6 Proof of Proposition 5.10

This proposition replaces T1’s original remark after Thm 5.4, which claimed that at most k roots for a metavariable makes identification fail forever. Verification refuted that claim for $k' \geq 2$, and part (b) is the counterexample.

Failure sets and specializations. For a metavariable x of σ_i and $f \in \Omega$ of arity r , let $\bar{z} = z_1, \dots, z_r$ be fresh distinct metavariables. Then $\text{inst}(\sigma_i[x \mapsto f(\bar{z})]) = \{\sigma_i\theta : \text{root}(\theta x) = f\}$ and $\text{inst}(\sigma_i[y \mapsto x]) = \{\sigma_i\theta : \theta x = \theta y\}$. So covering substitutions by failure sets is the same as covering the corresponding instances by these specializations.

(a) Let $\sigma'_1, \dots, \sigma'_{k-k'+1}$ be the specializations in the hypothesis, and let

$$R := \bigcup_{j \neq i} \text{inst}(\sigma_j) \cup \bigcup_{l \leq k-k'+1} \text{inst}(\sigma'_l).$$

This is a union of at most k schema instance sets, so $R \in H_k$. Every datum lies in R^* . A datum outside $\text{inst}(\sigma_i)$ lies in some $\text{inst}(\sigma_j)$ with $j \neq i$. A datum in $\text{inst}(\sigma_i)$ lies in another rule or, by hypothesis, in some $\text{inst}(\sigma'_l)$. So $D \subseteq R$, $R \in \text{VS}(D)$ and $\hat{R}_{H_k}(D) \subseteq R$. The instance of σ_i assumed to lie outside R is in $R^* \setminus \hat{R}_{H_k}(D)$.

The case $k' = 1$. If x is instantiated in the data with roots among $f_1, \dots, f_k$ only, the k specializations $\sigma_1[x \mapsto f_j(\bar{z})]$ cover the data. Any instance whose x has another root lies outside them. So identification fails for as long as this persists.

(b) Let $R = \text{inst}(\tau_1) \cup \text{inst}(\tau_2) \in \text{VS}_{H_2}(D)$, with $D = \{p(a), p(b), q(a), q(b)\}$; possibly $\tau_1 = \tau_2$. If some τ_l covers a p -term and a q -term, then $\tau_l \succeq \text{lbg}(p(\cdot), q(\cdot))$, which is a metavariable because the roots differ. So τ_l is a metavariable and R is everything. Otherwise each τ_l covers only p -terms or only q -terms of D . All four terms are covered, so one schema covers both p -terms and

the other both q -terms. Then $\tau_1 \succeq \text{lgg}(p(a), p(b)) = p(x)$ and $\tau_2 \succeq q(y)$ (up to the order of the two schemas). In both cases $R \supseteq R^*$. Since $R^* \in \text{VS}_{H_2}(D)$, $\hat{R}_{H_2}(D) = R^*$. If the substitution law of x is supported on $\{a, b\}$, the two failure sets $\{\text{root}(\theta x) = a\}$ and $\{\text{root}(\theta x) = b\}$ have full measure, so $\zeta_1 = 0$. $\square$

D.7 Proofs for Remark 5.12

Let Φ be finite and let the target be $G^* := \text{inst}(\sigma^*, \Psi^*) \neq \emptyset$. Write $\text{cl}(P) := \text{inst}(\text{lgg}(P), \Psi_P)$ for nonempty P , the least guarded instance set containing P (Lemma 2.8).

Anchors exist. Ascending generalization chains are finite (Lemma 2.6), so the lgg of the (possibly infinite) set G^* exists and equals $\text{lgg}(T_0)$ for some finite $T_0 \subseteq G^*$. For each $\phi \in \Phi$ that is not true on all of G^* , pick $t_\phi \in G^*$ with $\phi(t_\phi) = 0$. Let $T := T_0 \cup \{t_\phi\}$. Then $\text{lgg}(T) = \text{lgg}(G^*)$, because $\text{lgg}(T_0) \preceq \text{lgg}(T) \preceq \text{lgg}(G^*) = \text{lgg}(T_0)$. Also Ψ_T is exactly the set of predicates true on all of G^* . Hence $G^* \subseteq \text{inst}(\text{lgg}(T), \Psi_T) = \text{cl}(T)$. Conversely, $\text{cl}(T) \subseteq G^*$, since G^* is a guarded instance set containing T . So $\text{cl}(T) = G^*$, and every guarded instance set containing T contains G^* : T is an anchor.

Chains. Let $\text{cl}(P_0) = C_0 \subsetneq C_1 \subsetneq \dots \subsetneq C_m \subseteq G^*$ be closures (the chain of closures of the data plus escalated steps, for nonempty initial data $P_0 \subseteq G^*$), $C_j = \text{inst}(g_j, \Psi_j)$ with $g_j = \text{lgg}(C_j)$ and $\Psi_j = \Psi_{C_j}$. Inclusion gives $g_{j-1} \preceq g_j$ and $\Psi_{j-1} \supseteq \Psi_j$. If both were equalities, then $C_{j-1} = C_j$. So each step either strictly generalizes the schema (at most $\mu(g_0) - \mu(\sigma^*)$ times, by Lemma 2.6, since $g_m \preceq \text{lgg}(G^*) \preceq \sigma^*$) or strictly shrinks the guard set (at most $|\Phi|$ times). By T1 Prop. 3.3 and Thm 3.2 (Section 4) this bounds the escalations by $\mu(\text{lgg}(P_0)) - \mu(\sigma^*) + |\Phi|$. $\square$

D.8 Proof of Theorem 5.14

(a) If D has at most e invalid steps, then $|D \setminus R^*| \leq e$, so $R^* \in \text{VS}_e(D)$ and $\bigcap \text{VS}_e(D) \subseteq R^*$. The per-tag case is identical. Soundness against every prover follows by Lemma 3.1, as in Lemma 5.2.

(b) By (i), $R^* \in \text{VS}_{(e_i)}(D)$, so $\bigcap \text{VS}_{(e_i)}(D) \subseteq R^*$. Conversely, let $R \in \text{VS}_{(e_i)}(D)$ and fix i . Its tag- i component is $\text{inst}(\tau)$ for a schema τ , or empty. Let V_i be the multiset of valid rule- i samples and $V'_i := V_i \cap \text{inst}(\tau)$. Then $|V_i \setminus V'_i| \leq |D^{(i)} \setminus R^{(i)}| \leq e_i$.

- $V'_i \neq \emptyset$, because $|V_i| > e_i$. In particular the tag- i component is not empty.
- For each x and f , more than e_i members of V_i have $\text{root}(\Theta x) \neq f$, and at most e_i were removed. So some member of V'_i has $\text{root}(\Theta x) \neq f$. Since this holds for every f , the roots at x in V'_i are not all equal, and (R) holds for V'_i .
- Likewise, for each $x \neq y$ some member of V'_i has $\Theta x \neq \Theta y$, and (D) holds.

By Lemma 2.7, $\text{lgg}(V'_i) \equiv \sigma_i$. Since $V'_i \subseteq \text{inst}(\tau)$, $\tau \succeq \text{lgg}(V'_i) \equiv \sigma_i$, so $\text{inst}(\tau) \supseteq \text{inst}(\sigma_i)$. For a ground rule, V'_i is a nonempty multiset of copies of σ_i , so $\tau \succeq \sigma_i$ directly. Hence $R \supseteq R^*$ for every $R \in \text{VS}_{(e_i)}(D)$, and $\bigcap \text{VS}_{(e_i)}(D) = R^*$. $\square$

Why the clause “more than e_i valid samples”. This is the repair of verification item B-16. Take a ground rule $\sigma_i = c$, one valid sample (i, c) , no invalid samples and $e_i = 1$. The rest of (ii) holds vacuously. But a hypothesis whose tag- i component is $\emptyset$ or $\text{inst}(d)$ with $d \neq c$ misses one tag- i datum and lies in $\text{VS}_{(e_i)}$. So $c \notin \bigcap \text{VS}_{(e_i)}$.

Why per-tag budgets. Take two tags with $e_1 = e_2 = 1$ and the global budget $e = 2$. Let rule 1 be $s(x)$ with valid samples $s(a), s(b), s(c)$. These are 1-robustly generic: for every f , at least two of them have a root $\neq f$. Let rule 2 be a ground t with samples t, t . The hypothesis with components $\text{inst}(s(a))$ and $\text{inst}(t)$ misses exactly two data, so it lies in $\text{VS}_2(D)$. Hence $s(d) \notin \bigcap \text{VS}_2(D)$ for any constant $d \neq a$, and $\bigcap \text{VS}_2(D) \neq R^*$. Under per-tag budgets, the tag-1 component must contain at least two of the three samples. These satisfy (R), so the component generalizes $s(x)$.

D.9 Proof of Theorem 5.15

We use Hoeffding's inequality for $X \sim \text{Bin}(N, p)$ and $s > 0$: $\mathbb{P}[X - Np \geq sN] \leq e^{-2Ns^2}$ and $\mathbb{P}[X - Np \leq -sN] \leq e^{-2Ns^2}$. Fix a rule i and write $\alpha = \alpha_i$, $\beta = \beta_i$, $\rho = \rho_i$ and $\Delta = \Delta_i$, so that $\rho\beta = \alpha + 2\Delta$ and $e_i = \lfloor (\alpha + \Delta)N \rfloor$.

Noise count. The number X of invalid steps tagged i is $\text{Bin}(N, \alpha)$. Since $e_i + 1 > (\alpha + \Delta)N$,

$$\mathbb{P}[X > e_i] = \mathbb{P}[X \geq e_i + 1] \leq \mathbb{P}[X - \alpha N \geq \Delta N] \leq e^{-2N\Delta^2}.$$

Witness counts, $v_i \geq 1$. For each metavariable x , let G_x attain $\rho_{i,x}$ (Appendix D.3). Let C_x^+ and C_x^- count the valid rule- i data with $\text{root}(\Theta x) \in G_x$ and $\notin G_x$, respectively. For each pair $x \neq y$, let C_{xy} count those with $\Theta x \neq \Theta y$. These are $2v_i + \binom{v_i}{2} = c_i$ counts. Each is $\text{Bin}(N, p)$ with $p \geq \beta\rho_{i,x} \geq \beta\rho$, resp. $p = \beta r_{i,xy} \geq \beta\rho$. So $pN \geq (\alpha + 2\Delta)N$ and $(\alpha + \Delta)N \leq pN - \Delta N$. Hence, for each count C ,

$$\mathbb{P}[C \leq e_i] \leq \mathbb{P}[C \leq (\alpha + \Delta)N] \leq \mathbb{P}[C - pN \leq -\Delta N] \leq e^{-2N\Delta^2}.$$

Suppose all c_i counts exceed e_i . For any f , the side of the split G_x that does not contain f consists of samples with root $\neq f$. So $\#\{\text{root}(\Theta x) \neq f\} \geq \min(C_x^+, C_x^-) > e_i$. The pair conditions hold directly, and the number of valid rule- i samples is at least $C_x^+ > e_i$. So the valid rule- i samples are e_i -robustly generic.

Ground rules. With $\rho = 1$, the single required event is that the number of valid rule- i samples, which is $\text{Bin}(N, \beta)$ with $\beta = \rho\beta$, exceeds e_i . Its failure probability is at most $e^{-2N\Delta^2}$ by the same computation; here $c_i = 1$. This covers the gap noted in verification (B-17).

Conclusion. Outside an event of probability at most $(1 + c_i)e^{-2N\Delta_i^2}$ for each i , hypotheses (i) and (ii) of Theorem 5.14(b) hold for all rules, and the trimmed verifier accepts exactly R^* . Take a union bound over i . $\square$

D.10 Proof of Lemma 5.18

Let F be a probability law on the countable set Ω , $m = \max_f F(f)$ (attained, because only finitely many atoms exceed any positive mass), $f_{\max}$ a maximizer, and $\rho = \max_G \min(F(G), 1 - F(G))$.

$\rho \leq 1 - m$. For any G , the side of the split that does not contain $f_{\max}$ has mass at most $1 - m$, so the minimum is at most $1 - m$.

$1 - m \leq 2\rho$. If $m \geq 1/2$, the split $G = \{f_{\max}\}$ gives $\min(m, 1 - m) = 1 - m$, so $\rho \geq 1 - m$. If $m < 1/2$, enumerate Ω and add symbols to G one at a time until $F(G) \geq (1 - m)/2$. This happens at a finite stage, since the partial sums tend to 1. Before the last addition $F(G) < (1 - m)/2$, and the last symbol has mass at most m . So finally $(1 - m)/2 \leq F(G) < (1 + m)/2$, hence $1 - F(G) > (1 - m)/2$, and $\rho \geq (1 - m)/2$.

Tightness. For the uniform law on $2r + 1$ symbols, $m = 1/(2r + 1)$, and the best split has r against $r + 1$ symbols, so $\rho = r/(2r + 1)$. Then $1 - m = 2r/(2r + 1) = 2\rho$. $\square$

T1 originally stated the constant 3; verification sharpened it to 2 and found it tight (B-18; random laws gave ratios up to 1.974).

D.11 Proof of Corollary 5.19

Suppose $\beta_i \mathbb{P}[\text{root}(\Theta x) \neq f] \leq \alpha$ for some x, f . Let R'' be R^* with its tag- i component replaced by $\text{inst}(\sigma_i[x \mapsto f(\bar{z})])$, which lies in H_k^{tag} . By (F1), $R'' \subsetneq R^*$. Q is a law on R^* , and the Q -mass of $R^* \setminus R''$ is the probability that a datum is a tag- i instance with $\text{root}(\Theta x) \neq f$, namely $\beta_i \mathbb{P}[\text{root}(\Theta x) \neq f] \leq \alpha$. Apply Theorem 5.17 to $R'' \subsetneq R^*$. Every $q \in R^* \setminus R'' \neq \emptyset$ has $\mathbb{P}_Q[\exists t : q \in \text{Acc}_t] \leq \delta$. So $\mathbb{P}_Q[R^* \subseteq \text{Acc}_t] \leq \delta$ for all t , which contradicts the hypothesis. The pair case uses $\sigma_i[y \mapsto x]$ in the same way. All of the alternative’s “noise” sits on tag i , so robust soundness is only needed for laws whose noise on tag i is at most α ; this is the per-tag reading. $\square$

The factor β_i in the displayed conditions was added in verification (B-18). α is a fraction of all data, while the probabilities are under Λ_i .

D.12 Sketch for Remark 5.22

(proof sketch) The claims are elementary. They have not been through the project’s adversarial verification, so we record only the argument.

(i) An equation whose monomials have exponent vectors $a_1, \dots, a_r$ is g -homogeneous iff $g(a_j - a_1) = 0$ for all j . So all of E is homogeneous iff g vanishes on the generators of $K(E)$, iff it vanishes on $K(E)$. The projection $\pi : \mathbb{Z}^n \rightarrow \mathbb{Z}^n / K(E)$ is such a grading. By the homomorphism theorem every such g factors uniquely as $\bar{g} \circ \pi$. An equation is homogeneous under every consistent grading iff its differences lie in $\bigcap_g \ker g = \ker \pi = K(E)$.

Smith normal form. If M is an integer matrix whose rows generate $K(E)$ and $UMV = \text{diag}(d_1, \dots, d_r, 0, \dots)$ is its Smith normal form, with U, V unimodular, then $\mathbb{Z}^n / K(E) \cong \bigoplus_j \mathbb{Z} / d_j \mathbb{Z} \oplus \mathbb{Z}^{n-r}$. If the target group is torsion-free, $g(cv) = 0$ with $c \neq 0$ forces $g(v) = 0$. So g vanishes on the saturation $\{v : cv \in K(E) \text{ for some } c \neq 0\}$, and the finest torsion-free grading is $(\mathbb{Z}^n / K(E)) / \text{torsion}$.

(ii) Let $\text{Adm}(L)$ be the set of equations whose differences lie in $L \leq \mathbb{Z}^n$. Then $\text{Adm}(L) \cap \text{Adm}(L') = \text{Adm}(L \cap L')$, and Adm determines L : the equation $1 = q^v$ (Laurent monomials) is in $\text{Adm}(L)$ iff $v \in L$. The version space of E is $\{\text{Adm}(L) : K(E) \leq L\}$, so $\hat{R}(E) = \text{Adm}(K(E))$. For the target $L^* = \ker g^*$, choose generators $v_1, \dots, v_s$ (subgroups of $\mathbb{Z}^n$ are finitely generated). The equations $1 = q^{v_j}$ form an anchor. More generally, a finite $E \subseteq \text{Adm}(L^*)$ is an anchor iff $K(E) = L^*$: if $K(E) \subsetneq L^*$, then $\text{Adm}(K(E))$ is consistent with E and misses $\text{Adm}(L^*)$. Theorem 5.4 then gives the stated identification. Claims (iii)–(iv) of the remark are direct readings of these facts.

E Proofs for Section 6

Lemma 6.2, Proposition 6.3(a),(c), Theorem 6.5, Theorem 6.7, Theorem 6.9, Proposition 6.10, Corollary 6.11, Lemma 6.19, Theorem 6.21, Theorem 6.23, Theorem 6.27 (given Lemma 6.26), Theorem 6.28, Proposition 6.31, Theorem 6.34, Proposition 6.37 (immediate), Corollary 6.35 and Lemma 6.41 are proved in the main text, and Theorem 6.20 is cited. This appendix proves the remaining results. The proofs follow T2 in the form repaired after adversarial verification;

where a statement or proof was changed by the repair, we say so. Notation is as in Section 6. For formulas, $\equiv_2$ is classical equivalence.

E.1 Proof of Proposition 6.6

($\Leftarrow$) Let $\hat{R} \subseteq \bigcap S$ with $w(S) \geq \frac{1}{2}w(\text{VS})$. A detection has $\text{Steps}(\pi) = P \subseteq \hat{R}$, so every $R \in S$ contains P and is deleted; the deleted weight is at least $w(S)$.

($\Rightarrow$) The set of steps is countable, because Fm is. If $\hat{R} = \emptyset$, take $S = \text{VS}$. Otherwise enumerate $\hat{R} = \{s_1, s_2, \dots\}$ (finite or infinite), and let $P_k = \{s_1, \dots, s_k\}$ and $S_k = \{R \in \text{VS} : P_k \subseteq R\}$. If $P_k \not\subseteq \bigcap \text{VS}$, then by the worst-case assumption P_k is a possible detection, and it deletes exactly S_k ; so $w(S_k) \geq \frac{1}{2}w(\text{VS})$. If $P_k \subseteq \bigcap \text{VS}$, then $S_k = \text{VS}$. In both cases $w(S_k) \geq \frac{1}{2}w(\text{VS})$. The sets S_k decrease, and $\bigcap_k S_k = \{R \in \text{VS} : \hat{R} \subseteq R\} =: S$. Since w is a finite measure on the countable set VS , continuity from above gives $w(S) = \lim_k w(S_k) \geq \frac{1}{2}w(\text{VS}) > 0$, and $\hat{R} \subseteq \bigcap S$ by definition. (For finite VS one may instead take one step $s_R \in \hat{R} \setminus R$ for each R with $\hat{R} \not\subseteq R$; this is the version formalized in Lean.) $\square$

Repair note. The first version assumed VS finite and allowed every finite $P \subseteq \hat{R}$ as a detection, including $P \subseteq \bigcap \text{VS}$, which would delete the target and contradicts Lemma 6.2; the hypothesis is now restricted to legal detections and VS may be countable.

E.2 Proofs of Proposition 6.3(b) and Proposition 6.8

Proposition 6.3(b). Let $s_c := \bigvee_i (a_i \rightarrow \ell_i^{1-c_i})$ for $c \in \{0, 1\}^k$. If $b_i \neq c_i$ for some i , then $\ell_i^{1-c_i} = \ell_i^{b_i}$, so the disjunct $a_i \rightarrow \ell_i^{b_i}$ of s_c is an axiom of h_b and $\triangleright s_c \in h_b$. If $b = c$, the valuation with every a_i true and $p_i = c_i$ (so $\ell_i^{c_i}$ true and $\ell_i^{1-c_i}$ false) satisfies the axioms of h_c and falsifies s_c . So $\triangleright s_c \in h_b$ iff $b \neq c$. Every h_b is consistent, so $\mathcal{A} = \{\emptyset\}$ is coherent for all of them. A certified-sound learner accepts only steps in $\bigcap \text{VS}_t$, so it rejects $\triangleright s_c$ while $h_c \in \text{VS}_t$. For target h_{b^*} , present $\triangleright s_c$ for each $c \neq b^*$ in any order: each is valid in the target, each deletes exactly h_c , and at the moment it is presented h_c has not yet been deleted, so each is an incompleteness error. That makes $2^k - 1$ errors, although $\log_2 |\mathcal{H}| = k$. $\square$

Proposition 6.8. (a) As just shown, $\triangleright s_c \in h_b$ iff $b \neq c$. Fix a learner that in every round announces $\hat{R}_t \subseteq R$ for some $R \in \text{VS}_t$; we may assume it is deterministic, and the environment sees $\hat{R}_t$ before it moves. While $|\text{VS}_t| \geq 2$, the adversary picks $h_c \in \text{VS}_t$ with $\hat{R}_t \subseteq h_c$ and presents $\triangleright s_c$. Since $\triangleright s_c \notin h_c \supseteq \hat{R}_t$, this is an incompleteness error (for any target other than h_c). Since $\triangleright s_c \in h_b$ for every $b \neq c$, it deletes exactly h_c . After $2^k - 1$ rounds a single hypothesis h_{b^*} remains; take it as the target. Every presented datum $\triangleright s_c$ has $c \neq b^*$, hence lies in h_{b^*} , so the run is a prefix of a text for h_{b^*} ; extend it arbitrarily. The target must be chosen at the end: if $\text{VS}_t = \{h_c, h_{b^*}\}$ and the learner's coalition is $\{h_{b^*}\}$, a target fixed in advance as h_{b^*} would not be charged the last error.

The union learner. If $R^* \in \text{VS}_t$ then $R^* \subseteq \bigcup \text{VS}_t = \hat{R}_t$, so no incompleteness error occurs; and R^* is never deleted, by Lemma 6.2. Let $h_- = \{\Gamma \triangleright \varphi : \Gamma \cup \{\neg a_1, \dots, \neg a_k\} \models \varphi\}$. Since $\neg a_i \models a_i \rightarrow \ell$, every $h_b \subseteq h_-$, so $\hat{R}_t \subseteq h_-$ and $\text{Cl}_{\hat{R}_t}(\emptyset) \subseteq \text{Cn}_2(\{\neg a_1, \dots, \neg a_k\})$, which does not contain $\perp$. So no argument from $\emptyset$ to $\perp$ uses only accepted steps: there are no detections. For target h_{b^*} , as long as VS_t contains some $h_b \neq h_{b^*}$ the union learner accepts $\triangleright a_i \rightarrow \ell_i^{b_i}$ for an i with $b_i \neq b_i^*$, which is not in h_{b^*} (make a_i and $\ell_i^{b_i^*}$ true); so it is unsound, and it is not oligarchic, so Theorem 6.5 does not apply to it.

(b) Let $\mathcal{H} = \{h_1, h_2, h_3\}$ be the class of Theorem 3.12, each with weight $\frac{1}{3}$. Any step $s \in \bigcup \text{VS} \setminus \hat{R}$ can be presented as a positive datum for a target containing it, and then it is an incompleteness error that deletes $\{h_i : s \notin h_i\}$. This weighs at least $\frac{1}{2}$ only if at least two h_i omit s , i.e. only if s lies in at most one h_i . So if every possible incompleteness error deletes at least half the weight, $\hat{R} \supseteq M := \{s : s \in h_i \text{ for at least two } i\}$. But the argument π of Theorem 3.12 has $\text{Steps}(\pi) \subseteq M \subseteq \hat{R}$ and $\text{Steps}(\pi) \not\subseteq h_i$ for each i . So it is a legal detection for every target, and it deletes nothing. Equivalently, by Proposition 6.6, detection-halving needs $\hat{R} \subseteq h_i \cap h_j$ for some $i \neq j$, and $M \not\subseteq h_i \cap h_j$ for every pair: $h_1 \cap h_2$ lacks $\triangleright q$, while $h_1 \cap h_3$ and $h_2 \cap h_3$ lack $\triangleright p$. $\square$

Repair note. Both parts were added after verification. The first version concluded from Theorem 6.5 that “coherence bounds the price of boldness while nothing bounds the price of caution”; part (a) shows that on Proposition 6.3(b)’s own class every oligarchic learner pays the price of caution. Membership facts were checked by truth tables for $k \leq 4$ (T2 verification log).

E.3 Proof of Proposition 6.12

(a) For the target, $\Gamma \not\vdash^* \varphi$ iff $\Gamma \cup \{\neg\varphi\}$ is target-coherent, by classical reductio. So the designations eventually enumerate (codes for) every finite-premise sequent not in $\vdash^*$, and the text enumerates $\vdash^*$: together they form an informant. Let (h_i) enumerate $\mathcal{H}$, and let the learner output the least i such that h_i contains every sequent seen in the text and is coherent on every designated context seen. Let i^* be the least index of the target and $j < i^*$, so $h_j \neq \vdash^*$. If $\vdash^* \not\subseteq h_j$, the text eventually shows a sequent of $\vdash^* \setminus h_j$, and j is rejected from then on. Otherwise pick $\Gamma \triangleright \varphi \in h_j \setminus \vdash^*$. Then $\Gamma \cup \{\neg\varphi\}$ is target-coherent, hence eventually designated, and it is h_j -incoherent by reductio in h_j ; so j is rejected from then on. The target’s index i^* is never rejected. As there are finitely many $j < i^*$, the output is eventually i^* forever. If membership in the h_i is uniformly decidable, both tests are decidable (coherence of h_i on A is non-membership of $A \triangleright \perp$), so the learner is computable.

(b) $\{p_j\}$ is h_n -coherent iff $\{\neg p_i : i < n\} \cup \{p_j\}$ is consistent iff $j \geq n$, and it is h_ω -incoherent for every j . The chain is strictly increasing with union h_ω , and all members are $\emptyset$ -coherent. Consider the computable learner that outputs h_j for the least j such that $\{p_j\}$ has been designated, and h_ω if there is none (the enumeration learner of (a) also works). On target h_n , exactly the singletons $\{p_j\}$ with $j \geq n$ can be designated, and $\{p_n\}$ eventually is; from then on the least designated j is n , and the learner outputs h_n . On target h_ω no singleton $\{p_j\}$ is ever designated, and the learner outputs h_ω throughout. $\square$

Repair note. This proposition was added after verification; the referee’s counterexample (b) showed that the first version of Corollary 6.11, stated without “ $\mathcal{A}$ fixed”, was false.

E.4 Proofs for Section 6.2: Post-completeness

Theorem 6.13, pure $\{\rightarrow\}$. Let $\varphi \in C(X) \setminus \text{Cn}_2(X)$ and v a Boolean valuation with $v[X] = 1$, $v(\varphi) = 0$. Put $\sigma_v(p) = p_0 \rightarrow p_0$ if $v(p) = 1$ and $\sigma_v(p) = p_0$ otherwise. By induction on ψ , $\sigma_v\psi \equiv_2 p_0 \rightarrow p_0$ if $v(\psi) = 1$ and $\sigma_v\psi \equiv_2 p_0$ if $v(\psi) = 0$. For atoms this is the definition. For $\psi = \alpha \rightarrow \beta$: if $v(\alpha) = 0$, then $\sigma_v\psi \equiv_2 p_0 \rightarrow (\cdot)$, and both $p_0 \rightarrow (p_0 \rightarrow p_0)$ and $p_0 \rightarrow p_0$ are tautologies; if $v(\alpha) = v(\beta) = 1$, then $\sigma_v\psi \equiv_2 \top \rightarrow \top \equiv_2 \top$; if $v(\alpha) = 1$, $v(\beta) = 0$, then $\sigma_v\psi \equiv_2 \top \rightarrow p_0 \equiv_2 p_0$. Hence $\sigma_v X \subseteq \text{Taut} \subseteq C(\emptyset)$ and, by structurality, $\sigma_v\varphi \in C(\sigma_v X) \subseteq C(\emptyset)$, with $\sigma_v\varphi \equiv_2 p_0$. So $p_0 \in \text{Cn}_2(\{\sigma_v\varphi\}) \subseteq C(C(\emptyset)) = C(\emptyset)$. Substituting any ψ for p_0 and using $\sigma C(\emptyset) \subseteq C(\emptyset)$ gives $C(\emptyset) = \text{Fm}$. $\square$

Proposition 6.14. C_{ai} is a closure operator, and it is structural because σX is non-empty iff X is. In the $\{\wedge, \vee\}$ language $\text{Cn}_2(\emptyset) = \emptyset$, since the all-false valuation falsifies every formula; so $C_{\text{ai}} \supseteq \text{Cn}_2$. Now let $C \supsetneq \text{Cn}_2$ be structural. *Case* $C(\emptyset) \neq \emptyset$. Pick $\varphi \in C(\emptyset)$ and substitute p_0 for every atom. Every one-variable lattice term is $\equiv_2 p_0$, so $p_0 \in \text{Cn}_2(\{\sigma\varphi\}) \subseteq C(C(\emptyset)) = C(\emptyset)$, and by structurality $C(\emptyset) = \text{Fm}$: $C = C_{\text{Fm}}$. *Case* $C(\emptyset) = \emptyset$. Take $\varphi \in C(X) \setminus \text{Cn}_2(X)$ (so $X \neq \emptyset$) and a Boolean v with $v[X] = 1$, $v(\varphi) = 0$. Let σ send the atoms true under v to q and the false ones to r (distinct atoms). Each $\sigma\psi$ is a lattice term in q, r , and its value at $(q, r) = (1, 0)$ is $v(\psi)$. Up to $\equiv_2$, the lattice terms in q, r are $q, r, q \wedge r$ and $q \vee r$. So $\sigma\psi \in \{q, q \vee r\}$ up to $\equiv_2$ for $\psi \in X$, and $\sigma\varphi \in \{r, q \wedge r\}$. Hence $\sigma X \subseteq \text{Cn}_2(\{q\})$ and $r \in \text{Cn}_2(\{\sigma\varphi\})$, so

$$r \in C(\{\sigma\varphi\}) \subseteq C(C(\sigma X)) = C(\sigma X) \subseteq C(\text{Cn}_2(\{q\})) \subseteq C(C(\{q\})) = C(\{q\}).$$

Substituting χ for q and ψ for r gives $\psi \in C(\{\chi\})$ for all χ, ψ , so $C(Y) = \text{Fm}$ for every non-empty Y : $C = C_{\text{ai}}$. $\square$

Theorem 6.15. (a) Suppose $\langle D \rangle = \text{Cn}_2$. Every $h \in \mathcal{S}(D, A_0)$ is structural and contains D , hence contains $\langle D \rangle = \text{Cn}_2$; by Theorem 6.13, $h \in \{\text{Cn}_2, C_{\text{Fm}}\}$, and A_0 excludes C_{Fm} . Conversely, suppose $\langle D \rangle \subsetneq \text{Cn}_2$ (note $\langle D \rangle \subseteq \text{Cn}_2$ because Cn_2 is structural and contains D). Then $\langle D \rangle$ is structural, contains D , and is A_0 -coherent because Cn_2 is; so $\langle D \rangle$ and Cn_2 are distinct members of $\mathcal{S}(D, A_0)$.

Explicit finite bases. For a language containing $\neg$ and $\rightarrow$ (possibly also $\wedge, \vee$), with distinct atoms p, q, r , take the axioms $\triangleright p \rightarrow (q \rightarrow p)$, $\triangleright(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$, $\triangleright(\neg p \rightarrow \neg q) \rightarrow (q \rightarrow p)$ and modus ponens $p, p \rightarrow q \triangleright q$; if $\wedge, \vee$ are present, add their clauses *as $\rightarrow$ -axioms*: $\triangleright p \wedge q \rightarrow p$, $\triangleright p \wedge q \rightarrow q$, $\triangleright p \rightarrow (q \rightarrow p \wedge q)$, $\triangleright p \rightarrow p \vee q$, $\triangleright q \rightarrow p \vee q$, $\triangleright(p \rightarrow r) \rightarrow ((q \rightarrow r) \rightarrow (p \vee q \rightarrow r))$. For pure $\{\rightarrow\}$ take the first two axioms, Peirce's law $\triangleright((p \rightarrow q) \rightarrow p) \rightarrow p$ and modus ponens. For languages without $\rightarrow$, such as $\{\neg, \wedge\}$ and $\{\neg, \vee\}$, finite bases exist because every two-element matrix logic is finitely based (Rautenberg, 1981, as we recall the result; we have not re-checked its exact statement).

The least structural consequence relation generated by a set of sequents is given by derivations from substitution instances of those sequents. With the axioms and modus ponens above, these are derivations in a Hilbert system with modus ponens as its only rule: Łukasiewicz's axioms for $\{\neg, \rightarrow\}$, extended by the usual axioms for $\wedge$ and $\vee$, and for $\{\rightarrow\}$ the axioms K , S and Peirce's law, a variant of the Tarski–Bernays axiomatization (K and S give intuitionistic implication, and Peirce's law makes it classical). These systems are complete for tautologies (Kleene, 1952; Mendelson, 1964); for the $\wedge, \vee$ axioms Kalmár's lemma goes through because the system proves $p, q \vdash p \wedge q$, $\neg p \vdash \neg(p \wedge q)$, $\neg q \vdash \neg(p \wedge q)$, $p \vdash p \vee q$, $q \vdash p \vee q$ and $\neg p, \neg q \vdash \neg(p \vee q)$. Since modus ponens is the only rule and the first two axioms are present, the deduction theorem holds. So if $X \models \varphi$, compactness gives a finite $\{x_1, \dots, x_n\} \subseteq X$ with $\models x_1 \rightarrow (\dots \rightarrow (x_n \rightarrow \varphi))$, which is provable, and n applications of modus ponens derive φ from X . Hence $\langle D \rangle = \text{Cn}_2$.

(b) Let $j < i^*$, so $h_j \neq \text{Cn}_2$. If $h_j \not\supseteq \text{Cn}_2$, some sequent of $\text{Cn}_2 \setminus h_j$ eventually appears in the text, and j is ineligible from then on. If $h_j \supseteq \text{Cn}_2$, then $h_j = C_{\text{Fm}}$ by Theorem 6.13, and $A_0 \vdash_{h_j} \perp$ makes j ineligible from the start. i^* is always eligible. So the learner's output is non-decreasing, never exceeds i^* , and is eventually i^* ; in particular it changes its mind at most i^* times. Uniform decidability makes the eligibility tests computable; no property of the ordering was used.

(c) $\text{Cn}_2 \oplus \{\triangleright p\}$ with p not in A_0 : its consequences of A_0 are $\text{Cn}_2(A_0 \cup \{p\})$, which is consistent, and it contains every sequent of Cn_2 . C_{Fm} contains every sequent. C_{ai} contains every sequent of Cn_2 in the $\{\wedge, \vee\}$ fragment and is non-trivial on $\emptyset$. $\square$

The sequent-rule basis fails (countermodel). Take the language $\{\neg, \rightarrow, \wedge\}$ and let D' consist of the three $\{\neg, \rightarrow\}$ axioms, modus ponens, and the sequent rules $p \wedge q \triangleright p$, $p \wedge q \triangleright q$, $p, q \triangleright p \wedge q$. We show $\triangleright p \rightarrow (q \rightarrow p \wedge q) \notin \langle D' \rangle$; this countermodel is due to the verifier of T2 and was re-checked by script. For every assignment s of truth values to atoms define v_s on all formulas, simultaneously for all s , by recursion on complexity: $v_s(p) = s(p)$; v_s treats $\neg$ and $\rightarrow$ classically; and $v_s(\alpha \wedge \beta) = 1$ iff $\alpha \in \Theta^*$ and $\beta \in \Theta^*$, where $\Theta^* = \{\varphi : v_s(\varphi) = 1 \text{ for all } s\}$. This is well defined, since membership of α, β in Θ^* depends only on values at lower complexity. Every substitution instance of a $\{\neg, \rightarrow\}$ tautology lies in Θ^* , because each v_s evaluates $\neg, \rightarrow$ classically and treats the substituted subformulas as atoms. Θ^* is closed under modus ponens (if $v_s(\alpha) = v_s(\alpha \rightarrow \beta) = 1$ for all s , then $v_s(\beta) = 1$ for all s), under $\wedge$ -introduction (if $\alpha, \beta \in \Theta^*$ then $v_s(\alpha \wedge \beta) = 1$ for all s) and under $\wedge$ -elimination (if $\alpha \wedge \beta \in \Theta^*$ then $\alpha, \beta \in \Theta^*$ by definition). So $\langle D' \rangle(\emptyset) \subseteq \Theta^*$. But $p \notin \Theta^*$, so $v_s(p \wedge q) = 0$ for every s ; at $s(p) = s(q) = 1$ this gives $v_s(p \rightarrow (q \rightarrow p \wedge q)) = 0$. Hence $p \rightarrow (q \rightarrow p \wedge q) \notin \Theta^*$, and $\langle D' \rangle \neq \text{Cn}_2$. $\square$

Renaming is not enough (L5 Prop 2.5). Let $C(X) = \text{Cn}_2(X)$ if $\text{Cn}_2(X)$ contains no atom, and $C(X) = \text{Cn}_2(X \cup \text{At})$ otherwise, where At is the set of atoms. C is extensive; it is monotone because $\text{Cn}_2(X) \subseteq \text{Cn}_2(Y)$ for $X \subseteq Y$; and it is idempotent, since in the first case $\text{Cn}_2(\text{Cn}_2(X))$ has no atom and in the second $C(X)$ already contains every atom. For a renaming ρ (atoms to atoms) we have $\rho C(X) \subseteq C(\rho X)$: in the first case because Cn_2 is structural, and in the second because $\text{Cn}_2(\rho X)$ contains ρp for an atom $p \in \text{Cn}_2(X)$, so $C(\rho X) = \text{Cn}_2(\rho X \cup \text{At}) \supseteq \rho \text{Cn}_2(X \cup \text{At})$. C contains $p \triangleright q$, and $C(\emptyset) = \text{Taut}$; so C is $\emptyset$ -coherent, has the classical theorems, and differs from Cn_2 . It is the least renaming-invariant closure operator above Cn_2 containing $p \triangleright q$: any such operator puts every atom into $C'(X)$ as soon as $\text{Cn}_2(X)$ contains one. $\square$

Proposition 6.16(a). *Extensive and monotone:* clear. *Idempotent:* if $\sigma X \subseteq \text{Th}$, then $\sigma C_{\text{adm}}(X) \subseteq \text{Th}$ by definition, so every $\varphi \in C_{\text{adm}}(C_{\text{adm}}(X))$ has $\sigma \varphi \in \text{Th}$. *Structural:* if $\varphi \in C_{\text{adm}}(X)$ and $\sigma(\tau X) \subseteq \text{Th}$, then applying the definition to $\sigma \circ \tau$ gives $\sigma(\tau \varphi) \in \text{Th}$; so $\tau \varphi \in C_{\text{adm}}(\tau X)$. $C_{\text{adm}}(\emptyset) = \text{Th}$ because Th is substitution-closed. *Maximal:* if C is structural with $C(\emptyset) = \text{Th}$, $\varphi \in C(X)$ and $\sigma X \subseteq \text{Th}$, then $\sigma \varphi \in C(\sigma X) \subseteq C(C(\emptyset)) = \text{Th}$. For finite X , the defining condition is admissibility of X/φ . $\square$

Proposition 6.16(b),(c). (b) Cn_2 is structural with theorems Taut , so $\text{Cn}_2 \subseteq C_{\text{adm}}$ by (a). If $\varphi \notin \text{Cn}_2(X)$, the substitution σ_v of the proof of Theorem 6.13 maps X into Taut and φ to a contradiction (to a formula $\equiv_2 p_0$ in the pure $\{\rightarrow\}$ case), which is not in Taut ; so $\varphi \notin C_{\text{adm}}(X)$. In the $\{\wedge, \vee\}$ fragment $\text{Th} = \emptyset$, so the condition “ $\sigma X \subseteq \text{Th}$ ” never holds for non-empty X : $C_{\text{adm}}(X) = \text{Fm}$ for $X \neq \emptyset$ and $C_{\text{adm}}(\emptyset) = \emptyset$, i.e. $C_{\text{adm}} = C_{\text{ai}}$.

(c) The admissibility and non-derivability of Harrop’s rule are cited (Harrop, 1960; Rybakov, 1997; Iemhoff, 2001). The two relations have the same theorems by (a). For the coherence profile, first note that IPC decides every variable-free formula according to its classical value: by induction, $\vdash_{\text{IPC}} \neg \perp$; and if α, β are each provable or refutable according to their classical values, so are $\alpha \wedge \beta$, $\alpha \vee \beta$, $\alpha \rightarrow \beta$ and $\neg \alpha$. Now if A is classically consistent, pick a Boolean $v \models A$ and substitute $\top := \perp \rightarrow \perp$ or $\perp$ for each atom according to v . Then $\sigma_v A$ consists of classically true variable-free formulas, all IPC-theorems, while $\sigma_v \perp = \perp$ is not one; so the rule $A/\perp$ is not admissible, i.e. $A \not\vdash_{\text{adm}}^{\text{IPC}} \perp$. If A is classically inconsistent, then $A \vdash_{\text{IPC}} \perp$ by Glivenko’s theorem (Theorem 6.17(a)), and a fortiori $A \vdash_{\text{adm}}^{\text{IPC}} \perp$. $\square$

E.5 Proof of Theorem 6.17

(a) Glivenko's theorem (Glivenko, 1929) gives $\vdash_{\text{CPC}} \neg\psi$ iff $\vdash_{\text{IPC}} \neg\psi$. With $\psi = \bigwedge A$ and the deduction theorem (in both logics), $A \vdash_{\text{CPC}} \perp$ iff $\vdash_{\text{CPC}} \neg\bigwedge A$ iff $\vdash_{\text{IPC}} \neg\bigwedge A$ iff $A \vdash_{\text{IPC}} \perp$ (for $A = \emptyset$ neither side holds). An intermediate L is sandwiched between IPC and CPC.

(b) CPC is a structural proper extension of IPC, coherent on every classically consistent context; by (a) it has the same coherence data. So IPC is never a maximal coherent structural hypothesis.

(c) We use the following facts about Jankov formulas (Jankov, 1968); see Chagrov and Zakharyashev (1997, ch. 9). There is an infinite family $B_1, B_2, \dots$ of finite subdirectly irreducible Heyting algebras with $B_i \notin \mathbf{SH}(B_j)$ for $i \neq j$, and each B_i has a formula $\chi(B_i)$ such that a Heyting algebra B refutes $\chi(B_i)$ iff $B_i \in \mathbf{SH}(B)$. At most one B_i is the two-element algebra $\mathbf{2}$ (two copies would violate the antichain condition); drop it, so $|B_i| > 2$ for all i . Let L_k be the least structural consequence relation containing $\vdash_{\text{IPC}}$ and the axioms $\triangleright\chi(B_i)$ for $i \leq k$, and $L_\omega = \bigcup_k L_k$.

- *Strictly increasing.* The matrix consequence $\models_{B_{k+1}}$ of $(B_{k+1}, \{1\})$ is structural and contains $\vdash_{\text{IPC}}$, by soundness of IPC for Heyting algebras. B_{k+1} validates $\chi(B_i)$ for $i \leq k$, since $B_i \notin \mathbf{SH}(B_{k+1})$. So $L_k \subseteq \models_{B_{k+1}}$, whereas $B_{k+1} \in \mathbf{SH}(B_{k+1})$ refutes $\chi(B_{k+1}) \in L_{k+1}(\emptyset)$.
- *Intermediate.* $\mathbf{SH}(\mathbf{2})$ consists of $\mathbf{2}$ and the trivial algebra, and B_i is neither, so $\mathbf{2}$ validates every $\chi(B_i)$. Hence every $L_k \subseteq \models_{\mathbf{2}} = \text{CPC}$. By Proposition 6.10, L_ω is a consequence relation; it is structural as a union of structural relations, and $\text{IPC} \subseteq L_\omega \subseteq \text{CPC}$.
- *Impossibility.* By (a), for every $k \leq \omega$ the set of L_k -coherent finite contexts is the set of classically consistent ones. So whether $\mathcal{A}$ is fixed, or designations are target-dependent data, the environment can present one and the same designation stream δ for every target in $\{L_k : k \leq \omega\}$. Folding δ into the learner turns any learner for this class into a text learner for the chain $L_1 \subsetneq L_2 \subsetneq \dots$ together with its union, which is impossible by the locking-sequence argument of Corollary 6.11.

(d) The target is $\langle G \rangle$ for a finite set G of sequents. Since D_n is contained in the structural target, so is $\langle D_n \rangle$. Once the text has shown every sequent of G , $\langle D_n \rangle \supseteq \langle G \rangle$. From then on every conjecture is correct (its syntactic index, e.g. D_n , may keep changing), which is BC-identification. $\square$

E.6 Proof of Proposition 6.18

(a) If a consistent deductively closed $T' \supsetneq T$ contains $\varphi \notin T$, then $\neg\varphi \in T \subseteq T'$ by completeness of T , so T' is inconsistent. For learning, run the learner of Theorem 6.15(b): for $j < i^*$, either $h_j \not\supseteq T$ and the text refutes it, or $h_j \supsetneq T$ is inconsistent and the datum $\emptyset$ rejects it at once.

(b) Let h be parameter-structural, $\exists$ -closed, $h \supseteq \vdash_T$ and $\emptyset \not\vdash_h \perp$. Suppose $\Gamma \vdash_h \varphi$ but $T \cup \Gamma \not\vdash \varphi$, with parameters $\bar{c} = c_1, \dots, c_n$. Since the parameters do not occur in T , $T \not\vdash \forall \bar{x} (\bigwedge \Gamma(\bar{x}) \rightarrow \varphi(\bar{x}))$, so by completeness $T \vdash \exists \bar{x} \theta(\bar{x})$ for $\theta(\bar{x}) := \bigwedge \Gamma(\bar{x}) \wedge \neg\varphi(\bar{x})$. Take fresh parameters $\bar{d}$. By parameter-structurality $\Gamma(\bar{d}) \vdash_h \varphi(\bar{d})$. Since $h \supseteq \vdash_T$ is closed under cut and $\theta(\bar{d}) \vdash_T \gamma$ for each $\gamma \in \Gamma(\bar{d})$, we get $\theta(\bar{d}) \vdash_h \varphi(\bar{d})$; also $\theta(\bar{d}) \vdash_T \neg\varphi(\bar{d})$ and $\varphi(\bar{d}), \neg\varphi(\bar{d}) \vdash_T \perp$, so $\theta(\bar{d}) \vdash_h \perp$. For $0 \leq i \leq n$ let $\theta_i := \exists x_{i+1} \dots \exists x_n \theta(d_1, \dots, d_i, x_{i+1}, \dots, x_n)$, so $\theta_n = \theta(\bar{d})$ and $\theta_{i-1} = \exists x_i \psi(x_i)$ with $\psi(d_i) = \theta_i$. If $\theta_i \vdash_h \perp$, then $\exists$ -elimination applied to $\theta_{i-1} \vdash_h \theta_{i-1}$ and $\theta_{i-1}, \psi(d_i) \vdash_h \perp$ (monotonicity), with d_i not in θ_{i-1} or $\perp$, gives $\theta_{i-1} \vdash_h \perp$. By induction $\theta_0 \vdash_h \perp$, and since

$\emptyset \vdash_T \theta_0$, cut gives $\emptyset \vdash_h \perp$, a contradiction. (With no parameters, $T \vdash \neg(\bigwedge \Gamma \rightarrow \varphi)$ and the last step is immediate.) So $h \subseteq \vdash_T$, i.e. $h = \vdash_T$. Learning is as in (a).

(c) Terms are integer polynomials in the parameters. For a set X of sentences, let $\text{cl}_i(X)$ be the least set containing X , closed under Cn_{RCF} and under the instances of rule i ($t \cdot t = 1 + 1 \triangleright t > 0$ for $i = 1$; $t^3 - 4t + 1 = 0 \triangleright t > 1$ for $i = 2$). Substituting terms for parameters maps Cn_{RCF} -inferences to Cn_{RCF} -inferences and rule instances to rule instances, so cl_i is parameter-structural, and it is the least parameter-structural consequence relation h_i containing $\vdash_{\text{RCF}}$ and the rule. Because the parameters do not occur in RCF and RCF is complete with model $\mathbb{R}$, $\text{RCF} \cup Y \vdash \psi$ for parameter sentences iff $\mathbb{R}$ satisfies the universal closure of $\bigwedge Y_0 \rightarrow \psi$ for some finite $Y_0 \subseteq Y$.

h_1 . An instance fires from $\text{Cn}_{\text{RCF}}(\emptyset)$ only if $t(\bar{x})^2 = 2$ identically on $\mathbb{R}$, which no integer polynomial satisfies (a constant integer n has $n^2 \neq 2$, and a non-constant t has non-constant t^2). So $h_1(\emptyset) = \text{Cn}_{\text{RCF}}(\emptyset)$, which is consistent. The instance with $t = c$ gives $c \cdot c = 1 + 1 \vdash_{h_1} c > 0$, which RCF does not prove ($c = -\sqrt{2}$). In the context $\{c \cdot c = 1 + 1\}$, RCF proves $(-c) \cdot (-c) = 1 + 1$, the rule gives $-c > 0$ and $c > 0$, and RCF derives $\perp$.

h_2 . The cubic $f(x) = x^3 - 4x + 1$ has no rational root ($f(\pm 1) \neq 0$), so it is irreducible over $\mathbb{Q}$. Its discriminant is $-4(-4)^3 - 27 = 229$, not a square, so its Galois group is S_3 . Its real roots are $r_1 \approx -2.115$, $r_2 \approx 0.254$, $r_3 \approx 1.861$. For a root r , $\mathbb{Q}(r)$ has degree 3 and is not normal (else the Galois group would have order 3), so it contains no other root (if it contained two roots it would contain the third, since the roots sum to 0, and be the splitting field). Now:

- From $\emptyset$, an instance fires only if $f(t(\bar{x})) = 0$ for all real $\bar{x}$. Then t is a continuous map from the connected $\mathbb{R}^m$ into the finite set of roots, hence constant, hence an integer root of f , which does not exist. So $h_2(\emptyset) = \text{Cn}_{\text{RCF}}(\emptyset)$.
- Let $X = \{f(c) = 0\}$ and $Y = \text{Cn}_{\text{RCF}}(X \cup \{c > 1\})$. The instance with $t = c$ shows $Y \subseteq h_2(X)$. Conversely, Y is closed under the rule: if $f(t(c, \bar{d})) = 0 \in Y$, then $f(t(r_3, \bar{d})) = 0$ for all real $\bar{d}$ (the only root > 1 is r_3); by continuity and connectedness $t(r_3, \bar{d})$ is constant in $\bar{d}$, equal to $t(r_3, 0) \in \mathbb{Z}[r_3] \subseteq \mathbb{Q}(r_3)$, hence equal to r_3 ; so $\mathbb{R}$ satisfies $\forall c \forall \bar{d} (f(c) = 0 \wedge c > 1 \rightarrow t(c, \bar{d}) > 1)$ and $t > 1 \in Y$. Hence $h_2(X) = Y$, which is consistent, while $X \vdash_{h_2} c > 1$ is not RCF-valid ($c = r_2$). So h_2 is unsound, and coherent on both $\emptyset$ and X .
- In the context $\{f(c) = 0, c < 0\}$ the rule gives $c > 1$, and RCF derives $\perp$. □

Repair note. The first version claimed, from (a), that complete theories are pinned “as cleanly as CPC” in the user’s rule-level setting. The verifier’s counterexamples (c) refuted that; (b) is the repair. The algebraic facts about f were also checked with a computer algebra system.

E.7 Proof of Theorem 6.22

(a) If π is true, $\text{PA} + \pi \subseteq \text{Th}(\mathbb{N})$ is consistent. If π is false, $\neg\pi$ is a true Σ_1 sentence, so $\text{PA} \vdash \neg\pi$ by Σ_1 -completeness, and $\text{PA} + \pi$ is inconsistent.

(b) A true σ gives $\text{PA} + \sigma \subseteq \text{Th}(\mathbb{N})$. By Gödel’s second theorem $\text{PA} + \neg\text{Con}(\text{PA})$ is consistent, although $\neg\text{Con}(\text{PA})$ is a false Σ_1 sentence.

(c) On a Π_1 sentence π the learner says “true” at stage t unless a PA-proof of $\neg\pi$ with code $\leq t$ exists. By (a) and soundness of PA, such a proof exists iff π is false; so the verdict changes at most once and converges to the truth. On a Σ_1 sentence σ it says “false” unless a PA-refutation of $\neg\sigma$ (equivalently a proof of σ) with code $\leq t$ exists; apply (a) to the Π_1 sentence $\neg\sigma$.

(d) At stage t the learner accepts φ_n iff no PA-proof with code $\leq t$ refutes the conjunction of φ_n with the sentences among $\varphi_0, \dots, \varphi_{n-1}$ accepted at stage t . By induction on n : once

the statuses of $\varphi_0, \dots, \varphi_{n-1}$ have stabilized, the status of φ_n is a Π_1 question (consistency of a fixed finite extension of PA), answered by proof search with at most one change. So all statuses converge. In the limit φ_n is accepted iff it is consistent with PA plus the accepted earlier sentences: this is Lindenbaum's construction along the enumeration, so the limit is a complete consistent extension of PA, and it depends on the enumeration. If $\varphi_0 = \neg \text{Con}(\text{PA})$, no refutation of $\text{PA} + \neg \text{Con}(\text{PA})$ exists (Gödel II), so φ_0 is accepted forever and the limit contains a false Σ_1 sentence. On the enumeration $\text{Con}(\text{PA}) \wedge 0=0, \neg \text{Con}(\text{PA}), \text{Con}(\text{PA}), \dots$, the first sentence is consistent with PA (it is true) and stays accepted; the second is inconsistent with it, which proof search finds, so it is rejected.

(e) On a computable presentation, the learner's guesses about each sentence form a computable function $g(\varphi, n)$. If they converged to the truth value on every Σ_2 sentence, then by Shoenfield's limit lemma (Shoenfield, 1959) the set of true Σ_2 sentences would be Δ_2 ; but it is Σ_2 -complete, hence not Δ_2 , by Post's hierarchy theorem. $\square$

Repair note. The first version of (d) said that the naive learner accepts whichever of $\text{Con}(\text{PA})$ and $\neg \text{Con}(\text{PA})$ comes first; the verifier's enumeration above refutes that, and (d) now states the Lindenbaum form.

E.8 Proof of Proposition 6.24

(a) If limits exist on C , then for $\varphi \in C$: $\lim_t P_t(\varphi) = 1$ iff $\forall q \in \mathbb{Q}^+ \forall s \exists t \geq s P_t(\varphi) > 1 - q$, a Π_2 condition. The set of true Σ_2 sentences is Σ_2 -complete, so it is not Π_2 (else it would be Δ_2), and no convergent computable credences have it as their success set.

(b) By the convention on empty quantifier blocks every Π_1 sentence is Π_2 . If weakly coherent credences gradually verified all Π_2 sentences, they would be Π_1 -convergent, and Theorem 6.23 would give a true Π_2 sentence with $\liminf_t P_t = 0$, contradicting $P_t \rightarrow 1$ on true Π_2 sentences.

(c) Enumerate the sentences of Σ_2 form, $\sigma = \exists x \forall y R(x, y)$ with R in Δ_0 (quantifier blocks coded as single variables; an empty block is a dummy variable), with indices $\text{ind}(\sigma)$. At stage t :

- $x_t(\sigma)$ is the least $x \leq t$ with $R(x, y)$ for all $y \leq t$, or $t + 1$ if there is none;
- $c_t(\sigma)$ is the last stage $s \leq t$ with $x_s(\sigma) \neq x_{s-1}(\sigma)$, or -1 if there is none;
- $a_t(\sigma) := t - \max(\text{ind}(\sigma), c_t(\sigma))$, the age of the current witness, for $t \geq \text{ind}(\sigma)$;
- I_t is the symmetric set of pairs (σ, σ') of Σ_2 sentences with indices $\leq t$ that have a PA-proof of $\neg(\sigma \wedge \sigma')$ with code $\leq t$ (including $\sigma' = \sigma$);
- $P_t(\sigma) := \min(1 - 2^{-a_t(\sigma)}, \min\{2^{-a_t(\sigma')} : (\sigma, \sigma') \in I_t, a_t(\sigma') \geq a_t(\sigma)\})$.

Set $P_t := 0$ on sentences not of Σ_2 form, and on σ before stage $\text{ind}(\sigma)$.

Weak coherence. If $\text{PA} \vdash \psi \rightarrow \neg \varphi$ and one of them is not of Σ_2 form, it has credence 0 and the sum is at most 1. Otherwise $(\psi, \varphi) \in I_t$ from some stage on. At such a stage, say $a_t(\psi) \leq a_t(\varphi)$; then $P_t(\psi) \leq 2^{-a_t(\varphi)}$ and $P_t(\varphi) \leq 1 - 2^{-a_t(\varphi)}$, so the sum is at most 1. *False σ .* Every x is eventually refuted, so $x_t(\sigma) \rightarrow \infty$ and changes infinitely often. At each change $a_t(\sigma) = 0$ and $P_t(\sigma) \leq 1 - 2^0 = 0$. So $P_t(\sigma) \not\rightarrow 1$. *True σ .* Let x^* be its least witness. Once every $x < x^*$ is refuted and $t \geq x^*$, $x_t(\sigma) = x^*$ forever. Let $T_0 = \max(\text{ind}(\sigma), \text{last change})$; for $t > T_0$, $a_t(\sigma) = t - T_0 \rightarrow \infty$. A competitor σ' with $(\sigma, \sigma') \in I_t$ is PA-incompatible with the true σ , hence false since PA is sound. If $\text{ind}(\sigma') > T_0$ then $a_t(\sigma') \leq t - \text{ind}(\sigma') < a_t(\sigma)$. The finitely many competitors with index $\leq T_0$ are false, so each changes its witness at some stage $c > T_0$, after which $a_t(\sigma') \leq t - c < a_t(\sigma)$. So from some stage on no competitor enters the inner minimum,

and $P_t(\sigma) = 1 - 2^{-a_t(\sigma)} \rightarrow 1$. *Non-convergence.* By (a), these credences cannot converge on every Σ_2 sentence. $\square$

The construction is due to L3's referee and was sanity-checked by script (`L3-scripts/sigma2_weak_coherence.py`). It assigns 0 to every sentence not of Σ_2 form, so it is dogmatic on the strictly Π_2 truths, consistent with Theorem 6.23.

E.9 Proofs for Section 6.4: Carnap's problem

For finite Γ, Δ write $U(\Gamma, \Delta) = \{v : v[\Gamma] = 1, v[\Delta] = 0\}$; these sets form a clopen basis of 2^{Fm} , and a valuation satisfies the sequent $\Gamma \triangleright \Delta$ iff it lies outside $U(\Gamma, \Delta)$. So $\text{Val}(S)$ is closed for every set S of sequents.

Lemma 6.26. (a) If $v \notin \bar{V}$, some basic neighbourhood $U(\Gamma, \Delta) \ni v$ misses V ; then $\Gamma \models_V^m \Delta$ and v violates it. If $v \in \bar{V}$ violates a valid $\Gamma \triangleright \Delta$, then $U(\Gamma, \Delta)$ is a neighbourhood of v , so it contains some $u \in V$, which violates the sequent too; contradiction.

(b) *Finitary:* if $\varphi \in C_V(X)$, the closed sets $V \cap \{v : v(\varphi) = 0\}$ and $\{v : v(x) = 1\}$, $x \in X$, have empty intersection; by compactness a finite subfamily already does, so $\varphi \in C_V(X_0)$ for a finite $X_0 \subseteq X$. Hence a valuation respects every valid finite-premise single-conclusion sequent iff $C_V(v) = v$ (identifying v with its true-set). $\text{Fix}(C_V) = V^\cap$: a fixed point X equals $\bigcap \{v \in V : X \subseteq v\}$, an intersection of members of V (the empty intersection being Fm); conversely, for $X = \bigcap W$ with $W \subseteq V$, $C_V(X) \subseteq \bigcap W = X$.

(c) An intersection of classical theories is a theory, and Fm is the empty intersection. Conversely, a consistent theory T is the intersection of the Boolean valuations extending it: if $\varphi \notin T$, completeness gives a Boolean $v \supseteq T$ with $v(\varphi) = 0$. $\square$

Theorem 6.29. (a) Each schema enforces one row condition. For $\neg$: $\varphi, \neg\varphi \triangleright$ excludes both true and $\triangleright\varphi, \neg\varphi$ excludes both false, so $v(\neg\varphi) = 1 - v(\varphi)$. For $\wedge$: the two eliminations give $v(\varphi \wedge \psi) \leq \min$, introduction gives $\geq$. For $\vee$: the introductions give $v(\varphi \vee \psi) \geq \max$, and $\varphi \vee \psi \triangleright \varphi, \psi$ gives $\leq$. For $\rightarrow$: $\triangleright\varphi, \varphi \rightarrow \psi$ gives $v(\varphi \rightarrow \psi) = 1$ when $v(\varphi) = 0$; $\psi \triangleright \varphi \rightarrow \psi$ gives it when $v(\psi) = 1$; modus ponens gives $v(\varphi \rightarrow \psi) = 0$ when $v(\varphi) = 1, v(\psi) = 0$. So v satisfies all instances iff it is a homomorphism into the two-element Boolean algebra. (With $\perp, \top$ in the language, $\perp \triangleright$ and $\triangleright \top$ fix their values.) $\text{BV} = \text{Val}(\text{TT})$ is closed, and $\models_{\text{BV}}^m$ determines BV by Lemma 6.26(a).

(b) For any substitution σ and sequent $s = \Gamma \triangleright \Delta$, v satisfies σs iff $v \circ \sigma$ satisfies s . Let $v \in V$. Every instance of a schema in TT is $\sigma(\text{TT}_j(p, q))$ for some σ and j , and $v \circ \sigma \in V$ satisfies $\text{TT}_j(p, q)$; so v satisfies all instances and $V \subseteq \text{Val}(\text{TT}) = \text{BV}$. For equality, take any $v \in V$ (there is one, as $V \neq \emptyset$) and any $u \in \text{BV}$. Let $\sigma(p) = p_0 \vee \neg p_0$ if $u(p) = 1$ and $\sigma(p) = \neg(p_0 \vee \neg p_0)$ otherwise. Then $v \circ \sigma \in V \subseteq \text{BV}$ is Boolean and agrees with u on atoms (v is Boolean, so $v(p_0 \vee \neg p_0) = 1$), hence $v \circ \sigma = u$ and $u \in V$.

(c) The learner that outputs BV once the eleven sequents $\text{TT}(p, q)$ and the coherence datum have appeared is correct by (b), and it never needs to change its mind. $V = \emptyset$ satisfies every sequent vacuously, so without the coherence datum no finite set of positive data excludes it.

(d) *Pointwise refutation.* A text for $L_{\text{BV}} := \models_{\text{BV}}^m$ eventually presents every instance of TT , and by (a) every non-Boolean valuation violates one. *No tell-tale.* Fix a Boolean u_0 , and for a formula χ let v'_χ agree with u_0 except at χ . Then $v'_\chi(\neg\chi) = u_0(\neg\chi) = 1 - u_0(\chi) = v'_\chi(\chi)$, so $v'_\chi \notin \text{BV} = \bar{\text{BV}}$. $\text{BV} \cup \{v'_\chi\}$ is closed, and by Lemma 6.26(a) its relation L_χ satisfies $\text{Val}(L_\chi) = \text{BV} \cup \{v'_\chi\} \neq \text{BV}$, so $L_\chi \subsetneq L_{\text{BV}}$. Given a finite $T \subseteq L_{\text{BV}}$, choose χ occurring in no

sequent of T ; then v'_χ agrees with u_0 on every formula T mentions and satisfies T , so $T \subseteq L_\chi$. *Non-learnability.* Suppose a learner, computable or not, BC-identifies (a fortiori if it EX-identifies) every meaning in a class containing BV and all $BV \cup \{v'_\chi\}$ from texts of their relations, plus the coherence datum $[\cdot]$, which is in bounds for every member and so carries no information. By the locking-sequence argument (Blum and Blum, 1975; Angluin, 1980), which needs no effectivity, there is a finite σ of L_{BV} -data such that the learner conjectures BV on every extension of σ by L_{BV} -data. Choose χ not mentioned in σ ; then σ consists of L_χ -data, and on a text for L_χ beginning with σ the learner conjectures $BV \neq BV \cup \{v'_\chi\}$ forever. *Density.* A Boolean valuation is determined by its values on atoms, and the map from atom assignments to BV is a homeomorphism from 2^ω , which has no isolated points. So $BV \setminus \{u_0\}$ is dense in BV; by Lemma 6.26(a) it has the same multiple-conclusion relation, and a basic open set meets BV iff it meets $BV \setminus \{u_0\}$, so the two meanings have the same in-bounds positions. Composing any Boolean v with the constant substitution along u_0 yields u_0 , so $BV \setminus \{u_0\}$ is not structural. $\square$

Repair note. The first version of (d) ended “identification in the limit still holds via the text”. That was false and has been retracted; (d) now proves non-identifiability. The Lean formalization proves (d) for BC-identification from texts with pauses and arbitrary learners.

Proposition 6.30. Row by row, with designated set $\{1\}$. $\neg$: explosion $p, \neg p \triangleright q$ at $p = 1$, $q = 0$ forces $f_\neg(1) = 0$; then $p \triangleright \neg \neg p$ at $p = 1$ forces $f_\neg(0) = 1$. $\wedge$: $p \wedge q \triangleright p$ forces $f_\wedge(0, b) = 0$, $p \wedge q \triangleright q$ forces $f_\wedge(a, 0) = 0$, and $p, q \triangleright p \wedge q$ forces $f_\wedge(1, 1) = 1$. $\vee$: the introductions force $f_\vee(a, 1) = f_\vee(1, b) = 1$, and disjunctive syllogism at $p = q = 0$ (where $\neg p = 1$) forces $f_\vee(0, 0) = 0$. $\rightarrow$: modus ponens forces $f_\rightarrow(1, 0) = 0$; $q \triangleright p \rightarrow q$ forces $f_\rightarrow(a, 1) = 1$; $\neg p \triangleright p \rightarrow q$ at $p = 0$ forces $f_\rightarrow(0, b) = 1$. So only the standard matrix can validate all twelve rules, and it does. A brute-force check over all $4 \cdot 16^3$ matrices confirms that exactly one matrix validates them (T2-checks/matrix_check.py). $\square$

E.10 Proofs for Section 6.5: tonk and conservativity

Proposition 6.32. Intuitionistic implication with $\neg$ -introduction, $\neg$ -elimination and $\neg\neg$ -elimination is a standard natural-deduction system for the $\{\rightarrow, \neg\}$ fragment of classical logic (Prawitz, 1965), so it is coherent and proves Peirce’s law $((p \rightarrow q) \rightarrow p) \rightarrow p$. Peirce’s law is not intuitionistically valid: in the two-world Kripke model $w \leq w'$ with p true only at w' and q true nowhere, $p \rightarrow q$ fails at both worlds, so $(p \rightarrow q) \rightarrow p$ holds at w vacuously, while p fails at w . So the extension is not conservative over $\text{IPC}_\rightarrow$. *$\neg\neg$ -elimination alone is conservative.* Let τ erase negations ($\tau(\neg\varphi) = \tau(\varphi)$, τ commuting with $\rightarrow$, identity on atoms). It maps each instance of $\neg\neg\varphi \triangleright \varphi$ to the identity step $\tau\varphi \triangleright \tau\varphi$ and each instance of an $\text{IPC}_\rightarrow$ rule to an instance of the same rule, and it fixes $\neg$ -free formulas. So a derivation of a $\neg$ -free sequent in the extension maps to an $\text{IPC}_\rightarrow$ -derivation of the same sequent. *Conservativity and coherence.* If the base is non-trivial and coherence means non-triviality on the old vocabulary, a conservative extension proves no new old-vocabulary sequent, so it is non-trivial; likewise if $\perp$ with ex falso is in the base. If $\perp$ is new and has no rules, $\text{IPC}_\rightarrow + \{\triangleright \perp\}$ adds no $\perp$ -free theorem and so is conservative, yet derives $\perp$. $\square$

Theorem 6.33. (a) “ $R \not\vdash \perp$ ” says that no finite derivation in R ends in $\perp$, a Π_1 condition. For hardness, given e build a finite system R_e that derives the successive configurations of the computation of $\varphi_e(e)$ (elementary formal systems simulate Turing machines; Post 1943; Smullyan

1961), and add the rule “halting configuration $\triangleright \perp$ ”. Then $R_e \not\vdash \perp$ iff $\varphi_e(e)$ diverges, which reduces the Π_1 -complete complement of the halting set.

(b) *Membership*: “for every old-vocabulary s and every derivation of s in $B + N$ there is a derivation of s in B ” is Π_2 . *Hardness*, from the Π_2 -complete set $\text{Tot} = \{e : \varphi_e \text{ total}\}$. Let B be a universal elementary formal system that derives $T(\bar{x}, \bar{e})$ iff $\varphi_e(x) \downarrow$ (unary numerals $\bar{x}, \bar{e}$) and derives $\text{Num}(\bar{n})$ for exactly the unary numerals, designed so that T -strings never occur as premises of its rules. Its productions must also match only predicate-tagged strings over the old alphabet; in pure Post-string format a projection such as $xy \triangleright y$ could otherwise strip a new symbol, while in an elementary formal system with predicates this is automatic. Let N_e add a new symbol c with the rules $\triangleright c\bar{e}$ and $c\bar{y}$, $\text{Num}(x) \triangleright T(x, y)$ (schematic in x, y). The only derivable c -string is $c\bar{e}$, so the new old-vocabulary theorems are exactly $T(\bar{n}, \bar{e})$ for $n \in \mathbb{N}$, and since T -strings are never premises nothing further follows. So the old-vocabulary theorems of $B + N_e$ are $\text{Thm}(B) \cup \{T(\bar{n}, \bar{e}) : n \in \mathbb{N}\}$, and $B + N_e$ is conservative over B iff $\varphi_e(n) \downarrow$ for every n , iff $e \in \text{Tot}$.

(c) A set is decidable in the limit by a computable learner iff it is Δ_2 (Shoenfield, 1959; Putnam, 1965; Gold, 1965); a Π_1 set is decided in the limit with at most one mind change; a Π_2 -complete set is not Δ_2 . Refutability in the limit corresponds to Π_2 (Kelly, 1996).

(d) If B is decidable, “every old-vocabulary theorem of $B + N$ is a theorem of B ” quantifies universally over derivations and then checks a decidable property: Π_1 . $\square$

E.11 Proof of Theorem 6.38

The characterization of Alt as the never-refuted $\mathcal{G}$ -closed hypotheses is Theorem 6.9(ii), restricted to $\mathcal{G}$ -closed h .

(i) A structural $h \supseteq \text{Cn}_2$ is Cn_2 or C_{Fm} by Theorem 6.13, and C_{Fm} is incoherent on every context, so on the non-empty $\mathcal{A}$.

(ii) $\supseteq$: if h is structural with $\text{IPC} \subseteq h \subseteq \text{CPC}$ and A is classically consistent, then $A \not\vdash_{\text{CPC}} \perp$, so $A \not\vdash_h \perp$. $\subseteq$: let $h \supseteq \text{IPC}$ be structural and suppose $\Gamma \vdash_h \varphi$ with Γ finite and $\Gamma \not\models \varphi$. Take a Boolean v with $v[\Gamma] = 1$, $v(\varphi) = 0$, and let σ_v map each atom to $\perp \rightarrow \perp$ or $\perp$ according to v . As shown in the proof of Proposition 6.16(c), IPC proves every classically true variable-free formula and refutes every classically false one. So IPC proves each member of $\sigma_v\Gamma$ and proves $\neg\sigma_v\varphi$. By structurality $\sigma_v\Gamma \vdash_h \sigma_v\varphi$; by cut with the IPC-theorems $\sigma_v\Gamma, \emptyset \vdash_h \sigma_v\varphi$; with $\emptyset \vdash_h \neg\sigma_v\varphi$ and $\sigma_v\varphi, \neg\sigma_v\varphi \vdash_{\text{IPC}} \perp$, $\emptyset \vdash_h \perp$. By monotonicity h is incoherent on every $A \in \mathcal{A}$, which is non-empty. So every coherent structural $h \supseteq \text{IPC}$ lies inside CPC.

(iii) With $\mathcal{G}$ deductive closure and $\mathcal{A} = \{\emptyset\}$, the hypotheses above the target are the deductively closed theories containing PA, and $\emptyset$ -coherence is consistency. Designating $\{\text{Con}(\text{PA})\}$ removes $\text{PA} + \neg\text{Con}(\text{PA})$. If every finite set of true sentences is designated, a theory containing a false φ is incoherent on $\{\neg\varphi\}$, and a theory $T \subseteq \text{Th}(\mathbb{N})$ is coherent on every true context; so exactly the unsound ones are removed. The theories T_k of Theorem 6.21 contain PA and are sound, so they remain, and by that theorem text plus true-context coherence cannot discriminate among them. $\square$

Repair note. Part (ii) was strengthened from “ $\supseteq$ ” to equality after verification, with the σ_v argument above; part (iii) was made precise ($\mathcal{G}$ and $\mathcal{A}$ specified).

F Proofs for Section 7

The proofs of Proposition 7.5, Lemma 7.11, Lemma 7.24 and Corollary 7.25 are complete in the main text; this appendix adds the details they cite. Throughout, (WS) is assumed for every designated position unless a statement says otherwise, V (or $V_{[A:D]}$) denotes a total valuation as in (WS), and “refutation” means a refutation in the sense of Definition 7.2, with size counting the symbols of distinct judgments. Two facts are used repeatedly.

- (F1) *Cleanness is closed downward.* If $B' \subseteq B$, every refutation of B' is a refutation of B , because $R_{B'} \subseteq R_B$. Likewise for step sets.
- (F2) *Transfer.* If every step of a set S' lies in $\text{Sound}(R)$ and π is a refutation with steps in $S' \cup R \cup T_{\text{trust}}$, then replacing each S' -step of π by an R -derivation of its conclusion from its premises, and compressing (Definition 7.2(a)), gives a refutation with steps in $R \cup T_{\text{trust}}$, from the same leaves to the same conclusion, of some finite size. In particular, if R is clean at every depth, then every refutation using S' transfers to a refutation of R , which is impossible.

F.1 Proof of Lemma 7.4

Fix a designated position $[A : D]$ and its valuation V , and let $X = V^{-1}(1)$. Since V falsifies no step of R^* , X is closed under R^* , so $\text{Cl}_{R^*}(X) = X$. Hence if $P \triangleright j \in \text{Sound}(R^*)$ and $P \subseteq X$, then $j \in \text{Cl}_{R^*}(P) \subseteq \text{Cl}_{R^*}(X) = X$: V falsifies no step of $\text{Sound}(R^*)$.

(a) Let $S \subseteq \text{Sound}(R^*)$, and let π be a derivation relative to $[A : D]$ whose leaves have e -value 1 and whose steps lie in $S \cup T_{\text{trust}}$. By induction along the sequence π , every judgment of π is V -true. A leaf is e -true, hence V -true because V extends e . A judgment produced by a step of $S \cup T_{\text{trust}}$ from earlier, V -true judgments is V -true, by the claim for S and by (WS) for T_{trust} . So the conclusion of π is V -true, and since V extends e it is not e -false. Hence π is not a refutation; as π , its size and the position were arbitrary, S is clean at every depth. For $B \subseteq \Sigma^*$, $R_B \subseteq R^* \subseteq \text{Sound}(R^*)$ (Lemma 2.3(b)).

(b) A member of Conf_d has a d -refutation, so by (a) it is not contained in Σ^* ; as it is contained in $\Sigma^P = \Sigma^* \sqcup F$, it meets F .

(c) The definition of a d -refutation of B mentions only e (which is determined by $\mathcal{A}$ and W), T_{trust} , d and B . So Conf_d is a function of these data. Both tie-breaking rules choose among the d -refutations of B using only their sizes, a fixed lexicographic order, and whether their descent is blocked; the last depends only on π and e , because $e_\pi^\perp$ is computed from e and the trusted steps of π .

(d) ($\Rightarrow$) The proof of (a) used (WS) only for the position of π . ($\Leftarrow$) Suppose $e = e_{[A:D]}$ is well defined and Σ^* is clean at every depth relative to $[A : D]$. Let $Y = \text{Cl}_{R^* \cup T_{\text{trust}}}(A \cup W^{-1}(1))$ and let V be the indicator of Y . Since Y is closed under $R^* \cup T_{\text{trust}}$, V falsifies no step of $R^* \cup T_{\text{trust}}$, and V is 1 on every e -true judgment. Let j be e -false and suppose $j \in Y$. By Lemma 2.3(a), j has a finite $(R^* \cup T_{\text{trust}})$ -derivation tree from $A \cup W^{-1}(1)$, that is, from e -true leaves. Compressing it gives a refutation of Σ^* relative to $[A : D]$ of some finite size, contradicting cleanness. So V is 0 on every e -false judgment, V extends e , and (WS) holds for $[A : D]$. Nothing in the argument used that Σ^* is the target. $\square$

F.2 Details for Proposition 7.5

The first two claims are Theorems 5.14 and 5.15 applied to the tagged practice, together with Definition 2.10, which gives every fallacy an instance outside $\text{Sound}(R^*)$. For the CPC claim, let

$\Pi_c \triangleright j_c$ be the closed falsified instance of the pure fallacy τ (Lemma 7.24). Each closed formula has the same value under every valuation, so every member of Π_c and $\neg j_c$ are tautologies, and a complete Σ^* derives them from $\emptyset$. The step adds j_c , and $\{j_c, \neg j_c\}$ classically entails every formula, which Σ^* derives. For the contrast: if Σ^* is complete and $s = (p \triangleright q)$ for atoms p, q , the set of tautologies is closed under s because p is not a tautology, so $\text{Cl}_{R^* \cup \{s\}}(\emptyset) = \text{Cl}_{R^*}(\emptyset)$. $\square$

F.3 Proof of Lemma 7.8

(a) $M \subseteq U$ is clean iff it contains no member of $\mathcal{K}$, iff (by upward closure and finiteness) it contains no member of $\mathcal{C}$, iff $U \setminus M$ meets every member of $\mathcal{C}$, i.e. is a transversal. Complementation reverses inclusion, so maximal clean sets correspond to minimal transversals.

(b, $\Leftarrow$) Let $x \in C \in \mathcal{C}$ and $T = (U \setminus C) \cup \{x\}$. Let $E \in \mathcal{C}$. If $E \not\subseteq C$, then E meets $U \setminus C$. If $E \subseteq C$, then $E = C$ by minimality of C , so $x \in E$. Hence T is a transversal, and it contains a minimal transversal T' . Since T' meets C and $T' \cap C \subseteq T \cap C = \{x\}$, we get $x \in T'$.

(b, $\Rightarrow$) If $x \in T$ for a minimal transversal T , then $T \setminus \{x\}$ is not a transversal, so some $E \in \mathcal{C}$ has $E \cap T = \{x\}$, and $x \in E$.

(b, second claim) By (a), x lies outside some maximal clean set iff x lies in some minimal transversal, iff $x \in \bigcup \mathcal{C}$. Every clean set is contained in a maximal one, and $\emptyset$ is clean, so there are maximal clean sets and their intersection is $U \setminus \bigcup \mathcal{C}$. It is clean because cleanness is closed downward.

(c, $\Leftarrow$) If $\mathcal{C} = \{\{x_1\}, \dots, \{x_r\}\}$, every transversal contains $\{x_1, \dots, x_r\}$, which is itself a transversal.

(c, $\Rightarrow$) Let T be the unique minimal transversal. By (b), $T = \bigcup \mathcal{C}$. For each $x \in T$, minimality gives some $E \in \mathcal{C}$ with $E \cap T = \{x\}$; since $E \subseteq \bigcup \mathcal{C} = T$, $E = \{x\}$. A member $C \in \mathcal{C}$ with $|C| \geq 2$ would then strictly contain the singleton member $\{x\}$ for any $x \in C$, contradicting minimality. $\square$

The Lean formalization (`InfLearn.Blame`) proves (a) as `isMaxClean_iff` and `setOf_isMaxClean_eq_image`, (b) as `exists_minTransversal_mem_iff`, `iInter_maxClean_eq` and `isClean_sdiff_blameSet`, and (c) as `existsUnique_minTransversal_iff`, for a finite universe; upward closure is needed only inside U , and $\emptyset \notin \mathcal{K}$ only for the intersection claims.

F.4 Proof of Proposition 7.6

(a) $\emptyset \subseteq \Sigma^*$ is d -clean by Lemma 7.4(a), so $\emptyset \in \mathcal{V}_d$ and $\bigcap_{\Sigma \in \mathcal{V}_d} R_\Sigma \subseteq R_\emptyset = \emptyset$. The version-space verifier (Definition 4.1) accepts exactly this intersection.

(b) Before any refutation is found, $\mathcal{V}'_t = \mathcal{V}$, whose unique maximal member is Σ^P ; so the verifier accepts R^P , which contains an instance outside $\text{Sound}(R^*)$ of every $\tau \in F$.

(c) *Schema level.* $\mathcal{V}_d$ is the family of clean sets for $U = \Sigma^P$ and $\mathcal{K} = \text{Conf}_d$, which is upward closed and does not contain $\emptyset$. By Lemma 7.8(b) the intersection of its maximal members is $\Sigma^P \setminus \bigcup \mathcal{C}_d$, and since $\Sigma^P = \Sigma^* \sqcup F$ this equals $(\Sigma^* \setminus \text{Coll}_d) \cup F_d^{\text{surv}}$. For the inclusion $F_d^{\text{surv}} \subseteq F_{\text{res}}^{(d)}$, let $\tau \in F \setminus F_{\text{res}}^{(d)}$. Then $\Sigma^* \cup \{\tau\} \in \text{Conf}_d$ contains some $C' \in \mathcal{C}_d$, and $C' \not\subseteq \Sigma^*$ by Lemma 7.4(a), so $\tau \in C'$ and $\tau \notin F_d^{\text{surv}}$. Hence the asserted set lies in $R^* \cup R_{F_{\text{res}}^{(d)}}$.

Instance level. Let $\mathcal{M}$ be the maximal members of $\mathcal{V}_d$. For each $\Sigma \in \mathcal{M}$, $\bigcap \mathcal{M} \subseteq \Sigma$ gives $R_{\bigcap \mathcal{M}} \subseteq R_\Sigma$; so the instance-level set contains the schema-level one. Since $\Sigma^* \in \mathcal{V}_d$ and Σ^P is finite, $\Sigma^* \subseteq \Sigma_{\text{max}}$ for some $\Sigma_{\text{max}} \in \mathcal{M}$. For $\tau \in \Sigma_{\text{max}} \cap F$, $\Sigma^* \cup \{\tau\} \subseteq \Sigma_{\text{max}}$ is d -clean by (F1), so $\tau \in F_{\text{res}}^{(d)}$. Hence $\bigcap_{\Sigma \in \mathcal{M}} R_\Sigma \subseteq R_{\Sigma_{\text{max}}} \subseteq R^* \cup R_{F_{\text{res}}^{(d)}}$. As a step set it is d -clean, since a d -refutation

of it would be a d -refutation of the schema set $\Sigma_{\max}$. The MP/AC example of the main text shows that the inclusion can be strict: $(A, A \rightarrow A \triangleright A)$ is the instance of MP with $B := A$ and the instance of AC with $B := A$, so its instances lie in $R_{\text{MP}} \cap R_{\text{AC}}$, while $R_{\{\text{MP}\} \cap \{\text{AC}\}} = \emptyset$. $\square$

F.5 Proof of Lemma 7.9

Agreement. We show by induction on the order in which the fixed-point computation assigns values that every value of e_π^+ agrees with V . Initial values come from e , which V extends. If the forward rule assigns 1 to the conclusion of a trusted step whose premises have value 1, those premises are V -true by induction, and since V falsifies no trusted step the conclusion is V -true. If the backward rule assigns 0 to a premise π_0 of a trusted step whose conclusion has value 0 and whose other premises have value 1, then by induction the conclusion is V -false and the other premises are V -true; if π_0 were V -true the step would be falsified by V , so π_0 is V -false. Hence e_π^+ never assigns both values to a judgment, and if it does, no total V as in (WS) exists for $[A : D]$.

(a) Each move of the descent goes from a node of value 0 to one of its premises, which occurs strictly earlier in the sequence π ; so the descent terminates. Leaves have e -value 1, hence e_π^+ -value 1, while every visited node has value 0; so every visited node is the conclusion of some step of π . Since the descent is unblocked, it ends by outputting a step s all of whose premises have value 1 and whose conclusion has value 0 (a step without premises is output at once). By agreement, the premises of s are V -true and its conclusion V -false. By (WS), $s \notin T_{\text{trust}}$ and $s \notin R^*$; as π is a refutation of B , $s \in R_B$.

(b) The worklist assigns each judgment of π a value at most once and re-examines a trusted step only when one of its judgments receives a value, so the fixed point costs $O(|\pi|\varphi)$ operations, with one evaluation of e per judgment. The descent looks at the premises of each visited node once, and visits at most $|\pi|$ nodes. The number of judgments is at most $|\pi| \leq d$.

(c) If some genuine σ had $s \in \text{inst}(\sigma)$, then $s \in R^*$, contradicting (a).

(d) Let $\sigma \in B$ with $s \in \text{inst}(\sigma)$. Let J_1 be the list of judgments of π with e_π^+ -value 1, in the order in which they received the value (those with initial e -value 1 first). Each member of J_1 is e -true, and then is treated as a leaf, or received value 1 by the forward rule as the conclusion of a trusted step of π whose premises precede it in J_1 . The premises of s lie in J_1 . Let c_0 be the conclusion of s . Define $c_0, c_1, \dots$: if c_i is e -false, stop; otherwise c_i received value 0 by the backward rule from a trusted step t_i of π of which c_i is a premise, whose conclusion c_{i+1} had received value 0 strictly earlier, and whose other premises lie in J_1 . Since assignment times strictly decrease along the chain, it terminates at some e -false c_k , and the c_i are pairwise distinct. They are not in J_1 , because e_π^+ is consistent.

Let π_σ be the list J_1 , followed by c_0 (justified by s), c_1 (justified by t_0), $\dots$, c_k (justified by t_{k-1}). Its judgments are distinct; each non-leaf judgment is the conclusion of a step whose premises occur earlier; its leaves are e -true; its steps are $s \in \text{inst}(\sigma)$ and trusted steps; and its last judgment c_k is e -false. So π_σ is a refutation of $\{\sigma\}$ relative to $[A : D]$. All its judgments are judgments of π , and size counts distinct judgments, so $|\pi_\sigma| \leq |\pi| \leq d$. Hence $\{\sigma\}$ is a d -conflict. (Under a tree-size convention this bound could fail, because the premise derivations of s and the side premises of the chain may share judgments.) $\square$

The Lean theorem `InfLearn.Blame.LineArg.descent` proves the descent for a line-list argument under a total evaluation, with the walk length bounded by any height function that decreases along dependencies, and `LineArg.descent_not_mem` proves that the step found lies outside every rule set whose steps preserve truth. Trusted-step propagation and part (d) are not

formalized.

F.6 Proof of Proposition 7.12

(a) At a good position p , every schema moved into Blamed_p is a fallacy, by Lemma 7.9(c) applied relative to p . So a genuine schema collects votes only from bad positions, of which there are at most m . The bonus for $[\emptyset : \emptyset]$ is used only when that position is known to be good, and then it blames only fallacies. So every genuine schema has at most m votes and lies in A .

(b) Every call at p either ends p 's loop or moves at least one schema out of B_p . Indeed, if the call returns a refutation r , e_p is well defined, $e_{p,r}^+$ is consistent and the descent is unblocked, then the descent outputs a step s whose premises have value 1 and whose conclusion has value 0, by the argument of Lemma 7.9(a), which used only consistency and unblockedness. If s were trusted, the forward rule would have given its conclusion value 1 as well, contradicting consistency. So s is untrusted and lies in R_{B_p} , and some $\sigma \in B_p$ with $s \in \text{inst}(\sigma)$ is moved. Hence the loop at p makes at most $|B_p| + 1 \leq |\Sigma^P| + 1$ calls, and at most $|F| + 1$ at a good position, where only fallacies move. With $m' \leq m$ bad positions the total is at most $(|\mathcal{A}| - m')(|F| + 1) + m'(|F| + |\Sigma^*| + 1) \leq |\mathcal{A}|(|F| + 1) + m|\Sigma^*|$.

(c) Let p be complete. Its loop ended with NONE, so its final B_p is d -clean relative to p ; and since p is good, only fallacies were moved, so $\Sigma^* \subseteq B_p$. If $\tau \in F \setminus \text{Blamed}_p$, then $\tau \in B_p$, and $\Sigma^* \cup \{\tau\} \subseteq B_p$ is d -clean relative to p by (F1). Contrapositively, each complete p relative to which $\Sigma^* \cup \{\tau\}$ is not d -clean gives τ a vote. A $\tau \in A$ has at most m votes, which gives the claim. For $m = 0$ with all positions complete, $A = \Sigma^P \setminus \bigcup_p \text{Blamed}_p$ lies inside every final B_p , so it is d -clean relative to every position; it contains Σ^* , and each of its fallacies is residual. This is the conclusion of Lemma 7.11(a). $\square$

Why per-position loops (computed; prop32_voting.py). An earlier version ran one smallest-refutation query per position and voted on the results. With one bad position $[a, a \rightarrow b : b]$, target $\{\text{MP}\}$ and $F = \emptyset$, it made one successful descent, against its claimed bound $(m+1)|F| = 0$. With $m = 1$, the truthful positions $p_1 = [q, p \rightarrow q, \neg p : p, \neg q]$ and $p_2 = [\neg r, r \rightarrow t : \neg t]$ and the practice $\{\text{MP}, \text{AC}, \text{DA}\}$, the smallest refutation at p_1 always blames AC and hides DA, so DA got one vote and survived although it is refutable at two positions. The loops above exclude already-blamed schemas, so DA is blamed at both positions and removed, while AC, refutable at one position only, may survive.

F.7 Proof of Proposition 7.13

(a) By Definition 7.10, A is computed at $t = N_1$ from $\hat{P}$, a function of the first N_1 human steps, and from answers of Ref_d , which are deterministic functions of $(B, d, \mathcal{A}, W, T_{\text{trust}})$ (Lemma 7.4(c)). Phase-1 acceptance is membership in R_A . The sandbox, the red team and the prover interact with the tier only through queries, which do not change A , and escalation answers are not added to the sample. A promoted κ is a macro, answered by expanding it into an R_A -derivation, so the primitive accepted steps remain R_A and Cl_{R_A} is unchanged. If promoted rules were accepted as primitive steps instead, then $\kappa \subseteq \text{Sound}(R_A) \subseteq \text{Sound}(R^* \cup R_{F_{\text{res}}^{(d)}})$, so the accepted set would satisfy only this weaker inclusion; the closure would be unchanged (Lemma 2.3(b)). But d -cleanness of the enlarged step set can fail. If some $\tau \in A$ is residual at depth d with a shortest refutation of size $2d$, a promoted κ whose instances package the first half of that refutation into single steps removes its intermediate judgments and yields a refutation of size at most d . Primitive promoted rules would also need truth maintenance under the depth schedule.

(b) Let $h_0 \in \mathcal{H}_S$ be clean at every depth with $w(h_0) > 0$. A detection exhibits a refutation π of the step set $B_t = \bigcap S_t$ and deletes every $h \supseteq \text{Steps}(\pi) \setminus T_{\text{trust}}$. If h_0 were deleted, the steps of π would lie in $h_0 \cup T_{\text{trust}}$ and π would be a refutation of h_0 , which is impossible. So h_0 is never deleted. In a detection, every $h \in S_t$ contains $B_t \supseteq \text{Steps}(\pi) \setminus T_{\text{trust}}$, so all of S_t is deleted and $w(\text{VS}_{t+1}) \leq \frac{1}{2}w(\text{VS}_t)$. After D detections $w(h_0) \leq w(\text{VS}) \leq 2^{-D}$, so $D \leq \log_2(1/w(h_0))$. For the robust bound, run the proof of Theorem 6.7 with h_0 in place of R^* : every detection multiplies by β the weight of a superset of S_t , so the total weight shrinks by a factor at most $\frac{1+\beta}{2}$; and h_0 is penalized only in false alarms, since a refutation of h_0 can only come from a position violating (WS). Hence $\beta^m w(h_0) \leq (\frac{1+\beta}{2})^D$. Deletion must test only the *untrusted* steps of π : if it tested all steps against hypotheses that do not contain T_{trust} , coalition members whose refutations use trusted steps would never be deleted, and halving would fail.

(c) For finite S_t , $s \in \bigcap_{h \in S_t} h$ iff for each h (a finite set of schemas) s matches some schema of h , which is decidable. For schema-valued hypotheses, $\bigcap_h \text{inst}(\sigma_h)$ is empty or the instance set of the most general common instance, obtained by unifying renamed-apart copies (Robinson, 1965). $\square$

F.8 Proof of Theorem 7.14 and Corollary 7.15

Step 1: identification (noisy mode). For each tag i consider the following events on the first N_1 samples: (I _{i}) at most e_i invalid steps are tagged i ; and (II _{i}) the valid tag- i samples are e_i -robustly generic (Theorem 5.14(b)). As in the proof of Theorem 5.15, (II _{i}) follows from c_i count events: for each metavariable x with best split $G_{i,x}$ (achieving $\rho_{i,x}$), more than e_i valid tag- i samples have root in $G_{i,x}$ and more than e_i have root outside it; and for each pair $x \neq y$, more than e_i have $\Theta x \neq \Theta y$. (For any f , the samples on the side of $G_{i,x}$ not containing f all have root $\neq f$.) For a ground tag the single event is that more than e_i valid samples occur; this is the convention $\rho_i = c_i = 1$. Let G be the intersection of all these events.

The number of invalid tag- i steps is $\text{Bin}(N_1, \alpha_i)$ with $\alpha_i \leq \bar{\alpha}_i$. Since $e_i = \lfloor (\bar{\alpha}_i + \Delta)N_1 \rfloor$, exceeding e_i means exceeding the mean by more than ΔN_1 , which by Hoeffding's inequality (Hoeffding, 1963) has probability at most $e^{-2N_1\Delta^2}$. Each count in (II _{i}) is $\text{Bin}(N_1, p)$ with $p \geq \rho_i \beta_i \geq \bar{\alpha}_i + 2\Delta$ by (Floor); being at most $e_i \leq (\bar{\alpha}_i + \Delta)N_1$ means falling short of the mean by at least ΔN_1 , again with probability at most $e^{-2N_1\Delta^2}$. A union bound over the $1 + c_i$ events per tag gives

$$\mathbb{P}(G^c) \leq \sum_i (1 + c_i) e^{-2N_1\Delta^2} \leq (1 + c) K e^{-2N_1\Delta^2} \leq \delta$$

by the choice of N_1 . On G , Theorem 5.14(b), applied to the realizable tagged practice, shows that the trimmed verifier of tag i accepts exactly $\text{inst}(\sigma_i)$.

It remains to see that $\hat{\sigma}_i$ computes this set. On G , $n_i > e_i$. The trimmed version space of tag i consists of the hypotheses $\text{inst}(\tau)$ (and $\emptyset$) with $|P_i \setminus \text{inst}(\tau)| \leq e_i$; $\emptyset$ is excluded because $n_i > e_i$. Each $\text{inst}(\text{lbg}(P_i \setminus E))$ with $|E| = e_i$ belongs to it. Conversely, if $|P_i \setminus \text{inst}(\tau)| \leq e_i$, then $P_i \setminus E \subseteq \text{inst}(\tau)$ for some E of size e_i , so $\tau \succeq \text{lbg}(P_i \setminus E)$ (Theorem 2.5). Hence the accepted set is $\bigcap_{|E|=e_i} \text{inst}(\text{lbg}(P_i \setminus E))$, a finite intersection of instance sets. It equals $\text{inst}(\sigma_i) \neq \emptyset$, and a finite intersection of instance sets with a common instance is the instance set of the most general common instance. So $\text{inst}(\hat{\sigma}_i) = \text{inst}(\sigma_i)$, hence $\hat{\sigma}_i \equiv \sigma_i$, and $\hat{P} = \Sigma^P$ up to renaming.

Step 1 (noise-free mode). Now $e_i = 0$ and $\hat{\sigma}_i = \text{lbg}(P_i)$. Let G be the event that every tag's samples satisfy the witness events (R) and (D) of Lemma 2.7 (for a ground tag: that it is sampled). By Theorem 5.7, with its convention $\rho_i = c_i = 1$ for ground rules, $\mathbb{P}(G^c) \leq \sum_i c_i e^{-N_1 \pi_i \rho_i} \leq c K e^{-N_1 \lambda} \leq \delta$, and on G , $\text{lbg}(P_i) \equiv \sigma_i$. The noisy budget must not be used here:

with $\bar{a}_i = 0$ it would still give $e_i = \lfloor \Delta N_1 \rfloor > 0$, and the event of Theorem 5.7 does not imply that trimmed lggs identify the rule (`noisefree.py`). In both modes G depends only on the first N_1 human samples; human answers to escalated queries are not part of the sample.

Step 2: audit. On G , $\hat{P} = \Sigma^P$, and Lemma 7.11 applies. It gives the two cases of (iii), the d -cleanness of A in (i), and the audit part of (ii), with the evaluation count from Lemma 7.9(b). For the completeness claim in (iii) under (SB_d) : every member of $\mathcal{C}_d$ is a singleton $\{\tau\}$, and $\tau \in F$ by Lemma 7.4(b), so $\text{Coll}_d = \emptyset$; and $\text{Coll}_d(B_0) \subseteq \text{Coll}_d$, because a minimal member of $\text{Conf}_d \cap 2^{B_0}$ is a minimal member of Conf_d (its proper subsets lie in 2^{B_0}).

Step 3: soundness. Before N_1 the tier accepts nothing. From N_1 on it accepts exactly R_A , and $A \subseteq \Sigma^* \cup F_{\text{res}}^{(d)}$ gives $R_A \subseteq R^* \cup R_{F_{\text{res}}^{(d)}} =: R'$. Since $R' \subseteq \text{Sound}(R')$, Lemma 3.1 with R' as target gives $\text{Cl}_{R_A}(B) \subseteq \text{Cl}_{R'}(B)$ for all B ; the version with trusted steps follows from monotonicity of Cl in the rule set (Lemma 2.3(a)). If $F_{\text{res}}^{(d)} = \emptyset$ then $R' = R^*$. The accepted set does not depend on the prover's queries, so the conclusions hold for every prover.

Step 4: sandbox. Proposition 7.13(b) gives the sandbox clause of (ii), and Proposition 7.13(a) shows that sandbox activity cannot affect (i).

Step 5: depth schedule. For $B \subseteq \Sigma^P$ let m_B be the least size of a refutation of B (∞ if none) and, for the prefer-unblocked rule, u_B the least size of a refutation with an unblocked descent. Under the default rule $\text{Ref}_d(B)$ is NONE for $d < m_B$ and the lexicographically least refutation of size m_B for $d \geq m_B$: it changes at most once. Under the prefer-unblocked rule it is NONE for $d < m_B$, the least smallest refutation for $m_B \leq d < u_B$ (all of them blocked), and the least smallest unblocked refutation for $d \geq u_B$: it changes at most twice. The audit at depth d is determined by the answers $\text{Ref}_d(B)$, $B \subseteq \Sigma^P$, and by $\bigcup \mathcal{C}_d(B_0)$, which is determined by $\text{Conf}_d = \{B : \text{Ref}_d(B) \neq \text{NONE}\}$. There are $2^{|\Sigma^P|}$ sets B , each with at most two change points, so the audit's output is piecewise constant in d with finitely many changes and constant for $d \geq d_1^*$. At each stage, Steps 2 and 3 apply with the current d ; truth maintenance ensures that cached lemmas and macros are derivations in the current R_A . For $d \geq d_1^*$ the cleanness of each $\Sigma^* \cup \{\tau\}$ no longer changes, so $F_{\text{res}}^{(d)} = F_{\text{res}}$, and the tier is sound relative to $R^* \cup R_{F_{\text{res}}}$. The example in the main text shows that A need not be monotone.

Uniformity (v). G is an event about the human samples. On G , the accepted set at each time is fixed ($\emptyset$ before N_1 , R_A after, at each depth of the schedule), whatever the prover and the red team do; so (i)–(iv) hold simultaneously for all of them. $\square$

Proof of Corollary 7.15. On G , every step the reasoner chains lies in $R_A \subseteq R^* \cup R_{F_{\text{res}}^{(d)}}$, in T_{trust} , or (for the extended reasoner) is a world-channel step. The first two kinds preserve truth in $\mathfrak{M}$ by hypothesis. A world-channel step has all its judgments in J_W and is accepted only if it is not W -falsified; if its premises are true in $\mathfrak{M}$, they are W -true, so its conclusion is not W -false, hence W -true, hence true in $\mathfrak{M}$. By induction along derivations, everything derived from $\mathfrak{M}$ -true premises is $\mathfrak{M}$ -true. $\square$

F.9 Proofs of Propositions 7.16 to 7.19

Proof of Proposition 7.16. Every subset of Σ_2 is clean at every depth, so Σ_1 and Σ_2 both satisfy (WS) (Lemma 7.4(d)), and the oracle answers are the same under both (Lemma 7.4(c)). The learner's view consists of these answers, the fixed prover's queries and its own coins, so it has the same law under both targets. Let the prover query some $q \in R_{\Sigma_2} \setminus \text{Sound}(R_{\Sigma_1})$ in every round. If the learner ever accepts q with probability above δ , it is not δ -sound for Σ_1 ; otherwise, under

Σ_2 , it fails with probability at least $1 - \delta$ to accept the genuine step q . In the example, $\forall I_1$ is not derivable from the conjunction rules, while $A \wedge B \triangleright B \wedge A$ is derivable by $\wedge E_2$, $\wedge E_1$ and $\wedge I$. $\square$

Proof of Proposition 7.17(a). The prover is fixed. The learner's inputs (human data, oracle answers, W , escalation answers) have the same law under every $\Sigma \in \mathcal{T}$: the data law and the practice are shared, oracle answers by Lemma 7.4(c), and escalation answers follow the shared practice. Coupling the learner's coins, the whole run up to and including the answer to q has the same law, so $p(h, q)$ does not depend on Σ . If $q \notin \text{Sound}(R_\Sigma \cup R_{F_{\text{res}}^{(d)}(\Sigma)})$ for some admissible Σ , the event “ h , then q accepted” is an unsound acceptance under Σ .

TTL passes. On G the audit runs on $\hat{P} = \Sigma^P$, identically in all admissible scenarios. Fix $\Sigma \in \mathcal{T}$, with valuations V_Σ from (WS). If a descent on π outputs s , then e_π^+ agrees with V_Σ , since the agreement argument of Lemma 7.9 needs only a total valuation that extends e and falsifies no trusted step. So s has V_Σ -true premises and a V_Σ -false conclusion, and $s \notin R_\Sigma$; hence no schema of Σ is removed by descent, and Σ is contained in the returned set (descent path) or in B_0 (fallback). Let $x \in A \setminus \Sigma$. On the descent path $\Sigma \cup \{x\} \subseteq A$ is d -clean by (F1). In the fallback, if $\Sigma \cup \{x\} \subseteq B_0$ were not d -clean it would contain some $C \in \mathcal{C}_d(B_0)$, with $C \not\subseteq \Sigma$ because Σ is clean, so $x \in C$, contradicting $x \in A$. Either way $x \in F_{\text{res}}^{(d)}(\Sigma)$, so $R_A \subseteq R_\Sigma \cup R_{F_{\text{res}}^{(d)}(\Sigma)}$. $\square$

Part (b) of Proposition 7.17 collects facts proved elsewhere: Proposition 7.6(b); Theorem 6.5(iii) for the coalition, whose members all contain a falsified fallacy instance under the stated priors (so the coalition-unanimous set does too), while a prior with more than half its mass on R^* forces $R^* \in S_t$ and hence soundness; and Proposition 3.9.

Proof of Proposition 7.18. Both worlds have the practice $\{\wedge I, \text{AC}\}$ with the same frequencies and instance laws, so a learner without refutation evidence has a view with the same law in both. Let the prover query an AC instance outside $\text{Sound}(R_{\wedge I})$ under W_1 , such as $q, p \rightarrow q \triangleright p$; it is genuine under W_2 . Accepting it with probability above δ is unsound under W_1 ; otherwise the learner is incomplete under W_2 . For the separation, under W_1 the instance $(\top, \perp \rightarrow \top \triangleright \perp)$ has true premises and a false conclusion, while under W_2 its premise $\perp \leftrightarrow \top$ is false. $\square$

Proof of Proposition 7.19. (a) By time t at most t refutations have been exhibited, each a finite derivation implicating finitely many instances; a fallacy with infinitely many invalid instances keeps one that is not refuted. (b) is Proposition 6.37. (c) With every finite set of steps a hypothesis, the intersection of the version space of data D is D itself (Section 5.4). $\square$

F.10 Proof of Theorem 7.20

Existence of s , and admissibility. If every τ -instance were in $\text{Sound}(R_{\{\sigma, \mu\}})$, the refutation of $\{\mu, \tau\}$ would transfer, by (F2), to a refutation of $\{\sigma, \mu\}$, which is clean at every depth. So some τ -instance s lies outside $\text{Sound}(R_{\{\sigma, \mu\}})$. The target $\{\sigma, \mu\}$ of scenario 1 is clean at every depth, hence satisfies (WS), and its fallacy τ has the instance s outside $\text{Sound}(R_{\{\sigma, \mu\}})$; the target $\{\sigma, \tau\}$ of scenario 2 is clean at every depth and has $F = \emptyset$. In scenario 2, s is a genuine instance.

The inequality. Let P_j be the law of one human sample (a tag and a step) in scenario j . On samples whose tag is not μ , $P_1 = (1 - \pi)P_2$ as measures: the frequencies are $(1 - \pi)q_\sigma$ and $(1 - \pi)q_\tau$ against q_σ and q_τ , and the per-tag laws coincide. Let E_t be the event that none of the first t samples has tag μ . Then $P_1^{\otimes t}$ restricted to E_t equals $(1 - \pi)^t P_2^{\otimes t}$, and $P_2^{\otimes t}(E_t) = 1$.

Human samples are i.i.d. and independent of everything generated before them. By (OE), every other input of round r is drawn from a kernel applied to the history, the same kernel in

both scenarios. By induction over rounds, the joint law of the samples $x = (x_1, \dots, x_t)$ and the other inputs y through time t is $P_j^{\otimes t}(dx) K_t(x, dy)$, with one kernel K_t for both j . Whether s is accepted at time t is a function of (x, y) ; let $\phi(x) = K_t(x, \{\text{accept } s \text{ at } t\}) \in [0, 1]$. Then

$$\mathbb{P}_1(\text{accept } s \text{ at } t) \geq \int_{E_t} \phi dP_1^{\otimes t} = (1 - \pi)^t \int_{E_t} \phi dP_2^{\otimes t} = (1 - \pi)^t \mathbb{P}_2(\text{accept } s \text{ at } t),$$

so $\mathbb{P}_2 \leq \delta(1 - \pi)^{-t}$ whenever $\mathbb{P}_1 \leq \delta$.

The bounds. $\mathbb{P}_2 \geq 1 - \delta'$ requires $(1 - \pi)^t \leq \delta/(1 - \delta')$, i.e. $t \ln \frac{1}{1 - \pi} \geq \ln \frac{1 - \delta'}{\delta}$, which is positive when $\delta + \delta' < 1$. Since $\frac{1}{1 - \pi} = 1 + \frac{\pi}{1 - \pi}$, $\ln \frac{1}{1 - \pi} \leq \frac{\pi}{1 - \pi}$, which gives the second inequality. It fails without $\delta + \delta' < 1$: for $\delta = \delta' = 0.6$ and $\pi = \frac{1}{2}$ the two sides are -0.585 and -0.405 .

Tightness. Let the learner accept τ -instances at time t iff no μ -tag occurs among the first t samples and an independent coin with bias $\min(1, \delta(1 - \pi)^{-t})$ comes up heads. It does not escalate, so it satisfies (OE). In scenario 1 it accepts s at time t with probability $(1 - \pi)^t \min(1, \delta(1 - \pi)^{-t}) \leq \delta$; in scenario 2, where μ never occurs, with probability $\min(1, \delta(1 - \pi)^{-t})$. Linear programming over all learners that depend only on the tag sequence, for $t \leq 6$, confirms that this is optimal in all 108 cases checked (computed; `two_point_bounds.py`).

Practice-following escalation. By (F2) with τ and μ exchanged, some μ -instance s_μ lies outside $\text{Sound}(R_{\{\sigma, \tau\}})$, hence outside the scenario-2 practice $R_{\{\sigma, \tau\}}$, while it lies in the scenario-1 practice. A learner that escalates s_μ in round 1 and accepts s in every later round iff the answer was “reject” accepts s with probability 0 in scenario 1 and 1 in scenario 2. So without (OE) the conclusion fails. $\square$

F.11 Proof of Theorem 7.21

(a) The practice $\Sigma_0 \cup \{\mu, \tau\}$ and its frequencies are the same in both scenarios, so the data and practice-following escalation answers have the same laws, and oracle answers are target-blind (Lemma 7.4(c)). Both targets are clean at every depth, hence satisfy (WS), and by (F2) each target’s fallacy has an instance outside the closure of the other’s calculus; in particular some μ -instance s lies outside $\text{Sound}(R_{\Sigma_0 \cup \{\tau\}})$. In scenario 2, μ is not residual, because $\Sigma_0 \cup \{\tau, \mu\}$ contains the conflict $\{\mu, \tau\}$; so $F_{\text{res}}^{(d)} = \emptyset$ there, and accepting s is unsound, which δ -soundness caps at probability δ . The fixed prover queries s , and the acceptance probability at any given time is the same in scenario 1, where s is genuine. Finally $\{\mu\}$ and $\{\tau\}$ are clean by (F1), so $\{\mu, \tau\} \in \mathcal{C}_d$ and $\mu \in \text{Coll}_d$ in scenario 1.

(b) Since $d \geq d_0$, a subset of Σ^P is d -clean iff it is clean at every depth.

Choice of M . Apply Lemma 7.8(b) with $U = B_0$ and $\mathcal{K} = \text{Conf}_d \cap 2^{B_0}$, whose minimal members are $\mathcal{C}_d(B_0)$: since $g \in \bigcup \mathcal{C}_d(B_0)$, some maximal d -clean $M \subseteq B_0$ omits g . M is clean at every depth, so it satisfies (WS) by Lemma 7.4(d) (e is well defined because (WS) holds for the true target).

Admissibility of M . Let $x \in \Sigma^P \setminus M$. If $x \in B_0 \setminus M$, then $M \cup \{x\}$ is not d -clean by maximality; if every x -instance were in $\text{Sound}(R_M)$, (F2) would turn a refutation of $M \cup \{x\}$ into one of M , which is impossible. If $x \in \Sigma^P \setminus B_0$, then x was removed by a descent on some refutation π , with output $s \in \text{inst}(x)$. The valuation V_M extends e and falsifies no trusted step, so e_π^+ agrees with it (agreement in Lemma 7.9); hence s has V_M -true premises and a V_M -false conclusion, and since V_M preserves truth along R_M -derivations, $s \notin \text{Sound}(R_M)$. Either way x has an instance outside $\text{Sound}(R_M)$.

$F_{\text{res}}^{(d)}(M) = \emptyset$. For $x \in B_0 \setminus M$ this is maximality. For $x \notin B_0$, $\{x\}$ is a d -conflict by Lemma 7.9(d), applied in the true scenario, so $M \cup \{x\}$ is not d -clean by (F1).

Witness and conclusion. $g \in B_0 \setminus M$, so the first case gives $s_g \in \text{inst}(g) \setminus \text{Sound}(R_M)$. All admissible scenarios share the data law and the oracle answers, the prover is fixed, and escalation answers follow the shared practice; so the probability that s_g is accepted at a given time is the same under every admissible target. Under M , δ -soundness relative to $R_M \cup R_{F_{\text{res}}^{(d)}(M)} = R_M$ bounds it by δ . For the schema-level statement, a schema x outside $A = B_0 \setminus \bigcup \mathcal{C}_d(B_0)$ lies in $\bigcup \mathcal{C}_d(B_0)$ (handled above) or outside B_0 , and then its descent output is a witness outside $\text{Sound}(R_M)$ for any maximal clean $M \subseteq B_0$, which is admissible as shown. A policy that includes x in A' accepts that witness when the prover queries it, so includes x with probability at most δ . On the descent path take $M = A$: at stable depth A is clean at every depth, every schema outside A has a descent witness outside $\text{Sound}(R_A)$ by the second case, and $F_{\text{res}}^{(d)}(A) = \emptyset$ by Lemma 7.9(d); so A is admissible and the same argument applies.

(c) The claims were computed by `thm56_finite_d.py` (well-founded derivations, size counting distinct judgments) and can be checked by hand. On A_1 , AC infers p from q and $p \rightarrow q$, and MP infers $\perp$ from p and $p \rightarrow \perp$; the distinct judgments $q, p \rightarrow q, p, p \rightarrow \perp, \perp$ have $1+3+1+3+1 = 9$ symbols. On A_2 , AC infers $\perp$ from φ and $\perp \rightarrow \varphi$, with $5+7+1 = 13$ symbols. The closure of A_1 under AC alone is $\{q, p \rightarrow q, p \rightarrow \perp, p\}$, so $\{\text{AC}\}$ has no refutation on A_1 ; and $\{\text{MP}\}$ is classically sound while both contexts are satisfied by $q = 1, p = 0$. So for $9 \leq d \leq 12$, $\mathcal{C}_d = \{\{\text{MP}, \text{AC}\}\}$. With no world and no trusted steps e is defined only on the assertions and on $\perp$; at $\perp$ the producing MP step has the unevaluated premise p and no premise of value 0, so the descent is blocked, and the fallback returns $\emptyset$. The subsets of the practice that are clean at every depth are $\emptyset$ and $\{\text{MP}\}$; both are admissible, since AC is classically invalid and MP has instances outside $\text{Sound}(\emptyset)$. Under $\emptyset$, MP is residual at every depth, so asserting R_{MP} is sound under both. For $d \geq 13$, $\{\text{AC}\}$ is a conflict, so $\{\text{MP}, \text{AC}\}$ is no longer minimal; with the default oracle the blocked refutation of size 9 triggers the fallback with $\mathcal{C}_d = \{\{\text{AC}\}\}$, and with the prefer-unblocked oracle the refutation on A_2 has an unblocked descent that removes AC; either way $A = \{\text{MP}\}$. $\square$

F.12 Proof of Theorem 7.22

Construction. Let Σ_0 be a finite elementary formal system (Post, 1943; Smullyan, 1961) with judgments start_e that simulates the universal machine on input e from start_e and contains the rule $\text{halt} \triangleright \perp$, as in Theorem 6.33(a). Σ_0 has no rules without premises, so without a start judgment nothing is derivable; in particular Σ_0 is clean on $[\emptyset : \emptyset]$, and the step $\triangleright \text{start}_e$ is not in $\text{Sound}(R_{\Sigma_0})$. Let $\tau_e = (\triangleright \text{start}_e)$ and $\Sigma_e^P = \Sigma_0 \cup \{\tau_e\}$, with fixed computable frequencies and instance laws. If $\varphi_e(e)$ diverges, Σ_e^P is clean and is the target, with $F = \emptyset$. If $\varphi_e(e)$ converges, $\Sigma_0 \cup \{\tau_e\}$ derives $\perp$ from $\emptyset$, the target is Σ_0 , $F = \{\tau_e\}$ and $\tau_e \notin F_{\text{res}}$; accepting $\triangleright \text{start}_e$ is then a step outside $R^* \cup R_{F_{\text{res}}} = R_{\Sigma_0}$.

Uniformity. For each e only one case occurs, so this is not a two-scenario argument; what is used is that the data laws and the oracle answers Ref_d (for every d , by brute force) are computable uniformly in e .

Reduction. Let $S = \{e : \exists t \mathbb{P}(\text{the learner accepts } \triangleright \text{start}_e \text{ by time } t) > \frac{1}{2}\}$. If $\varphi_e(e)$ diverges, the prover queries the genuine step $\triangleright \text{start}_e$ in every round, and (b) and continuity of measure give $e \in S$. If $\varphi_e(e)$ converges, (a) gives $\mathbb{P}(\text{ever accepts}) \leq \delta < \frac{1}{2}$, so $e \notin S$. Hence S is the complement of the halting set $\{e : \varphi_e(e) \text{ converges}\}$. But for a computable learner, with computable data laws, oracle and prover, the probability of acceptance by time t is lower semicomputable uniformly in (e, t) , so S is Σ_1 . This also holds if the learner escalates, because escalation answers follow the practice Σ_e^P , which is the same in both cases and decidable by matching. The complement of the

halting set is not Σ_1 : a contradiction. $\square$

F.13 Proofs for classical propositional logic

Proof of Proposition 7.26. (a) $\text{inst}(\tau)$ is substitution-closed and contains an unsound step, and R^* is complete for CPC, so by Proposition 3.5 $\text{Cl}_{R^* \cup \text{inst}(\tau)}(\emptyset)$ contains $\perp$. A derivation of $\perp$ from $\emptyset$ is a refutation of $\Sigma^* \cup \{\tau\}$ relative to $[\emptyset : \emptyset]$, so d_τ is finite. For $d \geq d_\tau$, $\Sigma^* \cup \{\tau\} \in \text{Conf}_d$ contains some $C' \in \mathcal{C}_d$, and $\tau \in C'$ by Lemma 7.4(a). Hence $F_{\text{res}}^{(d)} = \emptyset$ for $d \geq \max_\tau d_\tau$, and by Theorem 7.14(iii) and Corollary 7.15 TTL asserts only instances of Σ^* , which are classically valid.

(b) $\{\text{AC}, K\}$ is a conflict. $K = A \rightarrow (B \rightarrow A)$ gives $\perp \rightarrow (p \rightarrow \perp)$ (with $A = \perp$, $B = p$) and $(p \rightarrow \perp) \rightarrow (\perp \rightarrow (p \rightarrow \perp))$ (with $A = p \rightarrow \perp$, $B = \perp$). AC, B , $A \rightarrow B \triangleright A$, applied to these two gives $p \rightarrow \perp$; applied to $p \rightarrow \perp$ and $\perp \rightarrow (p \rightarrow \perp)$ it gives $\perp$. This is a refutation relative to $[\emptyset : \emptyset]$. It is minimal at every depth beyond its size. $\{K\} \subseteq \Sigma^*$ is clean (Lemma 7.4(a)). $\{\text{AC}\}$ is clean at every depth on these positions: from $\emptyset$ it derives nothing, since AC has premises, and the closure of A_1 under AC alone is $\{q, p \rightarrow q, p \rightarrow \perp, p\}$. Likewise, with A_1 designated, AC followed by MP derives $\perp$, and $\{\text{MP}\}$ is clean, so $\{\text{AC}, \text{MP}\}$ is a minimal conflict. Every descent is blocked. With no world and no trusted steps, e is defined only on the assertions of the position and on $\perp$. An output step would need premises among the assertions (or none) and conclusion $\perp$. No genuine step is such a step, because the positions are satisfiable and Σ^* is sound; and no AC step is, because AC with premises in A_1 yields only p , and AC needs premises. So the fallback runs at $B_0 = \Sigma^P$, K (and with A_1 also MP) lies in Coll_d for all large d , and Theorem 7.21(b) applies.

(c) τ derives the denied φ' from the asserted Π' in one step, a refutation of size at most $|\tau| \leq d$ (a closed $\top/\perp$ instance has at most the symbols of τ). So $\{\tau\} \in \text{Conf}_d$ for each $\tau \in F$, every conflict contains a fallacy and hence a singleton conflict, and the minimal conflicts are exactly these singletons. Then $\text{Coll}_d = \emptyset$, $F_{\text{res}}^{(d)} = \emptyset$, and Lemma 7.11 gives $A = \Sigma^*$ on the descent path, and also in the fallback, where $\Sigma^* \setminus \text{Coll}_d(B_0) \subseteq A \subseteq \Sigma^*$ and $\text{Coll}_d(B_0) \subseteq \text{Coll}_d = \emptyset$. $\square$

The bounded computations behind the Hilbert illustration (`hilbert_blame.py`, formulas with at most six leaves over $\{p, q, \perp\}$) agree: on $\{[\emptyset : \emptyset], [A_1 : \emptyset]\}$ the minimal conflicts are $\{\text{AC}, K\}$ and $\{\text{AC}, \text{MP}\}$ with diagnoses $\{\text{AC}\}$ and $\{K, \text{MP}\}$; with the bilateral position $[A_1 : p]$ the only minimal conflict is $\{\text{AC}\}$; and the closed instance $(\perp \rightarrow \perp, \perp \rightarrow (\perp \rightarrow \perp) \triangleright \perp)$ of AC is falsified. The search for a matrix witnessing that $\{S, \text{DN}, \text{AC}\}$ is clean at every depth (validating S and DN, with AC designation-preserving, $\perp$ undesignated and A_1 satisfiable) found none with at most four values (`matrix_witness.py`, `matrix4.py`).

F.14 Proofs for complete decidable theories

We take Σ^* to be a standard sequent calculus: besides being sound and complete for T -consequence, it has $\rightarrow$ -introduction, $\forall$ -introduction, $\neg$ -elimination (deriving $\vdash \perp$ from $\vdash \psi$ and $\vdash \neg\psi$), ex falso and weakening, as derived rules at least.

Proof of Lemma 7.27. (a) Let $\Pi \triangleright j$ be a step. All premises W -true. Each premise $\Gamma \vdash \varphi$ has a universal closure true in $\mathfrak{M}$; since T is complete, T proves it, and Σ^* derives the premise from $\emptyset$. Hence $\text{Cl}_{R^*}(\Pi) = \text{Cl}_{R^*}(\emptyset)$, which by soundness and completeness of Σ^* and completeness of T is the set of W -true sequents; so $j \in \text{Cl}_{R^*}(\Pi)$ iff $W(j) = 1$. Some premise $\Gamma \vdash \varphi$ W -false. By $\rightarrow$ -I and $\forall$ -I, Σ^* derives from it the sentence $\vdash \forall \bar{x}(\gamma_1 \rightarrow \dots \rightarrow \gamma_n \rightarrow \varphi)$, which is false in $\mathfrak{M}$. T

refutes it, so Σ^* derives its negation; then $\neg E$, ex falso and weakening derive every sequent, and $\text{Cl}_{R^*}(\Pi) = J$. So $\Pi \triangleright j \in \text{Sound}(R^*)$ iff some premise is W -false or j is W -true, i.e. iff the step preserves W -truth.

(b) A schema is a fallacy iff it has an instance outside $\text{Sound}(R^*)$, iff by (a) it has an instance with W -true premises and a W -false conclusion. The smallest such instance is a one-step refutation of $\{\tau\} \subseteq \Sigma^* \cup \{\tau\}$ relative to $[\emptyset : \emptyset]$, so $d_\tau \leq f_\tau$. $\square$

Proof of Corollary 7.28. Under the realizability hypothesis Theorem 7.14 applies; (WS) holds with $V = \text{truth}$ in $\mathfrak{M}$ (of universal closures), the positions being truthful. The oracle is exact: if B is not d -clean it contains a fallacy (Lemma 7.4(b)), whose smallest falsified instance has size $f_\tau \leq d$ and is found; if B is d -clean the search cannot find a falsified instance of size at most d , which would be a d -refutation. Each $\{\tau\}$, $\tau \in F$, is then a conflict, and the proof of Corollary 7.25 applies verbatim; truth-soundness follows from Corollary 7.15, since R^* preserves truth in $\mathfrak{M}$ by Lemma 7.27(a). $\square$

The restricted oracle. Suppose W evaluates only quantifier-free sentences with rational constants, and first-order logic is trusted. The fallacy $\tau = (\triangleright \neg \exists x (x \cdot x = 1 + 1))$ is a step without premises whose conclusion is not quantifier-free, so it has no instance that W can evaluate, and Lemma 7.27(b) fails for this oracle. The position $[\exists x (x \cdot x = 1 + 1) : \emptyset]$ is truthful in $\mathbb{R}$. In the refutation “ $\vdash \neg \exists x (x \cdot x = 1 + 1)$ (by τ); $\vdash \perp$ (by trusted $\neg E$, with the asserted $\exists x (x \cdot x = 1 + 1)$)”, the backward rule through $\neg E$ gives $\neg \exists x (\dots)$ the value 0, and the descent outputs τ ’s step: $\{\tau\}$ is a singleton conflict. Without trusted logic the conflict is the bag $\{\tau, \neg E\}$; with the bilateral position $[\emptyset : \vdash \neg \exists x (x \cdot x = 1 + 1)]$ it is $\{\tau\}$, by a one-step refutation. An earlier version designated $[x \cdot x = 1 + 1 : \emptyset]$, which under the universal-closure semantics asserts the false $\forall x (x \cdot x = 2)$ and so violates (WS).

F.15 Proof of Theorem 7.29

(a) In the refutation, τ ’s step has no premises and $\forall E$ is trusted. The conclusion $\theta(\bar{n})$ is a false Δ_0 sentence, so it has value 0; the backward rule through $\forall E$ (whose only premise is $\forall \bar{x} \theta$; for k variables, through each of the k instantiation steps) gives $\forall \bar{x} \theta$ the value 0; and the descent outputs τ ’s step, whose (empty) premise set has value 1 vacuously. The descent is unblocked, so by Lemma 7.9(c) only schemas having $\vdash \forall \bar{x} \theta$ as an instance are removed, and these are fallacies; no axiom of PA is one, since its instances are true. With the prefer-unblocked oracle, as long as τ remains in B there is an unblocked refutation of size at most d , so the oracle returns an unblocked one, every call removes at least one fallacy and no genuine schema, and the loop does not fall back before τ is removed. *Size.* For one variable, $|\forall x \theta| = |\theta| + 2$, and $\theta(\underline{n})$ replaces each of the m occurrences of x (one symbol) by the numeral $S^n 0$ ($n + 1$ symbols), so $|\theta(\underline{n})| = |\theta| + mn$. The total is $2|\theta| + mn + 2 \leq 2|\theta| + m(n + 1) + O(1)$. For k variables the refutation has $k + 1$ judgments, each of size at most $|\theta| + m(n + 1) + O(k)$. For $\theta(x) = x < S^5 0$ ($|\theta| = 8$, $m = 1$, $n = 5$) the sizes are 10 and 13, total 23. Under the default tie-breaking rule a smaller *blocked* refutation through learned axioms could trigger the fallback, and those axioms could become collateral; and if the refutation ran through Q’s axioms by coherence instead of through W , the bag would contain them. So computation is subsumed by coherence as a source of refutations (Lemma 6.19), but not as a source of blame.

(b) $\text{PA} \not\vdash \text{Con}(\text{PA})$ by Gödel’s second incompleteness theorem (Gödel, 1931), so $\text{PA} + \neg \text{Con}(\text{PA})$ is consistent. A refutation of $\Sigma^* \cup \{\tau\}$ relative to a position with Δ_0 assertions and denials would derive, in first-order logic, from $\text{PA} + \neg \text{Con}(\text{PA})$ and true Δ_0 leaves, either $\perp$ or a false Δ_0

sentence. $\mathbf{Q} \subseteq \mathbf{PA}$ proves every true Δ_0 sentence and refutes every false one (Lemma 6.19), so in either case $\mathbf{PA} + \neg \text{Con}(\mathbf{PA})$ would be inconsistent. Hence $\Sigma^* \cup \{\tau\}$ is clean at every depth, $\tau \in F_{\text{res}}$, and by Theorem 7.14(iii) $\tau \in A$ at every depth.

(c) A false sentence with Σ_1 normal form $\exists \bar{x} \theta$ has no witness, so W never verifies one and the instance is never asserted. A true one has a witness, which enumeration finds after finitely many W -evaluations, with no computable bound on its size. The disguise $\forall y \exists p (\text{Prf}_{\mathbf{PA}}(p, \ulcorner \perp \urcorner) \vee y \neq y)$ is logically equivalent to $\exists p \text{Prf}_{\mathbf{PA}}(p, \ulcorner \perp \urcorner)$, i.e. to $\neg \text{Con}(\mathbf{PA})$, because $y \neq y$ is refutable; its prenex normal form is Π_2 , so the policy does not apply; and $\mathbf{PA}$ plus it is consistent and Δ_0 -sound, so by the argument of (b) it is residual.

(d) *Cleanness.* Suppose $\Sigma^* \supseteq \mathbf{Q}$ and $\Sigma^* \cup \{\tau\} \cup A \cup \neg D$ is consistent for every designated $[A : D]$. A refutation relative to $[A : D]$ would derive, from $\Sigma^* \cup \{\tau\} \cup A$ and true Δ_0 leaves, either $\perp$, or some $\delta \in D$, or a false Δ_0 sentence. $\mathbf{Q}$ proves the true Δ_0 leaves and refutes false Δ_0 sentences, so in each case $\Sigma^* \cup \{\tau\} \cup A \cup \neg D$ would be inconsistent. *Impossibility.* Let (φ_e) enumerate the Σ_2 sentences, let $\Sigma^* = \mathbf{PA}$, and consider the practices $\mathbf{PA} \cup \{\vdash \varphi_e\}$, with the practice data given as a computable text. Suppose a computable learner, using the data, Ref_d at every depth and the Δ_0 oracle, eventually and permanently asserts $\vdash \varphi_e$ if φ_e is true, and eventually and permanently withholds it if φ_e is false and $\mathbf{PA} \cup \{\vdash \varphi_e\}$ is clean at every depth. Define $g(e, s)$: if a refutation of $\mathbf{PA} \cup \{\vdash \varphi_e\}$ has been found by stage s of a systematic search (computable, since each Ref_d is), output “false”; otherwise output the learner’s current verdict. A true φ_e makes the practice sound, hence clean (Lemma 7.4(a)), so the search never succeeds and g converges to “true”. A false φ_e either has a refutation, which the search eventually finds, or is clean, and then the learner converges to “false”. So $\lim_s g(e, s)$ exists and is the truth value of φ_e , which makes Σ_2 -truth Δ_2 by Shoenfield’s limit lemma (Shoenfield, 1959). But the set of true Σ_2 sentences is Σ_2 -complete and therefore not Π_2 , let alone Δ_2 , by Post’s hierarchy theorem (Theorem 6.22(e)). *Randomized learners (proof sketch).* If the learner is randomized and correct in the limit with probability above $\frac{1}{2}$, the sets $\{e : \mathbb{P}(\text{eventually and permanently withholds } \vdash \varphi_e) > \frac{1}{2}\}$ and the corresponding set for assertion are Σ_2 , and the same combination would make Σ_2 -truth Δ_2 .

(e) *Tag count.* For every tag, $\Delta \leq (\rho_i \beta_i - \bar{\alpha}_i)/2 \leq \rho_i \beta_i/2 \leq \pi_i/2$, since $\rho_i \leq 1$ and $\beta_i \leq \pi_i$. As $\sum_i \pi_i = 1$, there are at most $1/(2\Delta)$ tags. *Identification.* The axioms of $\mathbf{Q}$ and each $\text{Con}(T_j)$ are ground, and fall under the convention $\rho_i = c_i = 1$; the Sub-encoded induction rule is a first-order pattern whose lgg on generic instances recovers it (`ind_sub_lgg.py`). So Theorem 5.15 applies and Theorem 7.14 gives $\hat{P} = \Sigma^P$. All practice schemas are genuine (every $\text{Con}(T_j)$ is true), so $F = \emptyset$ and the audit, which never removes genuine schemas, returns $A = \Sigma^P$. Instances of the encoded rule with a false Sub premise are harmless: Sub judgments are concluded by no practice schema, so they enter derivations only as leaves, which must be W -true. *Without the induction encoding,* the lgg of induction instances is over-general and has false instances (`ind_lgg.py`); for the computed one, $\forall x(0 + 0 = x \rightarrow 0 + S0 = Sx)$ is derivable by trusted equality logic from the true Δ_0 leaf $0 + S0 = S(0 + 0)$, so backward propagation through the trusted $\rightarrow E$ and $\forall E$ gives an unblocked descent, and the audit removes the identified schema with singleton blame.

The floor bound. Accepting $\vdash \text{Con}(T_{k-1})$ is unsound under T_{k-1} : by Gödel’s second theorem $\text{Con}(T_{k-1})$ is not a theorem of the consistent r.e. theory T_{k-1} , so the step is not derivable in the target calculus with trusted logic, even with all true Sub judgments as premises, and it lies outside $R^* \cup R_{F_{\text{res}}} = R_{T_{k-1}}$. (It is true, so this is soundness for the target calculus, Theorem 7.14(i), not truth-soundness.) Let E_t be the event that the tag $\text{Con}(T_{k-1})$ does not occur among the first t samples. As in Appendix F.10, $P_k^{\otimes t}$ restricted to E_t is $(1 - \pi)^t P_{k-1}^{\otimes t}$, $P_{k-1}^{\otimes t}(E_t) = 1$, and under

(OE) the acceptance probability given the samples is the same function ϕ in both scenarios. So

$$\mathbb{P}_k(\text{accept}) = \int_{E_t} \phi dP_k^{\otimes t} + \int_{E_t^c} \phi dP_k^{\otimes t} \leq (1-\pi)^t \mathbb{P}_{k-1}(\text{accept}) + 1 - (1-\pi)^t \leq 1 - (1-\delta)(1-\pi)^t.$$

Requiring $\mathbb{P}_k \geq 1 - \delta'$ gives $(1-\pi)^t \leq \delta'/(1-\delta)$, i.e. $t \geq \ln \frac{1-\delta}{\delta'}/\ln \frac{1}{1-\pi}$, positive when $\delta + \delta' < 1$. *Tightness:* the learner that, by one coin at time 0, accepts the step from the start with probability δ and otherwise accepts it once its tag has occurred is uniformly δ -sound under T_{k-1} , where the tag never occurs, and under T_k accepts at time t with probability $\delta + (1-\delta)(1-(1-\pi)^t) = 1 - (1-\delta)(1-\pi)^t$. An earlier version transplanted the bound of Theorem 7.20 here; it is false: for $\pi = \frac{1}{2}$, $\delta = 10^{-6}$ and $\delta' = 0.1$ it demands $t \geq 19.8$, while the learner “accept iff the tag has occurred” is 0-sound under T_{k-1} and accepts under T_k at $t = 4$ with probability 0.9375 (computed; `two_point_bounds.py`).

(f) With $[\text{Con}(\text{PA}) : \emptyset]$ designated, $\tau = (\triangleright \neg \text{Con}(\text{PA}))$ and the trusted $\neg E$ derive $\perp$ from the asserted $\text{Con}(\text{PA})$. The backward rule gives $\neg \text{Con}(\text{PA})$ the value 0, and the descent outputs τ 's step, so $\{\tau\}$ is a singleton conflict (Theorem 6.21). $\square$

G Proofs for Section 8

Several results of Section 8 are proved in full in the main text: Propositions 8.4, 8.5 and 8.16, Theorem 8.9, and Corollary 8.10, Lemma 8.8(i)–(ii), and Theorem 8.15 outside its idealized case. The language-relativity construction is also given in full there. This appendix proves the remaining results, and states and proves the two lemmas on which the kink rests. The proofs follow T5 in the form repaired after verification, and the repairs are noted where they occur.

Conventions:

- Logarithms are base 2 unless written $\ln$.
- κ denotes a machine constant. It absorbs $O(1)$ terms and may grow from one result to the next, but it is fixed before d , n or m .
- p_h is a shortest program for h , and $K(p_h) = K(h) + O(1)$ (Li and Vitányi, 2008).
- Prefix complexities of concatenated self-delimiting descriptions add, up to $O(1)$.

G.1 Proof of Lemma 8.2

Both minimizers exist. By Kraft, $\{h : \ell(h) \leq L\}$ has at most 2^L elements. Fix h_0 . Losses lie in $[0, B]$, so $\mathcal{R}(h_0), \hat{\mathcal{R}}_N(h_0) \leq B$. Any h with $\ell(h) > \ell(h_0) + B/c$ therefore has

$$J_c(h), \hat{J}_c(h) \geq c\ell(h) > c\ell(h_0) + B \geq J_c(h_0), \hat{J}_c(h_0).$$

So both infima are minima over a finite set. In particular a population minimizer h_c exists, and both $\ell(h_c)$ and $\ell(\hat{h})$ are at most $\ell(h_0) + B/c$.

Uniform deviation. For a fixed h , $-\log h(y_i \mid x_i)$ are i.i.d. in $[0, B]$. By Hoeffding's inequality,

$$\mathbb{P}[|\hat{\mathcal{R}}_N(h) - \mathcal{R}(h)| \geq \varepsilon] \leq 2e^{-2N\varepsilon^2/B^2}.$$

Setting the right side equal to $\delta 2^{-\ell(h)}$ gives $\varepsilon = \varepsilon_N(h) := B\sqrt{(\ell(h) \ln 2 + \ln(2/\delta))/(2N)}$. A union bound over h , using $\sum_h 2^{-\ell(h)} \leq 1$, shows that with probability at least $1 - \delta$ all deviations are at most $\varepsilon_N(h)$ simultaneously.

Excess risk. On this event,

$$J_c(\hat{h}) \leq \hat{J}_c(\hat{h}) + \varepsilon_N(\hat{h}) \leq \hat{J}_c(h_c) + \varepsilon_N(\hat{h}) \leq \Phi(c) + \varepsilon_N(h_c) + \varepsilon_N(\hat{h}).$$

Both ε_N terms are at most $B\sqrt{((\ell(h_0) + B/c) \ln 2 + \ln(2/\delta))/(2N)}$.

Uniqueness. If $J_c(h) \geq \Phi(c) + \gamma$ for every $h \neq h_c$, then a bound below γ forces $\hat{h} = h_c$. $\square$

G.2 Proof of Theorem 8.3

The minimax value in (i). Let the distinct inputs be x , with multiplicities n_x . Write $L(h, f) = \sum_x n_x [-\log h(f(x) | x)]$. For a distribution Q on $\mathcal{C}$,

$$\mathbb{E}_{f \sim Q} L(h, f) = \sum_x n_x [H(Q_x) + \text{KL}(Q_x \| h(\cdot | x))] \geq \sum_x n_x H(Q_x) = \sum_i H(Q_{x_i}),$$

with equality iff $h(\cdot | x) = Q_x$ for every x . Hence $\min_h \max_f L \geq \max_Q \sum_i H(Q_{x_i})$, and the maximum over the compact simplex is attained, since the objective is continuous.

For the reverse inequality:

- *Restricting h .* If $h(y | x) = 0$ for some $y \in \mathcal{C}(x)$, then $\max_f L(h, f) = \infty$. Mass outside $\mathcal{C}(x)$ only hurts. The function $h \mapsto \max_f L(h, f)$ is lower semicontinuous, with compact sublevel sets on the product of simplices. So a minimizer h° exists, and it puts mass at least some $\tau_0 > 0$ on each $y \in \mathcal{C}(x)$.
- *Sion's theorem.* For $\tau \leq \tau_0$, let H_τ be the convex set of product hypotheses with $h(y | x) \geq \tau$ for $y \in \mathcal{C}(x)$. On $H_\tau \times \Delta(\mathcal{C})$ the function $(h, Q) \mapsto \mathbb{E}_Q L(h, f)$ is continuous, convex in h and linear in Q , and $\Delta(\mathcal{C})$ is compact. By Sion's minimax theorem (Sion, 1958),

$$\min_h \max_f L = \inf_{h \in H_\tau} \max_Q \mathbb{E}_Q L = \max_Q \inf_{h \in H_\tau} \mathbb{E}_Q L.$$

- *Letting $\tau \rightarrow 0$.* For each Q , the hypothesis $h(\cdot | x) = (1 - \tau')Q_x + \tau' \text{Unif}(\mathcal{C}(x))$ with $\tau' = \tau |Y|$ lies in H_τ . Since $\text{KL}(Q_x \| (1 - \tau')Q_x + \tau'U) \leq -\log(1 - \tau')$, we get $\inf_{H_\tau} \mathbb{E}_Q L \leq \sum_i H(Q_{x_i}) - N \log(1 - \tau')$, uniformly in Q . Letting $\tau \rightarrow 0$ gives $\min_h \max_f L \leq \max_Q \sum_i H(Q_{x_i})$.

When the bound $\sum_i \log |\mathcal{C}(x_i)|$ is attained. We have $H(Q_x) \leq \log |\mathcal{C}(x)|$, with equality iff Q_x is uniform on $\mathcal{C}(x)$. Since the maximum is attained, the value equals $\sum_i \log |\mathcal{C}(x_i)|$ iff some Q has uniform marginals on every $\mathcal{C}(x_i)$.

The group condition. Suppose G acts on $\mathcal{C}$ with $(g \cdot f)(x) = \sigma_{g,x}(f(x))$, transitively on each $\mathcal{C}(x)$. Let $Q = \text{Unif}(\mathcal{C})$, which is G -invariant. Then

$$\begin{aligned} Q_x(\sigma_{g,x}y) &= Q\{f : f(x) = \sigma_{g,x}y\} = Q\{g \cdot f' : \sigma_{g,x}(f'(x)) = \sigma_{g,x}(y)\} \\ &= Q\{g \cdot f' : f'(x) = y\} = Q\{f' : f'(x) = y\} = Q_x(y), \end{aligned}$$

where the second equality substitutes $f = g \cdot f'$ and the fourth uses the G -invariance of Q . By transitivity Q_x is uniform on $\mathcal{C}(x)$. For parities, $g_b \cdot f_a = f_{a \oplus b}$ and $\sigma_{g_b,x}(y) = y \oplus f_b(x)$. This is transitive on $\mathcal{C}(x) = \{0, 1\}$ for $x \neq 0$, and trivially so on $\mathcal{C}(0) = \{0\}$. A group acting transitively on $\mathcal{C}$ alone does not suffice: in the example of the main text, $\text{Sym}(\mathcal{C})$ acts transitively on $\mathcal{C}$, but not compatibly with the $\mathcal{C}(x)$.

(ii) The uniform distribution over the distinct labelings $\mathcal{L} = \{f(x_{1:N})\}$ pays $\log |\mathcal{L}|$ against every f . Conversely, the labelings are disjoint events, so every P gives some labeling probability at most $1/|\mathcal{L}|$ (Shtarkov, 1987).

(iii) We have $\xi_{\text{fun}}(y \mid x) \geq w_q q(y \mid x)$ and $\xi_{\text{seq}} \geq w_q \prod_i q(y_i \mid x_i)$; take logarithms. For tightness, take the models $\text{const } 0$ and $\text{const } 1$ with $w = \frac{1}{2}$, on N ones. Then $\xi_{\text{fun}}(1 \mid x) = \frac{1}{2}$ at every input, so it pays N bits. And $\xi_{\text{seq}}(1^N) = \frac{1}{2}$, so it pays one bit. The chain-rule conditionals of ξ_{seq} depend on past labels; that dependence is the shared randomness. $\square$

G.3 The kink

The setting is that of Section 8.3. For a product h put $s_h(x) = h(0 \mid x) - h(1 \mid x)$ and $\chi_{a'}(x) = (-1)^{\langle a', x \rangle}$, so that $\hat{s}_h(a') = \mathbb{E}_x[s_h \chi_{a'}]$. Put $\sigma_h := s_h \chi_a \in [-1, 1]$, so that $\hat{s}_h(a) = \mathbb{E}_x \sigma_h$.

Lemma G.1 (List decoding; standard). ^[T5 Lemma 2.5a] For $t \in \mathbb{N}$, $|\{a' : \hat{s}_h(a') \geq 2^{-t}\}| \leq 2^{2t}$. Hence, if $\hat{s}_h(a) \geq 2^{-t}$, then

$$K(a \mid p_h) \leq 2t + 2 \log_2(t+1) + 2 \log_2 d + O(1).$$

Proof. Parseval gives $\sum_{a'} \hat{s}_h(a')^2 = \mathbb{E}_x[s_h^2] \leq 1$, so at most 2^{2t} coefficients are at least 2^{-t} . Given p_h , compute every $\hat{s}_h(a')$ exactly (finite rational sums), and list those that are at least 2^{-t} . Then specify d and t self-delimitingly, followed by the $2t$ -bit index of a in the list. The $2 \log_2 d$ term is needed because a program for h need not determine d . For real t , apply the lemma to $\lceil t \rceil$. $\square$

The list-size bound is the standard Parseval (Johnson-type) bound for the Hadamard code, as used by Goldreich and Levin (1989) and Kushilevitz and Mansour (1993). Only its combination with the coding theorem is used here.

Lemma G.2 (Risk versus correlation). ^[T5 Lemma 2.5b] For every product h ,

$$\mathcal{R}(h) \geq -\log_2 \left(\frac{1 + \hat{s}_h(a)}{2} + p \right).$$

Proof. By Jensen, $\mathcal{R}(h) \geq -\log_2 \mathbb{E}_x h(y^*(x) \mid x)$. Since $h(0 \mid x) + h(1 \mid x) \leq 1$, we have $h(y \mid x) \leq \frac{1+s_h(x)(-1)^y}{2}$. So off E , $h(y^* \mid x) = h(f_a(x) \mid x) \leq \frac{1+\sigma_h(x)}{2}$; on E it is at most 1. As $\frac{1+\sigma_h}{2} \geq 0$, we get $\mathbb{E}_x h(y^* \mid x) \leq \frac{1+\hat{s}_h(a)}{2} + p$. $\square$

As a function of $\hat{s}_h(a)$ alone the bound is tight: an h that is right on E and has $\hat{s}_h(a) = 0$ attains $\mathbb{E}_x h(y^* \mid x) = \frac{1}{2} + p$. So removing the additive p requires that E be random relative to h , which the next lemma, added after verification, exploits.

Lemma G.3 (Randomness of E removes the $2p$ slack). ^[T5 Lemma 2.5c, added after verification] For every product h ,

$$\mathcal{R}(h) \geq 1 - \log_2 (1 + (1 - 2p) \hat{s}_h(a) + \varepsilon_h), \quad \varepsilon_h := \sqrt{32 \ln 2 \cdot p (\delta_E^+ + K(h) + \kappa_n) / n},$$

where $\delta_E^+ = \max(\delta_E, 0)$ and $\kappa_n = \log_2 n + 2 \log_2(\log_2 n + 2) + \kappa$.

Proof. An exact identity. Always $h(y \mid x) \leq \frac{1+s_h(x)(-1)^y}{2}$, with equality when $h(0 \mid x) + h(1 \mid x) = 1$; this is because $h(0) + h(1) \leq 1$. Also $(-1)^{y^*(x)} = \chi_a(x)(1 - 2 \cdot 1_E(x))$. By Jensen, $\mathcal{R}(h) \geq -\log_2 \mathbb{E}_x h(y^* \mid x)$, and

$$\mathbb{E}_x h(y^* \mid x) \leq \frac{1}{2} + \frac{1}{2} \mathbb{E}_x [\sigma_h(1 - 2 \cdot 1_E)] = \frac{1}{2} + \frac{1}{2} \hat{s}_h(a) - \frac{1}{n} \sum_{x \in E} \sigma_h(x).$$

Reduction. Let $\eta_0 := |\frac{1}{pn} \sum_{x \in E} \sigma_h(x) - \hat{s}_h(a)|$, the deviation of E 's average of σ_h from the population average. Then $\frac{1}{n} \sum_E \sigma_h \geq p(\hat{s}_h(a) - \eta_0)$, so

$$\mathbb{E}_x h(y^* | x) \leq \frac{1}{2} + \frac{1-2p}{2} \hat{s}_h(a) + p\eta_0 \quad \text{and} \quad \mathcal{R}(h) \geq 1 - \log_2(1 + (1-2p)\hat{s}_h(a) + 2p\eta_0).$$

It suffices to show $2p\eta_0 \leq \varepsilon_h$.

Small deviations. If $\eta_0 < 1/n$, then $2p\eta_0 < 2p/n$. And $(2p/n)^2 \leq 32 \ln 2 \cdot p\kappa_n/n \leq \varepsilon_h^2$, because $p < 1$ and $\kappa_n \geq 1$.

Counting. For $\eta > 0$, let B_η be the set of pn -subsets $E' \subseteq X$ with $|\frac{1}{pn} \sum_{E'} \sigma_h - \hat{s}_h(a)| \geq \eta$. A uniformly random pn -subset is a sample without replacement from the population $\{\sigma_h(x)\}_{x \in X}$, whose mean is $\hat{s}_h(a)$ and whose range has length at most 2. Hoeffding's inequality for sampling without replacement (Hoeffding, 1963, §6) gives $|B_\eta| \leq 2e^{-pn\eta^2/2} \binom{n}{pn}$.

Coding. Otherwise $\eta_0 \geq 1/n$. Since $\eta_0 \leq 2$, we may pick $\eta = 2^{-k}$ with $\eta \leq \eta_0 < 2\eta$ and $-1 \leq k \leq d$; then $E \in B_\eta$.

- B_η is computable from p_h , from a (whose length gives d), from pn (written in $d+1$ bits) and from k (written in $\log_2(d+2) + 1$ bits). Here $\sigma_h = s_h \chi_a$ is computed by running p_h on all of X . So

$$K(E | a, p_h) \leq \log_2 |B_\eta| + \kappa_n \leq L_E - \frac{pn\eta^2}{2 \ln 2} + \kappa_n.$$

- Also $L_E - \delta_E = K(E | a, K(a)) \leq K(E | a) + O(1) \leq K(p_h) + K(E | a, p_h) + O(1)$, and $K(p_h) = K(h) + O(1)$.
- Combining these, with κ absorbing the constants, gives $\frac{pn\eta^2}{2 \ln 2} \leq \delta_E + K(h) + \kappa_n$.

Hence $\eta_0^2 < 4\eta^2 \leq 8 \ln 2 (\delta_E^+ + K(h) + \kappa_n)/(pn)$, i.e. $(2p\eta_0)^2 < \varepsilon_h^2$. $\square$

A numerical check (computed; T5 repair_checks). The identity holds to machine precision. For random E with $d = 12$ and $p \in \{1/64, 0.15, 0.2, 0.24\}$, the observed η_0 is at most 0.08, and the bound $\frac{1}{2} + \frac{1-2p}{2} \hat{s} + p\eta_0$ is never violated.

Proof of Theorem 8.6. (a), upper bounds.

- P_u is computable from the $d-u$ known bits of a together with d, u and p . So $K(P_u) \leq d-u + \kappa'$, using $K(p) = O(\log d)$.
- The true parity is a component of weight 2^{-u} . Its likelihood is exactly $(1-p)^{n-pn} p^{pn} = 2^{-nH(p)}$, because $|E| = pn$. Hence $-\log P_u(y^*) \leq u + nH(p)$.

(a), lower bound on $-\log P_u(y^*)$. Every other component $f_{a'}$ disagrees with y^* on $w \geq n/2 - pn$ points. This is because $f_a \oplus f_{a'}$ has weight $n/2$, and flipping E changes at most pn of them. So the likelihood of $f_{a'}$ is

$$2^{-nH(p)} (p/(1-p))^{w-pn} \leq 2^{-nH(p)} (p/(1-p))^{n(1/2-2p)},$$

and therefore $-\log P_u(y^*) \geq u + nH(p) - \log_2(1 + 2^u (p/(1-p))^{n(1/2-2p)})$. The exponent is positive because $p < 1/4$. The last term is below 2^{-2800} already at $d = 10, p = 1/64$ (computed).

(a), lower bound on $K(P_u)$. Given a program for P_u and (d, u) , compute the a' that maximizes $P_u(f_{a'})$, the probability of the label vector $f_{a'}$.

- Every a' with the true $(d-u)$ -prefix has $P_u(f_{a'}) \geq 2^{-u}(1-p)^n$, from its own component.

- Every other a' is at Hamming distance exactly $n/2$ from every component, so $P_u(f_{a'}) = (p(1-p))^{n/2}$.
- The first is larger, because $\frac{n}{2} \log_2 \frac{1-p}{p} > \frac{n}{2} \geq d \geq u$.

So the argmax has the true prefix, and $d \leq K(a) \leq K(P_u) + u + O(\log d)$.

(a), *selection*. For every u ,

$$\lambda K(P_u) - \log P_u(y^*) \geq \lambda(d - u - \kappa') + u + nH(p) - o(1),$$

while the value at $u = d$ is at most $\lambda\kappa' + d + nH(p)$. A minimizer u^* therefore satisfies $(\lambda - 1)(d - u^*) \leq 2\lambda\kappa' + o(1)$. In the computable version the code length is exactly $d - u$. The objective at u then exceeds the objective at $u = d$ by $(\lambda - 1)(d - u) + (o_d - o_u)$, where $o_u := u + nH(p) + \log P_u(y^*) = \log_2(1 + \sum_{a' \neq a} q_{a'}/q_a) \geq 0$, the sum running over the other components of P_u with likelihoods $q_{a'}$, is non-decreasing in u because the families are nested. So for $\lambda > 1$ it is minimized only at $u = d$.

(b). Suppose $\hat{s}_h(a) \geq 2^{-t}$. Then, by Lemma G.1,

$$d \leq K(a) \leq K(h) + K(a | p_h) + O(1) \leq K(h) + 2t + 2\log_2(t+1) + 2\log_2 d + O(1).$$

This contradicts the hypothesis on $K(h)$ once κ exceeds the $O(1)$. So $\hat{s}_h(a) < 2^{-t}$.

- Lemma G.2 gives $\mathcal{R}(h) \geq -\log_2(\frac{1}{2} + 2^{-t-1} + p) = 1 - \log_2(1 + 2^{-t} + 2p)$.
- Lemma G.3, with $K(h) \leq d$ (so $\varepsilon_h \leq \varepsilon_E$) and $1 - 2p > 0$, gives $\mathcal{R}(h) \geq 1 - \log_2(1 + (1 - 2p)2^{-t} + \varepsilon_E)$.
- Finally, $\log_2(1 + z) \leq z/\ln 2$.

(c).

- $P_h(y) := \prod_x h(y_x | x)$ is a semi-distribution on $\{0, 1\}^n$, computable from p_h and d . A program for h need not determine d ; omitting this was the error corrected after verification.
- By the conditional coding theorem, $K(y^* | p_h, d) \leq -\log P_h(y^*) + O(1) = n\mathcal{R}(h) + O(1)$. So $K(y^*) \leq K(h) + n\mathcal{R}(h) + 2\log_2 d + O(1)$.
- From y^* one computes a . The correlation of $(-1)^{y^*}$ with χ_a is $1 - 2p$, and with any other $\chi_{a'}$ it is $|-2\mathbb{E}[\chi_{a \oplus a'} 1_E]| \leq 2p < 1 - 2p$. Then $E = \{y^* \neq f_a\}$. So $K(a, E) \leq K(y^*) + O(1)$.
- Symmetry of information gives $K(a, E) = K(a) + K(E | a, K(a)) + O(1) \geq d + L_E - \delta_E - O(1)$.

Combining the last three items proves (c). □

G.4 Proof of Lemma 8.8(iii)

Fix $z = (\ell_z, r_z) \in Z$ and $c > 0$, and write $\varphi_c(y) = \langle (c, 1), y \rangle$. Compare z with the other points of Z :

- *Points to the left*, $Z_{<} = \{z' : \ell' < \ell_z\}$, a finite set. Here $\varphi_c(z) \leq \varphi_c(z')$ iff $c \leq \sigma(z') := (r' - r_z)/(\ell_z - \ell')$, strictly iff $c < \sigma(z')$.
- *Points to the right*, $Z_{>} = \{z' : \ell' > \ell_z\}$. Here $\varphi_c(z) \leq \varphi_c(z')$ iff $c \geq \tau(z') := (r_z - r')/(\ell' - \ell_z)$, strictly iff $c > \tau(z')$.
- *Other points with the same ℓ* have a different r , and z beats them iff $r_z < r'$, for every c .

Let $s_- := \min_{Z_<} \sigma$, with $s_- = \infty$ if $Z_< = \emptyset$, and $s_+ := \max(0, \sup_{Z_>} \tau)$. If $\sup_{Z_>} \tau > 0$, the supremum is attained, because $\tau(z') \leq r_z/(\ell' - \ell_z)$ and only finitely many points have $\ell' \leq \ell_z + r_z/s$ for any $s > 0$.

So the set of $c > 0$ at which z is the *unique* minimizing point is the open interval (s_+, s_-) when $r_z < r'$ for all same- ℓ points, and is empty otherwise.

If $(s_+, s_-) \neq \emptyset$, then z is a vertex. Take c in the interval.

- On $Z + \mathbb{R}_{\geq 0}^2$, φ_c is uniquely minimized at z , because $\varphi_c(z' + v) > \varphi_c(z')$ for $v \geq 0$, $v \neq 0$.
- A point of $\hat{C}$ is a finite convex combination of points of $Z + \mathbb{R}_{\geq 0}^2$. Its φ_c -value equals $\varphi_c(z)$ only if every point in the combination is z . So z is the unique minimizer of φ_c over $\hat{C}$.
- Hence z is an extreme point of $\hat{C}$, supported by a line of slope $-c$, i.e. a vertex of $\text{hull}^-(Z)$.

If z is a vertex, then $(s_+, s_-) \neq \emptyset$. First, extreme points of $\text{conv}(S)$ lie in S , and a point $z' + v$ with $v \neq 0$ is the midpoint of $z' + v/2$ and $z' + 3v/2$. So vertices lie in Z .

- Since $z \in \text{hull}^-(Z)$, some $c_0 > 0$ has z minimizing φ_{c_0} . So $s_+ \leq c_0 \leq s_-$, and same- ℓ points have $r' > r_z$.
- Suppose $s_+ = s_- = c_0$. Then some $z_L \in Z_<$ has $\sigma(z_L) = c_0$, since $Z_<$ is finite, and some $z_R \in Z_>$ has $\tau(z_R) = c_0$, since the supremum is attained when positive.
- Then z_L , z and z_R lie on one line of slope $-c_0$, with $\ell_L < \ell_z < \ell_R$. So z is a proper convex combination of z_L and z_R , and not extreme. This is a contradiction.

The interval and the hypothesis-level claim. The interval is (s_+, s_-) , where $s_{\pm}$ are the absolute slopes of the edges to the adjacent vertices; at a risk-minimal vertex $s_+ = 0$. Hypotheses with the same point have equal J_c for every c , so a hypothesis is the unique minimizer on the interval iff no other hypothesis has its point. $\square$

A computed check (computed; T5 c3(ii), test repaired after verification): 300 random point sets and 400 values of c each gave no minimizer off the hull, no monotonicity violation, and every vertex selectable.

G.5 Proof of Theorem 8.11

Write $R_j(h) = \mathbb{E}_z[-\log h(y^*(j, z) \mid j, z)]$ for the normalized risk on region j , so that $\mathcal{R}(h) = \sum_j \pi_j R_j(h)$. Let $s(t) = 2t + 2\log_2(t+1)$, $u_j = 2^{1-d_j/4}$, and $C := \sum_j (2\log_2 d_j + 2\log_2 m + \kappa) + \kappa$.

Upper bound. Let h_S know a_j for $j \in S$ and flip a fair coin elsewhere. Describing j , d_j and a_j for each $j \in S$ gives $K(h_S) \leq \sum_{j \in S} (d_j + 2\log_2 d_j + 2\log_2 m) + \kappa$, and $\mathcal{R}(h_S) = \sum_{j \notin S} \pi_j$. With $S = \{j : c d_j < \pi_j\}$,

$$J_c(h_S) \leq \sum_j \min(c d_j, \pi_j) + cC \leq \sum_j \min(c d_j, \pi_j) + \Delta_0(c).$$

Lower bound: complexity. Fix h . Let t_j be the least $t \in \mathbb{N}$ with $\hat{s}_{h,j}(a_j) \geq 2^{-t}$, or ∞ if there is none. For $t_j \geq 1$ we have $\hat{s}_{h,j}(a_j) < 2^{1-t_j}$.

- Given p_h , j and d_j , we can either list-decode a_j (Lemma G.1, applied on region j) or write it literally. So $K(a_j \mid p_h) \leq \min\{s(t_j), d_j\} + 2\log_2 d_j + 2\log_2 m + \kappa$.
- Then $\sum_j d_j \leq K(a_1, \dots, a_m) \leq K(h) + \sum_j K(a_j \mid p_h) + \kappa$.
- Since $d - \min\{s, d\} = (d - s)^+$, this gives $K(h) \geq \sum_j (d_j - s(t_j))^+ - C$.

Lower bound: risk. Lemma G.2 with $p = 0$, applied on region j , gives

$$R_j(h) \geq -\log_2 \frac{1+\hat{s}_{h,j}(a_j)}{2} > \omega(t_j) \quad \text{for } t_j \geq 1,$$

and $R_j \geq 0$ always. For $t_j = \infty$ we have $\hat{s} \leq 0$, so $R_j \geq 1$. Hence $J_c(h) \geq \sum_j \varphi_j(t_j) - cC$, where

$$\varphi_j(0) := c d_j, \quad \varphi_j(t) := c(d_j - s(t))^+ + \pi_j \omega(t) \quad (t \geq 1), \quad \varphi_j(\infty) := \pi_j.$$

Since $s(t) \leq 4t$ for $t \geq 1$, we have $\varphi_j(t) \geq \psi_j(t) := c(d_j - 4t) + \pi_j \omega(t)$.

- *Correction after verification.* The original proof continued with $\omega(t) \geq 1 - 2^{1-t}$. That is false for $t \geq 2$, since $\log_2(1+u) \geq u$ on $[0, 1]$. The exact floor ω is used from here on.
- ω is increasing and concave: with $u = 2^{1-t}$, $\omega'(t) = u/(1+u)$, which is positive and decreasing in t .

Per-region bound. We claim $\min_t \varphi_j(t) \geq \min(c d_j, \pi_j) - 4c - \pi_j u_j$. The values at $t = 0$ and $t = \infty$ already satisfy it.

- If $d_j < 4$, the right side is negative and the claim is trivial.
- If $d_j \geq 4$: on $t \in [1, d_j/4]$ the bound ψ_j is concave, so it is at least $\min(\psi_j(1), \psi_j(d_j/4)) = \min(c(d_j - 4), \pi_j \omega(d_j/4))$. For $t > d_j/4$ we have $\varphi_j(t) \geq \pi_j \omega(t) \geq \pi_j \omega(d_j/4)$, since ω is increasing. Hence $\min_t \varphi_j \geq \min(c(d_j - 4), \pi_j(1 - \log_2(1 + u_j)))$.

Absorption. The first entry is at least $\min(c d_j, \pi_j) - 4c$. For the second, use $\log_2(1+u) - u \leq (\frac{1}{\ln 2} - 1)u \leq 0.443u$, together with

$$0.886 d 2^{-d/4} \leq 4 \quad \text{for all } d,$$

whose left side has maximum 1.88.

- If $c d_j \geq \pi_j$, we need $\pi_j(\log_2(1+u_j) - u_j) \leq 4c$. This follows from $0.443\pi_j u_j \leq 4\pi_j/d_j \leq 4c$.
- If $c d_j < \pi_j$, we need $c(d_j - 4) \leq \pi_j(1 - \log_2(1+u_j) + u_j)$. This follows from $c(d_j - 4) < \pi_j(1 - 4/d_j)$ and $\log_2(1+u_j) - u_j \leq 0.443u_j \leq 4/d_j$.

Summing over j gives $\Phi(c) \geq \sum_j \min(c d_j, \pi_j) - c(C + 4m) - \sum_j \pi_j u_j = \sum_j \min(c d_j, \pi_j) - \Delta_0(c)$.

A numerical check (computed; T5 repair_checks): the per-region bound was checked for $d \in [1, 63] \cup \{80, 100, 200, 400\}$, $c = 2^{e/4}$ with $e \in [-60, 9]$, and seven values of π , with both $s(t)$ and $4t$. There were no violations.

Identification. Let $J_c(h) \leq \Phi(c) + \xi$. For each j ,

$$\varphi_j(t_j) \leq J_c(h) + cC - \sum_{i \neq j} \varphi_i(t_i) \leq \Phi(c) + \xi + cC - \sum_{i \neq j} [\min(c d_i, \pi_i) - 4c - \pi_i u_i].$$

Using the upper bound on Φ , this gives $\varphi_j(t_j) \leq \min(c d_j, \pi_j) + 2\Delta_0(c) + \xi = \min(c d_j, \pi_j) + \Delta$.

- *Clause (a).* If $\hat{s}_{h,j}(a_j) \geq \frac{1}{2}$, then $t_j \leq 1$, and $\varphi_j(t_j) \geq c(d_j - 4)$, since $s(1) = 4$. Under the hypothesis of (a) this exceeds $\pi_j + \Delta \geq \min(c d_j, \pi_j) + \Delta$, a contradiction.
- *Clause (b).* If $\hat{s}_{h,j}(a_j) < \frac{1}{2}$, then $t_j \geq 2$. On $[2, d_j/4]$, which is nonempty because $d_j \geq 8$, the bound ψ_j is concave. For $t \geq d_j/4$, including $t = \infty$, the bound is at least $\pi_j \omega(d_j/4)$. So $\varphi_j(t_j) \geq \min(c(d_j - 8) + \pi_j \omega(2), \pi_j \omega(d_j/4))$. Under the two hypotheses of (b), both entries exceed $c d_j + \Delta \geq \min(c d_j, \pi_j) + \Delta$, a contradiction.

- The implied forms use $\omega(d/4) = 1 - \log_2(1 + u) \geq 1 - u/\ln 2$ and $1/\omega(2) = 2.4094$.

A numerical check (computed; T5 repair_checks): under the new conditions, $\varphi_j(t) > c d_j + \Delta$ for every integer $t \geq 2$ in 8474 random admissible instances. The counterexample to the original clause (b), given after Theorem 8.11, violates the new second condition by a factor of 1.01–1.06. $\square$

G.6 Proof of Corollary 8.12

The coin region. On the coin region every hypothesis has cross-entropy at least one bit per unit mass, with equality iff it is uniform there. So $\mathcal{R}(h) \geq \pi_0 + \sum_{j \geq 1} \pi_j R_j(h)$, and the region plays no role in the complexity argument. The lower bound of Theorem 8.11 therefore holds with π_0 added. The witnesses h_S are uniform on the coin region, at an extra $O(1)$ bits to recognize it. Hence $|\Phi(c) - \Psi(c)| \leq cC' + \epsilon$ with $\Psi(c) := \pi_0 + \sum_j \min(c d_j, \pi_j)$, for all $c \geq 0$. At $c = 0$ both sides equal π_0 .

Legendre duality. $\Phi(c) = \inf_{y \in \hat{C}} \langle (c, 1), y \rangle$ is the support function of $\hat{C}$. The lower boundary $\check{\rho}$ of its closure is convex and lower semicontinuous, so $\check{\rho}(k) = \sup_{c \geq 0} [\Phi(c) - ck]$. Likewise $P(k) := \sup_{c \geq 0} [\Psi(c) - ck]$.

- Ψ is concave and piecewise linear, with breakpoints at $c = s_j$ and slope $\sum_{j: s_j > c} d_j$ between them.
- Its conjugate P is the polygon described in the statement: it starts at $P(0) = \Psi(\infty) = 1$, has an edge of slope $-s_j$ and length d_j for each j in decreasing order of s_j , and is flat at π_0 afterwards. For $k < 0$, $P(k) = +\infty$.

The sandwich. Then

$$\check{\rho}(k) \geq \sup_c [\Psi(c) - c(k + C')] - \epsilon = P(k + C') - \epsilon, \quad \check{\rho}(k) \leq \sup_c [\Psi(c) - c(k - C')] + \epsilon = P(k - C') + \epsilon.$$

Rescaling. Replacing d_j by $L d_j$ and s_j by s_j/L keeps every π_j . It raises C' only logarithmically in L , so the band is $o(L)$, and $\epsilon = \sum_j \pi_j 2^{1-Ld_j/4} \rightarrow 0$. $\square$

G.7 Proof of Theorem 8.13

Upper bound. We list the three models.

- The coin has $K \leq \kappa$ and $\mathcal{R} = 1$.
- The good model has $K \leq d + \kappa'$ and $\mathcal{R} = H(p)$ exactly, since $|E| = pn$.
- The error-memorizing model y^* has $K \leq d + L_E + \kappa \log n$ (index E among the pn -subsets, and give pn) and $\mathcal{R} = 0$.

Lower bound. Fix h and t .

- If $K(h) \leq d - \tau_t$, then Theorem 8.6(b) gives $J_c(h) \geq \mathcal{R}(h) \geq 1 - \beta_t/\ln 2$.
- Otherwise, Theorem 8.6(c) and $\mathcal{R} \geq 0$ give $J_c(h) \geq cK + \max(0, (K_{\text{tot}} - K)/n)$ with $K = K(h) > d - \tau_t$. As a function of K this is piecewise linear: its slope is $c - 1/n$ below K_{tot} and c above. Note that $K_{\text{tot}} - d + \tau_t = L_E - \delta_E + s(t) \geq 0$, because $L_E - \delta_E = K(E \mid a, K(a)) \geq 0$.
 - If $c \geq 1/n$, the minimum over $K \geq d - \tau_t$ is at $K = d - \tau_t$, with value $c(d - \tau_t) + (L_E - \delta_E + s(t))/n \geq c(d - \tau_t) + (L_E - \delta_E - \kappa)/n$.

- If $c < 1/n$, the minimum is at $K = K_{\text{tot}}$, with value cK_{tot} .

In every case $J_c(h)$ is at least the minimum of the three lower-bound terms.

The $2 \log_2 d$ in K_{tot} comes from the corrected Theorem 8.6(c), and the coin term uses the sharper β_t . Neither changes the switch points beyond their $O(\log)$ slack. $\square$

Proof of Corollary 8.14. Correlation. By the upper bound of Theorem 8.13, $\mathcal{R}(h) \leq J_c(h) \leq c(d + \kappa') + H(p) + \xi$. Lemma G.2 gives $2^{-\mathcal{R}(h)} \leq \frac{1+\hat{s}_h(a)}{2} + p$, i.e. $\hat{s}_h(a) \geq 2^{1-\mathcal{R}(h)} - 1 - 2p$. Substitute the bound on $\mathcal{R}(h)$.

Memorization. By Theorem 8.6(c), $K(h) \geq K_{\text{tot}} - n\mathcal{R}(h)$, so

$$J_c(h) \geq cK_{\text{tot}} - (cn - 1)\mathcal{R}(h).$$

Combine this with $J_c(h) \leq c(d + \kappa') + H(p) + \xi$ and with $L_E \geq nH(p) - \log_2(n + 1)$. This gives

$$(cn - 1)\mathcal{R}(h) \geq c(L_E - \delta_E - 2 \log_2 d - \kappa - \kappa') - H(p) - \xi \geq (cn - 1)H(p) - c\Lambda - \xi.$$

Divide by $cn - 1 > 0$. $\square$

Numbers for illustration (computed; T5 repair_checks): at $d = 20$, $p = 2^{-10}$ and $\kappa = \kappa' = \xi = 0$, $c = 10^{-4}$ gives $\hat{s} \geq 0.980$ and $\mathcal{R} \geq H(p) - 2.8 \cdot 10^{-5}$, and $c = 10^{-3}$ gives $\hat{s} \geq 0.955$.

G.8 Proof of Theorem 8.15(ii) in the idealized setting

Let h be near-optimal, and let c be such that each of v and e satisfies the hypothesis of clause (a) or that of clause (b) of Theorem 8.11. Suppose v is learned, i.e. $\hat{s}_{h,v} \geq \frac{1}{2}$.

- Then v does not satisfy the hypothesis of (a), so it satisfies that of (b). In particular $\pi_v \geq \pi_v \omega(d_v/4) > c d_v$, i.e. $\pi_v/d_v > c$.
- If e satisfied the hypothesis of (a), then $\pi_e < c d_e$. That would give $\pi_e/d_e < c < \pi_v/d_v$, contradicting $\pi_v/d_v \leq \pi_e/d_e$.
- So e satisfies the hypothesis of (b), and is learned.

$\square$

G.9 Proof of Theorem 8.17

(i).

- *Monotone risk.* By Lemma 8.8(ii), for grid values $c_1 < c_i$ and any population minimizers h at c_1 and h' at c_i , $\mathcal{R}(h) \leq \mathcal{R}(h')$.
- *Eventually exact.* For each grid value, Lemma 8.2 gives $J_c(\hat{h}_c) \rightarrow \Phi(c)$ in probability. Only finitely many hypotheses have $\ell \leq \ell(h_0) + B/c$, so the gap between $\Phi(c)$ and the best non-minimizer is positive. Hence, with probability tending to one, $\hat{h}_c$ is a population minimizer for every grid value simultaneously.
- *Uniform validation.* Held-out log-losses of the finitely many selected hypotheses converge uniformly to their risks, since losses lie in $[0, B]$.

So, with probability tending to one, selection by held-out log-loss picks a grid value whose selected hypothesis has least risk. By the first item, the smallest grid value always attains this least risk, and it is the unique choice when its risk is strictly smallest. At finite N the second item may fail at small c , where the bound of Lemma 8.2 is weakest. That is the caveat in the statement.

(ii). Strict propriety holds pointwise in x (Gneiting and Raftery, 2007). Integrating over μ , the teacher channel strictly maximizes the expected score among channels that differ from it on a set of positive μ -mass. With a growing held-out set, empirical scores converge uniformly over finitely many candidates. So the criterion selects, among candidates, one with the largest expected score, i.e. the smallest divergence from P^* in the score's divergence.

(iii). Any statistic of imitation data has the same law in the two worlds of Theorem 8.15(iv), while the good c differs between them. $\square$

G.10 The rates in Proposition 8.18

Measure log-loss relative to the option *none*, which gives every candidate probability $1/M$.

- *G-contexts, unguarded or guarded.* The σ -model gains g per step, by the definition of the per-step gain. Mass π_G .
- *$\neg G$ -contexts, unguarded.* The teacher's step is a non- σ step. It receives probability ε/M instead of $1/M$, a loss of $\log_2(1/\varepsilon)$ per step. Mass $\pi_{\neg G}$.
- *$\neg G$ -contexts, guarded.* The teacher's step receives $1/(M-1)$ instead of the unguarded ε/M , a gain of $\Delta_G = \log_2(M/\varepsilon) - \log_2(M-1)$ per step over the unguarded option. Note $\Delta_G > \log_2(1/\varepsilon)$, by $\log_2 \frac{M}{M-1}$; this corrects the earlier bound “at most $\pi_{\neg G} \log_2(1/\varepsilon)/k_G$ ” for the guard's rate.

So the unguarded option lowers risk by $\pi_G g - \pi_{\neg G} \log_2(1/\varepsilon) = k_\sigma \rho_\sigma$ at cost k_σ . The guard lowers it by a further $\pi_{\neg G} \Delta_G = k_G \rho_G$ at cost k_G . These are the three points used in the main-text proof. If the guarded option also models the teacher's alternative step in $\neg G$ -contexts, its gain there only increases, and the argument is unchanged with the larger Δ_G .

For the case $\rho_G \geq \rho_\sigma$: the middle point lies on or above the chord from $(0,0)$ to the guarded point. The lower hull of the three points is then that chord, with slope $-(k_\sigma \rho_\sigma + k_G \rho_G)/(k_\sigma + k_G)$, and apart from ties the selection jumps from guarded to none at that c . In the second numerical instance of the main text the jump is at $c = (0.0022 + 0.0087)/34 \approx 3.2 \cdot 10^{-4}$. $\square$

G.11 Proofs for the division of labour

Throughout, the components are finite, $\text{val}(S) = \sum_{j \in S} \nu_j$, and by separability minimizing J_c over admissible sets is maximizing val over $\mathfrak{A}$. Two facts are used repeatedly.

- If $V_+ \cup \{e\} \notin \mathfrak{A}$, then e has a witness inside V_+ . Indeed, a minimal $W \subseteq V_+$ with $W \cup \{e\} \notin \mathfrak{A}$ is minimal among all subsets of V .
- Conversely, a witness inside V_+ makes $V_+ \cup \{e\}$ inadmissible, by down-closure.

Proof of Theorem 8.20. Step 0: $S^(c)$ is admissible.* Apply error-independence with $U = V_+$.

Step 1: optimal sets have no negative components. Let S be admissible and optimal. Removing the components with $\nu < 0$ keeps admissibility (down-closure) and raises the value. So $S \subseteq \{\nu \geq 0\}$.

Step 2: optimal sets contain no error inadmissible with V_+ . Let $E_1 := (S \cap F) \setminus \{e : V_+ \cup \{e\} \in \mathfrak{A}\}$, and suppose $E_1 \neq \emptyset$.

- Each $e \in E_1$ has a witness inside V_+ . Every such witness W is nonempty, because $\{e\} \subseteq S$ is admissible.
- Since $(S \cap V) \cup \{e\} \subseteq S$ is admissible, $S \cap V$ contains no witness of e . So $B := V_+ \setminus S$ meets every witness of every $e \in E_1$ inside V_+ . In particular $B \neq \emptyset$, and $\sum_B \nu > 0$.
- If E_1 contains some $e \in F_+$, then $\sum_B \nu \geq \beta(e) > \sum_{F_+} \nu \geq \sum_{E_1} \nu$, using valid dominance and $E_1 \subseteq \{\nu \geq 0\}$. Otherwise $E_1 \subseteq \{\nu = 0\}$, and $\sum_{E_1} \nu = 0 < \sum_B \nu$.
- Let $S' := V_+ \cup ((S \cap F) \setminus E_1)$. It is admissible by error-independence with $U = V_+$. Since $S \cap V \subseteq V_+ \cup \{\nu = 0\}$, we have $\sum_{S \cap V} \nu = \sum_{V_+} \nu - \sum_B \nu$, and therefore $\text{val}(S') = \text{val}(S) + \sum_B \nu - \sum_{E_1} \nu > \text{val}(S)$.

This contradicts optimality, so $E_1 = \emptyset$.

Step 3: comparison with $S^(c)$.* Hence $S \cap V \subseteq V_+ \cup \{\nu = 0\}$ and $S \cap F \subseteq \{e : V_+ \cup \{e\} \in \mathfrak{A}, \nu_e \geq 0\}$. So $\text{val}(S) \leq \text{val}(S^*(c))$. Since $S^*(c)$ is admissible, it is optimal, and $\text{val}(S) = \text{val}(S^*(c))$. Every element of $S^*(c)$ has positive value, and S is contained in $S^*(c) \cup \{\nu = 0\}$, so equality forces $S \supseteq S^*(c)$ and $S \setminus S^*(c) \subseteq \{\nu = 0\}$. Conversely, every admissible $S^*(c) \cup Z$ with $Z \subseteq \{\nu = 0\}$ has the optimal value.

Correction after verification. The original last step argued that “adding $V_+ \setminus S$ keeps admissibility”. That can fail when S contains zero-valued valid rules. For example, take $V = \{v_+, v_0\}$ with $\nu(v_0) = 0$, and an error e whose only witness is $\{v_+, v_0\}$. Then $S = \{v_0, e\}$ is admissible, but $S \cup \{v_+\}$ is not. The direct comparison in Step 3 avoids this.

The attributions in the statement. When $\mathfrak{A}$ is the intersection of the coherence, world and postulate families, $V_+ \cup \{e\} \notin \mathfrak{A}$ iff it fails at least one of them. An error with $\nu_e \leq 0$ (rate at most c) is excluded by simplicity, up to zero-value ties. A valid rule with $\nu_v \leq 0$ is not in V_+ , whatever the filters. $\square$

Proof of Corollary 8.21. Recall that $\max \emptyset = 0$ and $k_j > 0$.

If. Suppose $\max\{\rho_e : V \cup \{e\} \in \mathfrak{A}\} < c < \min_v \rho_v$.

- Then $V_+ = V$, and no valid rule has value 0.
- Every e with $V \cup \{e\} \in \mathfrak{A}$ has $\rho_e < c$, so $\nu_e < 0$. Hence $S^*(c) = V$.
- By Theorem 8.20, the optimal sets are the admissible $V \cup Z$ with $Z \subseteq \{\nu = 0\} \cap F$. A zero-valued error has $\rho_e = c$, so $V \cup \{e\} \notin \mathfrak{A}$. By down-closure $V \cup Z$ is then inadmissible unless $Z = \emptyset$.

So V is the unique optimum, and separation holds.

Only if. We show that separation fails if either inequality fails.

- *If $c \geq \rho_v$ for some v ,* then $\nu_v \leq 0$ and $v \notin S^*(c)$. Since $S^*(c)$ is optimal, some optimal set omits v .
- *If $c < \min_v \rho_v$, but some e with $V \cup \{e\} \in \mathfrak{A}$ has $\rho_e \geq c$:*
 - if $\rho_e > c$, then $e \in F_+$ and $e \in S^*(c)$, which is optimal;
 - if $\rho_e = c$, then $\nu_e = 0$, and $S^*(c) \cup \{e\}$ is admissible by error-independence with $U = V$, and optimal.

Either way some optimal set contains an error. $\square$

Proof of Proposition 8.23(a). The postulates enter only through $\mathfrak{A}$, so Theorem 8.20 applies at each c .

The characterization at each c . If $c \geq \rho_e$, then $\nu_e \leq 0$, and e is excluded up to zero-value ties. If $c < \rho_e$, then $e \in F_+$. In that case $e \in S^*(c)$ iff $V_+(c) \cup \{e\} \in \mathfrak{A}$, iff no witness of e lies inside $V_+(c)$, by the two facts at the head of this subsection. Moreover $\nu_e > 0$, so e lies in every optimal set iff it lies in $S^*(c)$.

Exclusion at every c .

- *If.* Suppose some witness W has $\min_{w \in W} \rho_w \geq \rho_e$. Then $W \subseteq V_+(c)$ for every $c < \rho_e$, so e is excluded at every such c , and also at every $c \geq \rho_e$. This covers the case $W = \emptyset$, i.e. $\{e\}$ already inadmissible.
- *Only if.* Suppose every witness has a member of rate below ρ_e . There are finitely many witnesses, so we can choose $c \in (\max_W \min_{w \in W} \rho_w, \rho_e)$, with $\max \emptyset = 0$; this uses $\rho_e > 0$. At that c no witness lies inside $V_+(c)$, so $e \in S^*(c)$, and e survives in every optimal set. $\square$

Proof of Proposition 8.22. The repaired set is admissible. First, $(V_+ \setminus B) \cup \{e\} \in \mathfrak{A}$. Otherwise a minimal $W \subseteq V_+ \setminus B$ with $W \cup \{e\} \notin \mathfrak{A}$ would be a witness of e inside V_+ that B misses. Second, for $e' \in S_{\text{clean}} \cap F$, $(V_+ \setminus B) \cup \{e'\} \subseteq V_+ \cup \{e'\} \in \mathfrak{A}$, so it is admissible by down-closure. By error-independence with $U = V_+ \setminus B$, the set $S' = (V_+ \setminus B) \cup \{e\} \cup (S_{\text{clean}} \cap F)$ is admissible.

Its value. Since e has a witness inside V_+ , $V_+ \cup \{e\} \notin \mathfrak{A}$, so $e \notin S_{\text{clean}}$. Hence $\text{val}(S') = \text{val}(S_{\text{clean}}) - \sum_B \nu + \nu_e = \text{val}(S_{\text{clean}}) + \nu_e - \beta(e) > \text{val}(S_{\text{clean}})$. $\square$

In 1882 random instances with $\beta(e) < \nu_e$, the clean set was strictly suboptimal every time (computed; verification).

The two counterexamples quoted in the main text were recomputed (computed; T5 repair_checks). One is to Corollary 8.21 without valid dominance. The other is to the original Step 3 of Theorem 8.20.

H Proofs for Section 9

The following results are proved completely in the main text: Propositions 9.3, 9.20, 9.27 and 9.31, Theorems 9.4, 9.14, 9.25, 9.29 and 9.35, and Corollary 9.23, and Theorem 9.15(a). In Example 9.13 the first claim is a one-line check (the union bound of $p \wedge q \triangleright p$ is $(1 - 0.9) + 0.2 = 0.3 < 1$), and the facet list is computed (`research/theory/T6-checks/prob_coherence.py`); it is also easily verified by hand from the four vertices $(0, 0, 1)$, $(0, 1, 1)$, $(1, 0, 0)$, $(1, 1, 1)$. Theorems 9.33, 9.34 and 9.37 and Beth's theorem in Proposition 9.36 are cited, not reproved; the non-standardness clause of Theorem 9.33 and the examples after Proposition 9.36 are argued in the main text. Everything else is proved below. Throughout, “theory” means closed set.

H.1 The abstract frame

Proof of Lemma 9.2. Both sides of $K \subseteq \text{Mod}(\Sigma) \iff \Sigma \subseteq \text{Th}(K)$ say that $m \models \sigma$ for all $m \in K$, $\sigma \in \Sigma$. Both maps are inclusion-reversing. Taking $K = \text{Mod}(\Sigma)$ gives $\Sigma \subseteq \text{ThMod}(\Sigma)$, and taking $\Sigma = \text{Th}(K)$ gives $K \subseteq \text{ModTh}(K)$; so both composites are extensive and monotone. Applying the inclusion-reversing Mod to $\Sigma \subseteq \text{ThMod}\Sigma$ gives $\text{ModThMod}\Sigma \subseteq \text{Mod}\Sigma$, and the converse is extensivity of ModTh at $\text{Mod}\Sigma$; so $\text{ModThMod} = \text{Mod}$, and dually $\text{ThModTh} = \text{Th}$.

Idempotence of both composites follows. The fixed points of $\text{Th} \circ \text{Mod}$ are exactly the sets $\text{Th}(K)$, those of $\text{Mod} \circ \text{Th}$ the sets $\text{Mod}(\Sigma)$, and the two identities show that Mod and Th are mutually inverse on them. Fixed-point sets of closure operators on power sets are closed under arbitrary intersections, so they are complete lattices. $\square$

Proof of Theorem 9.5. We first record a consequence of finitariness: *the union U of a nonempty chain of closed sets is closed*. Indeed $C(U) = \bigcup \{C(U_0) : U_0 \subseteq U \text{ finite}\}$, each finite U_0 lies in a single member T of the chain, and $C(U_0) \subseteq C(T) = T \subseteq U$.

(a) Let T be closed and $\sigma \notin T$. The family of closed $T' \supseteq T$ with $\sigma \notin T'$ is nonempty (it contains T) and, by the observation, closed under unions of nonempty chains. By Zorn's lemma it has a maximal member T_σ . Then $T_\sigma \neq S$, and every closed $T' \supsetneq T_\sigma$ contains σ by maximality; so T_σ is a point containing T . Hence $\bigcap \{P \in \text{Pt}(C) : P \supseteq T\} \subseteq \bigcap_{\sigma \notin T} T_\sigma \subseteq T$, since $\tau \notin T$ implies $\tau \notin T_\tau$. The reverse inclusion is trivial. If $T = S$ there is no point above T and the empty intersection is S .

(b) $(\Leftarrow)$ Soundness gives $C(X) \subseteq \text{Th}(\text{Mod}(X))$. Conversely, by (a), $C(X) = \bigcap_i P_i$ for the points $P_i \supseteq C(X)$. By hypothesis $P_i = \text{Th}(\{m_i\})$, and $m_i \in \text{Mod}(X)$ because $X \subseteq P_i$. So $\text{Th}(\text{Mod}(X)) \subseteq \bigcap_i \text{Th}(\{m_i\}) = C(X)$.

$(\Rightarrow)$ Let P be a point, with witness σ . By strong completeness, $P = C(P) = \text{Th}(\text{Mod}(P)) = \bigcap_{m \in \text{Mod}(P)} \text{Th}(\{m\})$. If $\text{Mod}(P) = \emptyset$ the right-hand side is $S \neq P$; so $\text{Mod}(P) \neq \emptyset$. Each $\text{Th}(\{m\})$ with $m \in \text{Mod}(P)$ is closed, by soundness, and contains P . If each of them strictly contained P , each would contain σ , and so would their intersection P , which is impossible. So $P = \text{Th}(\{m\})$ for some m .

(c) Proper closed sets are closed under unions of nonempty chains: the union U is closed by the observation, and if $U = S$ then the finite set F lies in a single member T of the chain, so $T \supseteq C(F) = S$, a contradiction. By Zorn, every proper closed set extends to a maximal proper closed set. $(\Leftarrow)$ If X is coherent, $C(X)$ is proper and extends to a maximal proper closed T ; by hypothesis $T = \text{Th}(\{m\})$, and $m \in \text{Mod}(X)$. $(\Rightarrow)$ Let T be maximal proper closed. Since $C(T) = T \neq S$, weak completeness gives $m \in \text{Mod}(T)$. Then $T \subseteq \text{Th}(\{m\}) \neq S$, and $\text{Th}(\{m\})$ is closed *by soundness*; maximality gives $T = \text{Th}(\{m\})$. *Soundness is needed for $(\Rightarrow)$* . Let $S = \{a, b, \perp\}$ with closed sets $\emptyset, \{a\}, S$ (so C is finitary and $C(\{\perp\}) = S$), and let $M = \{m\}$ with $\text{Th}(\{m\}) = \{a, b\} \neq S$. The C -coherent sets are $\emptyset$ and $\{a\}$, both satisfied by m , so C is weakly complete; $\{a\}$ is maximal proper closed but differs from $\text{Th}(\{m\}) = \{a, b\}$, which is not closed, so C is unsound ($C(\{b\}) = S \not\subseteq \text{ThMod}(\{b\}) = \{a, b\}$). `repair_checks.py` (A) reproduces it and finds no failure of (b) or (c) on 1,602 random sound finite frames. $\square$

Details for Proposition 9.6. Soundness. Every Boolean valuation is a model of IPC, so each $\text{Th}(\{v\})$ is IPC-closed. *Failure of strong completeness.* $p \vee \neg p$ is true in every Boolean valuation, and refuted in the three-element Heyting algebra by $h(p) = \frac{1}{2}$, so $\text{IPC} \not\models p \vee \neg p$. *Weak completeness.* The hypotheses of Theorem 9.5(c) hold with $F = \{\perp\}$, and $\text{Th}(\{v\}) \not\supseteq \perp$. Let T be a maximal proper IPC-closed set; proper means consistent, as $\{\perp\}$ is explosive. If $\varphi \notin T$, then $C_{\text{IPC}}(T \cup \{\varphi\})$ is closed and strictly larger than T , hence equals Fm ; so $T, \varphi \vdash_{\text{IPC}} \perp$ and, by the deduction theorem, $\neg\varphi = \varphi \rightarrow \perp \in T$. So $\varphi \in T$ or $\neg\varphi \in T$, whence $\varphi \vee \neg\varphi \in T$ for every φ . As consequence relations $\text{CPC} = \text{IPC} + \text{EM}$, where EM is the set of instances of excluded middle; so $C_{\text{CPC}}(T) = C_{\text{IPC}}(T \cup \text{EM}) = C_{\text{IPC}}(T) = T \not\supseteq \perp$. Thus T is CPC-consistent, and maximal CPC-consistent (any CPC-consistent superset is IPC-consistent), hence $T = v^{-1}(1)$ for a Boolean v . Theorem 9.5(c) concludes. $\square$

Proof of Lemma 9.7. Let T be maximal among Cn_2 -theories omitting φ , and suppose neither ψ nor $\neg\psi$ is in T . By maximality $\varphi \in \text{Cn}_2(T \cup \{\psi\})$ and $\varphi \in \text{Cn}_2(T \cup \{\neg\psi\})$. By proof by cases, $\text{Cn}_2(T \cup \{\psi\}) \cap \text{Cn}_2(T \cup \{\neg\psi\}) = \text{Cn}_2(T) = T$, so $\varphi \in T$, a contradiction. Hence T is complete (it contains ψ or $\neg\psi$ for every ψ). Since $T \neq \text{Fm}$ it is consistent, and a complete consistent theory is maximal consistent. $\square$

H.2 Positions

Proof of Theorem 9.8. (a) Order the coherent positions extending $[X : Y]$ componentwise. A union of a nonempty chain $[X_k : Y_k]$ is coherent: an incoherence witness $\Gamma \triangleright \Delta$ with $\Gamma \subseteq \bigcup X_k$, $\Delta \subseteq \bigcup Y_k$ involves finitely many formulas, so it lies in a single $[X_k : Y_k]$. Zorn gives a maximal coherent $[X^* : Y^*]$. *Disjoint:* if $\varphi \in X^* \cap Y^*$, then $\{\varphi\} \triangleright \{\varphi\} \in \vdash$ by (Ov) witnesses incoherence. *Exhaustive:* suppose $\varphi \notin X^* \cup Y^*$. By maximality both $[X^* \cup \{\varphi\} : Y^*]$ and $[X^* : Y^* \cup \{\varphi\}]$ are incoherent, so there are $\Gamma_1, \varphi \triangleright \Delta_1$ and $\Gamma_2 \triangleright \varphi, \Delta_2$ in $\vdash$ with $\Gamma_i \subseteq X^*$, $\Delta_i \subseteq Y^*$. (If φ does not occur in a witness, that witness already shows $[X^* : Y^*]$ incoherent.) By (Wk) both hold with context $\Gamma_1 \cup \Gamma_2$, $\Delta_1 \cup \Delta_2$; (Cut) on φ gives $\Gamma_1 \cup \Gamma_2 \triangleright \Delta_1 \cup \Delta_2 \in \vdash$, which witnesses incoherence of $[X^* : Y^*]$: a contradiction. So $v = \chi_{X^*}$ is a valuation with $v^{-1}(0) = Y^*$. It satisfies every $\Gamma \triangleright \Delta \in \vdash$, since otherwise $\Gamma \subseteq X^*$ and $\Delta \subseteq Y^*$. And $v[X] = 1$, $v[Y] = 0$.

(b) For $v \in \text{Val}(\vdash)$, the position $[v^{-1}(1) : v^{-1}(0)]$ is coherent, because v satisfies every sequent of $\vdash$. It is maximal: a proper extension puts some φ on both sides, which (Ov) makes incoherent. Conversely, a maximal coherent position is, by the proof of (a), of this form. $\square$

Proof of Corollary 9.9. *Soundness*, $\langle S \rangle \subseteq \text{Th}(\text{Mod}(S))$: the sequents valid in every $v \in \text{Mod}(S)$ contain S and form a Scott relation. Every valuation satisfies the (Ov) instances and preserves (Wk); for (Cut), if $v[\Gamma] = 1$ and $v[\Delta] = 0$, the premises force $v(\varphi) = 0$ and $v(\varphi) = 1$. *Completeness.* If $\Gamma_0 \triangleright \Delta_0 \notin \langle S \rangle$, then $[\Gamma_0 : \Delta_0]$ is $\langle S \rangle$ -coherent, since a witness inside it would yield $\Gamma_0 \triangleright \Delta_0$ by (Wk). Theorem 9.8 gives $v \in \text{Val}(\langle S \rangle) \subseteq \text{Mod}(S)$ realizing $[\Gamma_0 : \Delta_0]$, i.e. violating $\Gamma_0 \triangleright \Delta_0$. $\square$

Proof of Theorem 9.10. *Syntactic closure.* By Corollary 9.9, $\text{Th} \circ \text{Mod}$ on sets of sequents is $S \mapsto \langle S \rangle$, so its fixed points are exactly the Scott relations. *Semantic closure.* For $V \subseteq 2^{\text{Fm}}$, $\text{Mod}(\text{Th}(V)) = \overline{V}$. The set $\text{Mod}(\text{Th}(V))$ is an intersection of the sets $\{v : v \models \Gamma \triangleright \Delta\}$, each of which is clopen (it depends on finitely many coordinates); so it is closed, and it contains V ; hence it contains $\overline{V}$. Conversely, let $w \notin \overline{V}$. Some basic open neighbourhood of w , determined by finitely many coordinates, misses V ; it has the form $U(\Gamma, \Delta) = \{v : v[\Gamma] = 1, v[\Delta] = 0\}$ with $\Gamma = w^{-1}(1) \cap E$, $\Delta = w^{-1}(0) \cap E$ for a finite $E \subseteq \text{Fm}$. Since $U(\Gamma, \Delta) \cap V = \emptyset$, every $v \in V$ satisfies $\Gamma \triangleright \Delta$, so $\Gamma \triangleright \Delta \in \text{Th}(V)$; and $w \in U(\Gamma, \Delta)$ violates it. So $w \notin \text{Mod}(\text{Th}(V))$. This is Lemma 6.26(a), formalized as `Carnap.Val_mrel`. *Duality.* By Lemma 9.2, $\text{Mod} = \text{Val}$ and Th are mutually inverse inclusion-reversing bijections between the fixed points on the two sides, i.e. between Scott relations and closed subsets of 2^{Fm} . $\square$

The Stone-duality reading after Theorem 9.10 needs only Stone's theorem (Stone, 1936), the observation that the clopen set of the clause $\bigvee_{\gamma \in \Gamma} \neg \gamma \vee \bigvee_{\delta \in \Delta} \delta$ is $\{v : v \models \Gamma \triangleright \Delta\}$, and conjunctive normal form in B . `scott_duality.py` checks Corollary 9.9 on 300 random sequent sets over four formulas, and Theorem 9.10 exhaustively over three formulas (all 256 sets of valuations are Galois-closed, and $V \mapsto \text{Th}(V)$ is injective).

Proof of Theorem 9.11. For every substitution σ and valuation v , $(v \circ \sigma)(\varphi) = v(\sigma\varphi)$, so

$$v \circ \sigma \models \Gamma \triangleright \Delta \quad \text{iff} \quad v \models \sigma\Gamma \triangleright \sigma\Delta. \quad (2)$$

(a) $(\Rightarrow)$ Let $\vdash$ be structural and $v \in \text{Val}(\vdash)$. For $\Gamma \triangleright \Delta \in \vdash$ we have $\sigma\Gamma \triangleright \sigma\Delta \in \vdash$, so v satisfies it, so $v \circ \sigma \models \Gamma \triangleright \Delta$ by (2). Hence $v \circ \sigma \in \text{Val}(\vdash)$. $(\Leftarrow)$ Let $\text{Val}(\vdash)$ be substitution-invariant and $\Gamma \triangleright \Delta \in \vdash$. For every $v \in \text{Val}(\vdash)$, $v \circ \sigma \in \text{Val}(\vdash)$ satisfies $\Gamma \triangleright \Delta$, so $v \models \sigma\Gamma \triangleright \sigma\Delta$. Thus $\sigma\Gamma \triangleright \sigma\Delta \in \text{Th}(\text{Val}(\vdash)) = \vdash$, the last equality by Corollary 9.9 since $\vdash$ is a Scott relation. The duality statement is the restriction of Theorem 9.10: a closed substitution-invariant V equals $\text{Val}(\text{Th}(V))$, so $\text{Th}(V)$ is structural by $(\Leftarrow)$.

(b) An assignment into $L_v = \langle \text{Fm}, v^{-1}(1) \rangle$ is a homomorphism $\text{Fm} \rightarrow \text{Fm}$, i.e. a substitution σ , and it satisfies $\Gamma \triangleright \Delta$ in the matrix iff not $(\sigma\Gamma \subseteq v^{-1}(1) \text{ and } \sigma\Delta \cap v^{-1}(1) = \emptyset)$, iff $v \models \sigma\Gamma \triangleright \sigma\Delta$, iff $v \circ \sigma \models \Gamma \triangleright \Delta$. So L_v validates $\Gamma \triangleright \Delta$ iff $v \circ \sigma \models \Gamma \triangleright \Delta$ for all σ . If $\vdash$ is structural and $\Gamma \triangleright \Delta \in \vdash$, this holds by (a). If $\Gamma \triangleright \Delta \notin \vdash$, the position $[\Gamma : \Delta]$ is coherent (a witness inside it would give $\Gamma \triangleright \Delta$ by (Wk)), so Theorem 9.8 gives $v \in \text{Val}(\vdash)$ with $v[\Gamma] = 1$, $v[\Delta] = 0$, and the identity assignment in L_v violates $\Gamma \triangleright \Delta$. Likewise, a coherent position $[X : Y]$ is realized by some $v \in \text{Val}(\vdash)$, and the identity assignment in L_v designates X and no member of Y . $\square$

H.3 Credences

Proof of Theorem 9.12. (i) $\Rightarrow$ (iii): if $P = \sum_v \mu(v)\bar{v}$ and $\lambda \cdot \bar{v} \geq c$ for all v , then $\lambda \cdot P = \sum_v \mu(v) \lambda \cdot \bar{v} \geq c$. (i) $\Rightarrow$ (ii): $\sum_v \mu(v) \lambda \cdot (\bar{v} - P) = \lambda \cdot P - \lambda \cdot P = 0$, so $\lambda \cdot (\bar{v} - P)$ cannot be negative for every v . (iii) $\Rightarrow$ (i) and (ii) $\Rightarrow$ (i): Π_F is the convex hull of finitely many 0/1 points, hence a polytope with rational vertices, hence (Weyl–Minkowski) $\Pi_F = \{x : a_i \cdot x \geq b_i, i \leq r\}$ with rational a_i, b_i . If $P \notin \Pi_F$, one of these inequalities, say $a \cdot x \geq b$ (a facet inequality, or half of an affine-hull equation), has $a \cdot P < b \leq a \cdot \bar{v}$ for all $v \in \text{BV}$. This violates (iii) with $\lambda = a$, $c = b$. And $\lambda = -a$ is a Dutch book: $\lambda \cdot (\bar{v} - P) = a \cdot P - a \cdot \bar{v} < 0$ for every v . (iii) $\Rightarrow$ (iv): for a valid (Φ, m) let λ_φ be the multiplicity of φ in Φ and $c = m$. Then $\lambda \cdot \bar{v}$ is the number of members of Φ true in v , which is $\geq m$, so (iii) gives $\sum_\Phi P = \lambda \cdot P \geq m$. (iv) $\Rightarrow$ (iii): let λ, c be rational with $\lambda \cdot \bar{v} \geq c$ for all v . Multiplying both by a positive common denominator preserves both inequalities, so assume $\lambda \in \mathbb{Z}^F$, $c \in \mathbb{Z}$. For each coordinate φ with $\lambda_\varphi < 0$ use $\lambda_\varphi v(\varphi) = |\lambda_\varphi| v(\varphi^*) - |\lambda_\varphi|$, valid for every $v \in \text{BV}$ since $v(\varphi^*) = 1 - v(\varphi)$, and the same identity for P , valid by negation coherence. Let Φ be the multiset containing λ_φ copies of φ for each $\lambda_\varphi > 0$ and $|\lambda_\varphi|$ copies of φ^* for each $\lambda_\varphi < 0$ (multiplicities add if φ^* also has a positive coefficient), and let $m = c + \sum_{\lambda_\varphi < 0} |\lambda_\varphi|$. Then for every v , $\lambda \cdot \bar{v} = \#_\Phi(v) - \sum_{\lambda_\varphi < 0} |\lambda_\varphi|$, where $\#_\Phi(v)$ counts the members of Φ true in v ; so (Φ, m) is valid. Likewise $\lambda \cdot P = \sum_\Phi P - \sum_{\lambda_\varphi < 0} |\lambda_\varphi|$, so (iv) gives $\lambda \cdot P \geq c$. Finally, $P \equiv 1$ satisfies every valid counting sequent, since validity at any world forces $m \leq |\Phi|$; but it is incoherent, as $\bar{v}$ never has $v(\varphi) = v(\varphi^*) = 1$. $\square$

Proof of Theorem 9.15(b). Let Φ contain a_t copies of A_t , b_t of $\neg A_t$, c_{ij} of $A_i \wedge A_j$ and n_{ij} of $\neg(A_i \wedge A_j)$. Put $N = \sum n_{ij}$ and $x_t = b_t - a_t$, and write $\#(S)$ for the number of members of Φ (with multiplicity) true in the world whose true atoms are S . *Two kinds of worlds.* $\#(\emptyset) = \sum_t b_t + N$. In the world $\{t\}$ all conjunctions are still false, A_t becomes true and $\neg A_t$ false, so $\#(\{t\}) = \#(\emptyset) - x_t$. *The credence sum.* Directly,

$$\sum_\Phi P_k = \frac{\sum_t a_t}{k-1} + \frac{(k-2)\sum_t b_t}{k-1} + N = \#(\emptyset) - \frac{\sum_t x_t}{k-1}.$$

Validity. Let $s = \#(\emptyset) - m$. Validity at $\emptyset$ gives $s \geq 0$, and validity at $\{t\}$ gives $x_t \leq s$. *Violation forces $m \geq k-1$.* Suppose (Φ, m) is valid and $\sum_\Phi P_k < m$, i.e. $\sum_t x_t > (k-1)s$. If $s = 0$ this contradicts $x_t \leq 0$; so $s \geq 1$. Since $a_t \geq 0$, $x_t \leq b_t$, so $\#(\emptyset) \geq \sum_t b_t \geq \sum_t x_t \geq (k-1)s + 1$ by integrality. Therefore $m = \#(\emptyset) - s \geq (k-2)s + 1 \geq k-1$. Contrapositively, every valid counting sequent with $m \leq k-2$ is satisfied. *The $m = 1$ case.* An ordinary sequent $\Gamma \triangleright \Delta$ over

F_k corresponds to $\Phi = \{\gamma^* : \gamma \in \Gamma\} \cup \Delta$, $m = 1$; it is valid iff the counting sequent is, and by negation coherence its union bound is $\sum_{\Phi} P_k \geq 1$. Since $1 \leq k - 2$ for $k \geq 3$, (b) covers it. $\square$

For $k = 3, 4, 5$, `repair_checks.py` (C) minimizes $\sum_{\Phi} P_k - m$ over valid counting sequents with multiplicities in $[0, 8]$ by exact MILP: the minimum is 0 for $m \leq k - 2$ and $-\frac{1}{k-1}$ for $m \leq k - 1$, attained by the repetition-free sequent of (a).

Proof of Proposition 9.16. ($\Rightarrow$) If $P = \sum_v \mu(v)\bar{v}$, the three conditions hold valuation by valuation: $v(\varphi) = v(\varphi \wedge \psi) + v(\varphi \wedge \neg\psi)$ for every Boolean v . ($\Leftarrow$) Taking $\varphi = \psi = \top$ gives $P(\top) = P(\top) + P(\perp)$, so $P(\perp) = 0$. Let $a_1, \dots, a_N$ be the atoms (minimal nonzero elements) of the finite Boolean algebra F ; distinct atoms are disjoint, and every nonzero element lies above some atom. Splitting successively,

$$P(\varphi) = P(\varphi \wedge a_1) + P(\varphi \wedge \neg a_1) = \dots = \sum_{j \leq N} P(\varphi \wedge a_j) + P\left(\varphi \wedge \bigwedge_{j \leq N} \neg a_j\right),$$

using $\varphi \wedge \neg a_1 \wedge \dots \wedge \neg a_{j-1} \wedge a_j \equiv \varphi \wedge a_j$; all formulas involved lie in F up to equivalence. The last term is $P(\perp) = 0$, and $\varphi \wedge a_j$ is a_j or $\perp$; so $P(\varphi) = \sum_{a_j \leq \varphi} P(a_j)$. Each a_j is consistent; pick $v_j \in \text{BV}$ with $v_j(a_j) = 1$. For $\varphi \in F$, $v_j(\varphi) = 1$ iff $a_j \leq \varphi$: if $a_j \not\leq \varphi$ then $a_j \wedge \varphi \equiv \perp$, so $a_j \leq \neg\varphi$. Put $\mu = \sum_j P(a_j)\delta_{v_j}$. It is nonnegative with total mass $\sum_j P(a_j) = P(\top) = 1$, and $\mu(\varphi) = \sum_{a_j \leq \varphi} P(a_j) = P(\varphi)$. The user's identity follows: $P(A \vee B) = P((A \vee B) \wedge A) + P((A \vee B) \wedge \neg A) = P(A) + P(B \wedge \neg A)$ and $P(B) = P(B \wedge A) + P(B \wedge \neg A)$. $\square$

Lemma H.1 (Elementary consequences of Gaifman coherence). *Let P satisfy (G1), (G2). Then:* (i) $P(\neg\varphi) = 1 - P(\varphi)$; (ii) if $\vdash \varphi \leftrightarrow \psi$ then $P(\varphi) = P(\psi)$; (iii) if $\vdash \varphi \rightarrow \psi$ then $P(\varphi) \leq P(\psi)$; (iv) $\max(P(\varphi), P(\psi)) \leq P(\varphi \vee \psi) \leq P(\varphi) + P(\psi)$; (v) $P(\varphi) + P(\psi) - 1 \leq P(\varphi \wedge \psi) \leq \min(P(\varphi), P(\psi))$.

Proof. (i) Apply (G2) and (G1) to $\varphi \vee \neg\varphi$. (ii) $\varphi \vee \neg\varphi$ and $\psi \vee \neg\psi$ are both provable, and both pairs are provably exclusive; so $P(\varphi) + P(\neg\varphi) = 1 = P(\psi) + P(\neg\psi)$. (iii) ψ is provably equivalent to $\varphi \vee (\psi \wedge \neg\varphi)$, a provably exclusive disjunction, so $P(\psi) = P(\varphi) + P(\psi \wedge \neg\varphi) \geq P(\varphi)$ by (ii) and (G2). (iv) The lower bound is (iii). For the upper bound, $P(\varphi \vee \psi) = P(\varphi) + P(\psi \wedge \neg\varphi) \leq P(\varphi) + P(\psi)$ by (ii), (G2) and (iii). (v) The upper bound is (iii). For the lower bound, $1 - P(\varphi \wedge \psi) = P(\neg\varphi \vee \neg\psi) \leq P(\neg\varphi) + P(\neg\psi)$ by (i), (ii), (iv). $\square$

Proof of Theorem 9.17. ($\Leftarrow$) If $P(\varphi) = \mu([\varphi])$: a provable φ lies in every complete consistent theory, so $[\varphi] = S_L$; and if $\vdash \neg(\varphi \wedge \psi)$, then $[\varphi] \cap [\psi] = \emptyset$ and $[\varphi \vee \psi] = [\varphi] \cup [\psi]$. ($\Rightarrow$) By the completeness theorem, $[\varphi] = [\psi]$ iff $\vdash \varphi \leftrightarrow \psi$, and $[\varphi] \cap [\psi] = \emptyset$ iff $\vdash \neg(\varphi \wedge \psi)$. Every clopen subset of S_L is compact, hence a finite union of basic clopens, hence of the form $[\varphi]$. So $\mu_0([\varphi]) := P(\varphi)$ is a well-defined function on the clopen algebra (Lemma H.1(ii)), with $\mu_0(S_L) = P(\top) = 1$, nonnegative, and finitely additive by (G2). It is countably additive on the clopen algebra: if $[\varphi] = \bigsqcup_n [\varphi_n]$, compactness of $[\varphi]$ and openness of the $[\varphi_n]$ give a finite subcover, and disjointness makes all other $[\varphi_n]$ empty. By Carathéodory's extension theorem μ_0 extends to a probability on the σ -algebra generated by the clopens. Since L is countable there are countably many clopens, and they form a basis, so every open set is a countable union of clopens and this σ -algebra is the Borel σ -algebra. Uniqueness: the clopens form a π -system generating it (π - λ theorem). $\square$

Proof of Theorem 9.18. Every sentence is provably equivalent to one built from atomic sentences by $\neg, \wedge, \exists$, and both P (Lemma H.1(ii)) and truth in $\mathbb{N}$ respect provable equivalence; so it suffices to treat such sentences. We show $P(\varphi) = [\mathbb{N} \models \varphi]$ by induction on the number of logical symbols,

simultaneously for all such sentences. *Atomic sentences* are quantifier-free. A true one has $P = 1$ by hypothesis. If σ is false, $\neg\sigma$ is a true quantifier-free sentence, so $P(\sigma) = 1 - P(\neg\sigma) = 0$. *Negation*: $P(\neg\varphi) = 1 - P(\varphi)$ (Lemma H.1(i)). *Conjunction*: if $P(\varphi), P(\psi) \in \{0, 1\}$, the bounds of Lemma H.1(v) coincide and give $P(\varphi \wedge \psi) = \min(P(\varphi), P(\psi))$. *Existential*: numerals $\underline{i} = S \cdots S0$ contain no logical symbols, so each instance $\varphi(\underline{i})$ has fewer logical symbols than $\exists x \varphi(x)$, and by induction $P(\varphi(\underline{i})) = [\mathbb{N} \models \varphi(\underline{i})]$. By Lemma H.1(iv), applied repeatedly, $\max_{i \leq n} P(\varphi(\underline{i})) \leq P(\bigvee_{i \leq n} \varphi(\underline{i})) \leq \sum_{i \leq n} P(\varphi(\underline{i}))$, so $P(\bigvee_{i \leq n} \varphi(\underline{i})) = \max_{i \leq n} P(\varphi(\underline{i})) \in \{0, 1\}$. By the Gaifman condition, $P(\exists x \varphi(x))$ is 1 iff some $\varphi(\underline{i})$ is true in $\mathbb{N}$, iff $\mathbb{N} \models \exists x \varphi(x)$, because every natural number is the value of a numeral. $\square$

Proof of Corollary 9.19. We use three properties from Garrabrant et al. (2016): a market is a *computable* sequence of rational pricings $(\mathbb{P}_n)$; by Limit Coherence the limit $\mathbb{P}_\infty(\varphi) = \lim_n \mathbb{P}_n(\varphi)$ exists and is a coherent probability relative to Γ ; and by Non-Dogmatism $\Gamma \not\models \neg\varphi$ implies $\mathbb{P}_\infty(\varphi) > 0$. (We cite these properties by name; we have confirmed the statements but not the theorem numbering of the arXiv version.) By Limit Coherence, $\mathbb{P}_\infty$ is Gaifman-coherent and gives 1 to every theorem of Γ , in particular to every true quantifier-free sentence, which $\mathbf{Q} \subseteq \Gamma$ proves. Suppose it satisfied the Gaifman condition. Then $\mathbb{P}_\infty = \delta_{\text{Th}(\mathbb{N})}$ by Theorem 9.18. Two arguments refute this. *Via Non-Dogmatism.* By the Gödel–Rosser theorem (Gödel, 1931; Rosser, 1936), the consistent r.e. $\Gamma \supseteq \mathbf{Q}$ is incomplete: some σ has $\Gamma \not\models \sigma$ and $\Gamma \not\models \neg\sigma$. Let ψ be whichever of $\sigma, \neg\sigma$ is false in $\mathbb{N}$. Then $\Gamma \not\models \neg\psi$ (as $\neg\sigma$ is equivalent to σ), so $\mathbb{P}_\infty(\psi) > 0 = \delta_{\text{Th}(\mathbb{N})}(\psi)$. *Via Δ_2 .* Since $(n, \varphi) \mapsto \mathbb{P}_n(\varphi)$ is computable, “ $\mathbb{P}_n(\varphi) > \frac{1}{2}$ ” is decidable. If the limit were $\delta_{\text{Th}(\mathbb{N})}$, then $\varphi \in \text{Th}(\mathbb{N})$ iff $\exists N \forall n \geq N \mathbb{P}_n(\varphi) > \frac{1}{2}$ (a Σ_2 condition) iff $\forall N \exists n \geq N \mathbb{P}_n(\varphi) > \frac{1}{2}$ (a Π_2 condition): if $\mathbb{P}_\infty(\varphi) = 1$ both hold, and if $\mathbb{P}_\infty(\varphi) = 0$ neither does. So $\text{Th}(\mathbb{N})$ would be Δ_2 , contradicting Tarski’s theorem that $\text{Th}(\mathbb{N})$ is not arithmetical. *The general statement.* The Δ_2 argument used only that the credence is the pointwise limit of a computable sequence of rational-valued functions, that it is coherent, and that it gives 1 to every true quantifier-free sentence; so no such credence satisfies the Gaifman condition. *The quantifier-free hypothesis is needed.* Let T be the theory of the one-element structure, in which $S0 = 0 = 0 + 0 = 0 \cdot 0$. T is decidable, so δ_T is computable and coherent; and it satisfies the Gaifman condition with respect to the numerals, because the only element is the value of a numeral. It fails the hypothesis only by giving $P(0 = S0) = 1$. $\square$

H.4 Contexts

Proof of Theorem 9.22. Note first that, since $C_i = \text{Th}_i \circ \text{Mod}_i$, every set of the form $\text{Th}_i(K)$ is C_i -closed (Lemma 9.2: $\text{Th}_i \text{Mod}_i \text{Th}_i = \text{Th}_i$), and $\text{Th}_i \text{Mod}_i(T) = T$ for every C_i -closed T . (a) For every i and φ : $c^\Gamma \models i:\varphi$ iff $\text{Mod}_i(T_i) \subseteq \text{Mod}_i(\varphi)$ iff $\varphi \in \text{Th}_i \text{Mod}_i(T_i) = T_i$. Hence c^Γ satisfies $K_i \cup \Gamma_i \subseteq T_i$. It is bridge-compatible: if $c^\Gamma \models j_k:\varphi_k$ for all premises of a bridge rule, then $\varphi_k \in T_{j_k}$, so $\psi \in T_i$ by (Br), so $c^\Gamma \models i:\psi$. (b) Let $c \models \Gamma$ be a model and $U_i = \text{Th}_i(c_i)$. Each U_i is C_i -closed and contains $K_i \cup \Gamma_i$. The family (U_i) is closed under bridges: if $\varphi_k \in U_{j_k}$ for all k , then $c \models j_k:\varphi_k$, so $c \models i:\psi$ by bridge-compatibility, i.e. $\psi \in U_i$. By leastness of $\text{Der}(\Gamma)$, $T_i \subseteq U_i$, so every $m \in c_i$ satisfies T_i : $c_i \subseteq \text{Mod}_i(T_i) = c_i^\Gamma$. (c) $(\Rightarrow)$ If $\varphi \in T_i$ and $c \models \Gamma$ is a model, then by (b) every $m \in c_i \subseteq c_i^\Gamma$ satisfies φ . $(\Leftarrow)$ If $\varphi \notin T_i$, then $c^\Gamma \not\models i:\varphi$ by (a), and c^Γ is a model of Γ . $\square$

Proof of Remark 9.24. With $M_i = \text{Fix}(C_i) \setminus \{L_i\}$ we have $\text{Th}_i \circ \text{Mod}_i = C_i$: if $C_i(X) \neq L_i$, then $C_i(X)$ is itself a model and $\text{Th}_i \text{Mod}_i(X) = \bigcap \{T \in M_i : X \subseteq T\} = C_i(X)$; otherwise

$\text{Mod}_i(X) = \emptyset$ and $\text{Th}_i(\emptyset) = L_i$. So Theorem 9.22 applies, and $c_i^\Gamma = \text{Mod}_i(T_i)$ is nonempty iff T_i is proper, since then $T_i \in c_i^\Gamma$, while no proper closed theory contains L_i . A falsum with $\text{Mod}_i(\perp_i) = \emptyset$ is exactly a C_i -explosive $\perp_i$, and then $T_i \neq L_i$ iff $\perp_i \notin T_i$. Without explosiveness the two readings come apart: let C_i be the identity closure (no learned rules), $K_i = \{\perp_i\}$ and no bridges into i ; then $\perp_i \in T_i = \{\perp_i\} \neq L_i$, yet $\{\perp_i\} \in c_i^\Gamma$. `repair_checks.py` (B) checks the equivalence on 3,000 random finite closure systems with an explosive falsum. $\square$

Proposition H.2 (Equilibria; non-monotone bridges). *[T6 Props 4.3, 4.4, as repaired] Call $S = (S_i)$ an equilibrium if $S_i = C_i(K_i \cup \Gamma_i \cup \{\psi : (\dots \Rightarrow i:\psi) \text{ has all premises in } S\})$ for all i (the monotone case of [Brewka and Eiter 2007](#), with each context accepting exactly the closure of its knowledge base). (a) $\text{Der}(\Gamma)$ is the least equilibrium, and there is an equilibrium with $\perp_i \notin S_i$ for every $i \in D$ iff the system is D -coherent. (b) Let context i have consistent K_i with $K_i \not\models p$ and no premises ($\Gamma_i = \emptyset$), and add the single non-monotone bridge “ $i:p$ if not $i:p$ ” (negation as failure). Then K_i is consistent, so the context is coherent without its bridge, yet there is no equilibrium. So existence for default-style bridges needs an extra condition, such as stratification or the absence of odd loops through negation as failure; stratification is sufficient, not necessary, and coherence alone does not supply it.*

Proof of Proposition H.2. Write $R(S)_i$ for the right-hand side of the equilibrium equation, and $T = \text{Der}(\Gamma)$. (a) $T \supseteq R(T)$: each T_i is C_i -closed and contains K_i , Γ_i and, by (Br), the head of every bridge rule whose premises lie in T . Conversely the family $R(T)$ contains $K \cup \Gamma$, is locally closed, and is bridge-closed: a rule whose premises lie in $R(T) \subseteq T$ has its premises in T , so its head lies in $R(T)$ by definition. By leastness of T , $T \subseteq R(T)$. So $T = R(T)$ is an equilibrium. Any equilibrium $S = R(S)$ is locally closed, contains $K \cup \Gamma$ and is bridge-closed, so $S \supseteq T$. Hence a D -consistent equilibrium forces T to be D -consistent, and conversely T itself is an equilibrium. (b) If $p \in S_i$, the rule is inapplicable, so $S_i = C_i(K_i) \not\models p$; if $p \notin S_i$, the rule fires and $p \in S_i$. Either way $S_i \neq R(S)_i$. $\square$

Proof of Theorem 9.26. Projection lemma. For a consistent closed theory T over atoms A and $B \subseteq A$: $\text{Mod}(T)|_B = \text{Mod}_B(T \cap L(B))$. The inclusion $\subseteq$ is clear. For $\supseteq$, let a B -valuation w satisfy $T \cap L(B)$, and suppose T together with the literals over B true in w were unsatisfiable. By compactness, $T \vdash \neg(l_1 \wedge \dots \wedge l_r)$ for some literals l_k true in w ; this sentence lies in $T \cap L(B)$ and is false in w , a contradiction. By the lemma, mutual conservativity on an edge $\{i, j\}$ says $\text{Mod}(T_i)|_{A_i \cap A_j} = \text{Mod}(T_j)|_{A_i \cap A_j}$. *Gluing.* Root the tree at i_0 and let $m_{i_0} \models T_{i_0}$ be given; assign the atoms of A_{i_0} accordingly. Visit the remaining vertices in breadth-first order. When c is visited, its parent p has been visited and its children have not, so p is the only visited neighbour of c and the tree path from c to any visited vertex passes through p . If an atom $a \in A_c$ is already assigned, it lies in A_d for some visited d ; by running intersection it lies in every A_e on the path from c to d , in particular in A_p . So the assigned atoms of A_c lie in $A_c \cap A_p$. After p was visited the global assignment restricted to A_p was a model $m_p \models T_p$, and assigned atoms never change. So the current assignment on $A_c \cap A_p$ is $m_p|_{A_c \cap A_p} \in \text{Mod}(T_p)|_{A_c \cap A_p} = \text{Mod}(T_c)|_{A_c \cap A_p}$, and some $m_c \models T_c$ agrees with it; extend the assignment by m_c . Atoms in no A_i are assigned arbitrarily. At the end the assignment restricted to each A_c is $m_c \models T_c$. $\square$

`contexts_local_global.py` checks the theorem on one fixed three-node path cover with 7,131 random local-theory systems and on 2,000 random tree covers with 2–5 nodes, with no failure; a negative control without running intersection fails in 437 of 2,000 cases.

Relation to the context-tree calculus of L7 §8 (proof sketch). L7's calculus has stipulations (local axioms), imports $\pi(c):\varphi \Rightarrow c:\varphi$ for φ in the import filter F_c , discharge $c:\psi \Rightarrow \pi(c):A \rightarrow \psi$, and side-conditioned export $c:\varphi_\beta(t)$, $\pi(c):\sigma_\beta(t) \Rightarrow \pi(c):\varepsilon_\beta(t)$. All are monotone bridge rules, so it is an MC system, and Theorem 9.22 is its completeness theorem for local-models semantics with bridges read as compatibility constraints; there an exported ε shrinks the parent's belief state. L7's own semantics differs: the models of the root are fixed by background and observations and do not depend on exports, and bridges are required to be sound; L7 proves soundness for it assuming every step and export instance is sound. The two semantics coincide when the local consequence is the complete base logic and every step and export instance is sound: then $T_c = \text{Th}(\text{Mod}_{L7}(c))$ for every c , where $\subseteq$ is L7's soundness proposition, and $\supseteq$ goes by induction down the tree (at the root by local completeness; an idealized child imports exactly $F_c \cap T_{\pi(c)}$; a supposition child imports all of $T_{\pi(c)}$ and its assumption). Without that hypothesis they differ: if an unsound export yields $@:Q = 5$ while background and observations prove $Q = 3$, MC derives $@:\perp$ and $c_{\mathbb{Q}}^\Gamma = \emptyset$, as Theorem 9.22 requires, while L7's model class of the root stays nonempty. For supposition contexts with full import, the canonical chain always satisfies $c_c^\Gamma = c_{\pi(c)}^\Gamma \cap \text{Mod}(A)$.

H.5 Learned calculi

Existence of the Leibniz congruence. The join of a family of congruences of $\mathbf{A}$ is the transitive closure of their union. If each congruence in the family is compatible with D , then a and b related by the join are connected by a chain $a = c_0 \theta_1 c_1 \cdots \theta_r c_r = b$ with each θ_k in the family; each step preserves membership in D , so the join is compatible. The join of all compatible congruences is therefore the largest one.

Proof of Theorem 9.30. (a) *Claim:* $\Omega(T) = h^{-1}(\Omega_{\mathbf{A}}D)$, where $h^{-1}(\theta) = \{(a, b) : (ha, hb) \in \theta\}$. The preimage of a congruence under a homomorphism is a congruence. It is compatible with T : if $(ha, hb) \in \Omega_{\mathbf{A}}D$ and $a \in T$, then $ha \in D$, so $hb \in D$ and $b \in T$. Maximality: let θ' be any congruence of Fm compatible with T . The kernel $\ker h$ is compatible with T , as $T = h^{-1}(D)$ is a union of $\ker h$ -classes, so $\theta'' = \theta' \vee \ker h$ is compatible with T . Let $\theta_{\mathbf{A}} = \{(ha, hb) : (a, b) \in \theta''\}$. It is reflexive because h is onto, and symmetric. It is transitive: if $(ha, hb), (hb', hc) \in \theta_{\mathbf{A}}$ come from pairs in θ'' with $hb = hb'$, then $b\theta''b'$ because $\ker h \subseteq \theta''$, so $a\theta''c$. It is compatible with the operations: if $(x_l, y_l) \in \theta_{\mathbf{A}}$ come from $(a_l, b_l) \in \theta''$, then for an operation f , $(f(\bar{x}), f(\bar{y})) = (hf(\bar{a}), hf(\bar{b}))$ with $f(\bar{a})\theta''f(\bar{b})$. And it is compatible with D : if $(ha, hb) \in \theta_{\mathbf{A}}$ comes from $(a, b) \in \theta''$ and $ha \in D$, then $a \in T$, so $b \in T$ and $hb \in D$. Hence $\theta_{\mathbf{A}} \subseteq \Omega_{\mathbf{A}}D$, and $\theta' \subseteq \theta'' \subseteq h^{-1}(\theta_{\mathbf{A}}) \subseteq h^{-1}(\Omega_{\mathbf{A}}D)$. *Isomorphism.* Define $[a]_{\Omega(T)} \mapsto [ha]_{\Omega_{\mathbf{A}}D}$. By the claim it is well defined and injective; it is onto because h is; it is a homomorphism because h is; and it maps $T/\Omega(T)$ onto $D/\Omega_{\mathbf{A}}D$ because $a \in T$ iff $ha \in D$. If $\langle \mathbf{A}, D \rangle$ is reduced, $\Omega_{\mathbf{A}}D$ is the identity and the right-hand side is $\langle \mathbf{A}, D \rangle$. (b) A maximal consistent theory of CPC is $v^{-1}(1)$ for a Boolean homomorphism $v : \text{Fm} \rightarrow \mathbf{2}$. This v is onto, since $v(p \vee \neg p) = 1$ and $v(p \wedge \neg p) = 0$. The matrix $\langle \mathbf{2}, \{1\} \rangle$ is reduced: the only congruence of $\mathbf{2}$ besides the identity identifies 0 with 1, which is not compatible with $\{1\}$. Apply (a). $\square$

`repair_checks.py` (D) checks a non-surjective example for the remark after Theorem 9.30: in the four-element Heyting chain $0 < a < b < 1$, $h(p) = a$ generates $\{0, a, 1\} \cong \mathbf{H}_3$, which is reduced, so the reduced Lindenbaum matrix of $h^{-1}(1)$ is $\langle \mathbf{H}_3, \{1\} \rangle$, not the intended four-element matrix.

Verification of Example 9.32. (i) $h : \mathbf{Fm} \rightarrow \mathbf{H}_3$ is a Heyting homomorphism with $h(p) = \frac{1}{2}$, $h(\perp) = 0$, $h(\top) = 1$, so it is onto. $T = h^{-1}(1)$ is an IPC-theory by soundness of IPC for Heyting algebras, and $p \notin T$. T is maximal among theories omitting p : let $\psi \notin T$, so $h(\psi) \in \{0, \frac{1}{2}\}$. If $h(\psi) = \frac{1}{2}$ then $h(\psi \leftrightarrow p) = 1$, so $\psi \leftrightarrow p \in T$ and any theory containing T and ψ contains p . If $h(\psi) = 0$ then $h(\neg\psi) = 1$, so $\neg\psi \in T$ and any theory containing T and ψ contains $\perp$, hence p . So every theory strictly above T contains p , and T is a point. It is not maximal consistent, since $h(p \vee \neg p) = \frac{1}{2} \vee 0 = \frac{1}{2}$ puts neither p nor $\neg p$ in T . $\langle \mathbf{H}_3, \{1\} \rangle$ is reduced: a congruence identifying $\frac{1}{2}$ with 1 is not compatible with $\{1\}$; one identifying 0 with $\frac{1}{2}$ also identifies $0 \rightarrow 0 = 1$ with $\frac{1}{2} \rightarrow 0 = 0$, hence everything, and is not compatible either. By Theorem 9.30(a) the reduced Lindenbaum matrix of T is $\langle \mathbf{H}_3, \{1\} \rangle$. (ii) In a one-element structure $\forall x \exists y \varphi$ and $\exists y \forall x \varphi$ are equivalent for every formula φ , so every instance of the schema is sound there; with equality, a structure in which every instance of the schema holds must have one element (take $R(x, y)$ to be $x = y$: $\forall x \exists y x = y$ holds in every structure, and $\exists y \forall x x = y$ says there is one element). So the models of first-order logic plus the schema are exactly the one-element structures, and a designated context asserting two distinct objects has none. $\square$

I Proofs for Section 10

Lemma 10.6, Proposition 10.7, Theorem 10.8, Proposition 10.11, Proposition 10.17, Lemma 10.20, Theorem 10.22(a), Theorem 10.24, Proposition 10.28 and Theorem 10.36 are proved in full in the main text. This appendix proves the remaining results of Section 10 and verifies Example 10.16. Notation is as in Section 10. Throughout, Cl_R is a closure operator (extensive, monotone, idempotent; Lemma 2.3), and for a first-order complete h and finite $\Gamma \cup \{y\} \subseteq D_h$ we use freely that $\Gamma \models_h y$ iff $T_h \cup \rho_h \Gamma \models \rho_h y$ iff $T_h \vdash \bigwedge \rho_h \Gamma \rightarrow \rho_h y$ (Gödel completeness).

I.1 Sound bounded-gap verification

Proof of Theorem 10.9. Practice data are steps, so they range over a countable set and each p_h is a probability mass function. Let $X_1, X_2, \dots$ be the practice data, i.i.d. from p_{h^*} and independent of everything the prover and the verifier do. Every other datum is a truthful constraint $c : \mathcal{H} \rightarrow \{0, 1\}$ with $c(h^*) = 1$. Moreover $c(h') = 1$ for every $h' \equiv_g h^*$: an escalation or link answer is a $\text{St}_{h^*}^g$ -label, and $\text{St}_{h'}^g = \text{St}_{h^*}^g$; a paradox exists only for hypotheses incoherent on some $A \in \mathcal{A}$, and h' is coherent on every A on which h^* is; an object datum is a restriction of a member of $\mathfrak{A}h^* = \mathfrak{A}h'$.

At time t , let $n(t)$ be the number of practice data seen and $c_1, \dots, c_{k(t)}$ the constraints received. Put

$$L_t(h) := \prod_{i \leq n(t)} p_h(X_i) \cdot \prod_{j \leq k(t)} c_j(h), \quad w_t(h) := \frac{w(h)L_t(h)}{\sum_{h'} w(h')L_t(h')}, \quad Z_t := \sum_h w(h) \frac{L_t(h)}{L_t(h^*)}.$$

Almost surely $L_t(h^*) > 0$, since $p_{h^*}(X_i) > 0$ and $c_j(h^*) = 1$, so $w_t(h) = w(h)L_t(h)/(L_t(h^*)Z_t)$. For $h \in [h^*]$, shadow-measurability gives $p_h = p_{h^*}$, and $c_j(h) = 1$ by the previous paragraph; so $L_t(h) = L_t(h^*)$ and

$$w_t([h^*]) = W^*/Z_t.$$

Since every $c_j(h) \leq 1$, pathwise $Z_t \leq Z_n^H$, where $Z_n^H := \sum_h w(h) \prod_{i \leq n} p_h(X_i)/p_{h^*}(X_i)$ depends

only on the practice data. Let $\mathcal{F}_n := \sigma(X_1, \dots, X_n)$. For every h ,

$$\mathbb{E}\left[\frac{p_h(X_{n+1})}{p_{h^*}(X_{n+1})} \mid \mathcal{F}_n\right] = \sum_{x: p_{h^*}(x) > 0} p_h(x) \leq 1,$$

so by Tonelli (all terms are nonnegative) $\mathbb{E}[Z_{n+1}^H \mid \mathcal{F}_n] \leq Z_n^H$: Z^H is a nonnegative supermartingale with $Z_0^H = \sum_h w(h) = 1$. Ville's inequality (Ville, 1939) gives $\mathbb{P}[\sup_n Z_n^H \geq 1/\delta'] \leq \delta'$. Let $G := \{\sup_n Z_n^H < 1/\delta'\}$; it depends only on the practice data, and $\mathbb{P}(G) \geq 1 - \delta'$. On G , for every prover and every t , $Z_t < 1/\delta'$, so $w_t([h^*]) > W^*\delta' \geq \delta$. If $q \notin \text{St}_{h^*}^g$, then $q \notin \text{St}_h^g$ for every $h \in [h^*]$, so the posterior mass of $\{h : q \notin \text{St}_h^g\}$ is at least $w_t([h^*]) > \delta$, and q is not accepted. $\square$

Proof of Theorem 10.10. (a) Let the oracle-answered escalations of an honest prover be $s_1, s_2, \dots$, in order; each s_i lies in $\text{St}_{h^*}^g$ and, when escalated, outside $\bigcap_{h \in \text{VS}} \text{St}_h^g$. After the oracle answers “yes” to s_j , every surviving h has $s_j \in \text{St}_h^g$. So when s_i is escalated, $\mathcal{F}_1 := \{\text{St}_h^g : h \in \text{VS}\}$ is contained in $\mathcal{F}_2 := \{S \in \mathcal{H}^g : S \supseteq \{s_1, \dots, s_{i-1}\}\}$, hence $\bigcap \mathcal{F}_2 \subseteq \bigcap \mathcal{F}_1 \not\ni s_i$. Thus $s_1, s_2, \dots$ is an elastic chain in $\text{St}_{h^*}^g$ for $\mathcal{H}^g$, and there are at most $\text{el}(\mathcal{H}^g, \text{St}_{h^*}^g)$ of them. For an elastic chain $s_1, \dots, s_m$, the families $F_i := \{S \in \mathcal{H}^g : S \supseteq \{s_1, \dots, s_i\}\}$ strictly decrease (some member of F_{i-1} lacks s_i) and $F_m \ni \text{St}_{h^*}^g$, so $m \leq |\mathcal{H}^g| - 1 \leq |\mathcal{H}/\equiv_g| - 1$, because $\equiv_g$ -equivalent hypotheses have the same g -step relation.

(b) Use the notation of the previous proof. An honest query q escalated at time u has posterior mass $\pi_u := w_{u-1}(\{h : q \notin \text{St}_h^g\}) \geq \delta$. The deterministic answer “yes” removes exactly these hypotheses, so it multiplies Z by $1 - \pi_u \leq 1 - \delta$. Other constraints multiply Z by at most 1. A practice datum X multiplies Z by $F := \sum_h w_{\text{prev}}(h) p_h(X)/p_{h^*}(X)$, where w_{prev} is the current posterior. Each practice datum is independent of everything that precedes it, including the prover's coins and choices (the prover does not foresee practice data), so $\mathbb{E}[F \mid \text{past}] \leq 1$. Let M_n be the product of the factors F over the first n practice data; it is a nonnegative supermartingale with $M_0 = 1$. Since the members of $[h^*]$ keep weight $w(h)$ in Z , $Z_t \geq W^*$. Hence $W^* \leq Z_t \leq M_{n(t)}(1 - \delta)^{\#\text{ESC}_t}$, and on the event $\{\sup_n M_n < 1/\delta''\}$, which has probability at least $1 - \delta''$ by Ville's inequality, $\#\text{ESC}_t \leq \ln(1/(W^*\delta''))/\ln(1/(1 - \delta)) \leq (\ln(1/W^*) + \ln(1/\delta''))/\delta$ for all t .

(c) Let the hypotheses be h_u , $u \in U$, with $\text{St}_{h_u}^g = B \cup (U \setminus \{u\})$, and target h_{u^*} . Any enumeration $s_1, \dots, s_{n-1}$ of $U \setminus \{u^*\}$ is an elastic chain: $\bigcap \{S \in \mathcal{H}^g : S \supseteq \{s_1, \dots, s_{i-1}\}\} = B \cup \{s_1, \dots, s_{i-1}\} \not\ni s_i$. So $\text{el} = n - 1$, the upper bound of (a). It is attained when the practice data lie in B : then VS consists exactly of the h_u with $u \notin \{s_1, \dots, s_{i-1}\}$ when s_i is queried, and s_i is escalated. For the lower bound, take practice distributions supported on B and identical for all hypotheses (shadow-measurable), and the honest prover that queries $s_1, \dots, s_{n-1}$ in order. Under target h_{s_i} the step s_i is invalid, and the answers to $s_1, \dots, s_{i-1}$ are the same (“yes”) under h_{u^*} and h_{s_i} . So the law of the transcript up to the verifier's answer in round i is the same under both targets, and δ' -soundness under h_{s_i} gives $\mathbb{P}_{h_{u^*}}[s_i \text{ accepted in round } i] \leq \delta'$. Summing, the expected number of these honest queries that are not accepted (escalated or rejected) under h_{u^*} is at least $(1 - \delta')(n - 1) = (1 - \delta')(1/W^* - 1)$ for the uniform prior; a verifier that never rejects, such as $V_t^{g, \delta}$, escalates them. With $\delta = W^*\delta' = \delta'/n$, the bound of (b) is $n(\ln n + \ln(1/\delta''))/\delta'$, and the ratio to the lower bound is at most $2(\ln n + \ln(1/\delta''))/(\delta'(1 - \delta')) = O((\ln(1/W^*) + \ln(1/\delta''))/\delta')$ for $n \geq 2$ and $\delta' \leq 1/2$. This is the argument of T1 Thm 3.2 and Cor 4.5 (Section 4). $\square$

I.2 Identification

Proof of Proposition 10.13. Write the interaction as a sequence of rounds. In round $k + 1$ the learner (and any prover whose strategy depends on the target only through its shadow) issues queries Q_{k+1} drawn from a kernel that depends only on the history H_k and its own coins; then the environment issues a datum D_{k+1} . By the assumptions, the conditional law of D_{k+1} given (H_k, Q_{k+1}) is $K(\text{sh}, H_k, Q_{k+1})$ for a kernel K and the shadow sh of the target: practice data depend on $\text{St}_{h^*}^g$ and the history; escalation and link answers are $\text{St}_{h^*}^g$ -labels; object data depend on $\mathfrak{A}h^*$ and the history; and paradoxes are computed by the learner from the fixed family $\mathcal{A}$ and the hypotheses themselves, independently of the target. (Since $h \equiv_g h'$ includes agreement on coherence, the same $\mathcal{A}$ satisfies the designation promise under both targets.) Let P_h be the law of the interaction under target h . By induction on k , P_h and $P_{h'}$ agree on (H_k, Q_{k+1}) : the step from k to $k + 1$ composes the same kernels. Hence they agree on all cylinder sets, and so on the whole sequence. Any learner's output is a function of the history and its coins, so it has the same law under h and h' . The closure-level version is the same argument with $\models$ in place of St^g . $\square$

Proof of Lemma 10.14. Write $D = D_h$ and $T = T_h$.

(i) $\subseteq$: let $v = v_M \circ \rho_h$ with $M \models T$. Since $T \vdash \rho_h(\sim x) \leftrightarrow \neg \rho_h(x)$, $v(\sim x) = 1 - v(x)$. If $\Gamma \subseteq v^{-1}(1)$ is finite and $\Gamma \models_h y$, then $M \models \rho_h \Gamma$ and $T \cup \rho_h \Gamma \models \rho_h y$, so $v(y) = 1$.

$\supseteq$: let v be a coherent valuation and $\Theta := T \cup \{\rho_h(x) : v(x) = 1\}$. Suppose Θ is inconsistent. By compactness some finite $\Gamma \subseteq v^{-1}(1)$ makes $T \cup \rho_h \Gamma$ inconsistent, so $\Gamma \models_h y$ for every $y \in D$. Pick $x \in D$ (here $D \neq \emptyset$ is used); one of $x, \sim x$ has v -value 0, call it y ; but $\Gamma \models_h y$ forces $v(y) = 1$, a contradiction. So Θ has a model M . If $v(x) = 1$ then $M \models \rho_h(x)$. If $v(x) = 0$ then $v(\sim x) = 1$, so $M \models \rho_h(\sim x)$ and hence $M \models \neg \rho_h(x)$. Thus $v = v_M \circ \rho_h \in \mathfrak{A}h$.

(ii) $(\Rightarrow)$ is (i) $\subseteq$, together with the domain clause in the definition of $\models_h$. ($\Leftarrow$) If $\Gamma \cup \{y\} \subseteq D$ and $\Gamma \not\models_h y$, then $T \cup \rho_h \Gamma \cup \{\neg \rho_h y\}$ is consistent, by completeness; a model M of it gives $v = v_M \circ \rho_h \in \mathfrak{A}h$ with $v(\Gamma) = 1$ and $v(y) = 0$.

(iii) A is h -coherent iff $\perp \notin \text{Cl}_{R_h}(\rho_h A)$ iff $T \cup \rho_h A$ is consistent iff it has a model M , iff $v_M \circ \rho_h$ is a member of $\mathfrak{A}h$ with $v(A) = 1$. For the second equivalence: if A is coherent, pick $x \in D$; in a model of $T \cup \rho_h A$ one of $x, \sim x$ is false, call it y , and then $A \not\models_h y$. If A is incoherent, $T \cup \rho_h A$ is inconsistent and $A \models_h y$ for every $y \in D$.

The converses. If $\mathfrak{A}h \neq \emptyset$, D is the common domain of its members and (ii) recovers $\models_h$. Coherence of finite subsets of D recovers $\models_h$, since for $\Gamma \cup \{y\} \subseteq D$, $\Gamma \models_h y$ iff $T \cup \rho_h \Gamma \cup \{\neg \rho_h y\}$ is inconsistent iff $\Gamma \cup \{\sim y\}$ is incoherent. That neither holds unconditionally: two hypotheses with inconsistent theories and different domains both have $\mathfrak{A} = \emptyset$; and two hypotheses that differ only on steps avoiding every member of a fixed $\mathcal{A}$ agree on coherence on $\mathcal{A}$. $\square$

The key step in Theorem 10.15 is the following lemma.

Lemma I.1 (Upper bracket). *Assume the hypotheses of Theorem 10.15, and let $h \in \mathcal{H}$ have $D_h \supseteq D^*$. Call an h^* -datum the restriction of some $u \in \mathfrak{A}h^*$ to a finite subset of D^* .*

(i) *If $\models_h|_{D^*} \not\subseteq \models_{h^*}$, some h^* -datum refutes h .*

(ii) *If $\models_h|_{D^*} \subseteq \models_{h^*}$, no h^* -datum refutes h , and h is coherent on every finite $A \subseteq D^*$ on which h^* is coherent.*

Proof. (i) Take $(\Gamma \triangleright y) \in \models_h \setminus \models_{h^*}$ with $\Gamma \cup \{y\} \subseteq D^*$. By Lemma 10.14(ii) for h^* , some $u \in \mathfrak{A}h^*$ has $u(\Gamma) = 1$ and $u(y) = 0$. Let $v := u|_{\Gamma \cup \{y\}}$, an h^* -datum with $\text{dom } v \subseteq D_h$. If v extended to some $v_M \circ \rho_h \in \mathfrak{A}h$, then $M \models T_h \cup \rho_h \Gamma$ and $M \not\models \rho_h y$, contradicting $\Gamma \models_h y$. So v refutes h .

(ii) Let $v = u|_F$ with $u \in \mathfrak{A}h^*$ and $F \subseteq D^*$ finite, and let $\Gamma_v := \{x \in F : v(x) = 1\} \cup \{\sim x : x \in F, v(x) = 0\}$, a finite subset of D^* (which is negation-closed). Suppose v extended to no member of $\mathfrak{A}h$. Then $T_h \cup \rho_h \Gamma_v$ is inconsistent, since a model of it would induce an extension of v (using $T_h \vdash \rho_h(\sim x) \leftrightarrow \neg \rho_h(x)$). Pick $x \in D^*$. Then $\Gamma_v \models_h x$ and $\Gamma_v \models_h \sim x$, and both steps have their occurrences in D^* , so $\Gamma_v \models_{h^*} x$ and $\Gamma_v \models_{h^*} \sim x$. But u makes Γ_v true, so by Lemma 10.14(ii) for h^* it makes both x and $\sim x$ true, which is impossible. For coherence: if h were incoherent on $A \subseteq D^*$, then $A \models_h x$ and $A \models_h \sim x$ for $x \in D^*$, hence $A \models_{h^*} x$ and $A \models_{h^*} \sim x$, and $T^* \cup \rho^* A$ would be inconsistent. $\square$

Proof of Theorem 10.15. (a) If $h \approx h'$, then $D_h = D_{h'}$ (both are recovered from the common relation), $\mathfrak{A}h = \mathfrak{A}h'$ by Lemma 10.14(i) (both are the coherent valuations of the same relation, for the same negation), and h, h' are coherent on the same sets by Lemma 10.14(iii), applied to $A \cap D_h$ (partial-reading convention). So $h \equiv_\infty h'$. The converse is part of the definition of $\equiv_\infty$. Nonempty domains are needed: with empty readings, a consistent and an inconsistent theory have the same (empty) $\models$ but different $\mathfrak{A}$.

(b) By (a), $\equiv_\infty$ and $\approx$ coincide; apply the closure-level version of Proposition 10.13.

(c) *Every $h \notin U^*$ is eventually refuted.* If $\models_{h^*} \not\subseteq \models_h$, some $s \in \models_{h^*} \setminus \models_h$ is presented in (P) and refutes h . Otherwise $\models_{h^*} \subseteq \models_h$; then $D_h \supseteq D^*$, because $\{x\} \models_{h^*} x$ for $x \in D^*$; and since all steps of $\models_{h^*}$ have occurrences in D^* , $\models_{h^*} \subseteq \models_h|_{D^*}$, with strict inclusion as $h \notin U^*$. So $\models_h|_{D^*} \not\subseteq \models_{h^*}$, and by Lemma I.1(i) some h^* -datum refutes h ; it is presented in (O). *No $h \in U^*$ is ever refuted.* Practice data lie in $\models_{h^*} = \models_h|_{D^*}$; $D_h \supseteq D^*$ as before; and by Lemma I.1(ii) no object datum refutes h and h is coherent on every designated context (which lies in D^* and is h^* -coherent). As $\mathcal{H}$ is finite, after finitely many data the survivors are exactly U^* , and the intersection of their relations, restricted to D^* , is $\models_{h^*}$. Refuting h by a practice datum s requires a certificate that $s \notin \models_h$, i.e. that $T_h \cup \rho_h \Gamma \not\models \rho_h y$; this is co-r.e. and in general not semi-decidable (for $T_h = \text{ZFC}$), so (c) describes the ideal version space.

(d) *Practice.* The data are the elements of $\text{St}_{h^*}^g$, and a datum s refutes h iff $s \notin \text{St}_h^g$. So complete presentation eventually refutes exactly the h with $\text{St}_{h^*}^g \not\subseteq \text{St}_h^g$. Since $\{x\} \triangleright x$ needs zero rule applications, $(\{x\} \triangleright x) \in \text{St}_{h^*}^g$ for each $x \in D^*$, and practice survivors have $D_h \supseteq D^*$. *Objects and paradoxes.* Among practice survivors, Lemma I.1 shows that some presented object datum refutes h iff $\models_h|_{D^*} \not\subseteq \models_{h^*}$, and that paradoxes refute none of those with $\models_h|_{D^*} \subseteq \models_{h^*}$. *Conclusion.* As $\mathcal{H}$ is finite, the survivors are eventually exactly S^g . $h^* \in S^g$, and every member of S^g contains $\text{St}_{h^*}^g$, so the intersection of the survivors' g -step relations is $\text{St}_{h^*}^g$. The upper bracket holds by definition of S^g . For the lower bracket, let $(\Gamma \triangleright y) \in \text{Ch}_{D^*}(\text{St}_{h^*}^g)$, witnessed by an argument with lines in $D^* \subseteq D_h$ and inferences and links in $\text{St}_{h^*}^g \subseteq \text{St}_h^g \subseteq \models_h$. Lemma 10.6 under h and monotonicity give $\Gamma \models_h y$. If h^* is step-expressive on D^* at gap g , the two brackets meet, so $\models_h|_{D^*} = \models_{h^*}$ and $h \in U^*$; conversely every $h \in U^*$ with $\text{St}_h^g \supseteq \text{St}_{h^*}^g$ is in S^g . *Effectivity.* Practice refutation is decidable because St_h^g is. An object datum v refutes h iff $T_h \cup \rho_h \Gamma_v$ is inconsistent (notation of Lemma I.1), which is semi-decidable since Cl_{R_h} is r.e.; paradoxes are found by search. Each non-member of S^g is refuted by a finite datum with a finite certificate, and $\mathcal{H}$ is finite, so the budgeted version space equals S^g after finitely many data and a finite budget. Example 10.16 shows that survivors can have strictly weaker meaning. $\square$

Verification of Example 10.16. Both hypotheses are propositional (a fragment of first-order logic) and share the reading; $D^* = D_h$ consists of occurrences read as $p_0, \neg p_0, p_3, \neg p_3$, and $\sim$ maps the occurrence read as p_i to the one read as $\neg p_i$ and back, so D^* is negation-closed. R^* consists of the zero-premise steps $\triangleright \tau$ for $\tau \in T^*$ and all steps $\Pi \triangleright \varphi$ with Π finite and Π tautologically

implying φ ; R_h consists of the tautological steps only. Then $\text{Cl}_{R^*}(\Gamma)$ is the set of tautological consequences of $\Gamma \cup T^*$ and $\text{Cl}_{R_h}(\Gamma)$ that of Γ , so both hypotheses are first-order complete, with $T_h = \emptyset$.

Let $1 \leq g \leq 3$. Since $R_h \subseteq R^*$, $\text{St}_h^g \subseteq \text{St}_{h^*}^g$. Conversely, let $\Gamma \triangleright y$ be in $\text{St}_{h^*}^g$, via an R^* -derivation π of $\rho(y)$ from $\rho\Gamma$ with at most g applications. If $\rho(y) \in \rho\Gamma$, the step is in St_h^g with zero applications. Otherwise the last step of π is tautological, because axiom steps conclude members of T^* , which are not readings of occurrences in D^* . So π uses at most $g - 1 \leq 2$ axiom steps, and $\rho(y)$ is a tautological consequence of $\rho\Gamma \cup T'$ for some $T' \subsetneq T^*$. We claim $\rho(y)$ is then a tautological consequence of $\rho\Gamma$ alone. Otherwise some valuation v of p_0, p_3 satisfies $\rho\Gamma$ and falsifies $\rho(y)$ (all four formulas mention only p_0, p_3). Let $p_i \rightarrow p_{i+1}$ be a chain axiom missing from T' , and extend v by $v(p_k) := v(p_0)$ for $k \leq i$ and $v(p_k) := v(p_3)$ for $k \geq i + 1$. Every axiom of T' is then of the form $p_k \rightarrow p_{k+1}$ with both sides equal, so v satisfies $T' \cup \rho\Gamma$ and falsifies $\rho(y)$, a contradiction. Hence the single tautological step $\rho\Gamma \triangleright \rho(y)$ is in R_h , and $\Gamma \triangleright y \in \text{St}_h^1 \subseteq \text{St}_h^g$. So $\text{St}_h^g = \text{St}_{h^*}^g$.

$(p_0 \triangleright p_3) \in \models_{h^*}$, since T^* implies $p_0 \rightarrow p_3$; it is not in $\models_h$, since $p_0 \not\models p_3$. So $\models_{D^*} \subsetneq \models_{h^*}$ and $h \notin U^*$. Every object datum extends to an h -admissible valuation, since $\text{Mod}(T^*) \subseteq \text{Mod}(\emptyset)$ gives $\mathfrak{A}h^* \subseteq \mathfrak{A}h$; and every A on which h^* is coherent is h -coherent, since $T_h \subseteq T^*$. So h survives every presentation in the sense of Theorem 10.15(d), although $h \notin U^*$. At $g = 4$, the derivation from p_0 by the three axiom steps and one tautological step puts $(p_0 \triangleright p_3)$ in $\text{St}_{h^*}^4 \setminus \text{St}_h^4$, and practice refutes h . $\square$

Proof of Proposition 10.18. Cited facts (Nelson, 1977). (N1) IST is a conservative extension of ZFC: an internal sentence provable in IST is provable in ZFC. (N2) Reduction: for every IST formula $A(\vec{t})$ there is an internal formula $B(\vec{t})$ with the same free variables such that $\text{IST} \vdash \forall^{\text{st}} \vec{t} (A(\vec{t}) \leftrightarrow B(\vec{t}))$. (T) Transfer: for internal $\varphi(\vec{x})$ with no other parameters, $\text{IST} \vdash \forall^{\text{st}} \vec{x} \varphi(\vec{x}) \rightarrow \forall \vec{x} \varphi(\vec{x})$.

Step 1. Let $x \in D_{\text{st}}$, so $\rho_{\text{IST}}(x) = A(\vec{c})$ with $\vec{c}$ contextual constants. By (N2) and $T_{\text{IST}} \vdash \text{st}(c_i)$ for each i , $T_{\text{IST}} \vdash A(\vec{c}) \leftrightarrow B(\vec{c})$, where $B(\vec{c}) = \rho_W(x)$.

Step 2. For internal $\varphi(\vec{c})$: $T_{\text{IST}} \vdash \varphi(\vec{c})$ iff $\text{ZFC} \vdash \varphi(\vec{c})$, where on the right the $\vec{c}$ are fresh constants. $(\Leftarrow)$ $\text{IST} \supseteq \text{ZFC}$. $(\Rightarrow)$ Finitely many axioms $\text{st}(c_i)$ are used, so $\text{IST} \vdash \bigwedge_i \text{st}(c_i) \rightarrow \varphi(\vec{c})$ by the deduction theorem. The c_i do not occur in the axioms of IST, so $\text{IST} \vdash \forall \vec{x} (\bigwedge_i \text{st}(x_i) \rightarrow \varphi(\vec{x}))$, i.e. $\text{IST} \vdash \forall^{\text{st}} \vec{x} \varphi(\vec{x})$. By (T), $\text{IST} \vdash \forall \vec{x} \varphi(\vec{x})$; this is an internal sentence, so $\text{ZFC} \vdash \forall \vec{x} \varphi(\vec{x})$ by (N1), and hence $\text{ZFC} \vdash \varphi(\vec{c})$.

Step 3. For finite $\Gamma \cup \{y\} \subseteq D_{\text{st}}$: $\Gamma \models_{\text{IST}} y$ iff $T_{\text{IST}} \vdash \bigwedge \rho_{\text{IST}} \Gamma \rightarrow \rho_{\text{IST}} y$ iff (Step 1) $T_{\text{IST}} \vdash \bigwedge \rho_W \Gamma \rightarrow \rho_W y$ iff (Step 2, the formula being internal) $\text{ZFC} \vdash \bigwedge \rho_W \Gamma \rightarrow \rho_W y$ iff $\Gamma \models_W y$. $\square$

Proof of Proposition 10.19. Let ω_{vN} be the set of von Neumann natural numbers, with successor $S_{\text{vN}}(x) = x \cup \{x\}$, and let ω_Z be the range of the function z on ω_{vN} defined by recursion, $z(0) = \emptyset$ and $z(S_{\text{vN}}n) = \{z(n)\}$ (Replacement), with successor $S_Z(x) = \{x\}$. ZF proves that both $(\omega_{\text{vN}}, \emptyset, S_{\text{vN}})$ and $(\omega_Z, \emptyset, S_Z)$ are Dedekind–Peano systems: the successor is injective, $\emptyset$ is not a successor, and every subset containing $\emptyset$ and closed under successor is everything (for ω_Z , by induction along z). Define $+, \times$ on each by the recursion equations. By Dedekind’s categoricity theorem (Dedekind, 1888), provable in ZF, there is a unique $f : \omega_{\text{vN}} \rightarrow \omega_Z$ with $f(0) = 0$ and $f(S_{\text{vN}}n) = S_Z f(n)$; it is a bijection, and by induction it preserves $+$ and $\times$.

For an arithmetic formula $\varphi(\vec{v})$ let φ^{vN} and φ^Z be its translations: quantifiers relativized to $\omega_{\bullet}$, and $0, S, +, \times$ interpreted as above. By induction on terms and then formulas, $\text{ZF} \vdash \forall \vec{n} \in \omega_{\text{vN}} (\varphi^{\text{vN}}(\vec{n}) \leftrightarrow \varphi^Z(f\vec{n}))$. Atomic formulas use that f commutes with the operations and is

injective; connectives are immediate; for $\exists m \psi$ use that f is surjective. So for closed arithmetic x , $\text{ZF} \vdash \rho_{\text{vN}}(x) \leftrightarrow \rho_{\text{Z}}(x)$, and for $\Gamma \cup \{y\} \subseteq D_{\text{ar}}$ closed, $\Gamma \models_{\text{vN}} y$ iff $\text{ZF} \vdash \bigwedge \rho_{\text{vN}} \Gamma \rightarrow \rho_{\text{vN}} y$ iff $\text{ZF} \vdash \bigwedge \rho_{\text{Z}} \Gamma \rightarrow \rho_{\text{Z}} y$ iff $\Gamma \models_{\text{Z}} y$.

Occurrences with contextual constants (“let n be a number”) are read with a fresh constant c and the axiom $c \in \omega_{\text{vN}}$, respectively $c \in \omega_{\text{Z}}$. They are not equivalent sentence by sentence, since c denotes different sets under the two readings, but the consequence relations still agree: $\Gamma \models_{\text{vN}} y$ iff $\text{ZF} \vdash \forall \vec{n} \in \omega_{\text{vN}} (\bigwedge \Gamma^{\text{vN}}(\vec{n}) \rightarrow y^{\text{vN}}(\vec{n}))$ (deduction theorem and generalization over the fresh constants), iff $\text{ZF} \vdash \forall \vec{m} \in \omega_{\text{Z}} (\bigwedge \Gamma^{\text{Z}}(\vec{m}) \rightarrow y^{\text{Z}}(\vec{m}))$ (transport along the bijection f), iff $\Gamma \models_{\text{Z}} y$. Finally, “ $1 \in 3$ ” is not an arithmetic sentence; under ρ_{vN} it is true, and under ρ_{Z} it is false, since $\{\{\emptyset\}\} \not\equiv \{\emptyset\}$. $\square$

I.3 Objects versus paradoxes

In the correction games, VS is the set of surviving hypotheses. A learner is deterministic; the environment may choose the target adaptively, as long as some target is consistent with all its answers, and an item is legal only if it is legal for such a target.

Proof of Theorem 10.22(b), (c). (b) Fix $r \geq 1$. A $(-\text{obj})$ item $s \in A_t \setminus h^*$ is also a legal $(-\text{bag}_r)$ item $B = \{s\}$, and it deletes the same hypotheses ($h \ni s$, i.e. $h \supseteq \{s\}$). $(+)$ items are common to both games. So against any learner, the object environment’s moves are a subset of the bag environment’s, and every bag-game strategy achieves in the object game at most its bag-game worst case. Hence $M_{\text{obj}} \leq M_{\text{bag}}^{(r)}$. At $r = 1$ a legal bag is a set $B \subseteq A_t$ with $|B| \leq 1$ and $B \not\subseteq h^*$, i.e. $B = \{s\}$ with $s \in A_t \setminus h^*$, so the games coincide.

(c) Let $\text{Ldim}(\emptyset) = -1$, and for nonempty $\mathcal{C} \subseteq 2^S$ let $\text{Ldim}(\mathcal{C})$ be the largest d such that there is a complete binary tree of depth d whose internal nodes are labelled by elements of S and such that every root-to-leaf path is realized by some $h \in \mathcal{C}$. A path is realized if, at each node labelled s , it goes right iff $s \in h$ (Littlestone, 1988). For $s \in S$ write $\mathcal{C}_s^1 := \{h \in \mathcal{C} : s \in h\}$ and $\mathcal{C}_s^0 := \mathcal{C} \setminus \mathcal{C}_s^1$. If $\text{Ldim}(\mathcal{C}) = d \geq 0$, then $\min(\text{Ldim}(\mathcal{C}_s^0), \text{Ldim}(\mathcal{C}_s^1)) < d$ for every s . Otherwise both subclasses would have trees of depth d , and a root labelled s above them would give a tree of depth $d + 1$ for $\mathcal{C}$.

Upper bound. Announce $A_t := \{s : \text{Ldim}(\text{VS}_s^1) \geq \text{Ldim}(\text{VS}_s^0)\}$. A $(-\text{obj})$ item $s \in A_t$ leaves VS_s^0 , and a $(+)$ item $s \notin A_t$ leaves VS_s^1 . In both cases the remaining class is the one of smaller Littlestone dimension, which is below $\text{Ldim}(\text{VS})$. The target always survives, so $\text{Ldim}(\text{VS}) \geq 0$, and there are at most $\text{Ldim}(\mathcal{H})$ corrections.

Lower bound. The environment maintains VS with a tree of depth $d = \text{Ldim}(\text{VS})$. While $d \geq 1$, let s label the root. If $s \in A_t$, it answers $(-\text{obj})$ s and commits to VS_s^0 . This is legal: any $h \in \text{VS}_s^0$ has $s \in A_t \setminus h$, and $\text{VS}_s^0 \neq \emptyset$ because it contains the hypotheses realizing the left subtree. If $s \notin A_t$, it answers $(+)$ s and commits to VS_s^1 . Either way the new version space contains a subtree of depth $d - 1$. So every learner suffers at least $\text{Ldim}(\mathcal{H})$ corrections, and $M_{\text{obj}}(\mathcal{H}) = \text{Ldim}(\mathcal{H})$. This is Littlestone’s theorem, transcribed. Equivalence queries with arbitrary hypotheses (Angluin, 1988) are the same game. $\square$

Proof of Theorem 10.23. (a) The cautious learner announces $A_t = \bigcap \text{VS}_t$. Since $h^* \in \text{VS}_t$, $A_t \subseteq h^*$, so it never receives a negative item; let $s_1, s_2, \dots$ be the $(+)$ items it receives. When s_i arrives, $\text{VS}_t = \{h : h \supseteq \{s_1, \dots, s_{i-1}\}\}$ and $s_i \notin A_t$, so (s_i) is an elastic chain in h^* , and there are at most $\text{el}(\mathcal{H}, h^*) \leq \text{el}^*(\mathcal{H})$ corrections. The bound $\text{el}^* \leq |\mathcal{H}| - 1$ is proved as in Theorem 10.10(a). Since $M_{\text{bag}}^{(r)}$ is the optimum over learners, $M_{\text{bag}}^{(r)} \leq \text{el}^*$.

(b) *Objects.* Announce S . Then $A_t \supseteq h^*$, so the only legal feedback is $(-obj)$ b_{j^*} , which deletes every h_j with $j \neq j^*$; at least one correction is needed since $|\mathcal{H}_n| \geq 2$.

Bags, upper bound. If $n > r$, first announce S . Only bags are legal; a bag B must contain b_{j^*} and deletes every h_j with $b_j \notin B$, leaving at most $|B| \leq r$ candidates. Then, while two or more candidates C remain, announce $S \setminus \{b_c\}$ for some $c \in C$. If $h^* = h_c$ there is no feedback. Otherwise a $(+)$ item must be b_c , which deletes h_c ; a bag $B \subseteq S \setminus \{b_c\}$ is contained in h_c , so it also deletes h_c . Each correction removes a candidate, so there are at most $1 + (r - 1) = r$ corrections. If $n \leq r$, skip the first step: at most $n - 1$. Hence $M_{\text{bag}}^{(r)} \leq \min(r, n - 1)$.

Bags, lower bound. The environment keeps the set C of culprits j consistent with its answers, initially all of $\{1, \dots, n\}$; while $|C| \geq 2$ every announcement differs from some consistent target, so it can always answer. Given A_t :

- if A_t rejects some b_i with $i \notin C$ (a step valid in every survivor), it answers $(+)$ b_i , which deletes nothing;
- otherwise, if A_t rejects some b_i with $i \in C$, it answers $(+)$ b_i , legal for any culprit in $C \setminus \{i\}$, and sets $C \leftarrow C \setminus \{i\}$;
- otherwise $A_t = S$. If $|C| > r$, it answers with a bag B of r candidates' steps and sets $C \leftarrow B$; if $|C| \leq r$, it answers $B = \{b_j : j \in C\}$, which deletes only hypotheses already deleted.

Let $f(c)$ be the number of corrections this environment forces from $|C| = c$ until $|C| = 1$. Then $f(1) = 0$, and against the best learner $f(c) = 1 + \min(f(c - 1), [c > r]f(r))$, where the second option is unavailable for $c \leq r$. So $f(c) = c - 1$ for $c \leq r$, and $f(c) = 1 + \min(f(c - 1), r - 1) = r$ for $c > r$, by induction. Hence $M_{\text{bag}}^{(r)}(\mathcal{H}_n) \geq \min(r, n - 1)$.

(c) *Coherence.* Under h_j (axioms $p_{i-1} \rightarrow p_i$ for $i \neq j$, with q_i read as p_i and $\sim q_n$ as $\neg p_n$), the valuation with p_i true iff $i < j$ satisfies every axiom together with p_0 and $\neg p_n$. So every h_j is coherent on $A = \{q_0, \sim q_n\}$, and the designation is truthful whatever the target. *Every paradox uses every b_i .* With p_0 true and p_n false, any valuation satisfying the axioms $p_{i-1} \rightarrow p_i$ for all $i \neq j$ falsifies the remaining one. So the set of all the b_i together with A is inconsistent, while A together with all b_i but one is consistent. A derivation of $\perp$ from A through suspect steps must therefore use all of them, and every bag is S . Such a bag deletes only hypotheses containing all of S , and there are none. *Lower bound.* As in (b) with bags restricted to S : an announcement that accepts all of S receives the useless bag S ; one that rejects a known-valid step receives a useless $(+)$ item; one that rejects a candidate receives $(+)$ on it and loses only it. So $n - 1$ corrections are forced. *The object.* Let $u \in \mathfrak{A}_{h_{j^*}}$ with $u(q_0) = 1$ and $u(q_n) = 0$. The axioms force $p_0, \dots, p_{j^*-1}$ true (forward along $p_{i-1} \rightarrow p_i$, $i < j^*$) and $p_{j^*}, \dots, p_n$ false (backward from $\neg p_n$ along $i > j^*$). So $u(q_i) = 1$ iff $i < j^*$, and descent (Lemma 10.20) along the chain argument $q_0 \triangleright q_1 \triangleright \dots \triangleright q_n$ stops exactly at b_{j^*} . $\square$

Proof of Proposition 10.25. Let the blocks be $X_1, \dots, X_k$, each of b steps, and let $\mathcal{H}$ consist of the sets that omit exactly one step from each block. Write C_X for the surviving candidate culprits of block X , and call X *unresolved* if $|C_X| \geq 2$.

$M_{\text{obj}} \leq k$. Announce S minus the culprits identified so far. Then $A_t \supseteq h^*$, so no $(+)$ item is legal, and each $(-obj)$ item is an unidentified culprit, which resolves its block.

$M_{\text{obj}} \geq k$. The environment always answers inside one unresolved block X : with $(+)$ on some $c \in C_X \setminus A_t$, legal for culprits in $C_X \setminus \{c\}$, or, if $C_X \subseteq A_t$, with $(-obj)$ on some $c \in C_X$, committing to culprit c . Each answer changes only one block and resolves at most one, and while

some block is unresolved an answer is available. Each of the k blocks must be resolved, so at least k corrections occur.

$M_{\text{bag}}^{(r)} \geq k \min(r, b - 1)$. Let $\Phi := \sum_X \min(r, |C_X| - 1)$, initially $k \min(r, b - 1)$. While $\Phi > 0$ some block is unresolved. The environment answers as follows.

- If some unresolved X has $c \in C_X \setminus A_t$: answer (+) c (legal, as the culprit can be another member of C_X). Then C_X loses one element, and Φ drops by at most 1.
- Otherwise every unresolved block has all its candidates accepted. Pick one, X . If $|C_X| > r$, answer with a bag B of r of its candidates (so $B \subseteq A_t$), committing to a culprit in B . Then $C_X \leftarrow B$, and $\min(r, |C_X| - 1)$ drops from r to $r - 1$. If $|C_X| \leq r$, answer $B = C_X$, which is legal and deletes no survivor; Φ is unchanged.

Rejections of known-valid steps and acceptances of identified culprits are ignored. Each correction lowers Φ by at most one, so at least $\Phi_0 = k \min(r, b - 1)$ corrections occur. With $b = r + 1$: $|\mathcal{H}| = (r + 1)^k$, so $k = \log |\mathcal{H}| / \log(r + 1)$, and $M_{\text{bag}}^{(r)} \geq kr$. $\square$

Proof of Proposition 10.26. $M_{\text{obj}}(\mathcal{H}) = 1$ means some first announcement A is such that every legal feedback leaves a single hypothesis. A feedback item s against A (either kind) leaves exactly the hypotheses h with $s \in A \triangle h$. The environment may return any $s \in A \triangle h^*$ for any $h^* \in \mathcal{H}$, so the sets $A \triangle h$, $h \in \mathcal{H}$, are pairwise disjoint. In the bag game, announce A . A (+) item $s \notin A$ lies in exactly one $A \triangle h$ and pins the target. A bag $B \subseteq A$ with $|B| \leq r$ leaves the h with $B \not\subseteq h$, i.e. with $B \cap (A \triangle h) \neq \emptyset$. Each element of B lies in at most one such set, so at most r hypotheses survive. Then announce the survivors one at a time. Any feedback against an announced $h \neq h^*$ deletes h : a (+) item $s \in h^* \setminus h$ deletes the hypotheses lacking s , and a bag $B \subseteq h$ with $B \not\subseteq h^*$ deletes the hypotheses containing B . So at most $r - 1$ further corrections occur, at most r in all. (If $|\mathcal{H}| \leq r$, plain elimination costs at most $r - 1$.) $\square$

I.4 Frege, Russell, Zermelo

Proof of Theorem 10.30. Refutations are sound, so a refuted set is incoherent. (a) I contains every support and is strictly shortest. (b) A coherent c has no refutation; an incoherent c has one of size $k(c)$, found by every search with budget $t \geq k(c)$. (c) For $t \geq t_0$, by (b), the unrefuted members of $\mathcal{R}$ are exactly the coherent ones. (d) Normalizing w_t does not change the arg max. For every $c \in \mathcal{R}$, $|\text{cov}_t(c) / \|w_t\|_1 - \text{cov}_\infty(c)| = |\sum_{i \in c} (w_t(i) / \|w_t\|_1 - w_\infty(i))| \leq \|w_t / \|w_t\|_1 - w_\infty\|_1 \rightarrow 0$, where $\text{cov}_\infty(c) := \sum_{i \in c} w_\infty(i)$. Let c^* be the unique coherent maximizer of cov_∞ , and $\gamma > 0$ its margin over the other coherent members ($\mathcal{R}$ is finite). Once $t \geq t_0$ and the ℓ_1 distance is below $\gamma/2$, c^* has strictly larger coverage than every other coherent member, so $\text{REP}_t = c^*$. Pointwise convergence alone would not suffice: the members of $\mathcal{R}$ are typically infinite, and mass escaping along instances inside some c breaks the convergence of coverages. (By Scheffé's lemma, pointwise convergence of probability vectors to a probability vector is enough.) (e) Coh is downward closed, so any superset of the incoherent $c_1 \cup c_2$ is incoherent. If $\text{supp}(w_t) \subseteq c_1 \cap c_2$, then $\text{cov}_t(c_1) = \text{cov}_t(c_2) = \|w_t\|_1$, and the lexicographic comparison reduces to ℓ . $\square$

Proof of Lemma 10.31. Membership of the menu formulas in the classes (Table 13):

- POS: V, S, INT, UNI, PAIR;
- STRAT $\cap \Phi$: V, EMP, INT, UNI, PAIR, DIFF, CMP;
- SEP: INT, DIFF, ZR;

- $Z \cap \Phi$: INT, DIFF, ZR, EMP, PAIR, UNI.

The formulas S, R and ZR contain $x \in x$ and so admit no stratification.

(F1) Comp(R) gives y with $\forall x (x \in y \leftrightarrow x \notin x)$; instantiating $x := y$ gives $y \in y \leftrightarrow y \notin y$, hence $\perp$.

(F2) Let $\{u\}$ be a one-element structure with $u \in u$. Every atomic formula ($x = y$ or $x \in y$) is true under every assignment, so every positive formula, built from atoms by $\wedge, \vee, \forall, \exists$, is true. Hence $\forall x (x \in u \leftrightarrow \varphi)$ holds for every positive φ and all parameters, and $y := u$ witnesses every positive comprehension instance. Extensionality holds trivially.

(F3) The finite and cofinite subsets of $\mathbb{N}$ form a countably infinite Boolean algebra FC; let $e : \mathbb{N} \rightarrow \text{FC}$ be a bijection and put $x \in y : \iff x \in e(y)$. For parameters a, b , the extensions of the stratified menu formulas are $\mathbb{N}$ (V), $\emptyset$ (EMP), $e(a) \cap e(b)$ (INT), $e(a) \cup e(b)$ (UNI), $\{a, b\}$ (PAIR), $e(a) \setminus e(b)$ (DIFF) and $\mathbb{N} \setminus e(a)$ (CMP). All are in FC, hence equal $e(y)$ for some y . Injectivity of e gives Extensionality.

(F4) In the hereditarily finite sets with true membership, INT, DIFF and ZR define subsets of $a \in \text{HF}$, which are in HF; EMP, PAIR and UNI define $\emptyset$, $\{a, b\}$ and $a \cup b$, which are in HF. Extensionality and Foundation hold.

(F5) $\text{POS} \cup Z$ and $\text{STRAT} \cup Z$ contain V and ZR. Comp(V) gives v with $\forall x (x \in v)$; Comp(ZR) with $a := v$ gives y with $\forall x (x \in y \leftrightarrow x \in v \wedge x \notin x)$, i.e. $\forall x (x \in y \leftrightarrow x \notin x)$, and (F1) applies. $\text{POS} \cup \text{STRAT}$ contains S and CMP. Comp(S) gives s with $\forall x (x \in s \leftrightarrow x \in x)$; Comp(CMP) with $a := s$ gives y with $\forall x (x \in y \leftrightarrow x \notin s)$, i.e. $\forall x (x \in y \leftrightarrow x \notin x)$, and (F1) applies.

(F6) $\text{SEP} \subsetneq Z$ by the lists. $S \in \text{POS} \setminus (\text{STRAT} \cup Z)$, $\text{CMP} \in \text{STRAT} \setminus (\text{POS} \cup Z)$, $V \in \text{STRAT} \setminus Z$, $\text{ZR} \in Z \setminus (\text{POS} \cup \text{STRAT})$, and $\text{DIFF} \in Z \setminus \text{POS}$. So the three classes are pairwise incomparable. $\square$

Proof of Theorem 10.32. (i) is Theorem 10.30(a), since NC contains every support and is shortest. (ii) is (F1); NC is the only incoherent member, so t_0 is the size of that refutation. (iii) NC is incoherent; POS, STRAT and Z are coherent (F2–F4); $\text{SEP} \subsetneq Z$ is coherent but not maximal; and POS, STRAT and Z are pairwise incomparable (F6), hence all maximal. They are pairwise jointly incoherent by (F5).

(iv) By Theorem 10.30(c), for $t \geq t_0$ the learner REP_t maximizes $(\text{cov}, -\ell)$ over the coherent classes POS, SEP, STRAT and Z, whose description lengths increase in that order. Let W be the total weight and supp the support, with all weights in the support positive; $\text{cov}(c) = W$ iff $\text{supp} \subseteq c$. *Support* $\subseteq \text{DC}$. STRAT and Z contain DC, so both have coverage W , and STRAT beats Z on ℓ . POS has coverage W iff the support avoids DIFF and EMP; SEP iff it avoids UNI, PAIR and EMP. The maximal coverage is W , and the shortest class achieving it is selected: POS if it qualifies, else SEP if it qualifies, else STRAT. So STRAT is selected iff the support meets both $\{\text{DIFF}, \text{EMP}\}$ and $\{\text{UNI}, \text{PAIR}, \text{EMP}\}$, as claimed; the full support DC selects STRAT. $\text{ZR} \in \text{supp} \subseteq \text{DC} \cup \{\text{ZR}\}$. POS and STRAT miss ZR, so their coverage is below W . Z has coverage W , and SEP does iff the support avoids UNI, PAIR and EMP. Since SEP is shorter than Z, SEP is selected in that case and Z otherwise. The full support selects Z. *Support* $\text{DC} \cup \{\text{ZR}, V\}$. $\text{cov}(\text{STRAT}) = W - w(\text{ZR})$; $\text{cov}(Z) = W - w(V)$; $\text{cov}(\text{POS}) = W - w(\text{DIFF}) - w(\text{EMP}) - w(\text{ZR}) < \text{cov}(\text{STRAT})$; $\text{cov}(\text{SEP}) = W - w(V) - w(\text{UNI}) - w(\text{PAIR}) - w(\text{EMP}) < \text{cov}(Z)$. So Z is selected iff $w(\text{ZR}) > w(V)$, and ties go to the shorter STRAT. *The partial example.* For support $\{\text{INT}, \text{UNI}, \text{PAIR}, \text{ZR}, V\}$ with $w(V) > w(\text{ZR})$: $\text{cov}(\text{POS}) = \text{cov}(\text{STRAT}) = W - w(\text{ZR}) > W - w(V) = \text{cov}(Z)$, and SEP misses UNI, PAIR and V. POS ties STRAT and is shorter, so POS is selected. All 93 sub-supports

(the nonempty subsets of DC, alone, with ZR, and with ZR and V) were checked against these conditions by `T4-checks/verification_checks.py` (V2), with no mismatch. $\square$

Proof of Proposition 10.33. For $n \geq 1$ let p_n be the sentence “some object has exactly n elements”. For a sentence q let $J(q) := \text{Comp}(x \notin x \wedge q)$, an instance of naive comprehension. For $A \subseteq \mathbb{N}_{\geq 1}$ let

$$I_A := \{J(p_n) : n \notin A\} \cup \{J(\neg p_n) : n \in A\} \cup \{\text{Comp}(x \neq x)\}.$$

$J(q) \wedge q \vdash \perp$: if q holds, $J(q)$ yields y with $\forall x (x \in y \leftrightarrow x \notin x)$, and (F1) applies. So $I_A \vdash \neg p_n$ for $n \notin A$ and $I_A \vdash p_n$ for $n \in A$, and $I_A \cup I_B$ is inconsistent whenever $A \neq B$. I_A is consistent. Let M_A have domain $\{e_0, e_1, \dots\} \cup \{a_n : n \in A\}$, where every e_i has no elements, a_n has exactly the elements $e_0, \dots, e_{n-1}$, and there is no other membership. Then p_n holds in M_A iff $n \in A$. Each member of I_A is either $J(q)$ with q false in M_A or $\text{Comp}(x \neq x)$; in both cases the defining formula has empty extension for all parameters, and e_0 witnesses it. *Maximal extensions.* Consistency has finite character, so by Zorn’s lemma each I_A extends to a maximal consistent set J_A of instances. If $A \neq B$, then $J_A \cup J_B \supseteq I_A \cup I_B$ is inconsistent, so $J_A \neq J_B$ and the two are incompatible. Every J_A contains $\text{Comp}(x \neq x)$. There are $2^{\aleph_0}$ sets A . There are only countably many instances, so there are at most $2^{\aleph_0}$ maximal consistent sets. The argument uses only pure logic. With Extensionality as background, the same argument works with the Zermelo numerals $z_0 = \emptyset$, $z_{k+1} = \{z_k\}$ in place of the e_i , $a_n = \{z_0, \dots, z_{n-1}\}$ and $A \subseteq \mathbb{N}_{\geq 2}$ (each z_k has at most one element, so p_1 always holds). $\square$

Limit approximation. Enumerate the instances $\iota_0, \iota_1, \dots$, and put $S_0 = \emptyset$, $S_{k+1} = S_k \cup \{\iota_k\}$ if this set is consistent and $S_{k+1} = S_k$ otherwise. Then $S = \bigcup_k S_k$ is consistent by compactness. It is maximal: if $\iota_k \notin S$, then $S_k \cup \{\iota_k\}$ is inconsistent, so $S \cup \{\iota_k\}$ is. Consistency of a finite set of sentences is Π_1 , so S is computable from the halting problem, i.e. Δ_2 , and by Shoenfield’s limit lemma it is the pointwise limit of a computable sequence of guesses. This does not conflict with [Incurvati and Murzi \(2017\)](#): being Δ_2 does not make S recursively axiomatizable.

I.5 The robust core

Proof of Theorem 10.35. (a) Every inference and link of the argument is a non-noise practice item, so by RCH_g it lies in $\text{St}_{\text{SV}}^g \subseteq \text{St}_\sigma^g \subseteq \models_\sigma$, or is a link valid under σ , for every σ . Validity puts every line used in D_{h_σ} , and unused premises are dropped. Lemma 10.6 for h_σ gives the conclusion in $\text{Cl}_{R_\sigma}(\rho_\sigma \text{Prem})$. If T_σ proves the premises, then since Cl_{R_σ} is consequence over T_σ , it proves the conclusion.

(b) Each datum is consistent with each h_σ :

- practice steps lie in $\text{St}_{\text{SV}}^g \subseteq \text{St}_\sigma^g$;
- a “yes” is given only on $\text{St}_{\text{SV}}^g \subseteq \text{St}_\sigma^g$, and a “no” only outside $\bigcup_{\sigma'} \text{St}_{\sigma'}^g \supseteq \text{St}_\sigma^g$, for steps and links alike;
- designated contexts are σ -coherent, so no paradox for h_σ exists;
- an SV-truthful datum at the precise object M is $v(x) = v_{M_\sigma}(\rho_\sigma(x))$ for super-determinate x , the same value for every σ ; so v is the restriction of $v_{M_\sigma} \circ \rho_\sigma \in \mathfrak{A}h_\sigma$.

Budgeted searches find only certificates, so no h_σ is ever refuted and $\{h_\sigma\} \subseteq \text{VS}_t$. Unanimity then gives acceptance $\subseteq \bigcap_\sigma \text{St}_\sigma^g = \text{St}_{\text{SV}}^g \subseteq \models_\sigma$ for every σ . *Convergence.* Let $\mathcal{H}$ be finite and every element of St_{SV}^g eventually presented. Any h lacking some element of St_{SV}^g is refuted when

that element is presented, so after finitely many data every survivor contains St_{SV}^g . The h_σ survive, so the intersection of the survivors' relations is contained in $\bigcap_\sigma \text{St}_\sigma^g$, and therefore equals St_{SV}^g . *Effectivity.* For infinite Σ the robust core can be undecidable. Let the vague sentences be $w_0, w_1, \dots$; let σ_n read w_m as the Δ_0 sentence “machine m does not halt within n steps” over a base theory proving all true Δ_0 sentences; and let each R_σ contain the zero-premise steps $\triangleright \delta$ for true Δ_0 sentences δ , a decidable set. Then $(\emptyset \triangleright w_m) \in \text{St}_{\sigma_n}^1$ iff machine m does not halt within n steps. A false Δ_0 sentence is refutable, hence undervivable in a sound calculus. So $(\emptyset \triangleright w_m) \in \text{St}_{\text{SV}}^1$ iff machine m never halts, which is Π_1 -complete. The idealized verifier is an algorithm only for finite Σ or under extra uniformity.

(c) By (b), under these data every h_σ remains a possible target forever. (c1) If an acceptance set A is contained in $\models_\sigma$ for every σ , then $A \subseteq \bigcap_\sigma \models_\sigma = \models_{\text{SV}}$. (c2) The same argument works with St_σ^g . If $\mathcal{H} = \{h_\sigma\}$, all members survive by (b), and the verifier's accepted set is exactly $\bigcap_\sigma \text{St}_\sigma^g = \text{St}_{\text{SV}}^g$ at all times. *The gap.* Let $\Sigma = \{\sigma\}$, let R_σ be natural deduction for classical propositional logic, and let $\rho(x) = q$ and $\rho(y) = q \wedge (q \wedge q)$. Then $(\{x\} \triangleright y) \in \models_\sigma$. No single rule application derives $q \wedge (q \wedge q)$ from q . $\wedge\text{I}$ would need the premise $q \wedge q$; $\rightarrow\text{E}$ would need an implication with that consequent; $\perp\text{E}$ would need $\perp$; and the classical reductio rule needs a subderivation of at least one further step. So $(\{x\} \triangleright y) \notin \text{St}_\sigma^1$, while two $\wedge\text{I}$ steps suffice.

(d) By (b), the accepted set is contained in St_{SV}^g , so steps outside it are never accepted. (d1) Let $s \notin \models_\sigma$ for some $\sigma \in \Sigma$. A “yes” on s would be false under σ , so no SV-truthful answer licenses s . Any library extension must keep R_σ sound for $\text{Mod}(T_\sigma)$, so it cannot make s valid under σ . Any expansion into σ -valid steps would make s σ -valid by Lemma 10.6. So while $\sigma \in \Sigma$ and s is unchanged, s stays outside St_{SV}^g ; only shrinking Σ (a definition) or changing the step (lemma-incorporation) removes it. (d2) Let $s = (\Gamma \triangleright y) \in \models_{\text{SV}}$. Adding the derived step $\rho_\sigma \Gamma \triangleright \rho_\sigma y$ to each library R_σ keeps R_σ sound, because the step is σ -valid. It puts s in $\text{St}_\sigma^1 \subseteq \text{St}_\sigma^g$ for every σ , so $s \in \text{St}_{\text{SV}}^g$. Alternatively, an expansion of s into steps of St_{SV}^g , when one exists, is accepted step by step (Proposition 10.11). $\square$

Proof of Proposition 10.37. (a) Each $\models_\sigma$, restricted to finite subsets of $D_\sigma := D_{h_\sigma}$, is reflexive ($\{x\} \models_\sigma x$ for $x \in D_\sigma$), monotone, and closed under cut ($\Gamma \models_\sigma x$ and $\Gamma \cup \{x\} \models_\sigma y$ give $\Gamma \models_\sigma y$), because Cl_{R_σ} is a closure operator. These properties are preserved by intersection over σ , on the common domain $\bigcap_\sigma D_\sigma$. (b) Let the argument have steps (inferences and links) $s_1, \dots, s_n$, and let $E_i := \{\sigma : s_i \in \models_\sigma\}$. Under every $\sigma \in \bigcap_i E_i$, Lemma 10.6 gives $(\text{Prem} \triangleright y_m) \in \models_\sigma$. So, by the union bound, the (measurable) event that the argument is valid has probability at least $\mu(\bigcap_i E_i) \geq 1 - \sum_i \mu(E_i^c) \geq 1 - n\varepsilon$. *Tightness.* Take a base theory deciding order between numerals; a vague predicate “heap”; sentences $q_k = “k \text{ grains make a heap}”$ for $k = 0, \dots, n$; and sharpenings σ_c , $c = 0, \dots, n$, defining “heap”(x) as $x \geq c$. The argument is $q_n \triangleright q_{n-1} \triangleright \dots \triangleright q_0$. Under σ_c , the step $q_k \triangleright q_{k-1}$ fails iff $k \geq c > k-1$, i.e. iff $c = k$, and the overall step $q_n \triangleright q_0$ fails iff $c \geq 1$. Give $\sigma_1, \dots, \sigma_n$ mass ε each and σ_0 mass $1 - n\varepsilon$ (for $\varepsilon \leq 1/n$). Each step is valid with probability exactly $1 - \varepsilon$, and the argument exactly with probability $1 - n\varepsilon$. At $\varepsilon = 1/n$, the argument is valid under no sharpening in the support of μ . `T4-checks/robust_core_and_md1.py` checks the cases $n = 3, 5, 10$. $\square$

Proof of Proposition 10.38. (a) A step $s = (\Gamma \triangleright y) \in A_0$ can be refuted only by a model of T in which $\rho^* \Gamma$ holds and $\rho^* y$ fails; such a model exists iff $s \notin \models_T$. By completeness of monster presentation, s survives iff $s \in \models_T$, so $A_\infty = A_0 \cap \models_T$. By Gödel completeness (Gödel, 1930), each surviving step has a derivation, in any complete calculus C for T , of $\rho^* y$ from $\rho^* \Gamma$. For an argument chained from surviving steps, replacing each step by such a derivation and concatenating

yields a C -derivation of the ρ^* -image of the conclusion from the images of the premises. This is Lemma 10.6 with the derivations made explicit.

(b) Here $A_\infty = \{\chi \triangleright \theta : \text{PA} \vdash \chi \rightarrow \theta\}$, which is r.e. but undecidable: with χ the sentence $0 = 0$, membership decides PA-provability of arithmetic sentences. Let C be a complete calculus for PA satisfying the effectivity assumptions, and suppose $A_\infty \subseteq \text{St}_C^g$ for some g . Since $\text{St}_C^g \subseteq \models_{\text{PA}}$ and $A_0 \cap \models_{\text{PA}} = A_\infty$, we get $A_\infty = A_0 \cap \text{St}_C^g$, which is decidable because A_0 and St_C^g are. Contradiction.

(c) Let $A_0 = \bigcup_{j \leq J} \text{inst}(\mathcal{S}_j)$. Under schema-level selection with complete monster presentation, $\mathcal{S}_j$ survives iff none of its instances is refuted, iff all its instances lie in $\models_T$. For each surviving $\mathcal{S}_j$, the template π_j instantiates to a derivation of each instance with at most g rule applications, and its formulas have size at most $|\pi_j|$ times the size of the instance. Choosing the size bound of Definition 10.2 to be at least $\max_j |\pi_j|$ times the input size, every surviving step is in St_C^g . The templates are finitely many derived rules of C , whose instances are the surviving practice. *Pure propositional schemas.* Let $\mathcal{S}$ have metavariables $A_1, \dots, A_k$ and mention no atoms, with all instances valid. Its generic instance $\mathcal{S}[A_i := p_i]$, for distinct atoms p_i , is valid, so it has a derivation π in a complete propositional calculus with schematic rules. Applying the substitution $p_i \mapsto \varphi_i$ to every formula of π yields a derivation, with the same number of applications, of $\mathcal{S}[A_i := \varphi_i]$, because schematic rules are closed under uniform substitution. So π is a uniform template. *Failure in PA.* Let $\theta(n)$ be $\neg \text{Prf}_{\text{PA}}(n, \ulcorner 0 = 1 \urcorner)$. Each instance $\theta(\bar{n})$ is a true statement about a numeral, decided by a primitive recursive relation, and is therefore provable in PA (by Σ_1 -completeness, since PA is consistent). Suppose a template with a schematic letter for the numeral instantiated to a derivation of $\theta(\bar{n})$ for every n . In a standard Hilbert calculus, substituting a fresh variable x for the schematic letter turns it into a derivation of $\theta(x)$, and generalization gives $\text{PA} \vdash \forall x \theta(x)$, i.e. $\text{PA} \vdash \text{Con}(\text{PA})$. This contradicts Gödel's second incompleteness theorem. $\square$

J Proofs for Section 11

The following results are proved completely in the main text: Propositions 11.3, 11.4, 11.10, 11.19, 11.22 and 11.40, Theorems 11.8, 11.16, 11.28 and 11.38, and Lemma 11.20. The minimal-repair drift stated after Theorem 11.28 (T3 Prop 4.2) is immediate: enlarging a tolerance enlarges an interval, so coherence is preserved upward. Proposition 11.33 and the numerical claims marked (computed) are computations, reproduced by the scripts in `research/theory/T3-checks/`; they are not proved here. Everything else is proved below. Throughout, w is a world, $\text{dom} = \text{dom } \mathfrak{M}_w$, and $\delta(F \times G) := \{\lambda[D:=\mu] : \lambda \in F, \mu \in G\}$.

J.1 Preliminaries on the filters of Definition 11.2

Lemma J.1 (Filter bases). (a) For $c = \text{DEF}(D, \lambda^\circ, \mathcal{G})$ with parent p , the family $\mathcal{B}_c = \{\delta(F \times G) \cap \text{dom} : F \in \mathcal{F}_p(w), G \in \mathcal{G}\}$ is a filter base: $X \in \mathcal{F}_c(w)$ iff $X \supseteq B$ for some $B \in \mathcal{B}_c$, and c is proper at w iff every member of $\mathcal{B}_c$ is nonempty. (b) For $c = \text{SUP}(A)$ with parent p , $X \in \mathcal{F}_c(w)$ iff $X \supseteq F \cap \|A\|_w$ for some $F \in \mathcal{F}_p(w)$. (c) If the parent filter is principal at a point λ , then in (a) one may take $F = \{\lambda\}$: $X \in \mathcal{F}_c(w)$ iff $X \supseteq \{\lambda[D:=\mu] : \mu \in G\} \cap \text{dom}$ for some $G \in \mathcal{G}$.

Proof. (a) $\delta(F_1 \times G_1) \cap \delta(F_2 \times G_2) \supseteq \delta((F_1 \cap F_2) \times (G_1 \cap G_2))$, and $F_1 \cap F_2 \in \mathcal{F}_p(w)$, $G_1 \cap G_2 \in \mathcal{G}$. So $\mathcal{B}_c$ is directed downward, and the filter it generates consists of the supersets (in dom) of its members. It contains $\emptyset$ iff some member is empty. (b) Likewise $\{F \cap \|A\| : F \in \mathcal{F}_p(w)\}$ is

closed under finite intersections. (c) If $\mathcal{F}_p(w)$ is principal at λ , every $F \in \mathcal{F}_p(w)$ contains λ , so $\delta(\{\lambda\} \times G) \subseteq \delta(F \times G)$, and $\{\lambda\} \in \mathcal{F}_p(w)$. $\square$

J.2 Proofs for §11.2

Proof of Proposition 11.5. (a) The root filter is principal at λ^* , where $p_{\text{atm}} = 1.0 \times 10^5$. The frame has no phases, and by hypothesis the model exists at $\lambda^*[p_{\text{atm}} := 0]$; by Lemma J.1(c) with the principal path the child filter is principal there; hence $c \models p_{\text{atm}} = 0$. Both filters are principal, hence proper, and by Proposition 11.3(b) each context's theory is consistent.

(b) Under limit semantics the only generator is $\{\lambda^*[m_r := 0]\} \cap \text{dom} = \emptyset$, so the limit child is improper. Under germ semantics the generators are $\{\lambda^*[m_r := \mu] : 0 < |\mu| < \eta\} \cap \text{dom} = \{\lambda^*[m_r := \mu] : 0 < \mu < \eta\}$, all nonempty, so the germ child is proper (Lemma J.1(a)). In the model at $m_r = \mu > 0$, $a = \Phi/\mu$, so $|a| > K$ whenever $\mu < |\Phi|/\max(K, 1)$; the generator with $\eta = |\Phi|/\max(K, 1)$ lies inside $\| |a| > K \|$.

(c) Let $E = \{\mu : \lambda^*[D := \mu] \in \text{dom}\}$. By Lemma J.1(c), $c_{\text{germ}} \models Q \in I$ iff there is $G \in \mathcal{G}$ with $q(\mu) \in I$ for all $\mu \in G \cap E$. “For every open $I \ni q(\lambda^\circ)$ there is $G \in \mathcal{G}$ with $q(G \cap E) \subseteq I$ ” is the definition of $q(\mu) \rightarrow q(\lambda^\circ)$ along the trace of $\mathcal{G}$ on E . (Intervals with endpoints named in L , e.g. rational ones, suffice for convergence.) The limit child is principal at $\lambda^*[D := \lambda^\circ]$, so its value of Q is $q(\lambda^\circ)$. $\square$

Proof of Lemma 11.7. (a) Let $F = \|\varphi\|_w \in \mathcal{F}_p(w)$ and $G \in \mathcal{G}$. Every element of $\delta(F \times G) \cap \text{dom}$ has the form $\lambda[D := \mu] \in \text{dom}$ with $\lambda \in F$; it differs from λ only in coordinates in D , so by D -stability it satisfies φ . Thus this generator lies inside $\|\varphi\|_w$, and $\|\varphi\|_w \in \mathcal{F}_c(w)$. Laws are D -stable because they hold on all of dom . An arithmetic formula in the constants p_i , $i \notin D$, is evaluated in the standard real sort with p_i interpreted as λ_i , so its truth value depends only on $(\lambda_i)_{i \notin D}$, which points differing only on D share.

(b) Let $\lambda, \lambda' \in \text{dom}$ differ only on D , with $\mathfrak{M}_w(\lambda) \models \varphi$. Take the actual point $\lambda^* = \lambda$ and the ideal value $\lambda^\circ = \lambda'_D$. Then $@ \models \varphi$, and the limit child's only generator is $\{\lambda[D := \lambda'_D]\} \cap \text{dom} = \{\lambda'\}$, so its filter is principal at λ' and $c \models \varphi$ says $\mathfrak{M}_w(\lambda') \models \varphi$. Exchanging the roles of λ and λ' gives the converse.

(c) Every generator $\{\lambda[k := 0] : \lambda \in F\} \cap \text{dom}$ of the child lies inside $\|k = 0\|$, so $c \models k = 0$ and, by Proposition 11.3(a), $c \models k \leq 1$, for every parent filter and every world (if c is improper it proves everything). So an import of φ into c never leads from a truth to a falsehood. But if dom contains λ and λ' differing only in k , with $k = 0.5$ and $k = 2$ respectively, then $\mathfrak{M}_w(\lambda) \models \varphi$ and $\mathfrak{M}_w(\lambda') \not\models \varphi$. $\square$

J.3 Proofs for §11.3: realizability

AUX contexts are identified with their parents throughout. For a set $M \subseteq \mathcal{K}$ and a sentence A , $\|A\|$ in (R1) denotes $\{N \in \mathcal{K} : N \models A\}$. Since SUP contexts have only SUP descendants, the parent of a DEF context is never a SUP context.

Proof of Theorem 11.12(a). Observation 1 (collapse). By (R1) and induction along SUP chains, $M_c = M_b \cap \bigcap_{A \in \text{Asm}(c)} \|A\|$ for every SUP context c with base b . Hence (c, Γ, φ) satisfies (R3) at M_c iff every $N \in M_b$ with $N \models \bigwedge \text{Asm}(c) \wedge \bigwedge \Gamma$ satisfies φ , i.e. iff $(b, \Gamma \cup \text{Asm}(c), \varphi)$ satisfies (R3) at M_b . So, given (R1), the judgments of J hold iff $M_b \subseteq \{N : N \models \text{Th}^*(b)\}$ for every non-SUP context b .

Observation 2 (anchoring lemma). In any realization, $M_c \neq \emptyset$ for every $c \in \text{Anc}$. We induct along the inductive definition of Anc . The root is nonempty by (R0) and certified contexts by (R5). Suppose c enters through an informative use (c, β, t) with $M_{\pi c} \neq \emptyset$, and suppose $M_c = \emptyset$. Then $M_c \models \phi_\beta(v)$ for every $v \in V$ (vacuously), so (R4) gives $M_{\pi c} \models \sigma_\beta \rightarrow \varepsilon_\beta(v)$ for all v . Since $(\pi c, \emptyset, \sigma_\beta) \in J$, (R3) gives $M_{\pi c} \models \sigma_\beta$. Any $N \in M_{\pi c}$ then satisfies $\{\sigma_\beta\} \cup \{\varepsilon_\beta(v) : v \in V_0\}$, contradicting informativeness, since $N \in \mathcal{K}$.

($\Rightarrow$) For $c \in \text{Anc}$, $M_c \neq \emptyset$ by Observation 2, and any $N \in M_c$ satisfies $\text{Th}^*(c)$ by Observation 1.

($\Leftarrow$) For each $c \in \text{Anc}$ choose $N_c \in \mathcal{K}$ with $N_c \models \text{Th}^*(c)$, and put $M_c = \{N_c\}$. Put $M_c = \emptyset$ for every non-SUP context outside Anc (these are DEF contexts, since $@ \in \text{Anc}$), and define SUP contexts by (R1). Then (R0), (R1), (R2) and (R5) hold by construction, since $@ \in \text{Anc}$ and $C_{\text{cert}} \subseteq \text{Anc}$. (R3) holds by Observation 1: at anchored b because $N_b \models \text{Th}^*(b)$, and at non-anchored b vacuously. For (R4), take a use (c, β, t) . If $\pi c \notin \text{Anc}$, then $M_{\pi c} = \emptyset$ and (R4) is vacuous. If $\pi c \in \text{Anc}$, then $c \in \text{Anc}$ because β is informative, so $M_c = \{N_c\}$. The judgment $(c, \emptyset, Q = \bar{t})$ lies in J^* unchanged (c is not a SUP context), so $Q^{N_c} = t$; by standard names, $M_c \models Q = \bar{v}$ only for $v = t$. Finally $(\pi c, \emptyset, \varepsilon_\beta(t)) \in J$ gives $N_{\pi c} \models \varepsilon_\beta(t)$, so $M_{\pi c} \models \sigma_\beta \rightarrow \varepsilon_\beta(t)$. $\square$

Proof of Theorem 11.12(b). Fix a frame realization at w .

Filter anchoring lemma: every $c \in \text{Anc}$ is proper at w . The root is proper, and certified contexts are proper by assumption. Suppose c enters through an informative use (c, β, t) with πc proper, and suppose c is improper. Then $c \models_w \phi_\beta(v)$ for every v , so bridge soundness gives $\pi c \models_w \sigma_\beta \rightarrow \varepsilon_\beta(v)$ for all v . The judgment $(\pi c, \emptyset, \sigma_\beta)$ holds, so $\pi c \models_w \varepsilon_\beta(v)$ for all v (Proposition 11.3(a)). By informativeness, $\{\sigma_\beta\} \cup \{\varepsilon_\beta(v) : v \in V_0\}$ has no model in $\mathcal{K}$ for some finite V_0 ; every $\mathfrak{M}_w(\lambda)$ is in $\mathcal{K}$, so the truth set of the finite conjunction is $\emptyset$. That set lies in $\mathcal{F}_{\pi c}(w)$, contradicting properness.

Conclusion. Let $c \in \text{Anc}$. For a SUP descendant $c' = \text{SUP}(A_k)$ of ... of $\text{SUP}(A_1)$ of c , iterating Proposition 11.4 and using Proposition 11.3(a) shows that $c' \models_w \chi$ iff $c \models_w \bigwedge_i A_i \rightarrow \chi$. Hence each judgment (c', Γ, φ) holds iff $c \models_w \bigwedge \text{Asm}(c') \wedge \bigwedge \Gamma \rightarrow \varphi$, and so $\text{Th}^*(c) \subseteq \text{Th}_w(c)$. Since $\text{Th}^*(c)$ is finite, the intersection of the truth sets of its members lies in $\mathcal{F}_c(w)$, and it is nonempty because c is proper. Any λ in it gives a model $\mathfrak{M}_w(\lambda) \in \mathcal{K}$ of $\text{Th}^*(c)$. No compactness is used. The argument does not depend on whether paths are principal or germs. $\square$

Proof of Theorem 11.12(c). *CE1.* $C_{\text{cert}} = \{c\}$, so $\text{Anc} = \{@, c\}$; $\text{Th}^*(@) = \{p_1 = 1\}$ and $\text{Th}^*(c) = \{p_1 = 0\}$ are each satisfiable in $\mathcal{K}$ (the constant p_1 may denote any real), so the criterion holds. In a frame realization, the root judgment forces $\lambda_1^* = 1$. The certified c is proper, so $\lambda^*[2:=0] \in \text{dom}$ and $\mathcal{F}_c$ is principal at that point, whose first coordinate is 1; hence $c \models p_1 = 1$, and $c \not\models p_1 = 0$ since $\text{Th}_w(c)$ is consistent. *CE2.* Two siblings with the same declaration have the same generators, hence the same filter. Both are certified, hence proper; the judgments $Q = 1$ and $Q = 2$ would put $\|Q = 1\| \cap \|Q = 2\| = \emptyset$ in this proper filter. The criterion holds because each of $\{Q = 1\}$, $\{Q = 2\}$ is satisfiable. Both counterexamples are also brute-forced over all frames on $\Lambda = \{0, 1\}^2$ (`realizability_frame.py`). $\square$

The exact frame criterion under limit semantics, summarized in the main text after Theorem 11.12, is the following.

Proposition J.2 (Frame realizability under limit semantics). *[T3 Prop 2.7, added after verification] (brute-force checked) Assume every DEF context in J uses limit semantics, every used bridge is an informative value-export bridge, and SUP contexts have only SUP descendants. For a candidate actual point λ , define the points of the non-SUP contexts by $\lambda_{@} = \lambda$ and $\lambda_c = \lambda_{\pi c}[D_c := \lambda_c^\circ]$.*

Let $S(\lambda)$ be the least set of non-SUP contexts that contains $@$ and C_{cert} and is closed under (i) parents: $c \in S \Rightarrow \pi c \in S$; (ii) informative bridges: $(c, \beta, t) \in U$ and $\pi c \in S \Rightarrow c \in S$; (iii) coincidence: $\pi c \in S$ and $\lambda_c = \lambda_{c'}$ for some $c' \in S \Rightarrow c \in S$. Then J is frame-realizable iff there is $\lambda \in \Lambda$ such that every point $x \in \{\lambda_c : c \in S(\lambda)\}$ lies in Λ and $\bigcup \{\text{Th}^*(c) : c \in S(\lambda), \lambda_c = x\}$ has a model $N \in \mathcal{K}$ with $p_i^N = x_i$ for all i .

The criterion is non-local in two ways: it quantifies over the actual parameter values, and it conjoins the theories of all forced-proper contexts that sit at one parameter point. CE1 fails it because c 's point shares p_1 with the root, and CE2 because the two siblings share a point. `realizability_frame.py` compares it with brute-force frame realizability over all frames on $\Lambda = \{0, 1\}^2$ (up to three DEF and two SUP contexts, 0–2 bridge uses per context): 8,000 instances, 0 mismatches.

Proof of Proposition J.2. Shape of the filters. Under limit semantics every non-SUP context's filter is improper or principal at its point. Indeed $\mathcal{F}_@$ is principal at $\lambda^* = \lambda_@$. If $\mathcal{F}_{\pi c}$ is principal at $\lambda_{\pi c}$, Lemma J.1(c) with $G = \{\lambda_c^\circ\}$ shows that $\mathcal{F}_c$ is generated by $\{\lambda_c\} \cap \text{dom}$: principal at λ_c if $\lambda_c \in \text{dom}$, improper otherwise. If $\mathcal{F}_{\pi c}$ is improper, so is $\mathcal{F}_c$, since $\delta(\emptyset \times G) = \emptyset$. A SUP context below a base principal at x is principal at x or improper (Lemma J.1(b)), and its judgments collapse into the base by Proposition 11.4.

($\Rightarrow$) Let $(\mathfrak{M}, \lambda^*)$ be a frame realization and let P be the set of proper non-SUP contexts. P contains $@$ and C_{cert} . It is closed under (i), since an improper parent has an improper child; under (ii), by the filter anchoring lemma in the proof of Theorem 11.12(b); and under (iii): if $\pi c \in P$ and $\lambda_c = \lambda_{c'}$ with $c' \in P$, then $\lambda_{c'} \in \text{dom}$ (a proper context is principal at its point), so $\mathcal{F}_c$ is principal at $\lambda_c \in \text{dom}$ and c is proper. Hence $S(\lambda^*) \subseteq P$. For each point $x = \lambda_c$ with $c \in S(\lambda^*)$ we have $x \in \text{dom} \subseteq \Lambda$; every $c' \in S(\lambda^*)$ at x is principal at x , so $N = \mathfrak{M}(x)$ satisfies all their collapsed theories, and $p_i^N = x_i$.

($\Leftarrow$) Fix such a λ and put $S = S(\lambda)$, $\lambda^* = \lambda$, $\text{dom} = \{\lambda_c : c \in S\} \subseteq \Lambda$, and let $\mathfrak{M}(x)$ be the given model at x ; this is a frame, since $p_i^{\mathfrak{M}(x)} = x_i$, and $\lambda^* = \lambda_@ \in \text{dom}$. *Contexts in S are proper*: all their ancestors are in S by (i), so all the ancestors' points are in dom , and by induction each is principal at its point. *Non-SUP contexts outside S are improper*: by induction on depth; if $\pi c \notin S$, then πc is improper and so is c ; if $\pi c \in S$, then (iii) gives $\lambda_c \neq \lambda_{c'}$ for all $c' \in S$, i.e. $\lambda_c \notin \text{dom}$. *Judgments hold at a proper $c \in S$* , because $\mathfrak{M}(\lambda_c)$ satisfies $\text{Th}^*(c)$, which by the collapse covers the judgments of c and of its SUP descendants; at improper contexts they hold vacuously. *Certified contexts lie in S* , hence are proper. *Bridge uses (c, β, t)* : if πc is improper, soundness is vacuous. Otherwise $\pi c \in S$, so $c \in S$ by (ii), and c is principal at λ_c . Then $c \models Q = \bar{v}$ iff $Q^{\mathfrak{M}(\lambda_c)} = v$, which by the judgment $(c, \emptyset, Q = \bar{t})$ and standard names happens only for $v = t$; and $\pi c \models \varepsilon_\beta(t)$ by the judgment $(\pi c, \emptyset, \varepsilon_\beta(t))$. $\square$

Proof of Corollary 11.13. (i) In the frame-free sense, by the collapse a judgment $(c, \Gamma, \perp)$ at a SUP context c with base b becomes $\neg(\bigwedge \text{Asm}(c) \wedge \bigwedge \Gamma) \in \text{Th}^*(b)$. In the frame semantics the same follows from Proposition 11.4, iterated as in the proof of Theorem 11.12(b). Adding this sentence to $\text{Th}^*(b)$ makes a satisfiable theory unsatisfiable only if the theory already entails $\bigwedge \text{Asm}(c) \wedge \bigwedge \Gamma$.

(ii) Let c be a DEF context with $(c, \emptyset, \perp) \in J$; then $\perp \in \text{Th}^*(c)$. If $c \in \text{Anc}$, $\text{Th}^*(c)$ is unsatisfiable, so J is not frame-free realizable by Theorem 11.12(a), and not frame-realizable by (b). If $c \notin \text{Anc}$: Anc depends only on C_{cert} and U , and removing $(c, \emptyset, \perp)$ does not affect U (bridge uses involve only judgments of the forms $Q = \bar{t}$, σ_β , $\varepsilon_\beta(t)$; we assume $\perp$ is none of these). So by (a), J is frame-free realizable iff $J \setminus \{(c, \emptyset, \perp)\}$ is: the judgment imposes no constraint. In

the frame semantics this second half can fail: a non-anchored context may still be forced proper. Example: $J = \{(\textcircled{A}, \emptyset, p_1 = 0), (c, \emptyset, \perp)\}$ with c an uncertified limit child deforming $D = \{1\}$ toward 0 and no bridges. Then $c \notin \text{Anc}$ and J is frame-free realizable, but the root forces $\lambda_1^* = 0$, so c sits at $\lambda^* \in \text{dom}$, is proper, and cannot satisfy $\perp$.

(iii) Let $N \in \mathcal{K}$ satisfy the stripped set. Put $M_c = \{N\}$ at every non-SUP context and define SUP contexts by (R1). (R0), (R2), (R5) hold. (R3): each judgment (c, Γ, φ) has $N \models \bigwedge \Gamma \rightarrow \varphi$, and $M_c \subseteq \{N\}$. (R4): for a use (c, β, t) , $N \models Q = \bar{t}$, so $M_c \models Q = \bar{v}$ only for $v = t$, and $N \models \varepsilon_\beta(t)$; here πc is not a SUP context, so $M_{\pi c} = \{N\}$. The same construction works if N satisfies $\bigcup_c \text{Th}^*(c)$ (the collapsed variant), using Observation 1. Conversely, the three judgment sets of Corollary 11.9 are realizable but flagged: (i) $\{(\text{SUP}(A), \emptyset, \perp)\}$ is realized by $M_{\textcircled{A}} = \{N\}$ with $N \models \neg A$; (ii) $\{(\textcircled{A}, \emptyset, p_{\text{atm}} = 10^5), (c, \emptyset, p_{\text{atm}} = 0)\}$ and (iii) $\{(c_1, \emptyset, Q = 1), (c_2, \emptyset, Q = 2)\}$ by one structure per context. Each stripped set is unsatisfiable. The collapsed variant does not flag (i), but flags (ii) and (iii). $\square$

J.4 Proofs for §11.3: blame localization and hygiene

Proof of Corollary 11.17. (a) Let w be a world at which d is proper. The root is proper at every $w \in W_\rho$, and $W_\rho \neq \emptyset$. If every Ax in the cone were true at w and every Step, Imp and Exp instance in it sound at w , Theorem 11.16 (which uses only the items of the derivation tree, i.e. the cone) would give $d \models_w \perp$, i.e. $\emptyset \in \mathcal{F}_d(w)$. (b) A derivation of $d \Vdash \perp$ contains only the items in its tree, and no rule of Definition 11.15 takes ist-claims, or a SUP context's undischarged conclusions, as premises. (c) Let πd be proper at w with $\pi d \models_w \omega_d$. Take a generator $\delta(F \times G) \cap \text{dom}$ of $\mathcal{F}_d(w)$. Since $F \cap \|\omega_d\| \in \mathcal{F}_{\pi d}(w)$ and πd is proper, there is $\lambda \in F \cap \|\omega_d\|$. Then $E_\lambda = \{\mu : \lambda[D := \mu] \in \text{dom}\} \in \mathcal{G}$, and since $\mathcal{G}$ is proper, $G \cap E_\lambda \neq \emptyset$. For μ in it, $\lambda[D := \mu] \in \delta(F \times G) \cap \text{dom}$. So every generator is nonempty, and d is proper by Lemma J.1(a). $\square$

Proposition J.3 (Propagation of $\perp$ along informative bridges). *[T3 Cor 2.6, propagation clause] Let c be a DEF context with an informative value-export bridge β cited for c , and suppose $c \Vdash \perp$, $\pi c \Vdash \sigma_\beta$ and $\pi c \Vdash \omega_c$ are derived. Suppose $\hat{R}$ contains ex falso ($\perp \triangleright \psi$) and the kernel of πc is refutation-complete for finite sets (it derives $\perp$ from any finite set of derived sentences that has no model in $\mathcal{K}$). Then $\pi c \Vdash \perp$ is derivable by adding finitely many synthetic Exp instances. Iterating along an anchoring chain reaches the root or a certified context. Moreover, fix a world w at which πc is proper: (i) if c is proper at w , some item of c 's own derivation of $\perp$ is unsound at w ; (ii) if c is improper at w , all of c 's own items are sound at w , and some item of the derivation of $\pi c \Vdash \omega_c$ is unsound at w ; (iii) without the certificate premise in Exp, in case (ii), if the derivation of $\pi c \Vdash \sigma_\beta$ is sound at w , some synthetic Exp instance is unsound at w : the bridge applied to an ill-posed context.*

Proof. Informativeness gives $t_1, \dots, t_k$ with $\{\sigma_\beta, \varepsilon_\beta(t_1), \dots, \varepsilon_\beta(t_k)\}$ unsatisfiable in $\mathcal{K}$. Ex falso gives $c \Vdash Q = \bar{t}_i$ for each i , and k Exp instances give $\pi c \Vdash \varepsilon_\beta(t_i)$. Refutation completeness gives $\pi c \Vdash \perp$. Iterate for πc along the chain by which c entered Anc.

(i) is the contrapositive of Theorem 11.16 in c . (ii) If c is improper at w , so is every context below it (SUP filters refine their parent's, and a DEF child of an improper context has the empty generator $\delta(\emptyset \times G) \cap \text{dom}$). Every set lies in an improper filter, so every Ax in c 's subtree is true at w , and every Step, Imp and Exp instance concluding in that subtree is sound at w . By Corollary 11.17(c), $\pi c \not\models_w \omega_c$, so by Theorem 11.16 the derivation of $\pi c \Vdash \omega_c$ contains an item unsound at w . (iii) Without ω_c , suppose the derivation of σ_β is sound at w , so $\pi c \models_w \sigma_\beta$. If

every synthetic instance were sound at w , then, as $c \models_w Q = \bar{t}_i$ trivially, $\pi c \models_w \varepsilon_\beta(t_i)$ for all i , and the finitely many truth sets would have empty intersection in the proper filter $\mathcal{F}_{\pi c}(w)$. $\square$

Proof of Theorem 11.18. (a) Let $c' = \text{SUP}(A)$ be the child of c in which the reductio is carried out. By Lemma J.1(b), $\mathcal{F}_{c'}(w) \supseteq \mathcal{F}_c(w)$, and $\|A\| \in \mathcal{F}_{c'}(w)$. So the premises K_c (true in c) are true in c' , A is true in c' , and every c -sound instance has its truth set in $\mathcal{F}_c(w) \subseteq \mathcal{F}_{c'}(w)$, i.e. is c' -sound. By induction along the derivation (filter closure, Proposition 11.3(a)) every line is true in c' , so $c' \models_w \perp$, and Proposition 11.4 gives $c \models_w A \rightarrow \perp$, i.e. $c \models_w \neg A$.

(b) A derivation of $\perp$ from K_c is also a derivation of $\perp$ from $K_c \cup \{B\}$ for every B , in particular for $B = A$ and $B = \neg A$. The set of refuted sentences is then all sentences, whatever the world.

(c) From $A \rightarrow q$ and $\neg A \rightarrow q$, case analysis gives q , contradicting $\neg q$; so K is unsatisfiable. The valuation $A = q = 0$ satisfies $\{A \rightarrow q, \neg q\}$, and $A = 1, q = 0$ satisfies $\{\neg A \rightarrow q, \neg q\}$. D_A derives q from A and $A \rightarrow q$ (modus ponens) and then $\perp$ from $q, \neg q$; $D_{\neg A}$ is the same with $\neg A$. As a derivation (used premises, steps, conclusion), D_A is identical to the reductio of A from the consistent set $K' = \{A \rightarrow q, \neg q\}$, and that reductio is sound because $K' \cup \{A\}$ is unsatisfiable. So a test that depends only on the derivation and accepts every sound reductio from a consistent premise set accepts D_A ; likewise $D_{\neg A}$ with $K'' = \{\neg A \rightarrow q, \neg q\}$. The accepted conclusions $\neg A$ and $\neg \neg A$ are jointly unsatisfiable. (`hygiene.py` checks these facts by truth tables.)

(d) Suppose $N \models A$. Every leaf of the reductio is in $K_c \cup \{A\}$, hence true in N ; every step has true premises and an instance true in N , hence a true conclusion. So $N \models \perp$, which is impossible; thus $N \models \neg A$. If both A and $\neg A$ were refuted by reductions whose instances N satisfies, then $N \models \neg A$ and $N \models \neg \neg A$. For the counterexample: if $c \models q$, the zero-premise instance $\triangleright q$ is c -sound; with $K_c = \{\neg q\}$, every $\text{SUP}(B)$ derives q and $\neg q$, hence $\perp$ by the sound instance $q, \neg q \triangleright \perp$, so every B is refuted; a model of $\neg q$ exists but violates the instance $\triangleright q$. For existence: if c is proper at w , $c \models_w K_c$ and the finitely many instances used are c -sound at w , then the truth sets of the members of K_c and of the instance conditionals $\bigwedge \varphi_i \rightarrow \psi$ all lie in $\mathcal{F}_c(w)$; their intersection lies in $\mathcal{F}_c(w)$ and is nonempty, and any λ in it gives $N = \mathfrak{M}_w(\lambda)$.

(e) By (b), each A yields both labels; they are jointly unsatisfiable, a fixed function of K_c , and the same in every world. If K_c is consistent and $\hat{R}$ logically sound, a model of K_c satisfies every instance, so by (d) no A and $\neg A$ are both refuted. $\square$

J.5 Proofs for §11.4

Proof of Theorem 11.21. In each case we exhibit f meeting the hypotheses of Lemma 11.20.

(a) Let $f(\lambda) = e^{-1/\lambda^2}$ for $\lambda > 0$ and $f(0) = 0$. It is C^∞ on $[0, \Lambda]$ with all derivatives 0 at 0, and $f(\lambda^*) > 0$. For $Q \in \mathcal{C}$ and $A \in \mathbb{R}$, $Af \in C^\infty$, so $Q + Af \in \mathcal{C}$ by the closure hypothesis, and $Q + Af$ has the same jet at 0 as Q .

(b) Since $\lambda^* \notin \bar{S}$, some interval $(\lambda^* - r, \lambda^* + r)$ misses S . Let f be a C^∞ bump supported in it with $f(\lambda^*) = 1$. Then $Af \in C^\infty$, $Q + Af \in \mathcal{C}$, and $(Q + Af)|_S = Q|_S$. The germ at 0 is the special case $S = [0, \lambda^*/2)$, since two functions that agree on $[0, \lambda^*/2)$ have the same germ.

(c) Let $f(\lambda) = \lambda^{N+1}$. It is entire, so $Q + Af$ is (the restriction of) an entire function; its derivatives of order $\leq N$ at 0 vanish; and $f(\lambda^*) \neq 0$.

(d) is (b) with the finite, hence closed, set $S = \{\lambda_1, \dots, \lambda_m\}$.

Weaker hypothesis. If only $\mathcal{C} \supseteq C^\infty$, a rule sound on $\mathcal{C}$ is sound on C^∞ , which satisfies $C^\infty + C^\infty \subseteq C^\infty$, so the rule abstains on every $Q \in C^\infty$. Outside C^∞ it need not: let $\mathcal{C} = C^\infty \cup \{\sqrt{\lambda}\}$, I the germ at 0, and let the rule output $[\sqrt{\lambda^*}, \sqrt{\lambda^*}]$ when shown the germ of

$\sqrt{\lambda}$ and abstain otherwise. No smooth function has the germ of $\sqrt{\lambda}$ (it is not differentiable at 0 from the right), so the rule is sound on $\mathcal{C}$, and it does not abstain on $\sqrt{\lambda}$. $\square$

Proof of Theorem 11.23. (a) At every world compatible with the parent's derivation, $\lambda^* \in U$, so $|Q(\mathfrak{M}(\lambda^*)) - q| \leq \epsilon$ by the frame theorem. (b) The negative half is Theorem 11.21(b) (the germ is information on a set whose closure omits λ^*); the positive half is (a) with U and the stated rate. (c) The root filter is the declared germ, so “ $Q \sim q$ ” is a root claim; the decidability statement is Gruntz's algorithm for limits of exp-log functions, modulo zero-equivalence of constants (Gruntz, 1996), which is undecidable for richer classes (Richardson, 1968). $\square$

Proof of Theorem 11.24. (a) Let $u(t) = |x(t) - y(t)|$. From $x(t) - y(t) = x_0 - y_0 + \int_0^t (f(x) - f(y) - g(s, y))ds$, the Lipschitz bound on K and $|g(s, y(s))| \leq \eta$ give $u(t) \leq u(0) + \int_0^t (Lu + \eta) ds =: v(t)$. Then v is C^1 with $v' = Lu + \eta \leq Lv + \eta$, so $\frac{d}{dt}(e^{-Lt}(v + \eta/L)) = e^{-Lt}(v' - Lv - \eta) \leq 0$, and $u \leq v \leq (u(0) + \eta/L)e^{Lt} - \eta/L = B(t)$.

(b) Let $t_1 = \sup\{t \in [0, T] : |x(s) - y(s)| \leq r \text{ for all } s \in [0, t]\}$. The set contains 0, because $|x_0 - y_0| = B(0) \leq B(T) < r$, and it is closed by continuity, so the supremum is attained. On $[0, t_1]$ both $x(s)$ and $y(s)$ lie in K_r , and $|g(s, y(s))| \leq \eta$ there; so (a) applies on $[0, t_1]$ with $K = K_r$ and gives $|x - y| \leq B(t) \leq B(T) < r$. If $t_1 < T$, continuity gives $|x - y| < r$ on a neighbourhood of t_1 , contradicting the definition of t_1 . Hence $t_1 = T$. $\square$

Proof of Theorem 11.25. Write $v = (v_x, v_y)$, $v(0) = v_0(\cos \theta, \sin \theta) = (v_{0x}, v_{0y})$, and $\dot{v} = -g\hat{y} - \kappa v$; existence of the solution up to landing is the root model's well-posedness. Let $T = \inf\{t > 0 : y(t) = 0\}$; since $\dot{y}(0) = v_{0y} > 0$, $y > 0$ on $(0, T)$.

Speed bound. On $[0, T]$, $\frac{d}{dt}(\frac{1}{2}|v|^2 + gy) = v \cdot \dot{v} + gv_y = -\kappa|v|^2 \leq 0$, and $y \geq 0$; so $|v| \leq v_0$ and $|a_n| = \kappa|v| \leq \kappa|v|^2 \leq \kappa v_0^2 = xg$.

Flight time. On $[0, T]$, $\ddot{y} = -g + a_{n,y} \in [-g(1+x), -g(1-x)]$, so $v_{0y}t - \frac{1}{2}g(1+x)t^2 \leq y(t) \leq v_{0y}t - \frac{1}{2}g(1-x)t^2$. The upper bound vanishes at $T_{hi} = 2v_{0y}/(g(1-x))$ and is negative after it; if $T > T_{hi}$ (or $T = \infty$), then $y(T_{hi}) \leq 0$ with $T_{hi} \in (0, T)$, a contradiction. So $T \leq T_{hi}$. The lower bound is positive on $(0, T_{lo})$ with $T_{lo} = 2v_{0y}/(g(1+x))$; if $T < T_{lo}$ then $y(T) > 0$, a contradiction. So $T \geq T_{lo}$.

Horizontal motion. $\dot{v}_x = -\kappa v_x$ gives $v_x = v_{0x}e^{-\int_0^t \kappa} > 0$, so X is increasing, and $\ddot{X} = -\kappa v_x \in [-\kappa|v|, 0] \subseteq [-xg, 0]$. Hence $R = X(T) \leq v_{0x}T \leq v_{0x}T_{hi} = 2v_{0x}v_{0y}/(g(1-x)) = R_0/(1-x)$, using $R_0 = 2v_{0x}v_{0y}/g$. And $R \geq X(T_{lo}) \geq v_{0x}T_{lo} - \frac{1}{2}xgT_{lo}^2 = \frac{R_0}{1+x} - \frac{2xv_{0y}^2}{g(1+x)^2}$; since $R_0 \tan \theta = 2v_{0y}^2/g$, this is the stated lower bound. $\square$

Proof of Theorem 11.26. (a) Triangle inequality. (b) The segment from q to Q lies in the convex box, on which F is differentiable; the mean value theorem gives $F(Q) - F(q) = \nabla F(\xi) \cdot (Q - q)$ for some ξ on the segment, and $|\nabla F(\xi) \cdot (Q - q)| \leq \sum_j L_j \delta_j$. (c) If the budget is at least $|F(q)|$, the certified interval $[F(q) - \sum L_j \delta_j, F(q) + \sum L_j \delta_j]$ contains 0. With $Q_1 \in [99.8, 100.2]$ and $Q_2 \in [99.7, 100.1]$, $Q_1 - Q_2 \in [-0.3, 0.5]$. For $F(q) = q^2$ on $[0.4, 1.6]$, $L = 3.2$ and the budget is 1.92, while $F([0.4, 1.6]) = [0.16, 2.56]$. $\square$

Proof of Theorem 11.27. (a) In a child with canonical stipulation $p_D = 0$ (e.g. $k = 0$), the smallness parameter (e.g. $s = \kappa v_0^2/g$) is 0, so $c \vdash s \leq \eta$ is derivable soundly in the child, whatever the world. An Exp rule that accepts it therefore exports $Q(\lambda^*) \in [Q(0) \pm \epsilon]$ from the value of Q at the ideal point alone. This is an export rule whose information is a single sample at $0 \neq \lambda^*$, so by Theorem 11.21(d) it is unsound on any class closed under adding smooth functions, for every ϵ . The ping-pong numbers are those of Theorem 11.25 (computed).

(b) With k and g both stipulated in the child, the child's range is $R_0(g') = v_0^2 \sin 2\theta/g'$, which takes any positive value as g' varies, while the parent condition $kv_0^2/g \leq \eta$ is evaluated at the actual g and holds for the steel ball (0.0276). The export $R \in [R_0(g') \pm \epsilon]$ is false for g' far from g .

(c) By Theorem 11.16 and Lemma 11.7(a), it suffices that each Exp instance is sound at every w . Let $p = \pi c$ and suppose $c \models_w Q = \bar{v}$, $p \models_w \sigma_\beta$ and $p \models_w \omega_c$. By Lemma J.1(a) there are $F_1 \in \mathcal{F}_p(w)$ and $G \in \mathcal{G}$ with $\delta(F_1 \times G) \cap \text{dom} \subseteq \|Q = \bar{v}\|$. Let $F = F_1 \cap \|\sigma_\beta\| \cap \|\omega_c\| \in \mathcal{F}_p(w)$ and fix $\lambda \in F$. Since $\lambda \in \|\omega_c\|$, $E_\lambda = \{\mu : \lambda[D:=\mu] \in \text{dom}\} \in \mathcal{G}$. For $\mu \in G \cap E_\lambda$, $\lambda[D:=\mu] \in \delta(F_1 \times G) \cap \text{dom}$, so $Q(\mathfrak{M}_w(\lambda[D:=\mu])) = v$. Hence $\{\mu : \lambda[D:=\mu] \in \text{dom}, Q(\mathfrak{M}_w(\lambda[D:=\mu])) = v\} \supseteq G \cap E_\lambda \in \mathcal{G}$, and the schema's antecedent holds at λ ; as $\mathfrak{M}_w(\lambda) \models \sigma_\beta$, the schema gives $\mathfrak{M}_w(\lambda) \models \varepsilon_\beta(v)$. So $F \subseteq \|\varepsilon_\beta(v)\|$, i.e. $p \models_w \varepsilon_\beta(v)$.

The rope. $\text{dom} = \{m_r > 0\}$, $D = \{m_r\}$, limit semantics. The child's generator $\{\lambda^*[m_r:=0]\} \cap \text{dom}$ is empty, so the child is improper and proves $a = 0$ by (vacuously) sound steps. The schema's antecedent requires $\lambda[m_r:=0] \in \text{dom}$, which never holds, so the schema is a theorem of the frame. The parent proves $m_r \leq \eta$ when the actual $m_r \leq \eta$, and Exp without ω_c would export $|a| \leq \epsilon$, whereas $|a| = |\Phi|/m_r \geq |\Phi|/\eta > \epsilon$ for $\eta < |\Phi|/\epsilon$. With the certificate requirement, $\|\omega_c\|$ must be empty (no λ has $\lambda[m_r:=0] \in \text{dom}$), so the proper root cannot soundly derive ω_c . $\square$

Lemma J.4 (Asymptotic schemas below a principal parent). *[T3 Remark after Thm 3.9] Read the child judgment “ $Q \simeq v$ ” as: $c \models_w |Q - \bar{v}| < \bar{r}$ for every rational $r > 0$. Call β an asymptotic schema for the declaration $(D, \lambda^\circ, \mathcal{G})$ if, for all w , all $\lambda \in \text{dom } \mathfrak{M}_w$ with $\mathfrak{M}_w(\lambda) \models \sigma_\beta$ and $E_\lambda = \{\mu : \lambda[D:=\mu] \in \text{dom } \mathfrak{M}_w\} \in \mathcal{G}$, and all v : if $Q(\mathfrak{M}_w(\lambda[D:=\mu])) \rightarrow v$ along the trace of $\mathcal{G}$ on E_λ , then $\mathfrak{M}_w(\lambda) \models \varepsilon_\beta(v)$. If πc is principal at w (the root, or a limit-semantics context below principal ancestors), the Exp instance from $c \Vdash Q \simeq v$, $\pi c \Vdash \sigma_\beta$ and $\pi c \Vdash \omega_c$ to $\pi c \Vdash \varepsilon_\beta(v)$ is sound at w .*

Proof. Let πc be principal at λ (if it is improper, soundness is vacuous). Then $\mathfrak{M}_w(\lambda) \models \sigma_\beta$ and $\lambda \in \|\omega_c\|$, so $E_\lambda \in \mathcal{G}$. By Lemma J.1(c), $c \models_w |Q - \bar{v}| < \bar{r}$ iff there is $G \in \mathcal{G}$ with $|Q(\mathfrak{M}_w(\lambda[D:=\mu])) - v| < r$ for all $\mu \in G \cap E_\lambda$. Holding for every rational $r > 0$, this is convergence to v along the trace of $\mathcal{G}$ on E_λ . The schema gives $\mathfrak{M}_w(\lambda) \models \varepsilon_\beta(v)$, i.e. $\pi c \models_w \varepsilon_\beta(v)$. Without ω_c the argument fails at $E_\lambda \in \mathcal{G}$: an improper child proves $Q \simeq v$ for every v . Under a non-principal parent the convergence would have to be uniform on a parent filter set, which is an additional side condition. $\square$

J.6 Proofs for §11.5

Proof of Theorem 11.29. (a) If $k > n$ then $\hat{e} = \infty$ and there is nothing to prove; note $k \geq 1$ for $\alpha < 1$. Call index $i \in \{1, \dots, n+1\}$ *high* if e_i is strictly greater than at least k of the other n values. If $e_{n+1} > \hat{e}$, the k -th smallest of $e_1, \dots, e_n$, then e_{n+1} exceeds at least k of them, so $n+1$ is high. In any realization at most $n+1-k$ indices are high (in sorted order, only positions $k+1, \dots, n+1$ can have k strictly smaller values). By exchangeability every index is high with the same probability p , so $(n+1)p = \mathbb{E}[\#\text{high}] \leq n+1-k$, and $\mathbb{P}(e_{n+1} > \hat{e}) \leq p \leq (n+1-k)/(n+1) \leq \alpha$ because $k \geq (n+1)(1-\alpha)$. Ties are handled by the strict inequality. (b) If the chosen instance has $e > \hat{e}$, it is not covered, and the solver can choose such an instance whenever one exists. $\square$

Proof of Theorem 11.30. We take an exact oracle (or any oracle whose answer is a fixed function of the value at the queried point); this only strengthens the certifier.

Deterministic. Fix $e \in \mathcal{C}$, let $\pi_1, \dots, \pi_q$ be the queries made on e and V the output. Suppose $\pi \in V$ is not queried. Let $f \geq 0$ be a continuous bump with $f(\pi) = 1$ whose support avoids the finite set $\{\pi_1, \dots, \pi_q\}$, and let $A > 0$. On $e' = e + Af \in \mathcal{C}$ the first query is π_1 , with the same answer since $f(\pi_1) = 0$; by induction the whole adaptive query sequence and the output coincide, so $\pi \in V(e')$. Soundness on e' requires $e(\pi) + A \leq \tau$ for all $A > 0$, which is impossible.

Randomized. Suppose that on some $e \in \mathcal{C}$ the event that V contains an unqueried point has positive probability (we assume the events below are measurable). For each outcome in this event there is an unqueried $\pi \in V$ and a ball B from a fixed countable base (rational centres and radii) with $\pi \in B$ and $\overline{B}$ disjoint from the finitely many queried points. Some fixed B occurs with positive probability. Let $f(\pi') = \text{dist}(\pi', \Pi \setminus B)$, a continuous bump, positive exactly on B . For outcomes in the event for B , the transcript on $e + Af$ equals that on e , so V still contains a point $\pi \in B$ with $f(\pi) > 0$, and $e(\pi) + Af(\pi) > \tau$ once A is large enough for that outcome. By monotone convergence, some fixed A makes the certifier unsound on $e + Af \in \mathcal{C}$ with positive probability. $\square$

Proof of Theorem 11.31. (a) If $\pi \in V$, then $\|\pi - \pi_j\|_\infty \leq (\tau - e_{hi}(\pi_j))/L$ for some centre π_j , so $e(\pi) \leq e(\pi_j) + L\|\pi - \pi_j\|_\infty \leq e_{hi}(\pi_j) + (\tau - e_{hi}(\pi_j)) = \tau$.

(b) Let $e(\pi) \leq \tau - \gamma$, and let π_j be the centre of a cell containing π , so $\|\pi - \pi_j\|_\infty \leq h/2$. Then $e_{hi}(\pi_j) \leq e(\pi_j) + \eta \leq e(\pi) + Lh/2 + \eta \leq \tau - \gamma + Lh/2 + \eta$, so the radius at π_j is at least $(\gamma - \eta)/L - h/2 \geq h - h/2 = h/2 \geq \|\pi - \pi_j\|_\infty$, and $\pi \in V$. With $h = (\gamma - \eta)/L$ there are $\lceil 1/h \rceil^d = \lceil L/(\gamma - \eta) \rceil^d$ centres.

(c) Let $e_0 \equiv \tau - \gamma$; completeness forces $V(e_0) = \Pi$. Put $\rho = \gamma'/L$ and pack $M = \lfloor 1/(2\rho) \rfloor^d$ disjoint open $\|\cdot\|_\infty$ -balls of radius ρ in Π . With fewer than M queries on e_0 , some ball $B(\pi_0, \rho)$ contains no query. Define $e_1 = e_0 + \max(0, \gamma' - L\|\pi - \pi_0\|_\infty)$. It is L -Lipschitz, equals e_0 outside the open ball, and $e_1(\pi_0) = \tau - \gamma + \gamma' > \tau$. All answers on e_1 agree with those on e_0 , so the deterministic certifier outputs $V(e_1) = \Pi \ni \pi_0$, which is unsound. $\square$

Proof of Theorem 11.32. (a) lo changes only to accepted points, so if $V = [0, lo]$ then $e_{hi}(lo) \leq \tau$ was observed, and for $\pi \leq lo$, $e(\pi) \leq e(lo) \leq e_{hi}(lo) \leq \tau$.

(b) Each query halves $hi - lo$, so $hi - lo = 2^{-k}$ at the end. With an exact oracle a rejection at hi means $e(hi) > \tau$, so by monotonicity $\{e \leq \tau\} \subseteq [0, hi]$; if nothing was rejected, $hi = 1$ and $\{e \leq \tau\} \subseteq [0, 1]$. If $V \neq \emptyset$, the missed part lies in $(lo, hi]$, of length 2^{-k} . If $V = \emptyset$, every query was rejected, so $hi = 2^{-k}$ and the missed part lies in $[0, hi)$. The endpoint variant spends two queries before bisecting, leaving $k - 2$ halvings.

(c) Let $\tau > 0$ and $e_t = 2\tau \mathbf{1}[\pi \geq t]$, non-decreasing with $\{e_t \leq \tau\} = [0, t]$. Exact answers on e_t take only the values 0 and 2τ , so a deterministic certifier with q queries has at most 2^q transcripts and hence at most 2^q outputs. Choose $M = \lceil 1/h \rceil$ thresholds in $(0, 1]$ pairwise more than h apart (possible since $(M - 1)h < 1$). If $2^q < M$, two thresholds $t < t'$ share a transcript and hence an output V . Soundness on e_t gives $V \subseteq [0, t)$, so on $e_{t'}$ the missed part contains $[t, t')$, of length more than h . Hence $2^q \geq M \geq 1/h$, i.e. $q \geq \lceil \log_2(1/h) \rceil$.

(d) *Sketch.* On a grid of side h , the cells meeting a hyperplane $\pi_1 + \dots + \pi_d = \text{const}$ form an antichain of size $\Theta(h^{1-d})$ for the coordinatewise order; down-sets can include or exclude each of them independently, so each must be probed. A staircase walk along the boundary of the down-set achieves the bound. $\square$

J.7 Proofs for §11.6

Proof of Theorem 11.37. Fix $w \in W_\rho$. We check that every Ax in the accepted derivation is true at w and every Step, Imp and Exp instance is sound at w ; then Theorem 11.16 gives $@ \models_w Q \in [q - \delta, q + \delta]$, and since $\mathcal{F}_@ (w)$ is principal at λ_w^* , $Q^{\mathfrak{M}_w(\lambda_w^*)} \in [q - \delta, q + \delta]$.

- *Step.* By (TB1) the instance conditional $\bigwedge \varphi_i \rightarrow \psi$ (and every cited law) is valid in every $\mathfrak{M}_w(\lambda)$, $\lambda \in \text{dom}$; its truth set is dom , which lies in every filter on dom . So the instance is c -sound for every context c .
- *Root axioms* lie in Ax_ρ , which is true at $\mathfrak{M}_w(\lambda_w^*)$ by the definition of a reading, hence in the principal root filter.
- *SUP axioms:* $\|A\| \in \mathcal{F}_{\text{SUP}(A)}(w)$ by definition.
- *DEF axioms* are canonical (Ax legality). Under limit semantics $\{\lambda^\circ\} \in \mathcal{G}$, and the generator $\delta(F \times \{\lambda^\circ\}) \cap \text{dom}$ lies in $\|p_D = \lambda^\circ\|$. Under germ semantics, for definable $U \in \mathcal{G}$ the generator $\delta(F \times U) \cap \text{dom}$ lies in $\|p_D \in U\|$. This check is what blocks Theorem 11.27(b).
- *Imports into DEF contexts* are catalogue laws (valid on dom by (TB1)) or undeformed data, hence D -stable at w , hence sound by Lemma 11.7(a). Imports into SUP children are sound because $\mathcal{F}_{\text{SUP}(A)}(w) \supseteq \mathcal{F}_p(w)$. AUX bridges are kernel-checked theorems, sound by (TB1).
- *Discharges* are sound by Proposition 11.4.
- *Exports.* Schema indexing ensures that the cited β is, by (TB2), a theorem for the context's actual declaration. In the pointwise form, the core of the proof of Theorem 11.27(c) shows that the instance is sound at w ; in the asymptotic form, cited only below a principal parent, Lemma J.4 does. Validated enclosures used in side conditions are correct by (TB3). The premises σ_β and ω_c are derived in the parent, so in the induction of Theorem 11.16 they are true there at w .

Nothing in this argument depends on how the derivation was produced, so it holds for every solver. For a germ root d with trusted declaration, the same argument yields $d \models_w \chi$. If $W_{\rho^*} \subseteq W_\rho$, the conclusion holds at every $w \in W_{\rho^*}$, i.e. it is supertrue for the intended problem; if the actual system is a world in W_{ρ^*} , the conclusion is true of it. $\square$

Proof of Theorem 11.39. (a) Supertruth of $Q \in [q \pm \delta]$ means $|Q^w - q| \leq \delta$ for every $w \in W_\rho$, i.e. $\sup_w |Q^w - q| \leq \delta$. If some q satisfies this, all Q^w lie in an interval of length 2δ , so $\text{osc } Q \leq 2\delta$. Conversely, if $\text{osc } Q \leq 2\delta$ (and $W_\rho \neq \emptyset$), the midrange $q = \frac{1}{2}(\sup Q^w + \inf Q^w)$ has $|Q^w - q| \leq \frac{1}{2} \text{osc } Q \leq \delta$. (b) is Theorem 11.37 applied to root derivations. (c) By Theorem 11.38(a), a checker sound for every $\rho^* \in \mathcal{R}$ accepts only claims holding on $\bigcup_{\rho \in \mathcal{R}} W_\rho$; and accepting exactly the claims supertrue on that union is sound for every $\rho^* \in \mathcal{R}$, since $W_{\rho^*} \subseteq \bigcup_{\rho \in \mathcal{R}} W_\rho$. $\square$

J.8 Proofs for §11.7

Proof of Proposition 11.41. (a) The incoming direction is $u = (\sin \alpha e(0), -\cos \alpha)$. Reflection in the vertical surface with outward normal $n = (e(\vartheta), 0)$ keeps the vertical component and maps the horizontal one to $\sin \alpha (e(0) - 2 \cos \vartheta e(\vartheta)) = \sin \alpha e(2\vartheta - \pi)$; the surface is lit where $u \cdot n < 0$, i.e. $\cos \vartheta < 0$. Descending a height h takes horizontal travel $h \tan \alpha = s$, so $P = a e(\vartheta) + s e(\psi)$. For the second form, $e(\vartheta) \cdot e(\psi) = \cos(\pi - \vartheta) = -\cos \vartheta = \sin \frac{\psi}{2}$ and $e(\vartheta) \cdot e(\psi + \frac{\pi}{2}) = \sin \vartheta = \cos \frac{\psi}{2}$.

(b) A direct computation (checked in SymPy) gives $\det \partial P / \partial(\vartheta, s) = a \cos \vartheta - 2s$, which is negative on the lit side, so $|\det| = 2s - a \cos \vartheta = 2s + ac$. For injectivity, note first that along one

reflected ray the coefficient $s + ac$ of $e(\psi)$ is strictly increasing in s , and distinct ϑ give distinct ψ . So it suffices that two distinct reflected rays never meet. By (a) the ray with parameter ψ is $\{te(\psi) + a \cos \frac{\psi}{2} e(\psi + \frac{\pi}{2}) : t \geq a \sin \frac{\psi}{2}\}$. Take $\psi_1 = 2x < \psi_2 = 2y$ with $x, y \in (0, \pi)$ and put $d = y - x \in (0, \pi)$. If $\sin 2d \neq 0$, solving the 2×2 linear system for the intersection of the two lines gives

$$t_1 = \frac{a(\cos x \cos 2d - \cos y)}{\sin 2d}, \quad t_2 = \frac{a(\cos x - \cos y \cos 2d)}{\sin 2d}$$

(checked in SymPy), and the rays meet only if $t_1 \geq a \sin x$ and $t_2 \geq a \sin y$. *Case $\sin 2d > 0$, i.e. $d < \pi/2$.* $t_2 \geq a \sin y$ is $\cos x \geq \cos(y - 2d) = \cos(d - x)$; as $x, |d - x| \in [0, \pi)$ and $\cos$ decreases there, $x \leq |d - x|$, which forces $d \geq 2x$. $t_1 \geq a \sin x$ is $\cos(x + 2d) \geq \cos(x + d)$; with $u = x + d \in (0, \pi)$ and $u < x + 2d < 2\pi$, this forces $x + 2d > \pi$ and $2\pi - (x + 2d) \leq u$, i.e. $2x + 3d \geq 2\pi$. Then $4d \geq 2x + 3d \geq 2\pi$, contradicting $d < \pi/2$. *Case $\sin 2d < 0$, i.e. $d > \pi/2$.* The inequalities reverse: $\cos x \leq \cos(d - x)$ gives $x \geq |d - x|$, hence $d \leq 2x$; and $\cos(x + 2d) \leq \cos(x + d)$, where $u = x + d \in (\pi/2, \pi)$ and $x + 2d \in (\pi, 2\pi)$, gives $2\pi - (x + 2d) \geq u$, i.e. $2x + 3d \leq 2\pi$. Then $4d \leq 2x + 3d \leq 2\pi$, contradicting $d > \pi/2$. *Case $d = \pi/2$.* Then $e(2y) = -e(2x)$, the lines are parallel, and they coincide only if $a \cos x = -a \cos y = a \sin x$, i.e. $x = \pi/4$; then $t_1 e(2x)$ and $-t_2 e(2x)$ with $t_1, t_2 > 0$ point apart, so the rays do not meet. (A numerical search over 2×10^6 random pairs also found no forward intersection.)

(c) The irradiance on a surface element is $I \sin \alpha |\cos \vartheta|$ and the element has area $a d\vartheta dh$ with $dh = ds / \tan \alpha$. So the reflected flux through $d\vartheta ds$ is $I \cos \alpha a |\cos \vartheta| d\vartheta ds$; dividing by the floor area $(2s + ac) d\vartheta ds$ and using $|\cos \vartheta| = c$, $I \cos \alpha = I_0$ gives $E = I_0 ac / (2s + ac)$. Floor points outside the image receive no reflected light, and points whose preimage has $s > L \tan \alpha$ would need reflection above the lit height. The total flux is $\int_0^L \int_{\pi/2}^{3\pi/2} I \sin \alpha |\cos \vartheta| a d\vartheta dh = 2aLI \sin \alpha$, the flux intercepted by the lit side.

(d) Neither $P(\vartheta, s)$ nor E as a function of (ϑ, s, I_0) contains α ; only the constraint $s \leq L \tan \alpha$ does. $\square$

Proof of Proposition 11.42. Since $\epsilon < 2/\sqrt{5} < 1$, $r > a$.

Step 0: the point is in the image and lit. On the ray with parameter ψ , the point at distance r from the axis is $ue(\psi) + a \cos \frac{\psi}{2} e(\psi + \frac{\pi}{2})$ with $u = \sqrt{r^2 - a^2 \cos^2 \frac{\psi}{2}} > 0$, and it lies on the forward ray ($s = u - ac \geq 0$) because $u^2 - a^2 c^2 = r^2 - a^2 \geq 0$. Its polar angle is $\varphi(\psi) = \psi + \delta(\psi)$ with $\tan \delta = a \cos \frac{\psi}{2} / u$, continuous in ψ , tending to $\arcsin \epsilon$ as $\psi \rightarrow 0^+$ and to $2\pi - \arcsin \epsilon$ as $\psi \rightarrow 2\pi^-$. By the intermediate value theorem every $\varphi \in (\arcsin \epsilon, 2\pi - \arcsin \epsilon)$ is attained. And $\sigma > m$ forces φ into this interval: for $\varphi \in [0, \arcsin \epsilon]$, $\sigma = \sin \frac{\varphi}{2} \leq \frac{\varphi}{2} \leq m$, and symmetrically near 2π . So the point has a preimage, unique by Proposition 11.41(b), with $s \leq u \leq r \leq L \tan \alpha$, so it is lit.

Step 1. With $u = s + ac \geq 0$, Proposition 11.41(a) gives $r^2 = u^2 + a^2 \cos^2 \frac{\psi}{2}$, so $u \in [r\sqrt{1 - \epsilon^2}, r]$.

Step 2. $\sin \delta = a \cos \frac{\psi}{2} / r$, so $|\sin \delta| \leq \epsilon$ and, as $\delta \in (-\pi/2, \pi/2)$, $|\delta| \leq \arcsin \epsilon$. Since $c = \sin \frac{\psi}{2} \geq 0$ and $\sin$ is 1-Lipschitz, $|\sigma - c| \leq |\sin \frac{\varphi}{2} - \sin \frac{\psi}{2}| \leq |\delta|/2 \leq m$.

Step 3. $E/E_0 = \frac{ac}{2s+ac} \cdot \frac{2r}{a\sigma} = \frac{c}{\sigma} \cdot \frac{2r}{2u-ac}$. The second factor is at least 1 because $2u - ac \leq 2r$, and at most $1/(\sqrt{1 - \epsilon^2} - \epsilon/2)$ because $2u - ac \geq 2r\sqrt{1 - \epsilon^2} - a$; the denominator is positive exactly when $\epsilon < 2/\sqrt{5}$.

Step 4. By Step 2, $c/\sigma \in [1 - m/\sigma, 1 + m/\sigma]$, and $1 - m/\sigma > 0$. Multiplying the two ranges of nonnegative factors gives the claim. $\square$

Proof of Proposition 11.43. Domain. Each direction d in the solar disc has zenith angle $\alpha_d \in [\alpha - \delta_s, \alpha + \delta_s] \subseteq [\alpha_{\min} - \delta_s, \pi/2]$, so $\cos \alpha_d \geq 0$ and $\tan$ is increasing there. The preimage of a floor point at distance r has $s \leq u \leq r$ (Step 0 above), and $r \leq L \tan(\alpha_{\min} - \delta_s) \leq L \tan \alpha_d$, so the point is lit by every direction, for every admissible α .

Convex combination. By Proposition 11.41(d) and rotational symmetry, direction d with azimuth offset β_d contributes $G(r, \varphi - \beta_d) \cos \alpha_d dI(d)$, where $G = E/I_0$ is the α -free geometric factor. The direct illuminance is $I_0 = \int \cos \alpha_d dI(d)$, so $E_{\odot}/I_0$ is a convex combination of the values $G(r, \varphi - \beta)$ over the azimuth spread, for any radiance profile over the disc.

Azimuth spread. In the right spherical triangle formed by the zenith, the sun’s centre and a tangent point of the disc, Napier’s rule gives $\sin \beta_{\max} = \sin \delta_s / \sin \alpha$. Since $\alpha \in [\alpha_{\min}, \pi/2]$ and $\delta_s < \alpha_{\min}$, this is at most $\sin \delta_s / \sin \alpha_{\min}$, so $|\beta| \leq \Delta$.

Combination. Let $\sigma' = |\sin \frac{\varphi - \beta}{2}|$. As $\sin$ is 1-Lipschitz, $|\sigma' - \sigma| \leq |\beta|/2 \leq \Delta/2$, so $\sigma' \geq \sigma - \Delta/2 > m$ and Proposition 11.42 applies at $\varphi - \beta$: $G(r, \varphi - \beta) \in [\text{lo}(\epsilon, \sigma'), \text{hi}(\epsilon, \sigma')] \cdot E_0(\varphi - \beta)/I_0$. Since lo increases and hi decreases in σ , these factors lie in $[\text{lo}(\epsilon, \sigma - \frac{\Delta}{2}), \text{hi}(\epsilon, \sigma - \frac{\Delta}{2})]$, and $E_0(\varphi - \beta)/E_0(\varphi) = \sigma'/\sigma \in [1 - \frac{\Delta}{2\sigma}, 1 + \frac{\Delta}{2\sigma}]$. All factors are nonnegative ($\sigma > \Delta/2 + m$), so multiplying and averaging over the convex combination gives the claim. $\square$

K Lean formalization

A subset of the discrete results is formalized in Lean 4 with Mathlib (tag v4.32.0). The formalization is in the directory `lean/`, in the library `InfLearn`. It has 17 modules and about 14,600 lines.

- There is no `sorry`, no axiom declaration and no `native_decide`.
- `lake build` reports no errors and no warnings.
- `lake build Audit` runs `#print axioms` on 483 main declarations. Each depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`, or on a subset of them. The output is in `lean/audit-output.txt`.

Theorems carrying the tag ^{Lean}:... in the text are formalized in the sense of Table 20. The column “fidelity” takes one of four values:

- *exact*: the paper statement, possibly in a more general form;
- *special case*: a strict special case, as described in `lean/README.md`;
- *weaker*: a weaker conclusion than the paper states;
- *variant*: a closely related statement, neither stronger nor weaker.

Modelling conventions. The following conventions apply throughout.

- **Language.** The object language is propositional, with atoms $p_0, p_1, \dots$ and the connectives $\perp, \top, \neg, \wedge, \vee, \rightarrow$. Results stated for arbitrary Boolean signatures are proved for every fragment of this set. As a consequence, the truth-table tell-tale has $13 + 1$ data rather than $11 + 1$, because $\perp$ and $\top$ are constants of the language. The bilateral completeness theorems are proved for sequents over an arbitrary type of formulas.
- **Consequence operators** are arbitrary closure operators. Finitarity is not assumed.

- **Learners, verifiers and certifiers** are arbitrary, non-computable functions. Interactive lower bounds are proved against deterministic agents only.
- **Probability** appears in two forms only. One is non-negative summable weights on a countable step set. The other is finite distributions, on the 2^k worlds or on a finite class of labellers.
- **Real analysis.** Smooth bumps are replaced by Lipschitz tents, because the compiled Mathlib subset does not include derivatives.

Not formalized. The following are outside the formalization:

- all probabilistic and anytime results, including Ville’s inequality and the Bayes-conservative verifiers (Section 4.4);
- anti-unification and the schema-specific escalation bounds;
- the noise results of Section 5;
- arithmetic (Section 6.3);
- the two-tier main theorem (Theorem 7.14);
- the kink theorem and the other Kolmogorov-complexity results of Section 8;
- first-order and real-parameter context semantics, with the exception of the export results listed below;
- the Structured Physics Solution checker theorem;
- the identification results of Section 10.

These rest on the paper proofs and on the adversarial verification described in Section 1.

Table 20: Map from paper results to Lean declarations. The namespace prefix `InfLearn.` is omitted.

Paper	Lean declaration(s)	Module	Fidelity
Lemma 2.3	<code>Cl_eq_sInter</code> , <code>ClOp</code> , <code>Cl_subset_Cl_iff_subset_Sound</code>	<code>Steps</code>	exact
Lemma 3.1	<code>StepSoundness.lemma_1_1</code>	<code>StepSoundness</code>	exact
Theorem 3.2	<code>StepSoundness.thm_2_1_i</code> , <code>thm_2_1_ii</code> , <code>thm_2_1_error_and_trivial</code>	<code>StepSoundness</code>	exact
Corollary 3.3	<code>StepSoundness.cor_2_2_trivializes</code>	<code>StepSoundness</code>	weaker
Theorem 3.12	<code>CoherenceGames.thm_2_4</code> , <code>thm_2_4_environment</code>	<code>CoherenceGames</code>	exact
Theorem 4.2 (a)	<code>StepSoundness.thm_3_1_a</code>	<code>StepSoundness</code>	exact
Theorem 4.2 (b)	<code>StepSoundness.thm_3_1_b</code> , <code>vsVerifier_optimal</code>	<code>StepSoundness</code>	special case
Theorem 4.4	<code>Unstructured.thm_3_2</code> , <code>vsVerifier_attains</code>	<code>Unstructured</code>	special case
Theorem 4.12	<code>Unstructured.thm_3_9_Esc_le_card</code> , <code>thm_3_9_coSingleton</code>	<code>Unstructured</code>	special case
Lemma 6.2	<code>CoherenceGames.negative_bag</code>	<code>CoherenceGames</code>	exact

Paper	Lean declaration(s)	Module	Fidelity
Theorem 6.5	CoherenceGames.thm_2_2, thm_2_2_iii_iv	CoherenceGames	special case (finite $\mathcal{H}$)
Proposition 6.6	CoherenceGames.prop_2_3	CoherenceGames	special case (finite)
Theorem 6.7	CoherenceGames.thm_2_5	CoherenceGames	special case (finite)
Theorem 6.13	Post.eq_Cn2_or_eq_trivial, structural_extensions_Cn2, frag_eq_Cn2F_or_eq_trivial, impFrag_postComplete	PostCompleteness	exact
Theorem 6.15	Post.eq_Cn2_iff_exists_coherent, learner_identifies, learner_mindChanges_le, versionSpace_eq_singleton_iff	PostCompleteness	exact except finite-basis clause
Lemma 6.26	Carnap.Val_mrel, lemma_4_1_b, intClosure_BV_eq	Carnap	exact
Theorem 6.27	Carnap.carnap_problem, carnap_single, carnap_not_learnable_single	Carnap	exact
Theorem 6.28	Carnap.denialRank_eq_zero_iff (and _one_iff, _two_iff, _top_iff)	Carnap	exact
Theorem 6.29	Carnap.structural_eq_BV, ttLearner_finite_identification, no_finite_telltale, BV_not_BC_learnable	Carnap	exact (13+1 data)
Lemma 7.8	Blame.isMaxClean_iff, iInter_maxClean_eq, existsUnique_minTransversal_iff	Blame	exact
Lemma 7.24	Post.schema_invalid_iff_closedRefutation, card_candidates_le	PostCompleteness	exact
Theorem 8.3 (product part)	NoAdaptation.thm_2_1_i_ennreal_maxF, unifHyp_loss, uniform_marginals_of_transitive	NoAdaptation	exact for the lower bound; the min
Lemma 8.8	RateThreshold.onLowerHull_iff, lemma_3_1_ii, lemma_3_1_iii_hyp	RateThreshold	special case (finite)
Theorem 8.9	RateThreshold.prod_isMin_iff, rate_threshold, thm_3_2	RateThreshold	exact
Corollary 8.10	RateThreshold.selectable_iff	RateThreshold	exact
Theorem 8.15	RateThreshold.rate_blindness	RateThreshold	special case (separable)
Proposition 8.16	RateThreshold.valid_not_minimizer, pareto_dominated	RateThreshold	exact
Theorem 9.8	Bilateral.thm_2_2_a, thm_2_2_b, coherent_iff_realizable	Bilateral	exact
Corollary 9.9	Bilateral.cor_2_3	Bilateral	exact
Theorem 9.10	Bilateral.scottClosedIso, isScott_iff_Th_Val_eq	Bilateral	exact (no filters)
Theorem 9.11	Bilateral.thm_2_6_a, structuralClosedIso, thm_2_6_b_realize	Bilateral	exact (fixed signature)
Theorem 9.14	Specker.thm_3_4, specker_parable	Specker	exact
Proposition 9.27 (probabilistic part)	Specker.frustrated_triangle	Specker	exact
Lemma 10.20	Blame.LineArg.descent_depth, ArgTree.descent_tree_not_mem	Blame	exact

Paper	Lean declaration(s)	Module	Fidelity
Theorem 10.22 (a),(b)	CoherenceGames.thm_4_3_a, ParadoxLowerBound.thm_4_3_b	CoherenceGames, Para- doxLower- Bound	exact
Theorem 10.23 (b),(c)	ParadoxLowerBound.thm_4_4_b, thm_4_4_c	ParadoxLowerBound	exact (deterministic learners; one c
Theorem 10.24	CoherenceGames.thm_4_5	CoherenceGames	exact
Proposition 11.3 (a),(b)	Contexts.cn2_closed_ThIn, consistent_ThIn_iff	Contexts	special case (propositional)
Proposition 11.4	Contexts.holdsIn_sup_iff	Contexts	special case (propositional)
Lemma 11.7	Contexts.import_sound	Contexts	special case (propositional)
Theorem 11.8	Contexts.not_exists_truthFunctional_reading	Contexts	special case
Proposition 11.10	Contexts.misdesignating_not_Cn2_le, atomic_paraconsistent	Contexts	exact
Theorem 11.18	Contexts.refutes_of_inconsistent, derivation_local_test_accepts_both, certificate_not_both	Contexts	exact for (b)–(d)
Lemma 11.20, Theorem 11.21 (b)–(d)	Export.invisible_perturbation, no_free_export_samples, no_free_export_finite_jet, no_free_export_germ	Export	exact for (d); variant for (b); speci
Proposition 11.22	Export.exists_interval_iff_radius_le	Export	exact
Theorem 11.28	Export.coherence_cannot_calibrate	Export	exact
Theorem 11.30	Export.no_certification_without_regularities	Export	weaker (deterministic)
Theorem 11.31	Export.lip_cert_sound, lip_cert_complete, lip_lower_bound_cube	Export	exact
Theorem 11.32	Export.lazy_bisection_sound, lazy_bisection_miss, mono_lower_bound	Export	exact

Signal	What it provably fixes	What it provably cannot fix (residue)
Imitation (P)	The <i>rules</i> actually followed. Exact identification of a schematic calculus at coupon-collector rates (Theorem 5.7), with systematic out-of-distribution generalization (Corollary 5.11).	Systematic errors, which are indistinguishable from rules at every rate (Corollary 5.20). Imitation converges to “competence” in the sense of Cohen (1981b), not to validity (Stich and Nisbett, 1980).
Cautious acceptance	Soundness against every adaptive arguer (Theorems 4.2 and 4.15).	Nothing new. It receives no coherence signal at all (Proposition 6.3), and its price is set by the structure of the class (Table 4).
Coherence (C)	Every error that is incoherent with the target on a designated context (Theorem 6.34). Exactly the target, for Post-complete targets: classical logic (Theorem 6.15), and complete theories at the level of theorems. The classical <i>meanings</i> of the connectives, once denials are allowed (Theorem 6.29).	Coherent uniform alternatives: the Kripkensteinian residue (Theorem 6.38). Intermediate logics (Theorem 6.17). Σ_1 -unsound extensions of arithmetic (Theorem 6.22). Which valuation is actual.
Simplicity	Expensive, idiosyncratic errors, via a rate threshold (Section 8).	Cheap, frequent fallacies. Rare valid rules are lost. Its strength cannot be calibrated from imitation data.
World (W)	Objects give step-level blame by descent, so $\log_2 \mathcal{H} $ corrections suffice, against $ \mathcal{H} - 1$ from paradoxes alone (Theorems 10.22 and 10.23). Computation with fresh randomness gives worst-case-valid stochastic checks (Lemma 3.11). Calibration of top models (Section 11).	Errors invisible on every presentable object. Effects outside the top model (Theorem 11.38).
Contexts	Reasoning under contradictory idealizations without explosion. Coherence where it belongs (Theorem 11.12).	The reading of a problem, and the adequacy of the top model. Both are ineliminable judgments (Theorem 11.38).

Table 18: The division of labour among the learning signals, with the residue each one provably leaves.

Question	Where and how it is answered
What is an appropriate learning algorithm for inference rules (learning from positive examples)?	Least generalization over schemas, with a trimmed version space for sporadic noise and cautious acceptance (Sections 4 and 5). MDL is the wrong bias for positive data (Proposition 3.9).
Does that plus coherence get formal math working?	Yes for classical propositional logic and complete decidable theories (Sections 6 and 7). Arithmetic is Popperian. There are sharp limits at Σ_2 and for Turing progressions.
Would a setup have worked before we knew how to formalize proofs?	A bounded-gap verifier over latent formalizations is sound against adaptive provers and identifies validity exactly up to informal-validity equivalence. Counterexample objects are exponentially more informative than paradoxes, and Frege $\rightarrow$ Russell $\rightarrow$ Zermelo comes out as constrained repair (Section 10). The hardest open problem is affordable soundness when the learner must invent language.
How should contexts be handled, and what about contradictory idealizations (“air pressure = 0”)?	Contexts are filters of models, and coherence is enforced only on the root and anchored contexts (Theorem 11.12). Designating “idealization + full background” as coherent forces a globally paraconsistent logic (Proposition 11.10). Reductio needs a consistency certificate (Theorem 11.18).
“Check that a claim is true in the context at hand. But wtf is that?” Can physics olympiad solutions be checked in a principled way?	Supertruth: true in every admissible completion of the intended model, to the demanded precision (Theorem 11.39). A structured-solution checker that is sound against adversarial solvers, relative to an explicit judgment layer that is provably ineliminable (Theorems 11.37 and 11.38). There is a worked EuPhO example (Section 11.7).
Does a steeper simplicity penalty recover intended rules rather than teacher errors?	Partly. The kink intuition is vindicated for product models. But selection is a rate threshold with provable pathologies, and the threshold is not identifiable from imitation data (Section 8).
Is there a “philosophical completeness theorem”?	Yes in the weak sense, and no in the intended sense (Section 9). Local coherence constraints of the kind used in consistency-based probing do not characterize coherence.
How come mathematicians ended up with such a nice notion of proof?	Through local checkability on objects, plus completeness, plus a bounded repertoire. Monster pressure leaves exactly the steps valid in every model of the theory, and Gödel completeness turns those into derivations (Proposition 10.38). Practice used only the robust core of its vague concepts (Theorem 10.35).

Table 19: The motivating questions and where they are addressed.